

The Cottages at Arcade Springs — Application for Rezoning R-1 to R-2

4535, 4537, 4539 and 4541 Arcade Road SW, Lilburn, Georgia 30047 — Land Lot 123, 6th District, Gwinnett County — PINs R6123 033, 015, 014 and 162 — 9.44 acres

Applicant and owner of record (PINs R6123 033 and 015): Mohammed Awad, 4541 Arcade Road SW, Lilburn, Georgia 30047.

Filed under: City of Lilburn Zoning Ordinance No. 2023-603, Article 10, section 1003, and the City of Lilburn 2026 Application Instructions — Rezoning / Special Use Permit / Change in Condition, items (1) through (11).

Request: rezone 9.44 acres from R-1 (Single-Family Residential) to R-2 (Medium-Density Residential) for 43 one-story detached cottage homes on fee-simple lots, restricted to housing for older persons under 42 U.S.C. section 3607(b)(2)(C) by voluntary condition.

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11	Site Context Brief	supporting
10	Submittal Checklist and Roadmap	supporting

> **Disclaimer:** This is an AI-generated analysis for preliminary planning purposes. All findings must be verified by a licensed professional before use in design, permitting, or regulatory submissions.

title	Transmittal and Index — Application for Rezoning, The Cottages at Arcade Springs
date	2026-09-03
to	Reid Turner, Planning Director, City of Lilburn, 340 Main Street, Lilburn, Georgia 30047
status	DRAFT — owner-prepared. Items marked NOT INCLUDED or DRAFT require a licensed professional before filing.
generator	tools/transmittal.py (index built by scanning drawings/ and docs/)

Transmittal and Index

To: Reid Turner, Planning Director — Department of Planning and Zoning, City of Lilburn, 340 Main Street, Lilburn, Georgia 30047 · returner@cityoflilburn.com · (770) 279-3715

From: Mohammed Awad, applicant and owner of record of PINs R6123 033 and R6123 015 — 4541 Arcade Road SW, Lilburn, Georgia 30047

Date: _____ **Zoning case no.:** RZ-2026-_____ (assigned by the City)

Re: Application to rezone 9.44 acres from R-1 to R-2 — The Cottages at Arcade Springs 4535, 4537, 4539 and 4541 Arcade Road SW · Land Lot 123, 6th District, Gwinnett County · PINs R6123 033, R6123 015, R6123 014 and R6123 162

1. The request

The applicant asks the City to rezone the 9.44-acre assemblage from **R-1, Single-Family Residential** to **R-2, Medium-Density Residential** (Lilburn Zoning Ordinance No. 2023-603, section 402) for **41 one-story detached cottage homes on fee-simple lots**, served by one private lane from Arcade Road, restricted to housing for older persons under 42 U.S.C. section 3607(b)(2)(C) by voluntary condition. The proposed use, "Single-family (cluster-cottage, creative lot configuration)", is permitted in R-2 by the section 602 Use Table; no special use permit is requested. The proposed density is **4.34 dwelling units per acre** against the 8 units per acre that Table 4.1 allows in R-2.

2. What is enclosed — the eleven required items

Item	Requirement (2026 Application Instructions)	Provided	Notes
1	Application Form	docs/01-application-form-data.md	Every field of the City form with its entry and its basis. The form itself is transcribed and signed by the applicant.
2	Application Fee	docs/01-application-form-data.md	Rezoning fee for a 5.0-9.9 acre property, per the fee schedule in force on the filing date. Check payable to City of Lilburn. CONFIRM THE CURRENT AMOUNT WITH THE PLANNING DEPARTMENT ON THE FILING DATE — the FY2026-2027 schedule (Resolution No. 2026-08, adopted 2026-08-31) supersedes the FY25-26 schedule still posted online.
3	Standards Governing the Exercise of the Zoning Power	docs/04-standards-governing-zoning-power.md	The six criteria of Ordinance 2023-603 section 1003-7, answered in full; "see attached" on the form.
4	Conflict of Interest Form	docs/07-conflict-of-interest-and-certifications.md	Campaign-contribution and gift disclosure under O.C.G.A. section 36-67A-3, filed WITH the application so the ten-day clock is never at issue. Only the signer can declare the contribution history.
5	Notarized Signatures	docs/07-conflict-of-interest-and-certifications.md	Applicant certification plus an owner certification for each owner of record, before a Georgia notary. Two ownership groups: PINs 033 and 015 (Awad) and PINs 014 and 162 (Mendez / Roblero de Leon).
6	Letter of Intent	docs/03-letter-of-intent.md	Describes the request and its justification, states whether any buffer reduction is requested, and offers the voluntary zoning conditions. The conditions are listed separately at data/voluntary-conditions.md and reproduced in the letter.
7	Legal Description	docs/05-legal-description-DRAFT.md	Typed metes and bounds of the property to be rezoned. DRAFT — derived from the Gwinnett County GIS parcel fabric. The sealed description from a Georgia registered land surveyor replaces it and governs.
8	Site Plan	(see the drawing set below)	One full-size copy drawn to scale, plus the PDF e-mailed to returner@cityoflilburn.com. The C-series sheets listed below answer this item; sheet C-2.0 is the site plan proper and the others support it.

Item	Requirement (2026 Application Instructions)	Provided	Notes
9	Boundary Survey	NOT INCLUDED	NOT INCLUDED — must be commissioned. A survey sealed by a Georgia registered land surveyor. Section 1003-4.4 permits the Director to accept a Tax Assessor map where no survey is available but does not oblige him to; the 236-246 ft site width that the entire 50 x 100-ft lot section depends on is unverified without it. This is the critical-path item.
10	List of Adjoining Property Owners	docs/06-adjoining-property-owners.md	All adjoining owners with mailing addresses, from the Gwinnett County Tax Assessor ownership file. Re-pull on the filing date.
11	Architectural Renderings and Elevations	docs/09-architectural-character-elevations-and-renderings.md	All four elevations of each dwelling type, the clubhouse, the mail kiosk and the monument entry sign, to scale, with the colour and material schedule; plus illustrative perspectives. No wall signs are proposed on any structure.

3. Drawing set

Sheet	Title	Size	Scale	Status
C-0.0	Cover Sheet, Sheet Index and Vicinity Map	ARCH D 36 x 24	As noted	Issued
C-1.0	Existing Conditions, Boundary and Topographic Survey	ARCH D 36 x 24	1" = 60'	Issued
C-1.1	Demolition and Tree Protection Plan	ARCH D 36 x 24	1" = 60'	NOT YET ISSUED
C-2.0	Master Concept Plan - Overall	ARCH D 36 x 24	1" = 60'	Issued
C-2.1	Master Concept Plan - Enlargement, Block A	ARCH D 36 x 24	1" = 30'	Issued
C-2.2	Master Concept Plan - Enlargement, Block B	ARCH D 36 x 24	1" = 30'	Issued
C-2.3	Entry, Frontage and Amenity Enlargement	ARCH C 24 x 18	1" = 20'	Issued
C-2.4	Fallback Lot-Depth Exhibit - 50 x 82-ft Lots	ARCH D 36 x 24	1" = 60'	Issued
C-3.0	Grading, Drainage and Stormwater Concept	ARCH D 36 x 24	1" = 60'	Issued
C-3.1	Stormwater Details and Sections	ARCH C 24 x 18	As noted	NOT YET ISSUED
C-4.0	Utility and Phasing Concept	ARCH D 36 x 24	1" = 60'	Issued
C-4.1	Sanitary Sewer Profiles	ARCH D 36 x 24	1" = 50' H / 5' V	NOT YET ISSUED
C-5.0	Private Lane Plan and Profile	ARCH D 36 x 24	1" = 50' H / 5' V	NOT YET ISSUED
C-6.0	Phasing, Block and Open Space Exhibit	ARCH D 36 x 24	1" = 60'	Issued
C-7.0	Landscape, Buffer and Tree Protection Concept	ARCH D 36 x 24	1" = 60'	Issued
C-8.0	Civil Details	ARCH D 36 x 24	As noted	Issued
A-1.1	Plan A "The Springbrook" - Floor Plan	ARCH C 24 x 18	3/16" = 1'-0"	Issued
A-1.2	Plan B "The Laurel" - Floor Plan	ARCH C 24 x 18	3/16" = 1'-0"	Issued
A-2.1	Plan A - Exterior Elevations (four sides)	ARCH C 24 x 18	3/16" = 1'-0"	Issued
A-2.2	Plan B - Exterior Elevations (four sides)	ARCH C 24 x 18	3/16" = 1'-0"	Issued
A-2.3	Exterior Colour Schemes and Materials	ARCH C 24 x 18	3/16" = 1'-0"	Issued
A-3.0	Clubhouse, Mail Kiosk and Entry Monument Sign	ARCH C 24 x 18	As noted	Issued
A-4.0	Typical Lot - Plan A	ARCH C 18 x 24 (portrait)	1" = 10'	Issued
A-4.1	Typical Lot - Plan B	ARCH C 18 x 24 (portrait)	1" = 10'	Issued

4. Supporting material (not required by the Instructions; provided so staff and the Commission have the

basis for every number)

Document	File	What it is
Development Summary	docs/02-development-summary.md	Site data in the order staff read it, with the Table 4.1 required-versus-provided matrix.
Technical Memoranda	docs/08-technical-memoranda.md	Traffic, sanitary sewer, water, stormwater, fire, schools and environmental screening.

Document	File	What it is
HOA, HOPA and Covenant Outline	docs/13-hoa-hopa-and-covenant-outline.md	How the 55+ restriction and the buffer easement are actually enforced, with a line-item association budget.
Outline Specifications	docs/12-outline-specifications.md	CSI-format outline specifications establishing the materials and colours shown on the elevations.
Site Context Brief	docs/11-site-context-brief.md	Environmental, mobility, demographic, market and precedent context for the site.
Submittal Checklist and Roadmap	docs/10-submittal-checklist-and-roadmap.md	Item-by-item status, the filing calendar, the legal-notice audit and the pre-application agenda.

5. Illustrative perspectives

- R-01_plan-a-front-scheme1.jpg
- R-02_plan-b-front-scheme2.jpg
- R-03_plan-b-rear-garden-scheme3.jpg
- R-04_lane-streetscape-nw.jpg
- R-05_hammerhead-pocket-green.jpg
- R-06_entrance-arcado-monument.jpg
- R-07_clubhouse-village-green.jpg
- R-08_pickleball-courts.jpg
- R-09_aerial-oblique-nw.jpg
- R-10_buffer-from-neighbour.jpg
- R-11_creek-woods-dry-basin.jpg
- R-12_materials-colour-board.jpg

Each is an illustrative perspective prepared by the owner-applicant from the scaled elevations and the master concept plan. Where an image and a drawing differ, the drawing governs. Item (11) permits imagery: "The drawings shall be to scale or in proper perspective ... Visual imagery may be used."

6. Copies and delivery

What	How delivered
One full-size copy of the site plan drawn to scale	Delivered to the counter at 340 Main Street, printed at ARCH D 36 x 24 in at 1" = 60'
PDF of the site plan	E-mailed to returmer@cityoflilburn.com on the filing date
Complete application in Instruction-item order	One bound copy at the counter, and submission/RZ-2026_cottages-at-arcado-springs.pdf by e-mail
Application fee	Check payable to "City of Lilburn", amount confirmed with the Department on the filing date
Conflict-of-interest disclosure	Filed herewith, so the ten-day period of O.C.G.A. section 36-67A-3(b) is not relied upon

7. Requests to the Department

1. **Completeness review.** The applicant asks the Director to confirm in writing that the application is complete under section 1003-3, or to identify what is missing, before the legal advertisement is placed.
2. **Confirmation on RZ-2025-01.** The applicant asks the Department to confirm in writing that RZ-2025-01 was **tabled** on 2025-08-28 because of a defect in the legal advertisement, was never heard and was **not denied**, so that the twelve-month bar of section 1003-11 does not apply; and to advise whether that application should be formally withdrawn before this one is accepted.
3. **Draft advertisement text.** Because RZ-2025-01 was lost to a notice defect, the applicant asks to review the draft legal advertisement and the sign copy before they are placed, and to be told the dates on which the advertisement runs, the letters are mailed and the sign is posted.
4. **Pre-application conference.** The applicant requests a conference at least one week before filing, on the agenda at docs/10-submittal-checklist-and-roadmap.md section 3.

8. Statements the applicant makes plainly

- **Nothing in this package is sealed.** The boundary survey and legal description require a Georgia registered land surveyor; the site, grading, utility and stormwater work require a Georgia professional engineer; the architectural drawings require a Georgia registered architect; the landscape plan requires a registered landscape architect. Every drawing and document is labelled DRAFT accordingly.
- **The boundary is from the Gwinnett County GIS parcel fabric, not a survey,** and the survey governs.
- Statements of consistency are written as "appears consistent with", never as a compliance certification.

***Disclaimer:** This is an AI-generated analysis for preliminary planning purposes. All findings must be verified by a licensed professional before use in design, permitting, or regulatory submissions.*

title	Rezoning Application Form — data sheet (Item 1)
date	2026-08-28
revised	2026-09-03
address	4535, 4537, 4539 & 4541 Arcado Rd SW, Lilburn, GA 30047 — Land Lot 123, 6th District, Gwinnett County (PINs R6123 033 / 015 / 014 / 162)
status	DRAFT — field-by-field content for the City of Lilburn Rezoning Application (Planning Dept. form, DocumentCenter/View/193); transcribe to the official form in black ink; owner-status items marked VERIFY must be settled before filing. Revised 2026-09-03 to the corrected site plan: the lot count is now 41 , not 43, and every field derived from it has been recomputed from data/layout.json.

Rezoning Application — form data sheet

Form: "Rezoning Application — An application to amend the official zoning map of the City of Lilburn, Georgia" (City of Lilburn Planning Department, 340 Main St., Lilburn, GA 30047, 770-279-3710; data/rezoning-application-form.txt, pp. 1–7).

Instructions: 2026 Application Instructions — Rezoning/SUP/CIC, items 1–11 (data/application-instructions-2026.txt).

The form says "Please type or print using BLACK ink" and "Be specific and fill in all pertinent and required information" (Instructions item 1) — every field below has an entry or an explicit "N/A".

Values in **bold** go on the form; text in brackets is a placeholder the applicant completes; VERIFY marks items not settled by the package data.

Numbers on this sheet come from data/layout.json **as corrected on 2026-09-03** (41 lots; 14 southwest, 27 northeast; Phase 1 = 24, Phase 2 = 17; gross density 4.34 du/ac on 9.44 ac) and data/plans.json. The filed site-plan sheet governs: if the drawing and this sheet ever disagree, the drawing is right and this sheet is wrong.

Page 1 — header

Form field	Entry	Basis / note
CASE NUMBER: RZ-	(leave blank — assigned by the Planning Department; 2026 cases are numbered RZ-2026-##)	Administrative
Date Received	(leave blank — city stamp)	Administrative

Page 1 — Applicant

Form field	Entry	Basis / note
Applicant	Mohammed Awad	Owner of record of R6123 033 and R6123 015 (2026 digest: "AWAD MOHAMMED")
Address	4541 Arcado Rd SW	Digest mailing address for 033/015
City, State & Zip	Lilburn, GA 30047-3968	Digest
Contact Person	Mohammed Awad	The applicant attends both hearings in person (Instructions p. 2 items 4 and 6: "The applicant must be represented")
Business Phone	[applicant to complete]	Not in package data
Email	[applicant to complete — use the address monitored daily; staff sends the PDF site-plan correspondence and the hearing schedule by email]	Not in package data
Cell Phone	[applicant to complete]	Not in package data
APPLICANT IS THE (check one)	■ Property Owner (Parcels R6123 033 and 015) and ■ Contract Purchaser (Parcels R6123 014 and 162, with the owners' written consent) — write "see attached" and explain on a rider	§1003-2: an application may be made by "the owner or owner's agent ... or by a contract purchaser with the owner's written consent". The form has one check-box; the two-status rider avoids a rejection under §1003-3.2. VERIFY: an executed purchase contract for 014/162 must exist at filing; if it does not, Santos C. Mendez and Lesvia Rosario Roblero de Leon sign as co-applicants instead

Page 1 — Property Owner

Form field	Entry	Basis / note
Property Owner	(1) Mohammed Awad — Parcels R6123 033 and R6123 015; (2) Santos C. Mendez and Lesvia Rosario Roblero de Leon — Parcels R6123 014 and R6123 162 (write "multiple owners — see attached" on the form and attach the owner schedule in docs/07 §3)	2026 digest owners of record; Instructions item 5 allows "an attachment, if multiple owners are involved"
Address	(1) 4541 Arcado Rd SW; (2) 4535 Arcado Rd SW	Digest mailing addresses
City, State & Zip	Lilburn, GA 30047-3968 (both)	Digest
Owner Contact	(1) Mohammed Awad; (2) [Santos C. Mendez — VERIFY preferred contact and language for notices]	—
Business Phone / Email / Cell Phone	[owners to complete]	Not in package data

Page 1 — Property

Form field	Entry	Basis / note
PROPERTY ADDRESS	4535, 4537, 4539 and 4541 Arcado Rd SW, Lilburn, GA 30047	Digest property addresses (4537 = R6123 014, 4539 = R6123 015; the GIS layer carries no house number for those two — the digest does)
LAND DISTRICT	6 (pre-printed)	Gwinnett 6th District
LAND LOT(S)	123	FACTS §1; PIN prefix R6123
PARCEL(S)	R6123 033, R6123 015, R6123 014, R6123 162	Gwinnett Tax Assessor PINs
ACRE(S)	9.44 (Tax Assessor deeded acreage: 0.99 + 2.00 + 2.00 + 4.45); note on the rider: "9.58 ac by Gwinnett GIS computation; the sealed boundary survey governs"	FACTS §1 ("Use 9.44 ac (deeded) as the application acreage")
CURRENT ZONING	R-1, Single-Family Residence (City of Lilburn)	Lilburn Zoning Ordinance 2023-603 §401; 2026 digest ZONEDESC
PROPOSED ZONING DISTRICT(S)	R-2, Medium-Density Residential	Lilburn Zoning Ordinance 2023-603 §402

Page 1 — Proposed development

Form field	Entry
PROPOSED DEVELOPMENT	"Cottages at Arcado Springs" — an age-restricted (55+, Housing for Older Persons Act, offered as a voluntary zoning condition) community of 41 one-story detached cottage homes on fee-simple lots (typical 50'-0" × 100'-0" = 5,000 SF), a permitted single-family cluster-cottage / creative-lot-configuration use in R-2 (Lilburn Zoning Ordinance 2023-603 §602 Use Table), served by one private lane from Arcado Road; with a ≈2,400-SF clubhouse and village green, two pickleball courts, mail kiosk, pocket greens and hammerhead turnarounds, two dry detention / water-quality ponds, a preserved stream-buffer "creek woods" tract, a 20-ft perimeter buffer to the adjoining R-1 lots, and 33.8 % common open space (3.24 ac); HOA-maintained yards; gross density 4.34 du/ac on 9.44 ac (R-2 maximum 8 du/ac); two phases. No commercial, attached, multi-family or mixed-use component and no Special Use Permit is requested.

Basis: data/layout.json metrics as corrected 2026-09-03 — **41 lots**; open space **141,089 SF = 3.24 ac = 33.8 % of the 9.58-ac GIS area / 34.3 % of 9.44 ac**; impervious 184,148 SF (44.1 %); density **4.34 du/ac** on 9.44 ac (4.28 on the GIS area) — with FACTS §4 program decisions. If the form's blank is too short, write "see Letter of Intent" and put this paragraph at the top of the letter.

The age restriction the paragraph offers is the one in voluntary condition 1: housing for older persons under 42 U.S.C. §3607(b)(2)(C) and 24 CFR §§100.300–.308, which requires at least 80 % of occupied units to have a resident aged 55 or older. At 41 units, $0.80 \times 41 = 32.8$, and because the fractional unit counts **into** the 80 % under 24 CFR §100.305(g), **at least 33 of the 41 units** must be occupied by at least one person aged 55 or older ($33 \div 41 = 80.5$ %; 32 units would be 78.0 % and would fail). Eight units carry no age requirement. The recorded declaration states that figure and recomputes it if the unit count changes.

Page 1 — Residential / non-residential development

Form field	Entry	Basis / note
RESIDENTIAL — Number of Lots/Dwelling Units	41 lots / 41 dwelling units	data/layout . json as corrected 2026-09-03: 41 lots — 14 southwesterly, 27 northeasterly; Phase 1 = 24 lots (u ≤ 980 ft), Phase 2 = 17 . The count fell from 43 on 2026-09-03: one lot was inside a stream buffer once every lot was screened against all three digitised reaches (that slot is now an unlotted open-space tract) and one was taken by the enlarged stormwater basins. FACTS §4's acceptable range is 40–47, so 41 is inside it but only one lot above the floor; voluntary condition 2 caps the development at 47. The figure entered on the form must equal the lot count on the filed site-plan sheet and in the Letter of Intent — re-check on the filing date
RESIDENTIAL — Dwelling Unit Size (sq. ft.)	1,406–1,492 SF conditioned (Plan A "The Springbrook", body 38'-0" × 51'-10", 2 BR / 2 BA, 1,406 SF conditioned + 458-SF garage + 102-SF covered front porch + a 12'-0" × 6'-0" = 72-SF uncovered rear patio , which is not under roof, so 1,967 SF under roof; Plan B "The Laurel", body 38'-0" × 57'-6", 2 BR + den / 2 BA, 1,492 SF conditioned + 504-SF garage + 102-SF covered front porch + a 14'-0" × 6'-0" = 84-SF covered rear porch = 2,182 SF under roof)	data/plans . json (the plans as drawn on Sheets A-1.1 and A-1.2, which supersede the "≈1,444 / ≈1,600 SF" targets of FACTS §4); minimum heated floor area for a cottage home in R-2 is 1,000 SF (Table 4.1), so both plans appear consistent with it
NON-RESIDENTIAL — Number of Buildings/Lots	None / N/A	No non-residential use is proposed (FACTS §4 "Not proposed"). The clubhouse (≈2,400 SF under roof, one story, on the common amenity tract 114.8 ft from the Arcade Rd right-of-way) is a residents-only accessory amenity building on a common tract, described under Proposed Development; VERIFY at pre-app whether staff wants it entered here as "1 accessory community building"
NON-RESIDENTIAL — Total Gross Square Feet	N/A (accessory clubhouse ≈2,400 SF, residents only)	As above
Has Applicant filed or intend to file any other variance, rezoning or waiver applications?	■ YES	The applicant filed RZ-2025-01 on this same land in July 2025. Answering "no" would be inaccurate on the face of the City's own file, and the case is discussed at length in the Letter of Intent (docs/03 §9) and in docs/04 criterion (E); the form answer must match
If YES, describe	Write: "(1) RZ-2025-01 — a rezoning of these same four parcels (R6123 033, 015, 014 and 162) from R-1 to Mixed-Use for 104 condominium units and 24,400 SF of commercial floor area, filed by this applicant in July 2025. The Planning Commission tabled it on August 28, 2025 because of a defect in the legal advertisement; it was never heard, no staff recommendation was issued and no denial was entered, so the twelve-month bar of §1003-11 is not engaged. That program is withdrawn and is replaced by this application. (2) A concurrent variance under §1005 on minimum lot depth (Table 4.1, 100 ft) will be filed only if staff requires the 20-ft R-1 buffer to be held in a separate tract rather than in a recorded buffer easement on the rear 20 ft of each perimeter lot under §313(1). No other variance, rezoning or waiver application is filed or intended."	Prior case: FACTS §2b; docs/03 §9; docs/10 Q10. VERIFY before filing : ask the Planning Director to confirm in writing that RZ-2025-01 was tabled rather than denied, and whether it should be formally withdrawn under §1003-10 first (docs/00 item 2). Fallback variance: FACTS §3 design basis (§313(1): buffers supersede the minimum yards) and fallback (50' × 82' lots + §1005 variance)

Page 1 — Required attachments checklist (form list, mapped to this package)

Form line	Status	Package item
STANDARDS GOVERNING EXERCISE OF THE ZONING POWER (attached) — criteria (A)–(F), form p. 3	To attach	docs/04-standards-governing-zoning-power.md
CONFLICT OF INTEREST CERTIFICATION / CAMPAIGN CONTRIBUTIONS (attached) — form pp. 4–6	To attach; file within 10 days of filing	docs/07-conflict-of-interest-and-certifications.md §1–2

Form line	Status	Package item
APPLICANT / PROPERTY OWNER NOTARIZED CERTIFICATIONS (attached) — form p. 7	To attach; three notarized owner signatures (one per owner)	docs/07-conflict-of-interest-and-certifications.md §3–4
TYPED LETTER OF INTENT	To attach	docs/03-letter-of-intent.md — Instructions item 6 requires the letter to state any buffer reduction requested, and one is requested : at the southwest curb return of the entrance the 20-ft perimeter buffer is reduced by 2.0 ft measured along the Arcado Road right-of-way , over about 0.1 SF of pavement , at the right-of-way corner only, leaving a residual undisturbed width of 18'-0" there and the full 20'-0" everywhere else. The return was already cut from 25'-0" to 15'-0" to get it that far, and the entrance cannot be shifted northeast because it is only 251 ft from the Arcadia Place centreline against a ~244-ft access-management standard. The letter and voluntary condition 4 must state the same figure as the drawing (data/layout.json metrics.sw_curb_return_buffer_encroachment_ft = 2.0). The letter also lists the voluntary conditions (55+/HOPA, 47-unit cap, one story, NFPA 13D sprinklers, no driveways to Arcado)
TYPED LEGAL DESCRIPTION OF PROPERTY	DRAFT attached; replace with the RLS description	docs/05-legal-description-DRAFT.md
SITE PLAN / BOUNDARY SURVEY — 1 full-size copy (to scale) and 5 reductions (8.5" × 11") or electronic file; PDF emailed to returner@cityoflilburn.com	Site plan in package; boundary survey to be commissioned (RLS)	Sheet C-2.0 , Master Concept Plan — Overall (drawings/mcp-sheet.svg / .png, 1" = 60'), with the balance of the sheet set listed in docs/00-transmittal-and-index.md; boundary survey — VERIFY / commission (§1003-4.4). The lot count, density, open space and impervious figures on this data sheet are read from that sheet's source data
LIST OF ADJOINING PROPERTY OWNERS — names and mailing addresses	Attached	docs/06-adjoining-property-owners.md
Application fee (Instructions item 2; not on the form's list)	\$1,250 — 5.0 to 9.9 acres, Rezoning; check payable to "City of Lilburn"; non-refundable	FY25-26 fee schedule; Instructions p. 1. VERIFY the amount with the Planning Department on the filing date : the FY2026-2027 schedule adopted 2026-08-31 supersedes the FY25-26 schedule still posted online (docs/00)
Architectural renderings and elevations (Instructions item 11)	To attach — each street-visible side, to scale or proper perspective, colors and materials, roofing, wall-sign location/size	renderings/ and the plan/elevation sheets (DRAFT until an architect seals them)

Page 3 — Standards Governing the Exercise of the Zoning Power

Criteria (A)–(F) are answered in docs/04-standards-governing-zoning-power.md. The form's preamble cites "Article 10, Section 1003 of the 2011 Zoning Resolution (as amended)" and "criteria in Section 1003-8"; cite the current ordinance as Lilburn Zoning Ordinance 2023-603 §1003 in the response.

Page 7 — Administrative use only (leave blank)

CASE NUMBER, DATE COMPLETE, RECEIVED BY, APPLICATION FEE, PAID BY/RECEIPT #, HEARING DATES PC / CC — completed by staff. Target cycle (Instructions p. 2 schedule): file by **Friday 2026-09-25** → legal ad 2026-09-29 / notices 2026-10-07 → **Planning Commission Thursday 2026-10-22, 7:30 PM** → **City Council Monday 2026-11-09, 7:30 PM**; fallback: file 2026-10-30 → PC *2026-11-26 (holiday week — date subject to change) → Council 2026-12-14. A pre-application conference with the Planning Director (Reid Turner, 770-279-3715, returner@cityoflilburn.com) one week before submittal is "highly recommended" (Instructions p. 1).

Filing-day checklist

1. Owner status settled: executed purchase contract for R6123 014/162 with the cooperation covenant (docs/07 §4), or the Mendez/Roblero owners as co-applicants.
2. Lot count on the form = site-plan sheet (C-2.0) = Letter of Intent = voluntary conditions: **41**, as corrected 2026-09-03. Re-read data/layout.json metrics.lots on the filing date rather than trusting this sheet.
3. Acreage 9.44 on the form; 9.58 GIS noted on the rider; survey governs.

4. Three notarized owner certifications + notarized applicant certification + conflict-of-interest certification (docs/07).
5. Fresh adjoining-owner extract (docs/06 — re-run on the filing date).
6. Fee \$1,250 (confirm the current schedule); one full-size site plan + five 8.5" × 11" reductions or electronic file; PDF emailed to the Planning Director.
7. Prior-applications field answered **YES** and RZ-2025-01 described; written confirmation from the Planning Director that it was tabled, not denied, in hand (or the withdrawal filed).
8. Diary: conflict-of-interest disclosure due within 10 days of filing (2026-10-05 if filed 2026-09-25); attendance at both hearings; no withdrawal after the legal ad (§1003-10).

Disclaimer: *This is an AI-generated analysis for preliminary planning purposes. All findings must be verified by a licensed professional before use in design, permitting, or regulatory submissions.*

Standards Governing Exercise of the Zoning Power — Applicant's Responses

City of Lilburn Rezoning Application, page 3 of 6 — "Standards Governing the Exercise of the Zoning Power" Case RZ-2026-___ · 4535, 4537, 4539 and 4541 Arcado Road SW · Parcels 6123 162, 6123 014, 6123 015, 6123 033 · 9.44 acres · R-1 to R-2 Applicant: Mohammed Awad · Date: 2026-09-___

DRAFT for owner and attorney review. The city form provides two lines per criterion; these responses are to be attached as "Applicant's Responses to Standards A–F (attached)" and each entry on the form should read "See attached." The criteria are quoted from the city form. Lilburn Zoning Ordinance 2023-603 §1003-7 states the same six standards in substantially the same words (its item 5 adds "including the future development map and future land use plan map"); the form's reference to "Section 1003-8" and the "2011 Zoning Resolution" appears to be a legacy citation — see the attorney-review note at the end. Items marked VERIFY are not yet confirmed against a primary source.

(A) Whether a proposed rezoning will permit a use that is suitable in view of the use and development of adjacent and nearby property:

Yes. The proposed use is 41 detached, one-story, single-family houses on fee-simple lots — the same use as all 31 privately owned parcels adjoining this land, every one zoned R-1 in the City of Lilburn and developed with a single-family house. The 32nd adjoiner is the Gwinnett County Arcado Road right-of-way. Along the northeast line are 13 King David Manor lots and 4531 Arcado Road (2026 digest values \$292,800 to \$412,500, median \$349,200); along the southwest line, 14 Legends at Parkview lots on Village Green Court and Fieldhouse Circle (\$579,100 to \$894,600, median \$693,500); at the northwest end, three Nantucket Drive lots.

Nearby property already includes the requested district: the Arcado Road Townhomes R-2 zoning lies about 106 feet northeast of the site's front corner. At 4.34 dwelling units per acre of detached houses, the proposal is less intense than that attached-townhome district and only 0.34 du/ac above the R-1 maximum of 4 du/ac. The City recently approved R-2 cottage lots beside R-1 neighborhoods at Annsbury Park (140 lots, under construction) and the Preserve at Killian Hill Phase 3 (24 lots, March 13, 2023).

Suitability is built into the plan: a 20-foot undisturbed buffer on all three shared property lines, reduced only where the entrance curb return meets the county right-of-way; one-story homes with ridges of 18 feet 6 inches and 19 feet 8 inches above finished grade; no lot fronting or driving onto Arcado Road; fiber-cement and brick or stone exteriors; and homes of 1,406 and 1,492 square feet conditioned, expected to price between the two adjoining medians [pricing: VERIFY]. Lilburn Zoning Ordinance 2023-603 §402-1 describes R-2 as intended "primarily" for "detached single-family uses," including "clustered and cottage housing types," located "where there is convenient access to collector streets" — which is this site.

(B) Whether a proposed rezoning will adversely affect the existing use or usability of adjacent or nearby property:

No. Nothing in the proposal changes what any adjoining owner may do, and each potential off-site effect has been designed out or limited.

Traffic. The site's only access is Arcado Road; nothing connects to King David Drive, Village Green Court, Fieldhouse Circle or Nantucket Drive, so the neighbors' streets receive no project traffic. The project adds roughly 12.3 trips in the evening peak hour (ITE Trip Generation Manual, 11th ed., LUC 251, 0.30 per unit × 41 units; 11–16 across published compilations).

Views, light and noise. One-story houses sit behind a 20-foot undisturbed buffer held in a recorded easement; the nearest rear structures — Plan A's patio and Plan B's covered porch — stand 20 feet 6 inches and 20 feet 10 inches from the rear line, outside the buffer. Evergreens will supplement the buffer where vegetation is thin. Lighting will be full-cutoff and directed away from adjoining lots, with none on the pickleball courts.

Stormwater. The 2025 petition raised runoff, erosion and flooding. The site is in FEMA Zone X. Two dry detention and water-quality ponds designed to the Gwinnett County Stormwater Management Manual will hold post-development peak discharge at or below pre-development levels, discharging only to existing drainage courses. The stream head, its 50-foot undisturbed buffer and its 75-foot impervious setback are preserved in a 20,000-square-foot common tract, and water-quality treatment is emphasized because the receiving stream is on Georgia's 2024 303(d) list.

Utilities. Phase 1 uses an 8-inch Gwinnett County sewer main already on the property and extends nothing. Phase 2 needs a short gravity connection and an easement from one Village Green Court owner; a condition offers that Phase 2 not be built without it.

Values and upkeep. Yards, lane, buffers and ponds are maintained by a mandatory homeowners' association [pricing: VERIFY].

(C) Whether the property to be affected by a proposed rezoning has a reasonable economic use as currently zoned:

The property has some economic use under R-1: the two existing houses at 4535 and 4541 Arcado Road can continue as residences, and a conventional R-1 subdivision is possible in principle. What R-1 cannot accommodate is this proposal — and the obstacle is dimensional, not a prohibition on the use.

Ordinance 2023-603 §602 lists "Single-family (cluster-cottage, creative lot configuration)" as a permitted use in R-1 as well as in R-2, so the use itself is already allowed on this land. What R-1 does not supply is a lot to put it on: Table 4.1 states no minimum lot size and no minimum lot width for a cottage home in the R-1 column, so no cottage lot can be created in that district. The 3,000-square-foot, 50-foot-wide cottage lot exists only in R-2. R-1's own dimensional standards — 9,000-square-foot lots at least 80 feet wide — are what forbid the proposal.

Those standards fit this land poorly. The assemblage is a strip about 236 feet wide with a single public frontage. After a street corridor [right-of-way width: VERIFY], roughly 93 feet of depth remains on each side, so an R-1 lot must be about 97 feet wide to reach 9,000 square feet — realistically 25 to 30 two-story houses on a 1,700-foot dead-end street. That layout would generate roughly three times the peak-hour traffic of the cottage plan (ITE 11th ed. LUC 210) and would add school-age children to a cluster in which the July 2025 School Impact Analysis reported Camp Creek Elementary and Parkview High above capacity [VERIFY]. It would not deliver the one-story, age-restricted, visitable housing that the 2024 Comprehensive Plan's work item H.03 (p. 47) asks the City to incentivize "within all medium and high density residential districts" — and R-2 is that district.

(D) Whether the proposed rezoning will result in a use which will or could cause an excessive or burdensome use of existing streets, transportation facilities, utilities, or schools:

No, on each count.

Streets. Under ITE Trip Generation Manual, 11th ed., LUC 251, 41 homes generate about 12.3 evening peak-hour trips at 0.30 per unit (11–16 across published compilations) and 152–177 daily. Arcado Road carried 14,800 vehicles per day at GDOT station 135-6689 in 2017, older than the two-year rule in Gwinnett DOT's Traffic Impact Study Guidelines (2023), so new 24-hour counts will be obtained. Those Guidelines require a study with every zoning application and exempt only residential projects of seven lots or fewer, so a Level 1 study is being prepared; turn-lane and operations analyses begin at Level 2. No left-turn lane is warranted; the threshold on a 2-lane, 35 mph road is 75 lots. The entrance sits 251 feet from the Arcadia Place centerline against the 244-foot minimum of Gwinnett UDO Table 900.3; 390 feet of sight distance is required (Table 900.2; survey to confirm).

Utilities. An 8-inch Gwinnett County gravity sewer main already crosses Parcel 6123 033 (manhole invert 927.13, about 270 feet in); Phase 1's 24 lots drain to it with no extension. Phase 2's 17 lots will be served only by a gravity connection of about 478 feet, 178 feet off site, to the Legends at Parkview main; a condition bars any private pump station, force main or grinder-pump system, which Gwinnett DWR's subdivision policy does not permit. The 1,754-foot dead-end lane goes to the Gwinnett County Fire Marshal for special approval under IFC 2024 as modified by Ga. Comp. R. & Regs. 120-3-3-.04, Appendix D103.4; sprinklers are offered voluntarily, Georgia's Appendix D107.1 requiring a second access road only above 120 units.

Schools. As housing for older persons under 42 U.S.C. §3607(b)(2)(C), the community generates zero public-school students in the Camp Creek, Trickum and Parkview cluster [VERIFY the 2025 figures].

(E) Whether the proposed rezoning is in conformity with the policy and intent of the Land Use Plan:

Yes, under the adopted plan and under the amendment now before the Department of Community Affairs.

Zoning ordinance link. Lilburn Zoning Ordinance 2023-603 §402-1 states that R-2 "generally corresponds to the ... Comprehensive Plan Character Area designation of 'Established Residential,'" and the §203 compatibility table maps Established Residential to R-1 and R-2. The site is within Established Residential, so R-2 is plan-consistent by the ordinance's own terms.

2024 Comprehensive Plan, printed page 35. Established Residential "is characterized by single-family houses on larger lots"; "infill parcels offer opportunities for some variety"; and "higher density development will generally not be encouraged in the area." The proposal is single-family houses at 4.34 du/ac on an infill strip: the "some variety" the plan contemplates, not the higher density it discourages. Work item H.03 (printed p. 47) directs the City to "incentivize accommodations for aging population and elements of visitability (accessible living, masters on main) within all medium and high density residential districts" — R-2 is the medium-density residential district, and every home here is one story with a zero-step entry.

2026 Comprehensive Plan Amendment — transmitted, not yet adopted. The City Council approved transmittal of the Amendment to the Georgia Department of Community Affairs at its public hearing on July 13, 2026; it has not been adopted, and adoption may occur before this case is heard [date: VERIFY]. Either way the proposal fits. The Amendment's Future Development Map places this land in "Suburban-Low," whose chart (p. 73) lists "Single-Family Cottages" and "Single-Family" as Appropriate and Mixed Use,

Multifamily, Office and Retail as Not Appropriate; the adjacent Suburban-Medium class rates both single-family categories Appropriate as well. Its policy "Do not further extend sewer in this area" is met in Phase 1, which uses the main already on the property.

(F) Whether there are other existing or changing conditions affecting the use and development of the property which give supporting grounds for either approval or disapproval of the proposed rezoning:

Several existing and changing conditions support approval.

An aging, under-served market. Census tract 504.61 is 35.5 percent aged 55 and older; ZIP 30047 has roughly 8,500 owner-householders 55 or older (ACS 2020–2024). Lilburn's only 55+ product, the Cottages at Noble Village, is sold out. Stakeholders in the 2024 plan's outreach recorded that "affordable housing for seniors is needed" (printed p. 100) and that housing is "pricing out young families and seniors are moving away" (printed p. 101). The same plan's survey found 64 percent of participants wanted more horizontal density such as cottage courts (p. 10) and, in the next sentence, that 59 percent thought Lilburn had the right amount of senior housing and needed less medium- and high-density housing. That 2023–24 survey is superseded on this point by the City's own 2026 amendment, which rates single-family cottages Appropriate here.

The City's recent decisions. The City denied a mixed-use rezoning at Arcade and Killian Hill Roads (RZ-2024-01, June 10, 2024) and approved R-2 cottage subdivisions at Annsbury Park and the Preserve at Killian Hill Phase 3. The 2025 mixed-use filing here (RZ-2025-01, 104 condominiums plus 24,400 square feet of commercial) was tabled August 28, 2025 for a defective legal advertisement, never heard, and is retired.

Infrastructure and land. An 8-inch sewer main already crosses the front of the property, and the site is in FEMA Zone X. The Killian Hill Road widening (Gwinnett SPLOST F-1262) remains at "Concept Complete" with no construction date [VERIFY]; this proposal adds under 1 percent of peak-hour traffic without relying on it. A 236-foot-wide strip with a single frontage is ill-suited to a conventional R-1 subdivision (see C) and well-suited to one lane of cottage lots behind the ordinance's widest buffer [VERIFY that the pending zoning-code rewrite does not alter R-2 cottage standards].

Attorney-review note

1. **Form versus ordinance numbering.** The city form cites "Article 10, Section 1003 of the 2011 Zoning Resolution" and "Section 1003-8"; in Lilburn Zoning Ordinance 2023-603 the rezoning criteria are §1003-7 and §1003-8 contains the Special Use Permit criteria. Confirm with the Planning Director that the six standards above are the ones staff will apply, and cite §1003-7 in the cover letter.
2. **Criterion C is conceded in part.** The response acknowledges a reasonable economic use under R-1 rather than asserting that the current zoning is confiscatory, and it rests the argument on Table 4.1's dimensional standards rather than on the §602 use table — because §602 lists the cluster-cottage use as permitted in R-1 as well as R-2, and an opponent reading the use table would otherwise say the applicant misstated the code. Confirm this is the intended posture: it forgoes a takings-type argument in exchange for accuracy on the face of the ordinance. If counsel wishes to preserve a constitutional challenge under Georgia law, the required written notice and specificity must be added to the application before the first hearing [VERIFY current Georgia requirements with counsel].
3. **Sworn statements.** The form is attached to notarized certifications. Every number above is drawn from FACTS.md, data/layout.json, data/plans.json, the 2026-09-03 re-verification audits in audit-2026-09-03/ and docs/11-site-context-brief.md; items marked VERIFY (2025 school-study figures, LUC 210 rate, right-of-way width, Killian Hill project status, zoning-code rewrite, Comprehensive Plan Amendment adoption date, pricing) should be confirmed or removed before signature.
4. **Consistency.** The unit count (41), density (4.34 du/ac on 9.44 acres), the buffer statement (20 feet on all three shared lines with the single reduction to 18 feet at the entrance curb return, held in a recorded easement within lots under §313(1)), and the voluntary conditions must match the Letter of Intent, the Development Summary and the site plan word for word; re-run tools/check_docs.py after any edit.
5. **The §313(1) reading is an interpretation, not settled law.** §313(1) speaks of "buffer requirements established by this article," and Article 3 sets no buffer widths. Counsel should be ready for staff to require the buffer in a separate tract, in which case the concurrent §1005 lot-depth variance and the 50-by-82-foot fallback layout are the answer.
6. **Conflict-of-interest disclosure** (O.C.G.A. §36-67A) must be filed within 10 days of filing; it is a separate form and is not addressed here.

Sources: Lilburn Zoning Ordinance 2023-603 §§203, 313, 402, 602, 1003, Tables 4.1 and 4.2; City of Lilburn Rezoning Application form, p. 3 of 6; Lilburn 2024 Comprehensive Plan pp. 10, 35, 47, 100, 101; 2026 Comprehensive Plan Amendment (draft of

6/16/2026 transmitted 7/13/2026) pp. 52–54, 71, 73; Gwinnett Tax Assessor 2026 digest (data/owners-found.json); Gwinnett County GIS zoning, sewer, stream and floodplain layers (2026-08-28); ITE Trip Generation Manual, 11th ed., LUC 251 and 210; Gwinnett DOT Traffic Impact Study Guidelines (2023) and Criteria and Guidelines for Left Turn Lanes (2007) Table 1; Gwinnett County Unified Development Ordinance §900-40.4 Table 900.2 and §900-40.5 Table 900.3; Gwinnett DWR Standard Policy for Private Developments: Condominiums, Townhomes and Subdivisions (rev. 9/2018); IFC 2024 as adopted and modified by Ga. Comp. R. & Regs. 120-3-3-.04, Appendix D103.4 and D107.1; 42 U.S.C. §3607(b)(2)(C) and 24 CFR §100.300–.308; ACS 2020–2024; RZ-2025-01 packet (Letter of Intent 2025-07-29, School Impact Analysis July 2025, PC minutes 2025-08-28); RZ-2024-01 Council minutes 2024-06-10; Gwinnett DOT SPLOST program (May 2024); site-plan data data/layout.json (41 lots, corrected 2026-09-03) and data/plans.json.

Disclaimer: This is an AI-generated analysis for preliminary planning purposes. All findings must be verified by a licensed professional before use in design, permitting, or regulatory submissions.

title	Conflict of Interest Disclosure and Notarized Certifications (Items 4 and 5)
date	2026-08-28
address	4535, 4537, 4539 & 4541 Arcado Rd SW, Lilburn, GA 30047 — Land Lot 123, 6th District, Gwinnett County (PINs R6123 033 / 015 / 014 / 162)
status	DRAFT — content and signing plan for form pp. 4–7; answers marked [to be completed by applicant] are declarations only the signers can make

Conflict of Interest Disclosure, Applicant Certification and Property Owner Certifications

Application items: Instructions item 4 — "Conflict of Interest Form ... Must be filed within ten (10) days of filing a rezoning ... application"; item 5 — "Notarized Signatures ... The form must have notarized signatures of both the property owner and applicant, or an attachment, if multiple owners are involved"; Lilburn Zoning Ordinance 2023-603 §1003-4.2 ("Application form ... including signed and notarized signature of property owner"). The city's form reproduces the statute on pp. 4–5 under the heading "Chapter 67A ... Section 36-37A-1...-4"; the correct citation is **O.C.G.A. §36-67A-1 through -4** ("Conflict of Interest in Zoning Actions", effective 1 July 1984) — the "36-37A" on the form is a typographical error in the city's document and should not be copied into anything the applicant writes.

1. What O.C.G.A. §36-67A requires of this applicant

Provision	Requirement	How it applies here
§36-67A-1(1) "Applicant"	The person who applies for a rezoning action and any attorney or other person representing or acting on behalf of that person	Mohammed Awad, plus any attorney, land planner, engineer or consultant who appears for the application; each such representative has the same disclosure duty
§36-67A-1(5) "Local government official"	Any member of the governing authority (Mayor and City Council) or any member of a planning or zoning commission	Lilburn Mayor, the City Council members and the Lilburn Planning Commission members serving during the two-year look-back — VERIFY the roster on the city website before signing
§36-67A-3(a) Disclosure of campaign contributions	If the applicant has, within the two years immediately preceding the filing , made campaign contributions aggregating \$250.00 or more , or gifts having an aggregate value of \$250.00 or more, to any such official, the applicant and the attorney representing the applicant must file a disclosure report showing the official's name, each contribution's amount and date, and each gift ≥ \$250	Look-back window: 2024-09-25 → 2026-09-25 if filed 2026-09-25 (or 2024-10-30 → 2026-10-30 for the fallback cycle). "Contribution" is as defined in O.C.G.A. §21-5-3(7) — money, in-kind services, loans and the like to a candidate or campaign committee
§36-67A-3(b) Timing	The disclosure "shall be filed within ten days after the application for the rezoning action is first filed "	Due 2026-10-05 if filed 2026-09-25 (2026-11-09 if filed 2026-10-30). Practice: file the signed form with the application so the 10-day clock is never at issue
§36-67A-2 Officials' own disclosure	An official with a property or financial interest in the affected property (or whose family member has one) must disclose and disqualify	Not the applicant's duty, but the applicant should know it: no Lilburn official is an owner of record of the four parcels or of the 31 adjoining parcels per the 2026 digest (docs/06) — family relationships cannot be checked from the digest (VERIFY if any name is recognized)
§36-67A-3(c)–(d) Opponents	An opponent who made ≥ \$250 in contributions in the two years must disclose at least five calendar days before the first hearing	Informational; if organized opposition appears (the 2025 petition — docs/11), the city, not the applicant, enforces this
§36-67A-4 Penalty	Knowing failure to comply is a misdemeanor	Sign only what is true; when in doubt, disclose

2. Disclosure of Campaign Contributions and/or Gifts (form p. 6) — content

Question on the form: "Have you, within the two years immediately preceding the filing of this application, made campaign contributions or gifts of an aggregate value that is \$250.00 or more to the Mayor and Council Members or a member of the Lilburn Planning Commission? YES NO."

Answer: [to be completed by applicant] — expected: NO (no contributions or gifts of \$250 or more in aggregate to any Lilburn Mayor, Council member or Planning Commission member in the prior two years). This is a personal declaration under a criminal-penalty statute; the package cannot make it for the signer. Before answering, the applicant (and each representative) should check bank/credit-card records and any campaign-finance filings for the look-back window against the current and former officials' names (Georgia Government Transparency and Campaign Finance Commission filings list contributors by name — **VERIFY**).

Table (Name of Official / Contribution-Gift / Description / Date): leave blank if the answer is NO; if YES, list every contribution and gift, one line each, and attach additional sheets as the form allows.

Who signs the disclosure/certification (form p. 5, "Conflict of Interest Certification for Rezoning"):

Signer	Capacity	Required?
Mohammed Awad	Applicant (owner of R6123 033/015; contract purchaser of R6123 014/162)	Yes — §36-67A-3(a)
Any attorney or representative who acts for the application before the city	Applicant's attorney/representative	Yes , each — §36-67A-1(1) and -3(a) ("the applicant and the attorney representing the applicant")
Santos C. Mendez; Lesvia Rosario Roblero de Leon	Owners of record of R6123 014/162 (sellers)	Not required by the statute unless they are applicants; recommended that each also sign a copy — it costs nothing, and some Georgia cities treat every signatory owner as an applicant. VERIFY staff preference at the pre-application conference

Certification text (form p. 5, verbatim): "The undersigned below, making application for rezoning, has complied with the Official Code of Georgia, Section 36-67A-1, et. seq, Conflict of Interest in Zoning Actions, and has submitted or attached the required information on the forms provided." Fields: Signature of Applicant/Applicant's Attorney or Representative; Date; Type or Print Name; Title (write "Applicant / Property Owner"); Notary Public; Date; (Seal).

3. Applicant Certification (form p. 7) — text and consequences

Verbatim: "The undersigned below is authorized to make this application and is aware that no application or reapplication affecting the same property shall be acted upon within twelve (12) months from the date of last action by the Mayor and Council, unless waived by the Mayor and Council. In no case shall an application or reapplication be acted upon in less than six (6) months from the date of last action by the Mayor and Council. Further, no application may be withdrawn once advertised and must receive final action by the Mayor and Council."

What the applicant is agreeing to (Lilburn Zoning Ordinance 2023-603):

- **§1003-10 Withdrawal.** The application may be withdrawn at the applicant's discretion, on notice to the Director, **only until the date of the legal advertisement of the City Council public hearing**; after that "all applications, having been advertised, shall be considered by the Mayor and Council ... and shall receive final action", and no part of the fee is refunded once a hearing is scheduled (§1003-10.2). On the 2026-09-25 cycle the Planning Commission ad runs about 2026-09-29 and the Council ad at least 30 days before 2026-11-09 — i.e. the practical last day to withdraw is **early October 2026**. A weak staff report or Planning Commission recommendation therefore has to be judged *before* the Council ad, not after.
- **§1003-11 Frequency of filing.** After a **denial** by the City Council, no application affecting the same land (or any portion of it) is accepted for **12 months** unless the Mayor and Council waive the period, and in no case sooner than 6 months. RZ-2025-01 on this site was **tabled on 2025-08-28 and never heard or denied** (FACTS §2b), so no bar is running now — but a denial of this application would burden all four parcels, including the sellers' land, until late 2027.
- "Authorized to make this application": for R6123 014/162 the authority is the owners' written consent (§1003-2) — the notarized Property Owner Certifications in §4 below and the purchase contract's cooperation covenant.

Signature block: Signature of Applicant — Mohammed Awad; Date; Type or Print Name — Mohammed Awad; Title — Applicant / Property Owner (R6123 033, 015) and Contract Purchaser (R6123 014, 162); Notary Public; Date; (Seal). Sign in the presence of a Georgia notary public with government photo ID; black ink.

4. Property Owner Certification (form p. 7) — who signs for which parcel

Verbatim: "The undersigned below, or as attached, is the record owner of the property considered in this application and is aware that if an application is denied by the Mayor and Council, no application or reapplication affecting the same land shall be acted upon within twelve (12) months from the date of last action by the Mayor and Council, unless waived by the Mayor and Council. In no case shall an application or reapplication be acted upon in less than six (6) months from the date of last action by the Mayor and Council. Further, no application may be withdrawn once advertised and must receive final action by the Mayor and Council."

The form carries one owner signature block; the Instructions (item 5) accept "an attachment, if multiple owners are involved". Attach **one Property Owner Certification page per owner** (three pages), each notarized:

#	Owner of record (2026 digest, exactly as vested)	Signs for	Digest deed reference (VERIFY vesting on the recorded deed)	Notes
1	AWAD MOHAMMED (Mohammed Awad)	R6123 033 (4541 Arcado Rd SW, 0.99 ac) and R6123 015 (4539 Arcado Rd SW, 2.00 ac)	Deed Book 61414, Pages 680 and 683 (2024-09-06/07)	Sole owner of record; signs as owner and as applicant (§3) — two separate notarized signatures
2	MENDEZ SANTOS C (Santos C. Mendez)	R6123 014 (4537 Arcado Rd SW, 2.00 ac) and R6123 162 (4535 Arcado Rd SW, 4.45 ac)	Deed Book 60727, Page 25 (2023-07-20)	Co-owner; both co-owners must sign — a certification by one of two record owners is not the owner's signature §1003-4.2 requires
3	ROBLERO DE LEON LESVIA ROSARIO (Lesvia Rosario Roblero de Leon)	R6123 014 and R6123 162	Deed Book 60727, Page 25 (2023-07-20)	Co-owner; as above. Print the name exactly as vested

Checklist for the owner pages:

1. **Title check before signing (VERIFY):** pull the vesting deeds (GSCCCA) for all four parcels and confirm (a) the grantee names match the digest, (b) no additional owner, estate, trust or lender consent is needed, (c) no recorded restriction or option affects the sellers' authority. If a parcel is held by an entity or estate, the signer's authority (officer resolution, letters testamentary) is attached.
2. **Power of attorney:** if either seller signs through an attorney-in-fact, attach the recorded POA; otherwise each signs personally before the notary with photo ID.
3. **Language:** if a signer prefers Spanish, provide a translation of the certification for reading, but the signed page is the English form; the notary acknowledges the signature, not the content (**VERIFY** with the notary that an interpreter is not required).
4. **Notary:** Georgia notary public; seal; date; the notary may not be a party to the transaction.
5. **Dates:** all signatures dated within 30 days before filing; re-execute if the application is amended in a way that changes the request (e.g. lot count or conditions) — **VERIFY** staff practice.
6. **Consent as contract purchaser (§1003-2):** the sellers' notarized certifications are the "owner's written consent"; keep a countersigned copy with the contract.

5. The Mendez / Roblero cooperation covenant — must be in the purchase contract

Because the sellers will still own R6123 014 and 162 through both hearings, and because §1003-10 and §1003-11 bind their land once the application is advertised, the purchase contract for 4535/4537 Arcado Rd SW must contain an express **cooperation covenant**. Minimum content (for the closing attorney to draft — this is not legal advice):

1. Sellers consent to the buyer filing, prosecuting and amending a rezoning application (R-1 → R-2) covering their parcels as part of the assemblage, as contract purchaser under Lilburn Zoning Ordinance 2023-603 §1003-2.
2. Sellers will execute, before a notary and within 5 business days of request, the Property Owner Certification, any owner consent, the conflict-of-interest form if the city requires owners to sign, and any re-execution required by an amendment or a re-filing.
3. Sellers permit the city's public-hearing sign on the frontage, site visits by staff, Planning Commission and Council members, the boundary survey, the sewer-invert survey, topographic survey, stream delineation, soil borings and rock probes (FACTS §2b), with reasonable notice and restoration.
4. Sellers will not encumber, subdivide, lease long-term or convey the parcels, apply for any zoning action, or oppose the application, during the contract term.
5. Sellers acknowledge that once the application is advertised it cannot be withdrawn (§1003-10) and that a denial bars a new application on their land for 12 months (§1003-11); the contract allocates that risk (e.g. buyer's option to terminate before the Council legal ad; seller's remedy limited to earnest money).
6. Zoning contingency: closing conditioned on R-2 approval with conditions acceptable to the buyer, final and unappealed (Georgia zoning appeals run 30 days from the decision — **VERIFY** the appeal period with counsel); the term covers filing 2026-09-25 → Council 2026-11-09 plus a tabling/continuance allowance and one re-hearing cycle.
7. Sellers will attend the hearings if requested, and will not make statements to the city inconsistent with the application.
8. Survival: the covenant survives an assignment of the contract to the buyer's development entity.

6. Signing schedule

When	Action
Pre-application conference (≈ 2026-09-18)	Confirm with the Planning Director: whether sellers also sign the conflict-of-interest form; one owner page per owner; whether the clubhouse is reported as non-residential (docs/01)

When	Action
Before filing	Title check on all four parcels; purchase contract with \$5 covenant executed; three notarized owner pages + notarized applicant certification + notarized conflict-of-interest certification
Filing day (2026-09-25)	File all certifications with the application; the 10-day disclosure clock starts
≤ 2026-10-05	Conflict-of-interest disclosure on file (already filed if included on filing day)
Before the Council legal ad (early October 2026)	Last opportunity to withdraw (§1003-10) — decide on the strength of the staff report / Planning Commission outcome

Disclaimer: This is an AI-generated analysis for preliminary planning purposes. All findings must be verified by a licensed professional before use in design, permitting, or regulatory submissions.

Letter of Intent — Rezoning R-1 to R-2, "The Cottages at Arcade Springs"

DRAFT for owner and attorney review — prepared to City of Lilburn 2026 Application Instructions item (6) and Lilburn Zoning Ordinance 2023-603 §1003-4(5). Bracketed items marked VERIFY must be settled before filing. Every drawing referenced is DRAFT and NOT SEALED; the boundary survey, legal description, engineering and landscape sheets must be superseded by the sealed work of a Georgia registered land surveyor, professional engineer, architect and landscape architect.

Mohammed Awad 4541 Arcade Road SW Lilburn, Georgia 30047-3968 [phone: VERIFY] · [email: VERIFY]

2026-09-__

City of Lilburn Planning & Economic Development Department Attn: Reid Turner, Planning Director 340 Main Street Lilburn, Georgia 30047

RE: Letter of Intent — Application to rezone 4535, 4537, 4539 and 4541 Arcade Road SW (Parcels 6123 162, 6123 014, 6123 015 and 6123 033; 9.44 acres, Land Lot 123, 6th District) from R-1 to R-2 for a 55+ detached cottage-home community of 41 one-story homes

Dear Mr. Turner:

I respectfully submit this Letter of Intent in support of my application to amend the official zoning map of the City of Lilburn by rezoning the four parcels listed below from R-1 (Single-Family Residential) to R-2 (Medium-Density Residential) for a single-family cluster-cottage community to be known as The Cottages at Arcade Springs. The site plan, the architectural elevations and renderings, the typed legal description, the adjoining-owner list and the required forms accompany this letter. The sealed boundary survey is in progress and will be filed as soon as the surveyor releases it [survey delivery date: VERIFY with the Director whether the application may be accepted with the DRAFT metes-and-bounds description and the survey supplied before the Planning Commission hearing].

1. The applicant and the property

I am the applicant. I am the owner of record of Parcels 6123 033 and 6123 015 and the contract purchaser of Parcels 6123 014 and 6123 162 [contract status: VERIFY before filing]. The owners of record of Parcels 6123 014 and 6123 162 have executed the notarized Property Owner Certification attached to the application form [VERIFY].

Parcel (Gwinnett R6123 ###)	Address	Deeded acreage (2026 digest)	Owner of record	Current zoning
6123 033	4541 Arcade Road SW	0.99 ac	Mohammed Awad	R-1 (Lilburn)
6123 015	4539 Arcade Road SW	2.00 ac	Mohammed Awad	R-1 (Lilburn)
6123 014	4537 Arcade Road SW	2.00 ac	Santos C. Mendez and Lesvia Rosario Roblero de Leon	R-1 (Lilburn)
6123 162	4535 Arcade Road SW	4.45 ac	Santos C. Mendez and Lesvia Rosario Roblero de Leon	R-1 (Lilburn)
Assemblage		9.44 ac	Land Lot 123, 6th District, Gwinnett County	R-1

The application acreage is the deeded total of 9.44 acres. The Gwinnett County GIS parcel fabric computes 9.58 acres (417,174 SF); the sealed boundary survey governs [DRAFT metes and bounds pending RLS seal].

The assemblage is a single strip with its only public frontage on Arcade Road (about 237 feet of curved frontage, measured as a chord; 237.6 feet measured along the right-of-way line). It is approximately 235 feet wide at the Arcade Road right-of-way, widening to approximately 246 feet at the northwest end, and 1,722 feet deep on the northeast line and 1,757 feet deep on the southwest line. Two older houses stand on the land, and only one of them is near the front. The dwelling at **4541 Arcade Road** stands about 200 feet in from the right-of-way (footprint about 2,396 SF; 2,053 SF heated in the 2026 digest). The dwelling at **4535 Arcade Road stands about 1,200 feet back from the right-of-way**, on the ridge, on the alignment of the proposed lane (footprint about 4,568 SF; 4,267 SF heated with a basement) — it is a Phase-2 demolition standing on the lane, not a frontage structure. Earlier drafts of this application described both houses as standing near the front; that was wrong, and the corrected positions are shown on Sheet C-1.0 [building locations are from aerial-imagery footprints fitted to the site coordinates and must be located by the boundary survey: VERIFY]. Accessory structures stand near each dwelling; the balance of the strip is wooded.

31 privately owned parcels adjoin the site, all zoned R-1 in the City of Lilburn and all developed with single-family houses — the King David Manor lots along the northeast line, the Legends at Parkview lots (Village Green Court and Fieldhouse Circle)

along the southwest line, and three Nantucket Drive lots at the northwest end. The Gwinnett County Arcado Road right-of-way is the 32nd abutting ownership, so 32 notices are required. A Lilburn R-2 district (the Arcado Road Townhomes site) lies approximately 106 feet northeast of the front corner across 4531 Arcado Road.

2. The request

I request that the 9.44-acre assemblage be rezoned from R-1 to R-2 for the use listed in Lilburn Zoning Ordinance 2023-603 §602 (Use Table) as "Single-family (cluster-cottage, creative lot configuration)," which is a permitted use (P) in R-2. **No special use permit is requested and no variance is requested. One buffer reduction is requested** — a reduction of the 20-foot buffer to 18'-0" at a single point, where the southwest curb return of the entrance meets the Arcado Road right-of-way — and it is described and requested in Section 5 of this letter, as Application Instructions item (6) requires. The application is for the R-2 district subject to the **18 voluntary zoning conditions in Section 11**, which I ask the Mayor and City Council to attach to any approval.

The proposed development:

- **41 detached one-story cottage homes** on fee-simple lots, restricted to households with at least one member aged 55 or older under the federal Housing for Older Persons Act (42 U.S.C. §3607(b)(2)(C); 24 CFR §100.300–.308), offered as a voluntary condition and enforced by a recorded declaration of covenants. At 41 units the 80-percent test and the fractional-unit rule (24 CFR §100.305(a) and (g); $0.80 \times 41 = 32.8$, and the fraction counts into the 80 percent) require **at least 33 of the 41 units** to be occupied by at least one person aged 55 or older.
- **Gross density 4.34 dwelling units per acre** (41 units ÷ 9.44 acres; 4.28 du/ac on the 9.58-acre GIS area), against the R-2 maximum of 8 du/ac in Table 4.1 and the R-1 maximum of 4 du/ac. The corrected layout is 0.34 du/ac above the R-1 ceiling and a little over half the R-2 ceiling.
- **Two floor plans**, both 2 bedroom / 2 bath with a two-car garage recessed 5 feet behind the front wall and a covered front porch 6 feet deep (17'-0" × 6'-0" = 102 SF on both plans). Plan A "The Springbrook": L-shaped body 38'-0" × 51'-10" overall with the garage as a front wing, **1,406 SF conditioned**, 458 SF garage, 102 SF covered front porch, 72 SF uncovered rear patio (12'-0" × 6'-0"), 1,966 SF total under roof, maximum ridge **18'-6" above finished grade** (21 lots). Plan B "The Laurel": body 38'-0" × 57'-6", **1,492 SF conditioned**, 504 SF garage, 102 SF covered front porch, 84 SF covered rear porch (14'-0" × 6'-0", recessed under the main roof), 2,182 SF total under roof, maximum ridge **19'-8" above finished grade** (20 lots). Ridge heights are plate 9'-0" plus a 6-inch raised heel plus half the span at 8:12, measured from a finished floor 8 inches above finished grade. Both plans are far above the 1,000-SF minimum heated floor area for a cottage home in Table 4.1 and far below the 40-foot height maximum. Every home has a zero-step entry, 36-inch doors, a 5-foot turning circle in a bathroom, and blocking for grab bars.
- **Lots 50'-0" × 100'-0" = 5,000 SF** (41 lots — 14 on the southwest side and 27 on the northeast side; minimum, average and maximum all 5,000 SF, 205,000 SF of lot area in all), against the Table 4.1 cottage-home minimum lot size of 3,000 SF, minimum width of 50 feet and minimum depth of 100 feet. Each lot has 50 feet of frontage on the private lane (§319 requires 30 feet on an approved private street).
- **Exterior character:** fiber-cement lap and board-and-batten siding, brick or stone water table and porch piers, architectural shingle roofs — 8:12 main and front-wing gables, a subordinate 6:12 gable over the garage, and the front porch roofed by carrying the front-wing gable forward rather than by a shed — and three color schemes (warm white, sage, clay). Floor plans are Sheets A-1.1 and A-1.2; elevations of every street-visible side of both house plans are Sheets A-2.1 and A-2.2; the colors and materials board is Sheet A-2.3; the clubhouse, mail kiosk and entry monument are Sheet A-3.0; typical lots are Sheets A-4.0 and A-4.1 (Application Instructions item 11). **No wall signs are proposed on any structure.**
- **Common open space of 141,089 SF (3.24 acres) in eight common tracts — 33.8 percent of the GIS area and 34.3 percent of the 9.44-acre deeded area** (basis: the sum of the tract areas tabulated on Sheet C-2.0 and Sheet C-6.0 — the 46,777-SF front amenity tract, two 10,000-SF pocket greens, the 9,312-SF terminus green, two 20,000-SF pond tracts, the 20,000-SF creek-woods tract and a 5,000-SF stream-setback tract on the southwest side), plus approximately 41,000 SF of recorded buffer easement on the rear 20 feet of the perimeter lots, which is reported separately and is not counted in that percentage. Of the common open space, about 13,960 SF is built or paved — the 2,400-SF clubhouse roof, 3,600 SF of court pads, the 160-SF mail kiosk, a 600-SF walk allowance and the 7,200 SF of hammerhead turnaround legs that fall inside the greens; the balance, 127,129 SF (30.5 percent of the GIS area), is green. **No open space is required of this development at all:** Table 4.1's open-space entry points to §5.9 of the Development Regulations, which is titled "Recreation Areas" and applies only to single-family detached subdivisions "having a gross area of 50 acres or more." At 9.44 acres this subdivision is not within it, so all of the open space offered here is voluntary.
- **Two phases:** Phase 1, 24 lots in the front 980 feet (the phase line stands at station 980, where the ground turns up toward the ridge), served by gravity to the sewer main that already crosses the property; Phase 2, 17 lots in the rear, which cannot be built until the sewer route described in Section 7 is settled.

The plan appears consistent with the dimensional standards of Table 4.1 (R-2), the §602 use table, the §319 frontage rule and the Table 8.1 parking standard, and with the design criteria of Table 4.2 subject to the two items identified in Sections 3 and 5. A

standard-by-standard tabulation is included in the Development Summary submitted with this application.

3. Access, the private lane, traffic and fire access

Access. The community will have one vehicular entrance on Arcado Road, a county-maintained two-lane minor collector posted at 35 mph. The entrance is placed in the southwest third of the frontage so that its centerline is approximately 251 feet from the centerline of Arcadia Place (251.2 feet measured along the Arcado Road centerline, 250.7 feet as a straight chord). Arcadia Place meets Arcado Road about 61 feet northeast of the site's northeast front corner and therefore cannot be aligned with our driveway. The 251-foot separation appears consistent with **Gwinnett County Unified Development Ordinance §900-40.5, Table 900.3**, which requires 244 feet centerline-to-centerline at a posted speed of 35 mph, and **§900-40.5(C)** independently limits this frontage to one driveway ("one for each 400 feet of property frontage, or fraction thereof"; the frontage is about 237 feet). **UDO §900-40.4, Table 900.2** requires intersection sight distance of 390 feet in each direction at 35 mph on a two-lane road; the sight lines are drawn on Sheet C-2.3 and will be certified by a Georgia Professional Engineer [field certification: VERIFY]. The entrance throat is 24 feet wide, with a 25-foot curb return on the northeast side and a 15-foot return on the southwest side (reduced to keep it out of the perimeter buffer — Section 5). No lot will have a driveway onto Arcado Road, and both existing residential drives are closed and removed — the 4541 drive at the frontage and the drive that runs back through the property to 4535 [alignments to be located by the boundary survey]. Under **Gwinnett DOT Criteria and Guidelines for Left Turn Lanes (June 20, 2007), Table 1**, a left-turn lane on a two-lane route posted at 30–35 mph with ADT of 6,000 or more is warranted at 75 lots; at 41 lots one is not warranted. A Gwinnett DOT driveway/encroachment permit will be obtained before land disturbance.

Private lane. A single private lane approximately 1,754 feet long (measured along the travelled way from the Arcado Road right-of-way, entry drive included) runs the length of the strip in a tract 35.9 to 46.1 feet wide (60 feet through the amenity block, where the guest and mail-kiosk bays sit in it), with 22 feet of pavement, a 2-foot strip and a 5-foot sidewalk on the northeast side, and sidewalks on both sides from about 530 feet in, where the tract reaches 37.3 feet and can carry the 36 feet that two walks need (22 + 2 + 5 + 2 + 5). Design speed is 15 mph. Parking is prohibited on the lane; each home provides 4 spaces (2 in the garage and 2 in the driveway), **164 in total against the Table 8.1 requirement of 2 per dwelling — 82 spaces (2 × 41) —** plus 12 guest and mail-kiosk spaces in bays (8 guest spaces, one of them accessible, and 4 short-term spaces at the kiosk). All components of the private lane will be built to the City's minimum standards for public streets, as the Site Development Plan Review Checklist requires.

Blocks and turnarounds. Table 4.2 asks for block widths and lengths not exceeding 600 feet and provides that "circular turnarounds shall not be permitted except in the most unusual circumstances." Because the strip has one public frontage, no abutting street at the rear and a width of only 235 to 246 feet, no cross street is possible; the lane therefore ends in hammerhead (T) turnarounds 20 feet wide with 60-foot legs rather than a circular cul-de-sac, and the turnarounds have been re-spaced so that the runs of lane between the Arcado Road entrance and the three T-legs are approximately 594, 570 and 580 feet — each within the 600-foot Table 4.2 figure. The three T-legs stand at stations 594.5, 1,164.5 and 1,744.5 measured along the travelled way (540, 1,110 and 1,690 feet along the strip), each 20 feet wide with 60-foot legs. I state plainly that this is the reading of "block" that the site allows: measured strictly between intersections with other *streets*, the lane is a single 1,754-foot block, and to that extent I ask the Council to find the strip's shape to be the "unusual circumstance" Table 4.2 contemplates. The reading used is stated on Sheet C-2.0.

Fire access. The lane is a dead-end fire apparatus access road longer than 750 feet. Under **IFC 2024 as adopted and modified by Ga. Comp. R. & Regs. 120-3-3-.04, Appendix D, Table D103.4**, a dead end "over 750" feet requires "special approval" by the fire code official, and the table pairs dead ends of 501 to 750 feet with a 26-foot width against the 22 feet proposed here. Special approval and width relief will therefore be sought from the Gwinnett County Fire Marshal (the City has no fire department), and Appendix D107.1 Exception 3 as modified by the same Georgia rule supplies the menu of alternatives the Fire Marshal may accept — enhanced turning radii, increased road widths, D103.6 fire-lane signage, an additional non-vehicular all-weather access road designed by a registered design professional to Appendix D loading, or the Fire Code Official's written statement — together with the waiver route of O.C.G.A. §25-2-12(e)(4), on which the State Fire Marshal must respond within 30 days.

A correction I want to make plainly: an earlier version of this proposal stated that a second access road is required because the development exceeds 30 dwelling units. That is the model IFC text, not the law in Georgia. **Ga. Comp. R. & Regs. 120-3-3-.04 replaces Appendix D107.1 with a threshold of more than 120 dwelling units:** "Developments of one- or two-family dwellings where the number of dwelling units exceeds 120 shall be provided with two separate and approved fire apparatus access roads." At 41 units this development **does not trigger the two-access requirement at all**. I therefore do not claim that sprinklers are a required mitigation. **NFPA 13D residential sprinklers in every home are offered voluntarily** (condition 5) as additional life-safety protection on a long single-access lane and because they halve the required fire flow under IFC 2024 Appendix B §B105.1 from 1,000 gpm to 500 gpm on a dead-end water main. Whether they buy the 22-foot width relief is the Fire Marshal's decision, and I will ask him for it in writing before the Planning Commission hearing [VERIFY].

Traffic. Using ITE *Trip Generation Manual*, 11th ed., Land Use Code 251 (Senior Adult Housing – Single-Family), 41 homes generate on the order of **12.3 trips in the PM peak hour** (0.30 trips per unit × 41) and about 177 trips per day (4.31 per unit × 41). Two fewer lots take about 0.6 of a trip off the evening peak; the conclusion below was already comfortable and is now more so. The published rate for this land use is not settled in the sources available to me: compilations in circulation give 0.20 to 0.34 trips per unit in the AM peak and 0.26 to 0.39 in the PM peak, and the 11th edition renamed and re-scoped the land use (the old "Senior Adult Housing – Detached" was merged into LUC 251 "Senior Adult Housing – Single-Family"), so several tables that carry an "11th edition" label are in fact keyed to earlier data. The traffic engineer will read the rate off the manual itself. **The conclusion does not turn on which rate is used:** across the published range the 41 homes generate about 11 to 16 PM peak-hour trips (0.26 and 0.39 per unit) and 8 to 14 in the AM peak, and even at the 47-unit cap of condition 2 and the highest rate the figure is about 18 — inside Gwinnett DOT's lowest band (0–20 peak-hour trips) in every case.

A traffic impact study is required and will be submitted. Gwinnett DOT's *Traffic Impact Study Guidelines* (2023) state that "a TIS must be submitted concurrently with all zoning applications and land disturbance development," and the only residential exemption is for "single-family detached and townhome residential development consisting of no more than seven lots or units." At 41 lots this project is not exempt. Because the site generates 0 to 20 peak-hour trips, the study is a **Level 1** study: location description, existing and proposed land use, trip-generation estimate, access-management review, adjacent access spacing upstream and downstream, intersection sight distance, connectivity and circulation review, functional classification, posted speed and future programmed projects. Operations analysis, turn-lane warrant analysis and signal warrants begin at Level 2 and are not part of it. I also record that the 2017 GDOT count for Arcado Road that earlier drafts of this application cited (station 135-6689, 14,800 AADT) **does not satisfy Gwinnett DOT's own currency rule**, which accepts counts "less than two years old" and otherwise makes "the developer responsible for obtaining new, 24-hour counts, from an approved contractor." New 24-hour counts have therefore been commissioned and will be the basis of the study [count date: VERIFY]. The guidelines are written for unincorporated Gwinnett County; I ask staff to confirm the City's own expectation for a city zoning case, and Gwinnett DOT will apply them at the driveway-permit stage on its own road in any event. This commitment is offered as condition 17.

Sidewalks and transit. A 5-foot sidewalk and 10-foot landscape strip will be built along the Arcado Road frontage. Ride Gwinnett Route 25 stops at Killian Hill Road and Arcado Road, 0.30 mile from the site; I ask that the City consider sidewalk gap closure between the site and that stop as part of its Arcado Road pedestrian improvements.

4. Amenity block and community design

The front of the site, behind the 50-foot Arcado Road building setback, is a common amenity tract of 46,777 SF containing a clubhouse of approximately 2,400 SF (40'-0" × 60'-0"), a village green, two pickleball courts (each a 20'-0" × 44'-0" court on a 30'-0" × 60'-0" fenced pad, unlighted), a cluster-mailbox kiosk with 4 short-term spaces, 8 guest spaces, a monument entry sign of not more than 32 SF standing 19.0 feet from the Arcado Road right-of-way, and the 10-foot landscape strip along Arcado Road. No lot fronts Arcado Road; the nearest structure to the Arcado Road right-of-way is the clubhouse at **114.8 feet measured perpendicular to the right-of-way line**, well beyond the 50-foot collector setback of Table 4.1. (An earlier draft of this letter reported that distance as 110 feet and the site-plan data as 120 feet; both were measured along the axis of the strip rather than perpendicular to the right-of-way, and 114.8 feet is the correct figure.) Two pocket greens of 10,000 SF each and a 9,312-SF terminus green at the northwest end break up the lane. The community will not be gated.

5. Buffers — one reduction requested, and how the buffer is held

The standard offered. Table 4.1 sets the buffer abutting R-1 property at 0 feet for detached single-family dwellings in R-2 and 20 feet for "all other allowed dwelling types." Because Table 4.1 lists the cottage home separately from detached single-family, I have designed to the more protective 20-foot standard and offer it as condition 4: a 20-foot undisturbed buffer along every property line except the Arcado Road frontage — that is, along the King David Manor line, the Legends at Parkview line and the Nantucket Drive line.

The reduction requested. Application Instructions item (6) requires this letter to state any buffer reduction requested, and there is one. Where the single entrance meets Arcado Road, the **southwest curb return** clips the corner of the 20-foot buffer band at the right-of-way. The return radius has already been reduced from 25'-0" to 15'-0" to minimize it, and the entrance cannot be shifted northeast because doing so would erode the 251-foot separation from Arcadia Place against the 244-foot Gwinnett standard. What remains is a reduction of the buffer to a **minimum of 18'-0" instead of 20'-0" — an encroachment of 2.0 feet at the tangent point of that curb return on the right-of-way line**, as dimensioned on Sheet C-2.3. The corrected drawing measures the area of curb return that actually falls inside the buffer band at about 0.1 SF, because the return only grazes the band at the right-of-way and pulls clear of it immediately behind. **I request that reduction, stated as a residual buffer width of 18'-0" at that one point.** It occurs at the point on the property farthest from any dwelling on the abutting lot (303 Fieldhouse Circle SW), it is in the plane of an existing county road, and the buffer stands at its full 20-foot width everywhere else on all three interior lines. Apart from that one point, no buffer reduction is requested. If final engineering of the entrance clears the buffer band entirely, the reduction will not be taken.

How the buffer is held — an interpretation I ask the Director to confirm. Lilburn Zoning Ordinance 2023-603 §313(1) provides that buffer requirements "supersede these minimum required yards." The site plan reads that provision to allow the 20-foot buffer to lie within the 20-foot rear yard of each perimeter lot, held in a buffer/conservation easement recorded on the final plat in favor of the homeowners' association and the City, with no structure, grading or clearing permitted within it. That reading is what allows 100-foot-deep lots on both sides of one lane inside a strip 235 to 246 feet wide. At the first pair of lots, 230 feet in, the strip measures 235.9 feet across: two 100-foot lots leave 35.9 feet for the lane tract, and the section is spent exactly — the 35.9-foot tract is the narrowest on the lane and holds the 22-foot pavement, the 2-foot strip and the 5-foot sidewalk with 6.9 feet to spare for grading. The tract widens to 46.1 feet at the northwest end as the strip widens. The rear-most structure on each lot stands 20'-6" (Plan A patio edge) to 20'-10" (Plan B rear porch) from the rear lot line, so no structure lies within the buffer easement.

I present that as an interpretation, not as settled law. §313(1) speaks of "buffer requirements established by *this article*" — Article 3 — and Article 3 sets no buffer widths; the district buffers of Table 4.1 are imported by §318. Nothing in the ordinance expressly authorizes holding a required buffer inside a private lot. I ask the Planning Director and the City Attorney to confirm the reading at the pre-application conference, and I have prepared the fallback so that the answer does not have to be discovered at a hearing.

The fallback if staff reads the buffer differently. If the buffer must be held in a separate common tract rather than in an easement on the lots, the same plan is built with lots of 50'-0" × 82'-0" (4,100 SF, above the 3,000-SF cottage-home minimum) and a concurrent variance under §1005 from the 100-foot minimum lot depth of Table 4.1, on the ground of the strip's exceptional narrowness (§1005-3.2 criteria a–g). That layout is drawn as Sheet C-2.4 so the Commission and Council can see exactly what the alternative looks like. I would prefer to resolve the interpretation before filing so that the application is heard without a variance, and I ask staff's guidance on that point.

6. Stormwater and environmental features

- **Floodplain.** The site is in FEMA Zone X (no special flood hazard area) per Gwinnett County GIS; no floodplain or floodway development is proposed (Table 4.2) [FIRM panel and effective date: VERIFY].
- **Stream buffer.** An unnamed headwater of Jackson Creek reaches the southwest line about 1,356 to 1,392 feet from Arcade Road and ends about 30 feet inside the site. The site plan shows the 25-foot state-waters buffer (O.C.G.A. §12-7-6(b)(15), measured from the point where vegetation has been wrested by normal stream flow) and the City's 50-foot undisturbed buffer and additional 25-foot impervious setback (75 feet total, measured from the top of the bank) as separate lines, because they are measured from different data. A 20,000-SF creek-woods common tract preserves the stream head and its buffers, and a further 5,000-SF unlotted tract on the southwest side (station 1,230 to 1,280) holds the ground that a lot occupied until the layout was corrected: on 2026-09-03 every lot was re-screened against all three digitized reaches of the stream, and one lot fell inside the buffer and was withdrawn. A field study will establish **first whether and where a regulated stream begins** — the ordinance's definition begins a stream at a spring or seep that sustains streamflow, or at the point where the drainage area reaches 25 acres, and the on-site reach drains far less than 25 acres — and then the top of bank. The rear lot count and the Pond 2 footprint depend on that study, and it will be done before the land-disturbance permit [VERIFY]. The layout carries a 5-foot allowance for the top-of-bank location, so the plan is measured to 80 feet from the mapped channel rather than 75. On that measure the nearest lot line stands 80.6 feet from the digitized channel, the nearest dwelling 106.2 feet, the nearest driveway 102.6 feet, the nearest pond top of bank 80.2 feet and the lane pavement 80.5 feet — that is, every impervious surface on the plan is at least 80.2 feet from the mapped channel, or 75.2 feet after the top-of-bank allowance is taken off. The receiving stream is on Georgia's 2024 303(d) list (bacteria and biota, urban runoff), so the design emphasizes runoff reduction and water-quality treatment, with no direct discharge into the buffer.
- **Detention and water quality.** Two dry detention and water-quality ponds are shown on the southwest side, one in the mid-site swale and one near the rear low pocket outside the 75-foot stream setback. Sizing is on the basis used in the 2025 hydrology study for this property: 10,000 cubic feet of detention per **disturbed** acre, less any runoff-reduction volume credited. The disturbed area has been recomputed from the plan as 328,484 SF = **7.541 acres** (the site raster less the 20-foot perimeter buffer bands, the creek woods and the undisturbed stream buffer, unioned so nothing is counted twice), not the 4.4 acres an earlier draft assumed. At 10,000 cubic feet per disturbed acre that is a screening requirement of **75,410 cubic feet** with no runoff-reduction credit taken, against **77,863 cubic feet provided** — Pond 1 at 33,129 cubic feet (52'-0" × 180'-0" at top of bank) and Pond 2 at 44,734 cubic feet (64'-0" × 180'-0"), both 6 feet deep on 3:1 side slopes, in 20,000-SF tracts. Provided storage is therefore about 103 percent of the requirement, and a runoff-reduction volume of about 15,500 cubic feet ($R_v = 0.05 + 0.009 \times 44.1 = 0.447$; $0.447 \times 1.0 \text{ in} \div 12 \times 417,174 \text{ SF}$) would reduce the requirement further if the credit is earned. **This is the second of the two corrections that took the plan from 43 lots to 41:** the basins had been sized on 4.4 disturbed acres, the real figure is 7.541 acres, and the enlargement needed to hold 77,863 cubic feet took the land one lot had occupied. The lot count follows the drainage design, not the other way round. Final sizing, the runoff-reduction credit and the water-quality volume ($WQ_v = 1.2 \times (0.05 + 0.009 \times 44.1) \div 12 \times 417,174 \text{ SF}$ = about **18,659 cubic feet**) will be by a Georgia PE under the Gwinnett County Stormwater Management Manual at the land-development-permit stage [DRAFT — to be superseded by

sealed work]. Discharge will be only to the existing drainage courses, not onto the yards of adjoining lots.

- **Impervious cover. 184,148 SF — 44.1 percent of the GIS site area, 44.8 percent of the deeded area** (basis: 86,448 SF of dwellings, garages, porches and patios; 31,390 SF of lot driveways and entry walks; 34,111 SF of lane pavement; 4,865 SF of entry drive; 184 SF of curb returns; 7,200 SF of hammerhead legs; 2,070 SF of guest and kiosk bays; 14,619 SF of sidewalks; 6,760 SF of clubhouse roof, court pads, kiosk and amenity walks; less 3,500 SF of driveway-apron and sidewalk overlap counted twice — the per-lot figures are the actual footprints from the architectural plan set, 2,752 SF for Plan A and 2,898 SF for Plan B, not a flat allowance). Earlier drafts of this application reported 38.4 percent; that figure rested on a flat 2,200-SF-per-lot allowance in the site-plan generator and understated the real area by about 24,000 SF. The corrected figure is stated here because the stormwater design depends on it, and because it is still far below the 50 to 57 percent per basin shown on the 2025 mixed-use hydrology sheet for this same land.
- **Soils and topography.** NRCS mapping shows Piedmont clays (Appling, Madison, Pacolet and Cecil-type units), Helena sandy loam (hydrologic group D, seasonal high water table about 24 inches) at the front southwest corner under the amenity block and sewer tie-in, and Ashlar (gneiss bedrock at 22 to 40 inches) under both pond sites; rock probes and an undercut allowance will precede final pond design. No mapped unit on the site is rated hydric. Elevations range from about 922 to 958 feet (USGS 3DEP; VERIFY by field survey), with a ridge about 1,080 to 1,200 feet in and the low pocket near the stream head. Existing ground along the lane reaches 10.9 percent at about 1,494 feet in, below the 12 percent maximum for street grades, but the finished profile is a PE item and is not claimed here.

7. Sanitary sewer and water

This is the part of the application where I want to be exact, because the answer changed when I read Gwinnett County's own policies rather than a summary of them.

Existing sewer on the property. An 8-inch Gwinnett County gravity main — the "Arcado Road Townhomes sewer outfall" — already crosses the property. It runs from Arcado Road along the southwest edge of Parcel 6123 033, 7 to 13 feet inside the property line, for about 314 feet, to a manhole about 272 feet in with a **927.13-foot invert**, and then leaves the site to the southwest. This is the "EX. 20' SSE" shown on the 2025 hydrology sheet. The inverts are GIS attributes and are unsurveyed; a field invert survey is the first money I intend to spend [VERIFY].

Phase 1 (24 lots, the front 980 feet): gravity, and no sewer extension of any kind. The front reach of the lane drains by gravity to that on-site manhole. The controlling reach is the lane sag about 780 feet in to the manhole: 615 feet of pipe falling from a 931.3 invert to 927.13, a slope of **0.68 percent**, against a 0.40 percent minimum. **Phase 1 requires no extension of the public sewer system.** It connects to a main that is already inside the property line. Phase 1's average daily flow is about 6,420 gpd (24 lots × 250 gpd per unit plus a 420-gpd allowance for the clubhouse).

Phase 2 (17 lots, the rear pocket): one route, and it is not the one earlier drafts of this application described. The rear low point of the lane is about 1,479 feet in at elevation 931.5; an invert there with 5 feet of cover is at 926.5, which is already 0.6 feet *below* the 927.13 manhole before any pipe slope is applied, and any pipe to that manhole would have to pass beneath the ridge at 1,080 to 1,200 feet at roughly 28 feet of cut. The rear pocket cannot be served by gravity to the on-site main.

Earlier drafts presented an in-tract homeowners'-association lift station, or individual grinder pumps, as the primary answer for Phase 2. **That option is not available, and I withdraw it.** Gwinnett DWR's *Standard Policy for Private Developments: Condominiums, Townhomes and Subdivisions* (rev. 9/2018) — the policy variant whose cover names subdivisions — states:

"PRIVATE FACILITIES: Private pump stations, private force mains and private gravity sewers are only allowed for commercial properties under single ownership within a development. Gwinnett County is not responsible for operation and maintenance of these facilities."

and immediately above it:

"OWNERSHIP OF MAINS AND APPURTENANCES: Absolute title and ownership of all water and sanitary sewer mains within the private development shall be conveyed to Gwinnett County WSA at the time of acceptance by the County."

A 41-lot fee-simple subdivision is not a commercial property under single ownership, so a homeowners'-association station, force main or private gravity sewer is not permitted here. A **county-owned** station is equally unavailable: Gwinnett DWR's *Developer Pump Station Standards* (January 2014, posted as WSR-24) §1.3.1(A) provides that

"Generally, pump stations will only be permitted when gravity sewer is unavailable to the property. Unavailable shall generally be interpreted to mean more than 5,000 feet down gradient, but this distance can be increased or decreased by GCDWR Division Directors based upon actual field conditions and the size of the project involved."

Gravity is 178 feet away, not 5,000. And there is no Gwinnett standard for grinder pumps or a low-pressure sewer system at all — the words do not appear in the County's *Water and Sewer Standards* (April 2016) or in WSR-24 — so such a system could only be private, which the first policy forecloses.

The only viable Phase-2 route is therefore a gravity connection. From the rear low pocket, an 8-inch gravity line runs southwest to the existing **Legends at Parkview 8-inch main at invert 919.58**: about 478 feet in total, of which approximately **178 feet is off-site**, falling from 926.5 to 919.58 at about **1.45 percent** — a comfortable gravity slope. Phase 2 adds about 4,250 gpd (17 lots × 250 gpd), so the completed community's average daily flow is about **10,670 gpd** (41 units × 250 gpd plus the clubhouse), roughly 7.4 gpm, with a peak of about 42,680 gpd at a peaking factor of 4 [unit flow and peaking factor to be confirmed with Gwinnett DWR: VERIFY]. It needs three things, and I ask the City to treat them as conditions of approval rather than as a barrier to the rezoning:

1. **A recorded permanent sewer easement** across one Village Green Court lot or along the stream corridor, which I must negotiate and which the City cannot grant.
2. **A Gwinnett DWR sewer capacity certification** covering that route. The request is made on DWR's *Sewer Capacity Certification Request* form (Rev. 07/2023), which has a "Pre-Rezoning" request type, must be signed and sealed by a design professional, and takes 10 business days, or 30 if flow monitoring is required. The written question I am submitting on that form is set out in the Technical Memoranda, Memo B.
3. **A reading of the comprehensive-plan sewer policy.** The 2026 Comprehensive Plan Amendment, which has been transmitted to the Department of Community Affairs but is not yet adopted (Section 9), carries for the Suburban-Low area the policy "Do not further extend sewer in this area" (p. 73). Phase 1 does not implicate that policy at all: it extends nothing and uses a main already on the property. Phase 2 is a 178-foot connection between two existing county mains inside the same sewer basin and adds no service area beyond this property. Whether the City reads the policy as reaching such a connection is a question of policy interpretation for staff and Council, and I ask for that reading in writing.

And if the answer is no. Phase 1 stands on its own. It is 24 lots, the amenity block, Pond 1 and the entrance, all served by gravity to the existing main, and it needs nothing from Phase 2. I ask for R-2 on the whole tract with condition 15 (phasing) and condition 16 (sewer) attached: Phase 2 cannot begin until the easement is recorded and DWR has certified the route, and if either fails the rear of the property is not built. That is a stronger commitment than a lift station would have been, and it is one the City can enforce.

Water. Ductile-iron and asbestos-cement mains lie in Arcade Road 47 to 107 feet from the front line, and the nearest hydrant is about 61 feet from the front corner. An 8-inch main is proposed for the full length of the lane. Because there is no second frontage, it cannot be looped; DWR may require an automatic flushing device or a terminal blow-off, and the hydraulic check is DWR's flow test at the Arcade Road hydrant [static and residual pressures: VERIFY]. Fire flow is 1,000 gpm for one hour under IFC 2024 Appendix B, Table B105.1(1), reduced by half to 500 gpm if the voluntary NFPA 13D sprinklers of condition 5 are installed. Hydrants are spaced at about 400 feet per Gwinnett DWR Standards (2016) §2.2.15.

8. Schools, services and fiscal effect

As an age-restricted community under the Housing for Older Persons Act, the development generates no public-school students; the Gwinnett County Public Schools cluster for the site is Camp Creek Elementary, Trickum Middle and Parkview High. By comparison, the July 2025 School Impact Analysis filed with the 2025 mixed-use case projected students from 104 condominiums at a time when Camp Creek Elementary and Parkview High were reported above capacity [VERIFY the 2025 study's figures before citing them]. I will ask GCPS Planning for a fresh letter confirming zero students for this program and stating the by-right R-1 baseline for comparison. Fire protection is by Gwinnett County Fire and Emergency Services; police protection by the City of Lilburn. Forty-one owner-occupied homes will add to the City and County digests; the adjoining 2026 digest medians are \$349,200 (King David Manor) and \$693,500 (Legends at Parkview), and the cottage product is expected to be priced between those figures — at \$420,000 to \$460,000 a home, 41 homes are roughly **\$17.2 to \$18.9 million** of new fair market value against the \$1.27 million the four parcels carry today [pricing and digest treatment: VERIFY against the current pro forma].

9. Why R-2 is the compatible district, and how this differs from the 2025 case

Zoning ordinance. Lilburn Zoning Ordinance 2023-603 §402-1 states that the R-2 district is intended for "moderate density uses consisting primarily of detached single-family uses," that "clustered and cottage housing types and other creative, flexible small lot development designs" may be appropriate, that R-2 is to be located "where there is convenient access to collector streets," and that the district "generally corresponds to the ... Comprehensive Plan Character Area designation of 'Established Residential.'" The §203 compatibility table maps the Established Residential character area to R-1 and R-2. The site is on a collector, is in Established Residential, and adjoins an existing R-2 district about 106 feet away.

Why the map amendment is necessary. I want to state this accurately, because an earlier draft of this application overstated it. §602 lists **"Single-family (cluster-cottage, creative lot configuration)" as a permitted use in R-1 as well as in R-2.** The bar in R-1 is dimensional, not a use prohibition: Table 4.1 provides no cottage-home minimum lot size and no cottage-home minimum lot width in R-1 ("N/P"), so the 3,000-SF, 50-foot cottage-home lot standard exists only in the R-2 column, and R-1's 9,000-SF, 80-foot standard cannot produce this product on a strip 235 feet wide. That is the reason for the request.

Comprehensive Plan — the adopted plan. The **2024 Comprehensive Plan is the adopted plan and governs today.** It places the site in "Established Residential," which it describes as single-family and double-family houses where "higher density development will generally not be encouraged" and where infill parcels "offer opportunities for some variety" (printed p. 35, PDF p. 37). At 4.34 du/ac of detached one-story houses, this proposal is the modest variety the plan describes — below the R-2 maximum of 8 du/ac and far below the 11 du/ac of the 2025 filing — and §203 of the zoning ordinance maps Established Residential to R-2, so **the request is consistent with the adopted plan as it stands.** The 2024 plan also carries work item H.03: "Incentivize accommodations for aging population and elements of visitability (accessible living, masters on main) within all medium and high density residential districts" (printed p. 47). R-2 is the medium-density residential district, and every home in this proposal is one story with a zero-step entry, 36-inch doors and grab-bar blocking.

Comprehensive Plan — the 2026 Amendment, and its status. The 2026 Comprehensive Plan Amendment is **not yet adopted.** The City Council held the public hearing and approved transmittal to the Georgia Department of Community Affairs on **2026-07-13**; the Atlanta Regional Commission's report window under Ga. Comp. R. & Regs. 110-12-1 closed on or about 2026-08-23, so adoption may be calendared at a regular Council meeting — 2026-09-14, 2026-10-12 or 2026-11-09 are the next opportunities [adoption date: VERIFY with the City Clerk]. It may well be adopted before this case is heard. **The request is consistent with the amendment either way.** The transmitted draft (dated 6/16/2026) places the residential properties south of the railroad in "Suburban-Low," whose land-use chart (p. 73) lists **Single-Family Cottages and Single-Family as "Appropriate"** and lists Mixed Use (vertical and horizontal), Multifamily, Office, Retail/Entertainment, Hotel/Motel, Flex/Light Industrial and Auto-oriented Retail as **"Not Appropriate,"** and whose zoning strategies call for allowing "a modest variety of housing types, including single-family cottages and infill," requiring sidewalks and improving street and block connectivity. Under the draft's own vocabulary (p. 54), a "Single-Family Cottage" is a house "typically 1,200 square feet or less" and "Single-Family" is a house "typically exceeding 1,200 square feet"; at 1,406 and 1,492 SF these homes are "Single-Family," which is likewise "Appropriate." The neighboring Suburban-Medium class (p. 71) also rates both categories "Appropriate." The Future Development Map is at p. 52 and is a raster without parcel labels; I ask staff to state in writing which class these four parcels fall in, because the two classes carry opposite sewer policies (Section 7). Whichever class applies, and whether the amendment is adopted before or after this hearing, the use proposed here is one the plan calls appropriate.

The 2025 case. In July 2025 I filed RZ-2025-01 for these parcels, requesting a Mixed-Use district for 104 condominium units and 13 commercial buildings totaling 24,400 SF. The Planning Commission tabled that case on August 28, 2025 because of a defective legal advertisement; it was never heard, no staff report or recommendation was issued, and no denial occurred, so the 12-month bar of §1003-11 does not apply [formal disposition of RZ-2025-01: VERIFY in writing with the Planning Director]. That program is retired — and the 2026 draft's own chart would rate it "Not Appropriate" here. I heard the neighborhood's objections to it (traffic, school and utility capacity, stormwater, and loss of single-family character) and this application is a different proposal in kind: single-family detached houses only, 41 rather than 104 units, no commercial floor area, one story, age-restricted, and buffered from its R-1 neighbors by the most protective buffer the table provides.

One point I do not want to be seen to have skipped. The 2024 plan's public survey is cited in support of horizontal density — "a majority of participants (64%) indicated a desire to see an increase in the number of housing units through a Horizontal Density Increase (Accessory Dwelling Units, Cottage Courts, Minimum Lot Requirement Reductions, etc.)" (p. 10). The very next sentence on that page reports that "a majority of participants (59% each) indicated Lilburn has the Right Amount of Senior Housing and Need(s) Less Medium- and High-Density Housing." Both sentences are in the record and both will be read at the hearing. My answer is that the 2023–24 opinion survey has been overtaken on this point by the City's own 2026 amendment, whose stoplight chart rates Single-Family Cottages and Single-Family "Appropriate" in this exact area, and by the market: the Cottages at Noble Village, Lilburn's only detached 55+ product, is sold out.

10. Community engagement

Since 2025 I have maintained a public project website (arcado-springs.vercel.app) with a browser-based three-dimensional walkthrough of the site and a moderated comment board on which neighbors can post questions and comments. That site presented the 2025 mixed-use concept; in August 2026 I added a cottage concept-study page with the two floor plans and renderings, and the public site will be updated to this cottage proposal before the application is advertised [VERIFY that the update is complete]. I am aware of the 2025 petition opposing the mixed-use case and its stated concerns, and Sections 3 through 8 of this letter respond to each of them with numbers. I will meet with the King David Manor, Legends at Parkview and Nantucket neighbors before the Planning Commission hearing and will attend both the Planning Commission and City Council hearings.

11. Voluntary zoning conditions

Under Lilburn Zoning Ordinance 2023-603 §1003-4(5), I voluntarily offer the following **18 conditions** and ask that they be made conditions of any approval. The list is identical to `data/voluntary-conditions.md` and to Section 9 of the Development Summary. The site plan shows 41 lots; condition 2 states a maximum of 47 so that the sealed boundary survey and final engineering can be absorbed without a change in conditions.

1. **Age restriction.** The development shall be operated as housing for older persons (55+) under 42 U.S.C. §3607(b)(2)(C) and 24 CFR §100.300–.308, with a recorded declaration, published policies and biennial verification (24 CFR §100.307(c)), enforceable by the City of Lilburn. At the 41 units shown, the 80-percent occupancy test of 24 CFR §100.305(a) together with the fractional-unit rule of §100.305(g) ($0.80 \times 41 = 32.8$, and the fraction counts into the 80 percent) requires **at least 33 of the 41 units** to be occupied by at least one person aged 55 or older, leaving at most 8 units that need not be; the declaration shall state that figure and shall recompute it if the unit count changes. Advertising and community documents shall not use "adult living," "adult community" or similar phrases (24 CFR §100.306(b)).
2. **Maximum units.** Not more than **47** detached single-family cottage homes (**41 shown**); no attached, multifamily, commercial or non-residential use other than the clubhouse and amenities serving residents.
3. **Height.** All dwellings one story; maximum ridge height 24 ft above finished grade (the plans as drawn on Sheets A-2.1 and A-2.2 are 17'-10" for Plan A and 19'-0" for Plan B above finished floor, i.e. 18'-6" and 19'-8" above finished grade at the front, finished floor being 8 in above grade); clubhouse one story.
4. **Buffers.** A 20-ft undisturbed buffer along all property lines abutting R-1 property (SW, NE and NW lines), held in a recorded buffer/conservation easement where it lies within lots and in common tracts elsewhere; supplemental evergreen planting where existing vegetation is sparse; no structures, fences or grading in the buffer except (a) approved pond and utility encroachments not exceeding 50 ft in perpendicular width and (b) **the single buffer reduction requested in the Letter of Intent §5 and dimensioned on Sheet C-2.3 (entry enlargement, to be issued with the filed set) — at the tangent point of the southwest curb return of the entrance on the Arcado Road right-of-way, where the buffer narrows to a minimum of 18'-0" instead of 20'-0", a reduction of 2.0 ft measured along the right-of-way affecting 0.1 SF of pavement.** The buffer is at its full 20-ft width everywhere else on the southwest, northeast and northwest lines. No other buffer reduction is requested, and none shall be taken without an amendment to these conditions.
5. **Fire sprinklers — voluntary offer.** Every dwelling shall be equipped with an NFPA 13D residential sprinkler system and the clubhouse sprinklered as the building code requires, with systems maintained under the HOA covenants. This is offered **voluntarily**: at 41 units the two-access requirement of IFC 2024 Appendix D107.1 as modified by Ga. Comp. R. & Regs. 120-3-3-.04 (which raises the trigger to more than 120 dwelling units) is not engaged, and sprinklers are therefore additional protection and a 50-percent fire-flow reduction (IFC 2024 Appendix B §B105.1), not a required mitigation. The applicant asks that this condition be accepted as offered; if the Gwinnett County Fire Marshal states in writing that sprinklers are unnecessary, the condition may be released by the Fire Marshal's written concurrence and City staff's approval without a further public hearing.
6. **Single access.** One driveway on Arcado Rd, located in the SW third of the frontage; no gate.
7. **No Arcado Road driveways.** No individual lot shall have direct vehicular access to Arcado Rd.
8. **Architectural standards.** Exterior walls of fiber-cement lap and board-and-batten siding with brick or stone water table/piers; architectural-shingle roofs with 8:12 main and front-wing gables, the front-wing gable carried forward to roof the covered front porch, and a 6:12 gable over the garage; front-loaded garages recessed ≥ 5 ft behind the front wall; covered front porches ≥ 6 ft deep; zero-step entries; floor plans, elevations and materials substantially as shown on the sheets submitted with the application — Sheets A-1.1 and A-1.2 (floor plans, Plans A and B), A-2.1 and A-2.2 (exterior elevations, Plans A and B), A-2.3 (exterior colour schemes and materials), A-3.0 (clubhouse, mail kiosk and entry monument sign) and A-4.0 and A-4.1 (typical lots); three approved colour schemes; no wall signs on any dwelling.
9. **Stream buffer.** The stream head and its 50-ft undisturbed buffer and 75-ft impervious setback preserved in a common "creek woods" tract, and the southwest area at u 1,230–1,280 that does not clear the stream screen preserved unlotted as a common stream-setback tract with no impervious cover; no lot, structure or pavement within 75 ft of the top of bank as field-delineated. The field study shall establish first whether and where a regulated stream begins (spring/seep or the 25-acre drainage-area point) and then the top of bank, and the plat shall show the 25-ft state buffer (O.C.G.A. §12-7-6(b)(15)) and the 50-ft/75-ft city lines separately.
10. **HOA maintenance.** A mandatory HOA owning and maintaining the private lane, sidewalks, stormwater facilities, buffers, common tracts, entry sign and lighting, with yard maintenance for all lots.
11. **Lighting.** Full-cutoff fixtures ≤ 3000 K; lane lighting mounted ≤ 14 ft with house-side shields facing R-1 lots; no court lighting for the pickleball courts.
12. **Construction hours.** Exterior construction limited to 7:00 AM – 7:00 PM Monday–Friday and 8:00 AM – 5:00 PM Saturday; none on Sundays or federal holidays (city ordinance governs if stricter).
13. **Tree survey and preservation.** A tree survey and preservation plan meeting Code Ch. 109 Art. V (16 TDU/ac) submitted with the LDP; preserved trees ≥ 12 -in DBH in the buffers, creek woods and stream-setback tract credited; phased clearing with immediate stabilization.
14. **Entry sign.** One monument sign ≤ 32 sf, ≤ 6 ft high, externally lit, standing about 19 ft from the Arcado Road right-of-way within the front amenity tract and outside the sight triangle; permitted separately under the sign regulations.

15. **Phasing.** Development in two phases at the **u ≈ 980-ft line** as shown on Sheet C-6.0 (phasing, block and open space exhibit, to be issued with the filed set); Phase 1 (24 lots) shall include the amenity block, Pond 1 and the first hammerhead turnaround shown on Sheet C-2.0. Phase 2 (17 lots) shall not commence until condition 16 is satisfied, and the existing dwelling at 4535 Arcade Road, which stands on the proposed lane alignment about 1,200 ft back from the right-of-way, shall be removed under permit before any Phase 2 land disturbance.
16. **Sanitary sewer.** Phase 1 shall be served by gravity to the existing 8-in Gwinnett County main already on the property (the Arcade Road Townhomes outfall, manhole invert 927.13), and requires **no sewer extension**. Phase 2 shall be served **only by a gravity connection**; no private pump station, private force main, private gravity sewer, grinder pump or low-pressure sewer system shall be proposed or built (Gwinnett DWR *Standard Policy for Private Developments: Condominiums, Townhomes and Subdivisions*, rev. 9/2018, allows private facilities "only ... for commercial properties under single ownership within a development"). No land-disturbance permit shall issue for Phase 2 until (a) a permanent off-site sanitary sewer easement for the gravity connection is recorded and (b) Gwinnett DWR has issued a sewer capacity certification covering the Phase-2 route. If either is not obtained, Phase 2 shall not be built and the Phase-2 area shall remain in its existing wooded condition or be re-platted as open space.
17. **Traffic impact study and access management.** The applicant shall submit a **Level-1 traffic impact study** prepared to Gwinnett DOT *Traffic Impact Study Guidelines* (2023) with the zoning application, based on **new 24-hour traffic counts** on Arcade Rd obtained from an approved contractor (the 2017 GDOT count does not satisfy the guidelines' two-year currency rule); shall furnish an intersection-sight-distance certification by a Georgia Professional Engineer for the single entrance (390 ft each way at 35 mph, Gwinnett UDO §900-40.4, Table 900.2); and shall obtain a Gwinnett County DOT driveway/encroachment permit before any land disturbance.
18. **Sidewalks and transit.** A 5-ft sidewalk along the full Arcade Rd frontage and 5-ft sidewalks within the lane tract as shown — on the northeast side for the full length of the lane and on the southwest side from u 530, where the tract first reaches the 36 ft that two walks require; the applicant will cooperate with the county and city on closing the sidewalk gap to the Killian Hill Rd bus stop.

12. Closing

The Cottages at Arcade Springs puts 41 one-story, owner-occupied, age-restricted houses on a strip of land that the zoning ordinance, the adopted 2024 Comprehensive Plan and the transmitted 2026 amendment all describe as the place for exactly this product: single-family, modest in density, on a collector, next to an existing R-2 district, and buffered from its R-1 neighbors by the most protective buffer the table provides. It adds no students, about 12 peak-hour trips, and no commercial floor area, and its first phase is served by a sewer main that is already inside the property line.

I have addressed the six criteria of §1003-7 in the attached Standards Governing Exercise of the Zoning Power form. I would welcome a pre-application conference to resolve, before the filing deadline: the §313(1) buffer interpretation and the buffer reduction requested in Section 5; the Phase-2 sewer route, the comprehensive-plan sewer policy and the Future Development Map class in Section 7 and Section 9; the fire-access package and the 22-foot lane width in Section 3; the traffic-study level in Section 3; and the disposition of RZ-2025-01 in Section 9. The filing deadline for the October 22, 2026 Planning Commission and November 9, 2026 City Council hearings is September 25, 2026.

I respectfully request approval of the rezoning to R-2 subject to the conditions in Section 11. Thank you for your consideration.

Respectfully,

Mohammed Awad Applicant; owner of Parcels 6123 033 and 6123 015; contract purchaser of Parcels 6123 014 and 6123 162 [contract status: VERIFY before filing]

Attachments: Rezoning Application Form; Standards Governing Exercise of the Zoning Power; Conflict of Interest Certification and Campaign Contribution Disclosure; Notarized Applicant and Property Owner Certifications; Typed Legal Description (DRAFT — pending RLS seal); Site Plan set, Sheets C-0.0 through C-8.0 (DRAFT — NOT SEALED; to be superseded by sealed work); Architectural Sheets A-1.1 through A-4.1 — floor plans, exterior elevations of every street-visible side, the colors and materials board, the clubhouse, mail kiosk and entry monument, and the typical lots; Development Summary; Technical Memoranda A–G; List of Adjoining Property Owners (31 parcels plus the Gwinnett County Arcade Road right-of-way). **Boundary Survey — to be filed when the Georgia registered land surveyor releases it; not attached to this submittal [VERIFY the Director's requirement].**

Sources: Lilburn Zoning Ordinance 2023-603 §§203, 313, 318, 319, 402, 602, 1003, 1005, Tables 4.1, 4.2, 8.1; Lilburn Development Regulations App. B Art. 5 §5.9; City of Lilburn 2026 Application Instructions — Rezoning/SUP/CIC; Lilburn 2024

Comprehensive Plan, printed pp. 10, 35 (PDF p. 37), 47; 2026 Comprehensive Plan Amendment, draft of 6/16/2026 transmitted to DCA 2026-07-13, pp. 52, 54, 71, 73; Ga. Comp. R. & Regs. 110-12-1; Gwinnett County GIS parcel, zoning, sewer, stream, floodplain and soils layers (queried 2026-08-28); Gwinnett Tax Assessor 2026 digest; USGS 3DEP; ITE Trip Generation Manual, 11th ed., LUC 251; Gwinnett DOT Traffic Impact Study Guidelines (2023); Gwinnett DOT Criteria and Guidelines for Left Turn Lanes (2007-06-20); Gwinnett County UDO §900-40.4 Table 900.2 and §900-40.5 Table 900.3; Gwinnett DWR Standard Policy for Private Developments: Condominiums, Townhomes and Subdivisions (rev. 9/2018), Developer Pump Station Standards (Jan 2014, WSR-24) §1.3.1(A), Water and Sewer Standards (April 2016), Sewer Capacity Certification Request (Rev. 07/2023); IFC 2024 as adopted and modified by Ga. Comp. R. & Regs. 120-3-3-.04, Appendices B and D; O.C.G.A. §§12-7-6(b)(15), 25-2-12(e)(4); 42 U.S.C. §3607(b)(2)(C), 24 CFR §§100.305–.307; RZ-2025-01 packet; site-plan data file data/layout.json and architectural data file data/plans.json.

Disclaimer: *This is an AI-generated analysis for preliminary planning purposes. All findings must be verified by a licensed professional before use in design, permitting, or regulatory submissions.*

Voluntary zoning conditions offered with the rezoning application

Governing text: docs/02-development-summary.md section 9. This file is generated by tools/sync_conditions.py; edit the development summary, not this file.

1. **Age restriction.** The development shall be operated as housing for older persons (55+) under 42 U.S.C. §3607(b)(2)(C) and 24 CFR §100.300–308, with a recorded declaration, published policies and biennial verification (24 CFR §100.307(c)), enforceable by the City of Lilburn. At the 41 units shown, the 80-percent occupancy test of 24 CFR §100.305(a) together with the fractional-unit rule of §100.305(g) ($0.80 \times 41 = 32.8$, and the fraction counts into the 80 percent) requires **at least 33 of the 41 units** to be occupied by at least one person aged 55 or older, leaving at most 8 units that need not be; the declaration shall state that figure and shall recompute it if the unit count changes. Advertising and community documents shall not use "adult living," "adult community" or similar phrases (24 CFR §100.306(b)).
2. **Maximum units.** Not more than **47** detached single-family cottage homes (**41 shown**); no attached, multifamily, commercial or non-residential use other than the clubhouse and amenities serving residents.
3. **Height.** All dwellings one story; maximum ridge height 24 ft above finished grade (the plans as drawn on Sheets A-2.1 and A-2.2 are 17'-10" for Plan A and 19'-0" for Plan B above finished floor, i.e. 18'-6" and 19'-8" above finished grade at the front, finished floor being 8 in above grade); clubhouse one story.
4. **Buffers.** A 20-ft undisturbed buffer along all property lines abutting R-1 property (SW, NE and NW lines), held in a recorded buffer/conservation easement where it lies within lots and in common tracts elsewhere; supplemental evergreen planting where existing vegetation is sparse; no structures, fences or grading in the buffer except (a) approved pond and utility encroachments not exceeding 50 ft in perpendicular width and (b) **the single buffer reduction requested in the Letter of Intent §5 and dimensioned on Sheet C-2.3 (entry enlargement, to be issued with the filed set) — at the tangent point of the southwest curb return of the entrance on the Arcado Road right-of-way, where the buffer narrows to a minimum of 18'-0" instead of 20'-0", a reduction of 2.0 ft measured along the right-of-way affecting 0.1 SF of pavement.** The buffer is at its full 20-ft width everywhere else on the southwest, northeast and northwest lines. No other buffer reduction is requested, and none shall be taken without an amendment to these conditions.
5. **Fire sprinklers — voluntary offer.** Every dwelling shall be equipped with an NFPA 13D residential sprinkler system and the clubhouse sprinklered as the building code requires, with systems maintained under the HOA covenants. This is offered **voluntarily**: at 41 units the two-access requirement of IFC 2024 Appendix D107.1 as modified by Ga. Comp. R. & Regs. 120-3-3-.04 (which raises the trigger to more than 120 dwelling units) is not engaged, and sprinklers are therefore additional protection and a 50-percent fire-flow reduction (IFC 2024 Appendix B §B105.1), not a required mitigation. The applicant asks that this condition be accepted as offered; if the Gwinnett County Fire Marshal states in writing that sprinklers are unnecessary, the condition may be released by the Fire Marshal's written concurrence and City staff's approval without a further public hearing.
6. **Single access.** One driveway on Arcado Rd, located in the SW third of the frontage; no gate.
7. **No Arcado Road driveways.** No individual lot shall have direct vehicular access to Arcado Rd.
8. **Architectural standards.** Exterior walls of fiber-cement lap and board-and-batten siding with brick or stone water table/piers; architectural-shingle roofs with 8:12 main and front-wing gables, the front-wing gable carried forward to roof the covered front porch, and a 6:12 gable over the garage; front-loaded garages recessed ≥ 5 ft behind the front wall; covered front porches ≥ 6 ft deep; zero-step entries; floor plans, elevations and materials substantially as shown on the sheets submitted with the application — Sheets A-1.1 and A-1.2 (floor plans, Plans A and B), A-2.1 and A-2.2 (exterior elevations, Plans A and B), A-2.3 (exterior colour schemes and materials), A-3.0 (clubhouse, mail kiosk and entry monument sign) and A-4.0 and A-4.1 (typical lots); three approved colour schemes; no wall signs on any dwelling.
9. **Stream buffer.** The stream head and its 50-ft undisturbed buffer and 75-ft impervious setback preserved in a common "creek woods" tract, and the southwest area at u 1,230–1,280 that does not clear the stream screen preserved unlotted as a common stream-setback tract with no impervious cover; no lot, structure or pavement within 75 ft of the top of bank as field-delineated. The field study shall establish first whether and where a regulated stream begins (spring/seep or the 25-acre drainage-area point) and then the top of bank, and the plat shall show the 25-ft state buffer (O.C.G.A. §12-7-6(b)(15)) and the 50-ft/75-ft city lines separately.
10. **HOA maintenance.** A mandatory HOA owning and maintaining the private lane, sidewalks, stormwater facilities, buffers, common tracts, entry sign and lighting, with yard maintenance for all lots.
11. **Lighting.** Full-cutoff fixtures ≤ 3000 K; lane lighting mounted ≤ 14 ft with house-side shields facing R-1 lots; no court lighting for the pickleball courts.
12. **Construction hours.** Exterior construction limited to 7:00 AM – 7:00 PM Monday–Friday and 8:00 AM – 5:00 PM Saturday; none on Sundays or federal holidays (city ordinance governs if stricter).

13. **Tree survey and preservation.** A tree survey and preservation plan meeting Code Ch. 109 Art. V (16 TDU/ac) submitted with the LDP; preserved trees ≥ 12 -in DBH in the buffers, creek woods and stream-setback tract credited; phased clearing with immediate stabilization.
14. **Entry sign.** One monument sign ≤ 32 sf, ≤ 6 ft high, externally lit, standing about 19 ft from the Arcade Road right-of-way within the front amenity tract and outside the sight triangle; permitted separately under the sign regulations.
15. **Phasing.** Development in two phases at the **u $\approx$ 980-ft line** as shown on Sheet C-6.0 (phasing, block and open space exhibit, to be issued with the filed set); Phase 1 (24 lots) shall include the amenity block, Pond 1 and the first hammerhead turnaround shown on Sheet C-2.0. Phase 2 (17 lots) shall not commence until condition 16 is satisfied, and the existing dwelling at 4535 Arcade Road, which stands on the proposed lane alignment about 1,200 ft back from the right-of-way, shall be removed under permit before any Phase 2 land disturbance.
16. **Sanitary sewer.** Phase 1 shall be served by gravity to the existing 8-in Gwinnett County main already on the property (the Arcade Road Townhomes outfall, manhole invert 927.13), and requires **no sewer extension**. Phase 2 shall be served **only by a gravity connection**; no private pump station, private force main, private gravity sewer, grinder pump or low-pressure sewer system shall be proposed or built (Gwinnett DWR *Standard Policy for Private Developments: Condominiums, Townhomes and Subdivisions*, rev. 9/2018, allows private facilities "only ... for commercial properties under single ownership within a development"). No land-disturbance permit shall issue for Phase 2 until (a) a permanent off-site sanitary sewer easement for the gravity connection is recorded and (b) Gwinnett DWR has issued a sewer capacity certification covering the Phase-2 route. If either is not obtained, Phase 2 shall not be built and the Phase-2 area shall remain in its existing wooded condition or be re-platted as open space.
17. **Traffic impact study and access management.** The applicant shall submit a **Level-1 traffic impact study** prepared to Gwinnett DOT *Traffic Impact Study Guidelines* (2023) with the zoning application, based on **new 24-hour traffic counts** on Arcade Rd obtained from an approved contractor (the 2017 GDOT count does not satisfy the guidelines' two-year currency rule); shall furnish an intersection-sight-distance certification by a Georgia Professional Engineer for the single entrance (390 ft each way at 35 mph, Gwinnett UDO §900-40.4, Table 900.2); and shall obtain a Gwinnett County DOT driveway/encroachment permit before any land disturbance.
18. **Sidewalks and transit.** A 5-ft sidewalk along the full Arcade Rd frontage and 5-ft sidewalks within the lane tract as shown — on the northeast side for the full length of the lane and on the southwest side from u 530, where the tract first reaches the 36 ft that two walks require; the applicant will cooperate with the county and city on closing the sidewalk gap to the Killian Hill Rd bus stop.

Disclaimer: *This is an AI-generated analysis for preliminary planning purposes. All findings must be verified by a licensed professional before use in design, permitting, or regulatory submissions.*

title	Legal Description (Item 7) — Cottages at Arcado Springs assemblage
date	2026-08-28
address	4535, 4537, 4539 & 4541 Arcado Rd SW, Lilburn, GA 30047 — Land Lot 123, 6th District, Gwinnett County (PINs R6123 033 / 015 / 014 / 162)
status	DRAFT — derived from Gwinnett County GIS parcel fabric; to be replaced by a metes-and-bounds description from a boundary survey sealed by a Georgia Registered Land Surveyor (§1003-4.3–4)
generator	tools/legal.py (output verified 2026-08-28; see verify/legal-output.md)

DRAFT — derived from Gwinnett County GIS parcel fabric; to be replaced by a metes-and-bounds description from a boundary survey sealed by a Georgia Registered Land Surveyor (§1003-4.3–4)

Application item: Lilburn 2026 Application Instructions item 7 ("Legal Description: ... typed and describe, by metes and bounds, only the property to be Rezoned") and Lilburn Zoning Ordinance 2023-603 §1003-4.3 ("Metes and bounds legal description of the property") together with §1003-4.4 ("Boundary survey plat of the property; provided, however, that where no survey is available the Director may, but is not obligated to, accept a map of the subject property from the Gwinnett County Tax Assessors or other reliable source").

What this is. A typed metes-and-bounds description of the four-parcel assemblage as **one tract**, computed from the Gwinnett County GIS parcel polygons (data/site-parcels-2240.json, queried 2026-08-28) by dropping the four interior lines (033/015, 015/014, 015/162, 014/162) and walking the outer boundary clockwise from the Arcado Road right-of-way corner. It exists so that the application can be assembled and checked now; it is **not a survey**, and the city should not be asked to accept it as one. Commission the boundary survey (FACTS §0: first spend, together with the sewer-invert survey) and replace this document with the surveyor's description before filing. Everything a Georgia RLS must seal is marked DRAFT.

1. Property identification

Item	Value	Source
Land lot / district	Land Lot 123, 6th District , City of Lilburn, Gwinnett County, Georgia	Gwinnett parcel numbering (R6123); FACTS §1
Tract	Assemblage of four contiguous tax parcels fronting the northwesterly side of Arcado Road SW	Gwinnett GIS parcel layer
Area — GIS-calculated	417,177.6 sf = 9.577 ac (sum of the four GIS polygon areas; outer-boundary shoelace gives the same figure)	tools/legal.py
Area — deeded (Tax Assessor digest)	9.44 ac (0.99 + 2.00 + 2.00 + 4.45) — use on the application form ; the sealed survey governs	2026 digest DEEDEDACREAGE; FACTS §1
Coordinate basis	Georgia State Plane, West Zone (EPSG/SR 2240), NAD83, US survey feet ; bearings are grid bearings	Gwinnett GIS service

PIN (Tax Assessor)	Property address	Deeded ac	GIS ac	Owner of record (2026 digest)	Digest deed reference (VERIFY against recorded deed)
R6123 033	4541 Arcado Rd SW	0.99	0.985	AWAD MOHAMMED	Deed Book 61414, Page 680 (2024-09-07)
R6123 015	4539 Arcado Rd SW	2.00	1.985	AWAD MOHAMMED	Deed Book 61414, Page 683 (2024-09-06)
R6123 014	4537 Arcado Rd SW	2.00	1.994	MENDEZ SANTOS C & ROBLERO DE LEON LESVIA ROSARIO	Deed Book 60727, Page 25 (2023-07-20)
R6123 162	4535 Arcado Rd SW	4.45	4.614	MENDEZ SANTOS C & ROBLERO DE LEON LESVIA ROSARIO	Deed Book 60727, Page 25 (2023-07-20)
Assemblage		9.44	9.577		

The deed references are the digest's DOC1REF field (book/page) and are listed only to help the surveyor pull the vesting deeds; they are **not** verified against the Clerk of Superior Court record (GSCCCA).

2. Basis of bearings, distances and the point of beginning

- **Datum / projection:** NAD83 Georgia West Zone (SR 2240), US survey feet. Bearings are **grid** bearings from the GIS coordinates; no convergence or combined-scale-factor correction has been applied (both are small at this longitude but must be

stated by the surveyor — **VERIFY**). Distances are grid distances to 0.01 ft; the surveyor will report ground distances.

- **Point of beginning (POB):** the corner common to Tax Parcel R6123 162 and the northwesterly right-of-way line of Arcado Road (the tract's easterly corner), grid **N 1,411,431.50 / E 2,310,954.00**. This is also the origin of the site-local (u, v) system used on every drawing in this package (u along N 28°43' W, v toward the northeasterly line — FACTS §1), so each course below carries its local end-point too.
- **Direction of travel:** clockwise — southwesterly along the Arcado Road right-of-way, northwesterly up the southwesterly line (Legends at Parkview), northeasterly across the rear (Nantucket), southeasterly down the northeasterly line (King David Manor and 4531 Arcado Rd) back to the POB.
- **Adjoiner calls** are derived from the GIS rings of the 31 adjoining parcels and the right-of-way polygon (data/site-context-local.json, data/owners-found.json); every course was matched to exactly one adjoiner ring within 0.04 ft. Lot and block numbers are the digest legal descriptions (LEGAL1) and the GIS LOT field — the subdivision plats (King David Manor Unit 1, plat book S page 159; Nantucket; Legends at Parkview, plat book 118 page 187) must be pulled by the surveyor.
- **Precision:** Gwinnett's parcel fabric is a compiled GIS layer, not a survey; positional accuracy of a few feet should be assumed. The 9.577-ac GIS figure versus the 9.44-ac deeded figure (a 1.4 % / 0.14-ac difference) will be resolved only by the survey.

3. Description (DRAFT)

ALL THAT TRACT OR PARCEL OF LAND lying and being in Land Lot 123 of the 6th District, City of Lilburn, Gwinnett County, Georgia, being the whole of Gwinnett County Tax Parcels R6123 033 (4541 Arcado Road SW), R6123 015 (4539 Arcado Road SW), R6123 014 (4537 Arcado Road SW) and R6123 162 (4535 Arcado Road SW), taken together as one tract, containing **9.577 acres (417,178 square feet) by Gwinnett County GIS computation (9.44 acres by the Tax Assessor's deeded acreage)**, and being more particularly described as follows:

TO FIND THE POINT OF BEGINNING, commence at the corner common to Tax Parcel R6123 162 and Lot 1, Block A, King David Manor Unit 1 (4531 Arcado Road SW, PIN R6123 017) on the northwesterly right-of-way line of Arcado Road (right-of-way width varies; county-maintained collector), said corner having Georgia West Zone (NAD83, US survey feet) grid coordinates **North 1,411,431.50, East 2,310,954.00**, and being the **POINT OF BEGINNING**;

thence along the northwesterly right-of-way line of Arcado Road (R/W width varies) the following 14 courses: S 48°41'35" W a distance of 21.97 feet; thence S 50°35'58" W a distance of 18.12 feet; thence S 51°04'08" W a distance of 4.18 feet; thence S 50°41'21" W a distance of 22.30 feet; thence S 50°56'17" W a distance of 22.22 feet; thence S 51°41'45" W a distance of 21.98 feet; thence S 52°28'47" W a distance of 8.83 feet; thence S 52°27'51" W a distance of 12.93 feet; thence S 53°31'16" W a distance of 22.08 feet; thence S 54°09'44" W a distance of 22.20 feet; thence S 54°57'19" W a distance of 21.99 feet; thence S 55°19'22" W a distance of 22.19 feet; thence S 55°42'53" W a distance of 16.64 feet; thence N 28°43'03" W a distance of 10.04 feet to a point, being a corner common to Lot 79, Block A, Legends at Parkview — 303 Fieldhouse Cir SW (PIN R6123 312);

thence along the common line with Lot 79, Block A, Legends at Parkview — 303 Fieldhouse Cir SW (PIN R6123 312) N 28°43'03" W a distance of 199.43 feet to a point, being a corner common to Lot 78, Block A, Legends at Parkview — 260 Village Green Ct SW (PIN R6123 311);

thence along the common line with Lot 78, Block A, Legends at Parkview — 260 Village Green Ct SW (PIN R6123 311) N 28°43'03" W a distance of 98.94 feet to a point, being a corner common to Lot 77, Block A, Legends at Parkview — 250 Village Green Ct SW (PIN R6123 310);

thence along the common line with Lot 77, Block A, Legends at Parkview — 250 Village Green Ct SW (PIN R6123 310) the following 2 courses: N 28°43'02" W a distance of 65.16 feet; thence N 28°50'16" W a distance of 34.90 feet to a point, being a corner common to Lot 76, Block A, Legends at Parkview — 240 Village Green Ct SW (PIN R6123 309);

thence along the common line with Lot 76, Block A, Legends at Parkview — 240 Village Green Ct SW (PIN R6123 309) N 28°39'26" W a distance of 99.27 feet to a point, being a corner common to Lot 75, Block A, Legends at Parkview — 230 Village Green Ct SW (PIN R6123 308);

thence along the common line with Lot 75, Block A, Legends at Parkview — 230 Village Green Ct SW (PIN R6123 308) N 28°42'16" W a distance of 100.83 feet to a point, being a corner common to Lot 74, Block A, Legends at Parkview — 220 Village Green Ct SW (PIN R6123 307);

thence along the common line with Lot 74, Block A, Legends at Parkview — 220 Village Green Ct SW (PIN R6123 307) the following 3 courses: N 28°42'15" W a distance of 64.60 feet; thence N 28°42'14" W a distance of 17.92 feet; thence N 28°45'55" W a distance of 17.29 feet to a point, being a corner common to Lot 73, Block A, Legends at Parkview — 210 Village Green Ct SW (PIN R6123 306);

thence along the common line with Lot 73, Block A, Legends at Parkview — 210 Village Green Ct SW (PIN R6123 306) N 28°45'58" W a distance of 100.82 feet to a point, being a corner common to Lot 72, Block A, Legends at Parkview — 200 Village Green Ct SW (PIN R6123 305);

thence along the common line with Lot 72, Block A, Legends at Parkview — 200 Village Green Ct SW (PIN R6123 305) N 28°45'57" W a distance of 201.77 feet to a point, being a corner common to Lot 71, Block A, Legends at Parkview — 190 Village Green Ct SW (PIN R6123 304);

thence along the common line with Lot 71, Block A, Legends at Parkview — 190 Village Green Ct SW (PIN R6123 304) the following 2 courses: N 28°45'57" W a distance of 106.63 feet; thence N 28°47'46" W a distance of 52.82 feet to a point, being a corner common to Lot 70, Block A, Legends at Parkview — 180 Village Green Ct SW (PIN R6123 303);

thence along the common line with Lot 70, Block A, Legends at Parkview — 180 Village Green Ct SW (PIN R6123 303) the following 2 courses: N 28°47'42" W a distance of 79.26 feet; thence N 28°47'44" W a distance of 81.46 feet to a point, being a corner common to Lot 69, Block A, Legends at Parkview — 170 Village Green Ct SW (PIN R6123 302);

thence along the common line with Lot 69, Block A, Legends at Parkview — 170 Village Green Ct SW (PIN R6123 302) the following 2 courses: N 28°47'44" W a distance of 52.20 feet; thence N 28°50'50" W a distance of 101.93 feet to a point, being a corner common to Lot 68, Block A, Legends at Parkview — 160 Village Green Ct SW (PIN R6123 301);

thence along the common line with Lot 68, Block A, Legends at Parkview — 160 Village Green Ct SW (PIN R6123 301) the following 3 courses: N 28°50'50" W a distance of 65.88 feet; thence N 28°50'49" W a distance of 39.97 feet; thence N 29°19'37" W a distance of 6.06 feet to a point, being a corner common to Lot 67, Block A, Legends at Parkview — 150 Village Green Ct SW (PIN R6123 300);

thence along the common line with Lot 67, Block A, Legends at Parkview — 150 Village Green Ct SW (PIN R6123 300) N 29°19'37" W a distance of 159.83 feet to a point at the westerly corner of the tract herein described, being a corner common to Lot 3, Block A, Nantucket — 4540 Nantucket Dr SW (PIN R6123 038);

thence along the common line with Lot 3, Block A, Nantucket — 4540 Nantucket Dr SW (PIN R6123 038) N 60°38'26" E a distance of 4.59 feet to a point, being a corner common to Lot 2, Block A, Nantucket — 4530 Nantucket Dr SW (PIN R6123 037);

thence along the common line with Lot 2, Block A, Nantucket — 4530 Nantucket Dr SW (PIN R6123 037) the following 2 courses: N 60°32'43" E a distance of 88.72 feet; thence N 61°35'48" E a distance of 13.93 feet to a point, being a corner common to Lot 1, Block A, Nantucket — 4520 Nantucket Dr SW (PIN R6123 036);

thence along the common line with Lot 1, Block A, Nantucket — 4520 Nantucket Dr SW (PIN R6123 036) N 61°27'20" E a distance of 139.17 feet to a point at the northerly corner of the tract herein described, being a corner common to Lot 16, Block A, King David Manor Unit 1 — 145 King David Dr SW (PIN R6123 032);

thence along the common line with Lot 16, Block A, King David Manor Unit 1 — 145 King David Dr SW (PIN R6123 032) the following 2 courses: S 28°25'25" E a distance of 113.42 feet; thence S 28°31'02" E a distance of 0.48 feet to a point, being a corner common to Lot 15, Block A, King David Manor Unit 1 — 155 King David Dr SW (PIN R6123 031);

thence along the common line with Lot 15, Block A, King David Manor Unit 1 — 155 King David Dr SW (PIN R6123 031) the following 2 courses: S 28°25'25" E a distance of 118.95 feet; thence S 28°24'09" E a distance of 0.75 feet to a point, being a corner common to Lot 14, Block A, King David Manor Unit 1 — 165 King David Dr SW (PIN R6123 030);

thence along the common line with Lot 14, Block A, King David Manor Unit 1 — 165 King David Dr SW (PIN R6123 030) S 28°25'27" E a distance of 99.79 feet to a point, being a corner common to Lot 13, Block A, King David Manor Unit 1 — 175 King David Dr SW (PIN R6123 029);

thence along the common line with Lot 13, Block A, King David Manor Unit 1 — 175 King David Dr SW (PIN R6123 029) S 28°25'26" E a distance of 101.08 feet to a point, being a corner common to Lot 12, Block A, King David Manor Unit 1 — 185 King David Dr SW (PIN R6123 028);

thence along the common line with Lot 12, Block A, King David Manor Unit 1 — 185 King David Dr SW (PIN R6123 028) S 28°25'27" E a distance of 100.67 feet to a point, being a corner common to Lot 11, Block A, King David Manor Unit 1 — 195 King David Dr SW (PIN R6123 027);

thence along the common line with Lot 11, Block A, King David Manor Unit 1 — 195 King David Dr SW (PIN R6123 027) S 28°25'25" E a distance of 101.45 feet to a point, being a corner common to Lot 10, Block A, King David Manor Unit 1 — 129 King David Dr SW (PIN R6123 026);

thence along the common line with Lot 10, Block A, King David Manor Unit 1 — 129 King David Dr SW (PIN R6123 026) S 28°25'26" E a distance of 98.61 feet to a point, being a corner common to Lot 9, Block A, King David Manor Unit 1 — 215 King David Dr SW (PIN R6123 025);

thence along the common line with Lot 9, Block A, King David Manor Unit 1 — 215 King David Dr SW (PIN R6123 025) S 28°25'26" E a distance of 101.95 feet to a point, being a corner common to Lot 8, Block A, King David Manor Unit 1 — 225 King David Dr SW (PIN R6123 024);

thence along the common line with Lot 8, Block A, King David Manor Unit 1 — 225 King David Dr SW (PIN R6123 024) S 28°25'26" E a distance of 100.18 feet to a point, being a corner common to Lot 7, Block A, King David Manor Unit 1 — 235 King David Dr SW (PIN R6123 023);

thence along the common line with Lot 7, Block A, King David Manor Unit 1 — 235 King David Dr SW (PIN R6123 023) S 28°25'26" E a distance of 101.04 feet to a point, being a corner common to Lot 6, Block A, King David Manor Unit 1 — 245 King David Dr SW (PIN R6123 022);

thence along the common line with Lot 6, Block A, King David Manor Unit 1 — 245 King David Dr SW (PIN R6123 022) S 28°25'25" E a distance of 99.38 feet to a point, being a corner common to Lot 5, Block A, King David Manor Unit 1 — 255 King David Dr SW (PIN R6123 021);

thence along the common line with Lot 5, Block A, King David Manor Unit 1 — 255 King David Dr SW (PIN R6123 021) S 28°25'26" E a distance of 154.31 feet to a point, being a corner common to Lot 4, Block A, King David Manor Unit 1 — 265 King David Dr SW (PIN R6123 020);

thence along the common line with Lot 4, Block A, King David Manor Unit 1 — 265 King David Dr SW (PIN R6123 020) S 28°25'27" E a distance of 175.47 feet to a point, being a corner common to Lot 1, Block A, King David Manor Unit 1 — 4531 Arcado Rd SW (PIN R6123 017);

thence along the common line with Lot 1, Block A, King David Manor Unit 1 — 4531 Arcado Rd SW (PIN R6123 017) S 28°25'27" E a distance of 254.15 feet to the northwesterly right-of-way line of Arcado Road and the POINT OF BEGINNING.

Said tract containing 417,178 square feet or 9.577 acres, more or less, by GIS coordinate computation (9.44 acres by the Gwinnett County Tax Assessor's deeded acreage), and being the same property shown as Tax Parcels R6123 033, 015, 014 and 162 on the Gwinnett County GIS parcel map retrieved 2026-08-28.

Subject to the existing 8-inch Gwinnett County sanitary sewer main ("Arcado Road Townhomes Sewer Outfall") and its easement — the "EX. 20' SSE" of the 2025 hydrology sheet — running along the southwesterly side of Tax Parcel R6123 033 from Arcado Road to a manhole near local (u 272, v -224) (FACTS §2), the Arcado Road right-of-way and any other easements, restrictions and matters of record that the boundary survey and a title examination will disclose (**VERIFY**: recorded easement book/page).

4. Courses (output of `tools/legal.py`)

No.	Side	Bearing (grid)	Distance (ft)	To: E	To: N	To: u	To: v	GIS parcel	Along (GIS adjoiner)
1	frontage	S 48°41'35" W	21.97	2,310,93 7.50	1,411,41 7.00	-4.8	-21.4	6123 162	the northwesterly right-of-way line of Arcado Road (R/W width varies)
2	frontage	S 50°35'58" W	18.12	2,310,92 3.50	1,411,40 5.50	-8.1	-39.2	6123 162	the northwesterly right-of-way line of Arcado Road (R/W width varies)
3	frontage	S 51°04'08" W	4.18	2,310,92 0.25	1,411,40 2.87	-8.9	-43.4	6123 015	the northwesterly right-of-way line of Arcado Road (R/W width varies)
4	frontage	S 50°41'21" W	22.30	2,310,90 3.00	1,411,38 8.75	-13.0	-65.3	6123 015	the northwesterly right-of-way line of Arcado Road (R/W width varies)
5	frontage	S 50°56'17" W	22.22	2,310,88 5.75	1,411,37 4.75	-17.0	-87.1	6123 015	the northwesterly right-of-way line of Arcado Road (R/W width varies)
6	frontage	S 51°41'45" W	21.98	2,310,86 8.50	1,411,36 1.13	-20.6	-108.8	6123 015	the northwesterly right-of-way line of Arcado Road (R/W width varies)
7	frontage	S 52°28'47" W	8.83	2,310,86 1.50	1,411,35 5.75	-22.0	-117.5	6123 015	the northwesterly right-of-way line of Arcado Road (R/W width varies)
8	frontage	S 52°27'51" W	12.93	2,310,85 1.25	1,411,34 7.87	-24.0	-130.3	6123 033	the northwesterly right-of-way line of Arcado Road (R/W width varies)

No.	Side	Bearing (grid)	Distance (ft)	To: E	To: N	To: u	To: v	GIS parcel	Along (GIS adjoiner)
9	frontage	S 53°31'16" W	22.08	2,310,83 3.50	1,411,33 4.75	-26.9	-152.2	6123 033	the northwesterly right-of-way line of Arcade Road (R/W width varies)
10	frontage	S 54°09'44" W	22.20	2,310,81 5.50	1,411,32 1.75	-29.7	-174.2	6123 033	the northwesterly right-of-way line of Arcade Road (R/W width varies)
11	frontage	S 54°57'19" W	21.99	2,310,79 7.50	1,411,30 9.13	-32.1	-196.1	6123 033	the northwesterly right-of-way line of Arcade Road (R/W width varies)
12	frontage	S 55°19'22" W	22.19	2,310,77 9.25	1,411,29 6.50	-34.4	-218.1	6123 033	the northwesterly right-of-way line of Arcade Road (R/W width varies)
13	frontage	S 55°42'53" W	16.64	2,310,76 5.50	1,411,28 7.13	-36.0	-234.7	6123 033	the northwesterly right-of-way line of Arcade Road (R/W width varies)
14	SW	N 28°43'03" W	10.04	2,310,76 0.67	1,411,29 5.93	-26.0	-234.7	6123 033	the northwesterly right-of-way line of Arcade Road (R/W width varies)
15	SW	N 28°43'03" W	199.43	2,310,66 4.85	1,411,47 0.84	173.4	-234.7	6123 033	Lot 79, Block A, Legends at Parkview — 303 Fieldhouse Cir SW (PIN R6123 312)
16	SW	N 28°43'03" W	98.94	2,310,61 7.31	1,411,55 7.60	272.4	-234.7	6123 033	Lot 78, Block A, Legends at Parkview — 260 Village Green Ct SW (PIN R6123 311)
17	SW	N 28°43'02" W	65.16	2,310,58 6.00	1,411,61 4.75	337.5	-234.7	6123 033	Lot 77, Block A, Legends at Parkview — 250 Village Green Ct SW (PIN R6123 310)
18	SW	N 28°50'16" W	34.90	2,310,56 9.17	1,411,64 5.32	372.4	-234.7	6123 015	Lot 77, Block A, Legends at Parkview — 250 Village Green Ct SW (PIN R6123 310)
19	SW	N 28°39'26" W	99.27	2,310,52 1.56	1,411,73 2.43	471.7	-234.6	6123 015	Lot 76, Block A, Legends at Parkview — 240 Village Green Ct SW (PIN R6123 309)
20	SW	N 28°42'16" W	100.83	2,310,47 3.13	1,411,82 0.87	572.5	-234.6	6123 015	Lot 75, Block A, Legends at Parkview — 230 Village Green Ct SW (PIN R6123 308)
21	SW	N 28°42'15" W	64.60	2,310,44 2.11	1,411,87 7.53	637.1	-234.6	6123 015	Lot 74, Block A, Legends at Parkview — 220 Village Green Ct SW (PIN R6123 307)
22	SW	N 28°42'14" W	17.92	2,310,43 3.50	1,411,89 3.25	655.1	-234.6	6123 014	Lot 74, Block A, Legends at Parkview — 220 Village Green Ct SW (PIN R6123 307)
23	SW	N 28°45'55" W	17.29	2,310,42 5.18	1,411,90 8.41	672.4	-234.6	6123 014	Lot 74, Block A, Legends at Parkview — 220 Village Green Ct SW (PIN R6123 307)
24	SW	N 28°45'58" W	100.82	2,310,37 6.66	1,411,99 6.79	773.2	-234.7	6123 014	Lot 73, Block A, Legends at Parkview — 210 Village Green Ct SW (PIN R6123 306)
25	SW	N 28°45'57" W	201.77	2,310,27 9.56	1,412,17 3.65	974.9	-234.8	6123 014	Lot 72, Block A, Legends at Parkview — 200 Village Green Ct SW (PIN R6123 305)
26	SW	N 28°45'57" W	106.63	2,310,22 8.25	1,412,26 7.13	1081.6	-234.9	6123 014	Lot 71, Block A, Legends at Parkview — 190 Village Green Ct SW (PIN R6123 304)
27	SW	N 28°47'46" W	52.82	2,310,20 2.81	1,412,31 3.41	1134.4	-235.0	6123 162	Lot 71, Block A, Legends at Parkview — 190 Village Green Ct SW (PIN R6123 304)
28	SW	N 28°47'42" W	79.26	2,310,16 4.63	1,412,38 2.88	1213.7	-235.1	6123 162	Lot 70, Block A, Legends at Parkview — 180 Village Green Ct SW (PIN R6123 303)

No.	Side	Bearing (grid)	Distance (ft)	To: E	To: N	To: u	To: v	GIS parcel	Along (GIS adjoiner)
29	SW	N 28°47'44" W	81.46	2,310,12 5.39	1,412,45 4.26	1295.1	-235.2	6123 162	Lot 70, Block A, Legends at Parkview — 180 Village Green Ct SW (PIN R6123 303)
30	SW	N 28°47'44" W	52.20	2,310,10 0.25	1,412,50 0.00	1347.3	-235.3	6123 162	Lot 69, Block A, Legends at Parkview — 170 Village Green Ct SW (PIN R6123 302)
31	SW	N 28°50'50" W	101.93	2,310,05 1.07	1,412,58 9.28	1449.2	-235.5	6123 162	Lot 69, Block A, Legends at Parkview — 170 Village Green Ct SW (PIN R6123 302)
32	SW	N 28°50'50" W	65.88	2,310,01 9.28	1,412,64 6.99	1515.1	-235.6	6123 162	Lot 68, Block A, Legends at Parkview — 160 Village Green Ct SW (PIN R6123 301)
33	SW	N 28°50'49" W	39.97	2,310,00 0.00	1,412,68 2.00	1555.1	-235.7	6123 162	Lot 68, Block A, Legends at Parkview — 160 Village Green Ct SW (PIN R6123 301)
34	SW	N 29°19'37" W	6.06	2,309,99 7.03	1,412,68 7.28	1561.1	-235.8	6123 162	Lot 68, Block A, Legends at Parkview — 160 Village Green Ct SW (PIN R6123 301)
35	SW	N 29°19'37" W	159.83	2,309,91 8.75	1,412,82 6.63	1721.0	-237.5	6123 162	Lot 67, Block A, Legends at Parkview — 150 Village Green Ct SW (PIN R6123 300)
36	NW	N 60°38'26" E	4.59	2,309,92 2.75	1,412,82 8.88	1721.0	-232.9	6123 162	Lot 3, Block A, Nantucket — 4540 Nantucket Dr SW (PIN R6123 038)
37	NW	N 60°32'43" E	88.72	2,310,00 0.00	1,412,87 2.50	1722.2	-144.2	6123 162	Lot 2, Block A, Nantucket — 4530 Nantucket Dr SW (PIN R6123 037)
38	NW	N 61°35'48" E	13.93	2,310,01 2.25	1,412,87 9.13	1722.1	-130.3	6123 162	Lot 2, Block A, Nantucket — 4530 Nantucket Dr SW (PIN R6123 037)
39	NW	N 61°27'20" E	139.17	2,310,13 4.50	1,412,94 5.62	1721.6	8.9	6123 162	Lot 1, Block A, Nantucket — 4520 Nantucket Dr SW (PIN R6123 036)
40	NE	S 28°25'25" E	113.42	2,310,18 8.49	1,412,84 5.88	1608.2	8.3	6123 162	Lot 16, Block A, King David Manor Unit 1 — 145 King David Dr SW (PIN R6123 032)
41	NE	S 28°31'02" E	0.48	2,310,18 8.72	1,412,84 5.46	1607.8	8.3	6123 162	Lot 16, Block A, King David Manor Unit 1 — 145 King David Dr SW (PIN R6123 032)
42	NE	S 28°25'25" E	118.95	2,310,24 5.33	1,412,74 0.85	1488.8	7.7	6123 162	Lot 15, Block A, King David Manor Unit 1 — 155 King David Dr SW (PIN R6123 031)
43	NE	S 28°24'09" E	0.75	2,310,24 5.69	1,412,74 0.19	1488.1	7.7	6123 162	Lot 15, Block A, King David Manor Unit 1 — 155 King David Dr SW (PIN R6123 031)
44	NE	S 28°25'27" E	99.79	2,310,29 3.19	1,412,65 2.43	1388.3	7.2	6123 162	Lot 14, Block A, King David Manor Unit 1 — 165 King David Dr SW (PIN R6123 030)
45	NE	S 28°25'26" E	101.08	2,310,34 1.30	1,412,56 3.53	1287.2	6.7	6123 162	Lot 13, Block A, King David Manor Unit 1 — 175 King David Dr SW (PIN R6123 029)
46	NE	S 28°25'27" E	100.67	2,310,38 9.22	1,412,47 5.00	1186.5	6.1	6123 162	Lot 12, Block A, King David Manor Unit 1 — 185 King David Dr SW (PIN R6123 028)
47	NE	S 28°25'25" E	101.45	2,310,43 7.51	1,412,38 5.78	1085.1	5.6	6123 162	Lot 11, Block A, King David Manor Unit 1 — 195 King David Dr SW (PIN R6123 027)
48	NE	S 28°25'26" E	98.61	2,310,48 4.45	1,412,29 9.06	986.5	5.1	6123 162	Lot 10, Block A, King David Manor Unit 1 — 129 King David Dr SW (PIN R6123 026)

No.	Side	Bearing (grid)	Distance (ft)	To: E	To: N	To: u	To: v	GIS parcel	Along (GIS adjoiner)
49	NE	S 28°25'26" E	101.95	2,310,53 2.97	1,412,20 9.40	884.5	4.6	6123 162	Lot 9, Block A, King David Manor Unit 1 — 215 King David Dr SW (PIN R6123 025)
50	NE	S 28°25'26" E	100.18	2,310,58 0.66	1,412,12 1.30	784.3	4.1	6123 162	Lot 8, Block A, King David Manor Unit 1 — 225 King David Dr SW (PIN R6123 024)
51	NE	S 28°25'26" E	101.04	2,310,62 8.75	1,412,03 2.44	683.3	3.5	6123 162	Lot 7, Block A, King David Manor Unit 1 — 235 King David Dr SW (PIN R6123 023)
52	NE	S 28°25'25" E	99.38	2,310,67 6.06	1,411,94 5.03	583.9	3.0	6123 162	Lot 6, Block A, King David Manor Unit 1 — 245 King David Dr SW (PIN R6123 022)
53	NE	S 28°25'26" E	154.31	2,310,74 9.51	1,411,80 9.32	429.6	2.2	6123 162	Lot 5, Block A, King David Manor Unit 1 — 255 King David Dr SW (PIN R6123 021)
54	NE	S 28°25'27" E	175.47	2,310,83 3.03	1,411,65 5.01	254.1	1.3	6123 162	Lot 4, Block A, King David Manor Unit 1 — 265 King David Dr SW (PIN R6123 020)
55	NE	S 28°25'27" E	254.15	2,310,95 4.00	1,411,43 1.50	0.0	0.0	6123 162	Lot 1, Block A, King David Manor Unit 1 — 4531 Arcado Rd SW (PIN R6123 017)

"Side" is the boundary segment; "Along" is the GIS adjoiner whose ring coincides with the course. Local u, v are site-plan feet from the POB (FACTS §1). Courses shorter than 1 ft (Nos. 41 and 43) and the 4.59-ft and 6.06-ft courses (Nos. 36 and 34) are GIS-fabric artifacts at adjoining-lot corners; the surveyor will consolidate the near-collinear sub-courses (the southwesterly line is 22 courses within N 28°39'–29°20' W; the northeasterly line is 16 courses within S 28°24'–28°31' E).

5. Closure and area

- Side totals (ft): Arcado Rd frontage 237.63 | SW line 1757.01 | NW end 246.41 | NE line 1721.68 | perimeter 3962.73
- Chord widths (ft): front (POB → SW front corner) 237.44; rear (SW rear corner → NE rear corner) 246.39
- Closure of the typed courses: ΔN -0.015 ft, ΔE -0.007 ft, linear error 0.016 ft in 3962.73 ft = 1 : 244,281
- Area (coordinate/shoelace, GIS fabric): 417,177.6 sf = 9.577 ac; area from the typed courses: 417,198.1 sf = 9.578 ac
- Parcel check: 6123 033 42,891.7 + 6123 015 86,453.4 + 6123 014 86,840.4 + 6123 162 200,992.0 = 417,177.6 sf (GIS CALCULATEDACREAGE sum 9.577 ac; DEEDEDACREAGE sum 9.44 ac)

The closure is computed from the **typed** courses (bearings rounded to the second, distances to 0.01 ft), i.e. it is the error a reader would find by plotting this description, not the GIS polygon's own closure (which is exact by construction). 1 : 244,000 is far better than the 1 : 10,000 minimum for a Georgia plat because the courses are computed from one coordinate set, not measured; it says nothing about the accuracy of the GIS fabric itself.

Chord widths: 237.44 ft at Arcado Road (POB to the southwesterly front corner) and 246.39 ft across the rear; the two long lines diverge by about 0°20' over 1,720 ft, which is why the strip is ~236 ft wide at the road and ~246 ft at the northwesterly end (FACTS §1). Side totals: Arcado Road frontage 237.63 ft (13 courses, a chorded curve), southwesterly line 1,757.01 ft, northwesterly end 246.41 ft, northeasterly line 1,721.68 ft; perimeter 3,962.73 ft.

6. What the sealed survey must add or settle (VERIFY list for the RLS)

- Monumentation found or set at every corner; the record deeds and plats for the tract and the three adjoining subdivisions; resolution of the 9.44-ac deeded vs 9.577-ac GIS difference and of the 236/246-ft widths (the 2025 filing's 329-ft width is wrong — FACTS §1).
- The Arcado Road right-of-way line (width varies; centerline ≈ 28–30 ft from the front line — FACTS §1) and the 10.04-ft course No. 14, which the GIS right-of-way polygon shows wrapping around the southwesterly front corner; whether that 10 ft is Arcado Road right-of-way or part of the Legends at Parkview plat.
- The existing 20-ft sanitary sewer easement along the southwesterly side of Parcel 033 (book/page), any overhead/underground utility easements, and the Legends at Parkview sewer main near (u 1,349, v –405) that the Phase 2 tie-in would use (off-site; FACTS §2).
- State-waters/stream location and top-of-bank at the southwesterly line, u ≈ 1,356–1,392 (for the 25/50/75-ft buffers on the site plan), and the FEMA Zone X statement.

5. Grid-to-ground scale factor and convergence, so that the description reads as ground distances with a stated bearing basis.
6. A single consolidated description (fewer courses) and a plat that "includes legal description survey information" so that item 9 (boundary survey) and item 7 (legal description) are one sealed document, as the instructions allow.

7. Verification record

Script: tools/legal.py (stdlib only). Inputs: data/site-parcels-2240.json, data/site-context-local.json, data/owners-found.json, data/layout.json. Checks run 2026-08-28:

- - **union area 417177.6 sf vs parcel sum 417177.6 sf (Δ 0.00 sf)**
- clockwise from POB: True
- typed-course closure 0.016 ft in 3962.73 ft (1:244,281)
- side sequence: frontage → SW → NW → NE
- layout.json boundary_ring: 55 vertices vs 55 here; worst vertex offset 0.06 ft; layout boundary_sf 417173.6
- adjoiner identified for every course: True (unresolved: []); worst gap to adjoiner ring 0.04 ft
- **RESULT: PASS**

***Disclaimer:** This is an AI-generated analysis for preliminary planning purposes. All findings must be verified by a licensed professional before use in design, permitting, or regulatory submissions.*

THE COTTAGES AT ARCADE SPRINGS

SHEET C-0.0 — COVER SHEET, SHEET INDEX AND VICINITY MAP

APPLICATION TO REZONE 9.44 ACRES FROM R-1 TO R-2 — 41 ONE-STORY DETACHED 55+ (HOPA) COTTAGE HOMES ON FEE-SIMPLE LOTS, 4.34 DU/AC

Use: "Single-family (cluster-cottage, creative lot configuration)" — permitted in R-2 by Lilburn Zoning Ordinance 2023-603 §602 Use Table. No special use permit requested.

4535, 4537, 4539 and 4541 Arcado Road SW, Lilburn, Georgia 30047 — Land Lot 123, 6th District, Gwinnett County — PINs R6123 033, R6123 015, R6123 014 and R6123 162

City of Lilburn, Gwinnett County, Georgia — Department of Planning and Zoning, 340 Main Street. Filed under Ord. 2023-603 §1003 and the 2026 Application Instructions (11 items).

DRAFT — NOT SEALED

OWNER, CONTACTS AND DESIGN PROFESSIONALS

Site Development Plan Review Checklist §1.e.1 — a blank line means no such professional is engaged

ROLE	NAME	ADDRESS	PHONE / E-MAIL
Owner / applicant	Mohammed Awad — owner of record, PINs R6123 033 and R6123 015	4541 Arcado Rd SW, Lilburn, GA 30047-3968	
Co-applicant owner	Santos C. Mendez and Lesvia Rosario Roblero de Leon — owners of record, PINs R6123 014 and R6123 162	4535 Arcado Rd SW, Lilburn, GA 30047-3968	
24-hour contact	Mohammed Awad — 24-hour emergency contact for the site	4541 Arcado Rd SW, Lilburn, GA 30047-3968	
Designer / preparer	Owner-prepared, AI-assisted drafting. NOT A DESIGN PROFESSIONAL.	As above	
Land surveyor (RLS)	NOT YET ENGAGED		
Civil engineer (PE)	NOT YET ENGAGED		
Architect (GA RA)	NOT YET ENGAGED		
Landscape architect	NOT YET ENGAGED		
Geotech. engineer	NOT YET ENGAGED		
Stream delineation	NOT YET ENGAGED		
Traffic engineer	NOT YET ENGAGED		
Attorney	NOT YET ENGAGED		
City contact	Reid Turner, Planning Director, City of Lilburn	340 Main St, Lilburn, GA 30047	(770) 279-3715 returner@cityoflilburn.com

ZONING AND PROPOSED USE DATA

Checklist §1.e.1, §4.b and §4.c ("required versus provided"); Lilburn Zoning Ordinance 2023-603 Tables 4.1, 4.2 and 8.1

ITEM	REQUIRED / PROVIDED
Existing zoning	R-1, Single-Family Residential, City of Lilburn — all four parcels (Ord. 2023-603 §401)
Proposed zoning	R-2, Medium-Density Residential, City of Lilburn (Ord. 2023-603 §402)
Proposed use	"Single-family (cluster-cottage, creative lot configuration)" — P (permitted by right) in R-2, §602 Use Table. No special use permit; no commercial, attached or multi-family use.
Occupancy	55+ Housing for Older Persons (42 U.S.C. §3607(b)(2)(C); 24 CFR Part 100 Subpart E) by voluntary condition and recorded covenants. 0 students generated.
Gross site area	9.44 ac deeded (411,206 SF) = 0.99 + 2.00 + 2.00 + 4.45 ac; 9,577 ac (417,174 SF) GIS-calculated. THE SEALED RLS SURVEY GOVERNS.
Lots / dwelling units	41 one-story detached cottage homes on fee-simple lots; 2 plan types; HOA-maintained yards
Gross density	PROVIDED 4.34 du/ac deeded (4.28 GIS) — REQUIRED max. 8.0 du/ac (Table 4.1, R-2)
Lot area / width / depth	PROVIDED 5,000 SF typical, 50'-0" x 100'-0" — REQUIRED min. 3,000 SF, 50 ft, 100 ft (Table 4.1, cottage home)
Setbacks	PROVIDED / REQUIRED from 15'-0" from the lane tract and 50'-0" from the Arcado Rd RW; side 5'-0"; rear 20'-0"; accessory 5'-0" (Table 4.1)
Buffer abutting R-1	PROVIDED / REQUIRED 20'-0" undisturbed on every line but the Arcado Rd frontage, within the lots as a recorded easement (§313(1)) — see note 14
Height / floor area	PROVIDED 1 story, ridge 19.0 ft; 1,406 and 1,492 SF conditioned (38'-0" × 51'-10" and 38'-0" × 57'-6" bodies) — REQUIRED max. 40 ft, min. 1,000 SF heated
Parking	PROVIDED 164 on lot (2-car garage + 2-car drive) + 12 guest and kiosk spaces incl. 1 van-accessible — REQUIRED 82 (Table 8.1, 2 per DU, no maximum)
Common open space	PROVIDED 141,089 SF = 3.24 ac = 33.8% of gross, in common tracts. Development Regs. §5.9 does not apply below 50 ac (note 13), so this is voluntary.
Streets	One private lane 1,754 ft long in a private tract; all components to meet the minimum public-street standards (Checklist §4.s). None offered for dedication.
Acreage disturbed	7.54 ac on site. Off-site: the Phase-2 sewer corridor, about 178 ft, quantified on Sheet C-4.0 (Checklist §4.e).
Adjoining zoning	R-1, City of Lilburn, single-family, on every property line. Nearest R-2 about 106 ft NE; unincorporated Gwinnett R-100 about 314 ft. Owners: docs/06 and Sheet C-1.0.
Water and sewer	Gwinnett County Dept. of Water Resources — water and sanitary sewer only. All other utilities by private providers (note 11).
Frontage / access	237.4 ft on Arcado Rd SW (chord); one entrance in the SW third of the frontage, about 252 ft from the Arcadia Pl centreline. No lot takes access from Arcado Rd.

Quantities are read at plot time from data/layout.json (regenerated 2026-09-03) and data/plans.json — the quantities actually drawn on Sheet C-2.0. The full required-versus-provided table for Articles 4, 5 and 8 is on Sheet C-2.0.

REZONING SUBMITTAL INDEX — 2026 APPLICATION INSTRUCTIONS

ITEM	PROVIDED
1 Application Form	docs/01-application-form-data.md
2 Application Fee	\$1,250 — see APPLICATION DATA in the title banner above
3 Standards Governing the Zoning Power (§1003-7)	docs/04 — the six §1003-7 criteria
4 Conflict of Interest Form (§36-67A-3)	docs/07 — file with the application
5 Notarized owner signatures	docs/07 — two ownership groups
6 Letter of Intent	docs/03 — states the voluntary conditions offered
7 Legal Description (metes and bounds)	docs/05 — DRAFT; sealed RLS description governs
8 Site Plan	This drawing set; Sheet C-2.0 is the site plan proper
9 Boundary Survey	NOT INCLUDED — must be commissioned (Georgia RLS)
10 List of Adjoining Property Owners	docs/06 — re-pull on the filing date
11 Architectural Renderings and Elevations	docs/09 and Sheets A-1.1 to A-4.1

SHEET INDEX

20 of 24 sheets issued as of 2026-09-03. Built by listing drawings/*.svg at plot time — see general note 16.

SHEET	TITLE	SIZE / SCALE	STATUS	FILE (drawingsf...)	T/B
C-0.0	Cover Sheet, Sheet Index and Vicinity Map	ARCH D 36 x 24 · As noted	ISSUED	cover	=
C-1.0	Existing Conditions, Boundary and Topographic Survey	ARCH D 36 x 24 · 1" = 60'	ISSUED	existing-conditions	C-0
C-1.1	Demolition and Tree Protection Plan	ARCH D 36 x 24 · 1" = 60'	NOT ISSUED	--	--
C-2.0	Master Concept Plan - Overall	ARCH D 36 x 24 · 1" = 60'	ISSUED	mcp-sheet	=
C-2.1	Master Concept Plan - Enlargement, Block A	ARCH D 36 x 24 · 1" = 30'	ISSUED	mcp-enlargement-a	=
C-2.2	Master Concept Plan - Enlargement, Block B	ARCH D 36 x 24 · 1" = 30'	ISSUED	mcp-enlargement-b	=
C-2.3	Entry, Frontage and Amenity Enlargement	ARCH C 24 x 18 · 1" = 20'	ISSUED	entry-enlargement	=
C-2.4	Fallback Lot-Depth Exhibit - 50 x 82-ft Lots	ARCH D 36 x 24 · 1" = 60'	ISSUED	fallback-layout	=
C-3.0	Grading, Drainage and Stormwater Concept	ARCH D 36 x 24 · 1" = 60'	ISSUED	grading-drainage	=
C-3.1	Stormwater Details and Sections	ARCH C 24 x 18 · As noted	NOT ISSUED	--	--
C-4.0	Utility and Phasing Concept	ARCH D 36 x 24 · 1" = 60'	ISSUED	utility-phasing	=
C-4.1	Sanitary Sewer Profiles	ARCH D 36 x 24 · 1" = 50' H / 5' V	NOT ISSUED	--	--
C-5.0	Private Lane Plan and Profile	ARCH D 36 x 24 · 1" = 50' H / 5' V	NOT ISSUED	--	--
C-6.0	Phasing, Block and Open Space Exhibit	ARCH D 36 x 24 · 1" = 60'	ISSUED	phasing-openspace	=
C-7.0	Landscape, Buffer and Tree Protection Concept	ARCH D 36 x 24 · 1" = 60'	ISSUED	landscape-buffer	=
C-8.0	Civil Details	ARCH D 36 x 24 · As noted	ISSUED	civil-details	=
A-1.1	Plan A "The Springbrook" - Floor Plan	ARCH C 24 x 18 · 3/16" = 1'-0"	ISSUED	plan-a-sheet	A-1
A-1.2	Plan B "The Laurel" - Floor Plan	ARCH C 24 x 18 · 3/16" = 1'-0"	ISSUED	plan-b-sheet	A-2
A-2.1	Plan A - Exterior Elevations (four sides)	ARCH C 24 x 18 · 3/16" = 1'-0"	ISSUED	plan-a-elev	=
A-2.2	Plan B - Exterior Elevations (four sides)	ARCH C 24 x 18 · 3/16" = 1'-0"	ISSUED	plan-b-elev	=
A-2.3	Exterior Colour Schemes and Materials	ARCH C 24 x 18 · 3/16" = 1'-0"	ISSUED	plan-colors	=
A-3.0	Clubhouse, Mail Kiosk and Entry Monument Sign	ARCH C 24 x 18 · As noted	ISSUED	amenity-sheet	A-3
A-4.0	Typical Lot - Plan A	ARCH C 18 x 24 (portrait) · 1" = 10'	ISSUED	plan-a-lot	--
A-4.1	Typical Lot - Plan B	ARCH C 18 x 24 (portrait) · 1" = 10'	ISSUED	plan-b-lot	--

T/B is the sheet number printed in that sheet's own title block: "-" matches this index, "--" means the sheet declares none, and any other value is a provisional number to be corrected at the next issue (general note 16).

ALSO IN drawings/ — EXHIBITS, NOT SHEETS OF THIS SET

FILE	WHAT IT IS
mcp-web.svg	Master concept plan, screen version — public web exhibit, not a plot sheet
plan-a.svg	Plan A floor plan, tight crop — working exhibit for docs/09 and the web page
plan-b.svg	Plan B floor plan, tight crop — working exhibit for docs/09 and the web page

SEAL BLOCKS

Blank on purpose: no Georgia RLS, PE, architect or landscape architect is yet engaged, and no sheet in this set is sealed

GEORGIA REGISTERED LAND SURVEYOR	
SEAL (BLANK)	Boundary survey plat, metes-and-bounds legal description and topographic survey — Ord. 2023-603 §1003-4.3 and §1003-4.4, Checklist §1.e.2, Sheet C-1.0.
NOT YET ENGAGED — NO SHEET IN THIS SET IS SEALED.	
SIGNATURE	DATE

GEORGIA REGISTERED PROFESSIONAL ENGINEER	
SEAL (BLANK)	Grading, drainage, certified stormwater report, sewer, water, private street design and retaining walls — Checklist §1.e.7-9, §8.a, §11.a, Sheets C-3.0 to C-5.0 and C-8.0.
NOT YET ENGAGED — NO SHEET IN THIS SET IS SEALED.	
SIGNATURE	DATE

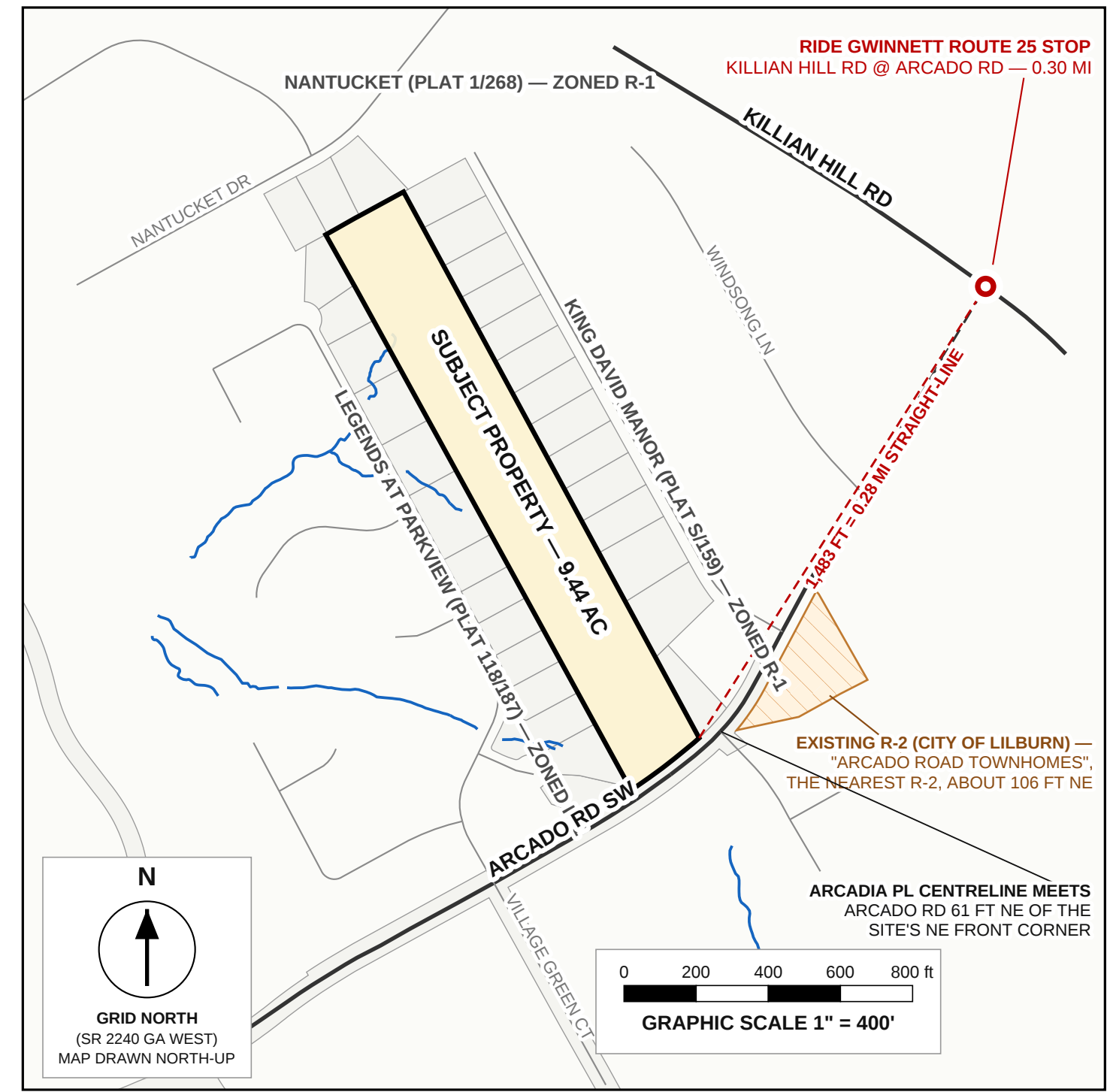
GEORGIA REGISTERED ARCHITECT	
SEAL (BLANK)	Dwelling and clubhouse plans and elevations — 2026 Application Instructions Item (11), Sheets A-1.1 to A-4.1.
NOT YET ENGAGED — NO SHEET IN THIS SET IS SEALED.	
SIGNATURE	DATE

GA REGISTERED LANDSCAPE ARCHITECT / FORESTER / ARBORIST	
SEAL (BLANK)	Buffer and landscape plan, tree protection and preservation — Checklist §1.e.10 and §6.a, Sheets C-1.1 and C-7.0.
NOT YET ENGAGED — NO SHEET IN THIS SET IS SEALED.	
SIGNATURE	DATE

Final plan sets must be sealed, signed and dated across the seal by the licensed registered party responsible for the contents of each sheet (Georgia Board Rule 180-12-.02; Checklist §1.e).

VICINITY MAP SCALE 1" = 400'

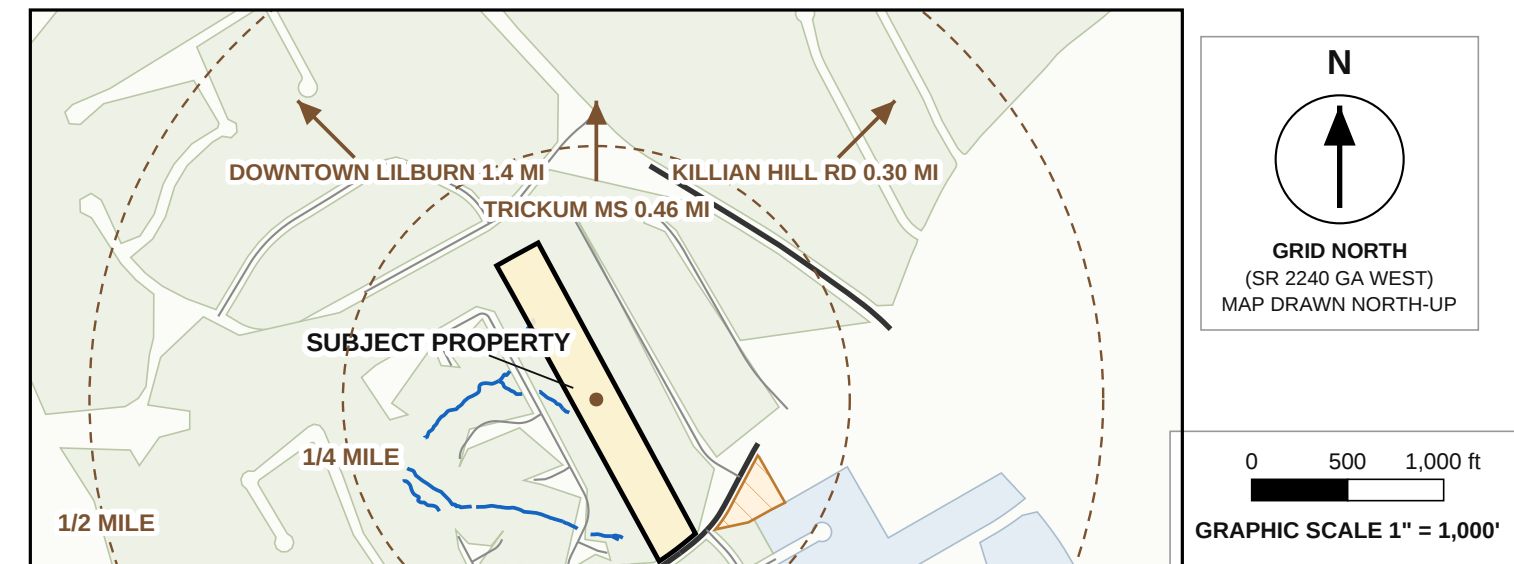
Arcado Rd, Killian Hill Rd, Arcadia Pl, the three adjoining subdivisions and the Ride Gwinnett Route 25 stop



Drawn north-up from the Gwinnett County GIS street centrelines, tax-parcel fabric, zoning polygons and hydrology carried in data/site-context-local.json and from the boundary ring in data/layout.json. Adjoining parcels are the 31 abutters plus the Arcado Rd right-of-way. Route 25 runs Gwinnett Place Transit Center to Stone Mountain; weekday headway about 50 minutes, Saturday hourly, no Sunday service. Neither centreline is mapped through the Killian Hill Rd / Arcado Rd intersection, so the intersection is shown where the mapped Arcado Rd centreline, produced on its last mapped bearing, cuts the mapped Killian Hill Rd centreline: it scales 1,483 ft (0.28 mi) against the 0.30-mi straight-line distance to stops 2528 and 2578 in the GTFS feed. Not a survey.

LOCATION MAP SCALE 1" = 1,000'

The site in the City of Lilburn zoning pattern, with distance rings on the site centroid



Zoning polygons and jurisdiction from the Gwinnett County GIS zoning layer; the R-1 (Lilburn) and R-100 (unincorporated Gwinnett) fills show the city limit. Off-map destinations are straight-line distances from docs/11 §6.3 and their compass directions are approximate.

LEGEND — VICINITY AND LOCATION MAPS

- Subject property — the 9.44-ac assemblage to be rezoned R-1 → R-2
- Adjoining tax parcel (Gwinnett GIS) — vicinity map
- Street centreline (Gwinnett GIS)
- Arcado Rd SW / Killian Hill Rd — county collectors
- Centreline produced — not mapped in the source data
- Stream — state waters (top of bank approximate)
- Ride Gwinnett Route 25 bus stop (GTFS feed 2026-07-02)
- Straight-line distance from the site
- Existing R-2 zoning, City of Lilburn — nearest R-2 precedent
- Existing R-1 zoning, City of Lilburn — location map
- R-100 zoning, unincorporated Gwinnett — location map
- Distance ring from the site, 1/4 and 1/2 mile — location map
- Direction and straight-line distance to an off-map destination

REVISIONS

NO.	DATE	DESCRIPTION	BY
0	2026-09-03	Issued for pre-application review — DRAFT, not sealed	M.A.

GENERAL NOTES

Notes the City of Lilburn Site Development Plan Review Checklist requires on the cover page are reproduced verbatim and identified as such

1. STATUS AND SEALS. This set is owner-prepared from public records for pre-application review and the rezoning hearing. DRAFT — NOT SEALED: no sheet is sealed by a Georgia registered land surveyor, professional engineer, architect or landscape architect, and the seal blocks are blank because none is yet engaged. Every statement of conformity reads "appears consistent with"; none is a certification of compliance. Final plan sets must be sealed, signed and dated across the seal by the licensed registered party responsible for the contents of the sheet (Georgia Board Rule 180-12-.02; Checklist §1.e).

2. APPROVAL NOTES — required on the cover by Checklist §4.a, verbatim: "APPROVAL OF THESE PLANS DOES NOT RELIEVE THE OWNER, DEVELOPER, AND/OR CONTRACTOR FROM COMPLYING WITH ALL APPLICABLE RULES, REGULATIONS, AND ORDINANCES." "DEVELOPER TO PROVIDE TO CITY CERTIFIED DETENTION POST-CONSTRUCTION (RECORD) DRAWINGS WITH THE SUBMITTAL OF THE FINAL PLAT OR ONE WEEK PRIOR TO REQUESTING A CERTIFICATE OF OCCUPANCY, SO THAT THE POST-CONSTRUCTION CONDITIONS MAY BE VERIFIED AND APPROVED. CERTIFIED RECORD DRAWINGS SHALL INCLUDE TOPOGRAPHY OF POND AND OUTLET STRUCTURE DETAIL USING POST-CONSTRUCTION SURVEY DATA. USING RECORD DRAWINGS, PROVIDE A CERTIFIED HYDROLOGY REPORT VERIFYING POND VOLUMES AND PEAK OUTFLOWS FROM REGULATED STORM EVENTS."

3. ZONING HISTORY AND CONDITIONS (Checklist §1.a, §4.b). THERE ARE NO ZONING CONDITIONS ON THIS PROPERTY. Zoning case no. RZ-2026-_____ (assigned by the City); date of approval _____; conditions _____. No special use permit, Zoning Board of Appeals variance or administrative variance is of record on any of the four parcels, and no overlay district applies. The only prior case is RZ-2025-01, a mixed-use request for 104 condominium units on this assemblage, which the Planning Commission TABLED on 2025-08-28 on a defective legal advertisement. It was never heard on its merits and never denied, so the twelve-month bar of §1003-11 does not appear to apply — confirm the file status with the Planning Director.

4. FLOODPLAIN (Checklist §9 Floodplain a, b). The property lies wholly in FEMA Zone X (area of minimal flood hazard), FIRM panel 13135C0114F, effective 2006-09-29 (FEMA National Flood Hazard Survey, Gwinnett GIS). NO SPECIAL FLOOD HAZARD AREA, FLOODWAY OR BASE FLOOD ELEVATION IS MAPPED ON THIS PROPERTY and no development is proposed in a floodplain. The Gwinnett future-conditions floodplain (www.gwinnettfloodplain.com) shall be checked before the Land Disturbance Permit; where one exists, finished floors shall be at least 1 ft above the future-conditions elevation and 3 ft above the 100-year base flood elevation (§9 Floodplain f). Lilburn Code Ch. 109 Art. III governs.

5. WETLANDS (Checklist §9 Wetlands a, b). No wetland is mapped on the property by the U.S. Fish and Wildlife Service National Wetlands Inventory. A field survey of any waters of the United States shall be made with the boundary and topographic survey; any delineation shall state its source and carry U.S. Army Corps of Engineers approval, and it is the owner's responsibility to obtain any Corps permit required.

6. STREAM BUFFERS — required by Checklist §9 Buffered State Waters e and f, verbatim: "STREAM BUFFERS ARE TO REMAIN IN A NATURAL AND UNDISTURBED CONDITION." "STREAM BUFFER SHALL BE STAKED AND PROTECTED PRIOR TO LAND DISTURBANCE." "An unnamed order-0 headwater of Jackson Creek (GAR030701030315; Georgia 2024 §303(d) list) reaches about 30 ft inside the SW property line and ends on the property. The 25-ft state buffer (O.C.G.A. §12-7-6(b)(15)), the 50-ft undisturbed buffer and the 75-ft impervious setback (Lilburn Code Ch. 109 Art. VII) are labelled on Sheets C-1.0, C-2.0 and C-2.2 from the top of bank. TOP OF BANK IS APPROXIMATE: the state waters shall be field-located and both banks delineated by a qualified professional, and that delineation governs. No buffer variance is requested with this rezoning.

7. SIGNS (Checklist §4.q). One monument entry sign not over 32 SF is proposed at the Arcado Rd entrance; its location, size and sight-triangle clearance are on Sheets C-2.3 and A-3.0. NO WALL SIGNS ARE PROPOSED ON ANY DWELLING OR ON THE CLUBHOUSE. ALL SIGNS SHALL BE PERMITTED SEPARATELY. A sign easement may be required.

8. MAIL DELIVERY (Checklist §4.r). Mail will be delivered to a cluster bus unit (CBU) kiosk in the amenity tract inside the entrance, with protection from the elements, an accessible route and four short-term spaces; detailed on Sheet A-3.0. THE LILBURN POST OFFICE GROWTH MANAGER SHALL BE CONTACTED FOR MAIL DELIVERY AND MAILBOX REQUIREMENTS. A cluster mailbox system may be required for the entire project and one street name may serve to assign all house numbers.

9. GRADING — required by Checklist §10.a on the grading plan and on all soil-erosion, sedimentation and pollution-control sheets, verbatim:

"MAXIMUM CUT OR FILL SLOPES IS 2H:1V" "CITY OF LILBURN/GWINNETT COUNTY ASSUMES NO RESPONSIBILITY FOR OVERFLOW OR EROSION OF NATURAL OR ARTIFICIAL DRAINS BEYOND THE EXTENT OF THE STREET RIGHT-OF-WAY, OR FOR THE EXTENSION OF CULVERTS BEYOND THE POINT SHOWN ON THE APPROVED AND RECORDED PLAN. THE CITY OF LILBURN/GWINNETT COUNTY DOES NOT ASSUME THE RESPONSIBILITY FOR THE MAINTENANCE OF PIPES IN DRAINAGE EASEMENTS BEYOND THE CITY/COUNTY RIGHT-OF-WAY." "STRUCTURES ARE NOT ALLOWED IN DRAINAGE EASEMENTS."

10. STREET GRADES. Maximum street grade 12%; grades of 12% to 15% require an "As Graded" survey before installation of the curb (Checklist §7.k). Existing ground on the lane alignment reaches 10.9% — see Sheet C-5.0.

11. UTILITIES (Checklist §4.d). Gwinnett County Department of Water Resources provides water and sanitary sewer only; electric, gas and telecommunication service is by the respective private providers, and new electric distribution within the development will be underground. PHASE 1 connects by gravity to the existing 8-in sewer already crossing this property (manhole invert 927.13) and needs no extension. PHASE 2 needs an off-site gravity extension of about 178 ft to the Legends at Parkview MS 8-in main at invert 919.58. A private pump station, force main or gravity sewer is NOT available: Gwinnett DWR's Standard Policy for Private Developments (Condominiums, Townhomes and Subdivisions, rev. 9/2018) allows private facilities "only ... for commercial properties under single ownership within a development", and a county station fails WSP-24 §1.3.1(A) because gravity is 1.78 ft away, not "more than 5,000 feet down gradient". A DVR sewer capacity certification is required. See Sheets C-4.0 and C-4.1.

12. FIRE ACCESS. One private lane about 1,754 ft long, dead-ended at the NW terminus, 22 ft of clear pavement, no parking, hammerhead turnarounds. SPECIAL APPROVAL OF A DEAD-END FIRE APPARATUS ACCESS ROAD OVER 750 FT IS REQUESTED under ICF 2024 (CA) Appendix D103.4, Table D103.4. Two access roads are NOT required: Ga. Comp. R. & Regs. 120-3-3-.04 replaces Appendix D107.1 and sets the trigger above 120 dwelling units, and 41 are proposed. NFPA 13D sprinklers in every dwelling are offered VOLUNTARILY and are not a code requirement. Codes: the Georgia State Minimum Standard Codes effective 1 January 2026 (2024 IRC / IBC / IFC / IPC / IMC / IFGC with the 2026 Georgia Amendments; 2023 NEC; 2015 IECC as amended).

13. OPEN SPACE AND RECREATION AREAS. Development Regulations §5.9.1 requires recreation land in single-family detached subdivisions "having a gross area of 50 acres or more". At 9.44 acres this development is below that threshold and §5.9 does not apply; the 33.8% of common open space shown is voluntary.

14. BUFFERS AND THE §313(1) READING. The 20-ft undisturbed buffer required abutting R-1 (Table 4.1, allowed dwelling types other than detached single-family) is held within the lots as a recorded buffer easement in reliance on §313(1), "Buffer requirements ... supersede these minimum required yards". THIS READING IS TO BE CONFIRMED WITH THE PLANNING DIRECTOR; Sheet C-2.4 carries the fallback layout, 50 x 82-ft lots with the buffer in a separate common tract, if a separate tract is required.

15. DIMENSIONS, DATUM AND COORDINATES. All dimensions are in US survey feet. Horizontal datum Georgia State Plane West Zone (SR 2240), NAD83, vertical NAVD88. Sheets C-1.0 to C-6.0 are drawn in a site-local system (+u along the strip on a bearing of N 28°43' W from the Arcado Rd RW corner at 2,310,931.00 E / 1,411,431.50 N; +v across the strip toward the NE line). THE TWO MAPS ON THIS SHEET ARE DRAWN NORTH-UP. Boundary geometry is Gwinnett GIS and is DRAFT; topography is USGS 3DEP and is approximate. THE SEALED SURVEY GOVERNS.

16. SHEET INDEX INTEGRITY. The sheet index is generated at plot time by listing the files present in the package drawings/ directory and matching them against the sheet table in tools/cover.py — the same table tools/transmittal.py uses for the filing transmittal. A row marked NOT ISSUED is not in this set and must not be relied on. Where the "T/B" column shows a number, that sheet's own title block still carries a provisional number from an earlier numbering; THE NUMBER IN THIS INDEX GOVERNS and the sheet will be renumbered at the next issue.

17. SCALES AND LETTERING. A graphic scale and a north arrow are shown on each map here and on every applicable sheet (Checklist §4.h); engineering scales on the C-sheets, architectural on the A-sheets. DO NOT SCALE A REDUCED PRINT — use the graphic scale. Lettering on this sheet is 10.5 pt in the general notes, 8.5 to 11 pt in the tables, legend and map annotation, and 7 to 8 pt in the title block and map furniture — a cap height of 0.068 to 0.107 in at full size. The set standard is a 1/8-in cap height; the content this checklist requires on a cover page does not fit an ARCH D sheet at that size, so this sheet carries the largest type that fits.

18. BASIS AND SOURCES. Gwinnett County GIS — parcels, zoning, streets, right-of-way, sewer, water, hydrology, floodplain, soils, queried 2026-08-28; Gwinnett Tax Assessor 2026 property-ownership file; USGS 3DEP; FEMA NFHL; Ride Gwinnett GTFS feed 2026-07-02; City of Lilburn Zoning Ordinance 2023-603, 2026 Application Instructions, Site Development Plan Review Checklist, 2024 Comprehensive Plan and the 2026 Comprehensive Plan Amendment (transmitted to DCA 2026-07-13; adoption pending as of 2026-09-03). Package files FACTS.md, data/layout.json and data/plans.json, all read at plot time.

Disclaimer: This is an AI-generated analysis for preliminary planning purposes. All findings must be verified by a licensed professional before use in design, permitting, or regulatory submissions.

Label: DRAFT / NOT SEALED — every item that must be sealed by a Georgia RLS, PE, architect or landscape architect is shown as a concept only. Statements of conformity read "appears consistent with".

THE COTTAGES AT ARCADE SPRINGS

COVER SHEET, SHEET INDEX AND VICINITY MAP — Rezoning R-1 → R-2 (City of Lilburn, Georgia)

4535 / 4537 / 4539 / 4541 Arcado Rd SW, Lilburn, GA 30047 — Land Lot 123, 6th District, Gwinnett County — PINs R6123 033, 015, 014, 162 — 9.44 ac (deeded)

Cover page per City of Lilburn Site Development Plan Review Checklist §1.e.1, §4.a and §4.b; sheet index generated from drawings/ at plot time

APPLICANT / OWNER
Mohammed Awad
4541 Arcado Rd SW, Lilburn GA 30047-3968 (parcels 033, 015); Mendez / Roblero de Leon (parcels 014, 162) — co-applicants, owner signatures required
PREPARED FOR
Pre-application conference, City of Lilburn Planning & Zoning (340 Main St) — rezoning application per Ord. 2023-603 §1003 (2026 Application Instructions)

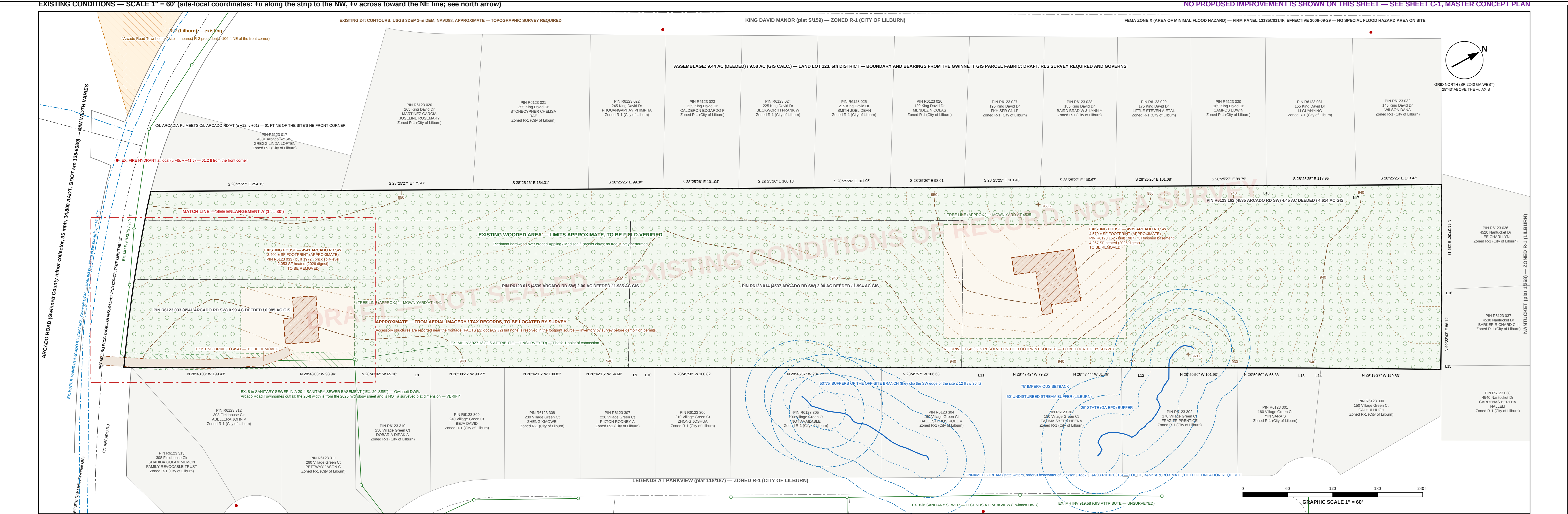
STATUS
DRAFT — NOT SEALED
Owner-prepared cover page for pre-application review. No sheet in this set is sealed; the seal blocks are blank because no Georgia RLS, PE, architect or landscape architect is yet engaged. To be superseded by sealed survey, civil, architectural and landscape drawings.
Generator: tools/cover.py (on tools/sitebase.py)

DATE	2026-09-03
SCALE	As noted — see each map
DATUM	SR 2240 GA WEST US-ft; NAVD88
DRAWN	owner-prepared (AI-assisted)
CHECKED	RLS / PE — pending

SHEET

C-0.0

COVER SHEET, SHEET INDEX AND VICINITY MAP



EXISTING SITE DATA — compiled from public records on 2026-09-03

EXISTING SITE DATA	RECORD VALUE (DRAFT — SURVEY GOVERNS)	SOURCE
Address (PINs)	4535, 4537, 4539, 4541 Arcado Rd SW, Lilburn GA 30047 — R6123 162, 014, 015, 033	Gwinnett 2026 digest
Land lot / district	Land lot 123, 6th District, Gwinnett County, City of Lilburn tax district	Gwinnett GIS
Area	9.44 ac (deeded (411,206 SF) = 0.99 ± 2.00 ± 2.00 ± 4.45; 9,577 ac (417,174 SF) GIS-calculated; RLS survey governs	2026 digest / GIS ring
Boundary lengths	Arcado Rd frontage 237.63' SW line 1797.01' NE line 1721.69' rear line 246.41' perimeter 3962.73'	Aerial imagery / 2026 digest
Width across the strip	224.7 ft at Arcado Rd, 246.4 ft at the NW end (measured NE line to SW line)	OSMIML footprints — APPROX.
Existing zoning	R-1, City of Lilburn (all four parcels). Adjoining: R-1 Lilburn on every line, nearest R-2 = 106 ft NE; unincorporated Gwinnett R-100 ± 314 ft	GIS ring
Existing use	Two single-family dwellings with drives near the frontage and at mid-strip, the balance wooded. No commercial or agricultural use.	Gwinnett GIS DRAFT
Existing structures	4541: 2,400 ± SF footprint at (u 203, v -174), built 1972, 2,063 SF heated, 4536, 4,570 ± SF footprint at (u 1202 ± 130, v 180), built 1987, 4,267 SF heated, area ± 250 ft long x 12 ft x 12 ft. All to be removed.	Aerial imagery / 2026 digest
Topography	Range 921.6 — 968.2 ft NAVD88; ridge 968.2 ft at (u 1184, v -18); low 921.6 ft at (u 1384, v -218); cross-slope falls NE — SW; contour interval 2 ft	OSMIML footprints — APPROX.
Floodplain	Zone X (area of minimal flood hazard) — FIRM panel 13135C0114F, effective 2006-09-29 — no SFHA, floodway or BFE on site	USGS 3DEP 1-m DEM (approx.)
Stream / buffers	One unnamed order-0 headwater (state waters) reaching ± 30 ft inside the SW line; 25' GA EPD / 50' undisturbed / 75' impervious setback shown from approximate top of bank	Stream — state waters (top of bank approx.)
Sanitary sewer	Existing 8-in gravity main in a 20-ft easement inside the SW line, u = -42 to 272; 24" MH invert 927.13 on site and 919.58 off site (Legends at Parkview) — GIS attributes, unsurveyed	25' state (GA EPD) buffer
Water	DIP / ACP mains in Arcado Rd 47–107 ft from the front line; nearest hydrant 61.2 ft from the front corner at (u -45, v +41.5)	50' undisturbed stream buffer (Lilburn)
Right-of-way	Arcado Rd, Gwinnett County minor collector, 35 mph, 14,800 AADT, RW WIDTH VARIES; 64.9 ft at v -40, 60.0 ft at v -120; 60.2 ft at v -200, 69.9 ft at v -235 — scaled perpendicular to the two GIS RW lines, not surveyed	75' impervious setback
Soils	AptB (Apping), AmC2 (Apping/Madison), P1C2 (Pacott), Gb2C (Cecil), Gc2C (Cecil, steep), ARE (Ashlar — gneiss at 22–40 in), HYB (Helena — HSG D)	USGS 3DEP 1-m DEM (approx.)
Vegetation	Existing wooded area over the balance of the site; limits approximate, no tree survey performed	Existing 8-in sanitary sewer + manhole
Adjoining owners	31 private tax parcels (24 NE, 14 SW, 3 NW) in the Gwinnett County right-of-way ± 32 notice recipients; PIN, owner and R-1 zoning labelled on the plan	Existing 20' sanitary sewer easement (VERIFY)

Existing building footprints and the 4541 drive: © OpenStreetMap contributors (ODN), ways 999395123 / 999395265 / 913015145 — aerial imagery footprints, approximate. Owner names and mailing addresses: Gwinnett County Tax Assessor 2026 property ownership file. Zoning: Gwinnett GIS zoning layer (City of Lilburn districts).

EXISTING SANITARY SEWER SCHEDULE — every value is a Gwinnett DWR GIS attribute, UNSURVEYED (an invert survey is required)

FACILITY ID	FROM (u, v) — TO (u, v)	SIZE / MAT.	LENGTH (ft)	SLOPE (in)	INVERT IN — OUT (ft)	PROJECT
3425385	(-2, -157) — (-28, -20)	8-in DIP	134.0	3.59	946.90 — 945.99	Arcado Road Townhomes sewer outfall
3425386	(-22, -50) — (-28, -125)	8-in DIP	75.1	4.42	943.79 — 940.47	Arcado Road Townhomes sewer outfall
3425387	(-28, -125) — (-42, -224)	8-in DIP	100.8	7.30	940.25 — 932.89	Arcado Road Townhomes sewer outfall
3425388	(-42, -224) — (-20, -229)	8-in DIP	62.1	3.98	932.69 — 930.84	Arcado Road Townhomes sewer outfall
3425389	(-20, -229) — (-272, -224)	8-in PVC	251.6	1.40	930.64 — 927.13	Arcado Road Townhomes sewer outfall
3425390	(-272, -224) — (-281, -392)	8-in PVC	167.9	1.05	926.93 — 925.17	Arcado Road Townhomes sewer outfall
2300318	(1340, -406) — (1146, -406)	8-in DIP	199.0	0.17	921.03 — 919.58	Legends At Parkview
2300316	(1160, -405) — (929, -406)	8-in DIP	231.2	0.67	919.38 — 917.83	Legends At Parkview

Slope is computed from the published invert pair and pipe length: (INV IN - INV OUT) / LENGTH. The 927.13 manhole is the Phase 1 point of connection; the 919.58 manhole is the Legends at Parkview tie-in studied in doc008 Memo B.

SOURCES AND ACCESS DATA

- Gwinnett County GIS — parcels, zoning, streets, right-of-way, sanitary sewer, water mains, hydrants, subdivisions, city limits (accolaslas, gwinnett, MapServer layers 2, 7, 21, 23, 25), streams, floodplain, soils (GISDataViewer/GC, Main layers 1, 2, 9), Quinet 2026-09-28.
- Gwinnett County Tax Assessor — 2026 Property Ownership quarterly file (owners, mailing addresses, deeded acreage, year built, heated area), Accessed 2026-09-28.
- USGS 3DEP elevation Imagery Server — 1-meter DEM sampled on a 100 × 50-ft grid, NAVD88 (data/shape-samples.json), Accessed 2026-09-28.
- FEMA National Flood Hazard Layer — flood zone and FIRM panel (doc011 S2), Accessed 2026-09-28.
- OpenStreetMap (© OpenStreetMap contributors, ODN) — aerial imagery building footprints and the 4541 drive, ways 999395123, 999395265, 913015145; Microsoft Bing ML Building Footprint Imagery, Accessed 2026-09-03.
- Public aerial imagery — visual check of the footprints and of the town-yard (tree-line) limits, 2026-09-03. Imagery is not reproduced on this sheet.
- City of Lilburn Zoning Ordinance 2023-603 §1003-4.6 (site-plan content) and the City of Lilburn Site Development Plan Review Checklist, FACTS.md §1, §2, §2b, doc005 §5, doc006, doc011 §2.

LEGEND

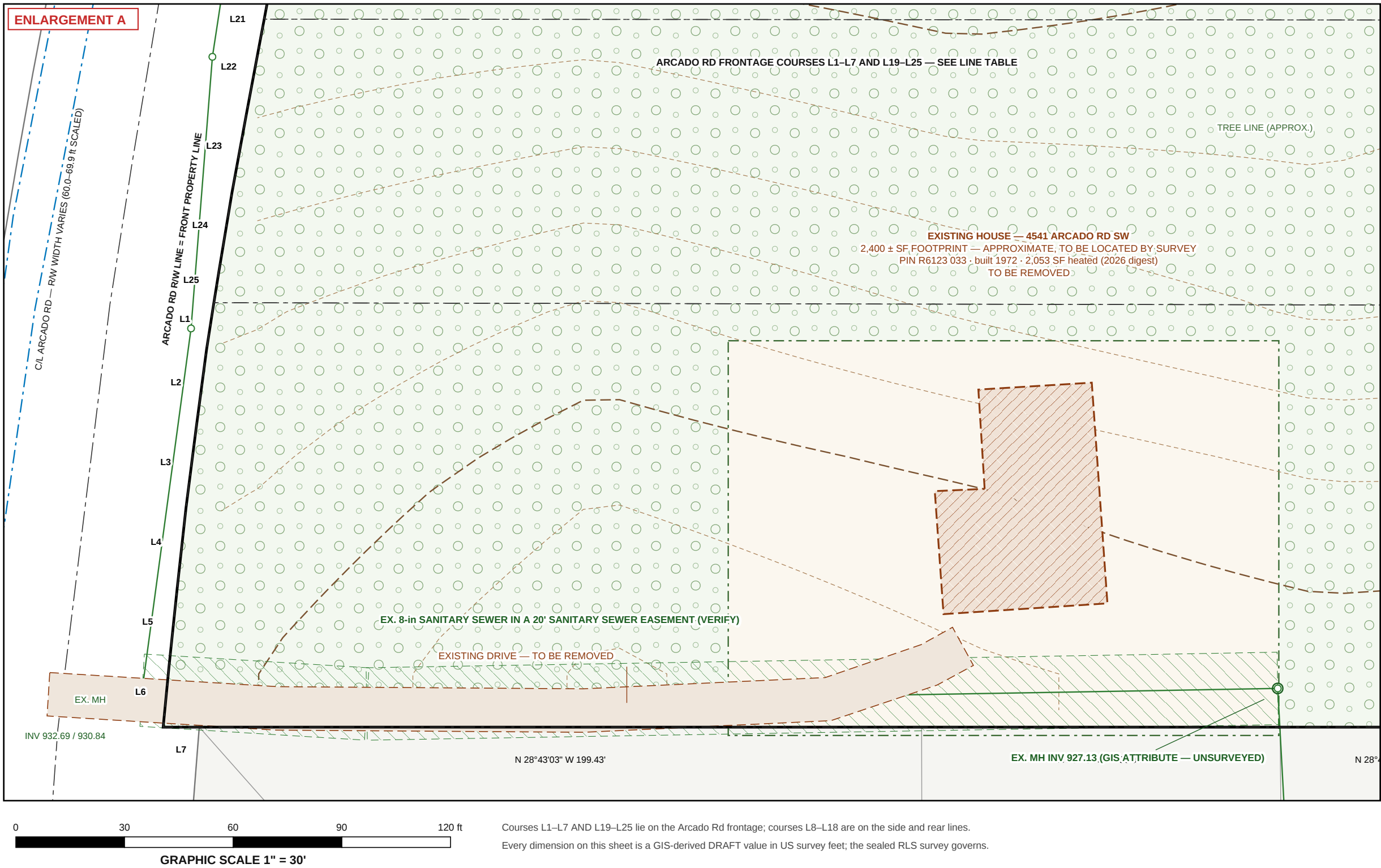
- Assemblage boundary (Gwinnett GIS — DRAFT)
- Interior parcel line (to be combined)
- Adjoining tax parcel (all zoned R-1, Lilburn)
- Arcado Rd right-of-way line (RW width varies)
- Existing road centerline
- Existing index contour, 10 ft (3DEP approx.)
- Existing contour, 2 ft (3DEP approx.)
- Stream — state waters (top of bank approx.)
- 25' state (GA EPD) buffer
- 50' undisturbed stream buffer (Lilburn)
- 75' impervious setback
- Existing 8-in sanitary sewer + manhole
- Existing 20' sanitary sewer easement (VERIFY)
- Existing water main (Gwinnett DWR)
- Existing fire hydrant
- Existing building — approximate, to be removed
- Existing drive — approximate, to be removed
- Existing wooded area (limits approximate)
- Tree line at mean year (approximate)
- Existing R-2 zoning (City of Lilburn)
- Match line — enlargement at 1" = 30'

LINE TABLE — courses under 60 ft (GIS-derived, DRAFT)

LINE	BEARING (SR 2240 GA WEST)	DISTANCE
L1	S 92°37'51" W	12.82
L2	S 53°31'16" W	22.08
L3	S 54°09'44" W	22.20
L4	S 54°51'18" W	21.98
L5	S 55°12'22" W	22.19
L6	S 55°42'53" W	16.64
L7	N 28°43'02" W	30.04
L8	N 28°50'12" W	34.90
L9	N 28°50'12" W	17.92
L10	N 28°45'55" W	17.59
L11	N 28°47'42" W	52.82
L12	N 28°47'42" W	52.20
L13	N 28°50'45" W	39.97
L14	N 29°19'57" W	6.86
L15	N 60°39'26" E	4.59
L16	N 61°35'48" E	13.93
L17	S 28°10'02" E	0.48
L18	S 28°50'02" E	0.75
L19	S 48°41'39" W	21.97
L20	S 50°35'58" W	38.12
L21	S 51°30'08" W	4.12
L22	S 50°41'21" W	22.30
L23	S 50°56'17" W	22.22
L24	S 51°41'48" W	21.96
L25	S 52°28'47" W	8.83

ENLARGEMENT A — ARCADO ROAD FRONTAGE, EXISTING HOUSE AT 4541 AND THE EXISTING SANITARY SEWER (SCALE 1" = 30')

Same data, same coordinates as the plan above, u = 80 to 300, v = 250 to -35. Frontage courses are tabulated in the line table at the left.



GENERAL NOTES — EXISTING CONDITIONS

- BOUNDARY AND PARCEL LINES: the assemblage boundary, the four interior parcel lines, the bearings, the distances and the acreages shown are computed from the Gwinnett County GIS parcel fabric; SR 2240 GA West, US survey feet) and are DRAFT. A boundary survey plat and metes and bounds legal description sealed by a Georgia registered land surveyor are required by Lilburn Zoning Ordinance 2023-603 §1003-4.3 and §1003-4.4 and govern over this sheet. The four parcels are to be combined, the interior lines are shown so the City can read the assemblage.
- TOPOGRAPHY: existing 2-ft contours are interpolated from the USGS 3DEP 1-meter digital elevation model sampled on a 100 × 50-ft grid (NAV88) and are APPROXIMATE. Elevations on site range 921.6 to 968.2 ft; the high point is on the ridge at local u 1384, v -18 and the low point is in the rear pocket at (u 1384, v -218). A topographic survey is required before any engineering design. Contour interval 2 ft; index contours at 10 ft.
- EXISTING BUILDINGS AND DRIVES §1003-4.4 (existing buildings and structures): the two dwellings and the drive shown are APPROXIMATE — FROM AERIAL IMAGERY / TAX RECORDS, TO BE LOCATED BY SURVEY. Footprint outlines are public aerial imagery building footprints published by OpenStreetMap (© OpenStreetMap contributors, ODN), Microsoft Bing ML Building Footprints Imagery — ways 999395123, 999395265 and 913015145, transformed into the site-local system and checked against public aerial imagery on 2026-09-03. Footprint areas scale at 2,400 SF (4541) and 4,570 SF (4535) against 2026 digest heated areas of 2,063 SF (split level, built 1972) and 4,267 SF (built 1987, full finished basement). BOTH DWELLINGS, ALL ACCESSORY STRUCTURES AND ALL EXISTING DRIVES ARE TO BE REMOVED. Accessory structures are reported near the frontage (FACTS §2, doc002 §2) but none is resolved in the footprint source; inventory them by survey before demolition permits. The septic-sewer status of 4535 is unknown — VERIFY. NOTE THE POSITION OF 4535: its dwelling stands about 1,025 ft back from the Arcado Rd right-of-way, on the ridge at mid-strip — NOT at the frontage. FACTS §2 and doc002 §2, which place both houses "near the front", should be corrected to match this sheet.
- EXISTING VEGETATION §1003-4.6 (areas of existing vegetation): EXISTING WOODED AREA — LIMITS APPROXIMATE, TO BE FIELD-VERIFIED. The wooded limit is the site boundary less the two mean yards at the existing dwellings, digitized from public aerial imagery on 2026-09-03 and consistent with doc011 (the rear two-thirds is Piedmont hardwood) and FACTS §2 (wooded rear). NO TREE SURVEY HAS BEEN PERFORMED; species, diameter, species, species and the actual inside the SW line shown. A tree survey and tree protection / replacement plan will be submitted with the Land Disturbance Permit; tree density units at 15 TD/acre are not required of a single-family residential subdivision (City of Lilburn Site Development Plan Review Checklist §6.1).
- STREAM AND BUFFERS: an unnamed order-0 headwater of Jackson Creek (GAR0307010303103; Georgia 2024 53036) list for bacteria and toxic, urban runoff reaches about 30 ft inside the SW line and terminates at local (u 1382, v -210). TOP OF BANK IS APPROXIMATE — a field delineation by a qualified professional is required and governs. Buffers shown: 25 ft state (GA EPD, O.C.G.A. §12-2-400(15)), 50 ft undisturbed and 25 ft additional impervious setback ± 75 ft total (City of Lilburn Code tit. 109 Art. VII). The 5075-ft buffers of a second branch that runs 38–58 ft outside the SW line clip the SW edge of the site.
- FLOODPLAIN: FEMA Zone X (area of minimal flood hazard) — FIRM panel 13135C0114F, effective 2006-09-29 (Gwinnett GIS floodplain layer; doc011 S2 Flood, FEMA NFHL, accessed 2026-09-28). No special flood hazard area, floodway or base flood elevation is mapped on the site; the headwater stream is not a FEMA-mapped watercourse, so the local stream buffer, not a BFE, governs the rear pocket.
- SANITARY SEWER: an existing 8-in gravity main (Arcado Road Townhomes sewer outfall) — Gwinnett DWR) enters the site at Arcado Rd and runs 7–13 ft inside the SW line to a manhole at local (u 272, v -224). Both invert elevations shown — 927.13 at that manhole and 919.58 at the Legends at Parkview manhole at (u 1349, v -407) — are GIS ATTRIBUTES, UNSURVEYED; an invert survey is the first field task. The 20-ft sanitary sewer easement shown is the TX, 20' SISE of the 2025 hydrology sheet and is NOT a surveyed plat dimension — VERIFY the recorded width and holder. A Gwinnett DWR sewer capacity certification is required for any connection.
- WATER: DIP and ACP mains are in the Arcado Rd right-of-way 47–107 ft from the front line; the public utility does not publish diameters — VERIFY with Gwinnett DWR. The nearest fire hydrant is at local (u -45, v +41.5), 51.2 ft from the front corner. Other hydrants shown are in the adjoining subdivisions.
- ARCADO ROAD RIGHT-OF-WAY: RW WIDTH VARIES. Measured perpendicular between the two Gwinnett GIS right-of-way lines it scales 64.9 ft at v -40, 60.0 ft at v -120, 60.2 ft at v -200, 69.9 ft at v -235 — a GIS-scaled dimension, not a surveyed one; the RW line, its width and any required dedicated right-of-way are established by the RLS survey. Arcado Rd is a Gwinnett County-maintained minor collector, 35 mph, 14,800 AADT (GDOT station 125+688, 2017). The centerline of Arcado Place meets the Arcado Rd centerline at local (u -12, v +61), 61 ft NE of the site's NE front corner.
- SOILS: PERCS SSURGO GA135, via Gwinnett GIS; AptB (Apping), AmC2 (Apping/Madison), P1C2 (Pacott), Gb2C (Cecil), Gc2C (Cecil, steep), ARE (Ashlar — gneiss at 22–40 in), HYB (Helena — HSG D) is hydrologic soil group D with a seasonal high water table near 24 in; ARE (Ashlar) carries gas, telecommunication and storm-drainage structures are not published in the layers used for this sheet and are NOT shown. Utility locates through Georgia B11 (Georgia Utility Facility Protection Act, O.C.G.A. §25-9) are required before any field work, and a full testing utility survey is required with the Land Disturbance Permit.
- STATUS: this sheet is a DRAFT record of existing conditions compiled from public data for pre-application review. It is NOT a survey and is NOT SEALED. All dimensions are used survey feet in the site-local system (u along the strip, v across). Every statement of conformity in this submittal reads "appears consistent with" and none is a compliance certification. THE SEALED SURVEY GOVERNS.

Disclaimer: This is an AI-generated analysis for preliminary planning purposes. All findings must be verified by a licensed professional before use in design, permitting, or regulatory submissions.

Label: DRAFT / NOT SEALED — every item that must be sealed by a Georgia RLS, PE, architect or landscape architect is shown as a concept only. Statements of conformity read "appears consistent with".

THE COTTAGES AT ARCADO SPRINGS

EXISTING CONDITIONS — Rezoning R-1 — R-2 (City of Lilburn, Georgia)

4535 / 4537 / 4539 / 4541 Arcado Rd SW, Lilburn, GA 30047 — Land Lot 123, 6th District, Gwinnett County — PINs R6123 033, 015, 014, 162 — 9.44 ac (deeded)

Existing conditions of record — Lilburn Zoning Ordinance 2023-603 §1003-4.6 (existing roads, streams, floodplain, buildings and structures, areas of existing vegetation)

APPLICANT / OWNER
Mohammed Awad

4541 Arcado Rd SW, Lilburn GA 30047-3968 (parcels 033, 015); Mendez / Roblero de Leon (parcels 014, 162) — co-applicants, owner signatures required

PREPARED FOR

Pre-application conference, City of Lilburn Planning & Zoning (340 Main St) — rezoning application per Ord. 2023-603 §1003 (2026 Application Instructions)

STATUS

DRAFT — NOT SEALED

Existing conditions compiled from public records for pre-application review. Boundary from Gwinnett GIS — RLS boundary survey required and governs. Topography USGS 3DEP (approximate). To be superseded by a sealed RLS boundary and topographic survey and PE civil design.

Generator: tools/sitebase.py + tools/existing_conditions.py

DATE
2026-09-03

SCALE
Scale 1" = 60' (ARCH D 36 × 24 in); enlargement 1" = 30'

DATUM
SR 2240 GA West US-ft; NAVD88

DRAWN
owner-prepared (AI-assisted)

CHECKED
RLS / PE — pending

SHEET

C-0

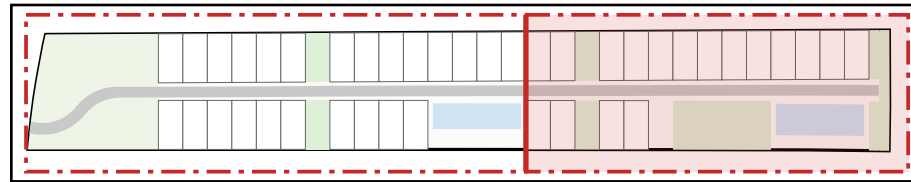
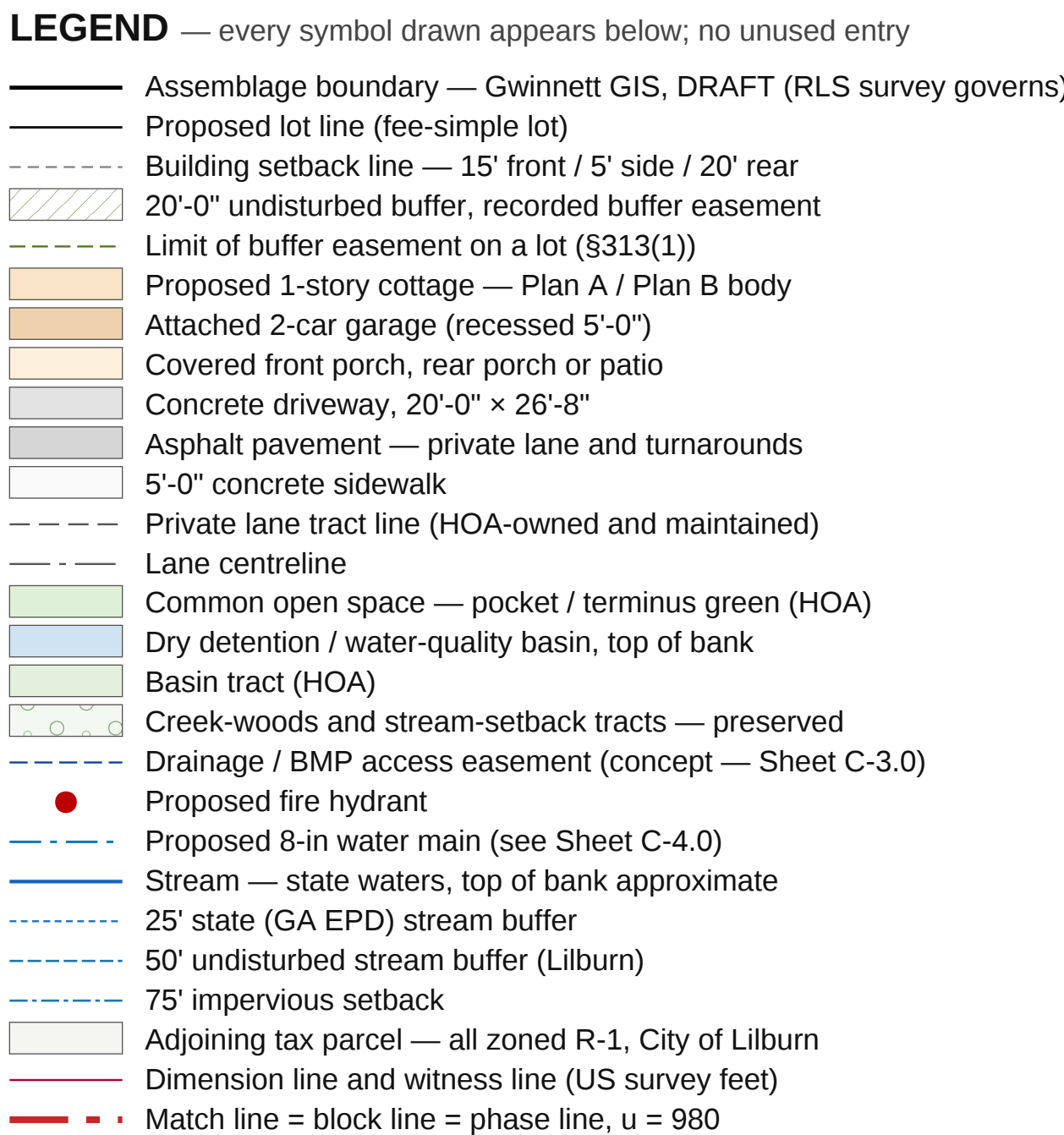
EXISTING CONDITIONS

DRAFT — NOT SEALED — CONCEPT FOR PRE-APPLICATION REVIEW — NOT FOR CONSTRUCTION, PERMITTING OR RECORDING

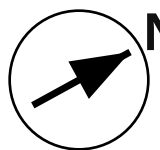
C-2.0

MASTER CONCEPT PLAN — OVERALL

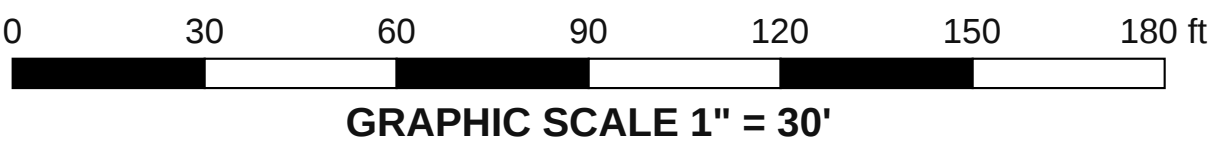
See also C-0.0 cover, C-1.0 existing conditions,
C-2.1 / C-2.2 enlargements, C-2.3 entry enlargements



BLOCK A = SHEET C-2.1 · BLOCK B = SHEET C-2.2
THIS SHEET IS SHADED · MATCH LINE u = 980



GRID NORTH (SR 2240 GA WEST)
= 28°43' ABOVE THE +u AXIS



DRAFT — NOT SEALED — CONCEPT PLAN FOR PRE-APPLICATION REVIEW

MATCH LINE u = 980 IS ALSO THE BLOCK LINE AND THE PHASE LINE — SEE THE KEY PLAN AND SHEET C-2.1

Every dimension is computed from data/layout.json at run time and is DRAFT — the sealed RLS survey and the PE design govern.

Lot designations read BLOCK-LOT. EG $\pm$ is EXISTING ground from the USGS 3DEP DEM, approximate; FFE is set on Sheet C-3.0. Boundary and adjoining parcels: Gwinnett County GIS parcel fabric.

GENERAL NOTES — SHEET C-2.2 (CONT.)

and the lane is the preserved CREEK-WOODS TRACT. No stream-buffer variance is requested (§9, BSW (g)).

6. POND 2 — DRY DETENTION / WATER QUALITY. Concept basin, top of bank 65'-5" x 180'-0", 6-ft depth, 3:1 slopes, about 44,734 cf. The two basins provide 77,863 cf against 75,410 cf required at the Gwinnett County Stormwater Management Manual rate of 10,000 cf per disturbed acre on 7.54 disturbed acres (RRV zero at concept). WQv = 18,659 cf at 44.1% impervious. The top of bank stands 30.0 ft from the south-west line — 10.0 ft clear of the 20-ft buffer — and 98 ft from the nearest stream reach, outside the 75-ft setback. The 10-ft drainage easement and the 30-ft BMP access easement with its 15-ft road at <= 20% are CONCEPT, to be set outside the 100-year ponding limit by the PE on Sheet C-3.0 with forebay, outlet structure, spillway and the note ACCESS EASEMENT TO BE CLEARED AND GRUBBED. Ashlar (ARE) soils map gneiss bedrock at 22-40 in here: rock probes and an undercut allowance are required before the hydrology is trusted.
7. PERIMETER BUFFER (Table 4.1; §313(1); Checklist §4.j, §6.b). A 20'-0" UNDISTURBED buffer runs the entire south-west, north-east and north-west lines — every line abutting R-1 — and is dimensioned on the plan. Table 4.1 gives 0 ft for detached single-family in R-2 and 20 ft for "all other allowed dwelling types"; cottage homes are listed separately, so the plan is designed to 20 ft. The buffer lies in the rear 20 ft of each perimeter lot in a RECORDED BUFFER EASEMENT, on the reading that §313(1) lets a buffer supersede and coincide with the 20-ft rear yard. THIS SINGLE READING CARRIES THE LOT MODULE — CONFIRM IT AT THE PRE-APPLICATION CONFERENCE. Fallback (buffer in a separate tract, 50' x 82' lots, §1005 variance): Sheet C-2.4.
8. GARAGES AND DRIVEWAYS (Table 4.2 R-2 "A"; Checklist §4.p, §4.t, §7.i). Front-loaded garages are RECESSED 5'-0" BEHIND THE FRONT WALL as Table 4.2 requires; the porch projects a further 6'-0". The garage door stands 26'-8" from the front lot line, so the driveway holds two cars clear of the sidewalk. Driveways are 20'-0" x 26'-8", 4-in concrete on min. 4-in compacted GAB (asphalt alternate 2 in GDOT Type E or F on 4-in GAB, §7.i). 2 garage + 2 driveway spaces per dwelling = 164 on Table 4.8.1's 2 per unit; the sidewalk crosses every apron with curb ramps to PROWAG / 2010 ADA Standards.
9. PRIVATE LANE (§319; Checklist §4.s). One private lane in an HOA tract 35'-11" to 46'-1" wide serves every lot; the tract is dimensioned at two stations. Typical section 2'-2" strip + 5'-0" walk + 2'-0" strip + 22'-0" asphalt + 2'-0" strip + 5'-0" walk + 2'-2" strip. Each lot has 50'-0" of frontage against the 30-ft minimum of §319. Labelled PRIVATE: "all components of private streets and alleys must meet minimum standards for public street" (§4.s). Pavement 2 in GDOT

GENERAL NOTES — SHEET C-2.2 (CONT.)

Type E or F on min. 4-in GAB; section on Sheet C-8.0. South-west walk from u = 530; travelled way 1754'-0". The 5-ft sidewalk and the 10-ft Arcado Rd landscape strip appear in the Checklist only under the Highway 29 Overlay comments (§5.b, §5.g) and are therefore provided VOLUNTARILY — confirm applicability at the pre-application conference.

10. FIRE HYDRANTS AND WATER. 5 hydrants on a new 8-in DIP main in the lane tract, 2 on this sheet at the stations shown; spacing 360 / 410 / 355 / 376 ft against Gwinnett DWR Water and Sewer Standards (April 2016) §2.2.15(b), "350 feet to an absolute maximum of 450 feet", with "a fire hydrant near the end of each main". Each stands in a common tract where the lane widens, so IFC 2024 (GA App. D103.1 (26 ft of width where a hydrant is on the access road) can be met. Existing hydrant at (u -45, v +41), 61 ft from the front corner — Sheet C-0. Fire flow 1,000 gpm / 1 hr (App. B Table B105.1(i)), 500 gpm with the voluntary NFPA 13D sprinklers. Main size, valving and the DWR flow test: Sheet C-4.0.
11. FIRE APPARATUS ACCESS. The lane is a 1754'-0" dead end; IFC 2024 (GA App. D103.4 SPECIAL APPROVAL of a dead end over 750 ft is requested of the Gwinnett County Fire Marshal. Three 20'-0" x 60'-0" hammerheads at U = 540, 1110, 1690, no interval over 594 ft, and no circular cut-de-sac (Table 4.2 R-2 "A"). A SECOND ACCESS IS NOT REQUIRED: Ga. Comp. R. & Regs. 120-3-3-.04 replaces App. D107.1 and sets the trigger at MORE THAN 120 DWELLING UNITS, not the model code's 30; this development has 41. NFPA 13D sprinklers are offered VOLUNTARILY, not as required mitigation. No on-lane parking; 22'-0" is kept clear.
12. FINISHED FLOOR ELEVATIONS (Checklist §4.g). FFE IS NOT SET AT THE REZONING STAGE. "EG ±" in each lot is the EXISTING ground at the centre of the pad (NAVd88), interpolated from the USGS 3DEP 1-m DEM on a 100 x 50-ft grid — APPROXIMATE, for reference only, NOT a proposed elevation; on this sheet it ranges 935 to 956 ft. Finished floors, proposed contours, pad grading, walls and the lane profile are set by the PE on Sheets C-3.0 and C-5.0 after a topographic survey. Each dwelling is 1 story, ridge 17'-10" (Plan A) / 19'-4" (Plan B) above finished floor against the 40-ft Table 4.1 maximum.
13. OPEN SPACE AND MAINTENANCE (Checklist §4.o). Every tract other than a numbered lot is common open space or private street tract, conveyed to and maintained by the HOA; none is offered for public dedication. Common open space 141,089 SF = 3.24 ac = 33.8% of the GIS area / 34.3% of 9.44 deeded ac, of which 13,960 SF is built or paved; green-only 127,129 SF = 30.5%. The 41,000 SF of buffer easement on lots is reported separately and is NOT counted. Development Regulations §5.9 "Recreation Areas" applies only at 50 acres or more, so all of this open space is voluntary.

GENERAL NOTES — SHEET C-2.2 (CONT.)

14. UTILITIES NOT DRAWN HERE. Storm drainage, sanitary sewer, gas, electric and telecommunications are on Sheets C-3.0 and C-4.0. PHASE 1 NEEDS NO SEWER EXTENSION — the existing 8-in Arcado Road Townhomes outfall crosses the property; the point of connection is the existing manhole at invert 927.13. Phase 2 is a roughly 178-ft OFF-SITE GRAVITY TIE to the Legends at Parkview 8-in main at invert 919.58. AN IN-TRAC PUMP STATION, PRIVATE FORCE MAIN, PRIVATE GRAVITY SEWER OR GRINDER SYSTEM IS NOT AVAILABLE: Gwinnett DWR's Standard Policy for Private Developments (rev. 9/2018) allows private facilities "only ... for commercial properties under single ownership", and WSR-24 §1.31(A) allows a county station only where gravity is "more than 5,000 feet down gradient" against an actual 178 ft. DWR capacity certification required; Georgia 811 locates (O.C.G.A. §25-9) precede field work.
15. ADJOINING PROPERTY. Every parcel adjoining this sheet is zoned R-1, City of Lilburn, in single-family use — King David Manor (plat §5/159) north-east, Legends at Parkview (plat 118/187) south-west, Nantucket at the north-west end. PINs, addresses, owners and zoning for all 31 adjoining parcels and the county right-of-way are labelled on Sheet C-0 and are not repeated here, so the lot dimensions stay legible.
16. STATUS, SIGNS AND MAIL. DRAFT — NOT SEALED. Concept information prepared by the owner for a rezoning application under Ord. 2023-603 §1003; not a survey, not a construction document, not a plat. SIGNS: the only sign proposed is one ground-mounted monument entry sign at the Arcado Rd entrance, face not over 32 SF and height not over 6'-0" above grade (VERIFY against the Lilburn sign regulations); NO WALL SIGNS ARE PROPOSED on any dwelling, the clubhouse or any accessory structure. ALL SIGNS SHALL BE PERMITTED SEPARATELY (§4.0). MAIL: cluster box units at the amenity tract; contact the Lilburn Post Office Growth Manager (§4.r).

Disclaimer: This is an AI-generated analysis for preliminary planning purposes. All findings must be verified by a licensed professional before use in design, permitting, or regulatory submissions.

Label: DRAFT / NOT SEALED — every item that must be sealed by a Georgia RLS, PE, architect or landscape architect is shown as a concept only. Statements of conformity read "appears consistent with".

THE COTTAGES AT ARCADE SPRINGS

MASTER CONCEPT PLAN — ENLARGEMENT, BLOCK B — Rezoning R-1 → R-2 (City of Lilburn, Georgia)

4535 / 4537 / 4539 / 4541 Arcado Rd SW, Lilburn, GA 30047 — Land Lot 123, 6th District, Gwinnett County — PINs R6123 033, 015, 014, 162 — 9.44 ac (deeded)

Lot-level enlargement of Sheet C-2.0 — Block B / Phase 2, u 960 to the north-west end · lots, creek-woods tract, stream buffers, Pond 2 and its easements, terminus green and hammerhead

APPLICANT / OWNER

Mohammed Awad

4541 Arcadio Rd SW, Lilburn GA 30047-3968 (parcels 033, 015); Mendez / Roblero de Leon (parcels 014, 162) — co-applicants, owner signatures required

PREPARED FOR

Pre-application conference, City of Lilburn Planning & Zoning (340 Main St) – rezoning application per Ord. 2023-603 §1003 (2026 Application Instructions)

STATUS

DRAFT — NOT SEALED

Concept lot layout for a rezoning application. Lot lines, setbacks, buffers, basins and easements are DRAFT. A sealed Georgia RLS boundary survey, PE civil design and a registered landscape architect's buffer and landscape plan are required and govern. Not a plat, not a survey. Generator: tools/sitebase.py + tools/enlargements.py

DATE 2026-09-03

SCALE Scale 1" = 30' (ARCH D 36 × 24 in)

DATUM SR 2240 GA West US-ft; NAVD88

DRAWN owner-prepared (AI-assisted)

CHECKED RLS / PE — pending

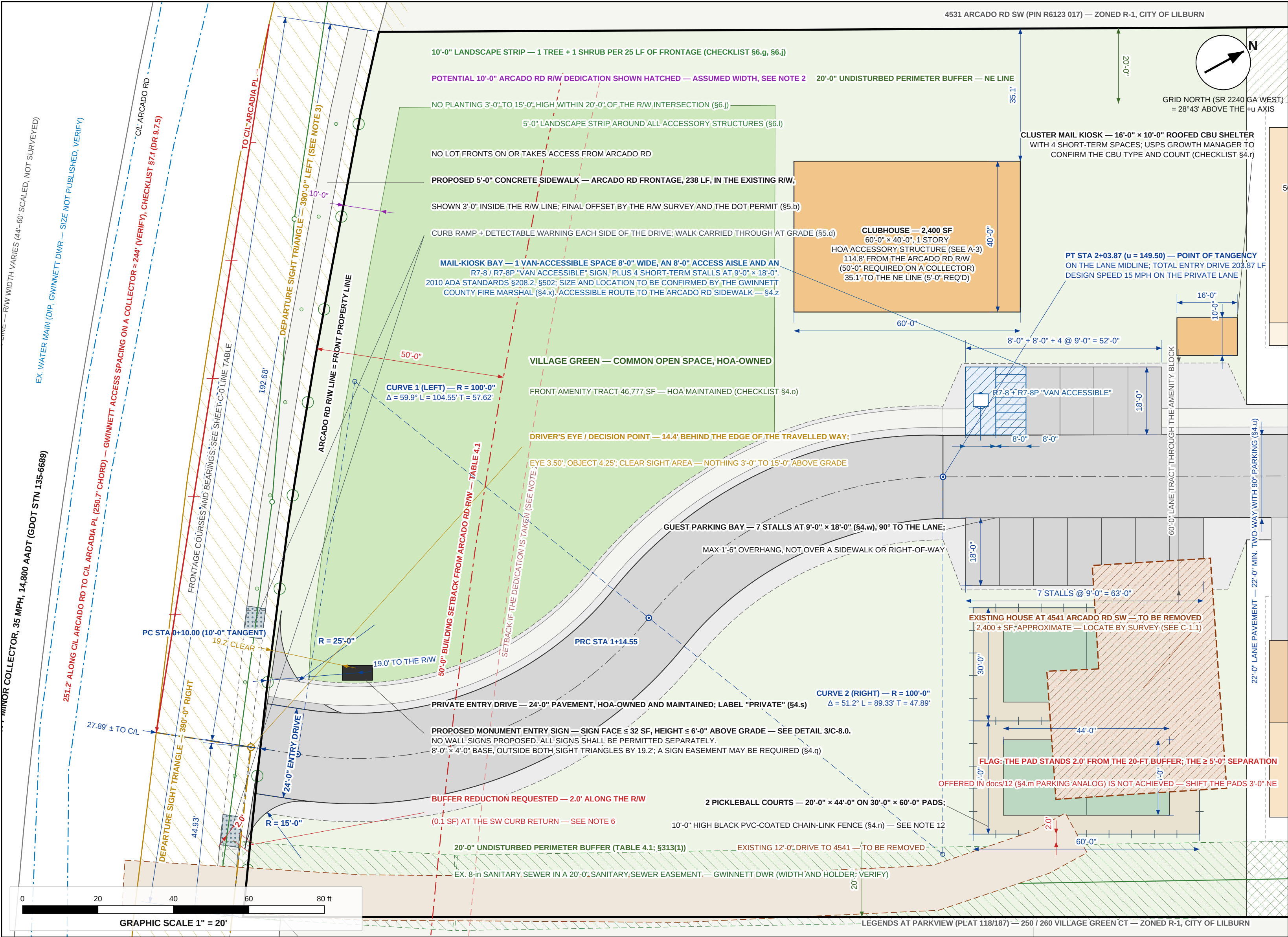
SHEET

C-2.2

MASTER CONCEPT PLAN — ENLARGEMENT, BLOCK E

ENTRY, FRONTAGE & AMENITY ENLARGEMENT — SCALE 1" = 20'

Site-local coordinates: +u along the strip to the NW, +v across toward the NE line (see north arrow). All dimensions in US survey feet, DRAFT — the sealed RLS boundary and right-of-way survey governs.



GENERAL NOTES — SHEET C-2.3

- SCOPE AND STATUS.** This sheet enlarges the frontage, entrance and front amenity block of Sheet C-2.0 at 1" = 20'. Coordinates are site-local (u, v) US survey feet from the Gwinnett GIS parcel fabric and data/layout.json. DRAFT, NOT SEALED; a Georgia RLS boundary and R/W survey, a PE sight-distance certificate and PE civil design are required and govern. Nothing here certifies compliance.
- RIGHT-OF-WAY AND DEDICATION.** R/W WIDTH VARIES — 44-60 ft scaled between the two Gwinnett GIS R/W lines. 27.89 ft from the entrance to the mapped centreline; scaled, not surveyed. A Gwinnett minor collector section is wider, so a dedication is likely: the 10'-0" strip is SHOWN HATCHED AS AN ASSUMPTION ONLY. If taken, the 50-ft setback line moves inboard 10 ft (dashed), the landscape strip moves with it and the monument sign, at 9.0 ft from the new R/W, must be re-set. Establish the R/W by survey and the Gwinnett DOT permit.
- INTERSECTION SIGHT DISTANCE.** 390'-0" required each way at 35 mph (Gwinnett UDO §900.40.4 Table 900.2; AASHTO case B1). Triangles are

drawn from a decision point on the drive centreline 14.4 ft behind the edge of the travelled way, eye 3.50 ft, object 4.25 ft, to points 390 ft each way on the mapped centreline. Edge of travelled way ASSUMED 11'-0" from the CIL — verify. THE AVAILABLE SIGHT DISTANCE IS NOT CERTIFIED HERE: the certificate is blank (Checklist §7.i, DR 9.7.4). Vertical (crest) sight distance must be field-measured too.

- CLEAR SIGHT AREA.** Nothing between 3'-0" and 15'-0" above the roadway grade may stand in either triangle — no sign, wall, fence, berm, cabinet, parked vehicle or landscape planting (Checklist §6.j bars planting 3-15 ft high within 20 ft of a R/W intersection). The monument sign is clear by 19.2 ft at its nearest corner. Keeping the area clear is an HOA covenant obligation.
- ACCESS.** One full-movement private entrance; no lot fronts Arcadio Rd or takes a driveway from it. The entrance CIL is 251.2 ft from the CIL of Arcadia Pl along the Arcadio Rd centreline (250.7 ft chord) against a collector spacing of about 244 ft — VERIFY with

Gwinnett DOT. Arcadia Pl cannot be aligned with the drive: it meets Arcadio Rd 61 ft NE of the NE front corner. Checklist §7.f (DR 9.7.5) offsets: no other access is mapped within 400 ft.

- BUFFER REDUCTION REQUESTED.** With the SW curb return cut from R = 25'-0" to R = 15'-0", 2.0 ft of it measured along the R/W (0.1 SF of pavement) still lies inside the 20-ft perimeter buffer at the R/W corner. It is requested as a buffer reduction under Ord. 2023-603 §1003-4; Checklist §6.i allows drives and utilities to encroach up to 50 ft of width and no further without a ZBA variance. The letter of intent and the voluntary conditions must match.
- PRIVATE STREET AND PAVING.** The entry drive and lane are private, HOA-owned and labelled "PRIVATE" per Checklist §4.s; all components must meet minimum public-street standards. Paving 4" GAB with 2" Type E or F asphalt minimum (§7.i), to be confirmed against the Lilburn Development Regulations and a Gwinnett Fire apparatus-load check. Materials and details on C-8.0 (§4.i).

- SIDEWALKS AND ACCESSIBLE ROUTE.** A 5'-0" concrete walk runs the full 238-LF frontage, shown 3'-0" inside the R/W line; the final offset (2'-0" min. behind back of curb, 6'-0" with street trees) is set by the R/W survey and the DOT permit. It is carried through the drive at grade with a curb ramp and detectable warning each side (§5.d). A 5'-0" walk on the NE side of the drive and lane links the clubhouse, kiosk and guest bay to the public walk (§4.2).
- LANDSCAPE.** 10'-0" landscape strip along the R/W: 1 tree + 1 shrub per 25 LF = 10 trees and 10 shrubs over 237.63 LF (§6.g, §6.j); parking lot planting 1 tree per 7 spaces with every space within 60 ft of a trunk (§6.k); 5'-0" landscape strip around every accessory structure (§6.i). The tree-protection, buffer and landscape plan must be sealed by a registered landscape architect, forester or arborist (§6.a) and is not part of this sheet.
- BUFFERS AND SETBACKS.** 20'-0" undisturbed perimeter buffer on the NE and SW lines (Table 4.1, R-2 abutting R-1, "all other allowed dwelling types"); §313(1) buffers supersede the minimum yards; none

on the Arcadio Rd frontage. Collector building setback 50'-0": the clubhouse, the nearest structure, is 114.8 ft from the R/W and 35.1 ft from the NE line against a 5'-0" accessory setback. Buffers in recorded easements (§4.i, §4.k, §4.l, §4.o).

- SIGNS.** One ground-mounted monument entry sign, face not exceeding 32 SF and height not exceeding 6'-0" above grade — see Detail 3/C-8.0 for the elevation and area computation. NO WALL SIGNS ARE PROPOSED on any dwelling, the clubhouse or any accessory structure. ALL SIGNS SHALL BE PERMITTED SEPARATELY (§4.q); a sign easement may be required. The sign regulations' area, height, setback and illumination limits to be confirmed at the pre-application.
- MAIL AND FENCING.** Cluster box units in a roofed 16'-0" x 10'-0" kiosk with four short-term spaces; the Lilburn Post Office Growth Manager confirms CBU type, count and the single street name used for house numbering (§4.r). The two pickleball pads are enclosed by a 10'-0" high black PVC-coated chain-link fence (§4.n); VERIFY the

residential fence-height limit for a 10-ft sport-court fence. No court lighting (voluntary condition 11).

- EXISTING CONDITIONS SHOWN.** The house and drive at 4541 Arcadio Rd SW are dashed and ARE TO BE REMOVED (C-1.1); the footprints are aerial-imagery approximations to be located by survey. The existing 8-in Gwinnett DWR gravity sewer and its 20-ft easement run inside the SW line through the buffer; the easement width and holder are from the 2025 hydrology sheet and are NOT surveyed. Georgia 811 locates (O.C.G.A. §25-9) are required before field work. Electric, gas, telecom and storm structures are NOT shown. The nearest existing fire hydrant is 61 ft from the front corner, NE of this sheet — see C-0.
- PERMITS STILL REQUIRED.** Gwinnett DOT driveway / encroachment permit and sight-distance certification; Gwinnett DWR sewer capacity certification; City of Lilburn Land Disturbance Permit; §1106 site and design review; a separate sign permit.

THE COTTAGES AT ARCADIO SPRINGS

ENTRY, FRONTAGE & AMENITY ENLARGEMENT — Rezoning R-1 → R-2 (City of Lilburn, Georgia)

4535 / 4537 / 4539 / 4541 Arcadio Rd SW, Lilburn, GA 30047
Land Lot 123, 6th District, Gwinnett County — PINs R6123 033, 015, 014, 162 — 9.44 ac (deeded)
Enlargement of Sheet C-2.0 at the Arcadio Rd frontage — right-of-way and dedication, sight triangles and ISD lines, curb returns and curve data, buffers, landscape strip, frontage sidewalk, clubhouse, courts, parking, sign, mail kiosk

APPLICANT / OWNER
Mohammed Awad
4541 Arcadio Rd SW, Lilburn GA 30047-3968 (parcels 033, 015); Mendez / Roblero de Leon (parcels 014, 162) — co-applicants, owner signatures required
PREPARED FOR
Pre-application conference, City of Lilburn Planning & Zoning (340 Main St) — rezoning per Ord. 2023-603 §1003

STATUS
DRAFT — NOT SEALED
Entry geometry, sight triangles and dimensions are DRAFT, from the Gwinnett GIS parcel fabric and data/layout.json. A sealed RLS boundary and right-of-way survey, a PE sight-distance certificate (blank at right) and a Gwinnett County DOT driveway permit are required and govern. Generator: tools/sitebase.py + tools/entry.py

DATE 2026-09-03
SCALE Scale 1" = 20' (ARCH C 24 x 18 in)
DATUM SR 2240 GA West US-ft; NAVD88
DRAWN owner-prepared (AI-assisted)
CHECKED RLS / PE — pending

SHEET
C-2.3
ENTRY, FRONTAGE & AMENITY ENLARGEMENT
ARCH C 24" x 18"

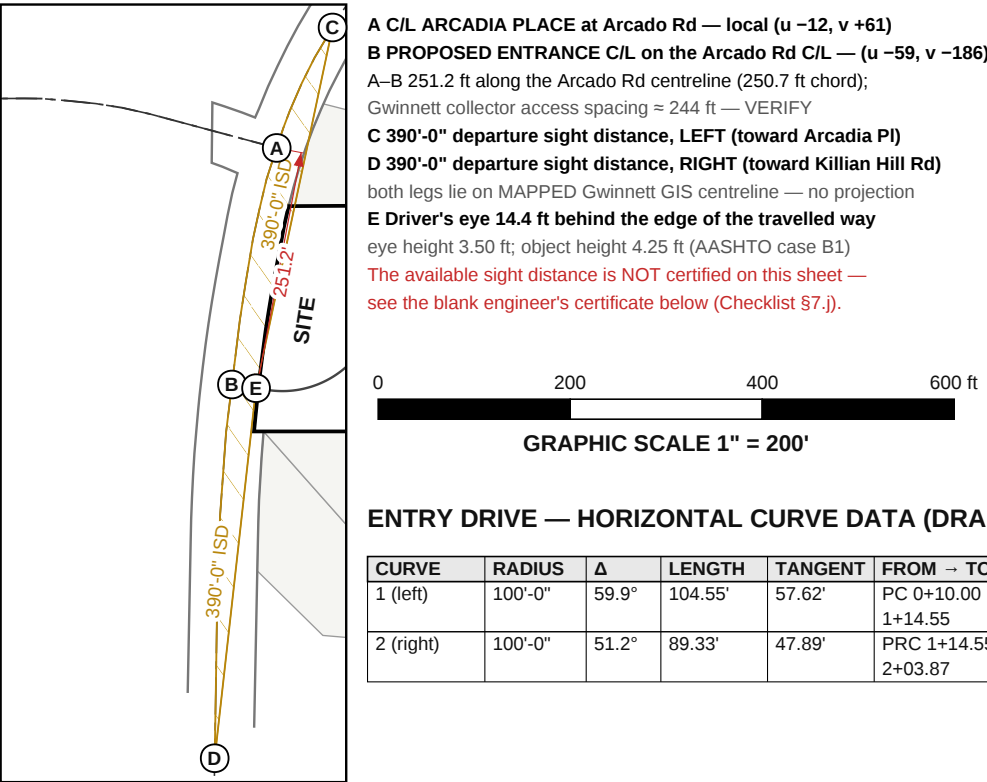
DRAFT — NOT SEALED — SEE SHEET C-2.0 FOR THE WHOLE SITE AND C-2.1 FOR THE LOTS

LEGEND — every symbol drawn on this sheet appears here

- Property line / Arcadio Rd R/W line
- Adjoining parcel — zoned R-1 (Lilburn)
- Opposite R/W line (R/W width varies)
- Existing road centreline
- Potential R/W dedication (10'-0" assumed)
- 10'-0" landscape strip + planting
- 20'-0" undisturbed perimeter buffer
- 50'-0" building setback from the R/W
- Setback line if the dedication is taken
- Proposed pavement (drive, lane, bays)
- Proposed 5'-0" concrete sidewalk
- Curb ramp + detectable warning
- HOA tract line (amenity / lane / drive)
- Front amenity tract — common open space
- Village green
- Clubhouse / mail kiosk (accessory)
- Pickleball pad 30'-0" x 40'-0"
- Court surface 20'-0" x 44'-0"
- 10'-0" black chain-link fence
- Parking stall 9'-0" x 18'-0"
- Van-accessible space + access aisle
- Sight triangle — clear sight area
- Sight line — 390' ISD leg
- Driver's eye / decision point
- Curve point — PC / PRC / PT
- Monument entry sign (face ≤ 32 SF)
- Proposed cottage, garage and porch
- Existing 8-in sanitary sewer + MH
- Existing 20'-0" sewer easement
- Existing water main (Gwinnett DWR)
- Existing building / drive — to be removed

SIGHT DISTANCE & ACCESS SPACING DIAGRAM

Both departure sight triangles at full length. Scale 1" = 200' — NOT the scale of the plan above.



ENGINEER'S CERTIFICATE — SIGHT DISTANCE (BLANK — TO BE COMPLETED AND SEALED)

City of Lilburn Site Development Plan Review Checklist §7.j; Gwinnett Dev. Regulations 9.7.4. To be completed, signed, dated and sealed across the seal by a Georgia registered professional engineer. NOT COMPLETED — this sheet makes no sight-distance claim.

I certify that I have field-measured the intersection sight distance available at the entrance shown on this sheet, from a driver's eye height of 3.50 ft located 14.4 ft behind the edge of the travelled way to an object height of 4.25 ft, and that the available sight distance is:

LEFT (toward Arcadia Pl) _____ ft RIGHT (toward Killian Hill Rd) _____ ft
REQUIRED _____ ft at _____ mph. Vertical (crest) sight distance: YES ___ NO ___
Clearing required in the R/W or the landscape strip: YES ___ NO ___
Signature _____ Printed _____ GA P.E. No. _____ Date _____

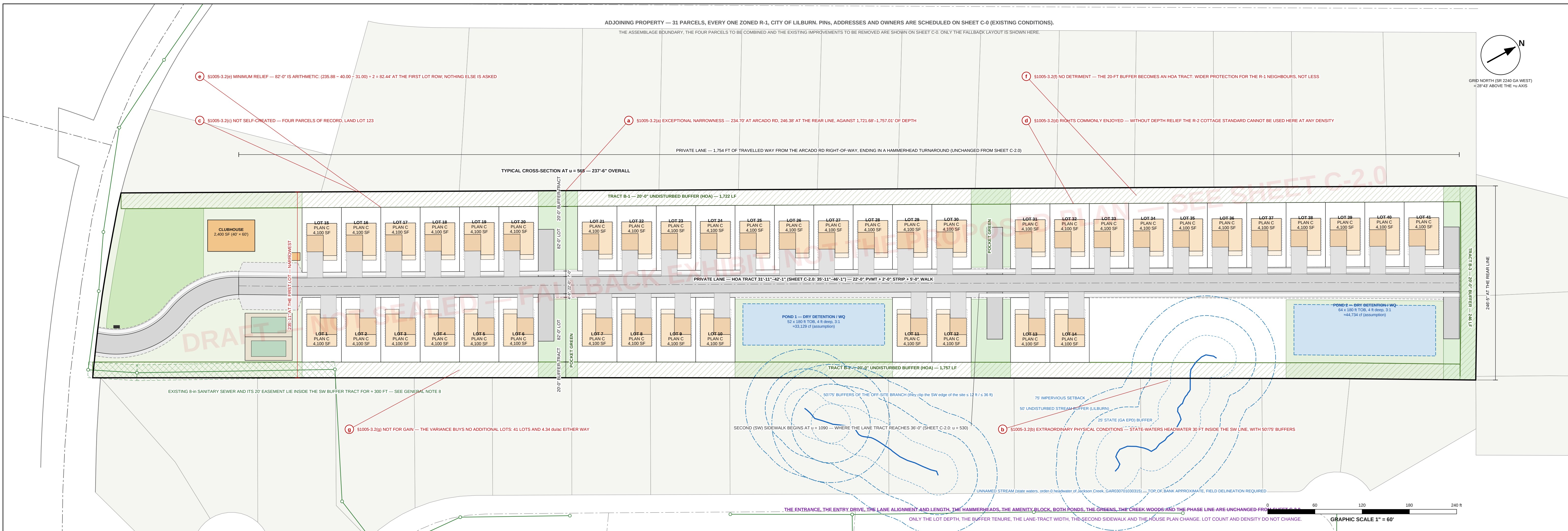
ENTRANCE AND ACCESS DATA (DRAFT — Checklist §4.p, §7.f, §7.j)

ITEM	VALUE / BASIS
Entrance on the R/W	u = 31.43, v = 190.00 — SW third of the frontage; 192.68' from the NE front corner, 44.93' from the SW corner, along the R/W
Entry drive	24'-0" pavement, private, two-way; 203.87 LF, R/W to the lane midline
Curb returns	NE R = 25'-0"; SW R = 15'-0", reduced from 25'-0" to clear the buffer
Access spacing	251.2' to the CIL of Arcadia Pl along the Arcadio Rd CIL (250.7' chord) against ≈ 244 ft — VERIFY (Checklist §7.i)
Sight distance	390'-0" required each way at 35 mph; drawn, NOT certified here (Checklist §7.j)
Left-turn lane	Not warranted below 75 lots (41 proposed) — VERIFY with Gwinnett DOT
Buffer reduction	2.0' along the R/W (0.1 SF) at the SW curb return — §1003-4, Checklist §6.f

PARKING AND ACCESSIBILITY (Checklist §4.v, §4.w, §4.x)

ITEM	REQUIRED / PROVIDED
Required, 41 dwellings	2 per dwelling unit, no maximum (Table 8.1) = 82 spaces
Provided on lot	164 — 2-car garage + 2-car driveway on every lot (Sheet C-2.1)
Guest bay	7 stalls at 9'-0" x 18'-0", 90° to the lane (Checklist §4.w)
Mail-kiosk bay	1 van-accessible 6'-0" stall + 8'-0" access aisle + R7-8 / R7-8P sign, and 4 stalls at 9'-0" x 18'-0"
Accessible required	1 of the 12 front-block spaces; 1 in 6 van-accessible (2010 ADA §208.2, §502.2) — 1 provided; size and location to be confirmed by the Gwinnett County Fire Marshal (Checklist §4.x)
Drive aisle	22'-0" minimum two-way with 90° parking (Checklist §4.u) — 22'-0" provided

Disclaimer: This is an AI-generated analysis for preliminary planning purposes. All findings must be verified by a licensed professional before use in design, permitting, or regulatory submissions.
DRAFT / NOT SEALED — every item that must be sealed by a Georgia RLS, PE, architect or landscape architect is shown as a concept only. Statements of conformity read "appears consistent with"; nothing on this sheet certifies compliance.



SIDE-BY-SIDE COMPARISON — the two readings of Ord. 2023-603 §313(1), priced

ITEM	SHEET C-2.0 — §313(1) BUFFER INSIDE THE LOT	THIS SHEET C-2.4 — BUFFER IN A SEPARATE TRACT
Buffer against R-1	20'-0" undisturbed buffer inside the rear 20 ft of every perimeter lot, in a recorded buffer easement (§313(1) reading)	20'-0" undisturbed buffer in separate HOA-owned Tracts B-1 / B-2 / B-3 — 73,697 SF
Typical lot	80'-0" × 100'-0" = 5,000 SF	50'-0" × 82'-0" = 4,100 SF
Table 4.1 lot area (3,000 SF min.)	5,000 SF — appears consistent	4,100 SF — appears consistent
Table 4.1 lot width (50 ft min.)	50'-0" — appears consistent	50'-0" — appears consistent
Table 4.1 lot depth (100 ft min.)	100'-0" — appears consistent	82'-0" — VARIANCE REQUIRED (§1005)
Loss	41 (14 SW + 27 NE)	41 (14 SW + 27 NE) — unchanged
Density (B dual max.)	4.34 du/acre (deeded) / 4.28 GIS	4.34 du/acre (deeded) / 4.28 GIS — unchanged
Buildable body depth	100 — 15 — 20 = 65'-0" less the 6-ft porch = 59'-0"	82 — 15 — 20 = 47'-0" less the 6-ft porch = 41'-0"
House plans that fit	Plan A 38'-0" × 51'-0"; Plan B 38'-0" × 57'-0"	NETHER — a third plan is required: Plan C 38'-0" × 41'-0"
Conditioned area	1,406 SF (Plan A) / 1,492 SF (Plan B)	1,032 SF (Plan C) — 12 SF above the 1,000-SF Table 4.1 minimum
Total under roof	1,966 SF / 7,182 SF	1,555 SF
Rear amenity	Plan A 6'-0" patio; Plan B 34'-0" covered rear porch	None within the 20-ft rear yard; uncovered patio shown dashed — VERIFY
Private lane tract	35'-11" — 46'-1"; 22-ft pavement, symmetrical	31'-11" — 42'-1"; 22-ft pavement set 18'-0" from the NE tract line
Second (SW) sidewalk	From u = 530 (≈ 1,170 ft)	From u = 1,090 (≈ 610 ft)
Common open space	141,089 SF = 33.8% of GIS area, plus 41,000 SF of buffer easement on lots reported separately	183,893 SF = 44.1% of GIS area, including the 73,697-SF buffer tracts; no buffer easement on any lot
Impervious area	184,148 SF = 44.1% of GIS area	157,158 SF = 37.7% of GIS area (Plan C is smaller)
Water-quality volume WQv	16,659 cf	16,230 cf
Disturbed area / detention	7.54 ac; 76,410 cf required; 77,863 cf provided	Unchanged — the preserved buffer bands and creek woods are identical
Fire access	41 units — under the 120-unit trigger of IFC 2024 (GA) App. D107.1 as amended by Ga. Comp. R. & Regs. 120-3-3-.04, D103.4 special approval for the dead end	Unchanged
Sanitary sewer	Phase 1 in the existing on-site 8-in main; Phase 2 by the = 178-ft off-site gravity tie	Unchanged
Approvals needed	Rezoning R-1 → R-2, plus a favorable §131(3) interpretation by the Planning Director	Rezoning R-1 → R-2, plus a concurrent §1005 variance on lot depth decided by Mayor and Council with the companion application
Filing cost	\$1,250 rezoning fee (City of Lilburn FY2026-2027 Fee Schedule, Res. 2026-08-31, 2026-08-31)	\$1,250 + \$500 variance hearing fee (+\$25 for each additional variance on the same property) — confirm the adopted amounts at filing
Principal risk	The §131(3) reading may be rejected. Article 3 sets no buffer widths and the ordinance nowhere authorizes a required buffer inside a private lot	A variance is discretionary; the applicant must carry every §1005-3.2 criterion (a) through (g) at the hearing

Every figure in the right-hand column is computed by tools/fallback.py from the same boundary, entrance and lane geometry as Sheet C-2.0.

LEGEND — every symbol drawn on this sheet

Assemblage boundary (GIS — DRAFT)	Front amenity tract (unchanged)
Adjoining lot parcel — zoned R-1	Village green (unchanged)
Arcado Rd right-of-way line	Chainage and mail kiosk (unchanged)
Existing road centerline	Pickelball pad (unchanged)
20'-0" perimeter buffer tract (HOA)	Pickelball court surface (unchanged)
Fallback lot line 50' × 82' = 4,100 SF	Monument entry sign (unchanged)
Building setback line 15 / 5 / 20	Stormwater pond tract (unchanged)
Plan C dwelling — SCHEMATIC	Dry detention / WQ basin (unchanged)
2-car garage, recessed 5'-0"	Stream — state waters (top of bank approx.)
Covered front porch 6'-0"	25' state (GA EPD) buffer
Driveway 20'-0" × 26'-0"	50' undisturbed stream buffer
Private lane / entry drive tract (HOA)	75' impervious setback
Pavement and hammerhead turnaround	Existing 8-in sanitary sewer and manhole
Guest / mail-kiosk parking stall 9' × 18'	Existing 20' sanitary sewer easement
Sidewalk 5'-0"	§1005-3.2 criterion key (a) — (g)
Line centerline (fallback alignment)	Dimension line and tick
Pocket / terminus green (unchanged)	

Existing contours, vegetation, structures, utilities, soils and the adjoining owner schedule are on Sheet C-0 and are not repeated here.

CONCURRENT VARIANCE — Ord. 2023-603 §1005, lot depth 100'-0" — 82'-0" (criteria §1005-3.2 (a)–(g), keyed on the plan)

KEY	CRITERION (SUBJECT — VERIFY)	APPLICANT'S RESPONSE
a	Exceptional narrowness.	The assemblage is a strip 234'-0" wide at the Arcado Rd right-of-way, 235'-11" at the first lot row (u = 230) and 246'-0" at the rear line, against 1,757'-0" of depth on the SW line and 1,721'-0" on the NE line — about 7.3 : 1. Take a 20-ft buffer tract off each side (40 ft) and the 31-ft minimum private-lane section that City of Lilburn Site Development Plan Review Checklist §4.4 ("all components of private streets... must meet minimum standards for public street") requires, and 164'-11" of width is left for two facing rows of lots — 82'-0" each. No arrangement of a 100-ft-deep lot fits. An unnamed order-0 headwater of Jackson Creek (state waters) reaches about 30 ft inside the SW line and carries 25-ft state, 50-ft undisturbed and 75-ft impervious setbacks, a ridge at u = 1,090–1,200 divides the sanitary basins; the only public frontage is 27.63 ft of Arcado Rd, a county collector on which no lot may be driveway-served. The developable envelope is fixed before any lot is drawn.
b	Extraordinary and exceptional physical conditions	R-1 lots "Single family (cluster-cottage, creative lot configurations)" as a permitted use (§602) and sets a 3,000-SF, 50-ft-cottage lot in Table 4.1. If the buffer must sit in its own tract, the 100-ft depth rule makes that permittee unworkable here at any density — the district standard the property is being rezoned to use cannot be used.
c	The condition is not the result of any action by the applicant	Relief is 18 ft of depth (100 ft — 82 ft, 18 ft), and 82 ft is arithmetic, not preference: (235.88 – 2 × 20.00 – 31.00) + 2 = 82.44 ft. Nothing else is asked: lot area 4,100 SF (min. 3,000), lot width 50 ft (min. 50), density 4.34 du/acre (max. 4), front 15 ft, side 5 ft, rear 20 ft, height one story, buffer 20 ft — every one met as drawn.
d	Strict application deprives the owner of rights commonly enjoyed by other property in the district	The variance moves protection outward, not inward: the 20-ft buffer stops being a private easement inside a back yard and becomes 73,697 SF of HOA-owned undisturbed tract, shown on the plat with a no-disturbance restriction and a maintenance obligation (Checklist §4.4, §6.3). Common open space rises from 33.8% to 44.1% of the site and impervious area falls to 37.7%. Density, use, height, setbacks and the 55+ (HOA) condition are unchanged.
e	The variance requested is the minimum necessary	The variance yields no additional lots and no additional density: 41 lots and 4.34 du/acre either. Its only effect is to let the required buffer be held in the separate tract staff may require, at the cost of 900 SF of private lot area per home and a smaller house. The applicant would prefer the §313(1) reading and no variance at all.
f	Relief may be granted without substantial detriment to the public good and without impairing the purposes and intent of the ordinance	Unchanged
g	The relief sought is not necessary for financial advantage to the applicant	Unchanged

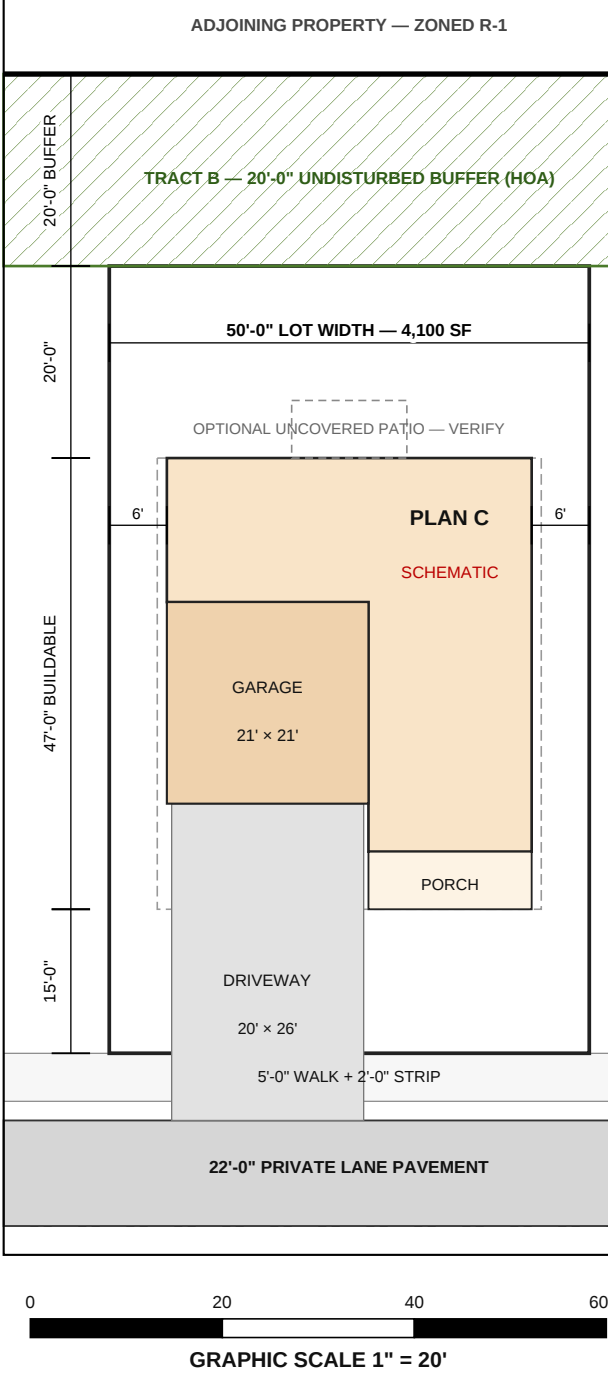
The letters (a) — (g) are keyed to the plan above in red. The adopted text of §1005-3.2 is NOT reproduced on this sheet — see General Note 5.

FALLBACK SITE DATA AND AREA TABULATION

ITEM	THIS SHEET (COMPUTED)	BASIS
Gross site area	417,174 SF = 9.577 ac (GIS); 9.44 ac: deeded	data/fallback.py boundary .ring, RLS survey geoids
Loss	41 × 4,100 SF = 168,100 SF (40.3%)	50'-0" × 82'-0", computed lot by lot (4,100–4,100 SF)
Private lane tract	57,725 SF	Residual between the two lot rows, 31'-11"–42'-1"
Entry drive tract	7,460 SF	Unchanged from Sheet C-2.0
Common open space (all HOA tracts)	183,893 SF = 4.22 ac = 44.1% of GIS / 44.7% of deeded	Residual: site less lots less the lane and entry-drive tracts
— of which perimeter buffer tracts	73,697 SF [B-1, NE 1,722 LF, B-2 SW 1,757 LF, B-3 rear 246 LF]	20'-0" mixed inward offset of the side and rear property lines
— of which amenity, greens, ponds, creek woods	110,190 SF	Unchanged tracts of Sheet C-2.0, less the area now inside the buffer tracts
Private lane tract	0 SF (Sheet C-2.0: 4,100 SF)	The point of this sheet
Impervious area	157,158 SF = 37.7% of GIS area	Computed from the Plan C footprint and the drawn pavement — see the impervious schedule
Water-quality volume WQv	16,230 cf	GC5WMM, 1.2 × (0.05 × 0.009 × 37.7) × 12 × 417,174 SF
Detention required / provided	75,810 cf / 77,863 cf	Unchanged — 7.54 disturbed ac at 10,000 cfs
Parking	82 required, 164 on lot + 12 guest	Table 8.1: 2 per dwelling unit; 2 garage + 2 driveway per lot
Lot frontage on the lane	50'-0" every lot (30 ft required)	Ord. 2023-603 §319
Clearance, lots to the stream centerline	99'-6" minimum (Sheet C-2.0: 80.6 ft)	All three disclosed setbacks, top of bank not yet delineated

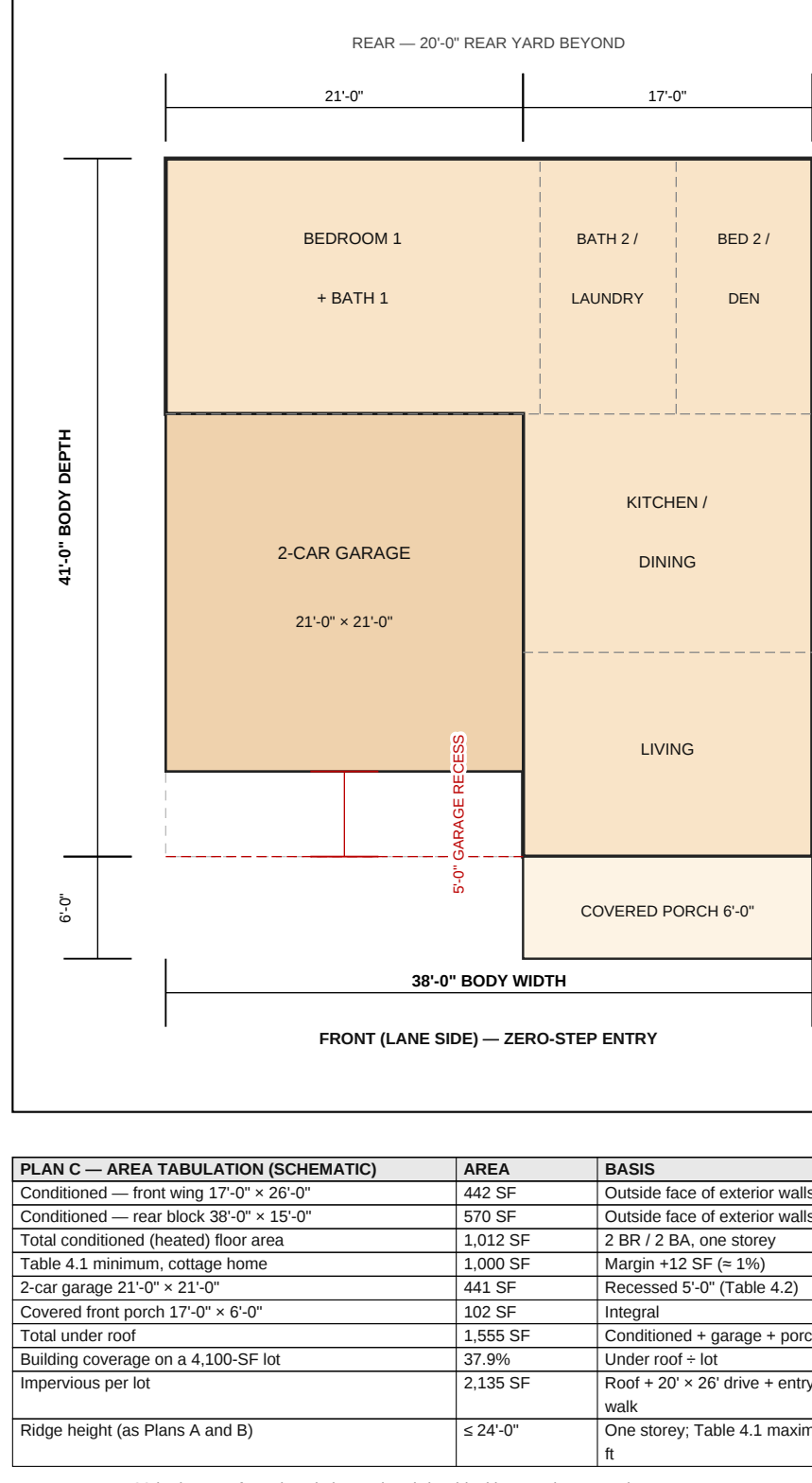
TYPICAL FALLBACK LOT — 50'-0" × 82'-0" (4,100 SF)

Scale 1" = 20'. Buffer in a separate HOA tract.



PLAN C — SCHEMATIC COTTAGE FOOTPRINT

Scale 3/32" = 1'-0". Block diagram only — SCHEMATIC, not a designed plan.



PLAN C — AREA TABULATION (SCHEMATIC)	AREA	BASIS
Conditioned — front wing 17'-0" × 26'-0"	442 SF	Outside face of exterior walls
Conditioned — rear block 36'-0" × 15'-0"	570 SF	Outside face of exterior walls
Total conditioned (related) floor area	1,012 SF	2 BR / 2 BA, one storey
Table 4.1 minimum, cottage home	1,000 SF	Margin +12 SF (≈ 1%)
2-car garage 21'-0" × 21'-0"	441 SF	Recessed 5'-0" (Table 4.2)
Covered front porch 17'-0" × 6'-0"	102 SF	Integral
Total under roof	1,555 SF	Conditioned + garage + porch
Building coverage on a 4,100-SF lot	37.9%	Under roof + lot
Impervious per lot	2,135 SF	Roof + 20' × 26' drive + entry
Ridge height (as Plans A and B)	≈ 24'-0"	One storey, Table 4.1, maximum 40 ft

Zero-step entry, 36-in doors, 5-ft turning circles and grab-bar blocking as Plans A and B.

GENERAL NOTES — FALLBACK LAYOUT

- STATUS OF THIS SHEET — IT IS NOT THE PROPOSED PLAN. The proposed layout is Sheet C-2.0, on which the 20-ft buffer required against the adjoining R-1 property is held inside the rear 20 ft of each perimeter lot in a recorded buffer easement, under Lilburn Zoning Ordinance 2023-603 §313(1) ("buffer requirements... supersede these minimum required yards"). This sheet is the alternative that is built if the Planning Director of the City Attorney reads §313(1) as not authorizing a required buffer inside a private lot. Only one of the two layouts will be constructed, both are shown so the Commission and the Council can price the interpretation before it is made.
- WHY THE FALLBACK IS CARRIED. Two independent readings put the §313(1) basis in doubt: (i) §313(1) speaks of "buffer requirements established by this article" — Article 3 — and Article 3 lists no buffer width; the 20-ft buffer abutting R-1 comes from Table 4.1 in Article 4, a zoning rule in Ord. 2023-603 expressly authorizing a required buffer inside a private lot; (ii) The City of Lilburn Site Development Plan Review Checklist §6.3 reads "If a portion of required buffer is within an easement, provide 25' additional buffer outside easement", which on its face would add 25 ft outside any buffer easement held on a lot. Both are questions for the pre-application conference (Planning Director, 340 Main St).
- WHAT CHANGES AND WHAT DOES NOT. Unchanged: lot depth 100'-0" — 82'-0", buffer tenure (easement on lot = HOA tracts), house plan (Plans A and B — Plan C), private-lane tract width, the station at which the second sidewalk begins. Unchanged: the boundary, the entrance at v = -190 and the 24-ft entry drive, the lane alignment and the 20-ft driveway, the 20-ft driveway, all three hammerhead turnarounds, the amenity block, both stormwater ponds, the pocket and terminus greens, the creek-woods tract, the stream setbacks, the phase line at u = 980, the perimeter buffer width (20 ft), every setback, the one-story height, the use, and the 55+ (HOA) condition. LOT COUNT AND DENSITY DO NOT CHANGE: 41 lots at 4.34 du/acre on 9.44 deeded acres, well under the 8 du/acre maximum of Table 4.1. Yield here is set by frontage on the single line, not by lot depth — the same 41 fifty-foot lots from the same travelled way in both layouts — so the shawler lot returns 900 SF per lot to common open space and takes nothing off the unit count.
- THE VARIANCE REQUESTED. Table 4.1 sets a minimum lot depth of 100 ft for ALL uses in R-2, with no cottage-home exception, so 82'-0" criteria a variance under Ord. 2023-603 §1005. It would be filed as a CONCURRENT variance with the rezoning and decided by Mayor and Council with the companion application (§1005-1); no variance may be approved before the Council's final action on the rezoning. NO OTHER RELIEF IS REQUESTED: lot area 4,100 SF (1,000 SF minimum), lot width 50'-0" (50 ft minimum), density 4.34 du/acre (8 maximum), front setback 15 ft, side 5 ft, rear 20 ft, height one story — each appears consistent with Table 4.1 as drawn. The applicant's responses to the §1005-3.2 criteria are tabulated at the left and keyed (a) — (g) on the plan.
- THE §1005-3.2 CRITERIA TEXT IS NOT REPRODUCED HERE — VERIFY. The excerpt set the package works from (disturbance-excepting) end inside §1005-3.1, so the adopted wording of the criteria at §1005-3.2(a) through (g) has not been transcribed from a primary source. The criterion subjects shown are the standard Georgia hardship-variance subjects and the letters are placeholders. Before filing, transcribe §1005-3.2(a)–(g) verbatim from Ord. 2023-603 and confirm each response — and its letter — to the adopted text.
- THE 82'-0" DIMENSION IS ARITHMETIC. With available at the first lot (u = 230) = 235.88 ft, less two 20'-0" buffer tracts (40.0 ft), less the 31'-0" minimum private-lane tract (22'-0" pavement + 2'-0" strip + 5'-0" sidewalk + 2'-0" strip) = 164.88 ft for two facing rows, i.e. 82.44 ft each. The lot module is set at 82'-0" and the surplus within that the strip picks up toward the rear (8 widens to 246'-5") is carried to the lane tract, exactly as it is on Sheet C-2.0. Lot depth is measured perpendicular to the front lot line; lot area is 4,100 SF on every lot computed lot by lot (4,100 – 4,100 SF).
- THE THIRD HOUSE PLAN. Body depth available on an 82-ft lot = 82 – 15 (front setback) = 20 (rear setback) = 47'-0", less the 6'-0" covered porch = 41'-0". Plan A "The Springfield" has a 31'-0" body and Plan B "The Laurel" 37'-0" (disabilities) on, so NEITHER FITS. Plan C is drawn as a SCHEMATIC one-story 38'-0" × 41'-0" body with the same front-facing garage, an integral 6'-0" covered porch and a 2-car garage matching 5'-0" behind the rear wall (Table 4.2 design criterion) — 1,012 SF conditioned, 441 SF garage, 102 SF covered porch, 1,555 SF total under roof. That is 12 SF above the 1,000-SF minimum heated floor area for a cottage home in Table 4.1 — about 1%. Plan C carries no covered rear porch or a grade-uncovered patio is shown dashed in the 20-ft rear yard and its permeability must be confirmed. Plan C is a BLOCK DIAGRAM, NOT A DESIGNED PLAN. It is drawn to show that a compliant cottage fits, and must be replaced by a Georgia-registered architect's drawings.
- THE PERIMETER BUFFER TRACTS. Tracts B-1 (NE line, 1,722 LF), B-2 (SW line, 1,757 LF) and B-3 (rear line, 246 LF) total 73,697 SF, are 20'-0" wide and extend perpendicular to the property line, are owned and maintained by the homeowners association, and are to be shown on the final plat labeled "20' UNDISTURBED BUFFER with a no-disturbance restriction and the HOA maintenance obligation (Checklist §4.4, §6.3). No dwelling, drive or accessory structure is drawn in them. Two encroachments carried over from Sheet C-2.0 must be disclosed: (i) the existing 8-in sanitary sewer and its 20-ft easement run inside Tract B-2 for about 300 ft from Arcado Rd to the manhole at (u 272, v -224) — Checklist §6.1 permits sanitary sewer conveyance facilities and easements to encroach a buffer "as near as perpendicular as possible, up to max 50' width", which this parallel run does not satisfy, and §6.4 may require 25 ft of additional buffer outside the easement; (ii) the SW curb return at the entrance reaches 2.0 ft into the buffer band (0.1 SF). Both need (i) the Director's reading, and (ii) exists on Sheet C-2.0 as well. The SW pickelball pad stands 2.0 ft clear of Tract B-2 (22.0 ft from the property line); Checklist §6.1's 5-ft landscape strip around accessory structures would move it about 1.0 ft to the NE — an amenity-block item inherited unchanged from Sheet C-2.0.

IMPERVIOUS SCHEDULE — FALLBACK LAYOUT	AREA
Dwellings, garages and covered porches (Plan C on every lot)	63,725
Lot driveways, aprons and entry walks	29,304
Private lane pavement (22'-0" unchanged)	34,111
Entry drive pavement and curb returns (unchanged)	5,049
Hammerhead turnaround (as unchanged)	7,200
Guest and mail-kiosk parking bays (unchanged)	2,070
Sidewalks (5'-0" — the SW walk begins at u = 1090)	11,869
Clubhouse roof, court pails, mail kiosk and amenity walks (unchanged)	6,760
less driveway impervious area counted twice	-2,900
TOTAL IMPERVIOUS — 37.7% of the 417,174-SF GIS site area (Sheet C-2.0: 184,148 SF = 44.1%)	157,158

THE COTTAGES AT ARCADE SPRINGS

FALLBACK LOT-DEPTH LAYOUT EXHIBIT — Rezoning R-1 → R-2 (City of Lilburn, Georgia)

4535 / 4537 / 4539 / 4541 Arcado Rd SW, Lilburn, GA 30047 — Land Lot 123, 6th District, Gwinnett County — PINs R6123 033, 015, 014, 162 — 9.44 ac (deeded)

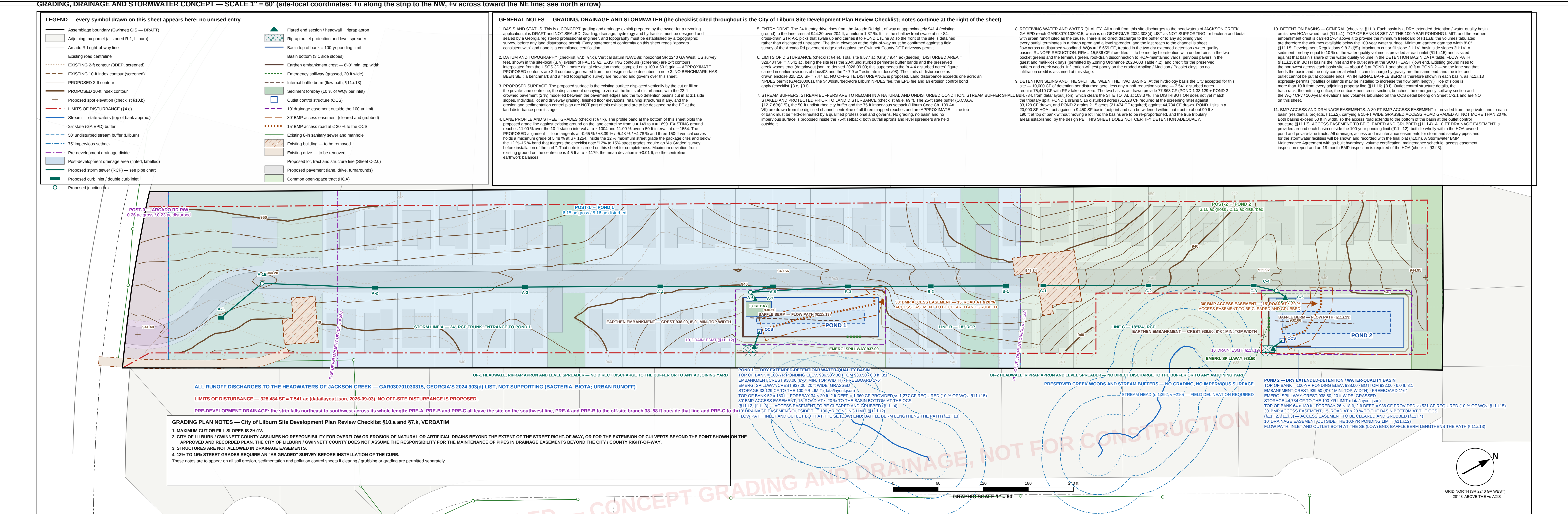
ALTERNATIVE LAYOUT — 50'-0" × 82'-0" lots, the 20-ft buffer in separate HOA tracts, and a concurrent §1005 lot-depth variance (100 ft — 82 ft)

APPLICANT / OWNER
Mohammed Awad
4541 Arcado Rd SW, Lilburn GA 30047-3968 (parcels 033, 015); Mendez / Roblero de Leon (parcels 014, 162) — co-applicants, owner signatures required
PREPARED FOR
Pre-application conference, City of Lilburn Planning & Zoning (340 Main St) — rezoning application per Ord. 2023-603 §1003 (2026 Application Instructions)

STATUS
DRAFT — NOT SEALED
ALTERNATIVE LAYOUT — NOT THE PROPOSED PLAN. The proposed plan is Sheet C-2.0 (Master Concept Plan). This exhibit applies only if staff require the 20-ft buffer in a separate tract; it then carries a concurrent §1005 variance on lot depth (100 ft — 82 ft) and a third house plan, Plan C. Generator: tools/sitebase.py + tools/fallback.py

DATE 2026-09-03
SCALE Scale 1" = 60' (ARCH D 36 × 24 in); details 1" = 20' and 3/32" = 1'-0"
DATUM SR 2240 GA West US-ft; NAVD88
DRAWN owner-prepared (AI-assisted)
CHECKED RLS / PE — pending

SHEET
C-2.4
FALLBACK LOT-DEPTH LAYOUT EXHIBIT



STORM STRUCTURE SCHEDULE — concept, checklist S10.c (rim elevations read from the proposed surface of this sheet)

STR.	TYPE	LOCATION (u, v)	RIM / GRADE	DESCRIPTION / LOCATION
A-1	DOUBLE CURB INLET	(95, -169)	941.58	entry drive at the existing front swale
A-1B	JUNCTION BOX	(150, -122)	944.12	entry drive / lane junction — bend
A-2	CURB INLET	(300, -128)	943.02	lane, southwest edge of pavement
A-3	CURB INLET	(800, -127)	941.72	lane, southwest edge of pavement
A-4	DOUBLE CURB INLET	(850, -126)	940.34	lane sag at PVI 2
A-6	JUNCTION BOX	(800, -134)	938.74	Pond 1 tract; also receives Line B
A-7	FLARED END SECTION	(806, -142)	936.07	into the Pond 1 forebay
B-1	CURB INLET	(1140, -129)	949.21	lane, southeast of the crest at PVI 3
B-2	CURB INLET	(1040, -126)	946.70	lane, southwest edge of pavement
B-3	CURB INLET	(930, -126)	941.40	lane, southwest edge of pavement
C-1	CURB INLET	(1190, -125)	948.54	lane, northwest of the crest at PVI 3
C-2	CURB INLET	(1330, -125)	941.46	lane, southwest edge of pavement
C-3	DOUBLE CURB INLET	(1470, -125)	935.70	lane sag at PVI 4
C-4	JUNCTION BOX	(1500, -132)	935.99	lane tract at the Pond 2 access
C-5	FLARED END SECTION	(1512, -140)	937.48	into the Pond 2 forebay
OF-1	HEADWALL + RIPRAP APRON	(805, -206)	932.74	Pond 1 outlet, thence level spreader
OF-2	HEADWALL + RIPRAP APRON	(1496, -206)	937.65	Pond 2 outlet, thence level spreader

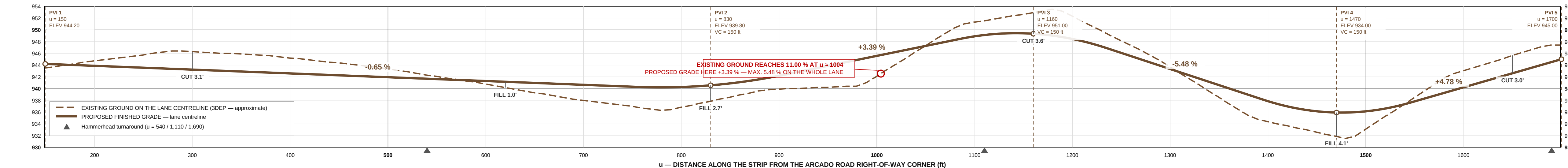
PIPE CHART — concept sizes and slopes; final design by the PE

REACH	PIPE	LENGTH	SLOPE %	INV. UP	INV. DN	COVER R
A-1 ~ A-1B	24" RCP	72	0.35	936.20	935.95	1.4
A-1B ~ A-2	24" RCP	150	0.35	935.95	935.42	6.2
A-2 ~ A-3	24" RCP	200	0.35	935.42	934.72	5.6
A-3 ~ A-4	24" RCP	180	0.35	934.72	934.09	5.0
A-4 ~ A-5	24" RCP	150	0.35	934.09	933.57	4.5
A-5 ~ A-6	24" RCP	31	0.50	933.57	933.41	4.8
A-6 ~ A-7	24" RCP	147	0.50	933.41	933.36	3.3
B-1 ~ B-2	18" RCP	100	2.50	945.20	942.70	2.5
B-2 ~ B-3	18" RCP	110	4.80	942.70	937.42	2.5
B-3 ~ A-6	18" RCP	130	3.00	937.42	933.51	2.5
C-1 ~ C-2	18" RCP	140	5.10	944.20	937.06	2.8
C-2 ~ C-3	18" RCP	140	4.20	937.06	931.18	2.9
C-3 ~ C-4	24" RCP	31	0.50	931.18	931.03	2.5
C-4 ~ C-5	24" RCP	147	0.50	931.03	930.56	3.0
POND 1 OCS ~ OF-1	24" RCP	21	1.40	930.50	930.20	—
POND 2 OCS ~ OF-2	24" RCP	16	1.40	932.00	931.78	—

Cover is measured from the structure rim to the top of the pipe barrel. All pipe reinforced concrete to Gwinnett County standards.

LANE PROFILE — PRIVATE LANE CENTRELINE, u = 149 TO u = 1699

HORIZONTAL 1" = 60' · VERTICAL 1" = 10' · EXAGGERATION 6:1 · DATUM ELEV. 930 · NAVD83 · DRAFT — NOT SEALED — grading and profile design must be sealed by a Georgia PE



Disclaimer: This is an AI-generated analysis for preliminary planning purposes. All findings must be verified by a licensed professional before use in design, permitting, or regulatory submissions.

Label: DRAFT / NOT SEALED — every item that must be sealed by a Georgia RLS, PE, architect or landscape architect is shown as a concept only. Statements of conformity read "appears consistent with".

THE COTTAGES AT ARCADE SPRINGS

GRADING, DRAINAGE AND STORMWATER CONCEPT — Rezoning R-1 → R-2 (City of Lilburn, Georgia)

4535 / 4537 / 4539 / 4541 Arcado Rd SW, Lilburn, GA 30047 — Land Lot 123, 6th District, Gwinnett County — PINs R6123 033, 015, 014, 162 — 9.44 ac (deeded)

Proposed grading, drainage-area map, detention concept and lane profile — 2026 Application Instructions item 8; City of Lilburn Site Development Plan Review Checklist S4.e, S7, S10, S11

APPLICANT / OWNER
Mohammed Awad

4541 Arcado Rd SW, Lilburn GA 30047-3968 (parcels 033, 015); Mendez / Robledo de Leon (parcels 014, 162) — co-applicants, owner signatures required

PREPARED FOR

Pre-application conference, City of Lilburn Planning & Zoning (340 Main St) — rezoning application per Ord. 2023-603 S1003 (2026 Application Instructions)

STATUS

DRAFT — NOT SEALED

Concept grading and drainage for pre-application review. Existing topography is USGS 3DEP (approximate) — a topographic survey is required and governs. Grading, hydrology, hydraulics and the detention design must be sealed by a Georgia PE. NOT FOR CONSTRUCTION.
Generator: tools/sitebase.py + tools/grading.py

DATE
2026-09-03

SCALE
Scale 1" = 60' (ARCH D 36 × 24 in); profile 1" = 60' H / 1" = 10' V

DATUM
SR 2240 GA West US-ft; NAVD88

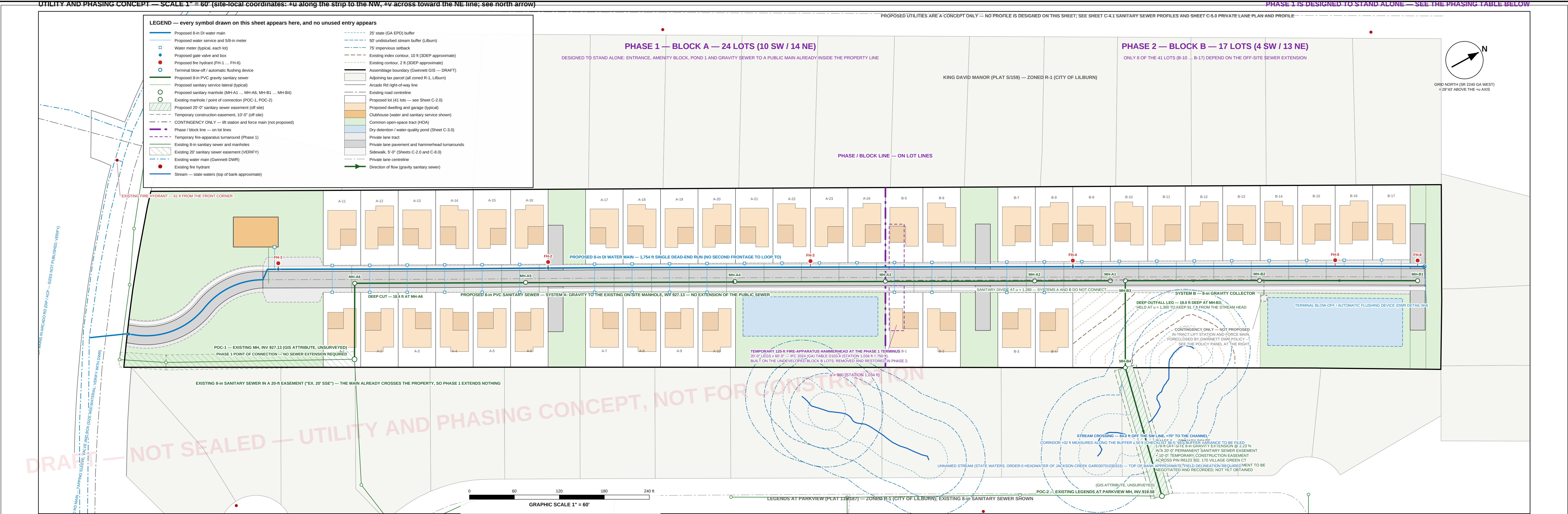
DRAWN
owner-prepared (AI-assisted)

CHECKED
RLS / PE — pending

SHEET

C-3.0

GRADING, DRAINAGE AND STORMWATER CONCEPT



WATER — PROPOSED SYSTEM DATA

ITEM	PROPOSED — CONCEPT ONLY; PE DESIGN AND GWINNETT DWR APPROVAL REQUIRED
Main	Proposed 8-in ductile iron, in the private lane tract 14 ft NE of the lane centreline, 3.5 ft cover, 8-in (not 6-in) no meet 504–1,000 gpm reaches the terminus at a 20 ps residual on a 1,754-ft dead-end.
Length	1,754 ft in the lane tract = 51 ft of connection in the Arcado Rd right-of-way = 1,805 ft.
Point of connection	DEAD-END MAIN — there is no second public frontage to loop to. Tapping sleeve, valve and box on the existing Arcado Rd main (DIP, diameter not published in the public layer) — VERIFY with Gwinnett DWR at local (u -82, v -185).
Domestic demand	4.1 DU x 250 gpd + clubhouse, 2,400 SF x 175 gpd/1,000 SF = 10,670 gpd average, max day (x 2.0) = 21,340 gpd; peak hour (x 4.0) 29,340 gpm.
Fire flow	Stated both ways because the sprinklers are voluntary, 1,000 gpm at 20 psi for 1 hour for one- and two-family dwellings @ 3,600 SF; reduced 50% (floor 500 gpm) where every dwelling is sprinklered to NFPA 13D — IFC 2024 (GA) Appendix B B102.1 and Table B102.1(1); clubhouse is a separate Table B102.2 check — VERIFY with the Gwinnett Fire Marshal.
Hydrants	6 proposed on the phase (FH-1 ... FH-4) + 1 existing on Arcado Rd 6 ft from the front corner. Maximum spacing 300 ft, average 304 ft; maximum distance from any point on the travelled way to a hydrant 224 ft.
Spacing standard	IFC 2024 (GA) Appendix C Table C102.1 — fire-flow requirement is 1,750 gpm; average spacing 500 ft, REDUCED 100 ft for a dead-end road > 400 ft; maximum distance from any point on the road frontage to a hydrant 250 ft; both are met (504 ft average, 224 ft maximum reach). Gwinnett DWR Standards (2016) §2.2.15(b) asks 350–450 ft; FH-1 to FH-5 are 350–360 ft apart; FH-6 sits 110 ft beyond FH-5 because §2.2.15(d) asks for “1 in the hydrant near the end of each main”; confirm with DWR that the 350-ft minimum does not govern a terminal hydrant.
Valves	3 at the connection (see assembly) + in-line valves at u 170, 530 and 1,230 — one every third hydrant and none more than 1,000 ft apart (DWR Standards (2016) §2.2.16(b)). Hydrant lead valves not separately shown.
Meters and services	41 individual 5/8-in x 3/4-in meters in the lane tract at each lot frontage + 1 clubhouse service + 1 HOA irrigation meter, SW-side services cross under the 22-ft pavement in casing or by bore. Backflow prevention per the DWR cross-connection policy.
Dead-end treatment	Terminal blow-off or automatic flushing device at the terminus hammerhead (DWR Standards (2016) Detail WA, “Typical Dead End Street Terminator”). The main holds a 4.710 gal against a 10.670 gpd of use — turnover under one day — but the device is DWR’s call.
Separation	Water main 14 ft NE and sanitary main 5 ft SW of the lane centreline = 19 ft horizontal separation, against a 10-ft minimum, 18-in vertical separation at every crossing.

HYDRANT SCHEDULE — stations along the travelled way

ID	STATION	LOCATION (u,v)	TO NEXT	PLACEMENT AND BASIS
EX		(u -45, v +11.5)		EXISTING hydrant in the Arcado Rd right-of-way, 61.2 ft from the front corner (Gwinnett DWR GIS). Covers the entrance apron.
FH-1	224	(u 170, v -100)	360	At the amenity block, opposite the guest parking bay.
FH-2	584	(u 530, v -102)	350	At pocket green, 1 and hammerhead 1 (u-540) — the lane is locally wider here, which suits IFC Appendix D §103.0.1.
FH-3	934	(u 880, v -100)	350	On the lot line at u 880, opposite the Pond 1 tract frontage.
FH-4	1,284	(u 1,230, v -100)	350	On the lot line at u 1,230, at pocket green 2 / hammerhead 2 (u 1,110–1,130) reach.
FH-5	1,634	(u 1,580, v -100)	110	On the lot line at u 1,580, opposite the Pond 2 tract frontage.
FH-6	1,744	(u 1,690, v -100)	— (end)	At the terminus hammerhead, with the terminal blow-off = 1 in the hydrant near the end of each main (DWR Standards (2016) §2.2.15(d)). Its 110-ft spacing from FH-5 is shorter than the 350-ft minimum of §2.2.15(b) — confirm with DWR.

Every hydrant is on the proposed 8-in main on the NE side of the lane, 3 ft behind the back of curb, on a lot line where one is allowed. Hydrant lead valves are required at each hydrant and are not separately symbolised; see Sheet C-6.0 for the hydrant assembly detail.

CONSTRUCTION SEQUENCE — CONCEPT; THE APPROVALS EACH PHASE DEPENDS ON

PHASE 1 — (1) Georgia RLS boundary and topographic survey, an invert survey of POC-1 and rock probes on the System A mark; (2) Gwinnett DWR sewer capacity certification (“Pre-Rezoning” request) and a DWR flow test at the Arcado Rd hydrant; (3) Gwinnett DOT driveway permit and the City land disturbance permit (NPDES NOI, erosion-control bond); (4) entrance, entry drive and lane to u 980 with the temporary hammerhead; System A sewer MH-A1, POC-1, 8-in water main to FH-A1, Pond 1, and the amenity block; (5) the Block A acceptance and the Block A final plat.

PHASE 2 — (6) a recorded off-site sanitary sewer easement across PIN R6123 302 and the Checklist §9.6 buffer variance for the stream crossing; (7) lane u 980 = 1,700, temporary hammerhead removed and the area restored; (8) System A extension MH-A2 — MH-A3, the System B collector and the 17-ft off-site extension to POC-2; 8-in water main to FH-B1 and the terminal blow-off; (9) Pond 2, 8-in culverts, DWR acceptance and the Block B final plat.

SANITARY MANHOLE SCHEDULE — SYSTEM A (PHASE 1 GRAVITY TO POC-1, INV 927.13)

STR.	LOCATION (u,v)	RIM	INVERT	DEPTH	RUN (ft)	SLOPE %	REMARKS
MH-A1	(1280, -119)	945.6	940.61	5.0	101.0	1.717	Upstream terminus of System A, sanitary divide (Block B lots up station of u 1,280 drain to System B). Phase 2 construction.
MH-A2	(1178, -119)	953.5	939.88	14.6	199.0	1.717	Under the u 1,178 ridge crest (DDEP 953.5) — the deepest structure on System A, Phase 2 construction.
MH-A3	(880, -120)	940.5	935.46	5.0	201.0	2.070	ON THE PHASE / BLOCK LINE. Upstream terminus of the Phase 1 contract; receives the Phase 2 reach MH-A1 — MH-A3 later.
MH-A4	(779, -120)	936.3	931.30	5.0	270.0	0.685	Lane sag (DDEP 936.3). START OF THE CONTROLLING REACH: 608.4 ft at 0.685 % to POC-1 against a 0.40 % minimum.
MH-A5	(600, -122)	943.4	929.39	14.0	298.0	0.685	Deep cut under the u 480–600 lane rise.
MH-A6	(272, -123)	946.3	927.82	18.4	101.3	0.685	DEEPEST STRUCTURE ON THE SHEET — the lane hump at u 272 (DDEP 946.3). Think Tunks SW to POC-1 across Lot A-1 in a 20-ft sanitary sewer easement (see note §9).
POC-1	(272, -224)	VERIFY	927.13	—	—	—	EXISTING manhole, “Arcado Road Townhomes sewer outfall” (Gwinnett DWR). Invert is a GIS attribute, UNSURVEYED. Not published — field survey.

Phase 1 constructs MH-A3 — POC-1 = 809 ft; Phase 2 extends MH-A1 — MH-A3 = 300 ft, all on site. RIM = existing ground interpolated from the USGS 3DEP 1-m DEM (APPROXIMATE). DEPTH = rim – invert. SLOPE = (invert – invert out) / run. The finished lane profile is not designed on this sheet; every rim and every slope must be re-cut to it — see Sheets C-4.1 and C-5.0.

SANITARY MANHOLE SCHEDULE — SYSTEM B (PHASE 2, THE 17-ft OFF-SITE EXTENSION TO POC-2, INV 919.58)

STR.	LOCATION (u,v)	RIM	INVERT	DEPTH	RUN (ft)	SLOPE %	REMARKS
MH-B1	(1690, -119)	947.4	942.40	5.0	211.0	7.536	Upstream terminus at the terminus hammerhead (u 1,690). Slope follows existing ground; < 10 %, so no concrete anchors are indicated at connect.
MH-B2	(1478, -119)	931.5	926.50	5.0	179.0	1.000	REAR LANE LOW POINT (DDEP 931.5) — 0.63 ft BELOW the POC-1 invert before any slope is applied, which is why the rear cannot reach POC-1 by gravity.
MH-B3	(1300, -119)	943.7	924.71	19.0	116.0	1.000	JUNCTION / OUTFALL. Deep: the alignment is held at u 1,300 so the SW leg stays 91.7 ft from the stream head and clear of the 5075-ft buffers (see note §9).
MH-B4	(1300, -235)	926.7	923.55	3.1	176.1	2.229	AT THE SW PROPERTY LINE — start of the >17-ft OFF-SITE extension. Rim EXTRAPOLATED 17.7 ft beyond the DDEP sample grid.
POC-2	(1349, -406)	VERIFY	919.58	—	—	—	EXISTING manhole on the Legends at Parkview 8-in main (Gwinnett DWR). Invert is a GIS attribute, UNSURVEYED. Doghouse / deep connection and manhole type per DWR.

System B: 506 ft on site + 176.1 ft of site = 684 ft, falling 22.82 ft from invert 942.40 at MH-B1 to invert 919.58 at POC-2 — an overall 3.34 % against a 0.40 % minimum. MH-B4 — POC-2 is the only reach outside the property. The MH-B2 — POC-2 sub-reach is 473.1 ft, of which 176 ft is off-site; that the route doc-008 Memo B 8.5 prices at “~479 ft, of which >176 ft off-site”.

ESTIMATED SANITARY FLOWS — 250 gpd/home, 175 gpd per 1,000 SF of clubhouse, peaking factor 4.0

BASIS (Gwinnett DWR Flow Estimation Guidelines, Rev. 10/2022)	AADF	PEAK (4.0)	NOTE
Whole development — 41 lots x 250 gpd + clubhouse 2,400 SF x 175 gpd/1,000 SF	10,670 gpd	42,680 gpd = 29.6 gpm	Governs the DWR capacity certification
On the lot line at u 1,230, at pocket green 2 / hammerhead 2 (u 1,110–1,130) reach.	2,000 gpd	8,000 gpd = 5.6 gpm	No public sewer is extended
Phase 1 — 24 lots + clubhouse (System A) to POC-1	6,420 gpd	25,680 gpd = 17.8 gpm	Split between System A and System B
Phase 2 — 17 lots, of which 9 also drain to POC-1 by gravity	4,250 gpd	17,000 gpd = 11.9 gpm	Inn on 8-in main where full-flow capacity at 0.40 % is = 344 gpm
System B only — the 8 lots that need the off-site extension (B-10 — B-17)	2,000 gpd	8,000 gpd = 5.6 gpm	

The constraint is the downstream county main, not the on-site pipe — which is exactly what the Gwinnett DWR sewer capacity certification tests. See question Q1 at the right.

DISTURBED AREA — SITE DEVELOPMENT PLAN REVIEW CHECKLIST §4.6

AREA (SF)	AREA (AC)
5,344	0.123
400	0.009
1,200	0.028
2,268	0.052
9,212	0.211
326,484	7.541
337,696	7.752

GOVERNING GWINNETT DWR POLICY

why the off-site gravity tie is the PRIMARY Phase 2 route and no pumping solution is proposed

Gwinnett County DWR, Developer Pump Station Standards (January 2014, posting WSR-24, §3.3A).

“Approval for the installation of a new pump station will only be granted by the GCDWR’s AMIS Division Director through consultation with the GCDWR. Generally, pump stations will only be permitted when gravity sewer is unavailable to the property. Unavailability shall generally be interpreted to mean more than 5,000 feet down gradient, but this distance can be increased or decreased by GCDWR Division Directors based upon actual field conditions and the size of the project involved.”

Gwinnett County DWR, Standard Policy for Private Developments: Condominiums, Townhomes and Subdivisions (rev. September 2018).

“PRIVATE FACILITIES. Private pump stations, private force mains and private gravity sewers are only allowed for commercial purposes under single ownership within a development. Gwinnett County is not responsible for operation and maintenance of these facilities.”

and, immediately above it, “OWNERSHIP OF MAINS AND APPURTENANCES. Absolute title and ownership of all water and sanitary sewers within the private development shall be conveyed to Gwinnett County WSA at the time of acceptance by the County.”

Gravity is 178 ft away, not more than 5,000 ft. A 41-lot FEE-SIMPLE subdivision is not “commercial purpose under single ownership.” Gwinnett DWR publishes no design standard for grinder pumps or low-pressure sewer — zero occurrences of “grinder” in the Water and Sewer Standards (April 2016) and in WSR-24. An in-tract lift station, private force main, private gravity sewer or grinder / low-pressure system is therefore NOT AVAILABLE to this project and is shown on the plan only as a labelled contingency.

QUESTIONS PUT TO GWINNETT DWR AND TO THE CITY ON THIS SHEET

Q1 — DWR, sewer capacity certification. Submitted on the Sewer Capacity Certification Request, Rev. 07/2023, “Pre-Rezoning” request type, sealed by a design professional, to DWR (sewer/Certification/GwinnettCounty.com) (after 10 business days, “>20 ft analysis or flow monitoring is required). Does downstream capacity exist for a 4.00 gpd AADF at POC-1, the Arcado Road Townhomes outfall, and 2,000 gpd of POC-2, Legends at Parkview? Please confirm both facility IDs.

Q2 — DWR, pumping. Does DWR concur that no pumping solution — county-owned station, HOA-owned station, force main, grinder pump or low-pressure sewer — is available to a fee-simple residential subdivision under the two policies quoted above, so that the 17-ft gravity tie is the only Phase 2 route?

Q3 — DWR, the dead-end main, LOOPING AND OVERSIZING. The lane is a single 1,754-ft dead-end main and there is no second public frontage to loop to. Does DWR accept an 8-in dead-end main at this length, or does it require an oversized main? (b) is a terminal blow-off sufficient, or is an automatic flushing device required (DWR Standards (2016) Detail WAF7 (c) is any logical alternative available — a connection through the backstreet or King David Manor systems at the rear, for example — that DWR would prefer? Please conduct a flow test at the existing Arcado Rd hydrant and the 8-in size can be confirmed against 1,000 gpm at 20 psi residual.

Q4 — DWR, easement, connection and ownership. What permanent easement width and manhole type does DWR require at POC-2, and will DWR accept a disjunctive connection on the existing main? Would DWR instead prefer a new manhole on the existing on-site main at u +220, inside the amenity tract, in place of the Lot A-1 crossing, POC-1 (note §9)? All manholes and the 8-in water main to be conveyed to Gwinnett County WSA on acceptance, per the ownership clause quoted above.

Q5 — City of Lilburn, the comprehensive-plan policy. The 2026 Comprehensive Plan Amendment (transmitted to OCA 2026-07-13; adoption pending as of 2026-09-03) carries the Suburban Low Policy “Do not further extend sewer in this area.” Phase 1 extends nothing — the existing 8-in main already crosses the property. Phase 2 is a 178-ft connection between two existing county mains in the same drainage basin that opens to an unsewered land. Does the City read the policy as reaching that connection? And which Future Development class do these two parcels fall in — Suburban Low (B, 73) or Suburban-Medium (B, 71)? This question IS UNSURVEYED AND IS DISCLOSED HERE RATHER THAN ASSUMED AWAY.

PHASING — 41 LOTS IN TWO BLOCKS

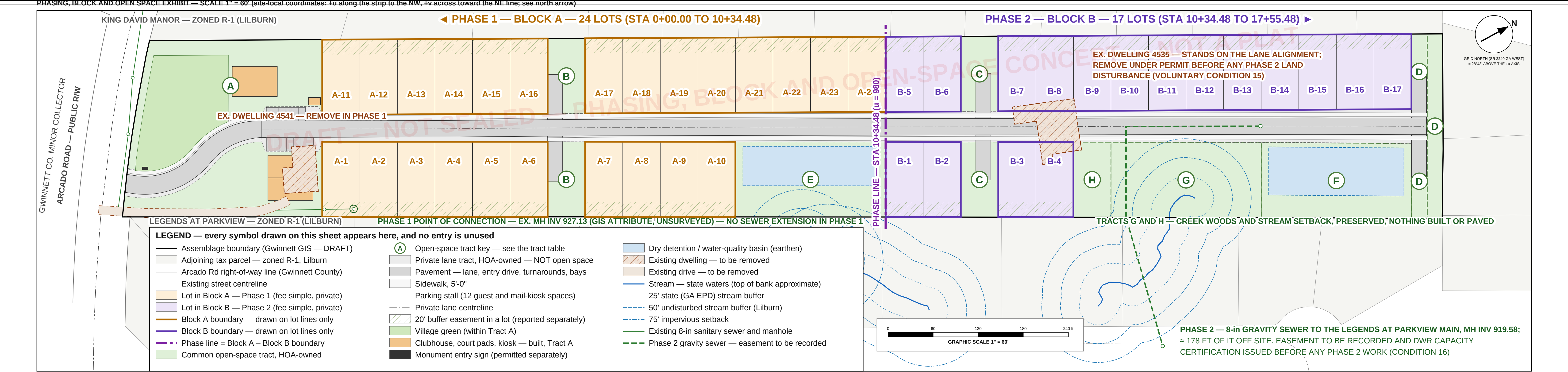
ITEM	PHASE 1 — BLOCK A	PHASE 2 — BLOCK B
Lots	24 (10 SW, 14 NE) — Block A	17 (4 SW, 13 NE) — Block B
Lot stations (u)	230 — 980	980 — 1,690
Cumulative density on 9.44 ac: decided	2.54 du/ac	4.34 du/ac
Private lane built	Arcado Rd RW + u 980; station 0 — 1,034 ft	u 980 — 1,700; station 1,034 — 1,754 ft
Fire-apparatus turnaround	Hammerhead at u 540, plus a TEMPORARY 120-ft hammerhead at u 950–970 while Phase 1 stands alone	Hammerheads at u 1,110 and u 1,690; the temporary turnaround is removed and restored
Amenity	Entrance, clubhouse, village green, courts, mail kiosk, guest bay, monument sign and frontage landscape strip at u 1,034 ft in Phase 1	None
Sanitary sewer	System A, MH-A3 — POC-1: 809 ft of 8-in gravity. ALL ON SITE, to the existing manhole at invert 927.13. NO EXTENSION OF THE PUBLIC SEWER.	System A extended MH-A1 — MH-A3: 300 ft on site, serving 9 lots. System B: 506 ft on site + 178 ft OFF SITE to POC-2 at invert 919.58, serving 8 lots.
Depends on a third party?	NO — no off-site easement, no off-site sewer, no second access	YES — a recorded off-site sanitary sewer easement, a DWR capacity certification and the City’s reading of the comprehensive-plan sewer policy

Site Development Plan Review Checklist §4.1: “Phasing is not permitted unless plotted in Blocks.” The phase line at u + 980 is drawn ON LOT LINES — is the common line between Lots A-10 and B-1 (SW), between Lots A-24 and B-5 (NE) and along the Pond 1 tract; no lot is divided by B-10 and B-9 to be plotted as such on the final plat, by a Georgia PLS.

PHASE 1 STANDS ALONE. It contains the entrance, the whole amenity block, P-1, a turnaround and gravity sewer to a public main that is already inside the property line. There are no off-site easement, no off-site sewer main and no second access. Phase 2 is the only part of this application that depends on a third party, and only 8 of the 41 lots depend on it.

SOURCES AND ACCESS DATES

SOURCES. dayalotay plan (boundary, use, lane, ground profile, ponds, phase line, disturbed area), dayalotay plan (existing sewer, water mains, hydrants, adjoining parcels) — Gwinnett County GIS, queried 2026-08-29; dayalotay samples (pin USGS 3DEP 1-m DEM); Gwinnett County Tax Assessor 2026 property ownership list; Gwinnett DWR Developer Pump Station Standards (Rev. 2018, WSR-24), Standard Policy for Private Developments: Condominiums, Townhomes and Subdivisions (rev. 9/2018), Water and Sewer Standards (April 2016), Wastewater Flow Control Guidelines (Rev. 10/2022) and Sewer Capacity Certification Request (Rev. 07/2023); IFC 2024 as adopted and modified by Ga. Comp. & Regs. R. 120-3-4.0, Appendix B, C and D; City of Lilburn Site Development Plan Review Checklist; Lilburn Zoning Ordinance 2023 §1002-4.6, FACTS and §2, §2b, §2c, §2d, §2e, §2f, §2g, §2h, §2i, §2j, §2k, §2l, §2m, §2n, §2o, §2p, §2q, §2r, §2s, §2t, §2u, §2v, §2w, §2x, §2y, §2z, §2aa, §2ab, §2ac, §2ad, §2ae, §2af, §2ag, §2ah, §2ai, §2aj, §2ak, §2al, §2am, §2an, §2ao, §2ap, §2aq, §2ar, §2as, §2at, §2au, §2av, §2aw, §2ax, §2ay, §2az, §2ba, §2bb, §2bc, §2bd, §2be, §2bf, §2bg, §2bh, §2bi, §2bj, §2bk, §2bl, §2bm, §2bn, §2bo, §2bp, §2bq, §2br, §2bs, §2bt, §2bu, §2bv, §2bw, §2bx, §2by, §2bz, §2ca, §2cb, §2cc, §2cd, §2ce, §2cf, §2cg, §2ch, §2ci, §2cj, §2ck, §2cl, §2cm, §2cn, §2co, §2cp, §2cq, §2cr, §2cs, §2ct, §2cu, §2cv, §2cw, §2cx, §2cy, §2cz, §2da, §2db, §2dc, §2dd, §2de, §2df, §2dg, §2dh, §2di, §2dj, §2dk, §2dl, §2dm, §2dn, §2do, §2dp, §2dq, §2dr, §2ds, §2dt, §2du, §2dv, §2dw, §2dx, §2dy, §2dz, §2ea, §2eb, §2ec, §2ed, §2ee, §2ef, §2eg, §2eh, §2ei, §2ej, §2ek, §2el, §2em, §2en, §2eo, §2ep, §2eq, §2er, §2es, §2et, §2eu, §2ev, §2ew, §2ex, §2ey, §2ez, §2fa, §2fb, §2fc, §2fd, §2fe, §2ff, §2fg, §2fh, §2fi, §2fj, §2fk, §2fl, §2fm, §2fn, §2fo, §2fp, §2fq, §2fr, §2fs, §2ft, §2fu, §2fv, §2fw, §2fx, §2fy, §2fz, §2ga, §2gb, §2gc, §2gd, §2ge, §2gf, §2gg, §2gh, §2gi, §2gj, §2gk, §2gl, §2gm, §2gn, §2go, §2gp, §2gq, §2gr, §2gs, §2gt, §2gu, §2gv, §2gw, §2gx, §2gy, §2gz, §2ha, §2hb, §2hc, §2hd, §2he, §2hf, §2hg, §2hi, §2hj, §2hk, §2hl, §2hm, §2hn, §2ho, §2hp, §2hq, §2hr, §2hs, §2ht, §2hu, §2hv, §2hw, §2hx, §2hy, §2hz, §2ia, §2ib, §2ic, §2id, §2ie, §2if, §2ig, §2ih, §2ii, §2ij, §2ik, §2il, §2im, §2in, §2io, §2ip, §2iq, §2ir, §2is, §2it, §2iu, §2iv, §2iw, §2ix, §2iy, §2iz, §2ja, §2jb, §2jc, §2jd, §2je, §2jf, §2jg, §2jh, §2ji, §2jj, §2jk, §2jl, §2jm, §2jn, §2jo, §2jp, §2jq, §2jr, §2js, §2jt, §2ju, §2jv, §2jw, §2jx, §2jy, §2jz, §2ka, §2kb, §2kc, §2kd, §2ke, §2kf, §2kg, §2kh, §2ki, §2kj, §2kl, §2km, §2kn, §2ko, §2kp, §2kq, §2kr, §2ks, §2kt, §2ku, §2kv, §2kw, §2kx, §2ky, §2kz, §2la, §2lb, §2lc, §2ld, §2le, §2lf, §2lg, §2lh, §2li, §2lj, §2lk, §2ll, §2lm, §2ln, §2lo, §2lp, §2lq, §2lr, §2ls, §2lt, §2lu, §2lv, §2lw, §2lx, §2ly, §2lz, §2ma, §2mb, §2mc, §2md, §2me, §2mf, §2mg, §2mh, §2mi, §2mj, §2mk, §2ml, §2mm, §2mn, §2mo, §2mp, §2mq, §2mr, §2ms, §2mt, §2mu, §2mv, §2mw, §2mx, §2my, §2mz, §2na, §2nb, §2nc, §2nd, §2ne, §2nf, §2ng, §2nh, §2ni, §2nj, §2nk, §2nl, §2nm, §2nn, §2no, §2np, §2nq, §2nr, §2ns, §2nt, §2nu, §2nv, §2nw, §2nx, §2ny, §2nz, §2oa, §2ob, §2oc, §2od, §2oe, §2of, §2og, §2oh, §2oi, §2oj, §2ok, §2ol, §2om, §2on, §2oo, §2op, §2oq, §2or, §2os, §2ot, §2ou, §2ov, §2ow, §2ox, §2oy, §2oz, §2pa, §2pb, §2pc, §2pd, §2pe, §2pf, §2pg, §2ph, §2pi, §2pj, §2pk, §2pl, §2pm, §2pn, §2po, §2pp, §2pq, §2pr, §2ps, §2pt, §2pu, §2pv, §2pw, §2px, §2py, §2pz, §2qa, §2qb, §2qc, §2qd, §2qe, §2qf, §2qg, §2qh, §2qi, §2qj, §2qk, §2ql, §2qm, §2qn, §2qo, §2qp, §2qq, §2qr, §2qs, §2qt, §2qu, §2qv, §2qw, §2qx, §2qy, §2qz, §2ra, §2rb, §2rc, §2rd, §2re, §2rf, §2rg, §2rh, §2ri, §2rj, §2rk, §2rl, §2rm, §2rn, §2ro, §2rp, §2rq, §2rr, §2rs, §2rt, §2ru, §2rv, §2rw, §2rx, §2ry, §2rz, §2sa, §2sb, §2sc, §2sd, §2se, §2sf, §2sg, §2sh, §2si, §2sj, §2sk, §2sl, §2sm, §2sn, §2so, §2sp, §2sq, §2sr, §2ss, §2st, §2su, §2sv, §2sw, §2sx, §2sy, §2sz, §2ta, §2tb, §2tc, §2td, §2te, §2tf, §2tg, §2th, §2ti, §2tj, §2tk, §2tl, §2tm, §2tn, §2to, §2tp, §2tq, §2tr, §2ts, §2tt, §2tu, §2tv, §2tw, §2tx, §2ty, §2tz, §2ua, §2ub, §2uc, §2ud, §2ue, §2uf, §2ug, §2uh, §2ui, §2uj, §2uk, §2ul, §2um, §2un, §2uo, §2up, §2uq, §2ur, §2us, §2ut, §2uu, §2uv, §2uw, §2ux, §2uy, §2uz, §2va, §2vb, §2vc, §2vd, §2ve, §2vf, §2vg, §2vh, §2vi, §2vj, §2vk, §2vl, §2vm, §2vn, §2vo, §2vp, §2vq, §2vr, §2vs, §2vt, §2vu, §2vv, §2vw, §2vx, §2vy, §2vz, §2wa, §2wb, §2wc, §2wd, §2we, §2wf, §2wg, §2wh, §2wi, §2wj, §2wk, §2wl, §2wm, §2wn, §2wo, §2wp, §2wq, §2wr, §2ws, §2wt, §2wu, §2wv, §2ww, §2wx, §2wy, §2wz, §2xa, §2xb, §2xc, §2xd, §2xe, §2xf, §2fg, §2fh, §2fi, §2fj, §2fk, §2fl, §2fm, §2fn, §2fo, §2fp, §2fq, §2fr, §2fs, §2ft, §2fu, §2fv, §2fw, §2fx, §2fy, §2fz, §2ga, §2gb, §2gc, §2gd, §2ge, §2gf, §2gg, §2gh, §2gi, §2gj, §2gk, §2gl, §2gm, §2gn, §2go, §2gp, §2gq, §2gr, §2gs, §2gt, §2gu, §2gv, §2gw, §2gx, §2gy, §2gz, §2ha, §2hb, §2hc, §2hd, §2he, §2hf, §2hg, §2hi, §2hj, §2hk, §2hl, §2hm, §2hn, §2ho, §2hp, §2hq, §2hr, §2hs, §2ht, §2hu, §2hv, §2hw, §2hx, §2hy, §2hz, §2ia, §2ib, §2ic, §2id, §2ie, §2if, §2ig, §2ih, §2ii, §2ij, §2ik, §2il, §2im, §2in, §2io, §2ip, §2iq, §2ir, §2is, §2it, §2iu, §2iv, §2iw, §2ix, §2iy, §2iz, §2ja, §2jb, §2jc, §2jd, §2je, §2jf, §2jg, §2jh, §2ji, §2jj, §2jk, §2jl, §2jm, §2jn, §2jo, §2jp, §2jq, §2jr, §2js, §2jt, §2ju, §2jv, §2jw, §2jx, §2jy, §2jz, §2ka, §2kb, §2kc, §2kd, §2ke, §2kf, §2kg, §2kh, §2ki, §2kj, §2kl, §2km, §2kn, §2ko, §2kp, §2kq, §2kr, §2ks, §2kt, §2ku, §2kv, §2kw, §2kx, §2ky, §2kz, §2la, §2lb, §2lc, §2ld, §2le, §2lf, §2lg, §2lh, §2li, §2lj, §2lk, §2ll, §2lm, §2ln, §2lo, §2lp, §2lq, §2lr, §2ls, §2lt, §2lu, §2lv, §2lw, §2lx, §2ly, §2lz, §2ma, §2mb, §2mc, §2md, §2me, §2mf, §2mg, §2mh, §2mi, §2mj, §2mk, §2ml, §2mm, §2mn, §2mo, §2mp, §2mq, §2mr, §2ms, §2mt, §2mu, §2mv, §2mw, §2mx, §2my, §2mz, §2na, §2nb, §2nc, §2nd, §2ne, §2nf, §2ng, §2nh, §2ni, §2nj, §2nk, §2nl, §2nm, §2nn, §2no, §2np, §2nq, §2nr, §2ns, §2nt, §2nu, §2nv, §2nw, §2nx, §2ny, §2nz, §2oa, §2ob, §2oc, §2od, §2oe, §2of, §2og, §2oh, §2oi, §2oj, §2ok, §2ol, §2om, §2on, §2oo, §2op, §2oq, §2or, §2os, §2ot, §2ou, §2ov, §2ow, §2ox, §2oy, §2oz, §2pa, §2pb, §2pc, §2pd, §2pe, §2pf, §2pg, §2ph, §2pi, §2pj, §2pk, §2pl, §2pm, §2pn, §2po, §2pp, §2pq, §2pr, §2ps, §2pt, §2pu, §2pv, §2pw, §2px, §2py, §2pz, §2qa, §2qb, §2qc, §2qd, §2qe, §2qf, §2qg, §2qh, §2qi, §2qj, §2qk, §2ql, §2qm, §2qn, §2qo, §2qp, §2qq, §2qr, §2qs, §2qt, §2qu, §2qv, §2qw, §2qx, §2qy, §2qz, §2ra, §2rb, §2rc, §2rd, §2re, §2rf, §2rg, §2rh, §2ri, §2rj, §2rk, §2rl, §2rm, §2rn, §2ro, §2rp, §2rq, §2rr, §2rs, §2rt, §2ru, §2rv, §2rw, §2rx, §2ry, §2rz, §2sa, §2sb, §2sc, §2sd, §2se, §2sf, §2sg, §2sh, §2si, §2sj, §2sk, §2sl, §2sm, §2sn, §2so, §2sp, §2sq, §2sr, §2ss, §2st, §2su, §2sv, §2sw, §2sx, §2sy, §2sz, §2ta, §2tb, §2tc, §2td, §2te, §2tf



LOT SCHEDULE — BLOCK A (PHASE 1), 24 LOTS

LOT	C-2.0	SIDE	STATION u, ft	AREA SF	WIDTH	DEPTH	PLAN
A-1	1	SW	230 – 280	5,000	50'-0"	100'-0"	A
A-2	2	SW	280 – 330	5,000	50'-0"	100'-0"	B
A-3	3	SW	330 – 380	5,000	50'-0"	100'-0"	A
A-4	4	SW	380 – 430	5,000	50'-0"	100'-0"	B
A-5	5	SW	430 – 480	5,000	50'-0"	100'-0"	A
A-6	6	SW	480 – 530	5,000	50'-0"	100'-0"	B
A-7	7	SW	580 – 630	5,000	50'-0"	100'-0"	A
A-8	8	SW	630 – 680	5,000	50'-0"	100'-0"	B
A-9	9	SW	680 – 730	5,000	50'-0"	100'-0"	A
A-10	10	SW	730 – 780	5,000	50'-0"	100'-0"	B
A-11	15	NE	230 – 280	5,000	50'-0"	100'-0"	A
A-12	16	NE	280 – 330	5,000	50'-0"	100'-0"	B
A-13	17	NE	330 – 380	5,000	50'-0"	100'-0"	A
A-14	18	NE	380 – 430	5,000	50'-0"	100'-0"	B
A-15	19	NE	430 – 480	5,000	50'-0"	100'-0"	A
A-16	20	NE	480 – 530	5,000	50'-0"	100'-0"	B
A-17	21	NE	580 – 630	5,000	50'-0"	100'-0"	A
A-18	22	NE	630 – 680	5,000	50'-0"	100'-0"	B
A-19	23	NE	680 – 730	5,000	50'-0"	100'-0"	A
A-20	24	NE	730 – 780	5,000	50'-0"	100'-0"	B
A-21	25	NE	780 – 830	5,000	50'-0"	100'-0"	A
A-22	26	NE	830 – 880	5,000	50'-0"	100'-0"	B
A-23	27	NE	880 – 930	5,000	50'-0"	100'-0"	A
A-24	28	NE	930 – 980	5,000	50'-0"	100'-0"	B

LOT SCHEDULE — BLOCK B (PHASE 2), 17 LOTS

LOT	C-2.0	SIDE	STATION u, ft	AREA SF	WIDTH	DEPTH	PLAN
B-1	11	SW	980 – 1,030	5,000	50'-0"	100'-0"	A
B-2	12	SW	1,030 – 1,080	5,000	50'-0"	100'-0"	B
B-3	13	SW	1,130 – 1,180	5,000	50'-0"	100'-0"	A
B-4	14	SW	1,180 – 1,230	5,000	50'-0"	100'-0"	B
B-5	29	NE	980 – 1,030	5,000	50'-0"	100'-0"	A
B-6	30	NE	1,030 – 1,080	5,000	50'-0"	100'-0"	B
B-7	31	NE	1,130 – 1,180	5,000	50'-0"	100'-0"	A
B-8	32	NE	1,180 – 1,230	5,000	50'-0"	100'-0"	B
B-9	33	NE	1,230 – 1,280	5,000	50'-0"	100'-0"	A
B-10	34	NE	1,280 – 1,330	5,000	50'-0"	100'-0"	B
B-11	35	NE	1,330 – 1,380	5,000	50'-0"	100'-0"	A
B-12	36	NE	1,380 – 1,430	5,000	50'-0"	100'-0"	B
B-13	37	NE	1,430 – 1,480	5,000	50'-0"	100'-0"	A
B-14	38	NE	1,480 – 1,530	5,000	50'-0"	100'-0"	B
B-15	39	NE	1,530 – 1,580	5,000	50'-0"	100'-0"	A
B-16	40	NE	1,580 – 1,630	5,000	50'-0"	100'-0"	B
B-17	41	NE	1,630 – 1,680	5,000	50'-0"	100'-0"	A

GENERAL NOTES — PHASING, BLOCKS AND OPEN SPACE

- BLOCKS. Site Development Plan Review Checklist §4.i: "Subdivision — add number of lots/units ... Lot and Block Numbers if applicable. Phasing is not permitted unless platted in Blocks." Block A (Phase 1, 24 lots) and Block B (Phase 2, 17 lots) are platted as blocks, and every block boundary drawn here lies on a lot line or a tract line. Numbers A-1...A-24 and B-1...B-17 carry to the plat; column 2 of each schedule gives the C-2.0 lot number.
- PHASE LINE. STA 10+34.48 along the travelled way from the Arcado Rd right-of-way (204-ft entry drive plus lane) = local station u 980. NE side: the common lot line of Lots A-24 and B-5. SW side: the tract line between Tract E (Pond 1) and Lot B-1. Between them it crosses only the private lane tract. IT DIVIDES 0 LOTS.
- PHASE 1 IS DESIGNED TO STAND ALONE. Block A carries the entrance, the entire amenity block (Tract A), Pond 1 (Tract E), the first turnaround (Tract B) and 24 of the 41 lots, served by gravity to the existing 8-in county main already crossing the property (MH inv 927.13). No sewer extension, no off-site easement, no second access; complete whether or not Phase 2 is built.
- PHASE 2 CONDITION. No Phase 2 work and no land-disturbance permit until (a) the off-site sewer easement for the ≈ 178-ft gravity tie to the Legends at Parkview main (MH inv 919.58) is RECORDED and (b) Gwinnett County DWR has issued its SEWER CAPACITY CERTIFICATION (condition 16). Gravity only: no private pump station, force main, private gravity sewer, grinder pump or low-pressure system (Gwinnett DWR Standard Policy for Private Developments, rev. 9/2018; WSR-24 §1.3.1(A)). If either fails, Phase 2 is not built, and 4535 Arcado Rd — which stands on the lane alignment — comes down under permit first (condition 15).
- OPEN SPACE IS VOLUNTARY. Table 4.1 refers the R-2 open-space requirement to §5.9 of the Development Regulations; §5.9.1 "Recreation Areas" reaches detached subdivisions "having a gross area of 50 acres or more." At 9.44 ac it is not engaged: none of the 141,089 SF is required.
- BUFFER EASEMENT — REPORTED SEPARATELY. The rear 20 ft of each of the 41 lots is undisturbed buffer in a recorded easement inside the private lot (Ord. 2023-603 §313(1)): 41,000 SF. It is not a common tract, is not in the 141,089 SF total, and is in no percentage on this sheet.
- AREA BASIS. 417,174 SF = 9.577 ac GIS; 411,206 SF = 9.44 ac deeded; percentages given against both. Tract areas are the published residual — GIS area less lots less the lane and entry-drive tracts — so the eight sum exactly. Tract A is NET: its outline encloses ≈ 58,956 SF, of which = 12,164 SF is entry drive and lane tract.
- BUILT AND PAVED INSIDE THE OPEN SPACE. 13,960 SF of 141,089 SF; the balance, 127,129 SF (30.5% of the GIS area), is green. Turnaround legs are counted at the full 1,200 SF each although only ≈ 6,041 SF of the 7,200 SF really falls inside the greens, so the figure is conservative. The 12 guest and kiosk spaces (2,070 SF of bays) lie in the lane tract and are not counted at all.
- MAINTENANCE (§4.o). A mandatory HOA owns and maintains Tracts A–H, the lane tract and its sidewalks, both ponds, the buffers and buffer easements, the entry sign and the lighting, and provides yard maintenance to every lot (condition 10). NOTHING IS OFFERED FOR DEDICATION TO THE CITY; only the sanitary mains and their easements become public, to Gwinnett DWR.
- NOT SHOWN, AND SOURCES. Dwellings, driveways and setback envelopes — Sheets C-2.0/2.1/2.2; bearings and topography — C-1.0; grading — C-3.0; utilities — C-4.0/4.1. Only the sewer reaches on or adjoining the property are drawn. Every area, count and station here is read at run time from data/layout.json (2026-09-03) and data/plans.json.

PHASE DATA — 41 LOTS IN TWO BLOCKS, 4.34 DU/AC ON THE DEEDED 9.44 AC
PHASE 1 · BLOCK A · STA 0+00.00–10+34.48 · 24 lots · 237,293 SF (5.45 ac) · 4.41 du/ac
Tracts A, B, E = 76,777 SF · Pond 1, 33,129 cf · turnaround 1 · gravity to the on-site main
PHASE 2 · BLOCK B · STA 10+34.48–17+55.48 · 17 lots · 179,880 SF (4.13 ac) · 4.12 du/ac
Tracts C, D, F, G, H = 64,312 SF · Pond 2, 44,734 cf · turnarounds 2, 3 · ≈ 178 ft off-site sewer tie
Every lot 5,000 SF · 50 ft x 100 ft · 50 ft of frontage on the private lane (§319 requires 30 ft)
PLAN A = "The Springbrook", 1,406 SF conditioned · PLAN B = "The Laurel", 1,492 SF conditioned (data/plans.json)

COMMON OPEN-SPACE TRACT TABLE — 141,089 SF = 3.239 AC = 33.8% OF THE 417,174-SF GIS AREA / 34.3% OF THE DEEDED 9.44 AC

TR.	NAME AND LOCATION	AREA SF	% GIS	% 9.44 AC	PH/BLK	WHAT THE TRACT CONTAINS — BUILT AND PAVED AREA STATED	MAINT.
A	Front amenity tract, Arcado Rd frontage to u 230	46,777	11.21%	11.38%	1 / A	Clubhouse 2,400 SF + two court pads 3,600 SF + mail kiosk 160 SF + walks 600 SF = 6,760 SF BUILT OR PAVED, 14.5% of the tract. Also the village green, landscape strip and monument sign. The 12 guest and kiosk spaces (2,070 SF of bays) are NOT in it — note 8.	HOA
B	Pocket green, both sides, u 530–580	10,000	2.40%	2.43%	1 / A	Lawn and plantings. Hammerhead turnaround 1 — 2,400 SF of pavement in two legs.	HOA
C	Pocket green, both sides, u 1,080–1,130	10,000	2.40%	2.43%	2 / B	Lawn and plantings. Hammerhead turnaround 2 — 2,400 SF of pavement in two legs.	HOA
D	Terminus green + rear buffer, u 1,680–1,722	9,312	2.23%	2.26%	2 / B	Lawn; rear 20-ft buffer in common ownership. Hammerhead turnaround 3 — 2,400 SF of pavement in two legs.	HOA
E	Pond 1 tract, SW side, u 780–980	20,000	4.79%	4.86%	1 / A	Dry detention / water-quality basin, 9,450 SF at top of bank — earthen, NOT impervious. Buffer, drainage and BMP access easements inside the tract.	HOA
F	Pond 2 tract, SW side, u 1,480–1,680	20,000	4.79%	4.86%	2 / B	Dry detention / water-quality basin, 11,520 SF at top of bank — earthen, NOT impervious. Buffer, drainage and BMP access easements inside the tract.	HOA
G	Creek woods, SW side, u 1,280–1,480	20,000	4.79%	4.86%	2 / B	Stream head, 50-ft undisturbed buffer and 75-ft impervious setback, preserved. NOTHING BUILT OR PAVED.	HOA
H	Stream-setback tract, SW, u 1,230–1,280	5,000	1.20%	1.22%	2 / B	The one 50 x 100-ft lot slot that fails the 75-ft screen, left unlotted. NOTHING BUILT OR PAVED.	HOA
TOTAL — EIGHT TRACTS		141,089	33.8%	34.3%	BOTH	Of which 13,960 SF is built or paved (note 8); 127,129 SF, 30.5% of the GIS area, is green. NO open space is required here — note 5.	HOA

TRACT A HOLDS 6,760 SF OF BUILT AND PAVED AREA — 14.5% OF THE TRACT — AND ITS 46,777 SF IS NET OF THE ≈ 12,164 SF OF ENTRY DRIVE AND LANE TRACT CROSSING IT; THE 12 GUEST AND MAIL-KIOSK PARKING SPACES (2,070 SF OF BAYS) ARE NOT IN IT — THEY LIE IN THE PRIVATE LANE TRACT. SEE NOTE 8.

OTHER COMMON PROPERTY AND EASEMENTS — MAINTENANCE RESPONSIBILITY (CHECKLIST §4.o). NONE OF IT IS COUNTED AS COMMON OPEN SPACE

ITEM	AREA / DIMENSION	PHASE	OWNERSHIP	MAINTENANCE — HOA OR PUBLIC
Private lane tract, including the entry drive	71,085 SF	1 and 2	HOA in fee	HOA — pavement, sidewalks, curb, drainage, lighting and street trees. Private street (§4.s), built to public-street standards.
20-ft buffer easement on the rear of all 41 lots	41,000 SF	1 and 2	Lot owner in fee, subject to a recorded buffer easement (§313(1))	HOA maintains and enforces it; the owner may not clear, grade, fence or build in it.
Existing 20-ft sanitary sewer easement (Arcado Rd Townhomes outfall)	20' wide, u –42 to 272; width to be confirmed from the recorded plat	existing	Gwinnett County (DWR)	PUBLIC — DWR maintains the main; HOA mows the surface.
Phase 1 sanitary sewer easement, on site, to MH inv 927.13	20' wide, ≈ 814 ft; area by survey	1	To be dedicated to Gwinnett County (DWR)	PUBLIC — DWR maintains the main after acceptance; HOA the surface.
Phase 2 sanitary sewer easement, off site, to the Legends at Parkview main (MH inv 919.58)	20' wide, ≈ 178 ft of it OFF SITE; to be acquired	2	To be dedicated to Gwinnett County (DWR)	PUBLIC — DWR maintains the main. MUST BE RECORDED BEFORE ANY PHASE 2 WORK (condition 16).
Drainage and 30-ft BMP access easements at each pond	Within Tracts E and F; Sheet C-3.0	1 and 2	HOA in fee	HOA under a recorded Storm Water BMP Maintenance Agreement with the City (§9.g.3).
Arcado Road right-of-way	R/W width varies — Sheet C-1.0	existing	Gwinnett County	PUBLIC — Gwinnett County DOT. Nothing here is offered for dedication to the City.

PHASE 1 STANDS ALONE. PHASE 2 SHALL NOT COMMENCE UNTIL THE OFF-SITE SANITARY SEWER EASEMENT IS RECORDED AND GWINNETT COUNTY DWR HAS ISSUED ITS SEWER CAPACITY CERTIFICATION (LETTER OF INTENT, VOLUNTARY CONDITIONS 15 AND 16 — SEE NOTES 3 AND 4).

Disclaimer: This is an AI-generated analysis for preliminary planning purposes. All findings must be verified by a licensed professional before use in design, permitting, or regulatory submissions.

Label: DRAFT / NOT SEALED — every item that must be sealed by a Georgia RLS, PE, architect or landscape architect is shown as a concept only. Statements of conformity read "appears consistent with".

THE COTTAGES AT ARCADE SPRINGS

PHASING, BLOCK AND OPEN SPACE EXHIBIT — Rezoning R-1 → R-2 (City of Lilburn, Georgia)

4535 / 4537 / 4539 / 4541 Arcado Rd SW, Lilburn, GA 30047 — Land Lot 123, 6th District, Gwinnett County — PINs R6123 033, 015, 014, 162 — 9.44 ac (deeded)

Blocks, phases, common open-space tracts, easements and maintenance responsibility — Site Development Plan Review Checklist §4.i and §4.o; Ord. 2023-603 §313(1)

APPLICANT / OWNER

Mohammed Awad

4541 Arcado Rd SW, Lilburn GA 30047-3968 (parcels 033, 015); Mendez / Roblero de Leon (parcels 014, 162) — co-applicants, owner signatures required

PREPARED FOR

Pre-application conference, City of Lilburn Planning & Zoning (340 Main St) — rezoning application per Ord. 2023-603 §1003 (2026 Application Instructions)

STATUS

DRAFT — NOT SEALED

Phasing, block and open-space CONCEPT compiled from public records for pre-application review. NOT A PLAT. Block and lot plating, tract and easement descriptions and every area shown are to be prepared and sealed by a Georgia RLS; utilities and stormwater by a Georgia PE. Generator: tools/sitebase.py + tools/phasing.py

DATE

2025-09-03

SCALE

Scale 1" = 60' (ARCH D 36 x 24 in)

DATUM

SR 2240 GA West US-ft; NAVD88

DRAWN

owner-prepared (AI-assisted)

CHECKED

RLS / PE — pending

SHEET

C-6.0

PHASING, BLOCK AND OPEN SPACE EXHIBIT



THE COTTAGES AT ARCADE SPRINGS CIVIL DETAILS — Rezoning R-1 → R-2 (City of Lilburn, Georgia)	APPLICANT / OWNER Mohammed Awad	STATUS DRAFT — NOT SEALED	DATE 2026-09-03	SHEET C-8.0
4535 / 4537 / 4539 / 4541 Arcade Rd SW, Lilburn, GA 30047 — Land Lot 123, 6th District, Gwinnett County — PINs R6123 033, 015, 014, 162 — 9.44 ac (deeded) Typical lot, private-lane sections, hammerhead turnaround, paving, accessible space, monument sign, mail kiosk, tree protection and the 20-ft buffer	4541 Arcade Rd SW, Lilburn GA 30047-3968 (parcels 033, 015); Mendez / Roblero de Leon (parcels 014, 162) — co-applicants, owner signatures required	Concept civil details for pre-application review. Every section, structure, footing and slope shown must be designed and sealed by a Georgia PE (and the tree / buffer details by a registered LA, forester or arborist). NOT CONSTRUCTION DOCUMENTS.	SCALE As noted per detail	DATUM SR 2240 GA West US-ft; NAVD88
	PREPARED FOR Pre-application conference, City of Lilburn Planning & Zoning (340 Main St) — rezoning application per Ord. 2023-603 §1003 (2026 Application Instructions)	Generator: tools/sitebase.py + tools/details.py	DRAWN owner-prepared (AI-assisted)	CHECKED PE / RLS / LA — pending
				CIVIL DETAILS

Development Summary — The Cottages at Arcade Springs

Rezoning application, R-1 → R-2 (Medium-Density Residential), City of Lilburn, Georgia **Application item:** Development Summary / site data report (companion to the Letter of Intent and Master Concept Plan) **Status:** DRAFT — owner-prepared, 2026-09-03. Boundary, topography, stream location, sewer inverts, stormwater volumes and traffic figures are preliminary and are to be superseded by sealed work of a Georgia RLS, PE, architect and landscape architect. **Basis documents:** FACTS.md (site geometry and code standards; §2 corrected 2026-09-03); data/layout.json (Master Concept Plan generator, **41 lots**, regenerated 2026-09-03); data/plans.json (architectural envelopes, regenerated 2026-09-03); the three re-verification audits at audit-2026-09-03/ (site-geometry.md, external-facts.md, docs-consistency.md); docs/11-site-context-brief.md; Lilburn Zoning Ordinance, Ord. 2023-603; Lilburn 2026 Application Instructions — Rezoning/SUP/CIC; RZ-2025-01 packet (status/rz-2025-01/). Derived numbers are computed in tools/dev_summary_calcs.py → data/dev-summary-derived.json, each one recomputed from the raw geometry and then cross-checked against the generator's own summary block. Since the generator was corrected on 2026-09-03 the two agree on every tracked quantity within 2 %; the three residual differences (all under 0.1 %) are stated where they occur rather than hidden.

Lilburn does not require the §736 Mixed-Use development report for an R-2 application; this summary is offered voluntarily so that staff has the same content in the same order.

Local coordinates used below: **u** = feet along the strip's long axis from the Arcade Rd right-of-way corner (bearing N 28°43' W toward the NW end); **v** = feet across the strip, positive toward the NE line, negative toward the SW line (FACTS §1).

Every finding in this document is stated as **"appears consistent with"** the cited standard. Nothing here determines that the project meets the code; that determination is the reviewing official's to make.

What changed on 2026-09-03, and what it cost

The site-plan generator was corrected against audit-2026-09-03/site-geometry.md. **The lot yield fell from 43 to 41**, for two reasons, both of them the applicant's own screening finding rather than a staff comment:

- 1. Every lot is now screened against all three digitised stream reaches, not only the on-site one.** The southwest slot at u 1,230–1,280 failed the screen and is no longer lotted; it is a stream-setback open-space tract of 5,000 SF. **Cost: one lot.**
- 2. The detention basins were sized on an assumed 4.4 disturbed acres. The real disturbed area is 7.541 ac**, so the required screening volume rose from ≈ 44,000 cf to **75,410 cf** and both basins had to be enlarged — Pond 2's tract from 15,000 to 20,000 SF. **Cost: one lot.**

Everything derived from the count moves with it: density, parking, trips, sewer flow, the tax base, the phase split, the HOPA 80-percent computation and the HOA budget. Two things got **better**: with the turnarounds re-spaced, the longest block is now **594 ft against the 600-ft standard of Table 4.2** — the block-length relief the earlier draft had to request is no longer needed — and no turnaround interval now approaches the 750-ft trigger of IFC 2024 (GA) Appendix D103.4. One thing got **worse**: the buffer, open-space and stream margins are all thinner than the earlier draft implied, and the required detention volume is now larger than the 2025 mixed-use filing's. Both are stated below at face value.

1. Property overview

1.1 Parcels

PIN (Gwinnett "R6123 ###")	Address	Deeded ac (2026 digest)	GIS-calculated ac	Owner of record (2026 digest)	2026 FMV (digest)
6123 033	4541 Arcade Rd SW	0.99	0.985	Awad, Mohammed	\$285,000
6123 015	4539 Arcade Rd SW	2.00	1.985	Awad, Mohammed	\$87,200
6123 014	4537 Arcade Rd SW	2.00	1.994	Mendez, Santos C. & Roblero de Leon, Lesvia Rosario	\$6,800
6123 162	4535 Arcade Rd SW	4.45	4.614	Mendez, Santos C. & Roblero de Leon, Lesvia Rosario	\$888,600
Assemblage		9.44	9.577	Land Lot 123, 6th District, Gwinnett County; City of Lilburn	\$1,267,600

- Application acreage: 9.44 ac (deeded) = 411,206 SF.** The GIS parcel fabric computes **417,174 SF = 9.577 ac** (shoelace of the 55-vertex boundary ring); the RLS boundary survey governs. The 2025 filing's "9.00 ac" was incorrect (FACTS §1).

- **Current zoning:** R-1, Single-Family Residence (City of Lilburn), all four parcels. **Requested zoning:** R-2, Medium-Density Residential (Lilburn Zoning Ordinance 2023-603 §402).
- **Comprehensive Plan (adopted):** 2024 Comprehensive Plan character area **"Established Residential"** (printed p. 35, PDF p. 37); Ord. 2023-603 §203 compatibility table maps Established Residential → R-1, R-2, so the request is plan-consistent under the ordinance's own terms.
- **Comprehensive Plan Amendment (pending, not adopted).** The City Council held the transmittal public hearing on **2026-07-13** and voted to transmit the Amendment to the Georgia Department of Community Affairs. Under Ga. Comp. R. & Regs. 110-12-1 the DCA findings go to the Regional Commission within 35 days and the Regional Commission's report to the City within 40 days of submittal, so that window closed on or about **2026-08-23**. As of **2026-09-03 the Amendment has not been adopted**; adoption can be calendared at the **2026-09-14, 2026-10-12 or 2026-11-09** Council meetings — i.e. it may well be adopted law on or before the Council hearing on this case. The application is written so that it stands either way: the Amendment's Future Development Map places the site in class **"Suburban-Low"** (FDM p. 52), whose stoplight chart (p. 73) rates **Single-Family Cottages** and **Single-Family Appropriate** and Mixed Use (vertical and horizontal), Multifamily, Office, Retail/Entertainment, Hotel/Motel, Flex and Auto-oriented Retail *Not Appropriate*. The adjacent class, Suburban-Medium (p. 71), also rates both single-family categories *Appropriate*. **VERIFY with the Planning Director which class the four parcels fall in** — the FDM is a raster with no parcel labels — and the adoption date.
- **Shape:** a single strip **~236 ft wide at Arcade Rd** (234.7 ft measured NE-line to SW-line at u = 0; 236.1-ft frontage chord) **widening to ~246 ft at the NW end**, ~1,720–1,760 ft deep (SW line 1,757.01 ft; NE line 1,721.68 ft; rear line 246.41 ft). Only public frontage: Arcade Rd, 237.63 ft (docs/05 §5).
- **Prior zoning action:** RZ-2025-01 (R-1 → MU; 104 condominiums + 24,400 SF commercial) was tabled by the Planning Commission on 2025-08-28 for a defective legal advertisement, was never heard, and received no denial; the §1003-11 twelve-month bar is therefore not apparently triggered (formal disposition VERIFY in writing with the Planning Director).

1.2 Surroundings

Side	Local extent	Adjoining use	Subdivision (plat)	Zoning	Notes
SE (front)	u ≈ -36 to 0	Arcade Rd, county-maintained 2-lane minor collector, 35 mph, 14,800 AADT (GDOT sta. 135-6689, 2017 — ineligible under Gwinnett DOT's two-year rule; see §6.2)	—	—	Arcadia Place centerline meets Arcade Rd at (u -12, v +61), 61 ft NE of the site's NE front corner
NE long side	v > 0	14 single-family lots: 4531 Arcade Rd (front corner) and 265–145 King David Dr, 0.45–0.76 ac each, 1970–71 brick split-levels/ranches, no HOA	King David Manor (plat S/159)	R-1 (Lilburn)	2026 digest FMV of abutting lots \$292,800–\$412,500, median \$349,200
SW long side	v < -237	14 single-family lots: 240–150 and 260, 250 Village Green Ct; 303 & 308 Fieldhouse Cir; 2007–2016 two-story homes 3,433–7,000 SF; HOA	Legends at Parkview (plat 118/187)	R-1 (Lilburn)	FMV of abutting lots \$579,100–\$894,600, median \$693,500
NW end (rear)	u > 1,700	3 single-family lots: 4520, 4530, 4540 Nantucket Dr, 1975–79	Nantucket (plats 1/268, 3/73, 4/157)	R-1 (Lilburn)	FMV \$308,400–\$417,000
Nearby precedent	~106 ft NE of the front corner, across 4531 Arcade	"Arcade Road Townhomes" R-2 polygon (Gwinnett zoning layer; case number VERIFY)	—	R-2 (Lilburn)	Nearest R-2; its sewer outfall crosses the subject site (\$2.5)
Nearby	~314 ft	Unincorporated Gwinnett R-100	—	R-100 (county)	—

Adjoining ownership: 31 privately owned tax parcels (14 NE, 14 SW, 3 NW), every one zoned R-1 in the City of Lilburn and developed with a single-family house, **plus the Gwinnett County Arcade Road right-of-way = 32 notice recipients**. The list is drawn from the Tax Assessor's 2026 ownership file (data/owners-found.json) and is reproduced at docs/06-adjoining-property-owners.md. Earlier drafts described "32 adjoining parcels ... developed with single-family houses"; the 32nd is a public road, not a house.

2. Existing conditions

2.1 Topography (USGS 3DEP 1-m DEM sampled on a 100 × 50-ft grid; NAVD88 ft; approximate — field survey required)

Location	Local u	Elevation (ft NAVD88)
Arcade Rd frontage	u ≈ 0	≈ 940 (SW corner) – 950 (NE corner); lane centerline 943.5 at the entry join (u 149.5)

Location	Local u	Elevation (ft NAVD88)
Mid-swale	u ≈ 700–1,000	≈ 931–940 (lane centerline low point of the front reach 936.3 at u 779 ; 940.4 at u 979)
Ridge	u ≈ 1,080–1,200	952–958 (lane crest 953.5 at u 1,179)
Rear low pocket	u ≈ 1,350–1,500, SW side	921.6 at (u 1,384, v -217) — site minimum; lane low point 931.5 at u 1,479
NW end	u ≈ 1,700–1,760	943–948

Site range 921.6–958.2 ft; cross-slope generally falls NE → SW. Data: data/topo-samples.json; 2-ft contours in data/layout.json. **The steepest existing ground on the lane centerline is 10.9 % at u ≈ 1,494** — on the climb out of the rear low pocket — against the 12 % maximum street grade in the City's Site Development Plan Review Checklist. The finished profile is not yet designed and will need cut/fill and possibly retaining at the lot fronts through that reach (PE).

2.2 Streams and buffers

- An unnamed order-0 headwater channel (Gwinnett hydrology layer) enters the site at the SW line around u ≈ 1,356–1,392 and terminates inside the site at (u 1,391.7, v -209.8), about 30 ft inside the line. Path (u, v): (1262.8, -353.4) → (1341.7, -296.6) → (1356.7, -252.5) → (1364.9, -215.4) → (1391.7, -209.8).
- A second branch runs 38–58 ft outside the SW line at u ≈ 868–1,038, and a third reach is digitized off-site to the southwest. **All three reaches are now screened against every lot, dwelling, driveway, pond and pavement** (\$4.3 item 40) — the earlier revision screened only the on-site reach, and that omission is what removed a lot.
- Buffers applied:** 25-ft state buffer (O.C.G.A. §12-7-6(b)(15), measured from the point where vegetation has been wrested by normal stream flow); Lilburn Code Ch. 109 Art. VII (Stream Buffer Protection Ordinance, adopting the Metro North Georgia Water Planning District model text): **50-ft undisturbed buffer measured from the top of the stream bank, plus a 25-ft setback beyond it in which all impervious cover is prohibited = 75 ft**. The state and local lines are measured from **different data** and both are shown separately on the concept plan. (Article and section numbers within Ch. 109 Art. VII are to be transcribed from the City's copy — Municode refuses automated retrieval.)
- Two open questions the concept plan must not pretend to have settled.** (a) *Datum*. Every clearance computed to date is measured to the **digitized channel centerline**, not to top of bank; with a 3–8-ft channel half-width the true clearances are 3–8 ft smaller, so the design carries a **5.0-ft top-of-bank allowance** and screens every element at 75 + 5 = **80 ft** for impervious cover and 50 + 5 = 55 ft for land disturbance. (b) *Whether the buffer attaches here at all*. The ordinance defines a "stream" as beginning at a spring or seep that sustains streamflow, or at a point with **25 acres or more** of drainage area. The reach that enters this property drains far less than 25 acres, so the regulated stream may begin **downstream, off the property**. A field study — GA EPD's *Field Guide for Determining the Presence of State Waters That Require a Buffer* (Sept 2017) for the state half — must establish **whether and where** a stream exists before top of bank is located. The outcome could return the u 1,230–1,280 slot to lotting, or take more.
- Receiving water: Jackson Creek headwaters (GAR030701030315), on Georgia's 2024 303(d) list for bacteria and fish-community impairment (urban runoff) — water-quality treatment is treated as substantive (\$5.3).

2.3 Floodplain

FEMA **Zone X** (not a special flood hazard area) for the entire site per Gwinnett GIS (FACTS §2). No floodway or 100-year floodplain is mapped on the property; Table 4.2 "Floodways and Floodplains" is therefore not engaged. **VERIFY** against the effective FIRM panel (13135C0114F, eff. 2006-09-29 per docs/11 §2) at LDP.

2.4 Soils (NRCS SSURGO survey area GA135 via the Gwinnett GIS soils layer; names per docs/11-site-context-brief.md §3.3 — **VERIFY** against a current Web Soil Survey AOI report; the Soil Data Access endpoint was not reachable on 2026-08-29)

Map unit	NRCS name (VERIFY)	Slope	HSG	Depth to rock	Notable limitation	Where on site
AmC2	Appling sandy loam, 6–10 %, moderately eroded	6–10 %	B	none	Low-strength subgrade for roads	Front half, u 0–800 (lane and NE side)
ApB	Appling–Hard Labor complex, 2–6 %	2–6 %	B / C	none	Hard Labor: seasonal water table ≈ 32 in	Band at u ≈ 400–500
PfC2	Pacolet sandy loam, 6–10 %, moderately eroded	6–10 %	B	none	—	Ridge and NE flank, u 800–1,400
GeB2	Gwinnett clay loam, 2–6 %, eroded	2–6 %	B	paralithic 40–60 in	—	NW corner (u ≈ 1,700, SW)
GgE2	Gwinnett loam, 10–25 %, eroded	10–25 %	B	paralithic ≈ 50 in	Slope	Rear, u 1,400–1,700

Map unit	NRCS name (VERIFY)	Slope	HSG	Depth to rock	Notable limitation	Where on site
ARE	Ashlar–Rion–Waterree complex, 10–25 %	10–25 %	B	Ashlar: lithic bedrock 22–40 in	Slope, shallow rock	Mid-swale u 800–1,100 (lane and Pond 1) and u ≈ 1,500 SW (Pond 2)
HYB	Helena sandy loam, 2–6 %	2–6 %	D	none	Seasonal high water table ≈ 24 in; moderate shrink-swell	Front SW corner, u 0–200 (amenity block, entry, sewer tie-in)

No map unit on site is rated hydric by NRCS. Design consequences: **rock probes and an undercut allowance at both pond sites and the mid-swale lane before the detention volumes are trusted** (§5.3) — and that matters more now that both basins are 6 ft deep; moisture-conditioned fill and pavement-subgrade care on HYB at the entrance; slab-on-grade foundations suit the HSG B uplands.

2.5 Existing improvements and utilities

- **Structures: two older houses, and only one of them is near the road (corrected 2026-09-03).** **4541 Arcadio Rd** (brick split-level, 1972; footprint ≈ 2,396 SF; 2026 digest heated area 2,053 SF) stands near the frontage, its footprint centred about (u 203, v –174) — inside the front amenity block, so it comes down in Phase 1. **4535 Arcadio Rd** (1987; footprint ≈ 4,568 SF; digest heated area 4,267 SF with a finished basement) does **not** stand near the frontage: it sits about **1,200 ft back from the right-of-way at about (u 1,202, v –114), on the u 1,080–1,200 ridge and directly on the proposed lane alignment**. It is therefore a **Phase-2 demolition standing on the lane**, not a frontage structure, and no Phase-2 land disturbance can begin until it is removed. Earlier revisions of this summary said "two older houses ... near the front"; that was wrong. Accessory structures and both existing driveways are likewise removed. 4535's septic-vs-sewer status is VERIFY. Footprints are affine-fitted from public aerial imagery by `tools/existing_conditions.py` and are approximate — the RLS survey locates them. Existing structures and the tree line are drawn on Sheet C-1.0 (Existing Conditions).
- **Sanitary sewer (existing):** an 8-in gravity main, Gwinnett DWR project "Arcadio Road Townhomes Sewer Outfall", runs from Arcadio Rd along the SW edge of parcel 033 (v ≈ –224 to –228, 7–13 ft inside the line) from u ≈ –42 to u ≈ 272 — a run of about **314 ft on the property** — then exits southwest toward King David Manor (the "EX. 20' SSE" on the 2025 hydro sheet). Inverts (ft): 943.79 / 940.47 at Arcadio → 932.69 / 930.84 (u ≈ –42 → 20) → 930.64 / **927.13 at the manhole near (u 272, v –224)**, i.e. about **272 ft in from Arcadio Rd** → 925.17 downstream. A downstream 8-in main in Legends at Parkview has invert **919.58** at the manhole near (u ≈ 1,349, v ≈ –405). Inverts are GIS attributes — **invert survey required**.
- **Water:** DIP/ACP mains in Arcadio Rd 47–107 ft from the front line; nearest hydrant ≈ 61 ft from the front corner (u –45, v +41).
- **Fire service:** Gwinnett County Fire & Emergency Services (the city has no fire department); first-due Station 3 ≈ 2.9 mi.
- **Environmental history:** no GA EPD Hazardous Site Inventory or UST records found for the parcels (brief §6.1); no industrial legacy.

3. Proposed development

3.1 Program

Item	Proposed	Basis
Use	Single-family detached cottage homes on fee-simple lots (Ord. 2023-603 §602 Use Table: "Single-family (cluster-cottage, creative lot configuration)", P in R-2, no SUP)	FACTS §3, §4
Age restriction	55+ under the Housing for Older Persons Act (42 U.S.C. §3607(b)(2)(C); 24 CFR §100.300–.308) — at least 80 % of occupied units with at least one occupant 55 or older. At 41 units the computation is $0.80 \times 41 = 32.8$, and 24 CFR §100.305(g) counts the fraction into the 80 percent, so 33 of the 41 units must have a 55+ occupant, leaving at most 8 that need not. Offered as a voluntary zoning condition and a recorded covenant	24 CFR §100.305(a), (g)
Dwelling units	41 (14 SW side, 27 NE side). FACTS §4's "≈44 lots, acceptable range 40–47" is a <i>target</i> , not a result; 41 is what the corrected concept plan draws, and it is still inside that range	data/layout.json
Density	4.34 du/ac on 9.44 ac deeded (4.28 du/ac on 9.577 ac GIS) vs. 8 du/ac R-2 maximum	$41 \div 9.44$; $41 \div 9.577$
Stories / height	1 story; ridge ≤ 24 ft offered as a condition (40 ft permitted by Table 4.1). As drawn on Sheets A-2.1 / A-2.2: Plan A 17'-10" and Plan B 19'-0" above finished floor = 18'-6" and 19'-8" above finished grade (finished floor is 8 in above grade at the front). data/plans.json still computes Plan B's main gable at a 29.5-ft span, i.e. 19'-4" (19.33 ft) above the floor — a 4-in difference to be conformed in the data file, not in the drawing (§10 item 4)	Sheets A-2.1 / A-2.2; data/plans.json
Tenure / maintenance	Fee-simple lots; mandatory HOA maintains the lane, common tracts, yards, ponds and buffers	FACTS §4
Not proposed	No commercial, attached units, mixed use, daycare, multifamily, or conservation-subdivision SUP. No wall signs on any structure; one monument entry sign	FACTS §4

3.2 Lot statistics (recomputed from the `data/layout.json` polygons, then cross-checked against the generator's summary block)

Statistic	Value
Lots	41 — 14 on the SW side, 27 on the NE side
Typical lot	50'-0" × 100'-0" = 5,000 SF (Table 4.1 R-2 cottage minimum 3,000 SF; detached-SF minimum 5,000 SF)
Lot area min / avg / max	5,000 / 5,000 / 5,000 SF (shoelace of every lot polygon)
Lot width min	50.00 ft every lot (Table 4.1 minimum 50 ft)
Lot depth min (perpendicular, front lot line to rear lot line)	100.00 ft every lot (Table 4.1 minimum 100 ft). The 0.006-ft shortfall reported in the earlier draft was a rounding artifact and is gone from the regenerated geometry
Lot frontage on the lane tract	50.00 ft every lot (\$319 minimum 30 ft)
Where the lots are	SW side u 230–1,280 (14 lots, with breaks at the u 530–580 green, the Pond 1 tract at u 780–980, the u 1,080–1,130 green and the u 1,230–1,280 stream-setback tract); NE side u 230–1,730 (27 lots, continuous but for the two greens). No lot fronts Arcade Rd — the nearest lot line is 230.0 ft from the right-of-way
Total lot area	205,000 SF (4.71 ac; 49.1 % of the GIS site)
Rear 20 ft of every lot	20-ft undisturbed buffer held in a recorded buffer/conservation easement — $41 \times 50 \text{ ft} \times 20 \text{ ft} = \mathbf{41,000 \text{ SF}}$, reported separately from common open space
Setback envelope per lot	front 15'-0" minimum to the porch face (drawn at 15'-8"), sides 5'-0" minimum (drawn at 6'-0"), rear 20'-0" (= the buffer easement); garage door 26'-8" from the front lot line , so the driveway holds two cars clear of the sidewalk
Plan mix	21 × Plan A, 20 × Plan B (alternating)
Phase 1 (u < 980) / Phase 2	24 lots / 17 lots

3.3 The two plans (DRAFT — the architect's sealed plans supersede; see Sheets A-1.1/A-2.1 (Plan A), A-1.2/A-2.2 (Plan B) and A-2.3 (colours and materials) in `drawings/`)

Areas below are read from `data/plans.json` — the file the architectural sheets are drawn from — and supersede the "38'-0" × 38'-0" / 40'-0" × 40'-0"" nominal bodies in FACTS §4. Both plans are **L-shaped**: the 2-car garage has to be a front wing to keep the body inside the 40'-0" buildable width (50-ft lot less two 5-ft side yards).

	Plan A "The Springbrook"	Plan B "The Laurel"
Overall body (outside face, conditioned + garage; porch and patio excluded)	38'-0" × 51'-10"	38'-0" × 57'-6"
Program	2 BR / 2 BA	2 BR + den / 2 BA, fireplace, walk-in pantry, laundry/mud room off the garage
Conditioned (heated) floor area	1,406 SF (program target 1,444 SF)	1,492 SF (program target 1,520 SF)
Garage (2-car, front-loaded, recessed 5'-0" behind the front wall)	$21'-0" \times 21'-10" = \mathbf{458 \text{ SF}}$	$21'-0" \times 24'-0" = \mathbf{504 \text{ SF}}$
Covered front porch	$17'-0" \times 6'-0" = \mathbf{102 \text{ SF}}$	$17'-0" \times 6'-0" = \mathbf{102 \text{ SF}}$
Rear porch / patio	uncovered patio $12'-0" \times 6'-0" = \mathbf{72 \text{ SF}}$ (no covered rear porch)	covered rear porch $14'-0" \times 6'-0" = \mathbf{84 \text{ SF}}$, recessed under the main roof
Total under roof (conditioned + garage + covered porches)	1,966 SF	2,182 SF
Building coverage on a 5,000-SF lot (under roof, before driveway)	39.3 %	43.6 %
Maximum ridge (as drawn on Sheets A-2.1 / A-2.2)	17'-10" above finished floor = 18'-6" above finished grade	19'-0" above finished floor = 19'-8" above finished grade
Rear structure to the rear lot line	patio edge 20'-6"	rear porch edge 20'-10"
Side yard as sited	6'-0" both sides	6'-0" both sides
Accessibility	zero-step entry, 36-in doors, 5-ft turning circles in baths, blocking for grab bars (voluntarily exceeds §734, which applies to attached dwellings only)	same

	Plan A "The Springbrook"	Plan B "The Laurel"
Exterior	fiber-cement lap and board-and-batten; brick or stone water table and porch piers; architectural-shingle roof; 8:12 main and front-wing gables, the front-wing gable carried 6 ft forward to roof the covered porch (there is no shed porch roof), and a subordinate 6:12 gable over the garage ; window sills at 2'-8"; three colour schemes (warm white / sage / clay)	same

- **Total conditioned area: $21 \times 1,406 + 20 \times 1,492 = 59,368$ SF, averaging 1,448 SF per unit.** Both plans clear the Table 4.1 minimum heated floor area of 1,000 SF for a cottage home with room to spare.
- The garage is a separate front wing, computed independently and **excluded** from conditioned area. Earlier drafts offered staff an alternative reading in which the garage sat inside the body ($\approx 1,024 / \approx 1,180$ SF heated); the filed drawings disprove it and it is withdrawn.
- Under the 2026 Comprehensive Plan Amendment's own vocabulary (land-use table p. 54) a "Single-Family Cottage" is a house "typically 1,200 square feet or less" and "Single-Family" is one "typically exceeding 1,200 square feet". At 1,406 and 1,492 SF these are **"Single-Family"** in the plan's terms — a category the Suburban-Low chart also rates *Appropriate*.
- **One drawing-versus-data difference is open on Plan B's ridge.** The issued elevation Sheet A-2.2 prints **19'-8" above finished grade** (19'-0" above the floor). data/plans.json computes the same gable at a 29.5-ft span and prints **19'-4" (19.33 ft) above the floor**, which would be 20'-0" above grade. The difference is 4 in and is below the 24-ft voluntary cap either way, but a recorded condition must not cite a dimension the filed sheet does not draw, so this document and voluntary condition 3 state the **drawn** figures and the data file is the one to conform (§10 item 4; docs/09 §5.3 item G-1). **VERIFY** with the architect.

3.4 Amenity block (front, u $\approx$ -36 to 230; common tract 46,777 SF)

- Clubhouse **2,400 SF (40'-0" $\times$ 60'-0")**, one story, **114.8 ft** (true perpendicular) from the Arcado Rd right-of-way — well beyond the 50-ft collector setback of Table 4.1 — with a village green.
- **2 pickleball courts** (20'-0" $\times$ 44'-0" courts on 30'-0" $\times$ 60'-0" fenced pads), set back from the R-1 lines; no court lighting (voluntary condition 11).
- Mail kiosk (USPS cluster-box unit) with **4 short-term spaces**; **8 guest spaces** (7 standard + 1 van-accessible) in a small bay. **Both bays now lie inside the lane tract, not the amenity tract:** the tract widens to 60 ft through the amenity block (u 149.5–230) so that the 18-ft-deep 90° bays sit wholly within the tract that owns and maintains them. Their 2,070 SF is therefore counted as lane pavement in §3.7, not as amenity paving.
- 10-ft landscape strip along the Arcado Rd right-of-way; one monument entry sign ≤ 32 SF and ≤ 6 ft high, standing **19.0 ft** from the right-of-way — inside the 50-ft principal-structure setback, as monument signs normally are, shown on the plan as an express exception and to be confirmed against the Lilburn sign regulations (**VERIFY**); a 5-ft sidewalk along the Arcado Rd frontage.
- **6,760 SF of the amenity tract is building or pavement** (clubhouse roof 2,400 + two court pads 3,600 + mail kiosk 160 + an amenity-walk allowance 600). Added to the **7,200 SF** of hammerhead turnaround legs that lie inside the pocket and terminus greens, **13,960 SF of the common open space is built or paved**. That is disclosed here and in §3.7 rather than left for a reviewer to discover inside the open-space number.

3.5 Access and private lane

Item	Value	Basis
Access	One full-movement driveway on Arcado Rd; no driveway from any lot to Arcado Rd; no gate. Gwinnett UDO §900-40.5(C) allows one driveway per 400 ft of frontage — at 237.63 ft of frontage, one is the maximum, which independently supports the single-entrance design	FACTS §4; Gwinnett UDO §900-40.5(C)
Entrance location	SW third of the frontage, centered at v = -190 , throat at u = -31.4 ; the entry drive curves to the lane midline and joins it at u 149.5 . Separation from the Arcadia Place centerline (v +61): 251.2 ft along the Arcado Rd centerline (250.7 ft chord) against the 244-ft minimum of Gwinnett UDO §900-40.5, Table 900.3 at 35 mph — 7.2 ft of margin	layout.json lane.entrance
Entrance geometry	24'-0" drive; curb returns R = 25'-0" northeast, R = 15'-0" southwest ; two reverse curves R = 100'-0" through a 59.9° deflection; drive length 204.0 ft (AASHTO minimum radius at 15 mph is 47 ft, so 100 ft is generous)	layout.json
Buffer at the SW curb return — a reduction is requested	Even with the southwest return cut from 25'-0" to 15'-0" , its tangent point on the right-of-way lies 2.0 ft inside the 20-ft perimeter buffer band, against 303 Fieldhouse Cir SW — so the buffer narrows to a minimum of 18'-0" instead of 20'-0" at that one corner, and the pavement actually inside the band is 0.1 SF . The entrance cannot be shifted northeast without spending the 7.2 ft of driveway-spacing margin. This is a buffer reduction; the Letter of Intent §5 requests it and voluntary condition 4 states it , as 2026 Application Instructions item 6 and Ord. 2023-603 §1003-4.5 require, and it is dimensioned on the entry enlargement sheet. The buffer is at its full 20 ft everywhere else on the SW, NE and NW lines	layout.json metrics sw_curb_return_buffer_encroachment_ft / _sf; audit SG-07; data/voluntary-condition s.md 4

Item	Value	Basis
Lane	Private, HOA-owned tract, 1,754 ft of travelled way from the Arcade Rd right-of-way to the end of pavement at u 1,700 (entry drive 204.0 ft + lane centreline 1,550.5 ft = 1,754.5 ft), dead-end; 22'-0" pavement , 2-ft strip, 5-ft sidewalk on the NE side for the whole length and on the entry drive, second 5-ft sidewalk on the SW side from u = 530; tract width 35.9 ft minimum to 46.1 ft maximum , widening to 60 ft through the amenity block; design speed 15 mph	layout . json
Lane and sidewalk areas	pavement 48,430 SF (lane 34,111 + entry drive 4,865 + hammerhead legs 7,200 + guest/kiosk bays 2,070 + curb returns 184); sidewalk 14,619 SF ; lane tract (lane + entry drive) 71,085 SF	layout . json; recomputed
Turnarounds	3 hammerhead (T) turnarounds, 20'-0" wide × 60'-0" legs, at u 540, 1,110 and 1,690 (terminus) — stations 594.5 / 1,164.5 / 1,744.5 ft from the right-of-way, spacing 594.5 / 570.0 / 580.0 ft. No circular cul-de-sac. Every interval is now well inside the 750-ft turnaround interval of IFC 2024 (GA) Appendix D103.4 — the earlier layout's 746.4-ft first interval, which cleared it by 3.6 ft, is gone	layout . json; audit SG-08
Blocks	Read between street intersections (the Arcade Rd right-of-way and the three hammerhead T-legs) the runs are 594.5 / 570.0 / 580.0 ft ; read as lot-frontage runs between the open-space breaks they are 300 / 500 / 550 ft. Table 4.2 asks for block lengths not exceeding 600 ft , and both readings are now inside it — the longest is 594 ft. A pocket green is an open-space tract, not a street, and cannot terminate a block, so the first reading is the honest one; it no longer needs relief	layout . json metrics.blocks_ft; audit SG-06
Lane half-section	The checklist section is 22'-0" of pavement plus, on each walked side, a 2-ft strip and a 5-ft walk: 29.0 ft for one walk, 36.0 ft for two . The tract is the residual after two 100-ft lot depths — 35.9 ft at the narrowest row , which carries the pavement and one walk with 6.9 ft to spare but is 0.1 ft short of two. The SW walk therefore starts at u 530, where the strip has widened; over u 230–530 the SW half-width is 16.94–18.31 ft	layout . json; \$4.3 item 36
Emergency access	No secondary access is physically possible — no street abuts the rear or either long side (Nantucket Dr lots close the NW end). Mitigation per \$5.4	FACTS \$4

3.6 Parking

Item	Count	Basis
Required — Table 8.1, single-family detached, 2 per DU	$41 \times 2 = \mathbf{82}$	Ord. 2023-603 Table 8.1
Provided on lots — 2 garage + 2 driveway per DU	$41 \times 4 = \mathbf{164}$	driveway 20'-0" wide × 26'-8" to the garage door
Guest spaces (bay at the amenity block)	8 (7 standard + 1 van-accessible with an 8-ft aisle)	layout . json
Mail-kiosk short-term spaces	4	layout . json
Total	176 (1 guest/kiosk space per 3.4 lots; FACTS target 1 per 4)	

3.7 Open space, impervious cover and site areas

Common open-space tract	Area (SF)
Front amenity tract (clubhouse, village green, courts, landscape strip)	46,777
Pocket green, u 530–580 (both sides of the lane)	10,000
Pocket green, u 1,080–1,130	10,000
Terminus green + rear buffer	9,312
Pond 1 tract (dry detention / water quality), SW side u 780–980	20,000
Pond 2 tract (dry detention / water quality), SW side u 1,480–1,680	20,000
Creek woods — stream head with 50/75-ft buffers, preserved	20,000
Stream-setback tract, SW side u 1,230–1,280 — unlotted, no impervious cover	5,000
Total common open space (enumerated)	141,089 SF = 3.24 ac = 33.8 % of the 417,174-SF GIS area (34.3 % of 9.44 deeded ac)

- **One number, enumerated, and it now closes exactly.** The tract-by-tract sum equals the residual (boundary – lots – lane tract) to the square foot: $417,174 - 205,000 - 71,085 = \mathbf{141,089 SF}$. The earlier draft carried 132,959 SF against a 133,159-SF residual with 200 SF of unallocated slivers; there are none now.
- **Of that total, 13,960 SF is building or pavement** — 6,760 SF in the amenity tract and 7,200 SF of hammerhead turnaround legs inside the greens (\$3.4). **Green-only open space is 127,129 SF = 30.5 % of the GIS area.**
- The **41,000 SF** of 20-ft buffer easement on the lots is **not** counted in the 33.8 %.
- **Open space is not a code requirement here.** Table 4.1 refers open space to Development Regulations §5.9, which has been retrieved: §5.9.1 "Recreation Areas" applies to "single-family detached subdivisions having a gross area of **50 acres or more**",

to duplex subdivisions of 50 acres or more, and to attached, townhouse and apartment developments of ten acres or more. At 9.44 acres of single-family detached, **\$5.9 is not triggered**, and the 33.8 % of common open space is entirely voluntary. (§5.9's text still names "Gwinnett County," "the Board of Commissioners" and "the Department" because Lilburn adopted the county regulations wholesale — confirm with staff who those actors are in a city case.)

- **Site-area accounting (GIS 417,174 SF):** lots 205,000 + lane tract 71,085 + common open space 141,089 = 417,174 SF, with nothing left over.
- **Impervious cover: 184,148 SF = 4.23 ac = 44.1 % of the GIS site (44.8 % of the deeded acreage).**

Impervious component	SF	Basis
Dwellings, garages, porches, rear porches and patios	86,448	21 × (1,966.5 under roof + 72 patio) + 20 × 2,182, from the data/plans.json footprints
Lot driveways and entry walks	31,390	the 41 drawn driveways (653–764 SF each, 706 average) + a 60-SF entry walk per lot
Private lane pavement	34,111	22'-0" × 1,550.5 ft
Entry drive pavement	4,865	the drawn polygon: 24'-0" wide on a 204.0-ft centreline (31 SF under the 4,896-SF nominal product, the arcs being drawn as chords)
Entrance curb returns	184	the two drawn fillets, R = 25 ft NE and R = 15 ft SW
Hammerhead turnaround legs	7,200	3 turnarounds × 2 legs × 20'-0" × 60'-0"
Guest and mail-kiosk parking bays	2,070	63 × 18 + 52 × 18, inside the lane tract
Sidewalks	14,619	shoelace of the three drawn walks
Clubhouse roof, court pads, mail kiosk and amenity walks	6,760	\$3.4
Less driveway apron / sidewalk overlap counted twice	–3,500	35 crossings × 20 ft × 5 ft
Total	184,148	44.1 % of GIS / 44.8 % of deeded

- The components as printed sum to 184,147 SF; the 1-SF difference from the stated total is rounding of the individual polygons. Independently, tools/dev_summary_calcs.py rebuilds the same quantity from the lot polygons and data/plans.json and gets **184,178 SF** — 0.02 % apart, the difference being how the curb-return fillets and the bay outlines are allowed for. Either figure rounds to 44.1 %.
- This supersedes the 160,167 SF / 38.4 % and 161,212 SF / 38.6 % figures carried in earlier drafts, both of which rest on a hard-coded allowance of 2,200 SF of impervious per lot. The real houses are larger than that allowance: **2,752 SF per Plan A lot and 2,898 SF per Plan B lot**, roof plus patio plus driveway plus walk.
- **Against the 2025 mixed-use filing** (hydrology sheet HYD-1: 235,388 SF impervious, 55.4 %), the cottage plan reduces impervious area by **21.8 %** — not the "–32 %" earlier drafts claimed on the understated figure.

3.8 Phasing

- **Phase 1** — $u < 980$: entrance, amenity block, lane to and beyond the first hammerhead turnaround (u 540, station 594.5 ft), Pond 1 (tract u 780–980), **24 lots**, gravity sewer to the existing on-site manhole. Phase 1 needs **no sewer extension of any kind**, and the only demolition it requires is 4541 Arcade Rd.
- **Phase 2** — $u \geq 980$: remaining lane, Pond 2, creek-woods and stream-setback preservation, **17 lots**, after the rear sewer route (§5.2) is settled with Gwinnett DWR. It also requires the demolition of **4535 Arcade Rd, which stands on the lane alignment at $u \approx 1,202$** (§2.5). The first turnaround and the tract layout allow Phase 1 to stand alone.

4. Required-vs-provided matrix

Status vocabulary: **appears consistent** / **interpretation requested** / **relief requested** / **VERIFY**. Citations are to the Lilburn Zoning Ordinance, Ord. 2023-603, unless noted. "Provided" is the current data/layout.json / data/plans.json geometry (regenerated 2026-09-03).

4.1 Use and dimensional standards

#	Requirement (citation)	Required (R-2)	Provided	Status
1	Use — "Single-family (cluster-cottage, creative lot configuration)" (§602 Use Table)	P — permitted, no SUP	41 detached cottage homes on fee-simple lots; no conservation-subdivision SUP, no multifamily	appears consistent
2	District intent (§402-1)	"detached single-family ... clustered and cottage housing types ... transition between lower density single-family districts and the MU District"	1-story detached cottages at 4.34 du/ac between two R-1 subdivisions	appears consistent
3	Minimum lot size, cottage home (Table 4.1)	3,000 SF	5,000 SF minimum, average and maximum (also meets the 5,000-SF detached-SF row)	appears consistent
4	Minimum lot width, cottage home (Table 4.1)	50 ft	50'-0" every lot	appears consistent
5	Minimum lot depth, all uses (Table 4.1)	100 ft	100.00 ft every lot	appears consistent
6	Maximum gross density (Table 4.1)	8 du/ac	4.34 du/ac (9.44 deeded ac); 4.28 du/ac (GIS)	appears consistent
7	Minimum heated floor area, cottage home (Table 4.1)	1,000 SF	Plan A 1,406 SF , Plan B 1,492 SF conditioned (garage excluded)	appears consistent
8	Maximum height (Table 4.1)	40 ft	1 story; ridge 17'-10" (A) and 19'-0" (B) above finished floor = 18'-6" and 19'-8" above finished grade as drawn; 24-ft cap offered as a condition	appears consistent
9	Buffer abutting R-1 property (Table 4.1)	0 ft for "detached single-family dwellings in R-2"; 20 ft for "all other allowed dwelling types"	20-ft undisturbed buffer along the SW, NE and NW lines (the stricter row), in recorded easements on the lots and within common tracts elsewhere; one 2.0-ft reduction requested at the SW curb return (item 41a)	interpretation requested (Q1) — is a detached cottage home a "detached single-family dwelling" (0 ft) or an "other allowed dwelling type" (20 ft)? The plan provides 20 ft either way
10	Buffers supersede minimum required yards (§313(1))	"Buffer requirements ... supersede these minimum required yards"	The 20-ft buffer coincides with the 20-ft rear yard of each perimeter lot. At the narrowest lot row the strip is 235.9 ft wide and the section is 100 + 35.9 + 100 = 235.9 ft — the lane tract is the residual and the arithmetic closes with nothing to spare	interpretation requested (Q2) — see the caution below. There is no slack: every foot the boundary survey removes from the width comes straight off the lane tract
10a	Fallback if Q2 is answered "separate tract required"	Lot depth 100 ft (Table 4.1)	Lots 50'-0" × 82'-0" (4,100 SF > 3,000 SF) with the 20-ft buffer in a separate tract, and a concurrent \$1005 variance on lot depth (criteria §1005-3.2 a–g; exceptional narrowness of a 236-ft-wide, 1,740-ft-deep strip as the hardship). A third, shallower house plan is required — the buildable depth on an 82-ft lot is 82 – 15 – 6 – 20 = 41'-0", and neither Plan A (51'-10") nor Plan B (57'-6") fits	relief requested (conditional fallback only)
11	Front setback, local street (Table 4.1)	15 ft from the right-of-way	15'-8" from the private-lane tract line to the porch face; the 8-in roof overhang is held outside the required yard	appears consistent — VERIFY that staff measures from the private-tract line
12	Front setback, collector/state road (Table 4.1)	50 ft from the Arcado Rd right-of-way	No lot fronts Arcado Rd; nearest lot line 230.0 ft ; nearest principal structure (clubhouse) 114.8 ft , true perpendicular. The monument entry sign stands 19.0 ft from the right-of-way and is not a principal structure — shown as an express exception (item 41b)	appears consistent — sign setback VERIFY
13	Side setback (Table 4.1)	5 ft	6'-0" both plans as sited (data/plans.json lot_siting.side_yard_ft = 6.0 for Plan A and Plan B alike)	appears consistent
14	Side setback, collector/state road (Table 4.1)	20 ft from the right-of-way	No lot has a side on Arcado Rd	not applicable

#	Requirement (citation)	Required (R-2)	Provided	Status
15	Rear setback (Table 4.1)	20 ft	Plan A patio edge 20'-6" ; Plan B rear porch edge 20'-10" ; no structure inside the rear 20-ft buffer easement	appears consistent — the clearance is 6 in and 10 in, and is to be re-checked when the architect seals the plans
16	Accessory building setbacks (Table 4.1)	5 ft side / 5 ft rear	No accessory buildings proposed on lots (HOA covenant)	appears consistent
17	Open space (Table 4.1 → Development Regulations §5.9, App. B)	§5.9.1 "Recreation Areas" applies only to single-family detached subdivisions of 50 acres or more	9.44 ac — §5.9 is not triggered . 141,089 SF = 33.8 % of the GIS area is provided voluntarily	resolved — not applicable ; provided voluntarily
18	Lot-size averaging / width reduction / setback adjustments (§309, §310, §314)	Available only in an Overlay District or redevelopment Character Area	Not used; not available here	not applicable
19	Street frontage (§319)	Each lot abuts ≥ 30 ft on a public street or an approved private street	50'-0" on the private lane; lane built to public-street standards per the Site Development Plan Review Checklist	appears consistent — VERIFY the lane's approval as an "approved private street" (Q4)
20	Floodplain (§325; Table 4.2)	No development in floodways; restrict development in floodplains	FEMA Zone X; no floodplain on site	appears consistent

Caution on item 10 — the reading that carries the lot yield. §313(1) says "Buffer requirements established by this article ... supersede these minimum required yards." "This article" is Article 3, and Article 3 sets no buffer widths (§318 imports the district buffers of Table 4.1 by reference). Nothing in the ordinance expressly authorizes holding a required buffer **inside a private lot** under an easement. The applicant therefore presents §313(1) as the **design basis and an interpretation to be confirmed**, not as settled law, and carries the item 10a fallback in the open. This is the single most consequential question at the pre-application conference.

4.2 Table 4.2 design criteria (R-2 column "A" = applicable)

#	Criterion (Table 4.2)	Provided	Status
21	Compatibility with existing character	1-story cottages, ridges 18'-6" and 19'-8" above grade — the lowest-massed buildings on the strip; fiber-cement and brick/stone; 20-ft undisturbed buffers on all three shared lines	appears consistent
22	Floodways and floodplains	None on site (Zone X)	appears consistent
23	Access to abutting public conservation lands	No public open space or greenway abuts the site	not applicable
24	Road and lot layout — grid, rectangular lots, block widths and lengths not exceeding 600 feet	Rectangular 50'-0" × 100'-0" lots; a single-spine lane is dictated by a 236-ft-wide strip with one frontage. Measured between street intersections the runs are 594.5 / 570.0 / 580.0 ft ; measured as lot-frontage runs between the open-space breaks, 300 / 500 / 550 ft. The longest block on either reading is 594 ft , inside the 600 ft the table asks for	appears consistent — relief no longer requested. The earlier layout's 746-ft first run needed a waiver; re-spacing the turnarounds removed the need
25	Culs-de-sac — "Circular turnarounds shall not be permitted except in the most unusual circumstances"	No circular turnaround anywhere; 3 hammerhead (T) turnarounds; the terminus is a T within a green	appears consistent
26	Non-pervious surfacing — pervious materials allowed	Pervious pavers proposed for the guest bay and mail-kiosk spaces; the lane in asphalt for fire-apparatus loading (PE to confirm)	appears consistent
27	Parking between the front façade and the street discouraged	Garages recessed 5'-0" behind the front wall and the porch projects 6 ft beyond that; driveway pairs are the only parking between façade and lane (needed for the two on-lot spaces beyond the garage); guest parking is off the lane in a bay	appears consistent — noted for staff
28	Alley access to parking encouraged / may be required	Not feasible: a rear alley would fall in the 20-ft buffer on a 236-ft-wide site	interpretation requested (waiver of the "may be required" clause)
29	Front-loaded garages recessed ≥ 5 ft behind the front wall	5'-0" recess on both plans, every lot	appears consistent

#	Criterion (Table 4.2)	Provided	Status
30	Land consumption minimized; perimeter buffers and open space preserved	5,000-SF lots (against 9,000 SF in R-1); 33.8 % common open space; continuous 20-ft perimeter buffer	appears consistent
31	Open-space design / landscape ecology	Creek woods preserved with the 50/75-ft buffers; the u 1,230–1,280 stream-setback tract left unlotted; ponds in the swale and the rear low pocket; greens on the lane	appears consistent
32	Greenbelts and wildlife corridors	Continuous 20-ft buffer on three sides plus the creek-woods and stream-setback tracts forms a perimeter greenbelt	appears consistent
33	Conservation principles for major conservation subdivisions	Not a conservation subdivision (§706 SUP not sought)	not applicable

4.3 Parking, streets, buffers, fire and other standards

#	Requirement (citation)	Required	Provided	Status
34	Parking, single-family detached (Table 8.1)	2 per DU → 82	164 on lots + 8 guest + 4 kiosk = 176	appears consistent
35	Interior drive width (Site Development Plan Review Checklist)	20 ft two-way, no parking; 22 ft with 90° parking	22'-0" pavement throughout, no on-lane parking; the tract widens to 60 ft through the amenity block so the 18-ft-deep 90° bays lie inside it	appears consistent
36	Sidewalks (checklist)	5 ft on all roads, 2 ft off back of curb (6 ft with street trees)	5-ft sidewalk on the NE side of the whole lane and on the entry drive; second 5-ft sidewalk on the SW side from u 530; 5-ft sidewalk along the Arcade Rd frontage	partial — VERIFY. Two walks need a 36.0-ft tract and the narrowest is 35.9 ft, so 6 of the 14 SW lots (u 230–530) front a reach with no walk on their side. Closing it costs either 0.5 ft of lot depth or a 1.5-ft strip. Ask at pre-application and record the answer
37	Landscape strip along a public right-of-way (checklist)	10 ft	10 ft along Arcade Rd, planted at 1 tree and 1 shrub per 25 lineal ft → 10 trees and 10 shrubs over 237.63 ft	appears consistent
38	Cul-de-sac / dead-end length (checklist)	≤ 2,000 ft	1,754 ft	appears consistent (fire-code review is separate — item 43)
39	Street grades (checklist)	≤ 12 %	Existing ground on the lane centerline reaches 10.9 % at u ≈ 1,494 ; the finished profile is not yet designed	VERIFY at engineering — a PE plan-and-profile is required before the section is claimed of the finished lane
40	Stream buffers (Lilburn Code Ch. 109 Art. VII, 50 ft + 25 ft; O.C.G.A. §12-7-6(b)(15), 25 ft)	Undisturbed 50 ft from top of bank; no impervious cover within 75 ft	All three digitized reaches are screened, not only the on-site one , and every element is held at ≥ 80 ft from a digitized centerline — 75 ft plus the 5.0-ft top-of-bank allowance: nearest lot 80.6 ft , nearest dwelling 106.2 ft , nearest driveway 102.6 ft , nearest pond top of bank 80.2 ft , lane pavement 80.5 ft . Worst impervious clearance 80.2 ft, or 75.2 ft net of the allowance. The SW slot at u 1,230–1,280 that failed the screen is not lotted ; it is a stream-setback open-space tract	appears consistent — top of bank to be field delineated ; the 5.0-ft allowance is a design assumption, not a survey. This screen is what took the yield from 43 to 41
41	Buffer encroachment by ponds and utilities (checklist)	≤ 50-ft width perpendicular; parking ≥ 5 ft from buffers	No pond now encroaches the perimeter buffer : Pond 1's top of bank is 41.5 ft and Pond 2's is 30.0 ft from the SW line. The guest bay is > 5 ft from any buffer	appears consistent — the earlier layout's 0.1-ft Pond 2 encroachment is gone
41a	Perimeter buffer at the entrance (Table 4.1; §1003-4 buffer reduction)	20-ft undisturbed buffer abutting R-1	The SW curb return is reduced from 25'-0" to 15'-0"; 2.0 ft measured along the right-of-way, and 0.1 SF of pavement, remain inside the band at the right-of-way corner — the buffer narrows to 18'-0" there. The entrance cannot be shifted NE: it is already only 251 ft from the Arcadia PI centerline against the ~244-ft spacing standard	relief requested — a buffer reduction of 2.0 ft over 0.1 SF is requested in the Letter of Intent §5, stated in voluntary condition 4 and dimensioned on the entry enlargement sheet

#	Requirement (citation)	Required	Provided	Status
41b	Monument entry sign in the 50-ft collector setback	50-ft principal-structure setback (Table 4.1)	One monument sign ≤ 32 SF, ≤ 6 ft high, 19.0 ft from the right-of-way; a sign is not a principal structure	VERIFY against the Lilburn sign regulations; shown as an express exception on the plan
42	Tree density (Code Ch. 109 Art. V, §109-168(3))	16 TDU/ac exclusive of buffer acreage	Tree survey and preservation plan at LDP; preserved creek woods, stream-setback tract and buffers credited. The Site Development Plan Review Checklist §6.i does not require TDU calculations for a single-family-residential subdivision	VERIFY at LDP
43	Dead-end fire apparatus road (IFC 2024 as adopted and modified by Ga. Comp. R. & Regs. 120-3-3-.04 , Appendix D103.4)	Table D103.4: over 750 ft → "Special approval required" ; the 501–750-ft band asks 26 ft of width	1,754-ft single access; 22'-0" clear lane; 3 hammerhead turnarounds at 594.5 / 570.0 / 580.0-ft intervals, every one inside the 750-ft trigger	special approval requested — Gwinnett County Fire Marshal. The 22-ft width against the table's 26 ft is the relief item
44	Two access roads for one- and two-family developments (Appendix D107.1 as modified in Georgia)	Ga. Comp. R. & Regs. 120-3-3-.04 replaces the model IFC threshold of 30 with 120 : two access roads are required where the number of dwelling units "exceeds 120"	41 units — the requirement is not triggered	not applicable . NFPA 13D sprinklers are therefore a voluntary offer , not the exception that relieves a requirement (§5.4)
45	Attached-dwelling visitability (§734)	Applies to attached dwellings only	Detached; 100 % zero-step entries offered voluntarily	not applicable (exceeded)
46	Traffic impact study (Gwinnett DOT <i>Traffic Impact Study Guidelines</i> , 2023)	"A TIS must be submitted concurrently with all zoning applications." The only residential exemption is "no more than seven lots or units." Study level by peak-hour trips: 0–20 → Level 1	41 lots — not exempt ; ≈ 11–16 PM peak trips → Level 1 (document-only) TIS is required	required — see §6.2. Earlier drafts said "no formal TIS expected"; that is wrong
47	Sewer capacity certification (Gwinnett DWR, form Rev. 07/2023)	Required; PE-sealed; the form carries a "Pre-Rezoning" request type	Request to be filed for 10,670 gpd average daily flow with the Phase 1 gravity route and the Phase 2 question (§5.2)	required — allow 10 business days, plus 20 more if analysis or flow monitoring is needed
48	Accessible parking (2010 ADA Standards §208.2, §502)	1 accessible space where 1–25 spaces are provided; van space 8 ft wide + 8-ft access aisle	1 van-accessible space with an 8-ft access aisle at the clubhouse end of the bay, on the shortest accessible route to the clubhouse entrance (§208.3.1), within the 12 common spaces	appears consistent

4.4 Staff-interpretation questions and the fallback (for the pre-application conference)

- **Q1 — Buffer row of Table 4.1.** Is a detached cottage home on its own fee-simple lot a "detached single-family dwelling in R-2" (0-ft buffer) or one of "all other allowed dwelling types" (20-ft buffer)? The application is designed to the 20-ft row and offers it as a condition either way; the answer affects only how the buffer is recorded.
- **Q2 — §313(1) buffer-within-lot.** May the 20-ft buffer occupy the 20-ft rear yard of each perimeter lot, held in a recorded buffer/conservation easement enforced by the HOA and the City? This is the design basis that lets 100-ft-deep lots sit on both sides of one lane in a 236-ft strip. The section closes exactly — 100 + 35.9 + 100 = 235.9 ft at the narrowest row — so there is no spare depth, and a survey width below about 229 ft leaves no room for a 22-ft lane with a 2-ft strip and a 5-ft walk. See the caution under item 10.
- **Q3 — Buffer reduction at the entrance.** With a 15-ft southwest curb-return radius, 2.0 ft of the return still lies inside the 20-ft band at the right-of-way corner and the buffer narrows to **18'-0"** over that short length, with 0.1 SF of pavement inside the band. The reduction is requested in the Letter of Intent §5 and stated in voluntary condition 4; is that the form of disclosure staff wants, and does Gwinnett DOT accept a 15-ft return on a minor collector?
- **Q4 — Approved private street.** Is the lane, built to the Site Development Plan Review Checklist's public-street section, an "approved private street" for §319 frontage purposes?
- **Fallback.** If staff requires the buffer in a separate common tract, the lots shorten to 50'-0" × 82'-0" (4,100 SF, above the 3,000-SF cottage minimum), a concurrent §1005 variance on the 100-ft lot depth is requested under §1005-3.2 (a)–(g), and a

third, shallower house plan is drawn. Lot count, density, buffers and open space are otherwise unchanged.

5. Utilities and infrastructure

5.1 Water

- Source: Gwinnett DWR mains in Arcade Rd (DIP/ACP, 47–107 ft from the front line); nearest hydrant ≈ 61 ft from the NE front corner.
- Proposed: an 8-in DIP main in the lane tract (≈ **1,754 ft**, dead-end) with hydrants at ≈ 400-ft spacing (Gwinnett DWR *Water and Sewer Standards* (April 2016) §2.2.15(b): 350–450 ft, one near the end of the main) — **6 hydrants**. Fire flow for dwellings of this size is 1,000 gpm for 1 hour under IFC 2024 (GA) Appendix B Table B105.1(1), reduced 50 % to **500 gpm** where NFPA 13D sprinklers are installed. DWR may require looping or oversizing the dead-end main; a second tie-in is not physically available, so a flushing hydrant at the terminus and DWR's dead-end criteria apply. Fire-flow test required.
- Individual meters per lot; an irrigation meter for HOA common areas. Average domestic demand **41 × 250 gpd = 10,250 gpd**.

5.2 Sanitary sewer (inverts are GIS attributes; invert survey required)

Phase 1 — gravity to the main already on the property. No extension of any kind.

Item	Value	Basis
Tie-in manhole (existing)	Invert 927.13 ft at (u 272, v –224), on the Arcade Road Townhomes outfall crossing parcel 033	FACTS §2
Phase 1 gravity line	u 980 → u 272 along the lane (708 ft) + a 106-ft lateral to the manhole = 814 ft of 8-in	layout . json path length
Controlling reach	the lane sag (936.3 at u 779) → the manhole, ≈ 613 ft with 5-ft cover at the sag: (936.3 – 5.0 – 927.13) ÷ 613 = 0.68 % available, above the 0.40 % minimum for 8-in	layout . json profile; docs/08 Memo B.3
The hard part	at the lane hump near u 280 (ground 946.4) the pipe is ≈ 17 ft deep . Either accept the deep cut — with rock possible on the ARE unit — or re-route the reach along the SW-edge easement corridor, where it runs at ≈ 10 ft	docs/08 Memo B.3 — to be re-run against the 41-lot alignment
Conclusion, Phase 1	The front 980 ft of lane (24 lots + clubhouse) drains by gravity into the existing 8-in main on the property . Nothing is extended off-site, so the Comprehensive Plan Amendment policy "Do not further extend sewer in this area" (Suburban-Low, p. 73) is not engaged by Phase 1 at all	

Phase 2 — one viable route, and it is not the one earlier drafts called "primary".

Item	Value	Basis
The problem	The lane low point (931.5 ft at u 1,479) lies behind the ridge (953.5 ft at u 1,179); a gravity line back to the Phase 1 system would have to run uphill	layout . json profile
~~In-tract lift station or grinder pumps~~ — foreclosed by policy	Gwinnett DWR <i>Standard Policy for Private Developments: Condominiums, Townhomes and Subdivisions</i> (rev. 9/2018): "Private pump stations, private force mains and private gravity sewers are only allowed for commercial properties under single ownership within a development." A 41-lot fee-simple subdivision is not that. A county-owned station is governed by DWR <i>Developer Pump Station Standards</i> (Jan 2014, WSR-24) §1.3.1(A) : stations "will only be permitted when gravity sewer is unavailable ... generally ... more than 5,000 feet down gradient, but this distance can be increased or decreased by GCDWR Division Directors based upon actual field conditions and the size of the project involved" — the Legends at Parkview main is 178 ft away. And there is no Gwinnett standard for grinder pumps or low-pressure sewer at all (zero occurrences in either the 2016 Water and Sewer Standards or WSR-24)	DWR documents
Off-site gravity tie — the only viable route	8-in from the rear low point via (u 1,300, v –114) → (u 1,300, v –235) → the Legends at Parkview manhole at (u ≈ 1,349, v ≈ –406), invert 919.58: ≈ 478 ft total , of which ≈ 178 ft lies off the property , at ≈ 1.45 % — comfortably above the 0.40 % minimum. It needs a recorded sewer easement across one Village Green Ct lot or along the stream corridor	layout . json; docs/08 Memo B
What must be settled	(1) the easement; (2) whether the City reads "Do not further extend sewer in this area" as barring a 178-ft tie to an existing main in the same basin — a policy-interpretation question for the Planning Director and Council; (3) DWR's own view, which should be asked on the "Pre-Rezoning" request type of the Rev. 07/2023 capacity form. Voluntary condition 16 binds this : Phase 2 may be served only by a gravity connection, no private pump station, force main, grinder pump or low-pressure system may be proposed or built, and no land-disturbance permit issues for Phase 2 until the off-site easement is recorded and DWR has certified capacity for the route — failing which the Phase 2 area stays wooded or is re-platted as open space. That is why Phase 1 is designed to stand alone	audit §4.1; data/voluntary-condition 16

Flows — Gwinnett DWR Wastewater Flow Estimation Guidelines (Rev. 10/2022): "Single Family Home 250 per unit" gpd; 175 gpd per 1,000 SF of clubhouse; peaking factor 4.0.

	Units	Average daily flow	Peak (× 4.0)
Whole project	41 DU + 2,400-SF clubhouse	10,670 gpd (41 × 250 + 420 = 10,670; 7.41 gpm)	42,680 gpd = 29.6 gpm

	Units	Average daily flow	Peak (× 4.0)
Phase 1	24 DU + clubhouse	6,420 gpd	25,680 gpd = 17.8 gpm
Phase 2	17 DU	4,250 gpd	17,000 gpd = 11.8 gpm

The "43 × 300 gpd = 12,900 gpd" figure in earlier drafts is retired twice over — the unit count is 41 and 300 gpd per unit appears in no Gwinnett source. The capacity-certification request would be returned on it.

5.3 Stormwater concept (Gwinnett County Stormwater Management Manual via the City; the PE's routing governs)

Method (2025 hydrology sheet HYD-1, Parker Land Design, 04-29-2025): required detention = 10,000 cf per disturbed acre less the required runoff-reduction volume **RRv**; $WQv = 1.2 \cdot (0.05 + 0.009 \cdot I) / 12 \cdot A$, with I = per-cent impervious and A = site area. Earlier drafts of this summary subtracted WQv instead of **RRv**; that is corrected here, and **RRv is carried at zero** at concept stage, as the concept plan states — any runoff-reduction credit the PE earns reduces the requirement.

Quantity	2025 MU filing (3 basins, HYD-1)	Cottage plan (this application)
Basin / site area	424,548 SF (9.75 ac)	417,174 SF (9.577 ac GIS)
Impervious area	235,388 SF = 5.41 ac (55.4 %)	184,148 SF = 4.23 ac (44.1 %)
$Rv = 0.05 + 0.009 \cdot I$	0.50–0.56	0.447
$WQr = 1.2 \cdot Rv$ (in)	0.60–0.67	0.537
WQv	23,293 cf	18,659 cf
RRv (1.0-in basis), available but not taken	—	15,552 cf
Disturbed area	whole basins	328,484 SF = 7.541 ac
Required detention (10,000 cf/ac, RRv = 0)	74,167 cf	75,410 cf
Provided detention	(3 basins)	77,863 cf — 103.3 % of the requirement

- **The disturbed-area correction is the single biggest arithmetic change in this package, and it is not in the applicant's favour.** The concept plan previously assumed ≈ **4.4 disturbed acres**; the real figure, rastered at 2 ft, is **7.541 ac**. Site raster area 418,228 SF less the preserved union of the 20-ft perimeter buffer bands (73,744 SF), the net creek-woods tract (16,000 SF) and the 50-ft undisturbed stream buffer (0 SF additional — it lies wholly inside the other two) = 89,744 SF preserved, leaving **328,484 SF disturbed**. The raster site area runs 0.25 % above the 417,174-SF shoelace because of 2-ft quantisation, so the disturbed area is if anything conservative. Required detention rose with it, from ≈ 44,000 cf to **75,410 cf — more than the 2025 mixed-use filing's 74,167 cf**, even though this plan is 21.8 % less impervious, because the 2025 sheet took its volume on three smaller drainage basins rather than on the whole disturbed site.
- **The basins were enlarged to meet it, and that cost a lot.** Pond 1 (SW side, basin u 790–970, top of bank **52 ft × 180 ft**) holds **33,129 cf**; Pond 2 (SW side, basin u 1,490–1,670, **64 ft × 180 ft**) holds **44,734 cf — 77,863 cf** total, each basin 6 ft deep with 3:1 side slopes, prismatic volume between top of bank and basin bottom. Pond 2's tract had to grow from 15,000 to 20,000 SF, which is where the second of the two lost lots went. An independent prismatic check in `tools/dev_summary_calcs.py` returns 33,696 and 45,360 cf for the same rectangles — about 1.5 % above the generator's figures, which is the margin of the end-slope treatment. The PE's routing governs either way.
- **Where the basins sit relative to the buffers.** Pond 1's top of bank is **41.5 ft** from the SW property line and Pond 2's is **30.0 ft**, so neither now encroaches the 20-ft perimeter buffer (the earlier layout put Pond 2's bank 0.1 ft inside it). The nearest pond top of bank is **80.2 ft** from a digitized stream centerline, i.e. 75.2 ft after the 5.0-ft top-of-bank allowance. **Rock probes come first:** both ponds sit on the ARE map unit (gneiss at 22–40 in), and a 6-ft basin is exactly where that matters.
- Runoff-reduction practices (bioretention with underdrains on clay subsoils, pervious guest parking) are favoured because the receiving Jackson Creek headwaters are 303(d)-listed for urban runoff; every cubic foot of **RRv** the PE can document comes straight off the 75,410-cf requirement (at the 1.0-in basis, 15,552 cf of it).
- The 2025 rule is used here only as a screening yardstick. **The PE's GCSWMM routing (WQv , CPv , $Qp25$, $Qf100$) governs**, and every disturbed acre carries the LDP fee (\$40/disturbed ac) and the NPDES fee (\$80/ac) — on 7.541 ac, not on 4.4.
- **Water quality:** $WQv = 1.2 \times 0.447 \div 12 \times 417,174 \text{ SF} = \mathbf{18,659 \text{ cf}}$, treated in the two basins. Recomputed independently from the lot polygons the same formula gives **18,662 cf**; the 3-cf difference is the 30-SF difference in the impervious take-off noted in §3.7.

5.4 Fire protection and emergency access (Gwinnett County Fire; IFC 2024 as adopted and modified by Ga. Comp. R. & Regs. 120-3-3-.04)

- **The one live issue is the dead end.** The lane is a single **1,754-ft** dead end. Appendix **D103.4** Table: 0–150 ft / 20 ft wide / no turnaround; 151–500 / 20 ft; 501–750 / **26 ft**; **over 750 ft** → **"Special approval required."** The applicant requests that special approval and discloses that the lane is 22'-0" wide against the 26 ft the table asks in the band below.
- **The turnaround intervals now have room.** The three hammerheads fall at stations 594.5, 1,164.5 and 1,744.5 ft, i.e. intervals of **594.5 / 570.0 / 580.0 ft** — the longest is 155 ft inside the 750-ft trigger. The earlier layout's first interval was 746.4 ft, 3.6 ft inside it, which no survey shift could be allowed to disturb; that fragility is gone.
- **The two-access requirement is not triggered.** The model IFC 2024 Appendix D107.1 sets the threshold at 30 dwelling units; **Georgia replaces it with 120** — Ga. Comp. R. & Regs. 120-3-3-.04, Modifications to Chapter 5, item 7: "Developments of one- or two-family dwellings where the number of dwelling units **exceeds 120** shall be provided with two separate and approved fire apparatus access roads." At **41 units** D107.1 does not apply, and a second access is in any case physically impossible — no street abuts the rear or either long side.
- **NFPA 13D sprinklers in every home are therefore a voluntary offer, not a code requirement** (voluntary condition 5). They are worth offering: they halve the required fire flow on a 1,754-ft dead-end main (1,000 → 500 gpm, IFC 2024 (GA) Appendix B105.1 exception), and they are the natural consideration to put on the table for the D103.4 special approval and the 22-ft width. At roughly \$3,000 per home they are a **≈ \$123,000** election at 41 units, and the Fire Marshal — not this document — should decide whether they buy the width relief.
- **Other mitigation offered:** a 22-ft clear lane with no on-lane parking (signed and enforced by the HOA); three 20 ft × 60 ft hammerhead turnarounds; hydrants in the lane; a sprinklered clubhouse as the building code requires (NFPA 13 or 13R — VERIFY).
- Georgia's D107.1 Exception 3 also supplies a named menu for developments with fewer than two entrances — enhanced turning radii, increased road widths, D103.6 fire-lane signage, an additional non-vehicular all-weather access designed to Appendix D loading, or a written statement by the Fire Code Official — and O.C.G.A. §25-2-12(e)(4) provides a waiver route on the Safety Fire Commissioner's form, on which "the State Fire Marshal or designated representative shall respond within 30 days" and plans are **deemed approved** if that clock is missed.
- **First-due:** Station 3, **≈ 2.9 mi / ≈ 6 min** free-flow (brief §4.6). Gwinnett Fire Marshal review is requested at the pre-application stage.

5.5 Refuse, electric, gas, communications

- Refuse: individual roll-carts collected curbside on the lane by the city's hauler; the 22-ft lane and T-turnarounds accommodate a rear-loader (turning template by the PE — VERIFY the hauler's vehicle). No dumpster pad.
- Electric, gas, telecommunications: underground in the lane tract (Jackson EMC / Georgia Power service territory VERIFY; Atlanta Gas Light); generator-ready clubhouse as a warming/cooling centre.
- Street lighting: full-cutoff, **≤ 3000 K**, house-side shields on fixtures facing R-1 lots (voluntary condition 11).

6. Traffic

6.1 Trip generation (ITE *Trip Generation Manual*, 11th ed.)

The 11th edition renamed and re-scoped **LUC 251** to "Senior Adult Housing – Single-Family" (merging the former detached and attached data sets), and the manual itself is paywalled. Four secondary compilations were reconciled on 2026-09-03 and none is demonstrably an 11th-edition source; the honest presentation is a **rate band**, with the note that the traffic engineer must pull the manual page before filing.

Scenario	Units	LUC	Daily	AM peak	PM peak
Proposed — 55+ detached cottages	41 DU	251, rate band 3.70–4.31 daily / 0.20–0.34 AM / 0.26–0.39 PM per DU	152–177	8.2–13.9	10.7–16.0
Same, at the FACTS \$3 rate (0.30 / 0.30)	41 DU	251	≈ 177	12.3	12.3
The same 41 lots, unrestricted single-family	41 DU	210 Single-Family Detached (9.43 / 0.70 / 0.94)	≈ 387	28.7	38.5
R-1 as currently zoned, theoretical maximum (4 du/ac × 9.44 ac = 37 lots; the real R-1 yield on a 236-ft strip would be lower)	37 DU	210	≈ 349	25.9	34.8
RZ-2025-01 MU filing (no TIS was filed; order-of-magnitude, before pass-by or internal capture)	104 condos + 24,400 SF retail/service/office	220 Multifamily Low-Rise + 822 Strip Retail Plaza	≈ 2,030	—	≈ 214

- **The conclusion does not turn on which rate is used.** At the highest rate found (0.34 AM / 0.39 PM) and at the top of the offered unit range (47 units — the cap in voluntary condition 2, six above the 41 now drawn), the PM peak is **18.3 trips** — still inside Gwinnett DOT's 0–20 band.
- The cottage program generates roughly **one-third** of the PM peak of the same lots without an age restriction (12.3 against 38.5) and about **5.8 %** of the 2025 mixed-use concept.

6.2 Findings

- **A Level 1 TIS is required, and this application will file one.** Gwinnett DOT *Traffic Impact Study Guidelines* (2023): "A TIS must be submitted concurrently with all zoning applications and land disturbance development," and the only residential exemption is "single-family detached and townhome residential development consisting of **no more than seven lots or units.**" At 41 lots the project is not exempt; at 0–20 peak-hour trips the study level is **Level 1 (document-only)** — location description, existing and proposed land use, trip generation, access-management review, adjacent access spacing upstream and downstream, intersection sight distance, connectivity and circulation, functional classification, posted speed and future identified projects. Operations analysis, turn-lane warrant analysis and signal warrants begin at Level 2 and are **not** required. Note that these guidelines are written for unincorporated Gwinnett County under UDO §270-20.8; the City's own requirement is to be confirmed at the pre-application conference, and Gwinnett DOT will apply them at the driveway-permit stage on its own road.
- **New traffic counts are required.** The 14,800 AADT figure comes from GDOT station 135-6689 in **2017**. The same Gwinnett DOT guidelines require counts "less than two years old" from GCDOT or GDOT and, failing that, that "the developer shall be responsible for obtaining new, 24-hour counts, from an approved contractor, as directed by GCDOT." A 24-hour count is budgeted.
- **Left-turn lane not warranted.** Gwinnett DOT *Criteria and Guidelines for Left Turn Lanes* (June 20, 2007), **Table 1**, residential developments, 2-lane route, posted 30–35 mph, ADT ≥ 6,000: the threshold is **75 lots**. At 41 lots no left-turn lane is warranted (to be confirmed with the new count).
- **Spacing and sight distance.** Gwinnett UDO §900-40.5, **Table 900.3**: minimum driveway separation at 35 mph is **244 ft**, centerline to centerline; provided **251.2 ft** from the Arcadia Place centerline (250.7 ft chord), a margin of 7.2 ft. Gwinnett UDO §900-40.4, **Table 900.2**: intersection sight distance at 35 mph on a 2-lane road is **390 ft** each way, on a curved frontage — **field survey required**. §900-40.5(C) limits the project to one driveway per 400 ft of frontage, i.e. one.
- **Pedestrian and transit.** 5-ft sidewalk along the Arcado Rd frontage; Ride Gwinnett Route 25 stops at Killian Hill Rd @ Arcado Rd, 0.30 mi away (no continuous sidewalk verified — the applicant requests county and city gap closure); the ADA paratransit band covers the site.
- **School buses:** none (HOPA 55+).

7. Schools (Gwinnett County Public Schools — Camp Creek ES / Trickum MS / Parkview HS cluster)

Measure	RZ-2025-01 (104 condos; GCPS planning estimate, July 2025, 4817_school-impact-analysis.pdf)	This application (41 HOPA 55+ cottages)
Camp Creek ES (capacity 1,325; forecast 1,414 → 1,383, over capacity by 89 → 58)	+25 students	0
Trickum MS (capacity 2,275; forecast 2,203 → 2,104, under capacity)	+13	0
Parkview HS (capacity 2,900; forecast 3,148 → 3,093, over capacity by 248 → 193)	+19	0
Total	+57	0

Housing for older persons under 42 U.S.C. §3607(b)(2)(C) — at least 80 % of occupied units with a resident 55 or older, which at 41 units means **33 units** ($0.80 \times 41 = 32.8$, rounded into the 80 percent by 24 CFR §100.305(g)) — generates no school-age children. The two over-capacity schools in the cluster receive no added enrolment, and the §1003-7(4) "burdensome use of ... schools" criterion is answered in the negative. VERIFY the covenant's minor-occupancy clause against HUD guidance; note also 24 CFR §100.306(b), which makes phrases such as "adult living" or "adult community" inconsistent with 55+ intent — the marketing language is bound in the covenants.

8. Fiscal and community effects

8.1 Tax base (illustrative; market figures VERIFY with a broker CMA and the Gwinnett Tax Commissioner once 2026 rates are adopted)

Item	Low	High	Basis
Sale price per cottage (assumed)	\$420,000	\$460,000	positioning between Cottages at Noble Village resales (\$342,500–\$414,900) and Snellville detached 55+ product (≈ \$447k Saddlebrook; mid-\$500s Soleil) — VERIFY
Completed value, 41 homes	\$17.22 M	\$18.86 M	41 × price
Existing 2026 FMV of the four parcels	\$1,267,600		2026 digest
Assessed value (40 %)	\$6.89 M	\$7.54 M	O.C.G.A. 40 % ratio
City of Lilburn tax at 5.43 mills (2026 rate set by Resolution No. 2026-11, calendared 2026-08-31)	≈ \$37,400 /yr	≈ \$41,000 /yr	before homestead exemptions; existing ≈ \$2,750/yr
City tax if every owner claimed S1R + L5 + the 2026 \$40,000 senior exemption (\$60,000 assessed each)	≈ \$24,000 /yr	—	illustrative floor: (6.89 M – 41 × \$60,000) × 5.43 ÷ 1,000
County + school levies at 31.6 mills (2025 rates: county 6.95 + fire 3.20 + recreation 1.00 + GCPS 18.70 + bond 1.45)	≈ \$218,000 /yr	≈ \$238,000 /yr	before exemptions; the school share (20.15 mills, ≈ \$139k–\$152k) is removed for owners 65+ under the LSA income cap (\$124,648 in 2026)

The community adds roughly **fourteen to fifteen times** the current taxable value of the assemblage (13.6× at the low price, 14.9× at the high) while sending **no children** to GCPS; the senior exemptions that trim the school levy are the same exemptions the neighbours already enjoy. Two fewer lots take about **\$840,000–\$920,000** off the completed value against the 43-lot draft — worth stating, because it is the fiscal cost of the stream screen and the basin enlargement.

8.2 HOA and maintenance

- Mandatory homeowners' association (Georgia Property Owners' Association Act, O.C.G.A. §44-3-220 et seq.) owning and maintaining the lane tract, sidewalks, ponds, buffers, creek woods, amenity tract, entry sign and lighting, and providing yard maintenance for every lot.
- The line-item budget is now built at 41 lots** (tools/hoa_budget.py → data/hoa-budget-derived.json, carried in docs/13): operating \$123,356 + reserves \$40,451 + contingency \$6,168 + bad-debt allowance \$3,399 = **\$173,374 a year, \$4,229 per lot per year = \$352 per lot per month** (operating \$251 + reserves \$82 + contingency and bad debt \$19), with curbside refuse a further \$22 if the association contracts it. **During Phase 1 alone, 24 lots carry nearly the same fixed plant, so the assessment is \$449 per lot per month** — a number the declarant must either subsidise or disclose to the first buyers.
- The "\$200–\$350/month" range quoted in earlier drafts is retired: it was not supported by a budget, and the real figure sits at the top of it at full build-out and above it during Phase 1. Comparables: Cottages at Noble Village \$93–\$280/mo; Soleil Summit Chase \$389/mo.

8.3 Demand references (market data — VERIFY; commercial sources cited in

docs/11-site-context-brief.md §5)

- Census tract 504.61 is 35.5 % aged 55+, 90 % owner-occupied, 67 % pre-1980 housing (ACS 2020–2024); ZCTA 30047 has ≈ 8,500 owner-householders 55+ (25.8 % of 70,121 residents are 55+; median household income \$88,652).
- Lilburn's only 55+ product, Cottages at Noble Village (1,225–1,450 SF), is sold out; no detached 55+ community exists inside the city. The nearest detached competitors are in Snellville.
- ARC Series 17 (2024) forecasts Gwinnett growing to 1.20 M by 2050 with residents 75+ rising to ≈ 12 % of the region.
- Lilburn ZHVI \$401,137 (June 2026, –2.6 % YoY); King David Manor median FMV \$349,200; Legends at Parkview \$693,500 — the cottages price between the two abutting subdivisions.

8.4 Comprehensive-plan fit

Single-family detached housing at **4.34 du/ac** inside the adopted "Established Residential" character area (2024 plan, printed p. 35) and, under the pending 2026 Amendment, the "Suburban-Low" class, where Single-Family Cottages and Single-Family are *Appropriate* and the 2025 mixed-use concept is *Not Appropriate*; below R-2's 8 du/ac maximum and far below the 2025 filing's ≈ 11 du/ac. The Amendment's stated policies are addressed directly: one-story massing answers "encourage new developments to match existing design character"; frontage and lane sidewalks answer "require sidewalks"; and **Phase 1 extends no sewer at all**, using the main already on the property, while Phase 2's route is left open for the City and DWR to settle (§5.2).

9. Voluntary zoning conditions offered (to be restated in the Letter of Intent and recorded in the declaration of covenants)

1. **Age restriction.** The development shall be operated as housing for older persons (55+) under 42 U.S.C. §3607(b)(2)(C) and 24 CFR §100.300–.308, with a recorded declaration, published policies and biennial verification (24 CFR §100.307(c)), enforceable by the City of Lilburn. At the 41 units shown, the 80-percent occupancy test of 24 CFR §100.305(a) together with the fractional-unit rule of §100.305(g) ($0.80 \times 41 = 32.8$, and the fraction counts into the 80 percent) requires **at least 33 of the 41 units** to be occupied by at least one person aged 55 or older, leaving at most 8 units that need not be; the declaration shall state that figure and shall recompute it if the unit count changes. Advertising and community documents shall not use "adult living," "adult community" or similar phrases (24 CFR §100.306(b)).
2. **Maximum units.** Not more than **47** detached single-family cottage homes (**41 shown**); no attached, multifamily, commercial or non-residential use other than the clubhouse and amenities serving residents.
3. **Height.** All dwellings one story; maximum ridge height 24 ft above finished grade (the plans as drawn on Sheets A-2.1 and A-2.2 are 17'-10" for Plan A and 19'-0" for Plan B above finished floor, i.e. 18'-6" and 19'-8" above finished grade at the front, finished floor being 8 in above grade); clubhouse one story.
4. **Buffers.** A 20-ft undisturbed buffer along all property lines abutting R-1 property (SW, NE and NW lines), held in a recorded buffer/conservation easement where it lies within lots and in common tracts elsewhere; supplemental evergreen planting where existing vegetation is sparse; no structures, fences or grading in the buffer except (a) approved pond and utility encroachments not exceeding 50 ft in perpendicular width and (b) **the single buffer reduction requested in the Letter of Intent §5 and dimensioned on Sheet C-2.3 (entry enlargement, to be issued with the filed set) — at the tangent point of the southwest curb return of the entrance on the Arcade Road right-of-way, where the buffer narrows to a minimum of 18'-0" instead of 20'-0", a reduction of 2.0 ft measured along the right-of-way affecting 0.1 SF of pavement.** The buffer is at its full 20-ft width everywhere else on the southwest, northeast and northwest lines. No other buffer reduction is requested, and none shall be taken without an amendment to these conditions.
5. **Fire sprinklers — voluntary offer.** Every dwelling shall be equipped with an NFPA 13D residential sprinkler system and the clubhouse sprinklered as the building code requires, with systems maintained under the HOA covenants. This is offered **voluntarily**: at 41 units the two-access requirement of IFC 2024 Appendix D107.1 as modified by Ga. Comp. R. & Regs. 120-3-3-.04 (which raises the trigger to more than 120 dwelling units) is not engaged, and sprinklers are therefore additional protection and a 50-percent fire-flow reduction (IFC 2024 Appendix B §B105.1), not a required mitigation. The applicant asks that this condition be accepted as offered; if the Gwinnett County Fire Marshal states in writing that sprinklers are unnecessary, the condition may be released by the Fire Marshal's written concurrence and City staff's approval without a further public hearing.
6. **Single access.** One driveway on Arcade Rd, located in the SW third of the frontage; no gate.
7. **No Arcade Road driveways.** No individual lot shall have direct vehicular access to Arcade Rd.
8. **Architectural standards.** Exterior walls of fiber-cement lap and board-and-batten siding with brick or stone water table/piers; architectural-shingle roofs with 8:12 main and front-wing gables, the front-wing gable carried forward to roof the covered front porch, and a 6:12 gable over the garage; front-loaded garages recessed ≥ 5 ft behind the front wall; covered front porches ≥ 6 ft deep; zero-step entries; floor plans, elevations and materials substantially as shown on the sheets submitted with the application — Sheets A-1.1 and A-1.2 (floor plans, Plans A and B), A-2.1 and A-2.2 (exterior elevations, Plans A and B), A-2.3 (exterior colour schemes and materials), A-3.0 (clubhouse, mail kiosk and entry monument sign) and A-4.0 and A-4.1 (typical lots); three approved colour schemes; no wall signs on any dwelling.
9. **Stream buffer.** The stream head and its 50-ft undisturbed buffer and 75-ft impervious setback preserved in a common "creek woods" tract, and the southwest area at u 1,230–1,280 that does not clear the stream screen preserved unlotted as a common stream-setback tract with no impervious cover; no lot, structure or pavement within 75 ft of the top of bank as field-delineated. The field study shall establish first whether and where a regulated stream begins (spring/seep or the 25-acre drainage-area point) and then the top of bank, and the plat shall show the 25-ft state buffer (O.C.G.A. §12-7-6(b)(15)) and the 50-ft/75-ft city lines separately.
10. **HOA maintenance.** A mandatory HOA owning and maintaining the private lane, sidewalks, stormwater facilities, buffers, common tracts, entry sign and lighting, with yard maintenance for all lots.
11. **Lighting.** Full-cutoff fixtures ≤ 3000 K; lane lighting mounted ≤ 14 ft with house-side shields facing R-1 lots; no court lighting for the pickleball courts.
12. **Construction hours.** Exterior construction limited to 7:00 AM – 7:00 PM Monday–Friday and 8:00 AM – 5:00 PM Saturday; none on Sundays or federal holidays (city ordinance governs if stricter).
13. **Tree survey and preservation.** A tree survey and preservation plan meeting Code Ch. 109 Art. V (16 TDU/ac) submitted with the LDP; preserved trees ≥ 12 -in DBH in the buffers, creek woods and stream-setback tract credited; phased clearing with immediate stabilization.

14. **Entry sign.** One monument sign ≤ 32 sf, ≤ 6 ft high, externally lit, standing about 19 ft from the Arcade Road right-of-way within the front amenity tract and outside the sight triangle; permitted separately under the sign regulations.
15. **Phasing.** Development in two phases at the **u $\approx$ 980-ft line** as shown on Sheet C-6.0 (phasing, block and open space exhibit, to be issued with the filed set); Phase 1 (24 lots) shall include the amenity block, Pond 1 and the first hammerhead turnaround shown on Sheet C-2.0. Phase 2 (17 lots) shall not commence until condition 16 is satisfied, and the existing dwelling at 4535 Arcade Road, which stands on the proposed lane alignment about 1,200 ft back from the right-of-way, shall be removed under permit before any Phase 2 land disturbance.
16. **Sanitary sewer.** Phase 1 shall be served by gravity to the existing 8-in Gwinnett County main already on the property (the Arcade Road Townhomes outfall, manhole invert 927.13), and requires **no sewer extension**. Phase 2 shall be served **only by a gravity connection**; no private pump station, private force main, private gravity sewer, grinder pump or low-pressure sewer system shall be proposed or built (Gwinnett DWR *Standard Policy for Private Developments: Condominiums, Townhomes and Subdivisions*, rev. 9/2018, allows private facilities "only ... for commercial properties under single ownership within a development"). No land-disturbance permit shall issue for Phase 2 until (a) a permanent off-site sanitary sewer easement for the gravity connection is recorded and (b) Gwinnett DWR has issued a sewer capacity certification covering the Phase-2 route. If either is not obtained, Phase 2 shall not be built and the Phase-2 area shall remain in its existing wooded condition or be re-platted as open space.
17. **Traffic impact study and access management.** The applicant shall submit a **Level-1 traffic impact study** prepared to Gwinnett DOT *Traffic Impact Study Guidelines* (2023) with the zoning application, based on **new 24-hour traffic counts** on Arcade Rd obtained from an approved contractor (the 2017 GDOT count does not satisfy the guidelines' two-year currency rule); shall furnish an intersection-sight-distance certification by a Georgia Professional Engineer for the single entrance (390 ft each way at 35 mph, Gwinnett UDO §900-40.4, Table 900.2); and shall obtain a Gwinnett County DOT driveway/encroachment permit before any land disturbance.
18. **Sidewalks and transit.** A 5-ft sidewalk along the full Arcade Rd frontage and 5-ft sidewalks within the lane tract as shown — on the northeast side for the full length of the lane and on the southwest side from u 530, where the tract first reaches the 36 ft that two walks require; the applicant will cooperate with the county and city on closing the sidewalk gap to the Killian Hill Rd bus stop.

10. Open items / VERIFY list

#	Item	What settles it
1	Boundary, acreage (9.44 deeded vs 9.577 GIS), metes-and-bounds legal description	Georgia RLS boundary survey and sealed legal description
2	Site width — there is no longer any margin in the section. 100-ft lots on both sides plus a 35.9-ft lane tract exactly consume the 235.9-ft width at the narrowest lot row, and the tract is the residual: every foot the survey removes comes off the lane. At 229 ft of width the tract is down to the bare 29.0 ft that pavement, one strip and one walk require	RLS survey (a 230-ft width forces the 50'-0" $\times$ 82'-0" fallback and a third house plan)
3	Q1 and Q2 buffer interpretations (Table 4.1 buffer row; §313(1) buffer-within-lot)	Pre-application conference with the Planning Director (Reid Turner, 770-279-3715). The Development Regulations §5.9 open-space question is closed — §5.9 is "Recreation Areas" and applies only at 50 acres or more
4	Plan B's ridge: the data and the drawing differ by 4 in. Issued Sheet A-2.2 draws 19'-8" above finished grade (19'-0" above the floor); data/plans.json computes 19'-4" (19.33 ft) above the floor from a 29.5-ft main-gable span, i.e. 20'-0" above grade. This summary and condition 3 state the drawn figures, because a recorded condition must not cite a dimension the filed sheet does not draw	Architect reconciles the roof model; conform data/plans.json (docs/09 §5.3 item G-1) or re-issue A-2.2, before filing . Plan A is consistent at 17'-10" / 18'-6"
5	Buffer reduction at the SW curb return — now consistent everywhere. The drawing, the Letter of Intent §5 and voluntary condition 4 all state the same figure: the buffer narrows to 18'-0" instead of 20'-0" at the tangent point, a 2.0-ft reduction affecting 0.1 SF of pavement. What remains is Gwinnett DOT's concurrence on the 15-ft return radius and the entry enlargement sheet	Gwinnett DOT driveway permit; Sheet C-2.3 to be issued with the filed set
6	Stream: whether the reach is a regulated "stream" at all (the 25-acre drainage-area test) and where its top of bank lies. The screen against all three reaches now holds every element at ≥ 80 ft from a digitized centerline (75.2 ft after the 5-ft allowance), and it cost the u 1,230–1,280 lot	Field study by a qualified professional (GA EPD <i>Field Guide</i> , Sept 2017, for the state-waters half) + City review authority. A favourable delineation could return that lot; an unfavourable one could take more
7	Sewer inverts (927.13 at the on-site MH; 919.58 at the Legends MH); the Phase 2 easement across a Village Green Ct lot; the City's reading of "Do not further extend sewer in this area"	Invert survey; Gwinnett DWR sewer capacity certification on the "Pre-Rezoning" request type at 10,670 gpd; Planning Director and Council on the policy question

#	Item	What settles it
8	Comprehensive Plan Amendment adoption date (available 9/14, 10/12 or 11/9/2026) and which Future Development class the four parcels fall in (Suburban-Low or Suburban-Medium — the sewer policies are opposite)	City Council adoption record; Planning Director in writing
9	Stormwater is now sized, but on a screening rule and on unprobed ground. 77,863 cf provided against 75,410 cf required on 7.541 disturbed acres — a 3.3 % margin, with RRv carried at zero. Both 6-ft basins sit on the ARE map unit (gneiss at 22–40 in)	PE hydrology per GCSWMM; rock probes / geotechnical report before the volumes are fixed ; any RRv credit reduces the requirement
10	Soil map-unit names and ratings (§2.4)	NRCS Web Soil Survey AOI report (Soil Data Access not reachable 2026-08-29)
11	ITE 11th-ed. LUC 251 rates (the band 0.20–0.34 AM / 0.26–0.39 PM); the Level 1 TIS itself; a new 24-hour count on Arcade Rd	The traffic engineer's copy of the manual; a count from a GCDOT-approved contractor; Gwinnett DOT pre-application
12	IFC 2024 (GA) Appendix D103.4 special approval for the 1,754-ft dead end at 22-ft width; hydrant spacing; refuse-vehicle turning	Gwinnett County Fire Marshal (Ga. Comp. R. & Regs. 120-3-3-.04; O.C.G.A. §25-2-12(e)(4) waiver route); city hauler
13	Water: dead-end main looping/oversizing, fire-flow test	Gwinnett DWR
14	Tax figures — 2026 city millage, GCPS 2026 rate and sale prices (\$420–460k). The HOA budget is no longer an open item: data/hoa-budget-derived.json carries a line-item 41-lot budget at \$352/lot/month full build-out and \$449 during Phase 1	Tax Commissioner; broker CMA; the declarant's decision on subsidising Phase-1 assessments
15	Formal disposition of RZ-2025-01 (tabled/dormant vs withdrawn)	Written confirmation from the Planning Director
16	Demolition permits: 4541 Arcade Rd in Phase 1 and 4535 Arcade Rd, which stands on the lane alignment at u ≈ 1,202 , in Phase 2; 4535's septic-vs-sewer status	Owner records; Gwinnett DWR; city permits
17	HOPA covenant terms (minor-occupancy clause, verification procedure, the 24 CFR §100.306(b) marketing-language bar); the declaration must state the 33-of-41 figure and recompute it if the count changes	Counsel; HUD 24 CFR Part 100 Subpart E
18	Sidewalk on the SW side for u < 530 (6 of the 14 SW lots) — a walk on both sides needs a 36.0-ft tract and the narrowest is 35.9 ft	Pre-application answer, recorded; 0.5 ft of lot depth or a 1.5-ft strip would close it
19	Tree-canopy inventory; USFWS IPaC screen	Arborist / landscape architect at LDP
20	The offered unit cap now sits 6 units above the drawing. Voluntary condition 2 caps the project at 47 units, a figure set when the plan showed 43; it shows 41. A cap that far above the plan invites the question of what a 47-lot version would look like, and the stream screen and basin sizes suggest it no longer exists	Owner's decision before filing — lower the cap to 43 or 44, or be ready to explain the difference at the hearing
21	Documents outside this summary that still carry 43-lot arithmetic must be re-run behind the regenerated layout — docs/03 §§2–6, docs/08 Memos A–E, docs/10 §4 and docs/13's budget tables. tools/dev_summary_calcs.py and tools/hoa_budget.py have been re-run; tools/memo_calcs.py has not been checked by this pass	Re-run each generator, then python3 verify/check_dev_summary.py, python3 tools/check_docs.py and python3 verify/check_sheet_refs.py

Closed since the previous draft: the block-length question (594 / 570 / 580 ft, all inside Table 4.2's 600 ft, so no waiver is sought); the 746-ft first turnaround interval (now 594.5 ft, with room against the 750-ft D103.4 trigger); Pond 2's 0.1-ft encroachment into the perimeter buffer (the basins were re-shaped and now stand 30.0 and 41.5 ft off the SW line); the "lot 10" stream conflict (that ground is now an unlotted stream-setback tract); the open-space residual (the enumerated tracts now close exactly on 141,089 SF with no unallocated slivers); the "two older houses near the front" description of the existing improvements (only 4541 is near the front — 4535 stands on the lane 1,200 ft back); the document-versus-drawing contradiction on the entrance buffer reduction (all three now say 18'-0" / 2.0 ft / 0.1 SF); the "≈ 4.4 disturbed acres" assumption (7.541 ac); the Development Regulations §5.9 open-space question; the "garage inside the body" floor-area alternative; the fire-code edition (IFC 2024 as modified by Ga. Comp. R. & Regs. 120-3-3-.04, never the 2018 edition); and the two-access requirement, which Georgia does not trigger at 41 units.

Prepared by: owner/applicant (self-prepared DRAFT per the Lilburn 2026 Application Instructions); to be reviewed and, where required, sealed by a Georgia RLS (boundary and legal description), PE (sewer, stormwater, lane, traffic), architect (plans and elevations) and landscape architect (tree and buffer plans) before filing.

Disclaimer: This is an AI-generated analysis for preliminary planning purposes. All findings must be verified by a licensed professional before use in design, permitting, or regulatory submissions.

08 — Technical Memoranda (Traffic, Sewer, Water, Stormwater, Fire, Schools, Environmental)

Project: The Cottages at Arcade Springs — R-1 → R-2 rezoning, 4541/4539/4537/4535 Arcade Rd SW, Lilburn, GA (PINs 6123 033, 015, 014, 162; Land Lot 123, 6th District; 9.44 ac deeded / 9.58 ac GIS) **Prepared:** 2026-08-29 · **Revised 2026-09-03** to the external-fact re-verification of that date · **Re-issued 2026-09-03** on the corrected site plan, at which the lot count fell from 43 to **41** (see both revision notes below) **Status:** DRAFT — NOT SEALED. To be superseded by the sealed work of a Georgia PE (memos A–E) and by GCPS / GA EPD correspondence (memos F–G). Nothing here is a design; every memo is a feasibility screen sized to the rezoning question "can this site be served?"

Revision note — what changed on 2026-09-03 and why. Four positions in the 2026-08-29 issue of this document were wrong on the primary sources and have been rewritten, not patched:

Memo	Was	Is
A	"at 12.9 peak-hour trips no formal study is expected"	A Level-1 TIS is required. Gwinnett DOT: "A TIS must be submitted concurrently with all zoning applications"; the only residential exemption is seven lots or units. The 2017 GDOT count is also ineligible under the county's own two-year currency rule
A	LUC 251 rates quoted from two "11th-edition" compilations	Both compilations carry pre-11th-edition land-use names. The rate is UNRESOLVED ; a range is carried and the conclusion is shown to be insensitive to it
B	Phase 2 "primary" route = in-tract HOA lift station or grinder pumps	Foreclosed by policy. Gwinnett DWR's <i>Standard Policy for Private Developments: Condominiums, Townhomes and Subdivisions</i> (rev. 9/2018) allows private pump stations, force mains and gravity sewers "only ... for commercial properties under single ownership"; WSR-24 §1.3.1(A) allows a county station only where gravity is "more than 5,000 feet down gradient" against an actual 178 ft; and Gwinnett has no grinder-pump or low-pressure-sewer standard. The 178-ft off-site gravity tie is the only viable Phase-2 route. Phase 1 needs no extension at all
E	"43 units (> 30) → two access roads unless sprinklered; NFPA 13D is the mitigation"	Ga. Comp. R. & Regs. 120-3-3-.04 raises the Appendix D107.1 trigger from 30 to 120 dwelling units. At 41 units the two-access requirement is not engaged . NFPA 13D sprinklers are a voluntary offer , not a required mitigation. The live fire-access item is the 1,754-ft dead end under Table D103.4

Two further corrections run through the whole document: the fire-code citation form (the operative Georgia modifications are issued by the Safety Fire Commissioner at **Ga. Comp. R. & Regs. 120-3-3-.04**, not as a DCA amendment packet), and the impervious area, which the 2026-08-29 issue understated by about 24,000 SF (160,167 SF reported against the 184,148 SF the corrected plan actually draws — see D.2).

Revision note — the corrected site plan (second issue of 2026-09-03), and what it does to these memos. After that revision was written, the site-plan generator was corrected against the geometry audit in audit-2026-09-03/site-geometry.md. Two of its findings cost lots. Every lot is now screened against **all three** digitised stream reaches rather than only the on-site one (SG-04), which put the southwest slot at u 1,230–1,280 inside a buffer and turned it into an unlotted open-space tract; and the stormwater basins, which had been sized on an assumed 4.4 disturbed acres against a real 7.5 (SG-01), had to be enlarged into ground that had held a lot. **The yield is 41, not 43.** Every per-unit and per-acre input in these memos moves with it, and so does everything computed from the basin sizes and the impervious area:

Quantity	2026-08-29 / first 09-03 issue (43 lots)	This issue (41 lots — data/layout.json)	Memo
Dwelling units	43	41 (14 southwest, 27 northeast)	all
Phase split	24 + 19 at u = 1,000 ft	24 + 17 at the drawn phase line u = 980 ft	B
PM peak-hour trips at 0.30/DU	12.9	12.3	A
Daily trips (mid rate)	185	177	A
Average daily sewer flow, DWR table	11,170 gpd	10,670 gpd	B
Average-day water demand	10,750 gpd	10,250 gpd	C
Impervious area	188,483 SF / 45.2 %	184,148 SF / 44.1 %	D
Disturbed area	327,280 SF / 7.513 ac	328,484 SF / 7.541 ac	D
Detention required / provided	75,133 cf / 44,218 cf (59 %)	75,410 cf / 77,863 cf (103.3 %)	D
Water-quality volume WQv	19,049 cf	18,659 cf	D
HOPA units needing a 55+ resident	35 of 43	33 of 41	F
Voluntary NFPA 13D offer	≈ \$129,000	≈ \$123,000	E
Off-site sewer, per Phase-2 lot	\$8,800–\$14,400 (19 lots)	\$9,900–\$16,000 (17 lots)	B

Three of these cut against the application and are stated plainly rather than buried. (i) The **off-site sewer cost per Phase-2 lot rises by about 12 %**, because the same ≈ 478-ft extension is now spread over 17 lots instead of 19. (ii) **41 lots is one above the**

bottom of the 40–47 design range of FACTS.md §4, so there is very little room left for a further lot loss at the boundary survey or the stream delineation before the program has to be re-thought. (iii) The corrected entrance geometry leaves **2.0 ft of the southwest curb return inside the 20-ft perimeter buffer band** (about 0.1 SF of pavement, at the right-of-way corner only, leaving a residual undisturbed width of 18'-0" there), so a buffer reduction is now requested in the letter of intent rather than disclaimed — Memo A.4 explains why the entrance cannot simply be shifted northeast. Working the other way, the enlarged basins now **over-provide** detention instead of falling 41 % short, and the impervious percentage falls slightly.

Program basis (data/layout.json, data/plans.json; **the site plan sheet governs over any number repeated here**): **N = 41** one-story detached cottage homes on fee-simple 50'-0" × 100'-0" lots (5,000 SF each — 205,000 SF of lots; acceptable design range 40–47, cap 47 by voluntary condition 2), 55+ (HOPA) as a voluntary condition, one private lane (22-ft pavement in a 35.9–46.1-ft tract, widening to 60 ft through the amenity block, **1,754 ft** long measured along the travelled way from the Arcado Rd right-of-way, dead-end with three hammerhead turnarounds at stations 594.5 / 1,164.5 / 1,744.5 ft), one driveway to Arcado Rd, a ≈ 2,400 SF clubhouse, two dry detention / water-quality ponds, common open space **141,089 SF (3.24 ac = 33.8 % of the GIS area, 34.3 % of 9.44 ac)**, of which 13,960 SF is built or paved, impervious **184,148 SF (44.1 % of the GIS area)**. Gross density **4.34 du/ac** on 9.44 ac (4.28 on the 9.58-ac GIS area). Two phases at the u = 980-ft phase line (**24 + 17 lots**; voluntary condition 15 describes the same line as "u ≈ 1,000 ft"). Site-local coordinates (u along the strip from Arcado Rd, v across; feet) are those of FACTS.md §1.

Editions cited: ITE *Trip Generation Manual*, 11th ed. (2021); Gwinnett DOT *Traffic Impact Study Guidelines* (2023); Gwinnett DOT *Criteria and Guidelines for Left Turn Lanes* (2007-06-20); Gwinnett County Unified Development Ordinance §900-40.4 Table 900.2 and §900-40.5 Table 900.3 (Ord. No. UDOA2014-00001, 7-22-14); Gwinnett DWR *Water Main and Sanitary Sewer Design and Construction Standards and Specifications* (April 2016); Gwinnett DWR *Wastewater Flow Estimation Guidelines* (Rev. 10/2022); Gwinnett DWR *Sewer Capacity Certification Request* (Rev. 07/2023); Gwinnett DWR *Developer Pump Station Standards* (January 2014, posted by Planning & Development as WSR-24); **Gwinnett DWR Standard Policy for Private Developments: Condominiums, Townhomes and Subdivisions (rev. September 2018)**; Gwinnett County Stormwater Management Manual (GCSWMM) / Georgia Stormwater Management Manual (GSMM 2016); **IFC 2024 as adopted and modified by Ga. Comp. R. & Regs. 120-3-3-.04, "State Minimum Fire Safety Standards with Modifications" (Safety Fire Commissioner; O.C.G.A. Title 25 Ch. 2), Appendices B and D**; NFPA 13D; Lilburn Zoning Ordinance 2023-603; Lilburn Development Regulations Appendix B Article 5 §5.9; Lilburn Code Ch. 109 Art. V and Art. VII; O.C.G.A. §§12-7-6, 25-2-12(e)(4).

A caution on the fire-code citation. The copy of Ga. Comp. R. & Regs. 120-3-3 posted at oci.georgia.gov is stale — it still recites the 2018 edition of the fire code. The Secretary of State's compiled version at rules.sos.ga.gov reflects the 2024 edition. Cite the SOS compilation and confirm the effective text with the Gwinnett County Fire Marshal [VERIFY].

Memo A — Traffic

Question. How many trips does the cottage program add to Arcado Rd, **what study does Gwinnett DOT require**, is a left-turn lane warranted, and does the access geometry meet the county's spacing and sight-distance standards?

A.1 The threshold question is settled: a study is required

Gwinnett DOT, *Traffic Impact Study Guidelines* (2023), on applicability:

"A TIS must be submitted concurrently with all zoning applications and land disturbance development."

and, in the exemptions:

"Exemptions — A TIS shall not be required under the following conditions: • Single-family detached and townhome residential development consisting of no more than seven lots or units ..."

At **41 lots** this project is not exempt ($41 > 7$). The 2026-08-29 issue of this memo already reached that conclusion; FACTS.md §3's "below 21 peak-hour trips → no formal TIS expected" and every document that repeated it are **wrong** and are being conformed. The trip count does not decide *whether* a study is required — it decides *which level*.

The guidelines are written for development "within unincorporated Gwinnett County" under Gwinnett UDO §270-20.8. The City of Lilburn is not bound by them for a city zoning case, but (a) Gwinnett DOT owns Arcado Rd and will apply them at the driveway-permit stage regardless, and (b) staff and neighbours will read them as the standard. **Ask the Planning Director at the pre-application conference whether the City adopts them for a city case and at what level** [VERIFY]. The applicant is preparing a Level-1 study either way, and offers it as voluntary condition 17.

A.2 Inputs

Input	Value	Source
Units	41 (cap 47 by voluntary condition 2)	data/layout.json metrics.lots
ITE LUC 251 rate — UNRESOLVED	AM 0.20–0.34, PM 0.26–0.39, daily 3.7–4.31 trips/DU	see A.3
LUC 210 <i>Single-Family Detached Housing</i> , 11th ed.	daily 9.43, AM 0.70, PM 0.94	Palm Beach County Engineering trip-rate sheet
LUC 220 <i>Multifamily Low-Rise</i> , 11th ed.	daily 6.74, AM 0.40, PM 0.51	same
LUC 822 <i>Strip Retail Plaza (< 40 ksf)</i> , 11th ed.	daily 54.45, PM 6.59 per 1,000 SF	same
As-zoned R-1 yield	4 du/ac × 9.44 ac = 37.8 → 37 lots	Lilburn Zoning Ordinance 2023-603 Table 4.1 (R-1 max 4 du/ac)
2025 MU program (RZ-2025-01)	104 condominiums + 24,400 SF commercial	status/rz-2025-01/4814_development-summary-report.pdf
Arcado Rd	county minor collector, 2 lanes, posted 35 mph	Gwinnett GIS; FACTS §2b
Arcado Rd volume	no eligible count exists — the 2017 GDOT count (station 135-6689, 14,800 AADT) is nine years old	see A.6
TIS study levels	0–20 peak-hour site trips = Level 1 ; 21–249 = Level 2; 250–499 = Level 3; > 500 or DRI = Level 4	Gwinnett DOT TIS Guidelines (2023), Tables 1 and 2
Left-turn-lane warrant	Table 1 (Residential Developments), 2-lane route, posted 30 to 35 mph , ADT ≥ 6,000 → 75 Lots	Gwinnett DOT <i>Criteria and Guidelines for Left Turn Lanes</i> (2007-06-20), Table 1
Driveway spacing	posted 35 mph → 244 ft , "measured from centerline to centerline of driveways, public roads or side streets"	Gwinnett UDO §900-40.5, Table 900.3
Maximum driveways	"one for each 400 feet of property frontage, or fraction thereof"	Gwinnett UDO §900-40.5(C)
Intersection sight distance	35 mph, 2-lane, SDL = SDR = 390 ft , on the AASHTO Case B1 basis (the distance travelled at the arterial speed during 7.5 seconds)	Gwinnett UDO §900-40.4, Table 900.2
Provided geometry	entrance centred at v = –190; 251 ft from the Arcadia Pl centreline measured along the Arcado Rd centreline (251.2 ft along the centreline, 250.7 ft chord); frontage 237.4 ft chord / 237.6 ft along the R/W; one driveway	data/layout.json lane.entrance, metrics

A.3 The LUC 251 rate is UNRESOLVED — and it does not matter to the conclusion

The ITE *Trip Generation Manual*, 11th ed. is a paywalled publication; no primary source was reachable on 2026-09-03. Four secondary compilations were retrieved and reconciled:

Compilation	Land-use name used	AM	PM	Daily
Palm Beach County Engineering trip-rate sheet	"Senior Adult Housing- Detached "	0.24	0.30	4.31
Phoenix, Oregon SDC table (labelled "ITE 11th Edition")	"Senior Adult Housing- Detached "	—	0.30	—
City of Corpus Christi "Common Trip Generation Rates – 11th Edition"	"Senior Adult Housing- Detached "	0.34	0.39	—
Falls Church VA <i>Railroad Cottages</i> study (states ITE 9th ed.)	"Senior Adult Housing Det."	0.20	0.30	3.70

None of these is an 11th-edition source. DelDOT's official *Trip Generation 11th Edition Updates* briefing records that the land use was renamed and re-scoped in the 11th edition: "New Land Uses: ... **Senior Adult Housing-Single Family (LUC 251) – Combination of detached and senior adult housing-attached**; Senior Adult Housing-Multifamily (LUC 252)." Every table above still uses the pre-11th name "Senior Adult Housing-**Detached**," and the Corpus Christi sheet's neighbouring rows ("220 Apartment," "230 Residential Condominium/Townhouse") are 9th-edition land-use names. The 2026-08-29 issue of this memo described two of these as "the two published 11th-ed. compilations"; that description was wrong and is withdrawn.

What would settle it: the LUC 251 data plots in the ITE *Trip Generation Manual*, 11th ed. itself, or an ITETripGen web-app query, read by the traffic engineer preparing the Level-1 study. That is a one-page task for anyone with an ITE subscription and it should be done before filing. Until then this memo carries the **range**, and states the rate source as UNRESOLVED wherever a single number is quoted elsewhere in the package (the package's headline figure, **12.3 PM peak-hour trips**, is 41×0.30 and sits in the middle of the range).

A.4 Results

Trips = $N \times \text{rate}$.

Scenario	Units	Daily	AM peak	PM peak
Proposed, low rate pair (0.20 AM / 0.30 PM / 3.70 daily)	41	152	8.2	12.3
Proposed, mid (0.24 / 0.30 / 4.31)	41	177	9.8	12.3
Proposed, high (0.34 / 0.39)	41	—	13.9	16.0
Proposed at the 47-unit cap, high rate pair	47	203	16.0	18.3
As-zoned R-1 by right, LUC 210	37	349	25.9	34.8
2025 MU program, raw (no internal capture or pass-by)	104 DU + 24.4 ksf	≈ 2,030	—	≈ 214 ($104 \times 0.51 = 53$, plus $24.4 \times 6.59 = 161$)

- **Study level: Level 1, at every rate pair and at the 47-unit cap.** The highest figure anywhere in the table for the proposed program is **18.3** PM peak-hour trips, inside the 0–20 band. The conclusion does not turn on the unresolved rate — that is the point to make to staff.
- Against the as-zoned R-1 subdivision the cottage program is about **–49 %** daily (177 against 349) and **–65 %** in the PM peak (12.3 against 34.8); against the 2025 mixed-use filing it is about **–94 %** in the PM peak (12.3 against ≈ 214). The two-lot loss improves each of these comparisons slightly; none of them turns on it.
- **Left-turn lane: not warranted.** 41 lots (and 47 at the cap) are below the **75-lot** threshold for a 2-lane route posted at 30–35 mph carrying ADT ≥ 6,000. A left-turn lane would also be physically difficult: the Arcado Rd centreline is only about 28–30 ft from the front property line.
- **Driveway spacing: appears consistent.** 251 ft provided against 244 ft required (UDO Table 900.3) — a margin of about 7 ft, which is why the entrance cannot be shifted northeast to cure the buffer clip at the southwest curb return. That clip is now 2.0 ft measured along the right-of-way — a residual buffer width of 18'-0" instead of 20'-0", over about 0.1 SF of pavement — after the southwest curb return was reduced from 25'-0" to 15'-0"; it is the buffer reduction requested in the Letter of Intent §5 and carried in voluntary condition 4.
- **Maximum driveways: one.** At ≈ 237 ft of frontage, §900-40.5(C) permits one driveway. The single-entrance plan is not a concession, it is the county standard.
- **Sight distance: 390 ft each way**, to be certified by a Georgia PE on the curved frontage; clearing within the 10-ft landscape strip and the right-of-way may be required [VERIFY by field survey].

A.5 Scope of the Level-1 study (the deliverable)

Table 1 of the guidelines marks these elements for a Level 1 study, and this memo already contains the substance of every one of them:

1. Location description and site access description.
2. Existing and proposed land use; zoning request.
3. Trip-generation estimate (A.3–A.4), with the ITE land-use code, edition and rate stated.
4. Access-management review against UDO §900-40.
5. **Adjacent access spacing, upstream and downstream** — Arcadia Pl at 251 ft; the next driveways on both sides of Arcado Rd to be measured centreline-to-centreline in the field.
6. **Intersection sight distance** — 390 ft, PE certificate.
7. Connectivity and circulation review — one entrance, 1,754-ft private lane, three hammerhead turnarounds, no through connection possible.
8. Functional classification and posted speed — minor collector, 35 mph.
9. Future identified projects on the corridor (Killian Hill Rd SPLOST project F-1262 status [VERIFY]).

Operations analysis, turn-lane warrant analysis and signal-warrant analysis are **Level 2** elements and are not required here (Table 2).

A.6 The traffic count must be new

Gwinnett DOT, *Criteria and Guidelines for Left Turn Lanes*:

*"GCDOT traffic counts shall be used **provided they are less than two years old** ... If GCDOT traffic counts are not available, or they are greater than two years old, Georgia DOT (GDOT) counts may be used **provided they are also less than two years old**. If counts are not available from either source, and/or both sources are greater than two years old, **the developer shall be responsible for obtaining new, 24-hour counts, from an approved contractor, as directed by GCDOT.**"*

The 2017 GDOT count (station 135-6689, 14,800 AADT, 2.2 % trucks) is nine years old and does not satisfy the rule. It is retained in this package only as historical context and must not be used as the study's base volume. **Budget a 24-hour tube count on Arcado Rd at the site frontage, by a GCDOT-approved contractor** — a small cost, and one that also refreshes the ADT $\geq 6,000$ input to the left-turn-lane test. Order it before the study is written.

A.7 Next steps

1. Traffic engineer: read LUC 251 (and 210 for the by-right comparison) from the ITE 11th-ed. manual; close A.3.
2. Commission the 24-hour count; obtain the sight-distance certificate.
3. Gwinnett DOT pre-submittal contact — the guidelines allow scope to be adjusted or waived at a pre-submittal meeting.
4. Ask Lilburn Planning, in writing, what study the **City** wants for a city zoning case.

Memo B — Sanitary Sewer

Question. Which lots drain by gravity to the existing "Arcado Road Townhomes Sewer Outfall" main on the property, why the rear pocket cannot, **what routes Gwinnett DWR's policies actually permit**, and what DWR needs before the zoning decision.

B.1 Inputs

Input	Value	Source
Existing 8-in gravity main on site	along the SW edge inside parcel 033 ($v \approx -224$ to -228), $u \approx -42 \rightarrow 272$ (≈ 314 ft on the property), then SW off site	FACTS §2; layout . json sewer.existing_on_site (Gwinnett DWR GIS)
Inverts (GIS attributes, unsurveyed)	943.79 / 940.47 (Arcado) $\rightarrow$ 932.69 / 930.84 ($u -42 \rightarrow 20$) $\rightarrow$ 930.64 / 927.13 at the MH near (u 272, v -224) $\rightarrow$ 925.17 downstream	FACTS §2
Lane ground profile (USGS 3DEP, approximate)	u 279: 946.4 (hump) · u 779: 936.3 (sag) · u 979: 940.4 · u 1,179: 953.5 (ridge) · u 1,479: 931.5 (rear low) · u 1,699: 947.4	layout . json lane.ground_profile (10-ft stations)
Legends at Parkview 8-in main	MH invert 919.58 near (u 1,349, $v -405$); 921.03 $\rightarrow$ 919.58 along u 1,160–1,349	FACTS §2
Phase-2 gravity route	rear low point (u 1,479) $\rightarrow$ lane u 1,300 $\rightarrow$ SW line $\rightarrow$ Legends MH: $\approx$ 478 ft total (179 + 121 + 178), of which the last $\approx$ 178 ft is off-site	layout . json sewer.phase2_alternative_extension
Design assumptions	8-in PVC/DIP; minimum slope 0.40 % ; Manning $n = 0.013$; 5-ft cover to invert at the upstream end (one-story slab houses, ~ 4 -ft laterals)	Gwinnett DWR Standards (2016) §2.3 [VERIFY Gwinnett's tabulated minimum slope for 8-in]
Flow basis	Single-family home 250 gpd/unit ; clubhouse 175 gpd per 1,000 SF; peaking factor 4.0 ("the peaking factor has been changed from 2.5 to 4.0 unless justification can be provided")	Gwinnett DWR <i>Wastewater Flow Estimation Guidelines</i> (Rev. 10/2022)
Private-facility policy	"Private pump stations, private force mains and private gravity sewers are only allowed for commercial properties under single ownership within a development "	Gwinnett DWR <i>Standard Policy for Private Developments: Condominiums, Townhomes and Subdivisions</i> (rev. 9/2018)
County pump-station policy	a station is permitted only where gravity is " more than 5,000 feet down gradient "	Gwinnett DWR <i>Developer Pump Station Standards</i> (Jan 2014, WSR-24) §1.3.1(A)
Grinder pumps / low-pressure sewer	no Gwinnett standard exists — zero occurrences of "grinder," and no low-pressure sewer system provisions, in the <i>Water and Sewer Standards</i> (April 2016, 117 pp.) or in WSR-24	full-text search, 2026-09-03
Comprehensive-plan policy	"Do not further extend sewer in this area" (Suburban-Low, 2026 CPA draft of 6/16/2026, p. 73) — transmitted to DCA 2026-07-13, not adopted as of 2026-09-03	FACTS §2b; audit external-facts.md §3.3

B.2 Formulas

Available slope $S = (\text{invert}_{\text{upstream}} - \text{invert}_{\text{downstream}}) / L$. Full-flow capacity $Q = (1.49/n) \cdot A \cdot R^{(2/3)} \cdot S^{(1/2)}$. AADF = $\Sigma(\text{units} \times \text{unit flow})$; peak = $4.0 \times \text{AADF}$; gpm = gpd $\div$ 1,440.

B.3 Phase 1 (24 lots, $u \leq 980$) — gravity, and no extension of the public sewer

- Route: the lane from the $u = 980$ -ft phase line back to u 272 (708 ft), then 106 ft across the amenity tract to the existing manhole at (271.7, -224): **$L \approx 814$ ft**.
- Upstream: invert at u 980 = $940.4 - 5.0 = \mathbf{935.4}$. From there to the lane sag at u 779 (invert $936.3 - 5.0 = \mathbf{931.3}$) the pipe falls 4.1 ft in 200 ft = **2.05 %**.

- **Controlling reach, sag → manhole:** $L = (779 - 272) + 106 = 613$ ft; $S = (931.3 - 927.13) \div 613 = 4.17 \div 613 = 0.68$ %, against the 0.40 % minimum. **Gravity works.** The two lots the plan lost do not touch this: the phase line and the controlling reach are fixed by ground elevation, not by yield.
- The buildability catch is the **lane hump at $u \approx 279$ (ground 946.4)**: a pipe leaving the sag at the 0.40 % minimum is at $931.3 - 0.004 \times 500 = 929.3$ there, about **17 ft to invert** — a deep-cut item (trench boxes, possible rock in the ARE map unit). Alternative alignment: run the front trunk in the **SW-edge easement corridor** ($v \approx -200$ to -225), where the 3DEP ground is about 939 at u 250 and the cut is roughly **10 ft**. The PE chooses; either way the answer to the comprehensive-plan sewer policy for Phase 1 is that **nothing is extended**. The main is already inside the property line.
- Capacity: an 8-in pipe at 0.40 % carries about 0.77 cfs = **344 gpm** full. The whole project's peak is 29.6 gpm (B.6), under 10 % of pipe capacity. The constraint is the downstream county system, not the on-site pipe — which is exactly what the capacity certification tests.

B.4 Phase 2 (17 lots, the rear pocket) — why gravity to the on-site main fails

- Rear lane low point u 1,479 at ground **931.5**; an invert there with 5 ft of cover is **926.5** — already **0.6 ft below** the 927.13 manhole invert before any slope is applied.
- At the 0.40 % minimum over $L = (1,479 - 272) + 106 = 1,313$ ft, the pipe would arrive at $926.5 - 5.25 = 921.2$, a **5.9-ft deficit** against 927.13; and to get there it would pass beneath the $u \approx 1,179$ ridge (953.5) at roughly **28 ft** of cut. **Not buildable by gravity to the on-site main.** The ridge is the reason the project is phased at all.

B.5 The two rear options, decided

Option 1 — 178-ft off-site gravity tie to the Legends at Parkview main: the only viable route

Item	Value	Basis
Total length	≈ 478 ft (179 ft along the lane + 121 ft across to the SW line + 178 ft off site), of which ≈ 178 ft off-site	layout . json sewer.ph ase2_alternative_extensi on
Fall	$926.5 \rightarrow 919.58 = 6.92$ ft	5-ft cover at the rear low point; Legends MH invert
Slope	$6.92 \div 478 = 1.45$ %	comfortable against the 0.40 % minimum
Depth — the weak point of the drawn alignment	the route as drawn runs back up the lane from the rear low point (931.5) to u 1,300, where existing ground is 943.8–944.7: the pipe is about 20 ft deep there, and about 10.5 ft deep where it reaches the SW property line at ($1,300, -235$), ground 932.7	layout . json lane.ground_profile; dat a/topo-samples . json
Depth, alternative alignment	leave the lane at the rear low point and run southwest through the low pocket (3DEP ground 921.6–929.5 at u 1,350–1,450, v -140 to -235 — the Pond 2 / creek-woods side); cover then stays 5 to 10 ft the whole way. The PE should price both. This memo carries the drawn route's length and slope because the drawing governs, and flags the 20-ft cut rather than hiding it	3DEP profile
Manholes	3 (rear low point; at the SW property line; connection at the Legends MH)	DWR Standards (2016) spacing
Capacity used	Phase-2 peak 11.8 gpm into an 8-in main	B.6

What it needs.

1. **A recorded permanent sanitary sewer easement** across one Village Green Court lot, or along the stream corridor, in the Legends at Parkview subdivision. This is a private negotiation with a private owner (and possibly the Legends HOA); the City cannot grant it and the applicant cannot promise it. Pull the current registered agent and officers of Legends at Parkview Homeowners Association, Inc. from ecorp.sos.ga.gov before approaching anyone [VERIFY — the "Liberty Community Management" attribution in docs/06 is unconfirmed].
2. **A buffer crossing that the stream ordinance permits.** The route crosses the stream corridor. The Site Development Plan Review Checklist allows utility encroachment into a buffer up to 50 ft of width measured perpendicular; the MNGWPD model ordinance that Lilburn adopted keeps sewer easements at least 25 ft from the top of bank where practicable. Design the crossing **perpendicular, ≤ 50 ft wide**, restored under the 2019 Lilburn Riparian Buffer Restoration Guidelines [VERIFY the Lilburn Code Ch. 109 Art. VII section numbers by hand from Municode — automated retrieval returns HTTP 403].
3. **A Gwinnett DWR sewer capacity certification** for the Legends basin (B.7).
4. **A staff reading of the comprehensive-plan sewer policy** (B.8).

Order-of-magnitude cost — planning estimate only, unit rates are assumptions to be replaced by contractor quotes [VERIFY]:

Item	Quantity	Assumed unit rate	Amount
8-in PVC gravity sewer, 5–20 ft cut, in a wooded easement	478 LF	\$150–\$225 / LF	\$71,700–\$107,600
Precast manholes, 4-ft dia., with drop/doghouse connection at the Legends MH	3 ea	\$7,000–\$10,000 ea	\$21,000–\$30,000
Rock excavation allowance (ARE map unit, gneiss at 22–40 in)	200 CY	\$150 / CY	\$30,000
Stream/buffer crossing, erosion control and restoration	lump	—	\$10,000–\$20,000
Off-site easement acquisition from one private owner	1	negotiated; no market source	\$15,000–\$50,000
Survey, PE design, DWR certification, inspection, as-builts	lump	—	\$20,000–\$35,000
Order-of-magnitude total			≈ \$168,000–\$273,000

That is roughly **\$9,900 to \$16,000 per Phase-2 lot** (17 lots) — up about 12 % per lot from the \$8,800–\$14,400 the 2026-08-29 issue carried, because the same extension is now spread over 17 Phase-2 lots instead of 19. It is material, it is worse than it was, and it is the honest number to carry into the pro forma. The three items that could move it are rock, the easement price and the 20-ft cut at the lane crossing, and none of them can be closed from a desk.

Option 2 — in-tract pump station, force main or grinder pumps: FORECLOSED BY POLICY

The 2026-08-29 issue of this memo, following FACTS.md §2b, presented this as the *primary* Phase-2 route. It is not available. Three independent bars:

(a) A private (HOA-owned) station or force main. Gwinnett DWR, *Standard Policy for Private Developments: Condominiums, Townhomes and Subdivisions* (rev. September 2018) — the policy variant whose cover carries a "Subdivisions" check-box, and therefore the one that governs a fee-simple lot subdivision:

"PRIVATE FACILITIES: Private pump stations, private force mains and private gravity sewers are only allowed for commercial properties under single ownership within a development. Gwinnett County is not responsible for operation and maintenance of these facilities."

and, immediately above it:

"OWNERSHIP OF MAINS AND APPURTENANCES: Absolute title and ownership of all water and sanitary sewer mains within the private development shall be conveyed to Gwinnett County WSA at the time of acceptance by the County."

A 41-lot fee-simple subdivision is not "commercial property under single ownership." (The 2026-08-29 memo's judgment cited the **Apartments** variant of the same policy; the Subdivisions variant is the applicable one and it carries the identical clause.)

(b) A county-owned station. Gwinnett DWR, *Developer Pump Station Standards* (January 2014, WSR-24) §1.3.1(A), in full:

"Approval for the installation of a new pump station will only be granted by the GCDWR's AMIS Division Director through consultation with the GCDP&D. Generally, pump stations will only be permitted when gravity sewer is unavailable to the property. Unavailable shall generally be interpreted to mean more than 5,000 feet down gradient, but this distance can be increased or decreased by GCDWR Division Directors based upon actual field conditions and the size of the project involved."

Gravity is **178 ft** away — two orders of magnitude inside the 5,000-ft test. The final clause is the only opening, it cuts both ways (the Director may decrease the distance as well as increase it), and it is exercised case by case. The January 2014 document is the current edition: the copy DWR posts as "current" is byte-identical (MD5 0fe478d40303252f12497f6996b7ee49).

(c) Grinder pumps or a low-pressure sewer system. There is **no Gwinnett design standard for either**. A full-text search of the *Water and Sewer Standards* (April 2016) returns zero occurrences of "grinder" and no low-pressure sewer system provisions (the only "low pressure" hits are the §5.7 low-pressure **air testing** procedure); WSR-24 returns zero occurrences of "grinder." A system with no county standard could only be private — which clause (a) forecloses.

Conclusion. Option 2 is withdrawn from the application. The Letter of Intent §7, the Development Summary §5.2, voluntary condition 16 and the site-plan general notes have been conformed. **Two artefacts of the withdrawn option are still in the drawing data and must come out before filing:** data/layout.json still carries sewer.phase2_primary describing "in-tract lift station / grinder pumps ... force main to the Phase 1 gravity main" as the primary Phase-2 route, and a lift_station_symbol polygon at (u 1,481–1,489, v –146.6 to –138.6). The same file's own screening table already states the opposite — that no pumping solution is available and that the 178-ft gravity tie is the Phase-2 route — so the file contradicts itself. Delete both, and delete the LS symbol from the site plan [VERIFY on the next regeneration].

Option 3 — Phase 1 alone (24 lots)

If the easement or the capacity certification cannot be obtained, Phase 1 stands by itself: 24 lots, the amenity block, Pond 1 and the entrance, all on gravity to the existing on-site main, at about 2.5 du/ac. Voluntary conditions 15 and 16 make that the

enforceable default, which is why the applicant can ask for R-2 on the whole tract without asking the City to take the Phase-2 risk.

B.6 Estimated flows

Basis	AADF	Peak (× 4.0)
DWR table (governs the certification): 41 × 250 gpd + 2,400 SF clubhouse × 175 gpd per 1,000 SF (420 gpd)	10,670 gpd = 7.41 gpm	42,680 gpd = 29.6 gpm
Phase 1: 24 lots + clubhouse (24 × 250 + 420)	6,420 gpd (4.5 gpm)	25,680 gpd (17.8 gpm)
Phase 2: 17 lots (17 × 250)	4,250 gpd (2.95 gpm)	17,000 gpd (11.8 gpm)
Cross-check, population basis: 41 × 2.2 persons × 70 gpd	6,314 gpd (4.4 gpm)	25,256 gpd (17.5 gpm)

The DWR table figures govern; the population-basis row is shown only to bracket them. The 300 gpd/unit figure used in earlier drafts of docs/02 and docs/10 is retired. For comparison, the 2025 mixed-use filing (104 condominiums plus 24,400 SF of commercial) would have requested roughly two to three times this average daily flow [VERIFY against 4819_sewer-capacity-request.pdf, an image scan].

B.7 The DWR capacity certification — and the exact question to ask

The request goes on the *Sewer Capacity Certification Request*, **Rev. 07/2023**, using its **"Pre-Rezoning"** request type. The form must be signed and sealed by a design professional (this is not owner-preparable), is submitted to **DWRCapacityCertification@GwinnettCounty.com**, and asks for the target zoning-submittal and zoning-approval dates — DWR expects to be engaged before the zoning decision. Turnaround: **"PLEASE ALLOW A MINIMUM OF 10 BUSINESS DAYS FOR PROCESSING,"** and **"AN ADDITIONAL 20 BUSINESS DAYS"** if analysis or flow monitoring is required — so **2 to 6 weeks** after a complete package. No fee is stated on the form or in the flow-estimation guidance [VERIFY with DWR].

Attach: a GIS map highlighting the site, surrounding utilities and the **tie-in manhole facility IDs** — the u ≈ 272 manhole (Arcado Road Townhomes outfall, invert 927.13) for Phase 1 and the Legends at Parkview manhole (invert 919.58) for Phase 2; the flow calculations of B.6; and the phasing.

Submit this question, in these words, with the form:

*"The project is a 41-lot fee-simple single-family detached subdivision in the City of Lilburn (PINs R6123 033, 015, 014, 162), phased. Phase 1 (24 lots, AADF 6,420 gpd) connects by gravity to the existing Arcado Road Townhomes outfall manhole on the property, invert 927.13, and requires no extension. Phase 2 (17 lots, AADF 4,250 gpd, peak 11.8 gpm) cannot reach that manhole by gravity because of an intervening ridge. We understand from the Standard Policy for Private Developments: Condominiums, Townhomes and Subdivisions (rev. 9/2018) that private pump stations, private force mains and private gravity sewers are allowed only for commercial properties under single ownership, and from Developer Pump Station Standards §1.3.1(A) that a county pump station is permitted only where gravity is unavailable — generally more than 5,000 feet down gradient — whereas gravity here is 178 feet away. We therefore propose the only route we believe the policies permit: an 8-inch gravity extension of approximately 478 feet, of which approximately 178 feet is off site, from the rear of the property southwest to the existing Legends at Parkview 8-inch main at invert 919.58, at approximately 1.45 percent, within an easement to be recorded. **Please confirm (1) that DWR concurs that no pumping solution — county-owned station, HOA-owned station, force main, grinder pump or low-pressure sewer — is available to this project under those policies; (2) whether downstream capacity exists in the Legends at Parkview basin at the stated flows for Phase 2 and in the Arcado Road Townhomes outfall for Phase 1; (3) the facility IDs and any conditions DWR would place on the off-site connection, including easement width and manhole type at the tie-in; and (4) whether DWR requires flow monitoring before certification. Target zoning submittal 2026-09-25; target zoning approval 2026-11-09.**"*

Then carry the certification number into the Lilburn application ("Water, sewer and fire review approved by Gwinnett County" on the Site Development Plan Review Checklist) and revise it at the land-development-permit stage if the unit count changes.

B.8 The comprehensive-plan sewer policy — how to ask

The Suburban-Low policy in the 2026 amendment reads "Do not further extend sewer in this area." Two facts should be put to staff together with the question, in writing:

- **Phase 1 extends nothing.** It connects to a main that already crosses the property.
- **Phase 2 is a 178-ft connection between two existing county mains in the same drainage basin.** It adds no service area beyond this property and opens no unsewered land.

The question for the Planning Director is therefore narrow: *does the City read "do not further extend sewer in this area" as reaching a 178-foot in-basin connection between two existing mains, on a property that is already crossed by a public sewer?* Ask it before filing, and ask the second question with it: **which Future Development class do these four parcels fall in** — Suburban-Low (p. 73, "do not further extend sewer") or Suburban-Medium (p. 71, "coordinate the expansion of sewer infrastructure to fill gaps")? The

Future Development Map (p. 52) is a raster without parcel labels and the answer cannot be read off the document. Both classes rate Single-Family Cottages and Single-Family "Appropriate," so the program survives either way; only the sewer policy differs.

B.9 First spend, before anything else

A **sewer-invert survey** of the $u \approx 272$ manhole and the Legends manhole. Every number in this memo rests on GIS attributes that were never surveyed, and a 1-ft error moves the Phase-1 slope from 0.68 % to about 0.5 % and changes the Phase-2 arithmetic. Add rock probes along the trunk alignment (ARE map unit, gneiss at 22–40 in).

Appears consistent with Gwinnett DWR Standards (2016) for Phase 1. Phase 2 is feasible **only** as the Option 1 gravity tie, and only with the easement and the capacity certification.

Memo C — Water

Question. Domestic demand, fire flow, hydrant layout and the dead-end-main issue.

C.1 Inputs

Input	Value	Source
Existing mains	DIP/ACP in Arcado Rd, 47–107 ft from the front line; nearest hydrant ≈ 61 ft from the front corner ($u -45$, $v +41$)	FACTS §2 (Gwinnett DWR GIS); sizes VERIFY
Unit demand	250 gpd/DU (the DWR wastewater unit flow used as the domestic proxy; HOA-irrigated yards add seasonal demand)	DWR Flow Guidelines (Rev. 10/2022) [VERIFY with DWR's water-model request]
Peaking	max day $2.0 \times$ average; peak hour $4.0 \times$ average	AWWA M22 (2014) practice [VERIFY against DWR model criteria]
Fire flow, one- and two-family dwellings $\leq 3,600$ SF	1,000 gpm for 1 hour; 50 % reduction, not below 500 gpm, where sprinklered per §903.3.1.3 (NFPA 13D)	IFC 2024 (Ga. Comp. R. & Regs. 120-3-3-.04) Appendix B §B105.1 and Table B105.1(1) [VERIFY that Gwinnett's fire-prevention ordinance adopts Appendix B]
Hydrant spacing, residential	"generally located every 400 feet ... spaced from a minimum of 350 feet to an absolute maximum of 450 feet ... on property lines where possible"; "place a fire hydrant near the end of each main"; in-line valves every 1,000 ft or every third hydrant	Gwinnett DWR Standards (2016) §2.2.15(b), (d); §2.2.16(b)
Main length	$\approx$ 1,754 ft single dead-end run in the lane tract	layout.json metrics.lane_length_ft

C.2 Formulas

$Q_{avg} = N \times 250$ gpd; $Q_{maxday} = 2.0 \times Q_{avg}$; $Q_{peakhr} = 4.0 \times Q_{avg} \div 1,440$. Fire flow per Table B105.1(1) with the §B105.1 sprinkler reduction. Hydrants $\approx L \div 400 + 1$.

C.3 Results

- Domestic: **10,250 gpd average** (41×250), 20,500 gpd max day, $\approx$ **28.5 gpm peak hour** ($4.0 \times 10,250 \div 1,440$) — negligible against a collector-road main. The design case is fire flow.
- Fire flow — stated both ways, because the sprinklers are voluntary.** Without sprinklers the requirement is **1,000 gpm at 20 psi residual for 1 hour**. With the NFPA 13D systems offered under voluntary condition 5 in every dwelling, §B105.1 halves it to **500 gpm**. The largest cottage (Plan B "The Laurel", 38'-0" $\times$ 57'-6" body, **1,492 SF conditioned, 2,182 SF under roof** — data/plans.json) is well under the 3,600-SF fire-flow area either way. The clubhouse ($\approx 2,400$ SF, Type V-B) is a separate Table B105.2 check — likely 1,500 gpm unsprinklered, so sprinkler the clubhouse (NFPA 13) to bring it into the same band [VERIFY with the Fire Marshal].
- Hydrants: $\blacksquare 1,754 \div 400 \blacksquare + 1 \rightarrow$ **6 hydrants** (about u 0, 350, 700, 1,050, 1,400 and one at the terminus hammerhead), on lot lines, plus the existing Arcado Rd hydrant at the entrance; 2 in-line valves. Where a hydrant sits on the fire apparatus access road, Appendix D §D103.1 asks for 26 ft of road width — place hydrants at the **hammerhead legs, pocket-green frontages and guest-parking bays**, where the lane locally widens (Memo E).
- Dead-end main:** there is no second frontage, so the main cannot be looped. (i) Use an **8-in main**, not 6-in, so 500–1,000 gpm arrives at the terminus with ≥ 20 psi residual — a 6-in, 1,754-ft dead end is the marginal case; the hydraulic check is DWR's flow test at the Arcado hydrant [static and residual pressures: VERIFY]. (ii) Water age: an 8-in main of this length holds about 4,580 gallons ($\pi \div 4 \times 0.667^2 \times 1,754 \times 7.48$) against about 10,250 gpd of use (turnover under one day), but DWR may still require an **automatic flushing device** or a terminal blow-off (DWR Standards (2016) Detail W4, "Typical Dead End Street Termination"). (iii) Sample taps, not hydrants, for bacteriological testing.

- Meters: individual 5/8-in meters at each lot in the lane tract; one irrigation meter for the HOA; backflow prevention per DWR's cross-connection policy.

Appears consistent with Gwinnett DWR Standards (2016) §2.2.15 and IFC 2024 (Ga. Comp. R. & Regs. 120-3-3-.04) Appendix B — **pending DWR's flow test and the Fire Marshal's concurrence (VERIFY).**

Memo D — Stormwater

Question. GCSWMM approach, impervious area, water-quality and runoff-reduction volumes, detention volumes and pond targets, and the stream-buffer and floodplain statements.

D.1 Inputs

Input	Value	Source
Regulatory framework	The City reviews stormwater under City Code and the Gwinnett County Stormwater Management Manual (GCSWMM) ; two certified stormwater reports at LDP; BMP maintenance agreement plus as-built volume certification	Lilburn Site Development Plan Review Checklist
Site area A	417,174 SF = 9.577 ac GIS (9.44 ac deeded)	layout . json
Impervious	184,148 SF = 4.23 ac = 44.1 % of the GIS area (44.8 % of 9.44 ac deeded)	layout . json metrics.impervious_sf — summed from the drawn footprints; see D.2
Disturbed area	328,484 SF = 7.541 ac (raster site area 418,228 SF less 73,744 SF of 20-ft perimeter buffer bands and 16,000 SF of net creek-woods tract outside those bands = 89,744 SF preserved)	layout . json stormwater.disturbed_area, 2-ft raster; supersedes the "≈ 4.4 ac" assumption the ponds were originally sized on, the "≈ 7.9 ac" upper estimate of 2026-08-29 and the 7.513 ac of the first 09-03 issue
WQv formula (2025 hydrology basis)	WQv = 1.2 · (0.05 + 0.009 · I) ÷ 12 · A (I in %, A in SF → cf)	FACTS §3; RZ-2025-01 hydrology sheet
Detention formula (2025 hydrology basis)	Detention = 10,000 cf per disturbed acre – required runoff-reduction volume (RRv)	same
RRv basis	RRv = 1.0 in · Rv · A ÷ 12, Rv = 0.05 + 0.009 · I	GSMM 2016 Vol. 2 runoff-reduction standard [VERIFY the 2025 sheet used this definition]
Ponds	two dry extended-detention / water-quality basins on the SW side, 6-ft depth, 3:1 side slopes , top of bank 52 × 180 ft (Pond 1) and 64 × 180 ft (Pond 2), in 20,000-SF tracts	layout . json ponds; Sheet C-3.0 (grading and stormwater concept — not yet drawn) will govern once issued
Soils	HSG B Appling/Pacolet-type clays (eroded phases, poor infiltration); ARE (Ashlar, rock at 22–40 in) under both pond sites ; HYB (Helena, HSG D, water table ≈ 24 in) at the front SW corner	FACTS §2b; docs/11 §3.3
Receiving water	Jackson Creek headwaters (GAR030701030315), 2024 303(d) Not Supporting (bacteria, biota; urban runoff)	GA EPD 2024 Integrated 305(b)/303(d) list
Floodplain	FEMA Zone X [FIRM panel and effective date VERIFY]	Gwinnett GIS

D.2 The impervious and disturbed-area corrections, and the arithmetic

Two errors were carried into the 2026-08-29 issue and both have now been rebuilt from the drawing rather than patched.

Impervious. The 2026-08-29 issue reported **160,167 SF (38.4 %)** and data/layout . json then reported 161,212 SF (38.6 %). Both rested on a flat **2,200 SF per lot** allowance hard-coded in the site-plan generator. The generator now sums the drawn footprints, at 41 lots:

Component	SF
Dwellings, garages, porches, rear porches and patios (41 footprints from data/plans . json)	86,448
Lot driveways and entry walks	31,390
Private lane pavement	34,111
Entry drive pavement	4,865
Entrance curb returns	184
Hammerhead turnaround legs	7,200
Guest and mail-kiosk parking bays	2,070

Component	SF
Sidewalks	14,619
Clubhouse roof, court pads, mail kiosk and amenity walks	6,760
less driveway apron / sidewalk overlap counted twice	-3,500
Total	184,148 SF = 4.23 ac = 44.1 % of 417,174 SF

Per lot that is **Plan A 2,752 SF** (1,967 under roof + 72 patio + 653 driveway + 60 walk, 21 lots) and **Plan B 2,898 SF** (2,182 under roof + 656 driveway + 60 walk, 20 lots); driveway areas vary a little with each lot's position on the curving lane. The first 09-03 issue's 188,483 SF / 45.2 % was the same computation at 43 lots and is superseded.

Disturbed area. The ponds were originally sized on an assumed **4.4 disturbed acres**. The real figure, taken off a 2-ft raster of the site less the preserved areas, is **7.541 ac**: raster site area 418,228 SF, less 73,744 SF of 20-ft perimeter buffer bands, less 16,000 SF of creek-woods tract lying outside those bands (the 50-ft undisturbed stream buffer is wholly inside the areas already deducted, so it adds nothing), = 89,744 SF preserved and **328,484 SF disturbed**. That understatement by a factor of about 1.7 is what hid the detention shortfall, and correcting it is what forced the basins to grow — which in turn cost one of the two lots.

Then, with $I = 44.14\%$ ($184,148 \div 417,174$) and $A = 417,174$ SF:

- $R_v = 0.05 + 0.009 \times 44.14 = \mathbf{0.4473}$; $WQ_r = 1.2 \times 0.4473 = \mathbf{0.5367}$ in.
- $WQ_v = \mathbf{0.5367} \div 12 \times 417,174 = \mathbf{18,659}$ cf (whole-site drainage area; the PE will use actual tributary areas, which exclude the undisturbed creek woods and buffers and will lower this). This supersedes the 16,503 cf of 2026-08-29 and the 19,049 cf of the first 09-03 issue.
- **RRv, if the full 1.0-in runoff-reduction credit were earned = $0.4473 \times 1.0 \div 12 \times 417,174 = 15,549$ cf.** The plan as drawn claims **none of it**: layout . json carries $RR_v = 0$ at concept stage, which is the conservative reading and the one this memo uses.
- **Detention required: $10,000$ cf/ac $\times 7.541$ ac = $75,410$ cf,** with RR_v taken as 0. If the PE earns the full 1.0-in credit the requirement falls to about **59,861 cf**.

D.3 Results, and the basins as now drawn

Item	Value	Inputs
Impervious	44.1 % of the GIS area (against 50–57 % per basin on the 2025 mixed-use hydrology sheet for this same land)	$184,148 \div 417,174$
WQv (treat the 1.2-in runoff)	18,659 cf	$1.2 \times (0.05 + 0.009 \times 44.14) \div 12 \times 417,174$
RRv, if the credit is earned (not claimed on the plan)	15,549 cf	$0.4473 \times 1.0 \div 12 \times 417,174$
Disturbed area	7.541 ac	328,484 SF, D.2
Detention required	75,410 cf	$10,000$ cf/ac $\times 7.541$ ac, $RR_v = 0$
Detention provided	77,863 cf — Pond 1 33,129 cf + Pond 2 44,734 cf	layout . json ponds
Provided $\div$ required	103.3 % — a margin of 2,453 cf	$77,863 - 75,410$

The basins were enlarged, and that is what cost the second lot. At the 4-ft depth previously drawn the two basins held 22,011 + 22,207 = **44,218 cf**, only 59 % of the requirement; the shortfall was hidden because the sizing assumed 4.4 disturbed acres rather than the 7.541 the plan actually disturbs. Both basins are now **6 ft deep with 3:1 side slopes**, with top of bank **52 $\times$ 180 ft** (Pond 1) and **64 $\times$ 180 ft** (Pond 2) in 20,000-SF tracts, and Pond 2's tract had to grow from 15,000 to 20,000 SF — into ground that had carried a lot. Arithmetic check, prismoidal volume $V = h/6 \cdot (A_{\text{top}} + 4 \cdot A_{\text{mid}} + A_{\text{bottom}})$: for a rectangular 52 $\times$ 180 ft basin, $A_{\text{top}} = 9,360$ SF, $A_{\text{mid}} = 34 \times 162 = 5,508$ SF, $A_{\text{bottom}} = 16 \times 144 = 2,304$ SF $\rightarrow V = 1.0 \times (9,360 + 22,032 + 2,304) = \mathbf{33,696}$ cf; for 64 $\times$ 180 ft, **45,360 cf**. The basins as drawn are trapezoidal rather than rectangular (they follow the tract and the property line), which is why layout . json reports the slightly smaller **33,129** and **44,734 cf**.

What that margin is worth. 3.3 % is not a design margin; it is a screening result on a screening rate. Two things can eat it and one can restore it: rock (the ARE map unit puts gneiss at 22–40 in **under both basins** — a shallow rock line means undercut, blasting, or shallower and longer basins), a stream delineation that moves Pond 2, and, in the other direction, any runoff-reduction credit the PE earns, which is worth up to about 15,500 cf. **Rock probes at both sites before any of this is trusted.** If rock is shallow the fallbacks are a third facility at the terminus green or shallower, longer basins — and either would be preferable to another lot loss.

- **Runoff reduction on clay.** Infiltration will test poorly, so meet RR_v with **bioretention with underdrains** in the pocket greens and lane-side swales, roof-drain disconnection to HOA-maintained yards, and pervious pavers in the guest bays (Table 4.2

permits pervious paving); credit the preserved creek woods and buffers as undisturbed area.

- **Full GCSWMM/GSMM stack for the PE at LDP:** water quality (WQv), runoff reduction (RRv), channel protection (CPv — 24-hour extended detention of the 1-year, 24-hour storm), overbank flood protection (25-year), extreme flood (100-year), downstream analysis to the 10 % point in the Jackson Creek basin, and an erosion, sedimentation and pollution control plan (GSWCC; NPDES GAR100001 — the site exceeds 1 acre and lies within 200 ft of a stream). The 303(d) listing makes **water-quality treatment and no direct discharge into the buffer** the exhibit the neighbours will read.
- **Stream setbacks.** Every lot, pond, pavement edge and structure is screened against **all three digitized stream reaches** — the on-site headwater and both off-site branches — at 75 ft **plus a 5-ft top-of-bank allowance**, i.e. 80 ft from the mapped channel centreline. That allowance exists because the package's distances are measured to a digitized *centreline* while the ordinance measures from *top of bank*; on a 6–16-ft channel the two differ by 3–8 ft. The 2026-08-29 statement that "no lot or pond lies within 75 ft" was true of the on-site reach only and false for one southwest lot against an off-site branch. The screen has been widened to all three reaches and the statement now holds by construction — but it holds because **that lot was removed**: the southwest slot at u 1,230–1,280 is now an unlotted 5,000-SF stream-setback open-space tract carrying no impervious surface, and it is one of the two lots that took the yield from 43 to 41. As drawn, the nearest lot stands 80.6 ft from a mapped channel centreline, the nearest dwelling 106.2 ft, the nearest driveway 102.6 ft, the nearest pond top of bank 80.2 ft and the lane pavement 80.5 ft. **Field delineation still governs** (Memo G).
- **Floodplain:** FEMA Zone X. Table 4.2's prohibition on floodway and floodplain development is met by absence [VERIFY the FIRM panel and effective date at LDP; Lilburn Ch. 109 §109-52 et seq.].
- **Open space is not a stormwater constraint here, and it is not a code requirement either.** The plan provides **141,089 SF = 3.24 ac = 33.8 % of the GIS area** (34.3 % of 9.44 ac) in named HOA tracts that sum exactly to that figure, of which 13,960 SF is built or paved (clubhouse roof, court pads, mail kiosk, amenity walks and the hammerhead legs inside the greens), leaving **127,129 SF = 30.5 %** of green-only open space; the 41,000 SF of buffer easement on lots is additional and is not counted in either figure. Table 4.1's open-space entry points to Development Regulations §5.9, which has now been retrieved: it is titled "**Recreation Areas**" and is triggered only "in single-family detached subdivisions having a **gross area of 50 acres or more** and a minimum lot size less than one acre" (and in duplex subdivisions of 50 acres or more, and attached/townhouse/apartment developments of ten acres or more), requiring "six percent of the gross land area." At **9.44 acres** of single-family detached, §5.9 does not apply. The 33.8 % common open space this plan provides is therefore entirely voluntary — and it rose, not fell, when the two lots came out. **This closes the VERIFY that FACTS §3 carried.** One caution: §5.9's text still refers to "Gwinnett County," "the Board of Commissioners" and "the Department" because Lilburn adopted the county regulations wholesale — confirm with staff who those actors are in a city case [VERIFY].

Appears consistent with the GCSWMM approach at the rezoning level; **all volumes are placeholders until a PE's hydrology (VERIFY the disturbed area, the tributary areas and the rock line).**

Memo E — Fire / Life Safety

Question. What a single 1,754-ft dead-end lane serving 41 one-story dwellings actually triggers under the Georgia fire code, and what to bring to the Gwinnett County Fire Marshal (the City has no fire department).

E.1 The correction that changes this memo

The 2026-08-29 issue stated that the development "exceeds the 30-unit threshold of Appendix D107.1 for a second access road unless the homes are sprinklered," and treated NFPA 13D sprinklers as the required mitigation. **That is the model IFC text, not the rule in force in Georgia.** Ga. Comp. R. & Regs. 120-3-3-.04, "State Minimum Fire Safety Standards with Modifications," Modifications to Chapter 5, item 7:

"Add a new section, Appendix D 107.1, as follows: Developments of one- or two-family dwellings where the number of dwelling units exceeds 120 shall be provided with two separate and approved fire apparatus access roads. Exceptions:
1. Where there are more than 120 dwelling units on a single public or private fire apparatus access road and all dwelling units are equipped throughout with an approved automatic sprinkler system in accordance with Section 903.3.1.1, 903.3.1.2 or 903.3.1.3, access from two directions shall not be required. ..."

The model IFC 2024 number is 30; **Georgia replaces it with 120.** At **41 dwelling units this development does not trigger the two-access requirement at all**, and no exception needs to be invoked to relieve it. The two lots lost on the corrected site plan move the project further below the threshold, not nearer it. Everything the package previously said about "43 units (> 30) → 13D exception" is withdrawn.

E.2 Inputs

Input	Value	Source
Code	IFC 2024 as adopted and modified by Ga. Comp. R. & Regs. 120-3-3-.04 , Appendices B and D, effective 2026-01-01 [Gwinnett's local adoption of Appendices B and D: VERIFY]	GA DCA <i>Current State Minimum Codes</i> ; SOS rules compilation
Dead-end length	1,754 ft (Arcado Rd right-of-way to the end of pavement, along the travelled way)	layout . json metrics
Units on one access	41 (cap 47 by voluntary condition 2)	same
Lane	22-ft pavement in a 35.9–46.1-ft tract (60 ft through the amenity block), no parking, three hammerhead turnarounds (two 60-ft × 20-ft legs each) at stations 594.5 / 1,164.5 / 1,744.5 ft , so the runs between intersections are 594.5 / 570.0 / 580.0 ft ; entrance radius 25 ft (NE) and 15 ft (SW); design speed 15 mph	layout . json lane, hammerheads, metrics; Sheet C-2.0 governs
Travelled distance to the farthest garage door	≈ 1,701 ft (204-ft entry drive + 1,497 ft of lane from the join at u 149.5 to the last driveway at u ≈ 1,646); end of pavement 1,754 ft	layout . json lots, lane
Buildings	1 story; ridge 18'-6" (Plan A) and 19'-8" (Plan B) above finished grade — 17'-10" and 19'-0" above the finished floor, which is 8 in above grade; clubhouse 1 story, ≈ 2,400 SF	Sheets A-2.1 and A-2.2; docs/09 §5 (data/plans . json still carries a superseded 19'-4" rear-block ridge for Plan B — VERIFY on regeneration)
Sprinklers	NFPA 13D in 100 % of dwellings — voluntary condition 5	data/voluntary-conditions.md

E.3 Code provisions

Provision	Requirement	This project
§503.1.1	fire apparatus access within 150 ft of all exterior walls	every dwelling is within about 100 ft of the lane
§503.2.1 / D102.1	width ≥ 20 ft, vertical clearance 13 ft 6 in, all-weather surface for 75,000 lb	22-ft asphalt lane; no overhead obstructions
Appendix D §D103.4, Table D103.4	0–150 ft: 20 ft wide, no turnaround · 151–500 ft: 20 ft + a 120-ft hammerhead, 60-ft "Y" or 96-ft-diameter cul-de-sac · 501–750 ft: 26 ft + turnaround · Over 750 ft: "Special approval required"	1,754 ft → special approval. The hammerheads match the 120-ft figure; the 22-ft width against the 26 ft the table pairs with long dead ends is the one true relief item
§D103.1	26-ft width where a hydrant is on the access road	hydrants placed at local widenings (Memo C)
§D103.2	grade ≤ 10 % unless approved	steepest existing ground grade on the lane centreline is 10.9 % at u ≈ 1,494 ft, with a second pitch of the same order near u ≈ 1,004; the finished profile must be designed to ≤ 10 % or the grade approved — a PE profile item, not yet designed [VERIFY]
§D103.3	turning radius as specified by the fire code official	25-ft entrance radius on the NE side; hammerhead legs to the D103.1 figure
§D103.6	"NO PARKING — FIRE LANE" signs on 20–26-ft roads	HOA covenant plus signs; 4 spaces per dwelling (164 on lot against the 82 that Table 8.1 requires at 2/DU × 41) and 12 guest and mail-kiosk spaces keep the lane clear
Appendix D §D107.1 (GA)	two access roads where dwelling units exceed 120	41 units — not triggered
§D105 (aerial access, buildings > 30 ft)	not applicable	one story
Gates	none	not gated

E.4 The one relief item, and the routes to it

The only live fire-access question is **special approval of a 1,754-ft dead end at 22 ft of pavement** under Table D103.4. Three routes exist and they are not exclusive:

1. **The Fire Marshal's special approval** under D103.4 itself. There are no published criteria; it is a discretionary decision, which is why it must be obtained in writing before the Planning Commission hearing.
2. **Appendix D107.1 Exception 3 as modified by Ga. Comp. R. & Regs. 120-3-3-.04**, which supplies an explicit menu for "residential developments having less than the minimum of two entrances": enhanced turning radii, **increased road widths**, D103.6 fire-lane signage, an additional **non-vehicular all-weather access road** designed by a registered design professional to Appendix D loading, or **a written statement by the Fire Code Official**.
3. **The statutory waiver, O.C.G.A. §25-2-12(e)(4)**: "the local fire official, building official, or developer may obtain a waiver ... on a waiver form designed and prescribed by the Safety Fire Commissioner. **The State Fire Marshal or designated representative shall respond within 30 days ... If the 30-day time frame is not met, the Plans submitted shall be**

deemed to be approved." The 30-day deemed-approval clock makes this worth filing early rather than late.

E.5 The package to put in front of the Fire Marshal

1. **A 22-ft clear lane with no on-street parking**, signed both sides under D103.6 and enforced by HOA covenant, with **26 ft of pavement offered wherever the strip allows it**. The lane tract widens from 35.9 ft at the entrance to 46.1 ft at the rear, passing 40 ft at about $u = 1,010$; from there north 26 ft of pavement fits alongside both 5-ft sidewalks and their 2-ft strips ($26 + 2 \times 7 = 40$ ft), and it is offered there. Between $u \approx 530$ and $u \approx 1,010$ the tract holds 37 to 40 ft, so 26 ft could be had only by giving up the southwest sidewalk — offered as a trade if the Fire Marshal prefers width to a second walk. In the narrow front reach the tract cannot hold 26 ft plus a 2-ft strip and a 5-ft sidewalk at all, which is precisely the relief being sought.
2. **Three hammerhead turnarounds** at 594.5, 570.0 and 580.0 ft of spacing, the third at the terminus, each two legs of 20 ft $\times$ 60 ft (the Figure D103.1 "120-foot hammerhead").
3. **Hydrants at about 400 ft** on an 8-in main (Memo C), located at the widened points so D103.1's 26-ft width is satisfied where a hydrant stands on the access road.
4. **NFPA 13D sprinklers in every dwelling — offered voluntarily**, and stated as such. They are worth offering for two reasons that survive the D107.1 correction: they halve the required fire flow from 1,000 gpm to 500 gpm on a 1,754-ft dead-end main (Appendix B §B105.1), and they are the strongest single argument for the 22-ft width relief. At roughly \$3,000 per home, 41 homes is about **\$123,000**, and the applicant should say plainly that this is an elective offer and ask the Fire Marshal whether it buys the width. If it does not, the offer should still stand as a life-safety measure on a single-access lane — but the package must not represent it as a code requirement, because it is not.
5. **Clubhouse sprinklered (NFPA 13)** and monitored; Knox box at the clubhouse.
6. **Addressing**: one street name; 4-in reflective numbers visible from the lane; cluster mailboxes at the kiosk.
7. **Grades**: the finished lane profile designed to $\leq 10\%$ (existing ground reaches 10.9 %), 13 ft 6 in vertical clearance, no gate.
8. **Construction sequencing**: the first hammerhead turnaround (station 594.5 ft) built before any certificate of occupancy beyond it, and a temporary turnaround at the Phase-1 limit ($u = 980$ ft). One item belongs on the sequencing sheet and on the demolition plan: **the existing dwelling at 4535 Arcado Rd stands about 1,200 ft back from the right-of-way, at about ($u = 1,202$, $v = -114$) on the ridge — directly on the proposed lane alignment**. It is a Phase-2 demolition. While it stands, the lane physically cannot continue past it, which is a further reason the Phase-1 temporary turnaround at $u = 980$ is not optional. Only 4541 Arcado Rd is near the frontage.
9. **Managed vegetation zone**: keep a 30-ft managed zone behind the rear lots within the buffer-easement rules at the wooded rear [VERIFY with the Fire Marshal; this is not a designated wildland-urban-interface area].

Special approval under D103.4 is a discretionary decision of the fire code official; nothing above may be represented as approval. The plan appears consistent with IFC 2024 (Ga. Comp. R. & Regs. 120-3-3-.04) Appendix D **only if items 1–3 and the D103.4 special approval are accepted in writing** — get the Fire Marshal's letter before the Planning Commission hearing [VERIFY].

Memo F — Schools (Gwinnett County Public Schools)

Question. Student generation of the cottage program against the 2025 mixed-use filing and the as-zoned R-1 baseline.

F.1 Inputs

Input	Value	Source
Cluster	Camp Creek ES / Trickum MS / Parkview HS	FACTS §2 (GIS school zones)
2025 GCPS letter for RZ-2025-01 (104 condominiums), July 2025	+25 ES / +13 MS / +19 HS ; forecasts against capacity 2025-26: Camp Creek 1,414 / 1,325 (+89 over), Trickum 2,203 / 2,275 (–72), Parkview 3,148 / 2,900 (+248 over)	status/rz-2025-01/4817_school-impact-analysis.pdf [VERIFY the figures before citing them]
HOPA 55+	at least 80 % of occupied units with a resident 55+ (24 CFR §100.305(a)), and the fractional unit counts into the 80 % (§100.305(g)): $0.80 \times 41 = 32.8 \rightarrow$ 33 of the 41 units must be occupied by at least one person aged 55 or older ($33 \div 41 = 80.5\%$; 32 would be 78.0 % and would fail). Eight units carry no age requirement. Recorded declaration; published policies; biennial verification (§100.307(c))	42 U.S.C. §3607(b)(2)(C); 24 CFR §§100.305–.307
GCPS per-unit multipliers (single-family detached, not age-restricted)	not in the record	VERIFY by requesting an updated GCPS Planning letter

F.2 Result

- **Proposed (41 HOPA cottages): 0 students** at each of the three schools. A 55+ community with a recorded age-restriction covenant is treated by GCPS as generating no K-12 enrolment; the 2025 letter's +57 total is superseded because the program that produced it — 104 non-age-restricted condominiums — is withdrawn.
- **As-zoned R-1 (37 lots, no age restriction — 4 du/ac × 9.44 ac)** would generate students at GCPS's single-family multipliers, into a cluster where Camp Creek ES and Parkview HS are already forecast over capacity. The cottage program's zero is therefore a net benefit against by-right development, and the §1003-7 "burden on schools" criterion can be answered with a GCPS letter rather than an estimate.
- **Action:** send GCPS Planning the R-2 concept plan and the HOPA condition and ask for a **new impact letter** stating zero students, with the R-1 baseline for comparison, before the Planning Commission hearing.

Memo G — Environmental Screen

Question. Regulated features and the studies the site will need.

G.1 Findings

Feature	Finding	Rule / source	Action
Stream (state waters)	An unnamed order-0 headwater reaches the SW line at u ≈ 1,356–1,392 and ends inside the site at (1,391.7, –209.8); a second branch runs 38–58 ft outside the SW line at u ≈ 868–1,038	Gwinnett GIS hydrology (FACTS §2)	See G.2 — the first question is whether a regulated stream exists here, not only where its bank is
State buffer	25 ft , measured "from the point where vegetation has been wrested by normal stream flow or wave action"; no land-disturbing activity within it	O.C.G.A. §12-7-6(b)(15) ; GA EPD <i>Field Guide for Determining the Presence of State Waters That Require a Buffer</i> (Sept 2017)	No encroachment proposed → no EPD buffer variance needed
City buffer	50-ft undisturbed buffer plus a 25-ft impervious setback (75 ft), measured from the top of the stream bank ; land development in the buffer or impervious cover in the setback "is prohibited unless a variance is granted"; no septic in either	Lilburn Code Ch. 109 Art. VII (MNGWPD model text; section numbers VERIFY — Municode returns HTTP 403 to automated retrieval, transcribe by hand)	Show the 25-ft state line and the 50/75-ft city lines separately on the plan: they are measured from different data
Whether the buffer applies at all	The ordinance begins a "stream" at (1) a spring, seep or groundwater outflow that sustains streamflow, (2) a point with a drainage area of 25 acres or more , or (3) where field studies show a stream in a smaller drainage area. The on-site reach drains far less than 25 acres , so under (2) the regulated stream would begin downstream, off the property; under (1) it begins at a spring or seep that must be located in the field	Lilburn Code Ch. 109 Art. VII / MNGWPD model definitions	This is an open question the package previously treated as settled. The field study must establish first <i>whether and where</i> a stream begins, then the top of bank. The rear lot count and the Pond 2 footprint both turn on the answer
Receiving water	Jackson Creek headwaters (GAR030701030315) — 2024 303(d) Not Supporting (bacteria, biota; urban runoff)	GA EPD 2024 Integrated 305(b)/303(d) list	Emphasise WQv/RRv treatment and no direct discharge into the buffer (Memo D)
Floodplain	FEMA Zone X	Gwinnett GIS [FIRM panel and effective date VERIFY]	Table 4.2 prohibition met by absence; note it on the plan
Wetlands	none mapped in the record (NWI not queried)	—	VERIFY: NWI check plus a wetland delineation at the stream head (hydric indicators are possible despite no hydric map unit)
Soils	AmC2 Appling sandy loam 6–10 % eroded (HSG B); ApB Appling–Hard Labor complex 2–6 %; PfC2 Pacolet-type sandy clay loam eroded; GeB2 / GgE2 Gwinnett/Cecil-type eroded phases (GgE2 at 10–25 % slopes at the rear); ARE Ashlar–Rion–Waterree complex 10–25 % (bedrock 22–40 in) under both pond sites and the mid-swale lane; HYB Helena sandy loam 2–6 % (HSG D, water table ≈ 24 in) at the front SW corner. No hydric units on site.	docs/11 §3.3 (NRCS SDA) [series names VERIFY]	Rock probes at the ponds and the trunk sewer; test pits and an undercut plan at HYB; phased clearing with immediate stabilisation
Trees	Residential subdivisions must file a tree preservation/replacement plan meeting 16 tree density units per acre (exclusive of zoning-buffer acreage); trees ≥ 3 in DBH credited, preserved trees ≥ 12 in earning up to 4× credit; plan by a landscape architect, forester or arborist	Lilburn Code Ch. 109 Art. V §§109-168(3), 109-169, 109-177	Tree survey of the wooded rear and the perimeter buffers; the creek woods, the 20-ft buffers and the terminus green should carry the requirement at multiplied credit

Feature	Finding	Rule / source	Action
Phase I ESA	Two pre-1980 houses with accessory structures. Only 4541 Arcado Rd is near the frontage (footprint centred about u 203, v -174, ≈ 2,396 SF). 4535 Arcado Rd sits about 1,200 ft back from the right-of-way, at about (u 1,202, v -114) on the u 1,080–1,200 ridge, directly on the proposed lane alignment (≈ 4,568 SF footprint) — a Phase-2 demolition standing on the lane , not a frontage structure. Likely septic tanks, possible wells, heating-oil tanks, asbestos and lead paint	ASTM E1527-21; footprints affine-fitted from aerial imagery by tools/existing_conditions.py, approximate, to be located by the RLS survey	Commission a Phase I ESA before closing on 4535/4537; septic abandonment per Gwinnett Environmental Health; asbestos survey before demolition (GA EPD Rule 391-3-14). Sequence the 4535 demolition with the Phase-2 lane construction (Memo E.5 item 8)
Species ; historic	not screened	USFWS IPaC; GA HPD GNAHRGIS	VERIFY: run IPaC and a GNAHRGIS check
Radon	Gwinnett is EPA Radon Zone 1	Georgia Radon Program (2024)	Passive radon stack in every slab (a specification item, not a zoning issue)
Erosion and sediment	site exceeds 1 ac and lies within 200 ft of a stream	GSWCC Manual (2016); NPDES GAR100001	ESPC plan by a certified design professional at LDP; show the 25-ft state buffer on it

G.2 Statement for the application

The site contains no floodplain (FEMA Zone X), no mapped wetlands and no mapped hydric soils. One headwater reach at the rear southwest corner is treated as regulated: the plan preserves it and its 25-ft state buffer, 50-ft undisturbed city buffer and 75-ft impervious setback inside a common "creek woods" tract, and every lot, pond and pavement edge is held **80 ft** from the mapped channel — the 75-ft requirement plus a 5-ft allowance for the top-of-bank location. A field study by the surveyor and the PE, using the GA EPD field guide, will establish whether and where a regulated stream begins and where its top of bank lies; the plan is drawn so that a stream origin anywhere within the mapped reach can be accommodated without moving lots. Subject to that study, a tree survey and a Phase I ESA, the plan appears consistent with Lilburn Code Ch. 109 Art. VII and Art. V.

Summary table (rezoning-level screen)

Memo	Headline	Status
A Traffic	152–203 daily / 8.2–16.0 AM / 12.3–18.3 PM trips at 41–47 units; 34.8 PM as-zoned R-1; ≈ 214 PM for the 2025 MU program	A Level-1 TIS is required (the exemption is 7 lots, not 21 trips); ITE LUC 251 rate UNRESOLVED ; new 24-hour counts required — the 2017 count is ineligible; no left-turn lane (< 75 lots); 251 ft > 244 ft spacing; ISD certificate needed
B Sewer	Phase 1 (24 lots): 0.68 % gravity to the existing on-site MH 927.13 — no extension . Phase 2 (17 lots): only the 478-ft / 178-ft-off-site gravity tie at 1.45 % to the Legends main 919.58, ≈ \$168k–\$273k = \$9,900–\$16,000 per Phase-2 lot, up 12 % on the smaller Phase 2	Pumping of every kind foreclosed by DWR policy ; needs a recorded easement, a DWR "Pre-Rezoning" capacity certification and a staff reading of the CPA sewer policy; the drawn alignment is ≈ 20 ft deep where it crosses the lane at u 1,300 — price the low-pocket alternative; invert survey is the first spend
C Water	10,250 gpd average (41 × 250); fire flow 1,000 gpm, or 500 gpm with the voluntary sprinklers ; 6 hydrants; 8-in dead-end main with flushing	DWR flow test
D Storm water	44.1 % impervious (184,148 SF, not 38.4 %); WQv 18,659 cf ; detention 75,410 cf required on 7.541 disturbed acres against 77,863 cf provided — 103.3 % , in two 6-ft basins (33,129 + 44,734 cf); RRv 15,549 cf available but not claimed	Rock probes under both basins; PE hydrology; a 3.3 % margin is a screening result, not a design margin; \$5.9 does not apply — the 33.8 % open space is voluntary
E Fire	1,754-ft dead end → D103.4 special approval ; 22 ft vs 26 ft is the relief item; D107.1 is 120 units in Georgia, so two access roads are not required at 41	Fire Marshal's letter; sprinklers are a voluntary offer (41 × ≈ \$3,000 ≈ \$123,000), not a required mitigation; 4535 Arcado Rd stands on the lane alignment and is a Phase-2 demolition
F Schools	0 students (HOPA; 33 of the 41 units must house a 55+ resident — 0.80 × 41 = 32.8, rounded into the 80 %)	New GCPS letter
G Environmental	25/50/75-ft stream buffers preserved with an 80-ft design screen; Zone X; no hydric soils; 16 TDU/ac tree plan; Phase I ESA	Field study must establish whether a regulated stream begins on site

Disclaimer: This is an AI-generated analysis for preliminary planning purposes. All findings must be verified by a licensed professional before use in design, permitting, or regulatory submissions.

title	List of Adjoining Property Owners (Item 10) — Cottages at Arcade Springs
date	2026-08-28
address	4535, 4537, 4539 & 4541 Arcade Rd SW, Lilburn, GA 30047 — Land Lot 123, 6th District, Gwinnett County (PINs R6123 033 / 015 / 014 / 162)
status	DRAFT — ownership as of the Gwinnett Tax Assessor 2026 ownership file retrieved 2026-08-28; re-pull on the filing date
generator	tools/adjoiners.py (see verify/adjoiners-output.md)

List of Adjoining Property Owners — names and mailing addresses

Application item: Lilburn 2026 Application Instructions item 10 — "Names and addresses of all adjoining property owners must be attached. Information may be obtained from the Gwinnett County Tax Assessor." The City mails the public-hearing letters to these addresses and posts the sign on the property at least 15 days before the Planning Commission hearing (Instructions p. 2, "Legal Notice"; Lilburn Zoning Ordinance 2023-603 §1003-6).

Data source and date. Gwinnett County Tax Assessor **2026 Property Ownership file** (quarterly extract; assessment date 2026-01-01), retrieved **2026-08-28**, joined to the Gwinnett GIS parcel layer (same date) to find every parcel that shares a boundary with the assemblage. Owner names and mailing addresses are reproduced exactly as the digest carries them (upper case, USPS abbreviations). Ownership changes; **re-run** tools/adjoiners.py **against a fresh extract on the filing date** (planned 2026-09-25 or 2026-10-30) so the mailed notices match the record.

Subject property (for reference — not adjoiners): R6123 033 and R6123 015, AWAD MOHAMMED, 4541 ARCADE RD SW, LILBURN, GA 30047-3968; R6123 014 and R6123 162, MENDEZ SANTOS C & ROBLERO DE LEON LESVIA ROSARIO, 4535 ARCADE RD SW, LILBURN, GA 30047-3968.

1. Adjoining parcels (31), grouped by side

All 31 adjoining parcels are single-family houses zoned **R-1 (City of Lilburn)** in the 2026 digest (ZONDESC = "R1-Single Family Residence LIL"). "Frontage on site" is the run of common boundary in site-local feet (u from the Arcade Road corner toward the rear; v across the strip), which is the order in which the lots appear on the site plan.

Northeasterly line (King David Manor side), front → rear

#	PIN	Property address	Owner(s) of record	Mailing address	Frontage on site (local ft)	Digest ac
1	R6123 017	4531 Arcade Rd SW, Lilburn, GA 30047	GREGG LINDA LOFTEN	4531 ARCADE RD SW, LILBURN, GA 30047-3968	u 0 to 254	0.57
2	R6123 020	265 King David Dr SW, Lilburn, GA 30047	MARTINEZ GARCIA JOSELINE ROSEMARY & ESQUIVEL MARGARITO GARCIA Sr	265 KING DAVID DR SW, LILBURN, GA 30047-3932	u 254 to 430	0.76
3	R6123 021	255 King David Dr SW, Lilburn, GA 30047	STONECYPHER CHELISA RAE	255 KING DAVID DR SW, LILBURN, GA 30047-3932	u 430 to 584	0.71
4	R6123 022	245 King David Dr SW, Lilburn, GA 30047	PHOUANGAPHAY PHIMPHA	2473 BELLVIEW AVE NW, ATLANTA, GA 30318-8347	u 584 to 683	0.47
5	R6123 023	235 King David Dr SW, Lilburn, GA 30047	CALDERON EDGARDO F & GARCIA GREGORIO GUERRA	235 KING DAVID DR SW, LILBURN, GA 30047-3932	u 683 to 784	0.47
6	R6123 024	225 King David Dr SW, Lilburn, GA 30047	BECKWORTH FRANK W	225 KING DAVID DR SW, LILBURN, GA 30047-3932	u 784 to 884	0.47
7	R6123 025	215 King David Dr SW, Lilburn, GA 30047	SMITH JOEL DEAN & SMITH FRANCES	215 KING DAVID DR SW, LILBURN, GA 30047-3932	u 884 to 986	0.47
8	R6123 026	129 King David Dr SW, Lilburn, GA 30047	MENDEZ NICOLAS & GONZALEZ MOISES RODRIGUEZ	3434 N CREEKVIEW DR, LAWRENCEVILLE, GA 30044-4143	u 986 to 1085	0.46
9	R6123 027	195 King David Dr SW, Lilburn, GA 30047	FKH SFR C1 LP	1850 PARKWAY PL SE STE 900, MARIETTA, GA 30067-8261	u 1085 to 1186	0.46
10	R6123 028	185 King David Dr SW, Lilburn, GA 30047	BAIRD BRAD W & LYNN Y	185 KING DAVID DR SW, LILBURN, GA 30047-3930	u 1186 to 1287	0.46
11	R6123 029	175 King David Dr SW, Lilburn, GA 30047	LITTLE STEVEN A ETAL	175 KING DAVID DR SW, LILBURN, GA 30047-3930	u 1287 to 1388	0.45

#	PIN	Property address	Owner(s) of record	Mailing address	Frontage on site (local ft)	Digest ac
12	R6123 030	165 King David Dr SW, Lilburn, GA 30047	CAMPOS EDWIN & MONTUFAR-MALDONADO AZUCESENA	165 KING DAVID DR SW, LILBURN, GA 30047-3930	u 1388 to 1488	0.45
13	R6123 031	155 King David Dr SW, Lilburn, GA 30047	LI GUANYING	4700 PARKVIEW WALK DR, LILBURN, GA 30047-8973	u 1488 to 1608	0.54
14	R6123 032	145 King David Dr SW, Lilburn, GA 30047	WILSON DANA & MORTON DEXTER C	145 KING DAVID DR SW, LILBURN, GA 30047-3930	u 1608 to 1722	0.53

Northwesterly end (Nantucket), SW → NE

#	PIN	Property address	Owner(s) of record	Mailing address	Frontage on site (local ft)	Digest ac
15	R6123 038	4540 Nantucket Dr SW, Lilburn, GA 30047	CARDENAS BERTHA NALLELI	4540 NANTUCKET DR SW, LILBURN, GA 30047-3940	v -334 to -232	0.42
16	R6123 037	4530 Nantucket Dr SW, Lilburn, GA 30047	BARKER RICHARD C II	4530 NANTUCKET DR SW, LILBURN, GA 30047-3940	v -233 to -122	0.44
17	R6123 036	4520 Nantucket Dr SW, Lilburn, GA 30047	LEE CHARI LYN & BARKER LORA GENE	4520 NANTUCKET DR SW, LILBURN, GA 30047-3940	v -130 to 23	0.54

Southwesterly line (Legends at Parkview side), front → rear

#	PIN	Property address	Owner(s) of record	Mailing address	Frontage on site (local ft)	Digest ac
18	R6123 313	308 Fieldhouse Cir SW, Lilburn, GA 30047	SHAHIDA GULAM MEMON FAMILY REVOCABLE TRUST	308 FIELD HOUSE CIR SW, LILBURN, GA 30047-5283	u -26 to -26	0.37
19	R6123 312	303 Fieldhouse Cir SW, Lilburn, GA 30047	ABELLERA JOHN P & KHAN RIZWANA	303 FIELD HOUSE CIR SW, LILBURN, GA 30047-5283	u -26 to 173	0.49
20	R6123 311	260 Village Green Ct SW, Lilburn, GA 30047	PETTWAY JASON G & PETTWAY FRANCES M	260 VILLAGE GREEN CT SW, LILBURN, GA 30047-4178	u 173 to 272	0.62
21	R6123 310	250 Village Green Ct SW, Lilburn, GA 30047	DOBARIA DIPAK A & DOBARIA PUSHPA D	250 VILLAGE GREEN CT SW, LILBURN, GA 30047-4178	u 272 to 372	0.42
22	R6123 309	240 Village Green Ct SW, Lilburn, GA 30047	BEJA DAVID & BEJA NANCY	240 VILLAGE GREEN CT SW, LILBURN, GA 30047-4178	u 372 to 472	0.35
23	R6123 308	230 Village Green Ct SW, Lilburn, GA 30047	ZHENG XIAOWEI	230 VILLAGE GREEN CT SW, LILBURN, GA 30047-4178	u 472 to 572	0.34
24	R6123 307	220 Village Green Ct SW, Lilburn, GA 30047	PIXTON RODNEY A & PIXTON MICHELLE P	220 VILLAGE GREEN CT SW, LILBURN, GA 30047-4178	u 572 to 672	0.34
25	R6123 306	210 Village Green Ct SW, Lilburn, GA 30047	ZHONG JOSHUA & WEN JING	210 VILLAGE GREEN CT SW, LILBURN, GA 30047-4178	u 672 to 773	0.34
26	R6123 305	200 Village Green Ct SW, Lilburn, GA 30047	NOT AVAILABLE	200 VILLAGE GREEN CT SW, LILBURN, GA 30047-4178	u 773 to 975	0.69
27	R6123 304	190 Village Green Ct SW, Lilburn, GA 30047	BALLESTEROS ROEL V	190 VILLAGE GREEN CT SW, LILBURN, GA 30047-4176	u 975 to 1134	0.55
28	R6123 303	180 Village Green Ct SW, Lilburn, GA 30047	FATIMA SYEDA HEENA & HABEEB TARAQ	180 VILLAGE GREEN CT SW, LILBURN, GA 30047-4176	u 1134 to 1295	0.55
29	R6123 302	170 Village Green Ct SW, Lilburn, GA 30047	FRAZIER PRENTICE & FRAZIER THERASA	170 VILLAGE GREEN CT SW, LILBURN, GA 30047-4176	u 1295 to 1449	0.53
30	R6123 301	160 Village Green Ct SW, Lilburn, GA 30047	YIN SARA S	160 VILLAGE GREEN CT SW, LILBURN, GA 30047-4176	u 1449 to 1561	0.34
31	R6123 300	150 Village Green Ct SW, Lilburn, GA 30047	CAI HUI HUGH	150 VILLAGE GREEN CT SW, LILBURN, GA 30047-4176	u 1561 to 1721	0.43

2. Right-of-way and courtesy notices (not tax parcels)

#	Entity	Interest	Mailing address	Basis
32	Gwinnett County Department of Transportation	Owner/maintainer of the Arcado Road right-of-way along the entire 237.6-ft frontage (the fourth "side" of the tract)	75 Langley Dr, Lawrenceville, GA 30046	Arcado Rd is a county-maintained collector (FACTS §1); the GIS right-of-way polygon abuts courses 1–14 of the legal description (docs/05)
—	Legends at Parkview Homeowners Association, Inc., c/o Liberty Community Management (registered agent)	<i>Courtesy only.</i> The HOA owns no parcel abutting the site (all 14 southwesterly adjoiners are individually owned lots), so it is not an "adjoining property owner"; it is the organized voice of the southwesterly neighbors (docs/11 §8)	VERIFY — pull the registered-agent address from the Georgia Secretary of State corporations search; not in the package data	docs/11-site-context-brief.md (registered agent identified; address not captured)

3. Notes and items to verify

- 200 Village Green Ct SW (R6123 305):** the digest shows the owner name as "NOT AVAILABLE" (mailing address 200 Village Green Ct SW; last transfer 2012-09-20, Deed Book 51675 Page 454). **VERIFY** the record owner on the Gwinnett Tax Assessor property record or the GSCCCA deed index before the notice is addressed; until then address it to "Owner of Record".
- 175 King David Dr SW (R6123 029):** vested "LITTLE STEVEN A ETAL" — there are additional owners of record not named in the digest; the notice goes to the digest mailing address, which is the property.
- Off-site mailing addresses (owner does not live on the lot):** 245 King David Dr (mail: Atlanta 30318), 129 King David Dr (mail: Lawrenceville 30044), 195 King David Dr (FKH SFR C1 LP — institutional single-family rental owner; mail: Marietta 30067), 155 King David Dr (mail: 4700 Parkview Walk Dr, Lilburn 30047). The tenants at those four houses will not receive the mailed notice; the sign on the property and the neighbor briefing (docs/11 §8) are the only channels that reach them.
- Spelling differences between the two county files:** the GIS street name is "FIELDHOUSE CIR" while the digest mailing address is "FIELD HOUSE CIR" (303 and 308); both are ZIP+4 30047-5283 and deliverable. 129 King David Dr sits between 215 and 195 in the street sequence (King David Manor's numbering jumps) — it is correct.
- Front-corner adjoiners:** the northeasterly front corner abuts 4531 Arcado Rd SW (Lot 1, Block A, King David Manor Unit 1; 254 ft of common line, u 0–254). At the southwesterly front corner the GIS right-of-way polygon wraps 10 ft around the corner (legal description course 14), so 308 Fieldhouse Cir SW (Lot 80) touches the tract only at a point (u –26) and 303 Fieldhouse Cir SW (Lot 79) runs u –26 to 173. Both are listed; the boundary survey will settle who owns that 10 ft.
- Properties across Arcado Road** (southeasterly side, and the Arcadia Place lots whose centerline meets Arcado at v +61 — FACTS §2b) are **not** in this list because they do not share a boundary with the tract. Some jurisdictions notify owners across a street; **ask the Planning Director at the pre-application conference** whether Lilburn wants them added, and if so pull them from the same extract.
- Count reconciliation:** FACTS §1 says "32 adjoining parcels"; the GIS query returns 31 tax parcels plus the Arcado Road right-of-way polygon = 32 adjoiners. The 31 parcels are 13 King David Manor lots + 4531 Arcado Rd (NE), 3 Nantucket lots (NW), and 14 Legends at Parkview lots (SW).

4. Verification record

Script: tools/adjoiners.py; inputs data/owners-found.json (2026 digest, 35 records = 4 subject + 31 adjoining) and data/site-context-local.json (adjoining parcel rings). Checks: 31 parcels found, each assigned to one side, each joined to exactly one digest record; one owner name unavailable (R6123 305). Output saved to verify/adjoiners-output.md.

Disclaimer: This is an AI-generated analysis for preliminary planning purposes. All findings must be verified by a licensed professional before use in design, permitting, or regulatory submissions.

09 — Architectural Character, Elevations and Renderings

Project: The Cottages at Arcade Springs — R-1 → R-2 rezoning, 4535 / 4537 / 4539 / 4541 Arcade Rd SW, Lilburn, GA 30047 (PINs R6123 033, 015, 014, 162; Land Lot 123, 6th District, Gwinnett County; 9.44 ac deeded / 9.577 ac GIS-calculated)

Answers: City of Lilburn 2026 *Application Instructions — Rezoning/Special Use Permit/Change In Condition*, **item (11)**

Architectural Renderings and Elevations; Lilburn Zoning Ordinance 2023-603 §1003-4 **Prepared:** 2026-09-03 (owner-applicant self-prepared; filing date 2026-09-__) **Status: DRAFT — ILLUSTRATIVE, NOT FOR CONSTRUCTION, NOT SEALED.** Every drawing and image named here is preliminary work of the owner-applicant, prepared with AI drafting tools, and is to be superseded by sealed drawings of a Georgia-registered architect before any building permit. See §8. **Geometry source:** data/plans.json (regenerated 2026-09-03 by tools/floorplans.py), data/layout.json (site-plan generator, corrected 2026-09-03 — **the lot count is now 41, not 43**; see §3), tools/elevations.py (Rev B, 2026-09-03) and tools/amenity.py (Sheet A-3.0, 2026-09-03). Every dimension below is read out of those files, not copied from the 2026-08-28 program brief in FACTS.md §4, which both plans have since departed from — see §5.2. **Sheet numbers** are those of the settled register in tools/transmittal.py SHEETS, and only those; python3 verify/check_sheet_refs.py fails on any number that is not in it.

1. What item (11) requires, and where each requirement is answered

Item (11) of the 2026 Application Instructions reads, verbatim (data/application-instructions-2026.txt, lines 49–52):

***(11) Architectural Renderings and Elevations:** Required for rezoning and special use permits where the applicant plans to erect or alter a building, sign, or other structure. An architectural rendering or elevation of each side of the structure visible from the street shall be submitted. The drawings shall be to scale or in proper perspective and shall include the color and materials of all structures and roofing and location and size of wall signs. Visual imagery may be used.*

The instruction names five separate obligations. They are answered as follows.

#	What item (11) requires	Where it is answered	Status
a	An elevation or rendering of each side of each structure visible from the street	§2 structure-by-side table; Sheets A-2.1 (Plan A), A-2.2 (Plan B) and A-3.0 (clubhouse, mail kiosk, monument sign)	Complete. All four elevations of both cottage plans and of the clubhouse; the kiosk and the sign are drawn on A-3.0 — §2.4
b	To scale or in proper perspective	Elevations at 3/16" = 1'-0" on ARCH C (24 × 18 in), drawings/plan-a-elev.svg, plan-b-elev.svg; perspectives in renderings/	Cottages complete; see §7
c	The colour and materials of all structures and roofing	§4 schedule; Sheet A-2.3 material and colour board; the material key printed on each elevation sheet; Sheet A-3.0 draws the clubhouse, kiosk and sign in Scheme 1	Complete for the cottages and for the amenity structures
d	Location and size of wall signs	§6 — no wall signs are proposed anywhere in the development ; one ground-mounted monument entry sign is proposed, scheduled in §6.1 and drawn on Sheet A-3.0 with its area computation	Complete
e	"Visual imagery may be used"	§7 rendering index — twelve AI-generated illustrative perspectives, the scaled drawings governing	All 12 generated (files renderings/R-01...R-12); the R-00 contact index remains to be laid out

Nothing in this document asserts that any element meets a code or ordinance requirement. Every regulatory statement is made as **"appears consistent with"** and each carries its ordinance and edition.

1.1 Sheet numbering — settled

The numbering conflict this section reported on 2026-09-03 is closed. **The register in tools/transmittal.py SHEETS governs**, the generated title blocks have been reconciled to it, and verify/check_sheet_refs.py tests every "Sheet X-N.N" reference in every document in docs/ against it. The architectural sheets are:

Sheet	Content	File	Issued?
A-1.1	Plan A "The Springbrook" — floor and roof plan, 3/16" = 1'-0"	drawings/plan-a-sheet.svg	Issued
A-1.2	Plan B "The Laurel" — floor and roof plan, 3/16" = 1'-0"	drawings/plan-b-sheet.svg	Issued
A-2.1	Plan A — exterior elevations, four sides, 3/16" = 1'-0"	drawings/plan-a-elev.svg	Issued
A-2.2	Plan B — exterior elevations, four sides, 3/16" = 1'-0"	drawings/plan-b-elev.svg	Issued
A-2.3	Exterior colour schemes and materials	drawings/plan-colors.svg	Issued
A-3.0	Clubhouse, mail kiosk and entry monument sign — plans, elevations and details	drawings/amenity-sheet.svg	Issued

Sheet	Content	File	Issued?
A-4.0 / A-4.1	Typical lot — Plan A / Plan B, 1" = 10'	drawings/plan-a-lot.svg, plan-b-lot.svg	Issued

Two consequences for the rest of the package:

1. **Voluntary condition 8 has been conformed** (2026-09-03). It now binds the applicant to "floor plans, elevations and materials substantially as shown on the sheets submitted with the application — Sheets A-1.1 and A-1.2 (floor plans, Plans A and B), A-2.1 and A-2.2 (exterior elevations, Plans A and B), A-2.3 (exterior colour schemes and materials), A-3.0 (clubhouse, mail kiosk and entry monument sign) and A-4.0 and A-4.1 (typical lots)" (data/voluntary-conditions.md, generated from docs/02 §9). That is the register, and it matches this document. The earlier citation — A-1.1 (Plan A), A-2.1 (Plan B), A-3.0, A-4.0 — pointed at Plan A's *floor plan*, Plan A's *elevations* and the *typical-lot* sheet, and is superseded.

2. **Two generated title blocks still print pre-register numbers — this remains a defect to close before filing.**
drawings/amenity-sheet.svg titles itself

"A-3" rather than A-3.0 and cross-refers to "Sheet C-1" for the site plan (the register makes that C-2.0, with the entrance enlargement C-2.3); drawings/existing-conditions.svg titles itself "C-0" and cross-refers to "C-1, MASTER CONCEPT PLAN" (the register makes those C-1.0 and C-2.0). Correct tools/amenity.py and tools/existing_conditions.py and regenerate.

This document uses the register throughout.

2. Structures the applicant plans to erect, and which of their sides are visible

2.1 What "the street" means on this site

The only public street is **Arcado Road SW** (Gwinnett County minor collector, 35 mph, two lanes). Every dwelling stands on a **private lane**, and Lilburn Zoning Ordinance 2023-603 §319 requires every lot to abut at least 30 ft on a public street **or an approved private street**, with the City's Site Development Plan Review Checklist §4.s adding that "All components of private streets and alleys must meet minimum standards for public street." The applicant therefore reads "visible from the street" in item (11) to include the private lane, and provides elevations on that basis rather than arguing that 41 of 41 dwellings face no street at all.

2.2 Why all four elevations of both cottage plans are provided

On the typical lot the arithmetic leaves no side hidden:

Input	Value	Source
Lot	50'-0" × 100'-0" = 5,000 SF	data/layout.json (all 41 lots identical)
Body width, both plans	38'-0"	data/plans.json overall_body_dims
Side yard, each side	6'-0" (50'-0" – 38'-0" = 12'-0", centred)	data/plans.json lot_siting.side_yard_ft
Roof-to-roof width	39'-4" (38'-0" body + 2 × 0'-8" side overhang)	tools/elevations.py check_width()
Clear gap, wall to wall, between adjacent cottages	12'-0" (6'-0" + 6'-0")	derived
Clear gap, roof edge to roof edge	10'-8" (12'-0" – 2 × 0'-8")	derived

A 12'-0" gap between one-story houses set 15'-8" back from a 22'-0"-wide lane is a direct sight line: from the lane a passer-by sees the front elevation of the house in front of them and the full side elevation of both its neighbours. Every one of the 41 lots is also a **perimeter lot** — the **fourteen** south-west lots (numbered 1–14) back to the south-west property line at $v \approx -234.7$ to -235.1 and the **twenty-seven** north-east lots (15–41) back to the north-east line at $v \approx +1.4$ to $+8.7$ (data/layout.json lot polygons) — so **every rear elevation faces the 20'-0" undisturbed buffer and, beyond it, an occupied R-1 back yard** in Legends at Parkview, King David Manor or Nantucket.

All four elevations of both cottage plans are therefore submitted, not only the front. That is more than item (11)'s literal minimum, and it is the only honest reading of "each side visible from the street" on a 236–246-ft-wide strip where nothing is screened by distance.

2.3 Structure-by-side table

Local coordinates are those of FACTS.md §1 (**u** along the strip from Arcado Rd, **v** across it, feet).

#	Structure	Qty	Location	Sides	Which sides are visible from a street or the private lane	Elevation of each such side is at
1	Plan A "The Springbrook" cottage — 1 story, 38'-0" × 51'-10" body	21	odd-numbered lots 1, 3, 5 ... 41, both sides of the lane	4	All four. Front faces the lane; both sides face 12'-0" gaps open to the lane; rear faces the 20-ft buffer and an abutting R-1 rear yard	A-2.1 , four panels: FRONT ELEVATION — FACES THE PRIVATE LANE; RIGHT SIDE; LEFT SIDE; REAR (drawings/plan-a-elev.svg / .png, 3/16" = 1'-0")
2	Plan B "The Laurel" cottage — 1 story, 38'-0" × 57'-6" body	20	even-numbered lots 2, 4, 6 ... 40, both sides of the lane	4	All four , same reasoning	A-2.2 , four panels, same arrangement (drawings/plan-b-elev.svg / .png, 3/16" = 1'-0")
3	Clubhouse — 1 story, 60'-0" (parallel to the lane) × 40'-0" = 2,400 SF under roof (2,000 SF enclosed + a 400-SF covered porch; 1,831 SF of net rooms)	1	amenity tract, north-east of the lane, u 110–170 / v –74.3 to –34.3; 114.8 ft from the Arcade Rd right-of-way and 15.0 ft clear of the 20-ft buffer (tools/amenity.py)	4	All four. The 40'-0" south-east face fronts the 14,354-SF village green and is the face seen from Arcade Road ; the 60'-0" south-west face fronts the lane and the entry drive; the 40'-0" north-west face closes the lane view; the 60'-0" north-east face is seen across the 20-ft buffer from King David Manor	A-3.0 — ISSUED (drawings/amenity-sheet.svg / .png). Four elevations plus floor plan and roof plan; plate 10'-0" + 0'-10" raised heel → eave 10'-10" above the finished floor; 40'-0" span at 6:12 → ridge 20'-10" above the finished floor / 21'-6" above finished grade , against the 24'-0" of voluntary condition 3
4	Mail kiosk — USPS cluster-box shelter, 16'-0" × 10'-0" = 160 SF, roofed and three-sided	1	north-east of the lane, u 211.5–227.5 / v –85.8 to –75.8, opposite a 4-space kiosk bay at u 155.5–207.5 / v –106.9 to –88.9	4	All four — a freestanding shelter with a closed north-east wall, open to the lane	A-3.0 — ISSUED. 4:12 hip, ridge 11'-7" above finished grade ; 3 cluster box units = 48 compartments + 6 parcel lockers; four 9'-0" × 18'-0" stalls and a 4'-0" walk. The sheet carries the Site Development Plan Review Checklist §4.r note ("accessory structure details with protection from the elements")
5	Monument entry sign — ground-mounted; an 8'-0"-long × 1'-4"-thick masonry wall with a 1'-4" wing / planter return at each end, 4'-0" deep overall	1	in the front amenity tract on the Arcade Rd frontage, u –9.7 to –1.7 / v –171.9 to –167.9 — 19.0 ft from the Arcade Rd right-of-way, outside it	2 faces + ends	The face toward Arcade Road and the entry drive is the item-(11) side; the reverse face is seen from the village green. Both faces carry copy	A-3.0 — ISSUED. Elevation, plan and the area computation: whole structure face 8'-0" × 3'-9" = 30.0 SF against the 32-SF voluntary limit, overall height 3'-9" against the 6'-0" voluntary limit, cast-stone copy panel 6'-0" × 1'-9" = 10.5 SF. The sheet shows the 390-ft departure sight triangle: the sign stands 20'-6" behind the driver eye and its nearest corner is 20.1 ft clear of the nearer sight line — no part of the sign or its bed lies inside the triangle
6	Pickleball court enclosure — 10'-0" black PVC-coated chain-link fence around two 60'-0" × 30'-0" pads carrying 44'-0" × 20'-0" courts	2 pads	south-west of the lane, u 157.5–217.5 / v –212.7 to –152.7	—	Visible from the lane and the entry drive. A fence, not a building; no wall, roof or sign	C-8.0 civil details — NOT YET ISSUED (fence height, mesh, colour, post spacing). No court lighting is proposed (voluntary condition 11)
7	Detention basin outlet structures, headwalls, retaining walls (if any)	—	Ponds 1 and 2, south-west side	—	Not visible from Arcade Rd; below the lane grade	C-3.0 — NOT YET ISSUED (grading, drainage and stormwater concept). Not item-(11) structures

Structures the applicant does NOT propose: no attached dwellings, no multifamily building, no commercial or office building, no gate house, no entry gate, no clubhouse pool building, no maintenance building, no freestanding utility building, no billboard, no changeable-copy sign, and **no wall signs of any kind** (§6.2).

2.4 What is complete, and what is left

Complete and to scale. All four exterior elevations of Plan A (**A-2.1**) and all four of Plan B (**A-2.2**) at 3/16" = 1'-0" on ARCH C 24 × 18 in, each with the eave, plate, ridge, finished-floor and finished-grade lines drawn, every opening placed from the same exterior_openings list that generates the floor plans, a keyed material legend, the height statement, the grade statement and the wall-sign statement printed on the sheet; the floor and roof plans at **A-1.1** and **A-1.2**; the colour and material board at **A-2.3**; the typical-lot sheets at **A-4.0** and **A-4.1**; and — new on 2026-09-03 — the **clubhouse, mail kiosk and monument entry sign on A-3.0**, which is the item that this section previously recorded as the package's blocking gap. It is closed. tools/elevations.py check_massing() and check_width() now pass for both plans.

What is left, and none of it is an item-(11) elevation:

1. **C-2.3 entry, frontage and amenity enlargement — NOT YET ISSUED.** Sheet A-3.0 already computes and draws the 390-ft departure sight triangle at the entrance and shows the sign clear of it by 20.1 ft, so the substantiation exists; but the civil enlargement that will be sealed by the PE has not been drawn, and the sign's plan footprint, the curb returns and the landscape strip belong on it.
2. **C-8.0 civil details — NOT YET ISSUED.** The pickleball fence section (height, mesh, colour, post spacing) is the only item-(11)-adjacent content on it.
3. **A-2.3 colour-key plan.** The distribution rule in §4.2 is deterministic but data/layout.json still carries no scheme field on the lot records, so the tinted key plan cannot yet be generated.
4. **Architect's seal.** Every sheet above is owner-prepared and illustrative (§8).

3. The architectural character of the community

The Cottages at Arcade Springs is a community of **41 one-story detached houses** on fee-simple 50'-0" × 100'-0" lots, all of them served by one private lane and all of them backing onto a 20'-0" undisturbed perimeter buffer. There is no second storey anywhere in the development, no attached unit, no dormer and no basement. The tallest thing proposed on the site is a roof ridge **18'-6" above finished grade** on Plan A and **19'-8"** on Plan B (§5) — lower than the two-storey houses that surround the property on three sides.

Massing. Each cottage reads as three simple gabled masses stepping back from the lane. Seen from the lane, left to right: a **two-car garage wing 21'-0" wide** under a front-facing **6:12** gable; then the **living wing 17'-0" wide** under a front-facing **8:12** gable; and behind both of them the **main body**, an 8:12 gable whose ridge runs left-to-right, parallel to the lane. The two front-facing gables and the long ridge behind them give the small house a hip-and-gable silhouette rather than the single blank ridge of a builder box, and each front-facing gable carries a single 2'-0" × 1'-4" louvered vent centred in the field with a cedar knee-brace bracket under each end of the living-wing rake.

The porch is the front of the house. The living-wing gable is **carried forward 6'-0" over the porch** — an integral gable porch, not an add-on. The porch is **17'-0" wide × 6'-0" deep = 102 SF**, with a flat beadboard ceiling at 9'-0", carried on three **8×8 nominal Western Red Cedar posts** standing on 20-in-square masonry piers 2'-0" tall with sloped cast-stone caps, with a painted header beam between them. It is a porch two people can sit on, which is the point of the product; a 6'-0" depth is also the minimum the applicant offers as voluntary condition 8.

The garage is subordinate, and can be shown to be. The garage door sits **26'-8" from the front lot line**, its face **5'-0" behind the front wall of the dwelling**, which appears consistent with Lilburn Zoning Ordinance 2023-603 Table 4.2 ("if they must be located in the front of the dwelling, garages must be recessed back at least five feet behind the front wall of the dwelling so that it is subordinate rather than dominant when viewed from the city street on which the dwelling fronts"). The recess alone would not have achieved that: at the 8:12 pitch first drawn, the 21'-0"-wide garage gable would have risen to **16'-6"** above the finished floor and stood **1'-4" above** the 15'-2" ridge of the house's own living-wing gable — dominant, exactly what Table 4.2 forbids in words. Dropping the garage roof to **6:12** brings its ridge to **14'-9"**, which is **0'-5" below** the living-wing gable. The garage is now subordinate in plan *and* in silhouette (§5.2).

Grade and entry. The finished floor sits **0'-8" above finished grade** at the front porch, not the 1'-6" a slab house usually carries. That is a deliberate accessibility decision: the 4'-0"-wide entry walk runs **15'-8"** from the lane to the porch slab, so a 0'-8" rise is a slope of **1:23.5** (4.3 %) — flatter than the 1:20 at which a walking surface becomes a ramp under ICC A117.1-2017 §403.3 (used voluntarily; the Georgia Accessibility Code, Ga. Comp. R. & Regs. 120-3-20, does not reach detached single-family dwellings) and short of a ramp under 2024 IRC (GA 2026) R311.8. At 0'-8" the porch is also far below the 30-in height at which 2024 IRC (GA 2026) R312.1.1 would require a guard, so the porch stays open to the street rather than being fenced by a code-driven railing. **Every cottage has a zero-step entry at the front porch and at the garage-to-house door**, and Plan B adds a third at the recessed rear porch.

Materials. Walls are fiber-cement — 7-1/4-in lap at a 6-in exposure, or vertical panel with 12-in-o.c. battens — over a **masonry water table 2'-6" (30 in) high**, adhered ledge stone or modular brick by scheme, capped with sloped cast stone. The porch piers are the same masonry as the water table, so the base of every house reads as one continuous plinth. Roofs are 30-year architectural asphalt shingles with a ridge vent, 5-in K-style aluminium gutters and round downspouts. Trim is 5/4 painted fiber-cement or cellular PVC: 4-in flat window casings, corner boards and a frieze. Windows are white vinyl single-hung units with a flat sill, 2-over-2 simulated divided lites on Scheme 1 and 1-over-1 with no grilles on Schemes 2 and 3. The garage door is a carriage-style insulated sectional, 16'-0" × 7'-0", with a single top row of lites. Full specifications are in docs/12-outline-specifications.md; the schedule keyed to it is §4 below.

Rhythm along the lane. The two plans alternate lot by lot (A-B-A-B) on both sides of the lane — **21 Plan A and 20 Plan B** (data/layout.json metrics.plan_mix) — and the three colour schemes rotate on a three-lot cycle over the top of that alternation. Because 2 and 3 have no common factor, the combined pattern repeats only every **six** lots and produces six visually distinct front elevations (§4.2). At a 50'-0" lot width that means a person walking the lane passes 300 ft of frontage before seeing the same house twice — the opposite of the repetition that the 2025 filing's neighbours objected to.

What the community is not. It is not a gated development (voluntary condition 6). It has no on-lane parking, no individual curbside mailboxes, no overhead utility poles or wires inside the site, and no cul-de-sac bulb — turnarounds are hammerheads, which Table 4.2 prefers over the cul-de-sac it says is "not permitted except in the most unusual circumstances."

4. Colour and material schedule

4.1 The reconciled schedule

Item (11) requires "the color and materials of all structures and roofing." The table below is the single reconciled schedule for the development. Each row names the CSI section in docs/12-outline-specifications.md that specifies the product, and gives, for each of the three approved schemes, the material, the colour and the **hex value actually drawn** on Sheets A-2.1, A-2.2 and A-2.3 by tools/elevations.py (SCHEMES). Hex values are illustrative placeholders for a manufacturer's standard colour, to be replaced by the manufacturer's name and number when the Owner and Architect select from dropdowns (docs/12 §09 91 13 1.04).

Element	CSI section (docs/12)	Scheme 1 — "Warm White"	Scheme 2 — "Sage"	Scheme 3 — "Clay"
Walls (field)	07 46 46 Fiber-Cement Siding	Fiber-cement vertical panel with 12-in-o.c. battens (board-and-batten), warm white — #F1EDE3	Fiber-cement lap, 7-1/4-in plank at 6-in exposure , sage green — #98A189	Fiber-cement lap, 6-in exposure , clay / beige — #D7C8AD
Trim, corner boards, frieze, fascia	06 20 00 Finish Carpentry; 07 46 46; 09 91 13	5/4 painted trim, warm white — #FFFFFF	5/4 painted trim, soft white — #F7F5EF	5/4 painted trim, cream — #F5EEDC
Water table (30 in) and porch piers	04 40 00 Stone Assemblies (schemes 1, 3); 04 20 00 Unit Masonry (scheme 2)	Adhered ledge stone , gray-tan "Appalachian" blend, dry-stack — #ABA495	Modular brick veneer , ASTM C216 Grade SW Type FBS, red-brown blend, running bond, tooled concave joints — #8C5A4A	Adhered ledge stone , buff / tan blend — #C4B594
Water-table and pier cap	04 40 00 §2.02 (ASTM C1364 cast stone, 2-1/2 in, sloped with drip); 04 20 00 §2.04 (rowlock alternate)	Cast stone, buff or gray	Brick rowlock or cast stone	Cast stone, buff
Roofing	07 31 00 Shingles — 30-year laminated architectural asphalt, Class A, ASTM D3462, algae-resistant, ridge vent	Charcoal — #4A4E52	Weathered wood — #6B6255	Driftwood / brown — #7A6E60
Front entry door	08 14 00 / 08 16 13 — insulated fiberglass, 3'-0" × 6'-8", glazed upper panel, ≤ 1/2-in bevelled sill	Deep green — #2E4A38	Oxblood (natural stain alternate) — #6E2B2B	Navy — #23364F
Garage door	08 36 13 Sectional Doors — carriage-style insulated steel, 16'-0" × 7'-0", R-12 min., top row of lites	Painted to match trim , warm white — #F4F1E9	Painted to match siding , sage — #98A189	Painted to match trim , cream — #F5EEDC
Windows (frame and sash)	08 53 13 Vinyl Windows — uPVC single-hung, low-E argon IGU, 4-in flat casing, sloped sill	White , 2-over-2 SDL — #FFFFFF	White , 1-over-1, no grilles — #FFFFFF	White , 1-over-1, no grilles — #FFFFFF
Porch posts and brackets	06 20 00 §2.02 — 8×8 nominal Western Red Cedar on standoff bases; 4×6 / 4×8 cedar brackets	Cedar, natural semi-transparent stain — #A98C5F	Cedar, natural stain — #A98C5F	Cedar, natural stain — #A98C5F
Porch ceiling	06 20 00 / 07 46 46 — 1/4-in fiber-cement beadboard, flat at 9'-0"	Pale sky #DCE5E8	Warm white	Cream
Gutters and downspouts	07 71 00 Roof Specialties — 5-in K-style aluminium, round downspouts	Match trim, warm white	Match trim, soft white	Match trim, cream

Distribution of materials, not just colour. Scheme 1 is the board-and-batten scheme and Schemes 2 and 3 are lap-siding schemes; Scheme 2 is the only brick scheme. The three schemes therefore differ in **texture and base material**, not merely in paint colour, which is what makes the six-lot rhythm read from the lane rather than from the paint chip.

Roofing across structures. The roofing product is identical on every dwelling, on the clubhouse and on the mail kiosk; only the colour changes with the scheme (note printed on Sheet A-2.3). The clubhouse, the mail kiosk and the monument sign **are drawn**

in **Scheme 1 "Warm White"** on Sheet A-3.0 (tools/amenity.py SCHEME, taken from tools/elevations.py SCHEMES['1']) with the Scheme 1 deep-charcoal door of docs/12 §09 91 13), so the amenity block reads as one building group at the entrance. The intent this section previously recorded as undrawn is now on the sheet.

4.2 How the three schemes are distributed

Voluntary condition 8 offers "three approved color schemes." Sheet A-2.3 carries the distribution rule: **no scheme repeats on adjacent lots, no scheme faces itself across the lane, and no scheme is used on more than 40 % of the 41 lots.** The rule is administered by the homeowners association under the recorded covenants and is proposed as a stated element of condition 8.

The rule had to be re-cut when the count fell to 41, and this is why. Lots are numbered 1–14 up the south-west side and 15–41 up the north-east side (data/layout.json). The single-sequence rule this section previously stated — $\text{scheme} = ((\text{lot number} - 1) \bmod 3) + 1$ across both sides, with one hand exception at old lot 16 — no longer works: the two sides are displaced by different amounts because the north-east side runs past three open-space breaks that the south-west side does not, so the across-lane pairs are (1,15) ... (10,24) and then **(11,29), (12,30), (13,31), (14,32)**. Under the single sequence the first ten pairs are clear and **all four of the last pairs collide**. Rotating the north-east side two steps removes every collision with no exception at all:

South-west side, lots 1–14: $\text{scheme} = ((n - 1) \bmod 3) + 1$ North-east side, lots 15–41: $\text{scheme} = ((n + 1) \bmod 3) + 1$ — the same three-lot cycle, offset by two

Checked against the lot polygons in data/layout.json:

- **No two abutting lots on the same side of the lane receive the same scheme** — consecutive numbers on a side always differ by one step of the cycle. Zero repeats on either side.

- **No cottage faces its own scheme across the lane.** Zero collisions across all fourteen facing pairs, and no exception is needed — an improvement on the 43-lot rule, which needed one.

- Resulting counts: **Scheme 1 on 14 lots, Scheme 2 on 14, Scheme 3 on 13.** The largest is 14 of 41 =

34.1 %, against the 40 % cap ($0.40 \times 41 = 16.4$, i.e. 16 lots). At 43 lots the same rule gave 14 / 15 / 14 and 34.9 %, so the margin against the cap has widened slightly.

- Crossed with the A-B-A-B plan alternation, the six plan-and-scheme combinations occur **6 to 8 times each**

(A-1 on 8 lots, A-2 on 7, A-3 on 6, B-1 on 6, B-2 on 7, B-3 on 7) and the streetscape pattern repeats only every six lots (300 ft of frontage).

- **Stated plainly, because a reviewer will notice it:** every across-lane pair is the *same* plan (the facing lot numbers differ by 14 or 18, both even, so the A-B alternation puts the same plan opposite itself). The colour offset is therefore the only thing distinguishing a facing pair, which is precisely what the two-step rotation is for.

MUST BUILD before filing: a colour-key plan — the master plan tinted by scheme, one colour per lot — belongs on Sheet A-2.3. data/layout.json carries no scheme field on the lot records today, so the key must be generated (the rule above is deterministic and can be written straight into tools/siteplan.py), not drawn by hand.

4.3 Reconciliation — where the specification and the drawn palette differ

docs/12-outline-specifications.md and the palette drawn by tools/elevations.py were written independently and do not agree everywhere. The table in §4.1 resolves each difference; the resolutions and the reasons are recorded here so the outline specification can be conformed rather than left in conflict. **The drawn palette governs**, because item (11) requires the *drawings* to carry the colour and because voluntary condition 8 binds the applicant to "elevations substantially as shown."

#	Element	docs/12 says	The drawings show	Resolution
1	Front door	§08 14 00 2.04 and §09 91 13 2.04: scheme 1 deep charcoal #3B3F44, scheme 2 near-black #2B2B2B, scheme 3 deep green #2F4F3E	Scheme 1 deep green #2E4A38, scheme 2 oxblood #6E2B2B, scheme 3 navy #23364F	Drawings govern. Conform docs/12 §08 14 00 2.04 and §09 91 13 2.04. Rendering R-01, already generated and in renderings/, carries the deep-green Scheme 1 door
2	Roof colour, schemes 2 and 3	§07 31 00 2.04: scheme 1 charcoal, scheme 2 weathered wood, scheme 3 driftwood/brown — agrees with the drawings	Same	No change to docs/12. But data/plans.json materials_by_scheme has schemes 2 and 3 reversed (it gives scheme 2 "charcoal" and scheme 3 "weathered wood"). Conform tools/floorplans.py materials_by_scheme
3	Window frame colour, scheme 3	§08 53 13 2.04 offers "bronze/black laminate (scheme 3)"	White on all three schemes	White on all three. One frame colour across the community simplifies the HOA's replacement stock; conform docs/12 §08 53 13 2.04

#	Element	docs/12 says	The drawings show	Resolution
4	Porch posts, scheme 2	\$06 20 00 2.04: cedar natural stain on schemes 1 and 3, "painted trim color on scheme 2 as selected"	Natural-stain cedar on all three	Natural stain on all three. The cedar post and bracket are the character element the three schemes share; conform docs/12 \$06 20 00 2.04
5	Field and trim hex values	\$09 91 13 2.04: scheme 1 field #F3EFE6 / trim #FAF8F3; scheme 2 field #8E9C86 / trim #F6F3EC; scheme 3 field #CDB89A / trim #F3EEE2	#F1EDE3 / #FFFFFF; #98A189 / #F7F5EF; #D7C8AD / #F5EEDC	Differences are within one drawdown step and are immaterial at hearing. Both files state the hex values are placeholders. Conform at colour selection, once, from the manufacturer's drawdowns
6	Batten spacing	\$07 46 46 2.02: battens at 12 in o.c., "16 in o.c. at the Architect's option"	12 in o.c.	12 in o.c. , as drawn and as the renderings were generated

5. Height

5.1 The numbers, and how each is derived

Every ridge height below is computed, not asserted. The formula used by both tools/floorplans.py and tools/elevations.py is:

ridge above finished floor = plate + heel + (span ÷ 2) × pitch, and ridge above finished grade = that + the finished-floor height above grade.

Inputs, from data/plans.json roof and tools/elevations.py:

Input	Value	Source
Wall plate	9'-0"	plans.json roof.plate_ft = 9.0; docs/12 \$06 10 00 1.01
Truss heel, as drawn	0'-6"	plans.json roof.heel_ft = 0.5
Eave height above finished floor	9'-6" (9'-0" + 0'-6")	derived
Finished floor above finished grade at the front porch	0'-8"	tools/elevations.py FF = 0.667
Main and living-wing pitch	8:12	plans.json roof.elements
Garage pitch	6:12	plans.json roof.elements (revised 2026-09-03 — \$5.2)
Voluntary maximum ridge	24'-0" above grade	data/voluntary-conditions.md condition 3
District maximum height	40'-0"	Lilburn Zoning Ordinance 2023-603 Table 4.1, R-2, "Maximum height (feet) — 40"

Plan A "The Springbrook"

Roof element	Span	Pitch	Rise	Ridge above finished floor	Ridge above finished grade
Main gable (rear block), ridge parallel to the lane	25'-0"	8:12	8'-4"	17'-10"	18'-6"
Front-wing (living) gable, front-facing, carried over the porch	17'-0"	8:12	5'-8"	15'-2"	15'-10"
Garage gable, front-facing	21'-0"	6:12	5'-3"	14'-9"	15'-5"
Governing ridge				17'-10"	18'-6"

Plan B "The Laurel"

Roof element	Span	Pitch	Rise	Ridge above finished floor	Ridge above finished grade
Main gable (rear block), ridge parallel to the lane	28'-6"	8:12	9'-6"	19'-0"	19'-8"
Front-wing (living) gable, front-facing, carried over the porch	17'-0"	8:12	5'-8"	15'-2"	15'-10"
Garage gable, front-facing	21'-0"	6:12	5'-3"	14'-9"	15'-5"
Governing ridge				19'-0"	19'-8"

The Sheet A-3.0 structures (tools/amenity.py, same formula)

Structure	Span	Pitch	Plate + heel	Ridge above finished floor	Ridge above finished grade
Clubhouse — main gable, ridge parallel to the lane	40'-0"	6:12	10'-0" + 0'-10" = 10'-10"	20'-10"	21'-6"

Structure	Span	Pitch	Plate + heel	Ridge above finished floor	Ridge above finished grade
Mail-kiosk shelter — hip	18'-0" (16'-0" + 2 × 1'-0" overhang)	4:12	—	—	11'-7"
Monument entry sign — masonry monument including cast-stone cap	—	—	—	—	3'-9"

Against the two limits. The tallest thing in the development is the **clubhouse ridge at 21'-6" above finished grade** (Sheet A-3.0) — the tallest *dwelling* ridge is Plan B's at **19'-8" above finished grade**. Both are under the 24'-0" voluntary maximum of condition 3 — the clubhouse by 2'-6", Plan B by 4'-4" — and both are under the 40'-0" maximum height of Ordinance 2023-603 Table 4.1 for the R-2 district by 18'-6" and 20'-4" respectively. All three appear consistent with both limits with substantial margin. **VERIFY:** Ordinance 2023-603 does not define, in the excerpts retrieved (data/ordinance-excerpts.md), how the 40'-0" is measured — from average finished grade, from the lowest adjacent grade, to the ridge or to the mean roof height. This document reports the ridge above finished floor **and** above finished grade so that the answer is available under any of those readings; the measurement datum should be confirmed at the pre-application conference.

5.2 What was revised on 2026-09-03 (Rev B), and why

Both plans were revised on 2026-09-03. The revision is recorded here because the elevations, the renderings and the outline specification all depend on it, and because several older numbers in the package (including FACTS.md §4) are now superseded.

- The front porch is now roofed by carrying the front-wing gable forward, not by a separate 4:12 shed.** The 2026-08-28 program assumed a shed roof over the porch. It cannot be built: a full-width shed cannot tie under a gable-end rake. At 4:12 across a 6'-0" porch the shed's high side lands **2'-0" above the plate**, and the 8:12 rake of the wing above it is *below* that height for the outer **3'-0" at each end** of the 17'-0" wall — the shed would have to cut through the gable it is leaning on. Carrying the gable forward 6'-0" solves it, and costs nothing: the porch keeps its full programmed **17'-0" × 6'-0" = 102 SF**, keeps a flat 9'-0" ceiling, and **adds no height** (the wing's own ridge, 15'-2", is unchanged). The gable end above the porch is carried on a header spanning between the outer posts. (tools/elevations.py, *FRONT PORCH note*.)
- Window sills were raised to 2'-8" (32 in) above finished floor.** The masonry water table is 2'-6" (30 in) high, with its cast-stone or rowlock cap set so that the **top of cap lands at 30 in** above finished floor (docs/12 §04 20 00 3.02). A sill at the earlier height fell below the cap, which would have driven the window sill, its casing and its sill flashing into the masonry. At **2'-8"** the sill clears the top of the cap by **0'-2"**, which is room for the Z-flashing and the 1/4-in clearance that fiber-cement requires above horizontal flashing (docs/12 §07 46 46 2.03). The raised sill also stays far below the **44-in maximum** sill height for an emergency escape and rescue opening under **2024 IRC (GA 2026) R310.2.2**: at 32 in there is 12 in of margin.
- The garage gable was lowered from 8:12 to 6:12.** At 8:12 the 21'-0"-wide garage gable reached **16'-6"** above the floor, standing **1'-4" above** the 15'-2" ridge of the house's living-wing gable — the garage would have been the dominant mass in the front elevation, which is precisely what Ordinance 2023-603 Table 4.2 tells a designer to avoid ("subordinate rather than dominant when viewed from the city street"). At **6:12** the garage ridge is **14'-9"**, **0'-5" below** the living-wing gable. The 5'-0" plan recess of Table 4.2 is now matched by a subordinate silhouette. Neither plan's governing ridge changes, because the main gable governs in both.
- Finished floor was set at 0'-8" above finished grade** instead of the 1'-6" a slab house usually carries, so that the 4'-0" entry walk can rise from the lane to a flush landing at **1:23.5** and remain a zero-step entry rather than becoming a ramp (§3, Grade and entry). The lower floor also keeps the porch below the 30-in guard threshold of 2024 IRC (GA 2026) R312.1.1.
- Plan B was re-proportioned from 40'-0" wide to 38'-0" wide.** A 50'-0" lot less two 5'-0" side yards (Table 4.1) leaves 40'-0" of buildable width — and the **roof** has to fit in it, not just the walls. At an 8-in side overhang a 40'-0" body measures **41'-4" roof to roof** and does not fit; Ordinance 2023-603 contains no provision permitting a projection into a required yard (**VERIFY** at pre-application). At 38'-0" the roof measures **39'-4"**, leaving **0'-4" of clearance to each side-yard line**. Depth was increased to recover floor area. Plan A was already 38'-0" wide.

Consequently superseded: the "Plan A 38'-0" × 38'-0", ~1,444 SF / Plan B 40'-0" × 40'-0", ~1,600 SF" and "8:12 main gable, 4:12 porch" of FACTS.md §4; the "40'-0" × 57'-6", 1,595 SF, ridge 19'-0", 114-SF porch, 96-SF rear porch" of audit-2026-09-03/docs-consistency.md §4.4; and the Plan B dimension block of renderings/PROMPTS.md / prompts.json (generated 13:47 on 2026-09-03, before the 14:16 regeneration of data/plans.json).

Current authoritative areas (data/plans.json areas, computed from the wall-face polygons):

	Plan A "The Springbrook"	Plan B "The Laurel"
Body (bounding box of conditioned + garage, outside face)	38'-0" × 51'-10"	38'-0" × 57'-6"
Roof to roof across the front	39'-4"	39'-4"

	Plan A "The Springbrook"	Plan B "The Laurel"
Conditioned area	1,406 SF	1,492 SF
Garage, gross / clear	458 SF / 20'-4" × 21'-2" = 430 SF	504 SF / 20'-4" × 23'-4" = 474 SF
Covered front porch	102 SF (17'-0" × 6'-0")	102 SF (17'-0" × 6'-0")
Rear	Uncovered patio 72 SF (12'-0" × 6'-0") , its edge 20'-6" from the rear lot line — 0'-6" outside the buffer easement	Covered rear porch 84 SF (14'-0" × 6'-0"), recessed under the main roof, its edge 20'-10" from the rear lot line
Total under roof	1,966 SF	2,182 SF
Building coverage on a 5,000-SF lot	39.3 %	43.6 %

Both plans exceed the R-2 minimum heated floor area for a cottage home of **1,000 SF**, and both exceed even the **1,400 SF** required of "all other dwelling types," under Ordinance 2023-603 Table 4.1 — which appears consistent with that table.

The plans did not change when the lot count did. The 2026-09-03 site-plan correction removed two lots (one inside a stream buffer once all three digitised reaches were screened, one to enlarge the basins for the real 7.54 disturbed acres); it did not alter either house. What it changed here is the *mix*: 21 Plan A and 20 Plan B instead of 22 and 21.

5.3 Geometry items — status at 2026-09-03

Eight inconsistencies between files were found on 2026-09-03. **Four are now closed.** None of them changes the answer to the 24'-0" or 40'-0" limits.

#	Item	Status
G-1	data/plans.json roof.elements[0].span_ft for Plan B still carries 29'-6" , the rear-block span of the superseded 58'-6" body, and the file therefore still prints max_ridge_ft = 19.33 ft (19'-4" above the floor, 20'-0" above grade). On the current 57'-6" body the rear block measures 28'-6" and the ridge is 19'-0" / 19'-8" , which is what tools/elevations.py computes and what Sheet A-2.2 draws	OPEN — a 0'-4" disagreement between data/plans.json and the issued sheet. Conform tools/floorplans.py PLAN_B['roof']['elements'] main-gable span 29.5 → 28.5 and regenerate. The sheet governs and this document uses 19'-0" / 19'-8" , which is also what voluntary condition 3 states. Note that docs/10 §1 item 14 has this backwards — it says the sheets draw 19'-4"; they do not
G-2	tools/elevations.py MASSING['B'] carried the 58'-6" rectangles and check_massing() failed	CLOSED — check_massing() and check_width() now pass for both plans; A-2.1 and A-2.2 were regenerated 2026-09-03
G-3	Truss heel drawn at 0'-6" ; docs/12 §06 10 00 2.03 specifies raised-heel trusses ≥ 12 in at exterior walls, to carry full-depth R-38 over the top plate	OPEN. Conforming raises every eave and ridge by 0'-6": Plan A to 18'-4" AFF / 19'-0" above grade, Plan B to 19'-6" AFF / 20'-2" . Still 3'-10" under the 24'-0" condition. Decide once — 12-in heel and redraw, or amend docs/12 §06 10 00 2.03 to a 6-in heel and re-check the ceiling R-value path
G-4	Plan A's rear patio edge sat 19'-6" from the rear lot line, 0'-6" inside the 20'-0" buffer easement	CLOSED — the patio was drawn at 12'-0" × 6'-0" = 72 SF , its edge 20'-6" from the rear lot line, 0'-6" clear of the easement (data/plans.json lot_siting.patio_edge_from_rear_line_ft = 20.5). It is 12 SF smaller than the 84-SF patio the program brief called for; that is the price of staying out of the easement
G-5	The 12-in rear eave on Plan B reaches 19'-10" from the rear lot line, 0'-2" inside the buffer easement; the 12-in porch-roof overhang on both plans reaches 14'-8" from the front lot line, 0'-4" inside the 15'-0" front setback	OPEN. Ordinance 2023-603 has no projections-into-yards provision either way. VERIFY at pre-application. If staff require the roof outside the line, hold Plan B's body at 57'-0" and use an 8-in rear eave
G-6	data/plans.json Plan A note 2 still reads "4:12 porch shed" and note 5 still describes NFPA 13D sprinklers as "mitigation for the single access road, IFC 2024 (GA) App. D107.1"; Plan B note 5 still reads "at 43 units"	OPEN. All three are superseded: the porch is under the wing gable (\$5.2); at 41 units Georgia's D107.1 trigger of <i>more than 120</i> dwelling units is not engaged, so sprinklers are a voluntary offer (voluntary condition 5; Ga. Comp. R. & Regs. 120-3-3-.04). Conform Plan A's notes to Plan B's, and conform Plan B's unit count, in tools/floorplans.py
G-7	tools/elevations.py CAP_RIDGE comment cites the 24'-0" ridge cap as "voluntary condition 4"	OPEN — it is condition 3 ; condition 4 is the buffer condition. Correct the comment
G-8	renderings/PROMPTS.md and prompts.json carried the superseded Plan B block	CLOSED — regenerated 2026-09-03 at 15:21. The locked design basis they now carry reads 38'-0" × 57'-6", 1,492 SF, ridge 19'-0" AFF / 19'-8" above grade, 41 lots , open space 141,089 SF = 33.8 %

A ninth item, found while conforming this document: drawings/amenity-sheet.svg (Sheet A-3.0) carries a clubhouse note reading "**the 43 dwellings** are sprinklered under NFPA 13D". Correct tools/amenity.py and regenerate — a superseded unit count printed on a filed sheet is exactly the kind of discrepancy the 2025 filing was tabled over.

6. Sign schedule

6.1 The one sign proposed — a ground-mounted monument entry sign

Item	Proposed	Source / basis
Number	One, for the whole development	Voluntary condition 14
Type	Ground-mounted monument sign — a low masonry wall with pin-mounted dimensional letters. Not a pole sign, not internally illuminated, not changeable-copy, not electronic	docs/12 §10 14 00 2.02–2.03
Location	Arcado Road frontage, in the 10'-0" landscape strip, at local coordinates u –9.7 to –1.7 / v –171.9 to –167.9 — about 18 ft north-east of the entrance drive centreline (v = –190), inside the front amenity tract and 19.0 ft from the Arcado Road right-of-way , outside it	data/layout.json amenity.entry_sign; tools/amenity.py; voluntary condition 14
Face area — as drawn on A-3.0	30.0 SF , against the voluntary limit of 32 SF. The 8'-0" × 4'-0" rectangle in data/layout.json is the bounding footprint of the monument in plan , not the face: the monument is an 8'-0"-long × 1'-4"-thick masonry wall with a 1'-4" wing / planter return at each end, 4'-0" deep overall. The 30.0 SF is the whole structure face (8'-0" × 3'-9"), the most conservative of the three measurement methods in §6.3; the cast-stone copy panel inside it is 6'-0" × 1'-9" = 10.5 SF . Both faces carry copy; the area above is one face	tools/amenity.py (SG_LEN, SG_H, SG_PANEL); Sheet A-3.0 notes
Overall height — as drawn on A-3.0	3'-9" above finished grade, cap included , against the voluntary limit of 6'-0" — 2'-3" of margin	tools/amenity.py SG_H; Sheet A-3.0
Structure	Reinforced CMU core on a 3,000-psi concrete footing 12 in below grade (frost), sized for wind. A Georgia PE designs the footing if the sign exceeds 6 ft or 32 SF — it is designed not to	docs/12 §10 14 00 1.05, 2.02
Materials and colours	Base and body : adhered ledge stone (04 40 00) in the Scheme 1 gray-tan blend, matching the cottage water tables — stone is the default at the entry; brick (04 20 00) if the Owner selects the Scheme 2 palette for the amenity block. Cap : cast stone, ASTM C1364, 2-1/2 in, sloped with a drip, buff or gray. Copy panel : cast stone or 15-lb HDU, painted to match the trim warm white. Letters : 1/2-in cast-aluminium or 1/4-in flat-cut aluminium, 8–10-in cap height , pin-mounted on 1/2-in standoffs, sat in black or dark bronze	docs/12 §10 14 00 2.02, 2.04; §4.1 above
Copy	"Cottages at Arcado Springs". No tenant panel, no leasing copy, no phone number, no changeable copy	docs/12 §10 14 00 1.01
Illumination	External only — two ground-mounted, fully shielded LED floods at ≤ 3000 K (2700 K preferred), aimed at the face only, ≤ 0.1 fc spill at the property line, on a photocell and timer. No internal illumination, no exposed lamps, no halo	docs/12 §10 14 00 2.03; voluntary condition 11 (full-cutoff, ≤ 3000 K)
Sight distance — computed and drawn	Located wholly outside the 390-ft departure sight triangle (35 mph, Gwinnett County UDO §900-40 Table 900.2, AASHTO Case B1), struck from a driver eye 14.5 ft behind the near edge of the travelled way on the drive centreline. The sign stands 20'-6" behind that point and its nearest corner is 20.1 ft clear of the nearer sight line; no part of the sign or its bed lies within the triangle . Bed planting held below 2'-6". A field sight-distance survey and the Gwinnett DOT driveway permit govern — VERIFY	Sheet A-3.0 (tools/amenity.py SIGHT_CLEAR, SIGN_BEHIND, SIGN_IN_TRI = False); docs/12 §10 14 00 1.03, 2.03; Checklist §6.j
Landscaping	Planted bed at the base with an irrigation stub, kept below 2'-6" within the sight triangle	docs/12 §10 14 00 2.05; §32 93 00
Maintenance	Owned and maintained by the mandatory homeowners association	Voluntary condition 10
Permitting	Permitted separately . The plan set carries the Checklist §4.q note verbatim: " <i>All signs shall be permitted separately.</i> " A sign easement may be required	Checklist §4.q; voluntary condition 14

Both cautions this section carried on 2026-09-03 are answered by the sheet as drawn:

- **The area is safe under every measurement method.** The sign was drawn to **30.0 SF measured the most conservative way** — the whole structure face, base and cap included — so it survives whichever of the three methods in §6.3 the ordinance turns out to use. The copy panel alone is 10.5 SF. It remains **VERIFY** until the governing article is identified, but the number no longer sits exactly on the limit.
- **The plan rectangle is understood.** The 8'-0" × 4'-0" rectangle in data/layout.json is the bounding footprint, and Sheet A-3.0's plan resolves it into a 1'-4"-thick wall with 1'-4" wing returns in a landscaped bed. That resolved footprint should be carried onto **C-2.3** when the entry enlargement is drawn.

6.2 No wall signs are proposed

No wall signs of any kind are proposed anywhere in the development — not on any of the 41 dwellings, not on the clubhouse, not on the mail-kiosk shelter, and not on any accessory structure, fence or wall. There is no wall sign to locate and no wall-sign area to state, which disposes of the fourth requirement of item (11).

This statement is printed as a general note **on each elevation sheet** (A-2.1, A-2.2 and A-3.0) and on the material and colour board (A-2.3), in this form:

WALL SIGNS: none are proposed on any dwelling, on the clubhouse or on any accessory structure. One ground-mounted monument entry sign not exceeding 32 SF in face area and 6'-0" in height is proposed at the Arcade Road entrance — see sheet A-3.0 for its elevation, dimensions, materials, colours and sight-triangle clearance.

It is also carried as **voluntary condition 8**, which now ends "...three approved color schemes; **no wall signs on any dwelling.**"

Two things that are not signs, and are proposed:

1. **Premises identification.** Address numerals on each dwelling, and a separate address-range sign at the entrance, 4-in minimum height in a contrasting colour, per the Gwinnett County Fire Marshal (docs/12 §10 14 00 2.05). The Site Development Plan Review Checklist §4.r notes that "one street name may serve to assign all house numbers." House numbers are premises identification required for emergency response, not advertising signage. **VERIFY** whether the ordinance's sign article excludes address numerals, as sign codes normally do.
2. **Traffic-control and regulatory markers** inside the private lane — stop sign at the Arcade Road entrance, "NO PARKING — FIRE LANE" signs to IFC 2024 (GA) Appendix D103.6 if the Fire Marshal requires them, and the accessible-space sign at the van-accessible stall in the amenity bay. These are regulatory devices, not on-premises advertising.

6.3 The ordinance question that must be asked at pre-application

data/ordinance-excerpts.md — the working extract of Lilburn Zoning Ordinance 2023-603 — covers §§308–319, §325, §§401–402, Tables 4.1 and 4.2, §602, §603, §§705–706, §734, Article 8 and §§1003–1005. **It contains no sign regulations.** The only occurrence of "sign" in a regulatory sense is §1003-6.2, the *public-hearing notice* sign the Director posts on the property 15 to 45 days before the hearing — which has nothing to do with a subdivision entrance sign. docs/12 §10 14 00 1.03 flags the same gap.

The applicant therefore cannot cite a section for the 32-SF area and 6'-0" height it has offered. Those figures are the applicant's own voluntary limits (condition 14), not a recitation of the code. This must be put to the Planning Director at the pre-application conference as a numbered question:

PRE-APPLICATION QUESTION (signs). Which article of Ordinance 2023-603 (or of the Lilburn Code of Ordinances) governs a residential subdivision entrance sign, and what does it set for (a) maximum sign area and **the method of measuring it** — copy panel, panel including border, or the entire monument face including base and cap; (b) maximum height above grade; (c) minimum setback from the right-of-way and from a side property line; (d) permitted illumination; (e) the number of entrance signs permitted per development; and (f) whether a sign easement is required (Checklist §4.q)? Are address numerals and traffic-control devices excluded from the definition of "sign"?

Until that is answered, every sign figure in this package is stated as a **voluntary limit the applicant offers**, not as a statement that a code requirement is met, and the sign is drawn to be defensible on either reading. **VERIFY.**

7. Rendering index

7.1 What the renderings are, and what they are not

Item (11) says "Visual imagery **may** be used." The renderings are the optional half of the requirement; the **scaled elevations are the submittal**, and the perspectives illustrate them.

Every image in renderings/ is an **illustrative AI-generated perspective prepared by the owner-applicant from the scaled elevations**. No image is a photograph, none depicts an existing building, none depicts a real neighbouring house or a recognisable person, and **none is offered as a representation of what will be built to the inch**. Where an image and a drawing disagree, **the scaled drawing governs** — that is stated on the sheet, in the caption under every image, and here.

The method is deliberate and is worth stating to the Commission: each perspective was generated against a **reference elevation exported from the same geometry file that generates the scaled drawings** (renderings/reference/ref-plan-*.png, written by tools/elevations.py), and every prompt states the real dimensions in architectural notation, so a returned image can be **measured against the plan set and rejected**. That is why voluntary condition 8 — "elevations substantially as shown" — can be offered at all: the images cannot show something the plan set does not permit without the discrepancy being visible.

7.2 Caption

Each image is captioned, burned into the sheet or set immediately beneath the image, in this form (plan name and scheme varying):

Illustrative concept rendering — AI-generated by the owner-applicant from the scaled elevations. Not a photograph; not an existing building. Plan A "The Springbrook", Scheme 1 "Warm White". Colours and materials per

docs/12-outline-specifications.md. **See Sheets A-2.1 / A-2.2 / A-3.0 for scaled elevations, which govern. DRAFT — to be superseded by sealed drawings by a Georgia-registered architect.**

(Sheet numbers in the caption are those of the settled register in tools/transmittal.py SHEETS: A-2.1 = Plan A elevations, A-2.2 = Plan B elevations, A-3.0 = clubhouse, mail kiosk and entry sign.)

7.3 The index

File names are those in renderings/prompts.json, generated by tools/prompts.py and re-generated on 2026-09-03 against the 41-lot layout. The eight images marked **core** are the item-(11) set — each side of each structure visible from a street, plus the sign, plus the colour and materials board. The remaining four answer specific objections raised against the 2025 filing on this property. **All twelve files are present in renderings/ as of 2026-09-03**, and each is listed below with the file name on disk.

Ref	File	What it shows	Item-(11) role	Aspect	Status (2026-09-03)
R-01	R-01_p lan-a- front- scheme 1.jpg	Plan A "The Springbrook" front elevation from the lane , Scheme 1 Warm White, on a 50'-0" × 100'-0" lot with the 20'-0" × 26'-8" driveway	core — the street-visible front of the more common dwelling; shows one story, the 5'-0" garage recess (Table 4.2), the colours and materials	3:2	Generated
R-02	R-02_p lan-b- front- scheme 2.jpg	Plan B "The Laurel" front elevation from the lane , Scheme 2 Sage lap over brick	core — the second dwelling type, the second scheme, the only brick base	3:2	Generated (2026-09-03 14:26, after G-2 and G-8 closed)
R-03	R-03_p lan-b- rear-g arden- scheme 3.jpg	Plan B rear / garden view , Scheme 3 Clay, the 14'-0" × 6'-0" covered rear porch recessed under the main roof, the rear yard running into the 20-ft undisturbed buffer	core — the rear elevation, which on this site faces an occupied R-1 back yard on every lot	3:2	Generated (2026-09-03 14:22, after G-2 and G-8 closed)
R-04	R-04_l ane-st reetsc ape-nw .jpg	Lane streetscape looking north-west: 22'-0" pavement, a 5'-0" sidewalk (on the north-east side for the lane's full length, and on the south-west side as well from u = 530 ft), the 50-ft lot rhythm with alternating Plan A / Plan B and rotating colour schemes	core — "proper perspective" for the street-visible fronts collectively; the answer to the "single-family character" objection	16:9	Generated — the prompt describes "the sidewalk on one side", which is true only of the front 530 ft; acceptable for a view down that reach, but say so if the image is presented as typical
R-05	R-05_h ammerh ead-po cket-g reen.j pg	A hammerhead (T) turnaround, two 20'-0" × 60'-0" legs, with the 50'-0" × 100'-0" pocket greens either side	supporting — shows no cul-de-sac bulb (Table 4.2) and the fire-apparatus turnaround	16:9	Generated
R-06	R-06_e ntranc e-arca do-mon ument. jpg	The Arcado Road entrance with the monument sign : 24'-0" drive, 25-ft north-east and 15-ft south-west curb returns, the sign, the 10'-0" landscape strip, the 5'-0" frontage sidewalk, the village green beyond	core — the sign is a "structure" and item (11) requires its location and size to be shown	16:9	Generated — re-check against Sheet A-3.0, which draws the sign at 8'-0" × 3'-9" (30.0 SF) and 3'-9" high , lower than the "not exceeding 6'-0"" the prompt described
R-07	R-07_c lubhou se-vil lage-g reen.j pg	The clubhouse and village green : the 60'-0" × 40'-0" one-story 2,400-SF clubhouse across the 14,354-SF green, with the 16'-0" × 10'-0" mail kiosk and its 4 short-term spaces in frame	core — clubhouse and kiosk are street-visible structures needing colour and material imagery	3:2	Generated — re-check against Sheet A-3.0 , which fixes the clubhouse ridge at 21'-6" above grade, a 6:12 roof and a 400-SF covered porch
R-08	R-08_p ickleb all-co urts.j pg	The two 20'-0" × 44'-0" courts on 30'-0" × 60'-0" fenced pads, 10-ft black chain link, no light poles	supporting — the amenity at its real size, honouring voluntary condition 11 (no court lighting)	3:2	Generated
R-09	R-09_a erial- obliqu e-nw.j pg	Aerial oblique of the whole strip from over Arcado Rd along the N 28°43' W axis: one entrance in the south-west third, 41 one-story roofs in two rows (27 north-east / 14 south-west), one lane, 3 hammerheads, 3 pocket greens, 2 dry basins, the creek woods, the 20-ft treed buffer all round	core — "proper perspective" at site scale; shows the true 236–246 ft × ~1,750 ft proportion and that nothing touches the neighbours' lines	16:9	Generated (2026-09-03 15:24, against drawings/mcp-sheet.png)

Ref	File	What it shows	Item-(11) role	Aspect	Status (2026-09-03)
R-10	R-10_b uffer- from-n eighbo ur.jpg	Eye-level view from an adjoining back yard , across the 20'-0" undisturbed buffer, to the rear of a cottage beyond the rear lot line	supporting — the exhibit that answers the "looming"/character objection	3:2	Generated — recheck the stated rear-wall setback (Plan A 26'-6", Plan B 20'-10" from the rear lot line) and ridge height against §5.1
R-11	R-11_c reek-w oods-d ry-bas in.jpg	The preserved creek-woods tract at the stream head with its 50-ft undisturbed buffer, and Pond 2 as a grassed dry basin	supporting — answers the stormwater / erosion objection and the Jackson Creek 303(d) water-quality point	16:9	Generated
R-12	R-12_m ateria ls-col our-bo ard.jp g	Flat-lay materials and colour board : three columns (Schemes 1 / 2 / 3), each with siding, trim, shingle, masonry base, cast-stone cap, cedar post, door and window colour	core — item (11)'s literal requirement, "the colour and materials of all structures and roofing," in one filable exhibit; labels typeset in post, never asked of the image model	4:3	Generated — must be checked against the reconciled schedule in §4.1 (door colours, scheme 2/3 roof colours)

Index sheet — TO BUILD. A one-page contact index, R-00_index.jpg, listing the twelve images and their subjects, is to be placed at the front of the exhibit set, which is bound **behind** the scaled elevation sheets — so that a reader reaches the drawings first and the pictures second. **It has not been laid out; renderings/ holds R-01 through R-12 and no R-00.**

Reference elevations. renderings/reference/ holds the twelve title-block-free coloured elevations exported by tools/elevations.py (ref-plan-a-front-scheme1/2/3, ref-plan-a-rear/right/left-scheme2, and the same six for Plan B). These are working files for image generation and are **not** part of the submittal.

7.4 Acceptance rule

An image goes into the package only after it is checked against its fidelity list in renderings/PROMPTS.md. If it fails, it is **re-generated** — the prompt's dimensions are never edited to match the image. **All twelve images now exist**, and the three that were outstanding on 2026-09-03 (R-02, R-03 and R-09) were generated after G-2 and G-8 closed, so R-02 and R-03 illustrate the 57'-6" Plan B drawn on Sheet A-2.2, and R-09 shows 41 roofs. **Two acceptance checks remain open:** R-06 and R-07 were generated before Sheet A-3.0 fixed the sign at 3'-9" / 30.0 SF and the clubhouse at a 6:12 roof with a 400-SF covered porch, and must be re-checked against that sheet and re-generated if they fail.

8. Supersession, seals and status

These are DRAFT illustrative drawings prepared by the owner-applicant. They were produced with AI drafting tools from a dimensioned geometry model, they are drawn to scale, and they are internally consistent with the floor plans and the site plan in this package. **They are not sealed, and they are not construction documents.**

- **They are to be superseded, in their entirety, by drawings prepared and sealed by a Georgia-registered architect before any building permit is issued** for any dwelling, the clubhouse, the mail kiosk or the monument sign. docs/10-submittal-checklist-and-roadmap.md carries the engagement and the budget for that work.
- Every sheet carries, in its title block: *"ILLUSTRATIVE — NOT FOR CONSTRUCTION · DRAFT, to be superseded by sealed drawings by a Georgia-registered architect,"* the AI-analysis disclaimer, and the preparer's name.
- Structural elements shown or implied — trusses, headers, porch beams and posts, the sign footing, the masonry veneer support and ties — are **presumptive** until designed by a Georgia Professional Engineer. Truss design drawings sealed by a Georgia PE are required (docs/12 §06 10 00 1.04).
- The site geometry these elevations sit on — lot lines, setbacks, the buffer easement, the sight triangles — is **DRAFT** until the boundary and topographic survey is sealed by a Georgia Registered Land Surveyor and the civil sheets are sealed by a Georgia PE.
- Nothing in this document is an opinion that a code or ordinance requirement is met. Every regulatory statement is "appears consistent with," and the interpretations this package rests on — chiefly the §313(1) reading that lets the 20-ft buffer coincide with the rear yard inside the lot — are offered as interpretations to be confirmed by the Planning Director, not as settled law.
- What the applicant asks the Commission and Council to bind is **voluntary condition 8**: one story; the materials and the three schemes scheduled in §4; front-loaded garages recessed at least 5'-0"; covered front porches at least 6'-0" deep; zero-step entries; elevations substantially as shown; and **no wall signs on any dwelling**. Together with condition 3 (one story, ridge not exceeding 24'-0" above finished grade — the plans are drawn at 18'-6" and 19'-8", the clubhouse at 21'-6") and condition 14 (one monument sign not exceeding 32 SF and 6'-0" — drawn at 30.0 SF and 3'-9"), those conditions are the enforceable content of this document; the drawings and images illustrate them. **Condition 8's sheet citations were conformed to the**

register on 2026-09-03 (§1.1).

9. Open items — what must be closed before this document is filed

The 2026-09-03 site-plan correction, the issue of Sheet A-3.0 and the conforming of the voluntary conditions closed six of the twelve items this section carried. The eleven that remain:

#	Item	Owner	Section
1	Re-number two generated title blocks — amenity-sheet.svg prints "A-3" and cross-refers to "Sheet C-1"; existing-conditions.svg titles itself C-0 and cross-refers to "C-1, MASTER CONCEPT PLAN". Correct tools/amenity.py and tools/existing_conditions.py and regenerate	Drawings / tools	§1.1
2	Correct the "43 dwellings" note on Sheet A-3.0 (tools/amenity.py clubhouse sprinkler note) to 41	Drawings / tools	§5.3
3	Draw the entry enlargement C-2.3 and carry onto it the resolved sign footprint (a 1'-4"-thick wall with 1'-4" wing returns, not the 8'-0" × 4'-0" bounding rectangle) and the sight triangle A-3.0 already computes	Civil	§2.4, §6.1
4	Generate the colour-key plan on A-2.3 from the deterministic rule in §4.2 (SW lots $((n-1) \bmod 3)+1$, NE lots $((n+1) \bmod 3)+1$); add a scheme field to the lot records in data/layout.json	Site plan	§4.2
5	Conform docs/12 on the six reconciliation points — door colours, window frame colour, porch-post finish, batten spacing, hex values; and conform tools/floorplans.py materials_by_scheme, whose scheme 2 and 3 roof colours are reversed	Specifications	§4.3
6	Close geometry items G-1, G-3, G-5, G-6 and G-7 ; G-2, G-4 and G-8 are closed	Drawings / tools	§5.3
7	Re-check R-06 and R-07 against Sheet A-3.0 — both were generated before the sheet fixed the sign at 3'-9" / 30.0 SF and the clubhouse at 6:12 with a 400-SF covered porch — and lay out the R-00 contact index , which does not exist	Renderings	§7.3, §7.4
8	Ask the sign question at the pre-application conference and record the answer. The sign is now drawn at 30.0 SF measured the most conservative way, so it is defensible on any of the three methods, but the governing article is still unidentified	Owner	§6.3
9	Confirm the height-measurement datum for the 40'-0" maximum of Table 4.1	Owner / Planning Director	§5.1
10	Architect's review and seal of A-1.1, A-1.2, A-2.1, A-2.2, A-2.3, A-3.0, A-4.0 and A-4.1. The scope is now review and seal, not origination	Architect	§8
11	Draw the pickleball fence section on C-8.0 — the only item-(11)-adjacent content not yet on a sheet	Civil	§2.3, §2.4

Closed since the earlier 2026-09-03 revision of this document: the sheet-numbering conflict, now settled in tools/transmittal.py SHEETS and carried into voluntary condition 8 (§1.1); the clubhouse, mail kiosk and monument sign, all now drawn on A-3.0 (§2.3, §2.4); the Plan B massing failure G-2 and the Plan A rear-patio buffer encroachment G-4 (§5.3); the stale rendering prompts G-8 and the three outstanding perspectives R-02, R-03 and R-09 (§7.3).

Disclaimer: This is an AI-generated analysis for preliminary planning purposes. All findings must be verified by a licensed professional before use in design, permitting, or regulatory submissions.

PLAN A — THE SPRINGBROOK

2 BR / 2 BA, 1 story, 2-car front-loaded garage recessed 5'-0", covered front porch, rear patio

AREA TABULATION — computed by tools/floorplans.py from the wall-face polygons

Conditioned (heated) area	1,406 SF	outside face of ext. walls
2-car garage	458 SF	outside face / centreline
Covered front porch	102 SF	slab
TOTAL UNDER ROOF	1,966 SF	cond. + garage + porch
Rear patio (uncovered, not under roof)	72 SF	slab

Program target 1,444 conditioned (FACTS §4 nominal 38'-0" × 38'-0"); Lilburn Table 4.1 R-2 cottage min heated 1,000 SF.

ROOM SCHEDULE (clear, inside face)

2-CAR GARAGE	20'-4" × 21'-2"	430 SF
HALL	3'-6" × 26'-4"	92 SF
BEDROOM 2	12'-2" × 10'-10"	148 SF
CLOSET	5'-6" × 2'-2"	12 SF
BATH 2	12'-2" × 7'-10"	95 SF
COAT	2'-8" × 4'-4"	12 SF
PANTRY	9'-2" × 4'-4"	40 SF
LAUNDRY / MUD	6'-8" × 10'-4"	69 SF
KITCHEN / DINING	16'-4" × 10'-6"	172 SF
W.I.C.	13'-4" × 5'-4"	71 SF
PRIMARY BATH	13'-4" × 8'-2"	109 SF
PRIMARY BEDROOM	13'-4" × 10'-2"	136 SF
SUITE HALL	4'-0" × 7'-2"	29 SF
GREAT ROOM	19'-4" × 13'-10"	289 SF

ROOF / HEIGHT

main gable (rear block): span 25'-0", 8:12 → ridge 17'-10" above FF (plate 9'-0" + heel + rise)

front-wing gable (extended over the porch): span 17'-0", 8:12 → ridge 15'-2" above FF (plate 9'-0" + heel + rise)

garage gable: span 21'-0", 6:12 → ridge 14'-9" above FF (plate 9'-0" + heel + rise)

Max ridge 17'-10" ≤ 24 ft program cap; Ord. 2023-603 Table 4.1 R-2 max height 40 ft — appears consistent.

ACCESSIBILITY CHECKLIST (voluntary)

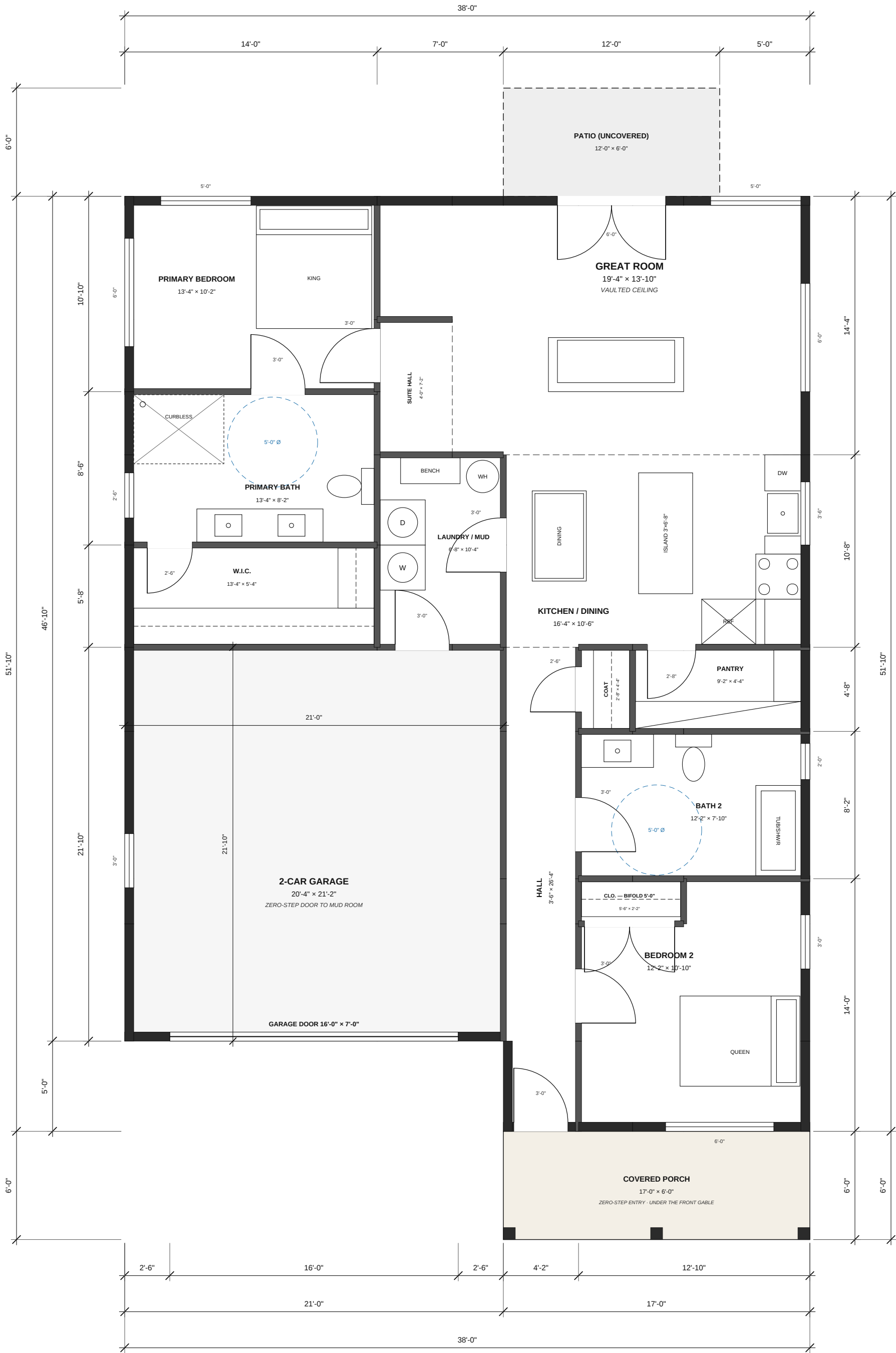
- Zero-step entries — front porch (slab flush with finished floor) and garage-to-mud-room door, patio door ≤ 1/2-in threshold [design intent]
- Door clear width — all passage doors 36 in (3'-0"); closets 30 in; french door 6'-0" [design intent]
- Hall width — main hall 3'-0" clear (42 in); suite hall 4'-0" clear [design intent]
- Turning circles — 5'-0" (60 in) circle drawn and checked clear in both baths [checked by script]
- Curbless shower — primary bath 5'-0" × 3'-10" curbless (recessed slab per 03 30 00 §2.05) [design intent]
- Grab-bar blocking — 2×10 solid blocking 33–36 in AFF at all toilets, tub and showers (2024 IRC (GA 2026) — voluntary; ANSI A117.1 Type C-type) [design intent — VERIFY blocking heights with architect]
- Plate height / vaulted — 9'-0" plate, great room vaulted [design intent]
- Single-level living — 1 story, no interior steps; laundry on the living level [design intent]
- Lever hardware, rocker switches, 48-in max reach — to be specified (Division 08 / 26) [later stage]
- Lilburn §734 visitability — applies to attached dwellings only; voluntarily exceeded (100% zero-step) [appears consistent with Ord. 2023-603 §734 (not applicable)]

LEGEND

- Exterior wall — 2×6 @ 6 in (poché)
- Interior partition — 2×4 @ 4 in
- Cased opening / ceiling beam (no wall)
- Window (triple line), width noted
- Door leaf + swing, width noted (36 in typ.)
- 5'-0" turning circle (ANSI A117.1-type)
- Closet shelf + rod
- Covered porch slab
- Uncovered patio slab

NOTES

- Illustrative architectural plan for the rezoning application (Lilburn Zoning Ordinance 2023-603 §1003-4 renderings/elevations item). NOT FOR CONSTRUCTION. DRAFT — to be superseded by sealed drawings by a Georgia-registered architect.
- Dimensions are to the outside face of exterior walls (2×6, 6 in nominal) and to the centreline of interior partitions (2×4, 4 in). Plate height 9'-0"; great room vaulted; roof 8:12 main gables, 4:12 porch shed, architectural shingles.
- Accessibility (voluntary) — Ord. 2023-603 §734 applies to attached dwellings only); zero-step entries at the porch and garage/mud room; 36-in passage doors; 42-in halls; 5'-0" turning circles in both baths; curbless primary shower; 2×10 blocking for grab bars at all toilets, tub and showers.
- Garage door 16'-0" × 7'-0" recessed 5'-0" behind the front wall of the dwelling — appears consistent with Ord. 2023-603 Table 4.2 (front-loaded garages recessed ≥ 5 ft). Garage clear 20'-4" × 21'-2" ≥ 20 × 21 program minimum.
- Life safety: NFPA 13D residential sprinklers (mitigation for the single access road, IFC 2024 (GA) App. D107.1), smoke/CO alarms per 2024 IRC (GA 2026) R314/R315; bedroom egress windows per 2024 IRC (GA 2026) R310 — VERIFY net clear opening at unit selection.
- Program body is nominally 38'-0" × 38'-0" (1,444 SF). With the 2-car garage as a front wing (the only way to keep the 40-ft width cap), the conditioned envelope is an L: 17'-0" × 25'-10" front wing + 38'-0" × 25'-0" rear block = 1,406 SF conditioned (~2.6%); overall body 38'-0" × 51'-10" plus porch and patio.
- Rear patio drawn 12'-0" × 7'-0"; the program 12 × 9 slab would end 10 in inside the 20-ft rear buffer easement on a 100-ft lot (15 + 6 + 5 + 21'-10" + 25 + 8 = 80'-10"). VERIFY with Planning staff whether an at-grade uncovered slab is a buffer encroachment; otherwise hold the patio at 7 ft.
- Areas are computed by tools/floorplans.py from the wall face polygons (conditioned = outside face of exterior walls; total under roof = conditioned + garage + covered porch; the uncovered patio is excluded).



▼ FRONT — TOWARD THE LANE (ENTRY, GARAGE, PORCH FACE AT THE 15-FT SETBACK LINE) ▼

PLAN A — THE SPRINGBROOK — ILLUSTRATIVE ARCHITECTURAL PLAN — NOT FOR CONSTRUCTION — 2026-08-28

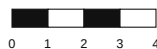
THE COTTAGES AT ARCADE SPRINGS — R-1 — R-2 REZONING (Lilburn Zoning Ordinance 2023-603 §1003-4) — 4535/4537/4539/4541 Arcade Rd SW, Lilburn GA 30047 — Land Lot 123, 6th District, Gwinnett County

Sheet A-1 - The Springbrook · Scale 3/16" = 1'-0" on ARCH C (18 × 24 in) · Prepared by the owner-applicant (Mohammed Awad) with AI drafting tools · DRAFT — to be superseded by sealed drawings (GA-registered architect)

Disclaimer: This is an AI-generated analysis for preliminary planning purposes. All findings must be verified by a licensed professional before use in design, permitting, or regulatory submissions.

<- architecture-studio/requires-disclaimer ->

GRAPHIC SCALE — 4 FT (1 FT DIVISIONS)



NORTH: see site plan (lots face the lane; N 28°43' W along the strip)

PLAN B — THE LAUREL

2 BR + den / 2 BA, 1 story, 2-car front-loaded garage recessed 5'-0", covered front porch, covered rear porch 14'x6' (recessed), fireplace, walk-in pantry, laundry/mud room off the garage

AREA TABULATION — computed by tools/floorplans.py from the wall-face polygons

Conditioned (heated) area	1,492 SF	outside face of ext. walls
2-car garage	504 SF	outside face / centreline
Covered front porch	102 SF	slab
Covered rear porch (recessed under main roof)	84 SF	slab
TOTAL UNDER ROOF	2,182 SF	cond. + garage + porches

Program target 1,520 conditioned (FACTS §4 nominal 38'-0" × 40'-0"); Lilburn Table 4.1 R-2 cottage min heated 1,000 SF.

ROOM SCHEDULE (clear, inside face)

2-CAR GARAGE	20'-4" × 23'-4"	474 SF
FOYER / HALL	3'-6" × 28'-6"	100 SF
DEN	12'-2" × 8'-10"	107 SF
BEDROOM 2	12'-2" × 11'-2"	124 SF
CLOSET	4'-6" × 2'-2"	10 SF
BATH 2	12'-2" × 7'-8"	93 SF
LAUNDRY / MUD	6'-8" × 10'-8"	71 SF
PANTRY	3'-10" × 5'-8"	22 SF
KITCHEN	12'-2" × 10'-10"	151 SF
W.I.C.	13'-4" × 5'-8"	76 SF
PRIMARY BATH	13'-4" × 8'-2"	109 SF
PRIMARY BEDROOM	13'-4" × 13'-4"	178 SF
GREAT ROOM	23'-4" × 11'-10" + 9'-4" × 6'-0"	309 SF

ROOF / HEIGHT

main gable (rear block): span 28'-6", 8:12 ~ ridge 19'-0" above FF (plate 9'-0" + heel + rise)
front-wing gable (extended over the porch): span 17'-0", 8:12 ~ ridge 15'-2" above FF (plate 9'-0" + heel + rise)
garage gable: span 21'-0", 6:12 ~ ridge 14'-9" above FF (plate 9'-0" + heel + rise)
Max ridge 19'-0" ≤ 24 ft program cap; Ord. 2023-603 Table 4.1 R-2 max height 40 ft — appears consistent.

ACCESSIBILITY CHECKLIST (voluntary)

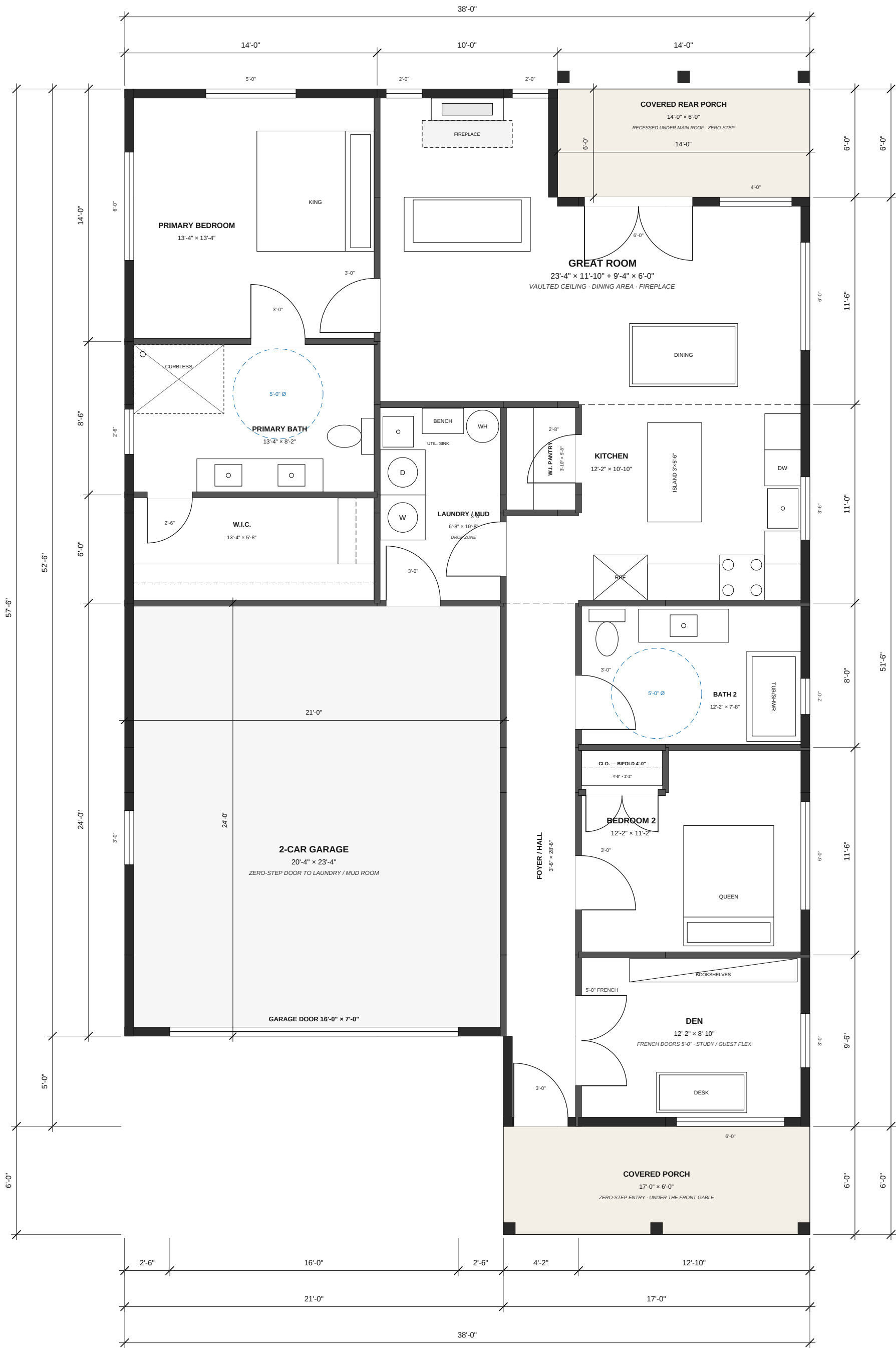
- Zero-step entries — front porch (slab flush with finished floor), garage-to-laundry/mud-room door and recessed rear porch (french door ≤ 1/2-in threshold) [design intent]
- Door clear width — all passage doors 36 in (3'-0"); closets/pantry 30–32 in; french doors 5'-0" (den) and 6'-0" (porch) [design intent]
- Hall width — foyer/hall 3'-6" clear (42 in), kitchen corridor 3'-10" clear [design intent]
- Turning circles — 5'-0" (60 in) circle drawn and checked clear in both baths [checked by script]
- Curbless shower — primary bath 5'-0" × 3'-10" curbless (recessed slab per 03 30 00 §2.05) [design intent]
- Out-swinging bath door — primary bath door swings into the bedroom (fall-rescue clearance) [design intent]
- Grab-bar blocking — ≥10 solid blocking 33–36 in AFF at all toilets, tub and showers (2024 IRC (GA 2026) — voluntary; ANSI A117.1 Type C-type) [design intent — VERIFY blocking heights with architect]
- Plate height / vaulted — 9'-0" plate; great room vaulted [design intent]
- Single-level living — 1 story; no interior steps; laundry/mud room on the living level off the garage [design intent]
- Lever hardware, rocker switches, 48-in max reach — to be specified (Division 08 / 26) [later stage]
- Lilburn §734 visitability — applies to attached dwellings only, voluntarily exceeded (100% zero-step) [appears consistent with Ord. 2023-603 §734 (not applicable)]

LEGEND

- Exterior wall — 2x6 @ 6 in (poché)
- Interior partition — 2x4 @ 4 in
- Cased opening / ceiling beam (no wall)
- Window (triple line), width noted
- Door leaf + swing, width noted (36 in typ.)
- 5'-0" turning circle (ANSI A117.1-type)
- Closet shelf + rod
- Covered porch slab
- Uncovered patio slab

NOTES

1. Illustrative architectural plan for the rezoning application (Lilburn Zoning Ordinance 2023-603 §1003-4 renderings/elevations item). NOT FOR CONSTRUCTION. DRAFT — to be superseded by sealed drawings by a Georgia-registered architect.
2. Dimensions are to the outside face of exterior walls (2x6, 6 in nominal) and to the centreline of interior partitions (2x4, 4 in). Plate height 9'-0"; great room vaulted; roof 8:12 main gables, 4:12 front-porch shed, architectural shingles; the rear porch is recessed under the main roof.
3. Accessibility (voluntary — Ord. 2023-603 §734 applies to attached dwellings only): zero-step entries at the front porch, garage/mud room and rear porch; 36-in passage doors; 3'-6" (42-in) clear foyer/hall; 5'-0" turning circles drawn and checked clear in both baths; curbless primary shower; out-swinging primary bath door; ≥10 blocking for grab bars at all toilets, tub and showers.
4. Garage door 16'-0" × 7'-0" recessed 5'-0" behind the front wall of the dwelling — appears consistent with Ord. 2023-603 Table 4.2 (front-loaded garages recessed ≥ 5 ft). Garage clear 20'-4" × 23'-4" ≥ the 20'-0" × 20'-0" program minimum (extra bay depth for storage / golf cart).
5. Life safety: smoke and CO alarms per 2024 IRC (GA 2026) R314/R315; bedroom emergency escape and rescue openings per 2024 IRC (GA 2026) R310 (note 9), NFPA 13D residential sprinklers are offered as a VOLUNTARY condition, not as code-required mitigation: IFC 2024 as adopted and modified by Ga. Comp. R. & Regs. R. 120-3-3-.04 replaces Appendix D107.1's 30-unit trigger with 120 dwelling units, so at 43 units a second fire-apparatus access road is not required and no sprinkler exception is being relied on. (The 1,754-ft dead-end lane still needs IFC 2024 (GA) Appendix D103.4 special approval — a site-plan issue, not a house-plan issue.)
6. BODY WIDTH — why 38'-0" and not the 40'-0" of the program brief. A 50'-0" lot less two 5'-0" side yards (Ord. 2023-603 Table 4.1, R-2) leaves 40'-0" of buildable width, and the ROOF has to fit inside it, not just the walls: at an 8-in overhang on each side wall a 40'-0" body measures 41'-4" roof to roof and does not fit. Ord. 2023-603 was searched and contains no provision permitting a projection or encroachment into a required yard (VERIFY at pre-application). The body is therefore 38'-0" wide — 38'-0" + 2 × 0'-8" = 39'-4" roof to roof, 4 in of clearance to each side-yard line — and the depth was increased 1'-0" to 58'-6" to recover floor area. Conditioned area 1,530 SF, against 1,595 SF for the superseded 40'-0" × 57'-6" version (~4.1%).
7. BODY DEPTH — why 58'-6" is the maximum. On the typical 50'-0" × 100'-0" lot the front porch face sits on the 15-ft front setback, the porch is 6'-0" deep and the rear 20'-0" of the lot is the undisturbed buffer easement (Ord. 2023-603 Table 4.1 rear yard; §313(1) buffer-in-lot reading): 100 – 15 – 6 – 20 = 59'-0" of available body depth. At 58'-6" the rear wall stands 20'-6" from the rear lot line — 6 in outside the buffer easement (the ROOF is a separate question: see the first VERIFY item — the drawn 12-in rear eave reaches 19'-6" from the rear line). Verified against data/layout.json (Plan B lot 2: lot 50 × 100, setback envelope 15 ft front / 20 ft rear, porch rect 6 ft deep).
8. Rear porch: recessing the 14'-0" × 6'-0" covered porch under the main roof keeps the roof edge on the line of the rear wall, so nothing projects past 58'-6" and the 20-ft buffer easement is not touched. No uncovered patio is proposed (an added at-grade slab would reach the buffer line — VERIFY with Planning staff).
9. EGRESS. Both bedrooms have an emergency escape and rescue opening in an exterior wall: BEDROOM 2 a 6'-0" × 5'-0" twin on the right side wall, the PRIMARY BEDROOM a 5'-0" × 5'-0" unit on the rear wall, sills at 2'-0" (22 in) above the finished floor, well under the 44-in maximum (2024 IRC (GA 2026) R310.2.2). The egress leaf of each unit is specified as a 3'-0" × 5'-0" CASEMENT: taking a 3-in allowance all round for frame, sash and hardware, net clear opening = 2'-6" × 4'-6" = 30 in × 54 in = 11.25 SF ≥ 5.7 SF, net clear width 30 in ≥ 20 in, net clear height 54 in ≥ 24 in ~ appears consistent with 2024 IRC (GA 2026) R310.2.1. A 3'-0" × 5'-0" single-hung is NOT relied on: its net clear height is roughly half the unit height and the resulting area is marginal against 5.7 SF. Companion leaves and all other windows may be single-hung.
10. Program body is nominally 38'-0" × 40'-0" = 1,520 SF (re-proportioned 2026-09-03 from the FACTS §4 nominal 40'-0" × 40'-0" = 1,600 SF — see note 6). With the 2-car garage as a front wing and the 14'-0" × 6'-0" rear porch recessed under the main roof, the conditioned envelope is a 17'-0" × 29'-0" front wing + a 38'-0" × 29'-6" rear block less the 84-SF porch notch = 1,530 SF conditioned (+0.7% on the 1,520-SF nominal); overall body 38'-0" × 58'-6" plus the front porch, 1,530 SF exceeds the ≥1,400-SF voluntary condition (docs/10 item 14) and the 1,000-SF minimum heated floor area for a cottage home in R-2 (Ord. 2023-603 Table 4.1).
11. Areas are computed by tools/floorplans.py from the wall-face polygons (conditioned = outside face of exterior walls; total under roof = conditioned + garage + both covered porches).



▼ FRONT — TOWARD THE LANE (ENTRY, GARAGE, PORCH FACE AT THE 15-FT SETBACK LINE) ▼

PLAN B — THE LAUREL — ILLUSTRATIVE ARCHITECTURAL PLAN — NOT FOR CONSTRUCTION — 2026-08-28

THE COTTAGES AT ARCADE SPRINGS — R-1 ~ R-2 REZONING (Lilburn Zoning Ordinance 2023-603 §1003-4) — 4535/4537/4539/4541 Arcade Rd SW, Lilburn GA 30047 — Land Lot 123, 6th District, Gwinnett County

Sheet A-2 : The Laurel - Scale 3/16" = 1'-0" on ARCH C (18 × 24 in) . Prepared by the owner-applciant (Mohammed Awad) with AI drafting tools . DRAFT — to be superseded by sealed drawings (GA-registered architect)

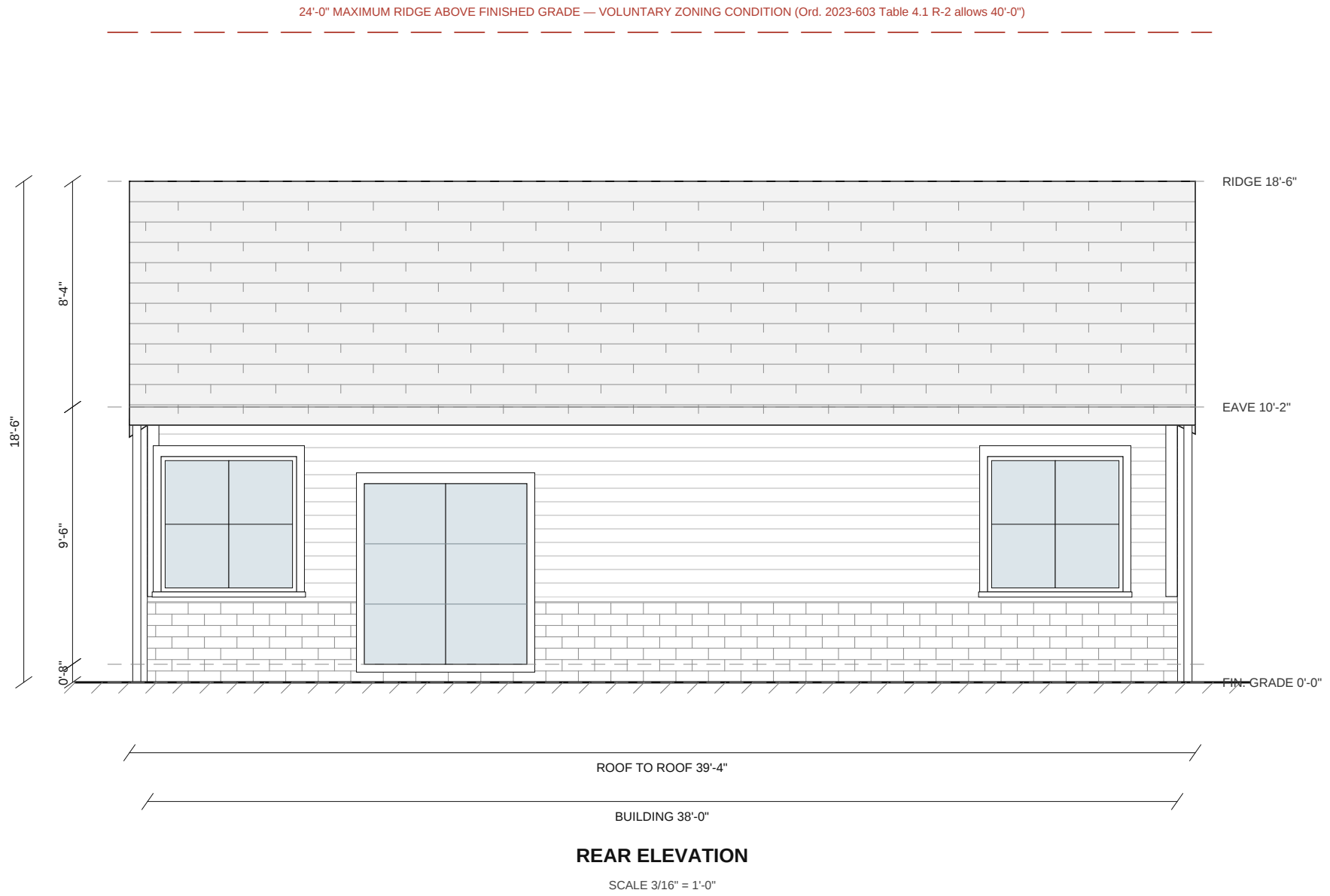
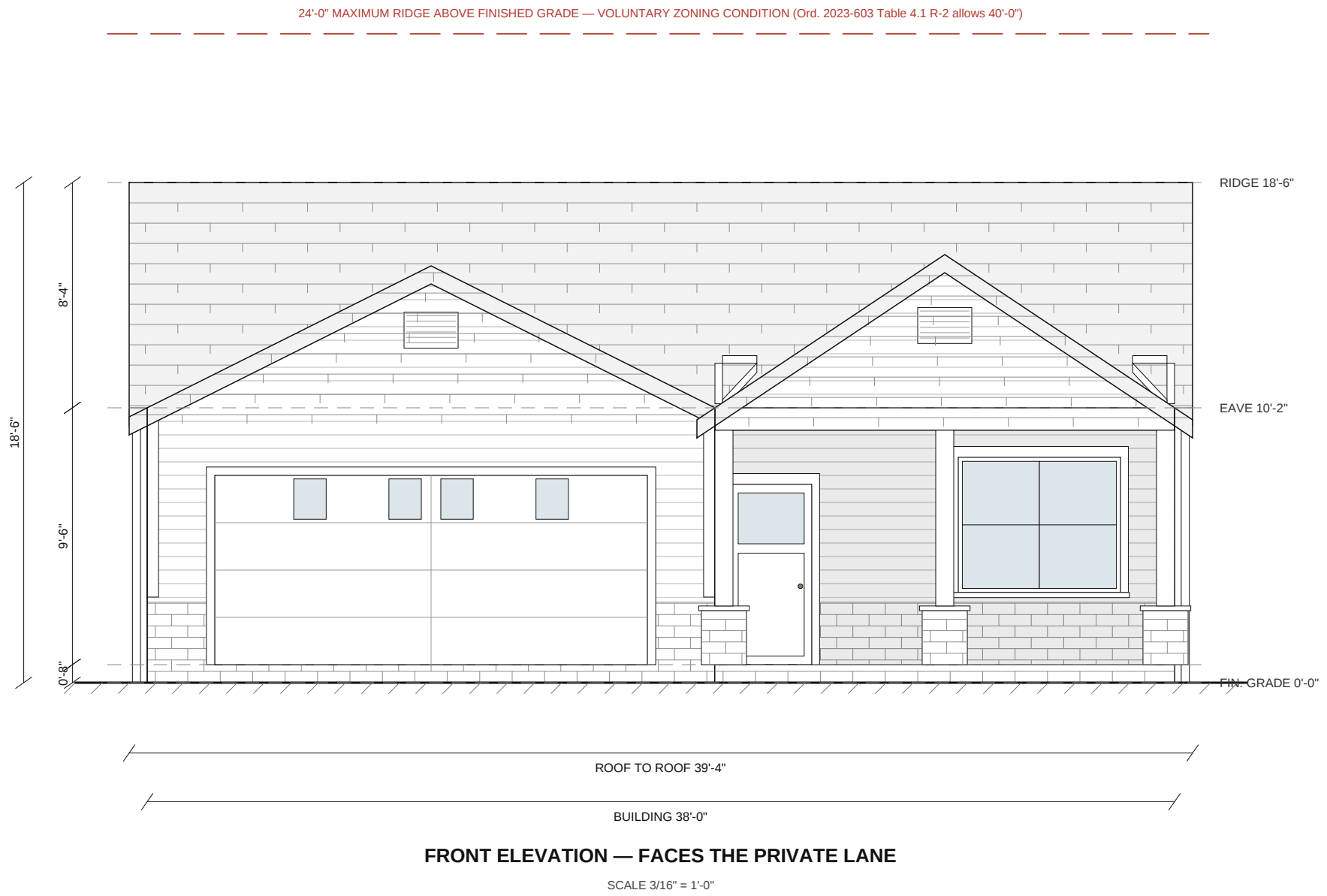
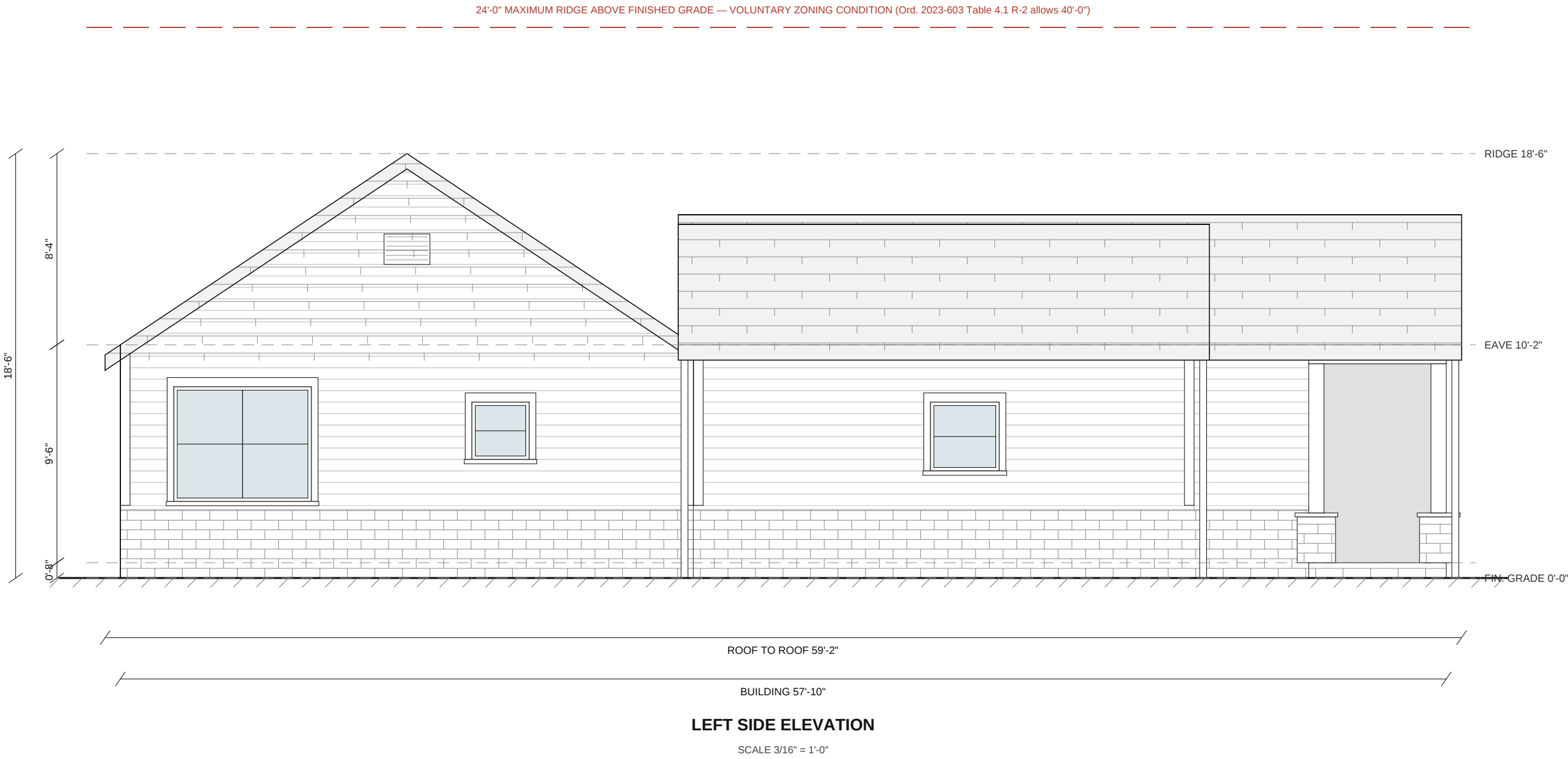
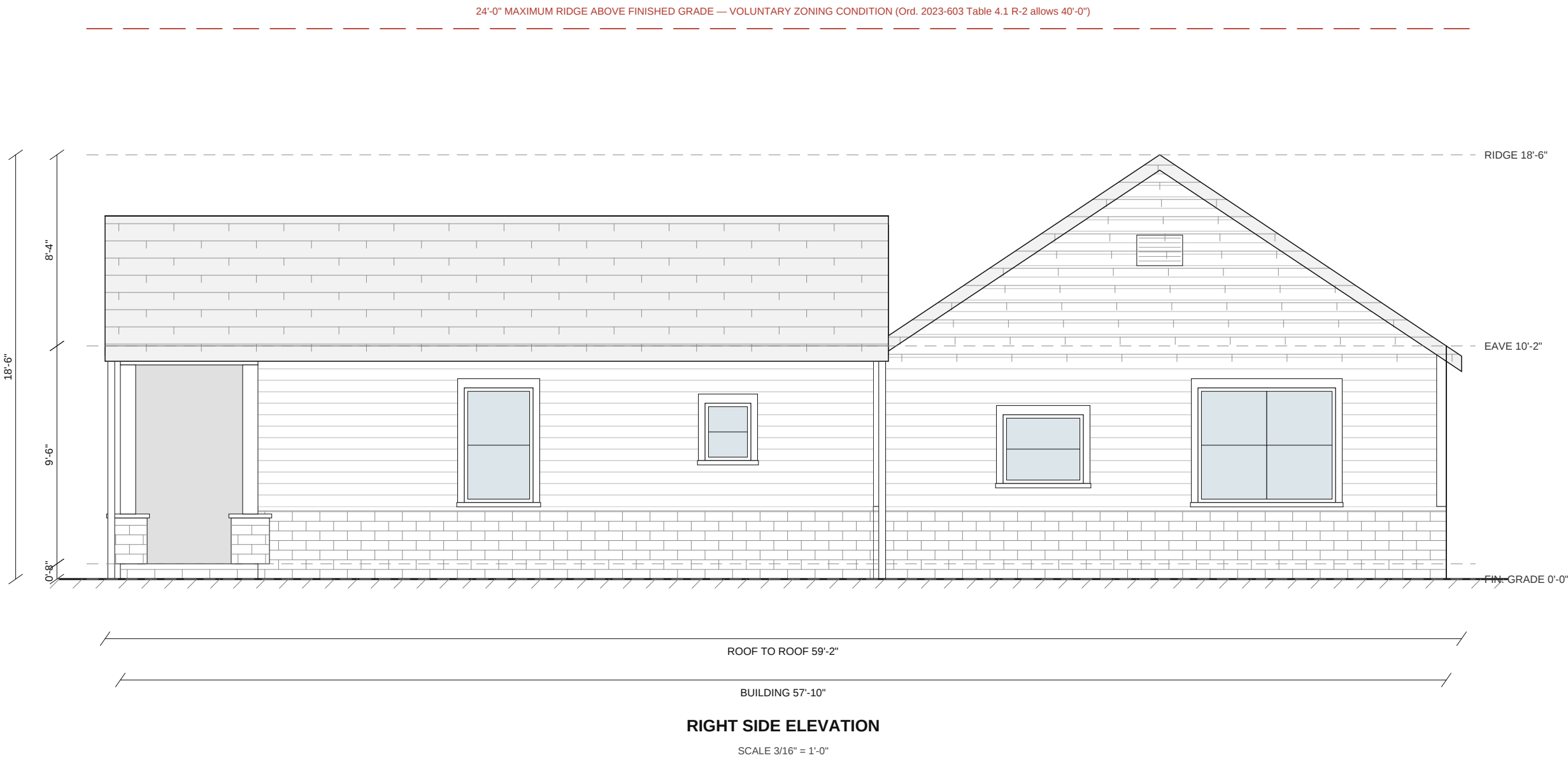
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GRAPHIC SCALE — 4 FT (1 FT DIVISIONS)



NORTH: see site plan (lots face the lane; N 28°43' W along the strip)



MATERIALS AND COLOURS — Application Instructions item (11)

- 1 **ROOF (07 31 00)** Architectural asphalt shingle, 30-year, over synthetic underlayment; ridge vent; 5-in K-style aluminium gutters and downspouts (07 71 00). Colour by scheme.
- 2 **WALL (07 46 46)** Fiber-cement siding: 7-1/4-in lap at 6-in exposure (schemes 2 and 3) or vertical panel with 12-in battens (scheme 1); factory finish. S/4 cellular-PVC or fiber-cement trim, corner boards and frieze (06 20 00).
- 3 **BASE (04 20 00 / 04 40 00)** Masonry water table 30 in above finished floor and porch piers: modular brick veneer (scheme 2) or adhered ledge stone (schemes 1 and 3), cast-stone cap sloped with drip.
- 4 **PORCH (06 20 00)** 8x8 nominal Western Red Cedar posts on 20-in square masonry piers with cast-stone caps; cedar brackets at the gable; beadboard porch ceiling, 9'-0" flat porch ceiling.
- 5 **WINDOWS (08 53 13)** Vinyl (fiberglass alternate) single-hung and fixed units, white; 4-in flat casing with a sloped sill; grilles at the front elevation only.
- 6 **DOORS (08 14 00 / 08 36 13)** Insulated fiberglass entry door 3'-0" x 6'-8" with a glazed upper panel; carriage-style sectional garage door 16'-0" x 7'-0" with a top row of lites.

WALL SIGNS: none proposed on any dwelling, the clubhouse or an accessory structure. One ground-mounted monument entry sign not exceeding 32 SF is proposed at the Arcado Road entrance — see sheet A-4 for its elevation, dimensions, materials and colours.

HEIGHT: maximum ridge 18'-6" above finished grade — within the 24'-0" voluntary condition and well below the 40'-0" maximum of Ord. 2023-603 Table 4.1 (R-2). One story throughout; no basement.

GRADE: elevations are drawn on level finished grade. Finished floor is 0'-3" above finished grade at the front porch so the entry walk stays at or under 1:20 (zero-step entry, voluntary accessibility condition); grade falls away at the sides and rear. Lot grading is shown on sheet C-3.

THE COTTAGES AT ARCADE SPRINGS — PLAN A "The Springbrook" — EXTERIOR ELEVATIONS

R-1 ~ R-2 REZONING (Lilburn Zoning Ordinance 2023-603 §1003-4; 2026 Application Instructions item (11)) — 4535 / 4537 / 4539 / 4541 Arcado Rd SW, Lilburn GA 30047 — Land Lot 123, 6th District, Gwinnett County

Scale 3/16" = 1'-0" on ARCH C (24 x 18 in) · Rev B — 2026-09-03 · Prepared by the owner-applicant (Mohammed Awad) with AI drafting tools · ILLUSTRATIVE — NOT FOR CONSTRUCTION · DRAFT, to be superseded by sealed drawings by a Georgia-registered architect

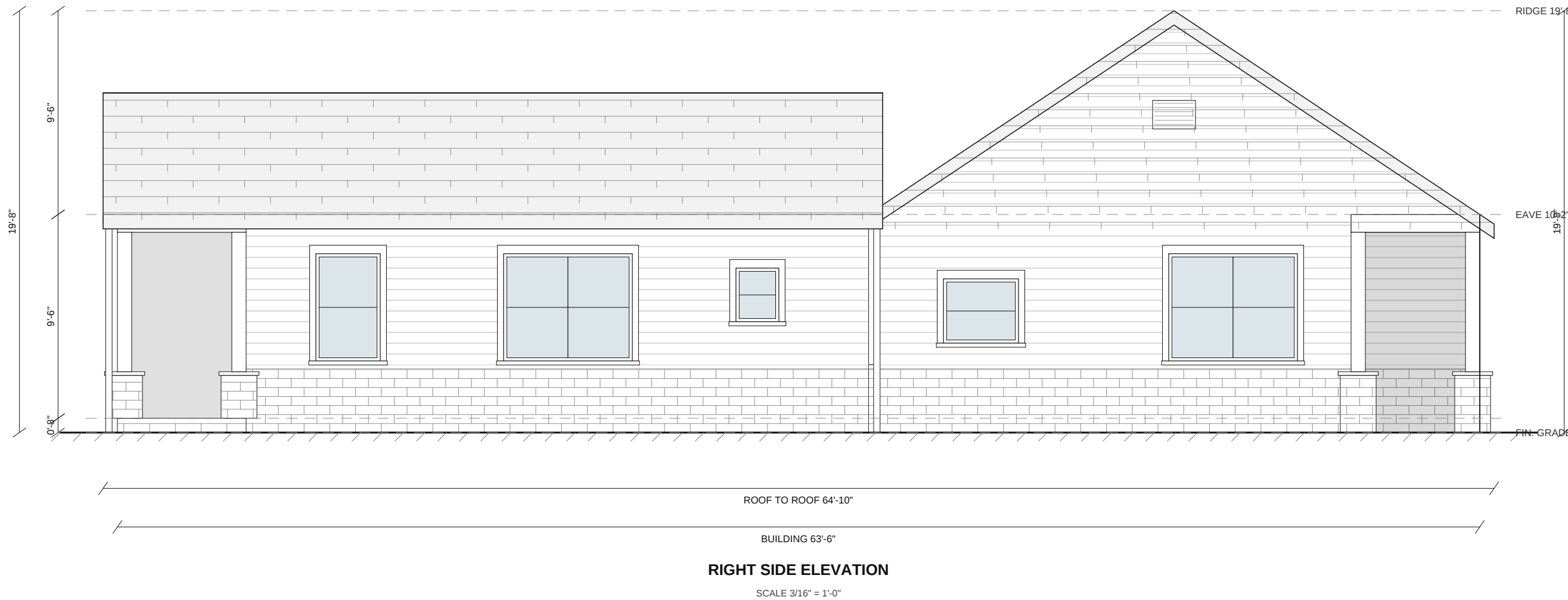
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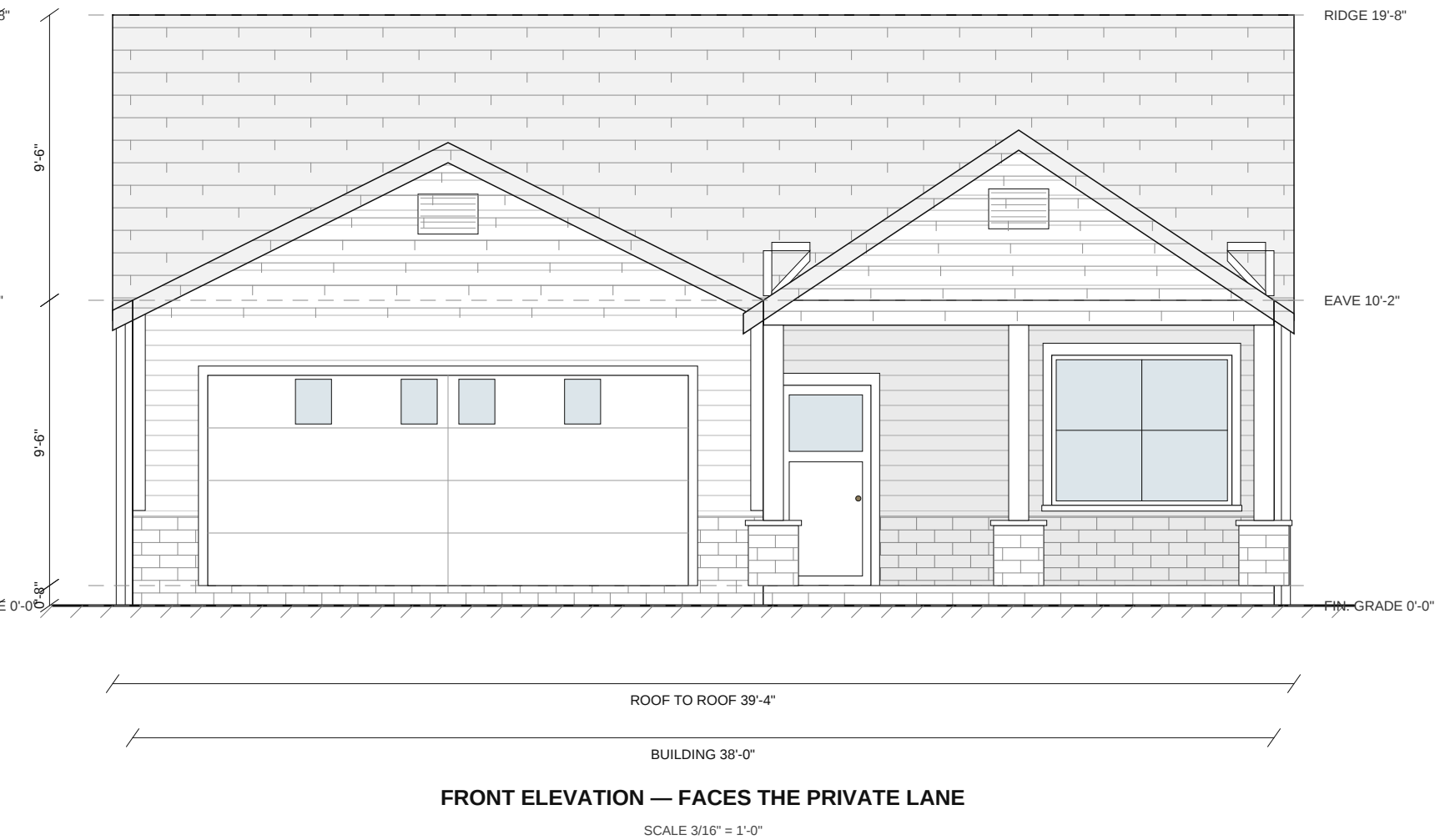
A-2.1

PLAN A "The Springbrook"
2026-09-03

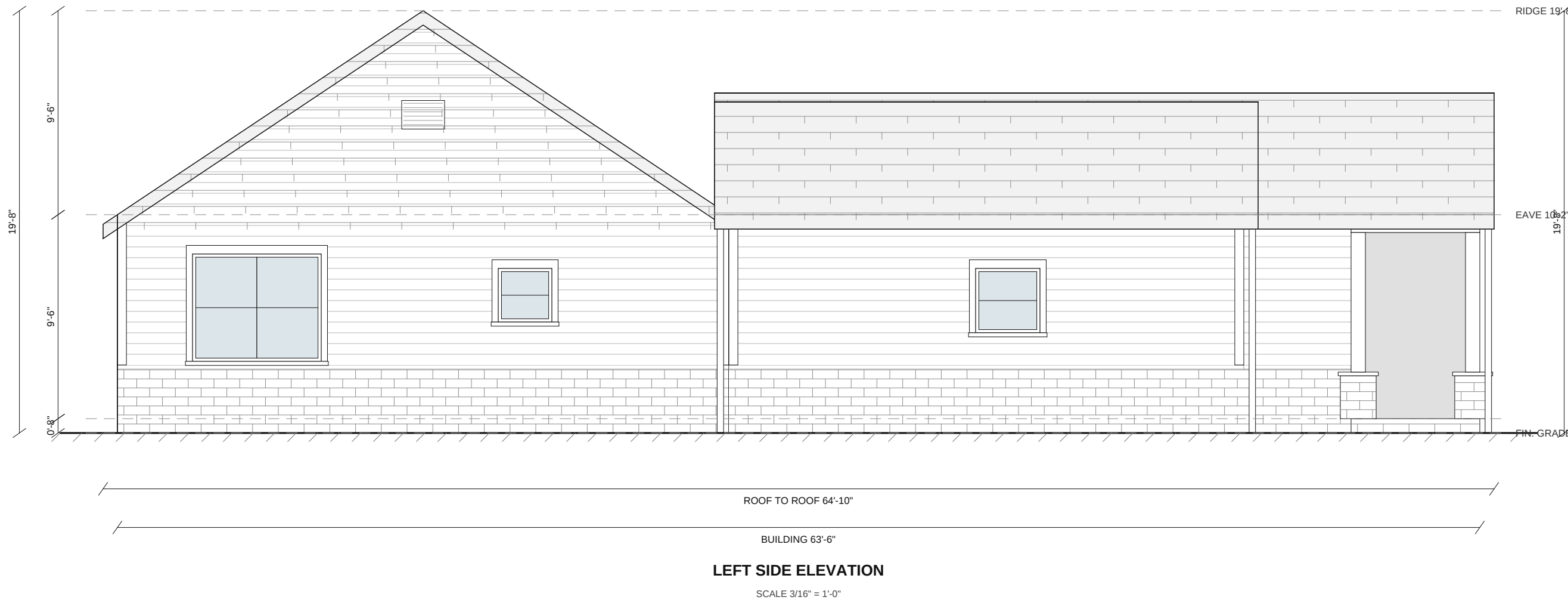
24'-0" MAXIMUM RIDGE ABOVE FINISHED GRADE — VOLUNTARY ZONING CONDITION (Ord. 2023-603 Table 4.1 R-2 allows 40'-0")



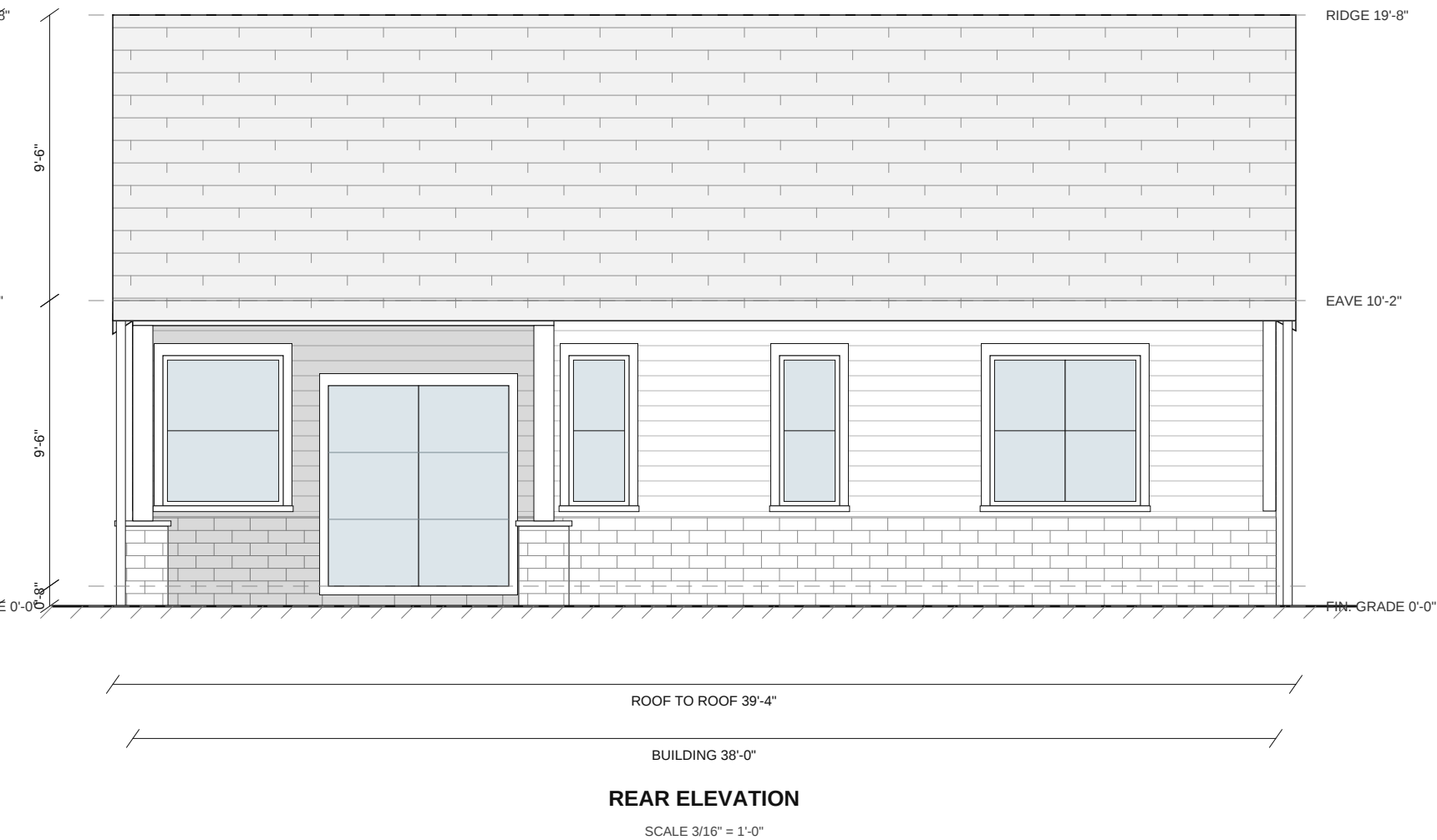
24'-0" MAXIMUM RIDGE ABOVE FINISHED GRADE — VOLUNTARY ZONING CONDITION (Ord. 2023-603 Table 4.1 R-2 allows 40'-0")



24'-0" MAXIMUM RIDGE ABOVE FINISHED GRADE — VOLUNTARY ZONING CONDITION (Ord. 2023-603 Table 4.1 R-2 allows 40'-0")



24'-0" MAXIMUM RIDGE ABOVE FINISHED GRADE — VOLUNTARY ZONING CONDITION (Ord. 2023-603 Table 4.1 R-2 allows 40'-0")



MATERIALS AND COLOURS — Application Instructions item (11)

- 1 ROOF (07 31 00)** Architectural asphalt shingle, 30-year, over synthetic underlayment; ridge vent; 5-in K-style aluminium gutters and downspouts (07 71 00). Colour by scheme.
- 2 WALL (07 46 46)** Fiber-cement siding: 7-1/4-in lap at 6-in exposure (schemes 2 and 3) or vertical panel with 12-in battens (scheme 1); factory finish. S/4 cellular-PVC or fiber-cement trim, corner boards and frieze (06 20 00).
- 3 BASE (04 20 00 / 04 40 00)** Masonry water table 30 in above finished floor and porch piers: modular brick veneer (scheme 2) or adhered ledge stone (schemes 1 and 3), cast-stone cap sloped with drip.
- 4 PORCH (06 20 00)** 8x8 nominal Western Red Cedar posts on 20-in square masonry piers with cast-stone caps; cedar brackets at the gable; beadboard porch ceiling, 9'-0" flat porch ceiling.
- 5 WINDOWS (08 53 13)** Vinyl (fiberglass alternate) single-hung and fixed units, white; 4-in flat casing with a sloped sill; grilles at the front elevation only.
- 6 DOORS (08 14 00 / 08 36 13)** Insulated fiberglass entry door 3'-0" x 6'-8" with a glazed upper panel; carriage-style sectional garage door 16'-0" x 7'-0" with a top row of lites.

WALL SIGNS: none proposed on any dwelling, the clubhouse or an accessory structure. One ground-mounted monument entry sign not exceeding 32 SF is proposed at the Arcado Road entrance — see sheet A-4 for its elevation, dimensions, materials and colours.

HEIGHT: maximum ridge 19'-8" above finished grade — within the 24'-0" voluntary condition and well below the 40'-0" maximum of Ord. 2023-603 Table 4.1 (R-2). One story throughout; no basement.

GRADE: elevations are drawn on level finished grade. Finished floor is 0'-3" above finished grade at the front porch so the entry walk stays at or under 1.20 (zero-step entry, voluntary accessibility condition); grade falls away at the sides and rear. Lot grading is shown on sheet C-3.

THE COTTAGES AT ARCADE SPRINGS — PLAN B "The Laurel" — EXTERIOR ELEVATIONS

R-1 ~ R-2 REZONING (Lilburn Zoning Ordinance 2023-603 §1003-4; 2026 Application Instructions item (11)) — 4535 / 4537 / 4539 / 4541 Arcado Rd SW, Lilburn GA 30047 — Land Lot 123, 6th District, Gwinnett County

Scale 3/16" = 1'-0" on ARCH C (24 x 18 in) · Rev B — 2026-09-03 · Prepared by the owner-applicant (Mohammed Awad) with AI drafting tools · ILLUSTRATIVE — NOT FOR CONSTRUCTION · DRAFT, to be superseded by sealed drawings by a Georgia-registered architect

Disclaimer: This is an AI-generated analysis for preliminary planning purposes. All findings must be verified by a licensed professional before use in design, permitting, or regulatory submissions.

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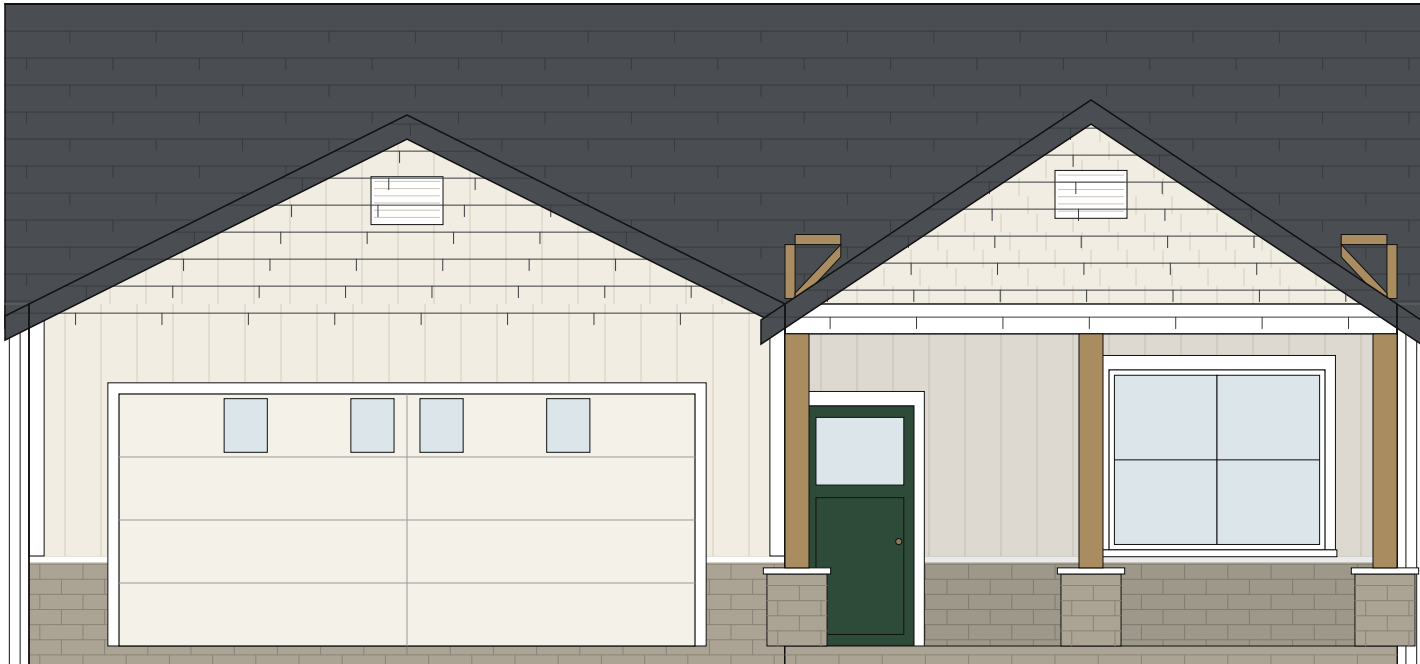
A-2.2

PLAN B "The Laurel"
2026-09-03

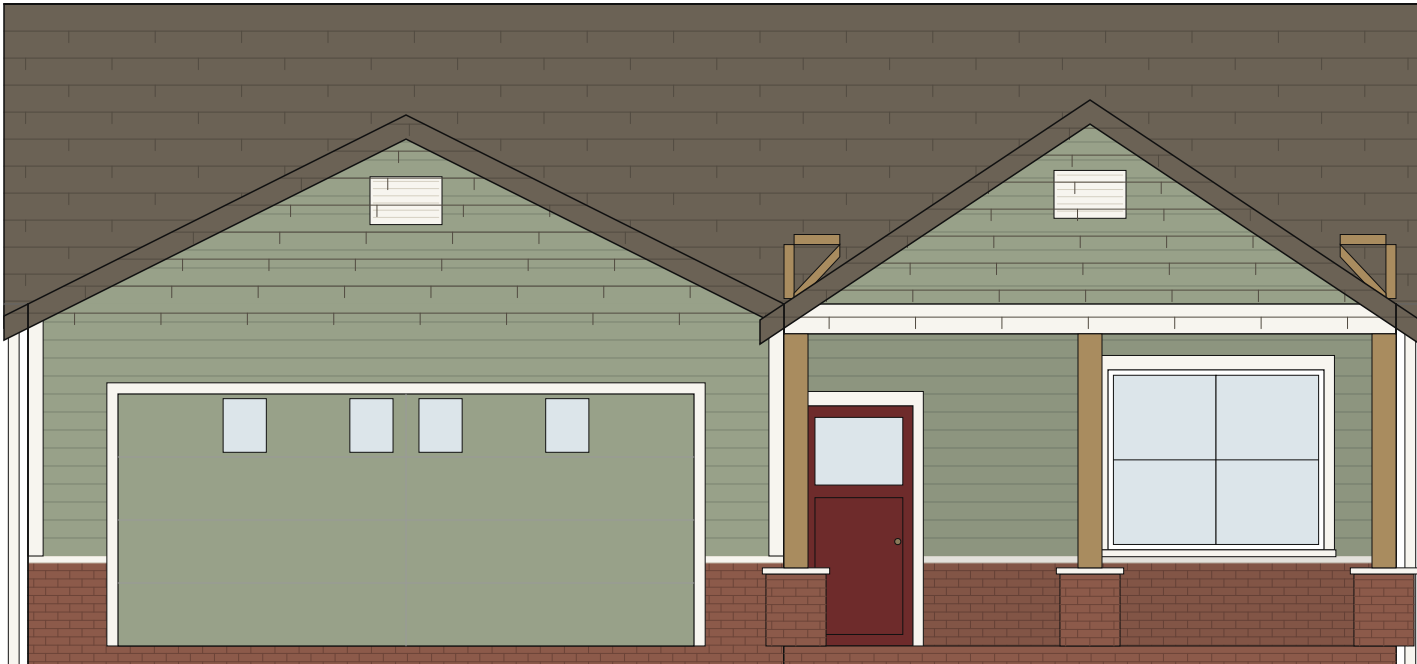
SCHEME 1 — WARM WHITE

SCHEME 2 — SAGE

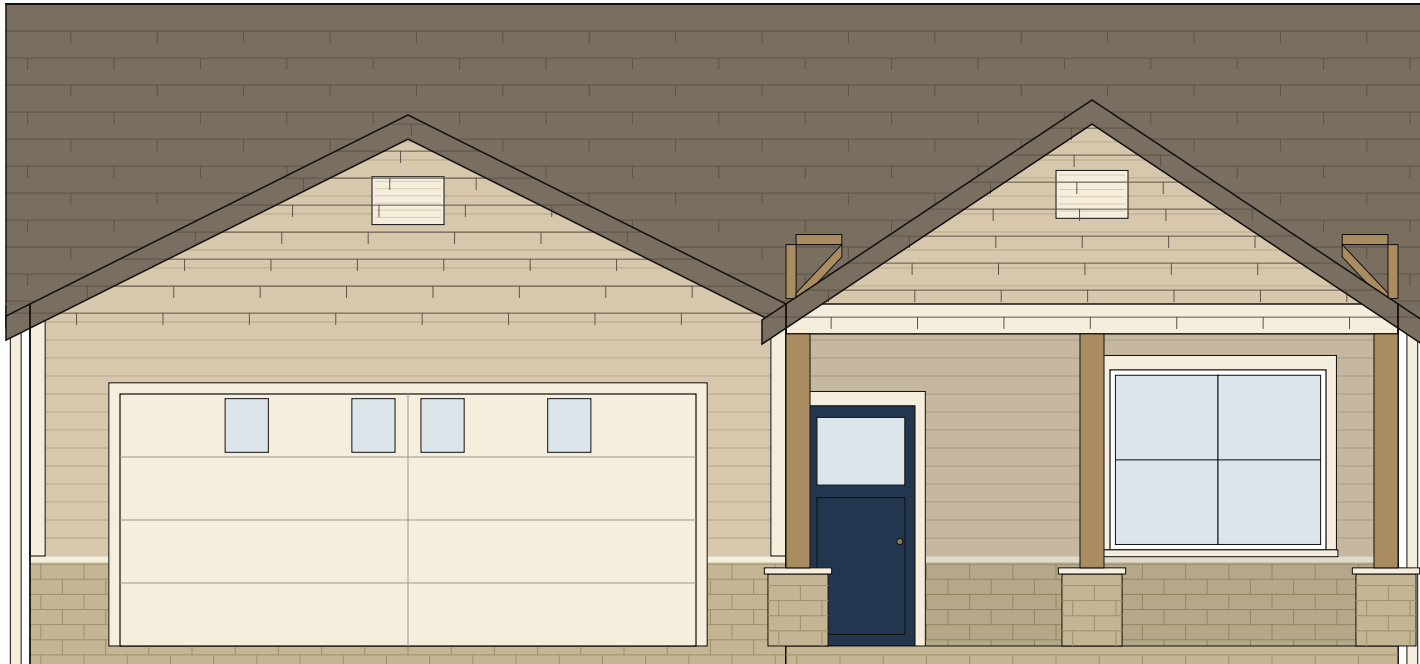
SCHEME 3 — CLAY



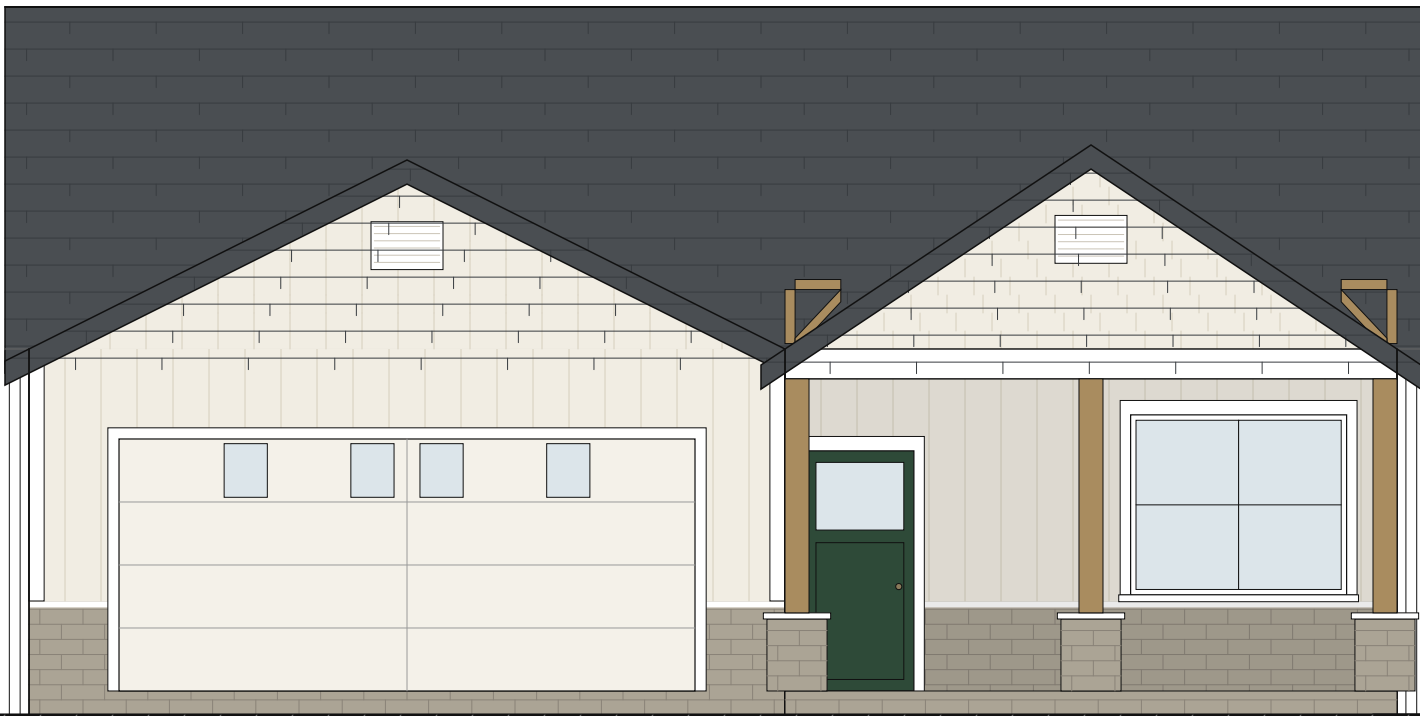
PLAN A "The Springbrook" — FRONT ELEVATION



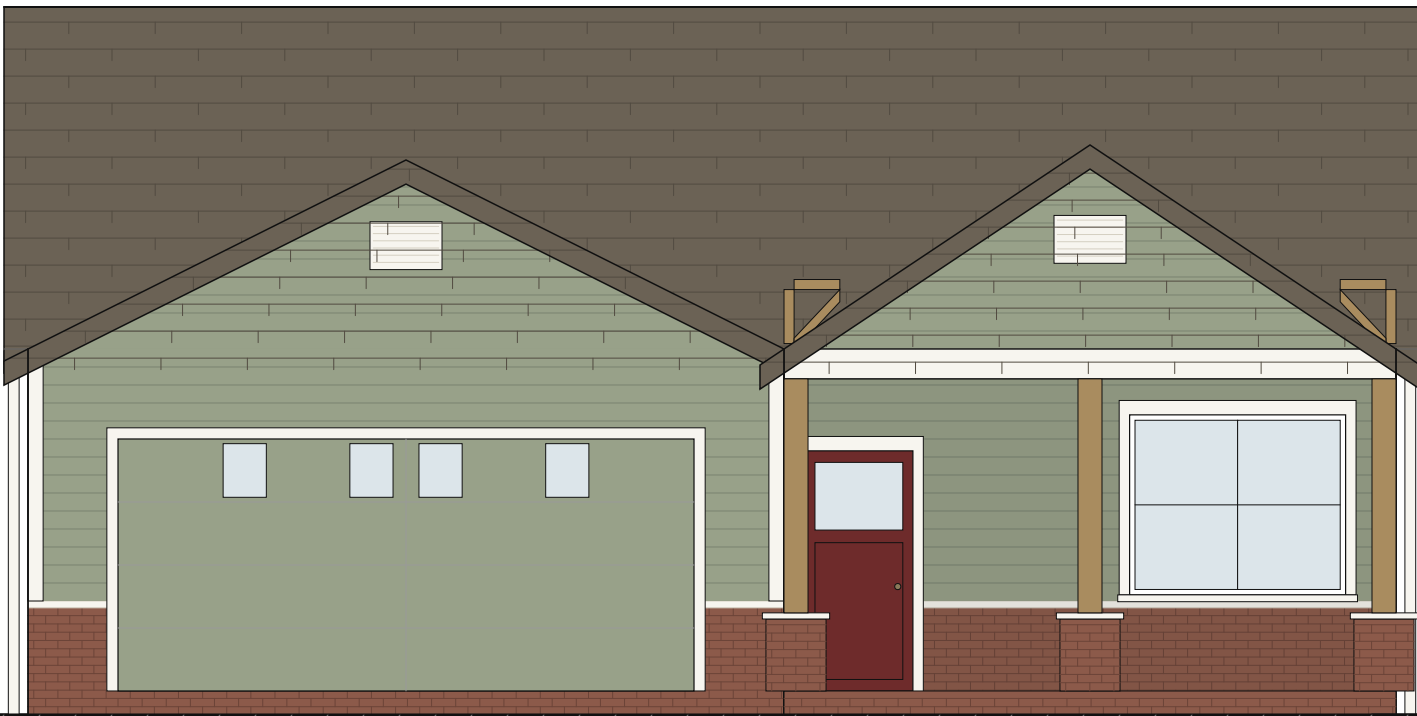
PLAN A "The Springbrook" — FRONT ELEVATION



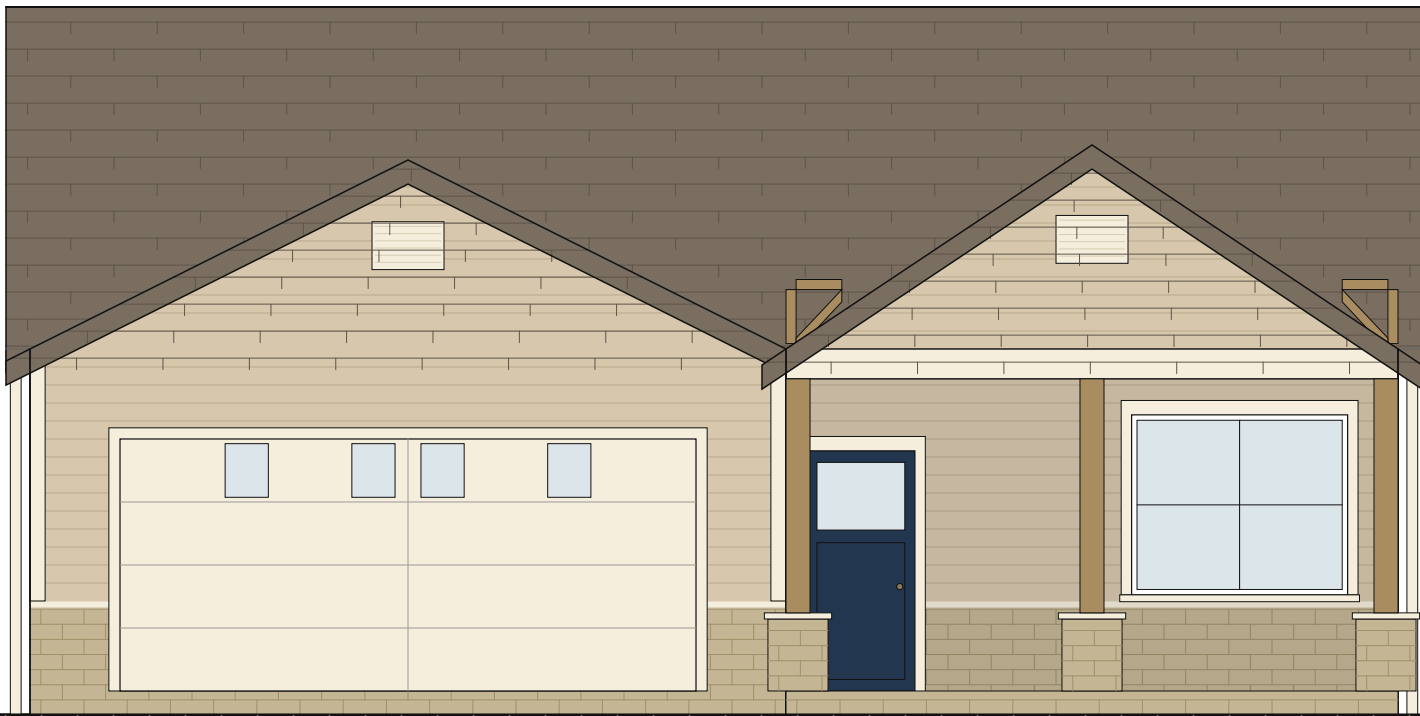
PLAN A "The Springbrook" — FRONT ELEVATION



PLAN B "The Laurel" — FRONT ELEVATION



PLAN B "The Laurel" — FRONT ELEVATION



PLAN B "The Laurel" — FRONT ELEVATION

MATERIAL AND COLOUR BOARD — basis of design per docs/12-outline-specifications.md; final selections by the Owner and Architect from the manufacturers' full ranges

SCHEME 1 — WARM WHITE		
SIDING (07 46 46)		Board-and-batten, warm white
WATER TABLE / PIERS (04 20 00, 04 40 00)		Ledge stone, gray-tan blend
ROOFING (07 31 00)		Architectural shingle, charcoal
TRIM / FASCIA / FRIEZE (06 20 00, 09 91 13)		Warm white
FRONT DOOR (08 14 00)		Deep green
GARAGE DOOR (08 36 13)		Painted to match trim
PORCH POSTS (06 20 00)		Western Red Cedar, natural stain

SCHEME 2 — SAGE		
		Lap 6-in exposure, sage
		Modular brick, red-brown blend
		Architectural shingle, weathered wood
		Soft white
		Oxblood
		Painted to match siding
		Western Red Cedar, natural stain

SCHEME 3 — CLAY		
		Lap 6-in exposure, clay/beige
		Ledge stone, buff blend
		Architectural shingle, driftwood
		Cream
		Navy
		Painted to match trim
		Western Red Cedar, natural stain

Every dwelling is one story. Colour schemes are distributed so that no scheme repeats on adjacent lots and no scheme exceeds 40 % of the 43 lots; the distribution is a voluntary condition and is administered by the homeowners association.

Roofing is the same product on every dwelling, the clubhouse and the mail kiosk; only the colour changes with the scheme.

WALL SIGNS: none are proposed on any dwelling, on the clubhouse or on any accessory structure. One ground-mounted monument entry sign not exceeding 32 SF in face area and 6'-0" in height is proposed at the Arcado Road entrance; see sheet A-4 for its elevation, dimensions, materials, colours and sight-triangle clearance.

These elevations are illustrative and are drawn to scale. They are DRAFT work by the owner-applicant and are to be superseded by sealed drawings by a Georgia-registered architect before any building permit.

THE COTTAGES AT ARCADE SPRINGS — EXTERIOR COLOUR SCHEMES AND MATERIALS

R-1 ~ R-2 REZONING (Liburn Zoning Ordinance 2023-603 §1003-4; 2026 Application Instructions item (11)) — 4535 / 4537 / 4539 / 4541 Arcado Rd SW, Lilburn GA 30047 — Land Lot 123, 6th District, Gwinnett County

Scale 3/16" = 1'-0" on ARCH C (24 x 18 in) · Rev B — 2026-09-03 · Prepared by the owner-applicant (Mohammed Awad) with AI drafting tools · ILLUSTRATIVE — NOT FOR CONSTRUCTION · DRAFT, to be superseded by sealed drawings by a Georgia-registered architect

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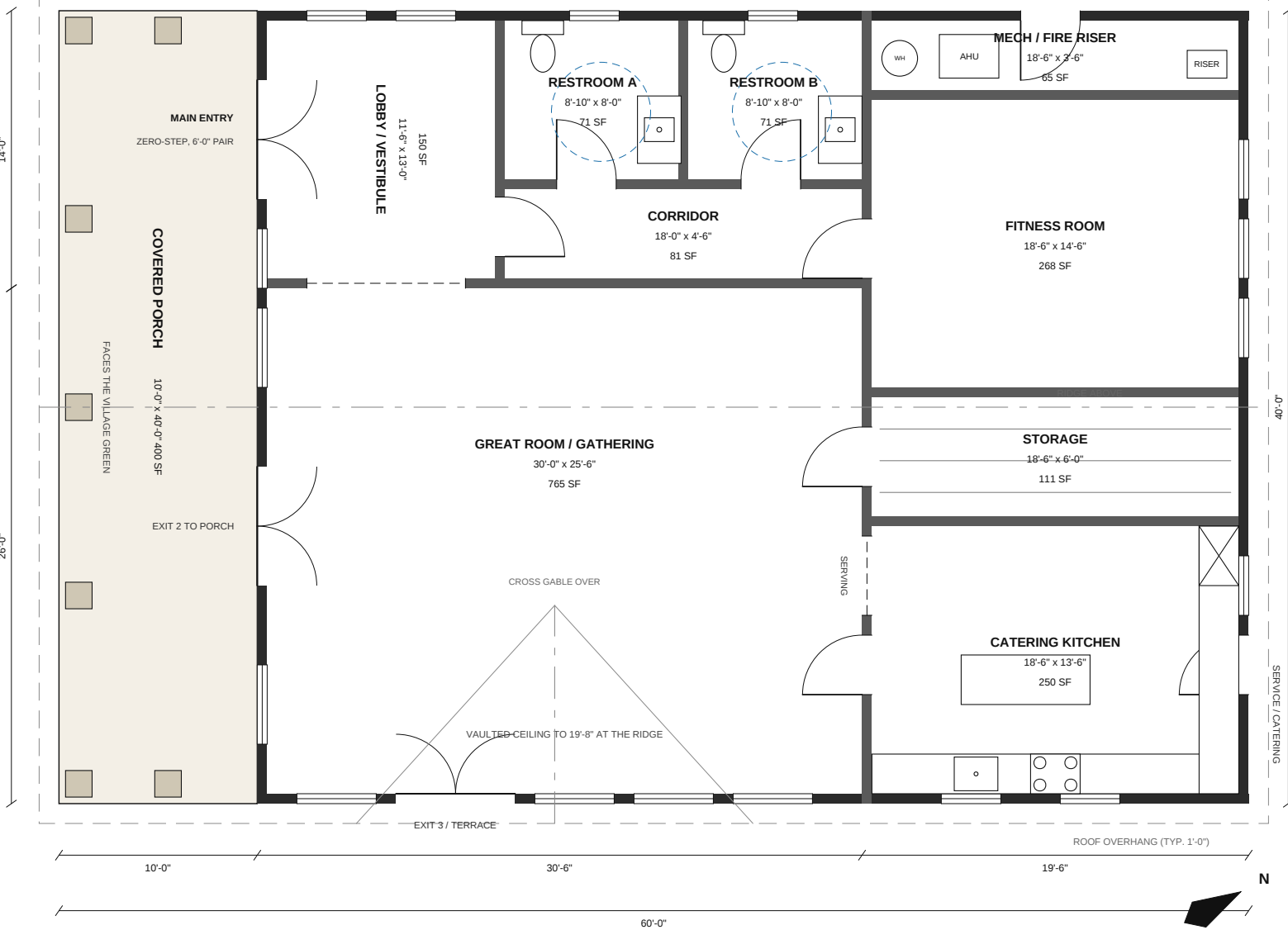
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A-2.3

EXTERIOR COLOUR SCHEMES AND MATERIALS
2026-09-03

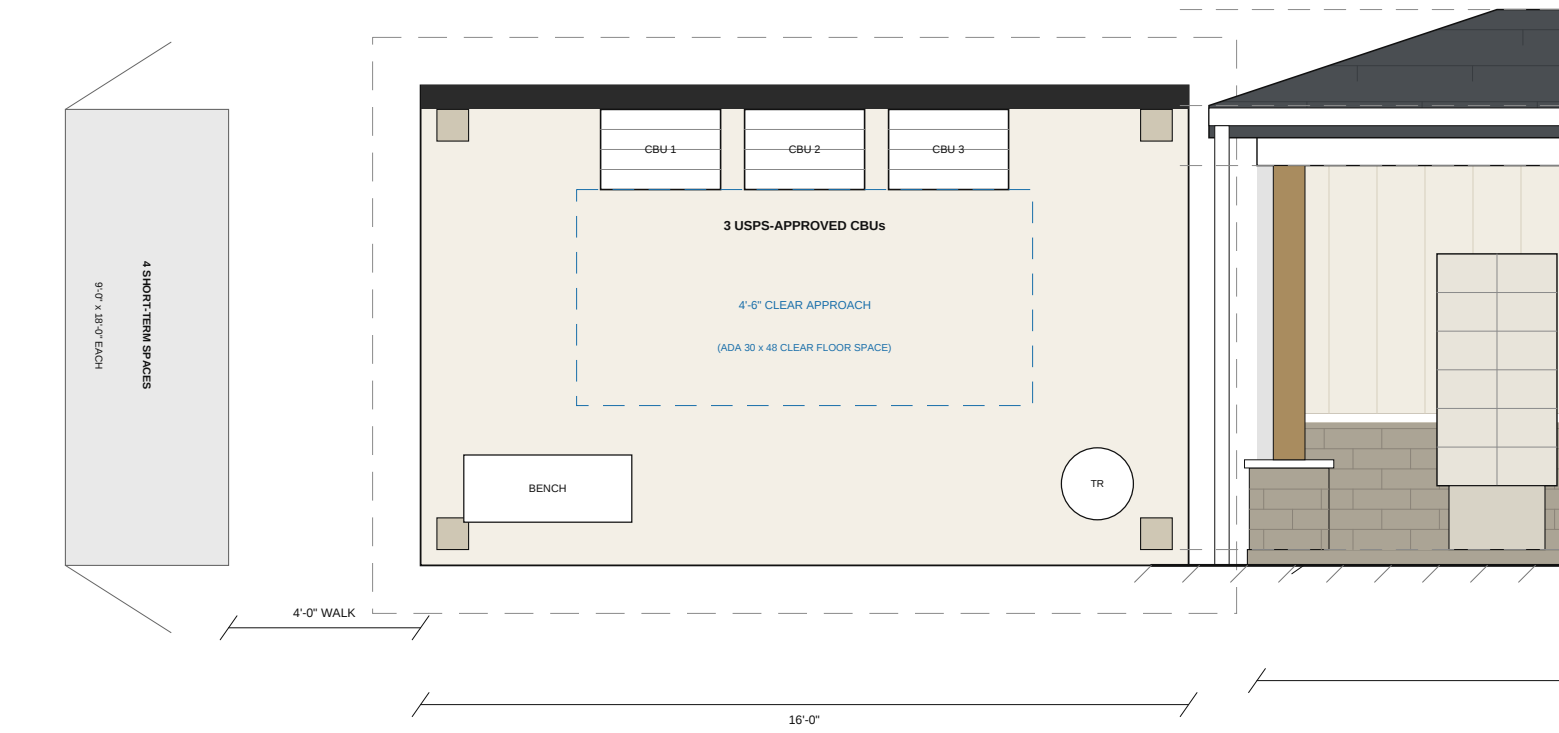
CLUBHOUSE — FLOOR PLAN

2,400 SF under roof on the 60'-0" x 40'-0" footprint of Sheet C-1 — SCALE 1/8" = 1'-0"



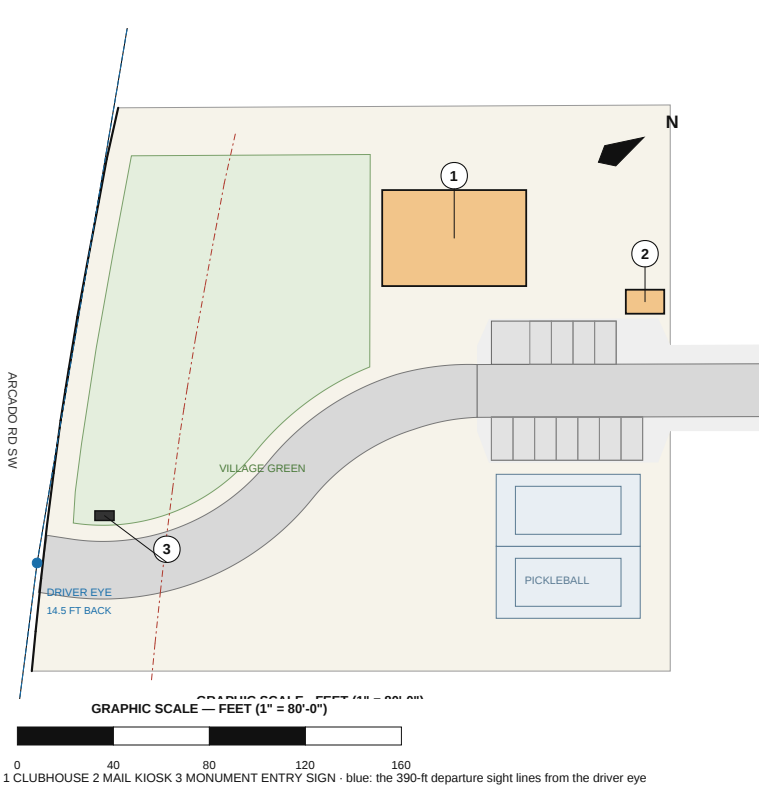
MAIL KIOSK — USPS CLUSTER BOX UNIT SHELTER

16'-0" x 10'-0" shelter and its four short-term spaces, from data/layout.json — SCALE 1/4" = 1'-0"



KEY PLAN — FRONT AMENITY BLOCK

The front block only — the full site is Sheet C-1 — SCALE 1" = 80'-0"



MATERIALS AND COLOURS — Application Instructions item (11)

All three non-dwelling structures are built in SCHEME 1 - WARM WHITE - the palette of Sheets A-2.1 / A-2.2 / A-2.3 and docs/12-outline-specifications.mxd. The three cottage schemes are distributed among the dwellings.

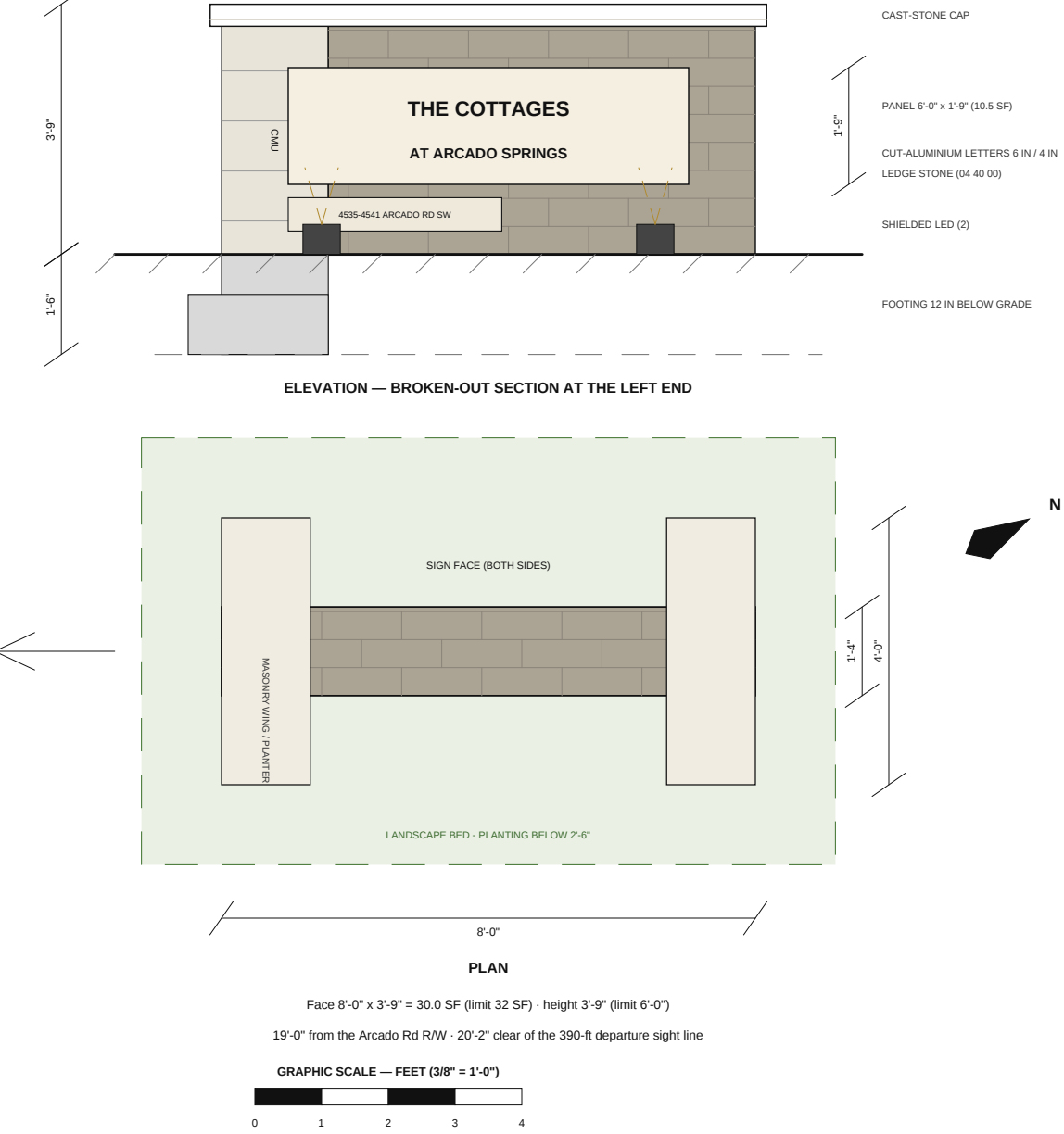
- ROOF (07 31 00)**
Architectural asphalt shingle, 30-year, over synthetic underlayment, ridge vent, 5-in K-style aluminum gutters and downspouts (07 31 00). Colour by scheme.
- WALL (07 46 46)**
Five-centim siding, 7-1/4-in lap at 6-in exposure (schemes 2 and 3) or vertical panel with 12-in battens (scheme 1); factory finish, S/4 cellular-PVC or fiber-cement trim, corner boards and frieze (06 20 00).
- BASE (04 20 00 / 04 40 00)**
Masonry water table 30 in above finished floor and porch piers: modular brick veneer (scheme 2) or adhered ledge stone (schemes 1 and 3), cast-stone cap sloped with drip.
- PORCH (06 20 00)**
8x6 nominal Western Red Cedar posts on 20-in square masonry piers with cast-stone caps; cedar brackets at the gable; beadboard porch ceiling; 9'-0" flat porch ceiling.
- WINDOWS (08 53 13)**
Vinyl (fiberglass alternate) single-hung and fixed units, white; 4-in flat casing with a sloped sill; grilles at the front elevation only.
- DOORS (08 14 00 / 08 36 13)**
Insulated fiberglass entry door 3'-0" x 6'-8" with a glazed upper panel; carriage-style sectional garage door 16'-0" x 7'-0" with a top row of tiles.

CLUBHOUSE — AREAS, HEIGHT AND OCCUPANT LOAD

Building footprint (Sheet C-1)	60'-0" x 40'-0" = 2,400 SF
Enclosed (gross, outside face)	2,000 SF
Covered porch (village green)	400 SF
Net room area (sum of rooms)	1,831 SF
Eave / ridge above finished grade	11'-6" / 21'-4"
Voluntary ridge condition / Table 4.1 max	24'-0" / 40'-0"
Setback from the Arcado Rd R/W (50 ft req.)	114.8 ft
Clear of the 20-ft NE buffer band	15.0 ft
Lane-facing wall to the lane pavement edge	29'-0"
Occupant load (IBC 2024 Table 1004.5)	71
Exits required / provided	2 / 3
Exit separation req. (1/3 diagonal) / provided	21'-4" / 35'-0"
Occupant load: Great Room 765 SF / 15 = 51; Lobby 150 SF / 15 = 10; Fitness Room 268 SF / 50 = 6; Catering Kitchen 250 SF / 200 = 2; Storage 111 SF / 300 = 1; Mech 65 SF / 300 = 1	

MONUMENT ENTRY SIGN

The only sign in the project — SCALE 3/8" = 1'-0"



GENERAL AND CODE NOTES

- ONE GROUND-MOUNTED MONUMENT SIGN. NO WALL SIGNS ARE PROPOSED ON ANY DWELLING, ON THE CLUBHOUSE OR ON ANY ACCESSORY STRUCTURE.**
- SIGN STANDARD** — VERIFY: the ordinance excerpts collected for this application (site/ordinance-excerpts.mxd — Liburn Zoning Ordinance 2023-603, Articles 3 and 4, §602, §734, §81003-1005) contain NO sign article, so the permitted area, height, setback and illumination for a residential subdivision entrance sign are NOT established here. The sign is drawn to a self-imposed limit of 32 SF of face and 6'-0" of height (it is built at 30.0 SF and 5'-0"). Confirm the governing article and the area-measurement method at the pre-application conference. All signs are permitted separately and a sign easement may be required (Site Development Plan Review Checklist §4.4).
- SIGN FOOTPRINT**: the 6'-0" x 4'-0" rectangle shown for the sign on Sheet C-1 is the bounding footprint of the monument, not a wall thickness: an 8'-0" long x 1'-4" thick masonry wall with a 1'-4" wide masonry wing/planter return at each end, 4'-0" deep, in a landscaped bed. Both faces carry copy, the area above is one face.
- SIGN ILLUMINATION**: external only — two ground-mounted, fully shielded LED sign lights with glare louvers set in the bed and aimed at the face, cut off at the top of the cap so no light is emitted above the horizontal; 3000K maximum, photocell and timer, spill 0.1 fc or less at any property line. A shielded, downward-directed cap-mounted LED downlight is an approved alternate. No internal illumination, no exposed lamps, no changeable copy (docs/12 10 14 00 2.05).
- SIGHT DISTANCE**: intersection sight distance 350 ft at 35 mph (Gwinnett County Unified Development Ordinance §900-40 Table §900.2, AASHTO Case B1; docs/08 Memo A, docs/11 59). The departure sight triangle is struck from a driver eye 14.5 ft behind the rear edge of the travelled way on the drive centreline. The sign stands 20'-0" behind that point and its nearest corner is 20'-2" clear of the nearer sight line; no part of the sign or its bed lies within the triangle. Planting in the bed is held below 2'-6". A field sight-distance survey and the Gwinnett DOT driveway permit govern — VERIFY.
- MAIL DELIVERY** (City of Liburn Site Development Plan Review Checklist §4.1): a cluster-box system is proposed for the entire project and one street name serves all house numbers. Contact the Liburn Post Office Growth Manager before construction drawings. The checklist directs "accessory structure details with protection from the elements" — the shelter above provides a roofed, three-sided enclosure over the boxes with a closed northeast wall, a clear approach and a bench.
- CLUBHOUSE OCCUPANCY AND SPRINKLERS**: Group A-2 with accessory Group B, Type V-B construction, one story, no basement (IBC 2024 (GA 2026) §303.4, Table 603). Sprinklered throughout under NFPA 13 and monitored; the 43 dwellings are sprinklered under NFPA 13D (voluntary condition 5, docs/0C Memo E). Knox box at the clubhouse. Fire-riser room with a direct exterior door on the northeast elevation. VERIFY the occupant load, exiting and panic hardware with the Gwinnett County Fire Marshal at plan review.
- ACCESSIBILITY**: accessible route from the van-accessible space in the guest bay across the lane at a marked crossing with flush ramps to the porch and a zero-step entry (finished floor 0'-0" above finished grade, entry walk 1.20 ft or flatter). Two single-user accessible toilet rooms, each with a 60-in turning circle and grab-bar blocking; 36-in doors throughout; a 34-in accessible counter segment in the kitchen. 2010 ADA Standards and ICC A117.1-2017 — final layouts by the Architect.
- PLUMBING**: as an occupant load of 71, IPC 2024 (GA 2026) Table 403.1 (A-3) calls for 1 water closet and 1 lavatory per sex, a drinking fountain and a service sink. Two single-user accessible toilet rooms, a bottle-filler/fountain at the corridor and a service sink in the kitchen appear consistent — VERIFY at plan review.
- HEIGHTS** (Checklist §4.4, height at the highest point): clubhouse ridge 21'-6", mail kiosk ridge 11'-7", monument sign 9'-9" above finished grade — each below the 24'-0" voluntary ridge condition and the 40'-0" maximum of Ord. 2023-603 Table 4.1 (R-2). The clubhouse gable is 6.12, not the cottages: 8.12, over the 40'-0" span an 8.12 roof puts the ridge 24'-10" above finished grade, above the voluntary condition. Materials, trim, porch and detailing are the cottages: Kiosk 4-12 to, subordinate.
- CLUSTER BOX UNITS**: 41 dwellings served by 3 USPS-approved 16-compartment CBUs = 48 tenant compartments (7 spare, 15%) and 6 parcel lockers, plus an outgoing-mail slot. Compartments assigned to residents needing accessible reach are to fall between 15 in and 48 in above the slab (ICC A117.1-2017 308, USPS-STD-4C). Model, dimensions and the final count are VERIFY items with USPS.
- PARKING AT THE KIOSK**: 4 short-term spaces, 9'-0" x 18'-0" each (Checklist §4.4), in an 36'-0" x 18'-0" bay 4'-0" southeast of the shelter slab; 8 guest spaces (one van-accessible) serve the clubhouse across a marked crossing of the lane (Sheet C-1). The shelter slab stands 21'-0" from the edge of the 22'-0" lane pavement and 14'-0" behind the back of the 5'-0" sidewalk (11'-0" pavement half-width + 2'-0" strip + 5'-0" walk).
- Accessory structures** are set back at least 5 ft from any lot line and any required buffer (Checklist §4.1 and §4.4); the clubhouse stands 114.8 ft from the Arcado Rd R/W against the 50-ft collector setback of Ord. 2023-603 Table 4.1, and 15.0 ft clear of the 20-ft northeast buffer band.
- DRAFT - NOT SEALED**. These are concept drawings by the owner-applicant for the rezoning hearing. The clubhouse and the kiosk must be designed and sealed by a Georgia-registered architect and engineer before permit; the sign requires a separate sign permit and, if it exceeds 6 ft or 32 SF, a Georgia PE footing design (docs/12 10 14 00 1.09).

CLUBHOUSE — EXTERIOR ELEVATIONS, ALL FOUR SIDES

Application Instructions item (11): an elevation of each side visible from the street — SCALE 1/8" = 1'-0" — SCHEME 1 - WARM WHITE



CLUBHOUSE ROOF AND HEIGHT — HOW THE NUMBERS ARE DERIVED

ROOF GEOMETRY, computed from this plan: plate 10'-0" above finished floor + 0'-10" raised heel = eave 10'-10" AFF; main gable span 40'-0" (eave plate to eave plate) at 6.12 gives a rise of 10'-0", so the ridge is 20'-10" AFF and, with the finished floor 0'-8" above finished grade, 21'-6" above finished grade. The cross gable over the great room spans 20'-0" at 6.12, 15'-10" AFF, and dies into the main roof 10'-0" in from the eave. Roof overhangs 1'-0" at the eaves and 1'-0" at the rakes; the 60'-0" x 40'-0" rectangle on Sheet C-1 is the wall and porch-slab footprint, not the roof edge.

THE COTTAGES AT ARCADE SPRINGS — CLUBHOUSE, MAIL KIOSK AND ENTRY MONUMENT SIGN — PLANS, ELEVATIONS AND DETAILS

R-1 ~ R-2 REZONING (Liburn Zoning Ordinance 2023-603 §1003-4; 2026 Application Instructions item (11)) — 4535 / 4537 / 4539 / 4541 Arcado Rd SW, Lilburn GA 30047 — Land Lot 123, 6th District, Gwinnett County

Scales as noted — clubhouse 1/8" = 1'-0"; mail kiosk 1/4" = 1'-0"; monument sign 3/8" = 1'-0"; key plan 1" = 80'-0" — on ARCH C (24 x 18 in) - Rev B — 2026-09-03 - Prepared by the owner-applicant (Mohammed Awad) with AI drafting tools - ILLUSTRATIVE — NOT FOR CONSTRUCTION - DRAFT, to be superseded by sealed drawings by a Georgia-registered architect

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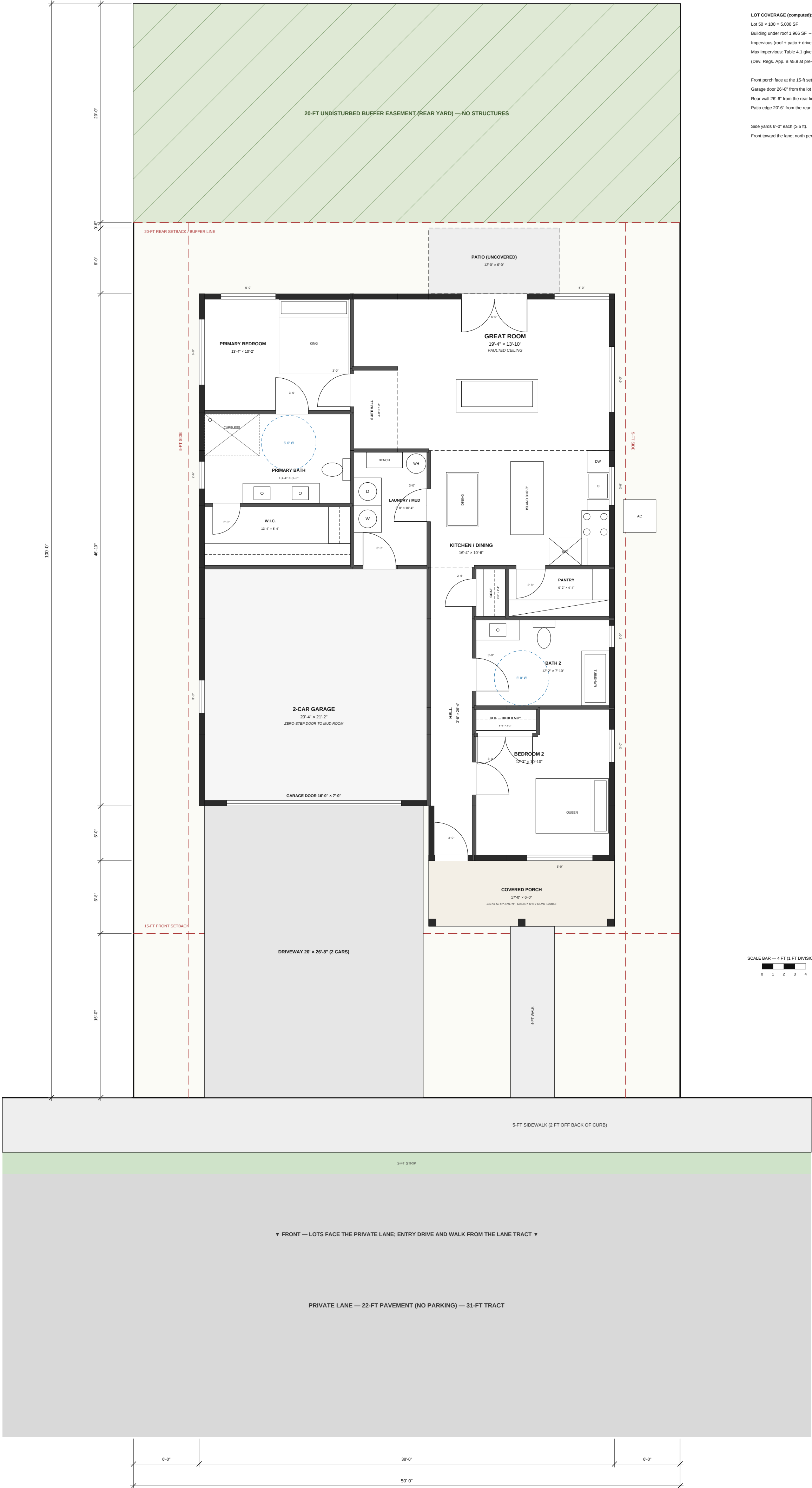
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A-3

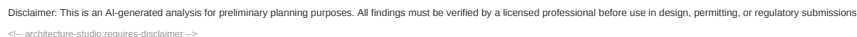
CLUBHOUSE, MAIL KIOSK AND ENTRY MONUMENT SIGN

2026-09-03

Setbacks per Lilburn Zoning Ordinance 2023-603 Table 4.1 (R-2): front 15 ft (local street), side 5 ft, rear 20 ft; rear 20 ft = undisturbed buffer easement (§313(1)); Table 4.2 garage recess ≥ 5 ft



Setbacks per Lilburn Zoning Ordinance 2023-603 Table 4.1 (R-2): front 15 ft (local street), side 5 ft, rear 20 ft; rear 20 ft = undisturbed buffer easement (§313(1)); Table 4.2 garage recess $\geq$ 5 ft



Illustrative Renderings — Application Instructions item (11)

These are illustrative perspectives prepared by the owner-applicant. They are generated from the scaled elevations in sheets A-2.1, A-2.2, A-2.3 and A-3.0 and from the master concept plan, and the scaled drawings govern wherever an image and a drawing differ. Item (11) permits imagery: "The drawings shall be to scale or in proper perspective... Visual imagery may be used."



R 01 — plan a front scheme1



R 02 — plan b front scheme2



R 03 — plan b rear garden scheme3



R 04 — lane streetscape nw



R 05 — hammerhead pocket green



R 06 — entrance arcado monument



R 07 — clubhouse village green



R 08 — pickleball courts



R 09 — aerial oblique nw



R 10 — buffer from neighbour



R 11 — creek woods dry basin



R 12 — materials colour board

Outline Specifications — Cottages at Arcade Springs

Generated: 2026-08-28 Project Type: 55+ (HOPA) detached single-family cottage-home community, fee-simple lots with HOA-maintained yards — R-2 rezoning submission, City of Lilburn, Gwinnett County, Georgia (4535–4541 Arcade Rd SW, Land Lot 123, 6th District; 9.44 ac deeded) Divisions: 11 | Sections: 30 Status: **DRAFT — outline level, prepared by the Owner for the rezoning package** (Lilburn Zoning Ordinance 2023-603 §1003: "architectural renderings/elevations ... colors and materials"). To be superseded by the Architect of Record's sealed construction documents and full-length specifications. Not for bidding or permit.

Basis of design (from FACTS.md §4, decided 2026-08-28)

- **Plans:** Plan A "The Springbrook" 38'-0" × 38'-0" body, 2 BR / 2 BA, ~1,444 SF conditioned; Plan B "The Laurel" 40'-0" × 40'-0" body, 2 BR + den / 2 BA, ~1,600 SF conditioned. One story, 9'-0" plate, ridge ≤ 24'-0"; 8:12 main gable, 4:12 porch; 6'-0"-deep covered front porch; rear porch/patio; 2-car garage (≥ 20'-0" × 21'-0") recessed ≥ 5'-0" behind the front wall (Table 4.2).
- **Lots:** typical 50'-0" × 100'-0" (5,000 SF); rear 20'-0" of perimeter lots held as undisturbed buffer easement (§313(1)). Private lane: 22'-0" pavement in a 31'-0" tract with 5'-0" sidewalk; hammerhead turnarounds; ~44 lots.
- **Universal design (voluntary — exceeds §734, which applies to attached dwellings only):** zero-step entries at porch and garage (finished floor within 1/2 in of the approach), 36-in doors, 60-in turning circles in baths, curbless showers, grab-bar blocking at every fixture, lever hardware, no interior thresholds, LED 3000K lighting, comfort-height toilets, hand-held showers. NFPA 13D sprinklers in every home (Division 21 — not in this outline).
- **Exterior character:** fiber-cement lap (6-in exposure) or board-and-batten; brick or stone water table (30 in) and porch piers; 8×8 cedar posts and brackets; architectural shingles. Three color schemes: (1) Warm White board-and-batten + stone; (2) Sage lap + brick; (3) Clay/Beige lap + stone.
- **Site elements:** 2 pickleball courts (each 20'-0" × 44'-0" court on a 30'-0" × 60'-0" fenced pad), village green, mail kiosk, monument entry sign ≤ 32 SF, 10-ft landscape strip along Arcade Rd, 20-ft buffer supplement, sod lawns.

Governing codes and standards (verified against Georgia DCA "Current State Minimum Codes for Construction", 2026-08-28)

- **2024 International Residential Code (IRC) with Georgia Amendments (2026)** — mandatory statewide 1 January 2026 (DCA memo, 9 Dec 2025). The "IRC 2018 w/ GA amendments" assumed in the task brief and in earlier project notes is **superseded**; IRC section numbers cited below follow the 2018–2024 numbering, which is unchanged for the provisions cited, but each shall be re-checked against the 2024 text and the 2026 Georgia amendments. Per the DCA Board's long-standing omission, the IRC's plumbing, electrical, and energy chapters are NOT used: the **2024 International Plumbing Code (GA Amendments 2026)**, **2023 National Electrical Code (GA Amendments 2026)**, and the IECC below govern instead.
- **2015 International Energy Conservation Code with Georgia Supplements and Amendments (2020, 2022, 2023)** — still the Georgia State Minimum Standard Energy Code; Georgia has not adopted the 2021 or 2024 IECC. Gwinnett County is **Climate Zone 3** (IECC Table R301.1). Georgia-amended Table R402.1.2, Zone 3: fenestration U-0.35 / SHGC-0.27; skylight U-0.55; **ceiling R-38** (R-30 accepted where full-height uncompressed R-30 extends over the top plate at the eaves, GA §R402.2.1); **wood-frame wall R-13** (Georgia amended Zones 2–4 to R-13 — the unamended 2015 IECC "R-20 or R-13 + R-5 ci" in the task brief does not apply); floor R-19; slab R-0; crawl 5/13; air leakage ≤ 5 ACH50 tested (GA §R402.4.1.2); ducts ≤ 4 cfm25/100 SF post-construction. The 2×6 R-20 wall specified here exceeds the Georgia minimum.
- **2024 International Fire Code** as adopted by the Georgia Safety Fire Commissioner (FACTS.md cites "IFC 2018 App. D" — update the citation); Gwinnett County Fire is the AHJ.
- **Accessibility:** the Georgia Accessibility Code (Ga. Comp. R. & Regs. 120-3-20) and the Fair Housing Act design standards do not apply to detached single-family dwellings; **ICC A117.1-2017** is used throughout as a voluntary design guide. HOPA 55+ per 42 U.S.C. §3607(b)(2)(C).
- **Local:** Lilburn Zoning Ordinance 2023-603 (Tables 4.1, 4.2; §313; §319; §1003); Lilburn Development Regulations (Code Appendix B — **not retrievable online; every citation marked VERIFY at the pre-application conference**); Lilburn Site Development Plan Review Checklist (DocumentCenter/1888); Gwinnett County UDO §900-70 (public-street pavement standard, used as the presumptive basis for the private lane); GDOT Standard Specifications, 2021 edition; Gwinnett County Stormwater Management Manual (2017).
- **Climatic and geographic design criteria (IRC Table R301.2, Gwinnett/Lilburn — VERIFY the locally adopted table):** ultimate design wind speed ≈ 115 mph, Exposure B; ground snow ≈ 5 psf; Seismic Design Category B; weathering "moderate"; frost line 12 in; termite "very heavy"; ice-barrier underlayment not required; FEMA Zone X (no SFHA on site).

- **Soils (NRCS via Gwinnett GIS):** Appling / Madison / Pacolet / Cecil-type Piedmont residual clays with a probable hydric unit near the stream head. A geotechnical report is required before foundation design (Division 31 — not in this outline).

Material mapping (pre-confirmed by the Owner, 2026-08-28)

Div	Section	Items
03	03 30 00 Cast-in-Place Concrete	monolithic slab-on-grade, turned-down footings, porch/garage slabs, zero-step approaches
04	04 20 00 Unit Masonry; 04 40 00 Stone Assemblies	brick veneer water table (scheme 2); stacked/ledge stone piers and water table (schemes 1, 3)
06	06 10 00 Rough Carpentry; 06 20 00 Finish Carpentry	2×6 walls, trusses, sheathing, blocking; cedar posts/brackets, painted trim
07	07 21 00; 07 27 00; 07 31 00; 07 46 46; 07 62 00; 07 71 00; 07 92 00	insulation; WRB/air barrier; shingles; fiber-cement siding; flashing; gutters; sealants
08	08 14 00 (with 08 16 13); 08 36 13; 08 53 13	entry and interior doors; carriage-style sectional garage door; vinyl (fiberglass alternate) windows
09	09 29 00; 09 30 13; 09 65 00; 09 91 13; 09 91 23	gypsum board; tile; LVP; exterior painting (three schemes); interior painting
10	10 14 00 Signage; 10 28 13 Toilet Accessories	monument entry sign (assigned here rather than 13 34 00 — it is a signage assembly, not a fabricated engineered structure); grab bars and accessories
12	12 36 00 Countertops	quartz / solid surface
22	22 40 00 Plumbing Fixtures	fixtures and trim only
26	26 50 00 Lighting	interior LED, porch and lane lighting (fixtures only)
32	32 12 16; 32 16 13; 32 31 13; 32 92 00; 32 93 00	lane paving; curb and sidewalks; pickleball fencing; sod; buffer and landscape plantings

General notes applying to every section: (0) Out of scope of this outline (listed for the roadmap): 21 13 00 NFPA 13D sprinklers; 23 00 00 HVAC; 26 00 00 wiring; 31 10 00 site clearing / 31 23 00 earthwork / 31 31 16 termite control; 32 18 23 pickleball court surfacing; 32 84 00 irrigation; 33 40 00 storm drainage and ponds. (a) Substitution requests shall be submitted in writing to the Architect a minimum of 10 days prior to bid date, with product data, samples, and a point-by-point comparison with the specified product. (b) "Appears consistent with" is used deliberately; nothing in this outline asserts code compliance. (c) Manufacturer lists are illustrative national/regional producers; Georgia-based producers are noted where they exist.

Division 03 — Concrete

Section 03 30 00 — Cast-in-Place Concrete

Part 1 — General

- **1.01 Section Includes:** Monolithic (turned-down) slab-on-grade foundations for each cottage; thickened slab edges and interior thickened slabs at bearing lines; garage slabs; porch and patio slabs; zero-step approach landings; brick/stone ledges; sign and pier footings.
- **1.02 Related Sections:** 04 20 00 / 04 40 00 (veneer ledges); 06 10 00 (anchor bolts, sill plates); 07 92 00 (isolation-joint sealant); 09 30 13 (recessed slab at curbless showers); 09 65 00 (slab moisture and flatness); 32 16 13 (site flatwork); 31 31 16 (termite pretreatment) and 31 23 00 (subgrade) — not in this outline.
- **1.03 References:** 2024 IRC (GA 2026) R401–R403.1 (footings), R403.1.4 (12-in minimum depth), R403.1.6 (anchorage), R404.1.6 / R317.1 (wood-to-grade clearances), R506 (slab-on-grade floors), R318 (termite protection); ACI 318-19 and ACI 332-20 (residential concrete); ASTM C94/C94M (ready-mixed concrete); ASTM C150/C150M (portland cement); ASTM C33 (aggregates); ASTM A615 (reinforcing bars); ASTM A1064 (welded wire reinforcement); ASTM E1745 (vapor retarders); ACI 302.1R (floor construction).
- **1.04 Submittals:** Mix designs; reinforcing shop drawings; vapor-retarder and admixture data; foundation plan sealed by a Georgia PE (DRAFT plans in this package are unsealed).
- **1.05 Quality Assurance:** Installer with 5 years' residential slab experience; geotechnical engineer to observe bearing at each footing excavation; slab elevations at every entry landing verified by survey before placement (zero-step tolerance $\pm 1/4$ in).
- **1.06 Delivery, Storage, and Handling:** Reinforcing stored off the ground; concrete placed within 90 minutes of batching; no water added on site beyond the approved mix.
- **1.07 Warranty:** Contractor's 2-year workmanship warranty; structural warranty per the Georgia new-home warranty offered by the builder (10-year structural typical).

Part 2 — Products

- **2.01 Manufacturers (ready-mix):** Thomas Concrete (Atlanta); Argos USA; Heidelberg Materials; or approved equal.
- **2.02 Materials:** Normal-weight concrete, 3,000 psi minimum at 28 days per IRC Table R402.2 (weathering "moderate"); **3,500 psi, air-entrained 5–7 percent, at garage slabs, porches, patios, and exterior landings;** maximum 4-in slump (before plasticizer); footings 2-#4 continuous; slab 4 in minimum (R506.1) with 6×6-W1.4×W1.4 WWR at mid-depth or synthetic macro-fiber per Engineer; 10-mil Class A vapor retarder (ASTM E1745) directly under the slab (R506.2.3) over 4 in of clean crushed stone (R506.2.2); 1/2-in anchor bolts at 6 ft o.c. maximum and within 12 in of plate ends (R403.1.6).
- **2.03 Performance Criteria:** Footing bearing 12 in minimum below finished grade (frost/R403.1.4) and on undisturbed residual soil or engineered fill with allowable bearing per the geotechnical report (assume 2,000 psf until confirmed); slab flatness FF 25 / FL 20 minimum, with 3/16 in in 10 ft local tolerance at LVP areas (09 65 00); interior slab surface **within 1/2 in of the porch landing and garage approach at every entry (zero-step);** exterior landings and garage aprons slope 1/8–1/4 in per ft (max 2 percent) away from the door for 3 ft minimum.
- **2.04 Finishes:** Interior slab steel-trowel; garage and exterior slabs medium broom; exposed edges 3/4-in chamfer; porch slab set 1/2 in below interior finished floor with beveled 1:2 transition at the door (ICC A117.1 §303).
- **2.05 Accessories:** Brick/stone ledge 4-1/2 in wide × 8 in deep formed into the turned-down edge at veneered walls; recessed shower area (slab dropped 1-1/2 in) at each bath; 1/2-in premolded isolation joint (ASTM D1751) at garage/house slab and at all flatwork abutting the foundation; sawcut control joints at ≤ 12 ft each way; termiticide pretreatment (R318) by the 31 31 16 applicator before the vapor retarder.

Part 3 — Execution

- **3.01 Examination:** Verify subgrade compaction (95 percent standard Proctor, ASTM D698, top 12 in) and geotechnical sign-off; verify plumbing/sprinkler sleeves, shower recess forms, and landing elevations against the approved zero-step entry detail.
- **3.02 Preparation:** Fine-grade stone base; lap vapor retarder 6 in and tape; set forms to ± 1/4 in; place and chair reinforcement; wet the subgrade in hot weather (ACI 305R).
- **3.03 Installation:** Place monolithically where practical; consolidate with internal vibration at turned-down edges; finish and cure 7 days (ASTM C309 curing compound compatible with LVP adhesives — or water cure); hold anchor bolts by template, not "wet set"; do not disturb termite-treated soil.
- **3.04 Field Quality Control:** Slump, air, and cylinder tests per ASTM C31/C39 for each 50 cy or each day's pour; survey check of finished-floor elevation at every entry before framing.
- **3.05 Cleaning and Protection:** Protect slab edges from backfill damage; keep slabs free of hydraulic-fluid and gypsum-mud staining; no vehicular loads on garage slabs for 7 days.

[REVIEW REQUIRED] Structural values (bearing, reinforcement, slab thickness at thickened lines) are assumed pending the geotechnical report and a Georgia PE's foundation design. The zero-step detail must reconcile IRC R404.1.6 / R317.1 wood clearances and siding manufacturers' 2-in clearance to hardscape with the 1/2-in maximum rise at the door — see 04 40 00 and 07 46 46 for the masonry-base solution.

Division 04 — Masonry

Section 04 20 00 — Unit Masonry (Brick Veneer Water Table — Scheme 2)

Part 1 — General

- **1.01 Section Includes:** Anchored brick veneer water table, 30 in high above finished floor, on scheme 2 ("Sage") cottages, including porch pier bases where brick is used in lieu of stone; rowlock sill cap; brick at the scheme 2 monument sign.
- **1.02 Related Sections:** 03 30 00 (brick ledge); 06 10 00 (sheathing and tie backing); 07 27 00 (WRB behind veneer); 07 62 00 (through-wall flashing); 07 46 46 (siding above the cap); 07 92 00 (sealant at cap-to-siding joint); 10 14 00 (sign base).
- **1.03 References:** 2024 IRC (GA 2026) R703.8 (anchored masonry veneer: support, ties, air space, flashing, weeps); TMS 402/602-22 (Building Code Requirements and Specification for Masonry Structures); ASTM C216 (facing brick); ASTM C270 (mortar); ASTM C1714 (preblended mortar); ASTM A153 (galvanized ties); ASTM C1405 (glazed brick — not used); BIA Technical Notes 28B and 7.
- **1.04 Submittals:** Brick samples (3 units per range) and full-size mortar color samples; tie and flashing data; mock-up approval before production.
- **1.05 Quality Assurance:** Mason with 5 years' veneer experience; mock-up 4 ft × 3 ft including the ledge, flashing, weeps, rowlock cap, and the siding/sealant transition; retain as the standard for the work.

- **1.06 Delivery, Storage, and Handling:** Brick on pallets, covered, off the ground; mortar materials dry; no frozen or wet units laid.
- **1.07 Warranty:** Manufacturer's 50-year (or lifetime) limited brick warranty; Contractor's 2-year workmanship warranty.

Part 2 — Products

- **2.01 Manufacturers:** Cherokee Brick (Macon, GA); General Shale (Meridian Brick); Glen-Gery; Pine Hall Brick; or approved equal.
- **2.02 Materials:** Modular face brick (3-5/8 × 2-1/4 × 7-5/8 in), ASTM C216 Grade SW, Type FBS, red-brown blend as selected per scheme 2; mortar ASTM C270 Type N by proportion, colored to match; 22-ga × 7/8-in corrugated ties (or adjustable two-piece) hot-dip galvanized ASTM A153 Class B-2 at ≤ 32 in horizontal / ≤ 24 in vertical, ≤ 2.67 SF per tie (R703.8.4).
- **2.03 Performance Criteria:** 1-in nominal air space (R703.8.4); veneer bears fully on the concrete ledge (projection ≤ 1/3 of the unit thickness per TMS 402; support per R703.8.2); through-wall flashing and weeps at ≤ 33 in o.c. at the base (R703.8.6); maximum height as a water table 30 in, so no lintels or expansion joints are expected beyond corner control.
- **2.04 Finishes:** Rowlock (or cast-stone) sill cap sloped 15 degrees with drip; tooled concave joints; scheme 2 field brick with matching or lighter mortar as selected by the Architect from the manufacturer's full range.
- **2.05 Accessories:** Flexible flashing ASTM D1970 or 40-mil self-adhered with stainless drip edge; cotton-rope or cell weep vents; mortar-net collection device; rowlock cap flashing turned up 8 in behind the siding WRB.

Part 3 — Execution

- **3.01 Examination:** Verify ledge elevation, WRB coverage, and flashing sequence with 07 27 00 before laying; verify porch-pier footings.
- **3.02 Preparation:** Install through-wall flashing lapped over the WRB and extended 1/2 in past the face; set weeps; establish coursing so the cap lands at 30 in above finished floor.
- **3.03 Installation:** Lay in running bond, full head and bed joints, per TMS 602 Article 3; keep the air space clean; tie to studs through sheathing; tool joints when thumbprint hard; install the rowlock cap with full mortar joints and a sealant joint (07 92 00) at the siding above.
- **3.04 Field Quality Control:** Verify tie spacing and embedment (1-1/2 in minimum) before enclosure; check weep continuity with water test at the mock-up.
- **3.05 Cleaning and Protection:** Clean with bucket-and-brush method and proprietary cleaner per BIA Technical Note 20; no muriatic acid on colored mortar; cover walls at end of day.

[REVIEW REQUIRED] Brick and mortar selections and the cap profile are generic pending scheme 2 samples; the veneer ledge and tie schedule shall be confirmed by the Engineer.

Section 04 40 00 — Stone Assemblies (Ledge/Stacked Stone Piers and Water Table — Schemes 1 and 3)

Part 1 — General

- **1.01 Section Includes:** Adhered ledge/stacked stone veneer (manufactured stone as the basis of design; quarried thin-veneer natural stone as an Architect-approved alternate) at the 30-in water table and at porch piers (nominal 20 × 20 in or 24 × 24 in) on schemes 1 ("Warm White") and 3 ("Clay/Beige"); stone-clad monument sign base for schemes 1 and 3; cast-stone pier caps and water-table sills.
- **1.02 Related Sections:** 03 30 00 (ledge, pier footings); 06 10 00 (pier framing or CMU cores); 06 20 00 (cedar posts bearing on pier caps); 07 27 00 (two WRB layers); 07 62 00 (flashing); 07 92 00 (sealants); 10 14 00 (sign).
- **1.03 References:** 2024 IRC (GA 2026) R703.12 (adhered masonry veneer installation and clearances) and R703.7 (lath/scratch coat by reference); TMS 402/602-22 Chapter 12 (adhered veneer); ASTM C1670 (adhered manufactured stone masonry veneer units); ASTM C1780 (installation practice for adhered manufactured stone veneer); ASTM C1063 (lath) and ASTM C847 (metal lath); ASTM C270 / C1714 (mortar); ASTM C1364 (architectural cast stone); MVMA Installation Guide (5th ed.); natural stone alternate: ASTM C1527 (adhered natural thin stone veneer).
- **1.04 Submittals:** 12 × 12 in stone boards (2 per scheme), cap samples, mortar color, lath/WRB data, ICC-ES report for manufactured stone; mock-up.
- **1.05 Quality Assurance:** Installer trained by the manufacturer; mock-up of one full pier and 4 ft of water table with cap, weep screed, and siding transition.
- **1.06 Delivery, Storage, and Handling:** Boxed units kept dry; blend from at least three boxes; no installation below 40 degrees F without protection.

- **1.07 Warranty:** Manufacturer's 50-year limited warranty (manufactured stone) or Contractor's 2-year workmanship warranty (natural stone).

Part 2 — Products

- **2.01 Manufacturers:** Eldorado Stone and Cultured Stone (Westlake Royal Building Products); ProVia (Hearth & Home); Coronado Stone Products; or approved equal. Natural thin-veneer alternate: Tennessee or North Georgia fieldstone ledge from an Architect-approved quarry.
- **2.02 Materials:** Ledge/stacked ("dry-stack" appearance, tight joint) profile, ≤ 15 psf installed (R703.12); scratch/setting mortar ASTM C270 Type S polymer-modified (ANSI A118.4-type "stone veneer mortar" acceptable); 2.5-lb self-furring galvanized lath (ASTM C847); two layers WRB per R703.12 (the outer layer may be the 07 27 00 housewrap; inner layer Grade D paper or a second housewrap); cast-stone caps ASTM C1364, 2-1/2 in thick, sloped with drip, buff or gray per scheme.
- **2.03 Performance Criteria:** Clearances per R703.12.1: 4 in above earth, 2 in above paved areas, **1/2 in above exterior walking surfaces supported by the same foundation** — which is the mechanism that lets the stone base meet the zero-step landing; weep screed at the base (ASTM C1063); stone freeze-thaw rated (ASTM C1670 §6).
- **2.04 Finishes:** Scheme 1: gray-tan "Appalachian"-type ledge blend; scheme 3: buff/tan ledge blend; caps as selected; joints raked/dry-stack; no sealer unless the manufacturer requires it.
- **2.05 Accessories:** Corner units; galvanized casing bead and weep screed; kick-out and cap flashings (07 62 00); 1/2-in setting-bed drainage mat where the manufacturer requires a rainscreen (recommended in Georgia's humid climate).

Part 3 — Execution

- **3.01 Examination:** Verify sheathing, WRB, flashing laps, and pier footings; verify landing elevations for the 1/2-in clearance.
- **3.02 Preparation:** Install second WRB layer, weep screed, lath (6-in laps, fasteners into studs at 6 in o.c.), and 1/2-in scratch coat; cure 48 hours.
- **3.03 Installation:** Per ASTM C1780 and the manufacturer's instructions: back-butter each unit, press with a slight twist, achieve 100 percent mortar contact; set caps in full bed with sealant at cap joints; keep stone 1/2 in clear of the landing and 2 in clear of walks.
- **3.04 Field Quality Control:** Random unit pull test (hand) at the mock-up; verify weep screed is open and unclogged after installation.
- **3.05 Cleaning and Protection:** Clean with soft brush and clean water while mortar is fresh; no acid; protect caps until Substantial Completion.

[REVIEW REQUIRED] Manufactured-vs-natural stone and cap profile are Owner/Architect selections pending samples; the pier core (CMU vs. framed) and post connection shall be detailed by the Engineer.

Division 06 — Wood, Plastics, and Composites

Section 06 10 00 — Rough Carpentry (including 06 17 53 Shop-Fabricated Wood Trusses)

Part 1 — General

- **1.01 Section Includes:** 2×6 exterior wall framing at 16 in o.c., 9'-0" plate; 2×4 interior partitions; pressure-treated sill plates; wall and roof sheathing; pre-engineered metal-plate-connected roof trusses (8:12 main gables, 4:12 porch, raised heels); porch beams and posts backing; garage/house separation framing; **grab-bar and accessory blocking**; fire blocking and draft stopping; rough hardware.
- **1.02 Related Sections:** 03 30 00 (anchorage); 06 20 00 (cedar posts and trim); 07 21 00 (cavity insulation, raised-heel truss); 07 27 00 (WRB/air barrier continuity at plates); 07 31 00 (roof sheathing); 08 series (rough openings); 09 29 00 (garage separation board); 10 28 13 (grab-bar locations).
- **1.03 References:** 2024 IRC (GA 2026) R301.2 (design criteria), R302.6 and R302.11 (garage separation, fireblocking), R502–R602 (floor/wall framing), R602.10 (wall bracing), R802 (roof framing, uplift), R806 (attic ventilation), Table R602.3(1) (fastening); AWC NDS-2024 and WFCM-2024 (as referenced by the 2024 IRC — VERIFY edition); ANSI/TPI 1-2022 (metal-plate-connected wood trusses); APA PRP-108 / DOC PS 2 (sheathing); AWPA U1 (preservative treatment, UC2 for sills on concrete); SPIB / WWPA grading rules; ICC A117.1-2017 §604.5, §608.3, §609 (grab-bar geometry for blocking layout).
- **1.04 Submittals:** Truss layout and truss design drawings sealed by a Georgia PE with reactions, uplift, and bracing; sheathing/nailing plan for wall bracing; blocking plan for each bath (grab bars, seats, slide bars, towel bars); treated-lumber certifications.

- **1.05 Quality Assurance:** Framers with 5 years' single-family experience; truss fabricator certified under the SBCA in-plant QC program; pre-drywall inspection of blocking, fireblocking, and air-sealing before insulation.
- **1.06 Delivery, Storage, and Handling:** Lumber stacked and covered; trusses stored flat and braced; treated lumber separated from untreated.
- **1.07 Warranty:** Contractor's 2-year workmanship; builder's 10-year structural warranty; truss manufacturer's design warranty.

Part 2 — Products

- **2.01 Manufacturers:** Trusses: Builders FirstSource, UFP Industries, 84 Lumber components (regional plants); engineered lumber: Weyerhaeuser Trus Joist, Boise Cascade, LP Building Solutions; sheathing: Georgia-Pacific (GA), Huber (ZIP System alternate with 07 27 00), LP; connectors: Simpson Strong-Tie, MiTek; or approved equal.
- **2.02 Materials:** Studs SPF or SYP No. 2, 2×6 at 16 in o.c. (104-5/8-in precut for 9'-0" plate); double top plates; SYP No. 2 headers or LVL; PT sill plate (AWPA UC2 minimum, UC4A at grade-level exposure) with foam sill sealer; wall sheathing 7/16-in OSB (or 15/32-in plywood) APA-rated, Exposure 1, continuous (CS-WSP bracing, R602.10.4); roof sheathing 15/32-in APA-rated with H-clips at 24 in o.c. trusses; trusses SYP with 20-ga plates per TPI 1; blocking 2×10 flat (or 3/4-in plywood 12 in wide) let in flush.
- **2.03 Performance Criteria:** Loads per IRC R301.2 and the Engineer: roof live 20 psf, ground snow 5 psf, wind 115 mph ultimate Exposure B (VERIFY), SDC B; uplift connectors at every truss bearing per truss reactions (R802.11); porch trusses/beams sized for the 6-ft porch and the 8×8 posts; attic ventilation 1/300 with balanced soffit/ridge (R806.2); raised-heel trusses ≥ 12 in at exterior walls to carry full-depth R-38 over the plate (07 21 00); garage ceiling trusses designed for the 1-hr-equivalent gypsum weight (not rated — R302.6 separation only).
- **2.04 Finishes:** Not applicable (concealed); exposed porch framing wrapped per 06 20 00.
- **2.05 Accessories:** Hot-dip galvanized fasteners at treated lumber and exterior (ASTM A153); hurricane ties; anchor-bolt plate washers 3 × 3 × 0.229 in; fireblocking mineral wool or wood per R302.11; foam gaskets at plates for air sealing (IECC Table R402.4.1.1).

Part 3 — Execution

- **3.01 Examination:** Verify foundation dimensions, anchor bolt positions, and slab elevations at every door (zero-step); verify truss bearing walls match the truss layout.
- **3.02 Preparation:** Snap layout; set sills level on sealer; shim ≤ 1/4 in; lay out bath blocking from the grab-bar plan: **continuous horizontal blocking centered 33–36 in above finished floor on all shower/tub and toilet walls, vertical blocking for 18-in vertical bars, and blocking at 30–48 in for slide bars, fold-down seats (rated 250 lb, A117.1 §609.8), and towel bars.**
- **3.03 Installation:** Frame per IRC Chapter 6 and Table R602.3(1); sheathe walls continuously with edge nailing 6 in / field 12 in (or per bracing plan); set trusses per the sealed layout with temporary and permanent bracing per BCSI; install uplift connectors before roof sheathing; frame garage/house wall and ceiling continuous for the 1/2-in gypsum separation (R302.6); fireblock at plates, soffits, and chases; leave 1/2-in gaps at slab edges for foam sealing.
- **3.04 Field Quality Control:** Third-party or AHJ framing inspection; truss bracing inspection by the fabricator's engineer where required; blocking photographed and logged per lot before drywall (HOA maintenance record for future grab-bar installation).
- **3.05 Cleaning and Protection:** Keep sheathing dry (cover roof within 7 days); remove debris from stud cavities before insulation; protect PT cut ends with field treatment (AWPA M4).

[REVIEW REQUIRED] All structural loads, bracing, and connections are presumptive until designed by the Georgia PE; confirm the 2024 IRC/2026 GA amendment section numbers and the NDS/WFCM editions referenced.

Section 06 20 00 — Finish Carpentry

Part 1 — General

- **1.01 Section Includes:** 8×8 nominal Western Red Cedar porch posts on stone/brick piers; cedar gable brackets/corbels; cedar-wrapped porch beams; painted exterior trim (cellular PVC or fiber-cement per 07 46 46); porch ceilings (beadboard); interior casing, base, window stools, closet shelving; door and cabinet blocking.
- **1.02 Related Sections:** 04 40 00 (pier caps); 06 10 00 (structural posts/beams); 07 46 46 (fiber-cement trim); 07 62 00 (post-base and beam flashing); 07 92 00; 09 91 13 / 09 91 23 (finishes).
- **1.03 References:** AWI/AWMAC/WI Architectural Woodwork Standards (AWS) 2nd ed., Custom grade; WRCLA (Western Red Cedar Lumber Association) grading and installation guides; WCLIB/WWPA grading rules; ASTM D4216 (rigid PVC compounds; cellular PVC trim per the manufacturer's ICC-ES report); 2024 IRC (GA 2026) R317.1 (protection of wood; post-to-pier separation), R407.3 (post-base connection); ANSI A208.2 (MDF).
- **1.04 Submittals:** Cedar grade certification and sample; bracket shop drawing (one per plan); PVC/fiber-cement trim profiles; interior trim profiles; post-base hardware data.

- **1.05 Quality Assurance:** Carpenter with 5 years' finish experience; first cottage porch serves as the mock-up for post, bracket, and beam wrap.
- **1.06 Delivery, Storage, and Handling:** Cedar stickered and covered, acclimated 7 days; interior trim delivered after the building is dried-in and conditioned.
- **1.07 Warranty:** Contractor's 2-year workmanship; cellular PVC manufacturer's 25-year (or lifetime) limited warranty.

Part 2 — Products

- **2.01 Manufacturers:** Cedar: WRCLA-member mills; cellular PVC trim: AZEK (Timbertech), Versatex, Kleer (Tapco); interior trim: Woodgrain, Metrie, Boise Cascade; or approved equal.
- **2.02 Materials:** Posts 8×8 nominal (7-1/2 in actual) Western Red Cedar, WRCLA "Appearance"/No. 1 grade, S4S or resawn, ends sealed; alternate: 6×6 PT structural post (06 10 00) wrapped in 1×10 clear cedar with mitered corners; brackets 4×6 and 4×8 cedar, glued and through-bolted; porch beam wrap 1× cedar or fiber-cement; exterior painted trim 5/4 × 4 and 5/4 × 6 cellular PVC (ASTM D4216 compound) or fiber-cement trim (07 46 46); porch ceiling 5/16-in fiber-cement beadboard or 1×6 T&G cedar; interior casing 3-1/2 in and base 5-1/4 in primed finger-jointed pine or MDF, shaker/flat profile; stools solid poplar.
- **2.03 Performance Criteria:** Posts separated from the masonry cap by a standoff base ≥ 1 in (R317.1.4) with a 1-in-diameter minimum uplift/lateral connection (R407.3) rated by the Engineer; brackets non-structural (decorative) unless noted.
- **2.04 Finishes:** Cedar: semi-transparent penetrating stain (09 91 13) "natural cedar" on schemes 1 and 3, painted trim color on scheme 2 as selected; painted trim per scheme (09 91 13); interior trim semi-gloss white (09 91 23).
- **2.05 Accessories:** Stainless or hot-dip galvanized post base/cap (Simpson ABU/CPS-type or approved equal); stainless finish nails at cedar; PVC cement and color-matched fasteners at PVC; construction adhesive.

Part 3 — Execution

- **3.01 Examination:** Verify pier cap elevations and beam heights; verify wall trim backing.
- **3.02 Preparation:** Back-prime all cedar and paint-grade trim; seal cut ends; lay out post centerlines from the porch beam.
- **3.03 Installation:** Set posts plumb on standoff bases; cut to bearing with full contact; install brackets with concealed bolts and caps; run exterior trim with scarfed joints over backing, 1/8-in sealant gap at siding; interior trim with tight miters, nails set and filled, caulked to wall.
- **3.04 Field Quality Control:** Post plumb ± 1/8 in over the height; reveal lines consistent ± 1/16 in across each porch.
- **3.05 Cleaning and Protection:** Protect cedar from mortar and paint splatter; protect interior trim until final painting.

[REVIEW REQUIRED] Post size versus structural demand (8×8 is architectural; a 6×6 PT wrapped post may be the economical structure) and bracket attachment shall be confirmed by the Engineer.

Division 07 — Thermal and Moisture Protection

Section 07 21 00 — Thermal Insulation

Part 1 — General

- **1.01 Section Includes:** Wall cavity insulation (2×6, R-20); attic insulation (R-38 blown); insulation at the garage/house wall and any conditioned-space ceiling over the garage; eave baffles; insulation certificate.
- **1.02 Related Sections:** 06 10 00 (raised-heel trusses, baffles backing); 07 27 00 (air barrier — insulation performance depends on it); 09 29 00 (interior air barrier at ceilings).
- **1.03 References:** 2015 IECC with Georgia Amendments (2020–2023) Table R402.1.2 (Zone 3: wall R-13, ceiling R-38, floor R-19, slab R-0), §R402.2.1 (R-30 uncompressed alternative), §R401.3 (certificate), §R303.1 (installation and marking); ASTM C665 (mineral-fiber batts), ASTM C764 (loose-fill mineral fiber), ASTM C739 (cellulose), ASTM C1029 (spray polyurethane, if used); ASTM E84 (surface burning); RESNET/ANSI 301 Grade I installation.
- **1.04 Submittals:** Product data with R-value per inch, bag-count chart for blown attic insulation, and the IECC R401.3 certificate posted at the electrical panel.
- **1.05 Quality Assurance:** Installer certified by the manufacturer; Grade I installation verified by the HERS/energy rater at the pre-drywall inspection.
- **1.06 Delivery, Storage, and Handling:** Keep dry and in packaging; do not install wet batts.
- **1.07 Warranty:** Manufacturer's limited lifetime (batts); Contractor's 2-year workmanship.

Part 2 — Products

- **2.01 Manufacturers:** Owens Corning; Johns Manville; Knauf Insulation; CertainTeed; cellulose alternate Greenfiber; or approved equal.
- **2.02 Materials:** Walls: R-20 unfaced fiberglass batts (ASTM C665 Type I) friction-fit in 2×6 cavities, or R-21 high-density, or blown-in-blanket fiberglass at R-23; attic: blown fiberglass (ASTM C764 Type I) or cellulose (ASTM C739) to R-38 minimum (R-49 optional upgrade); garage/house wall R-20; attic access hatch insulated R-38 with gasket; kraft-faced batts permitted only where a Class III vapor retarder is acceptable (Zone 3: no interior vapor retarder required, IRC R702.7).
- **2.03 Performance Criteria:** Whole-house target: exceed GA Table R402.1.2 by ≥ 50 percent at walls (R-20 vs R-13) and meet R-38 at 100 percent of the ceiling using raised-heel trusses; Grade I; ENERGY STAR Single-Family New Homes v3.2 optional (VERIFY Owner interest).
- **2.04 Finishes:** Not applicable.
- **2.05 Accessories:** Cardboard or plastic eave baffles at every truss bay with 2-in clear vent path; attic rulers (one per 300 SF); insulation dams at the hatch and at eaves; metal-faced fire-rated tape at spray foam if used.

Part 3 — Execution

- **3.01 Examination:** Verify air-sealing (07 27 00) is complete and inspected; verify baffles and dams; verify no wet framing (≤ 19 percent MC).
- **3.02 Preparation:** Install baffles and dams; mark attic depth on rulers; seal top plates and penetrations before blowing.
- **3.03 Installation:** Batts cut to fit around wiring/boxes without compression (Grade I); split around cables; blow attic to the settled depth on the bag chart; keep insulation 3 in from non-IC recessed fixtures (all fixtures IC-rated per 26 50 00, so not expected).
- **3.04 Field Quality Control:** Rater's Grade I inspection with photos; depth probe at 10 attic locations per house; certificate posted.
- **3.05 Cleaning and Protection:** Remove overspray/loose fill from soffit vents; protect batts from moisture until drywall.

[REVIEW REQUIRED] Georgia has not adopted the 2021/2024 IECC; if adoption occurs before permit, wall R-values and ACH50 limits change. Confirm the energy-code path (prescriptive vs. ERI) with the rater.

Section 07 27 00 — Air Barriers (Weather-Resistive Barrier and Air Barrier)

Part 1 — General

- **1.01 Section Includes:** Mechanically attached drainable housewrap as the water-resistive barrier (WRB) and exterior air barrier; self-adhered flashing at openings; sill pans; sealing of the ceiling plane, plates, and penetrations to form a continuous air barrier; blower-door testing.
- **1.02 Related Sections:** 04 20 00 / 04 40 00 (WRB behind veneer, two layers at stone); 06 10 00 (sheathing); 07 46 46; 07 62 00; 07 92 00; 08 53 13 / 08 14 00 (flashed openings); 09 29 00 (airtight drywall).
- **1.03 References:** 2024 IRC (GA 2026) R703.2 (WRB) and R703.4 (flashing); 2015 IECC (GA) §R402.4.1.1 and Table R402.4.1.1 (air barrier and insulation installation), §R402.4.1.2 (≤ 5 ACH50, tested); ASTM E1677 (Type I WRB); ICC-ES AC308; ASTM E2273 (drainage efficiency); ASTM E2112 (window/door installation); AAMA 711 (self-adhered flashing), AAMA 714 (sill pans); ASTM E779 / RESNET (blower door).
- **1.04 Submittals:** WRB and flashing data with ICC-ES reports; sequencing details at window sill/head, stone base, kick-out, and porch roof-to-wall; blower-door test reports per house.
- **1.05 Quality Assurance:** Installer trained by the WRB manufacturer; the first cottage is a full-envelope mock-up reviewed before the WRB is covered; independent RESNET/BPI rater performs testing.
- **1.06 Delivery, Storage, and Handling:** Rolls stored out of sunlight; WRB exposure limited to the manufacturer's UV limit (typically 120 days) before siding.
- **1.07 Warranty:** Manufacturer's 10-year limited (WRB) system warranty where offered; Contractor's 2-year workmanship.

Part 2 — Products

- **2.01 Manufacturers:** DuPont (Tyvek DrainWrap / HomeWrap); Kingspan (GreenGuard DrainWrap); Benjamin Obdyke (HydroGap); Henry/Carlisle (Blueskin VP100 self-adhered alternate); integrated-sheathing alternate: Huber ZIP System or Georgia-Pacific ForceField (coordinate with 06 10 00); or approved equal.
- **2.02 Materials:** Drainable housewrap, ASTM E1677 Type I, drainage efficiency ≥ 90 percent (ASTM E2273), UV-stable, perm ≥ 10 ; 4-in and 6-in self-adhered butyl or acrylic flashing (AAMA 711 Level 3); preformed or site-formed sill pans (AAMA 714); cap-nail fasteners; manufacturer's seam tape.
- **2.03 Performance Criteria:** Continuous air barrier at the exterior sheathing plane and the ceiling gypsum plane, tied at top plates; ≤ 5 ACH50 (Georgia) with a project target of ≤ 3 ACH50; water-shedding laps ≥ 6 in horizontal, ≥ 6 in vertical; every

opening flashed "shingle fashion" with sill pan, jamb, head, and the WRB lapped over the head flashing.

- **2.04 Finishes:** Not applicable (concealed).
- **2.05 Accessories:** Spray foam (low-expansion) and acoustical/air-seal sealant at plates, penetrations, and window shims; gaskets at electrical boxes; fire-rated sealant at chases (R302.11).

Part 3 — Execution

- **3.01 Examination:** Verify sheathing is fastened, dry, and free of gaps > 1/4 in; verify rough openings are square and sized for the sill pan.
- **3.02 Preparation:** Install sill pans first; cut the WRB with the modified "I" cut at openings; install stone-base weep screed and second WRB layer where 04 40 00 requires.
- **3.03 Installation:** Wrap from the bottom up, lapped shingle fashion, fastened at 12–18 in o.c. to studs with cap nails; tape all seams and penetrations; integrate with 07 62 00 kick-out and step flashing and with veneer through-wall flashing; seal top plates to sheathing and to ceiling gypsum (airtight-drywall approach) per Table R402.4.1.1; seal every wire, pipe, duct, and sprinkler penetration.
- **3.04 Field Quality Control:** Pre-siding inspection with photo log of every opening; blower-door test at completion (and optional mid-construction test at drywall) documented on the IECC certificate; failed houses re-sealed and re-tested.
- **3.05 Cleaning and Protection:** Repair tears with tape the same day; cover the WRB with siding within the exposure limit.

[REVIEW REQUIRED] Generic system pending Architect's selection between housewrap and integrated-sheathing WRB; the stone-base and zero-step flashing details require the Architect's drawn detail.

Section 07 31 00 — Shingles and Shakes (Architectural Asphalt Shingles)

Part 1 — General

- **1.01 Section Includes:** Laminated ("architectural"/dimensional) asphalt shingles, Class A, on 8:12 main roofs and 4:12 porch roofs; underlayment; self-adhered membrane at eaves, valleys, and penetrations; starter and hip/ridge shingles; ridge and soffit ventilation products; roof accessories.
- **1.02 Related Sections:** 06 10 00 (sheathing, ventilation area); 07 62 00 (drip edge, step and kick-out flashing, valley metal); 07 71 00 (gutters); 07 92 00.
- **1.03 References:** 2024 IRC (GA 2026) R905.1–R905.2 (asphalt shingles: slope, underlayment, fastening, flashing), R903.2 (flashing), R806 (ventilation); ASTM D3462 (asphalt shingles, glass felt); ASTM E108 / UL 790 Class A; ASTM D7158 Class H (wind, 150 mph) and ASTM D3161 Class F; ASTM D8257 (mechanically attached polymeric underlayment) or ASTM D226 Type II felt; ASTM D1970 (self-adhered underlayment); ASTM D4869; ARMA Residential Asphalt Roofing Manual; NRCA Roofing Manual — Steep-slope.
- **1.04 Submittals:** Shingle samples (3 colors per scheme), underlayment data, ventilation calculation (net free area per R806), warranty sample, fastening pattern.
- **1.05 Quality Assurance:** Roofer certified by the manufacturer (for the system warranty); mock-up: first cottage roof reviewed for starter, valley, and ridge details.
- **1.06 Delivery, Storage, and Handling:** Bundles flat, covered, ≤ 2 pallets high; no installation of shingles below 40 degrees F without hand-sealing.
- **1.07 Warranty:** Manufacturer's limited lifetime (minimum 30-year) shingle warranty with **10-year non-prorated period, 130-mph wind coverage (with specified starter/ridge), and 10-year algae-resistance**; Contractor's 5-year workmanship.

Part 2 — Products

- **2.01 Manufacturers:** GAF (Timberline HDZ); Owens Corning (Duration); CertainTeed (Landmark); Atlas Roofing (Pinnacle Pristine, Atlanta GA); or approved equal.
- **2.02 Materials:** Laminated fiberglass-mat asphalt shingles, ASTM D3462, Class A, algae-resistant (copper-bearing granules), with manufacturer's starter strip and matching hip/ridge cap; underlayment: synthetic ASTM D8257 (one layer at 8:12; two layers or self-adhered at 4:12 per R905.2.7 — VERIFY), self-adhered ASTM D1970 36 in at eaves, valleys, and around penetrations (ice barrier not required in Zone 3 but specified for durability); ridge vent shingle-over, ~18 in² NFA/ft, with external baffle.
- **2.03 Performance Criteria:** Class A; wind ASTM D7158 Class H; nail pattern per manufacturer's high-wind (6-nail) pattern; 12-ga galvanized or stainless roofing nails, 3/8-in heads, penetrating ≥ 3/4 in or through sheathing (R905.2.5); balanced ventilation 1/300 with ≥ 40 percent at the ridge (R806.2).
- **2.04 Finishes:** Scheme 1 charcoal; scheme 2 weathered wood; scheme 3 driftwood/brown — as selected by the Architect from the manufacturer's full range; cool-roof color not required.

- **2.05 Accessories:** Painted-steel or aluminum drip edge (07 62 00); lead-free flexible pipe boots (EPDM/aluminum) rated 20 years; roof-to-wall kick-out diverters; sprinkler-system and bath-fan roof jacks with backdraft dampers; snow guards not required.

Part 3 — Execution

- **3.01 Examination:** Verify sheathing is dry, fastened, with H-clips and 1/8-in gaps; verify baffles, ridge slot (2 in), and drip edge sequence (under underlayment at eaves, over at rakes — R905.2.8.5).
- **3.02 Preparation:** Install self-adhered membrane at eaves/valleys; install underlayment with 4-in horizontal / 6-in end laps, cap-nailed; install step/kick-out flashing bases with 07 62 00.
- **3.03 Installation:** Per R905.2, ARMA, and the manufacturer: starter strip at eaves and rakes; 5-5/8-in exposure (or per product); closed-cut or woven valleys over metal; 6 nails per shingle; hand-seal at rakes and at 4:12 pitches in cold weather; ridge vent continuous with end plugs; cap shingles fastened with 1-3/4-in nails.
- **3.04 Field Quality Control:** Manufacturer's inspection where required for the system warranty; visual check of nail placement (no exposed, high, or overdriven nails) on 10 percent of each roof.
- **3.05 Cleaning and Protection:** Remove loose granules and nails from gutters and grounds (magnetic sweep); no traffic on finished roofs above 80 degrees F.

[REVIEW REQUIRED] Life-safety adjacent (Class A rating, wind attachment); underlayment requirement at 4:12 and the locally adopted wind speed to be confirmed by the Architect against the 2024 IRC/GA amendments.

Section 07 46 46 — Fiber-Cement Siding

Part 1 — General

- **1.01 Section Includes:** Fiber-cement lap siding, 7-1/4-in plank at 6-in exposure (schemes 2 and 3); fiber-cement vertical panels with battens (board-and-batten, scheme 1); fiber-cement trim, corner boards, frieze, soffit, and porch-ceiling beadboard; factory or field finish.
- **1.02 Related Sections:** 04 20 00 / 04 40 00 (water-table cap transition); 06 10 00 (sheathing, stud backing at butt joints); 06 20 00 (cedar and PVC trim); 07 27 00 (WRB); 07 62 00 (Z- and head flashing); 07 92 00; 09 91 13 (paint).
- **1.03 References:** 2024 IRC (GA 2026) Table R703.3(1) (fiber-cement siding attachment) and R703.10 (fiber-cement siding), R703.4 (flashing); ASTM C1186 (Grade II, Type A fiber-cement flat sheets/siding); ASTM E84 (Class A); ASTM E136 (noncombustible); ICC-ES ESR-2290 (James Hardie) or equivalent evaluation report; manufacturer's installation requirements for the HZ10 (hot-humid) climate zone.
- **1.04 Submittals:** Product data, texture and profile samples (12 in), batten spacing elevation for scheme 1, ColorPlus or field-paint color chips, fastener schedule, flashing details at butt joints and the water-table cap.
- **1.05 Quality Assurance:** Installer with 3 years' fiber-cement experience and manufacturer training (James Hardie "Preferred" or equivalent); mock-up 8 ft × 8 ft per scheme with trim, corner, batten, and stone/brick base transition.
- **1.06 Delivery, Storage, and Handling:** Store flat, covered, off the ground; keep dry before installation (wet boards shrink after installation); carry planks on edge; use HEPA-vacuum dust-collecting saws (OSHA silica rule 29 CFR 1926.1153).
- **1.07 Warranty:** Manufacturer's 30-year non-prorated substrate warranty and 15-year finish warranty (factory finish); Contractor's 2-year workmanship.

Part 2 — Products

- **2.01 Manufacturers:** James Hardie Building Products (HardiePlank, HardiePanel, HardieTrim — HZ10); Nichiha USA (Johns Creek, GA; Sierra Premium lap and panels); Allura (Elementia); or approved equal.
- **2.02 Materials:** ASTM C1186 Grade II Type A; lap: 5/16 × 7-1/4-in plank, smooth or "cedarmill" texture as selected, installed at 6-in exposure (1-1/4-in overlap); board-and-batten: 5/16 × 48 × 108/120-in smooth vertical panels with 5/4 × 2-1/2-in fiber-cement battens at 12 in o.c. (16 in o.c. at the Architect's option); trim 5/4 × 4 (corners, 3-1/2-in exposed face) and 5/4 × 6 (windows/doors) with 1-in drip cap; frieze 5/4 × 8; vented soffit 1/4-in panels; porch ceiling 1/4-in beadboard.
- **2.03 Performance Criteria:** Class A (E84), noncombustible (E136); wind attachment per ESR and Table R703.3(1) for 115 mph Exposure B (VERIFY): blind-nailed 6d (2-in) hot-dip galvanized or stainless siding nails into studs at 16 in o.c., 1 in from top edge; clearances: **6 in to finished grade, 2 in to roofs/paving/decks, 1/4 in to horizontal flashing, 1/8 in to trim (sealed)** — at zero-step landings the siding stops on the masonry water table/cap so the 2-in hardscape clearance is honored by the stone (04 40 00), not by the siding.
- **2.04 Finishes:** Factory ColorPlus (preferred for the HOA's maintenance cycle) or field-painted per 09 91 13: scheme 1 warm white B&B; scheme 2 sage lap; scheme 3 clay/beige lap; trim warm white on all schemes unless the Architect selects otherwise.

- **2.05 Accessories:** Pre-formed galvanized or color-matched joint flashing (6 in wide) behind every butt joint; Z-flashing at horizontal trim and the water-table cap; color-matched touch-up kits; corner posts; sealant (07 92 00, paintable, ASTM C920 Class 25).

Part 3 — Execution

- **3.01 Examination:** Verify WRB is complete, taped, and flashed; verify the water-table cap and its flashing are installed; verify stud layout for panel edges and butt joints.
- **3.02 Preparation:** Snap level lines at each course; install Z-flashing over the cap and over all horizontal trim; install starter strip for the first lap course; lay out battens symmetrical about openings.
- **3.03 Installation:** Per the ESR and manufacturer's instructions: butt joints in moderate contact over studs with joint flashing (no caulk at flashed joints); panels vertical with edges on studs, joints under battens; 1/8-in gaps at trim, sealed; no face-nailing except at the top course and where required; factory-primed cut edges field-primed; keep required clearances; install trim before siding (Hardie "trim-first") with siding butted to trim.
- **3.04 Field Quality Control:** Check exposure uniformity ($\pm 1/16$ in per course), nail depth (flush, not overdriven), and clearances on every elevation before painting.
- **3.05 Cleaning and Protection:** Wash dust from the surface before painting; touch up cut edges; protect from mortar and stain splatter at stone piers and cedar posts.

[REVIEW REQUIRED] Fastener schedule and exposure/wind values depend on the adopted wind speed and product ESR; the batten spacing and trim widths are placeholders pending the Architect's elevations.

Section 07 62 00 — Sheet Metal Flashing and Trim

Part 1 — General

- **1.01 Section Includes:** Roof drip edge; step, apron, and kick-out flashing at porch and gable-return roof-to-wall conditions; valley metal; head/drip caps over windows, doors, and horizontal trim; Z-flashing; through-wall flashing drip at veneer; post-base and pier-cap flashings; garage-door head flashing.
- **1.02 Related Sections:** 04 20 00 / 04 40 00; 07 27 00 (self-adhered membrane flashing); 07 31 00; 07 46 46; 07 71 00; 07 92 00.
- **1.03 References:** 2024 IRC (GA 2026) R703.4 (wall flashing locations), R903.2 (roof flashing), R905.2.8 (asphalt shingle flashing: base, step, kick-out, drip edge); SMACNA Architectural Sheet Metal Manual, 8th ed.; ASTM A653/A653M (G90 galvanized steel); ASTM B209 (aluminum sheet); ASTM A240 (stainless); AAMA 2605 (fluoropolymer finish, where pre-finished).
- **1.04 Submittals:** Profiles and gauges; finish color chips; shop drawings for kick-out and pier-cap flashings.
- **1.05 Quality Assurance:** Installed by the roofer and the siding installer under one sequencing plan; reviewed at the 07 27 00 envelope mock-up.
- **1.06 Delivery, Storage, and Handling:** Keep pre-finished metal in protective film until installed; store dry to prevent wet-storage stain.
- **1.07 Warranty:** Finish warranty 20 years (AAMA 2605 fluoropolymer) or 10 years (polyester); Contractor's 5-year workmanship (with roofing).

Part 2 — Products

- **2.01 Manufacturers:** Berger Building Products; Amerimax Home Products; Petersen Aluminum (PAC-CLAD); Drexel Metals; or approved equal.
- **2.02 Materials:** Pre-finished aluminum 0.024 in (or 26-ga G90 galvanized steel, pre-finished) for exposed flashings; 0.032-in aluminum at kick-outs and pier caps; stainless steel Type 304, 26 ga, where in contact with treated lumber, mortar, or copper-bearing algae-resistant shingle runoff; galvanized steel step flashing 4 × 4 × 8 in minimum (R905.2.8.3); drip edge with 2-in roof leg and 1-1/2-in face.
- **2.03 Performance Criteria:** No dissimilar-metal contact (isolate aluminum from mortar, treated lumber, and steel fasteners); hemmed exposed edges; expansion joints at ≤ 10 ft runs of continuous trim; kick-out flashing ≥ 4 in high directing water into the gutter, never behind siding.
- **2.04 Finishes:** Color-matched to trim or roof per scheme; concealed flashings mill finish.
- **2.05 Accessories:** Compatible fasteners (stainless at aluminum), sealant per 07 92 00, butyl tape at laps.

Part 3 — Execution

- **3.01 Examination:** Verify WRB and underlayment sequence allows shingle-fashion laps; verify porch roof/wall intersections have backing for kick-outs.
- **3.02 Preparation:** Cut and form on site or shop; pre-bend kick-outs; isolate metals with tape or coating.
- **3.03 Installation:** Per SMACNA and the referenced IRC sections: drip edge under underlayment at eaves and over at rakes; step flashing one piece per course, counter-flashed by the siding with a 1-in gap to the roof; kick-out at every roof-to-wall eave termination; head flashing over every opening lapped by the WRB; Z-flashing over horizontal trim with end dams; pier caps sloped with drips over the stone.
- **3.04 Field Quality Control:** Water test the porch roof-to-wall and one kick-out at the mock-up house; inspect all head flashings before siding closes them.
- **3.05 Cleaning and Protection:** Remove strippable film; remove metal filings from roofs and gutters to prevent rust staining.
[REVIEW REQUIRED] Generic profiles; the Architect shall detail the kick-out, water-table cap, and pier-cap flashings for the drawings.

Section 07 71 00 — Roof Specialties (Gutters and Downspouts)

Part 1 — General

- **1.01 Section Includes:** Seamless aluminum K-style gutters, downspouts, hangers, outlets, and splash blocks or connections to lot drainage; gutter guards (optional HOA upgrade).
- **1.02 Related Sections:** 07 31 00 (drip edge into gutter); 07 62 00; 07 92 00; 32 16 13 (walk-side downspout discharge); 33 40 00 storm laterals (not in this outline — required where lots drain to the lane system).
- **1.03 References:** SMACNA Architectural Sheet Metal Manual, 8th ed.; ASTM B209 (aluminum alloy 3105-H24); ASTM B221 (extrusions); 2024 IRC (GA 2026) R801.3 (roof drainage discharged ≥ 5 ft from foundation or to an approved drainage system); Gwinnett County Stormwater Management Manual (lot drainage, 2017).
- **1.04 Submittals:** Gutter profile and gauge, color chip, downspout layout on each plan elevation, splash-block/lateral connection detail.
- **1.05 Quality Assurance:** Installer with a truck-mounted seamless machine and 3 years' experience.
- **1.06 Delivery, Storage, and Handling:** Roll-form on site; protect finished coil from scratching.
- **1.07 Warranty:** Coil finish 20-year; Contractor's 2-year workmanship (no leaks at outlets/miters).

Part 2 — Products

- **2.01 Manufacturers:** Coil: Amerimax (Euramax); Berger; Spectra Metals (Gutter Supply); Englert; or approved equal.
- **2.02 Materials:** 6-in K-style seamless aluminum, 0.027 in (0.032 in at runs > 40 ft or where guards are added), baked-enamel finish; 3 $\times$ 4-in corrugated rectangular downspouts, 0.019 in; hidden hangers with screws at ≤ 24 in o.c. (16 in o.c. where the roof is 8:12 and the eave is long); aluminum miters and outlets riveted and sealed.
- **2.03 Performance Criteria:** Sized for the 8:12 roof plane areas using SMACNA Table 1-2 for the Atlanta 5-minute rainfall intensity (≈ 8 in/hr, 10-yr — VERIFY); slope 1/16 in per ft to outlets; at least two downspouts per gutter run > 35 ft; discharge to splash blocks onto graded swales sloping away, or to solid 4-in PVC laterals to the lot drainage system where the lot grading plan requires.
- **2.04 Finishes:** Gutters and downspouts match trim color (warm white) on all schemes unless the Architect elects roof-matching color at the fascia.
- **2.05 Accessories:** Downspout straps at ≤ 6 ft; splash blocks (precast concrete) or lateral adaptors; optional micro-mesh gutter guards (stainless mesh, aluminum frame) as an HOA maintenance upgrade; 07 92 00 sealant at miters.

Part 3 — Execution

- **3.01 Examination:** Verify fascia is straight and drip edge extends into the gutter; verify downspout locations do not discharge onto walks, the zero-step landing, or a neighbor's lot.
- **3.02 Preparation:** Snap the slope line; set outlet positions at low points; pre-punch outlets.
- **3.03 Installation:** Install per SMACNA with the front lip below the roof plane extension; fasten hangers into rafter tails/sub-fascia; seal all joints with gutter sealant; strap downspouts plumb, 1 in off the siding on standoffs, terminating with an elbow to a splash block or into the lateral with a debris cleanout.
- **3.04 Field Quality Control:** Hose test each run for ponding ($< 1/8$ in standing water) and leaks before final painting.
- **3.05 Cleaning and Protection:** Remove sealant smears and debris; the HOA maintenance plan shall include semi-annual cleaning.

[REVIEW REQUIRED] Sizing rainfall intensity and the discharge method (splash block vs. lateral) depend on the civil engineer's lot grading and stormwater plan.

Section 07 92 00 — Joint Sealants

Part 1 — General

- **1.01 Section Includes:** Exterior sealants at siding-to-trim, trim-to-masonry, window and door perimeters, penetrations, and flashings; horizontal traffic sealant at slab isolation joints and the zero-step landing joint; interior sanitary sealants at tile, tubs, showers, countertops; acoustical/air-seal sealants (with 07 27 00); interior painter's caulk.
- **1.02 Related Sections:** 03 30 00; 04 20 00 / 04 40 00; 07 27 00; 07 46 46; 08 series; 09 30 13; 12 36 00; 22 40 00.
- **1.03 References:** ASTM C920 (elastomeric joint sealants); ASTM C834 (latex sealants); ASTM C1193 (guide for use of joint sealants); ASTM C1330 (backer rod); ASTM C1521 (field adhesion); ASTM C1311 (solvent-release sealants — not used); SWRI Sealants Validation Program; siding manufacturer's sealant requirements (paintable, ASTM C920 Class 25 minimum, permanently flexible).
- **1.04 Submittals:** Product data and color charts; joint schedule (location, sealant type, backer, primer); adhesion test log.
- **1.05 Quality Assurance:** Installer with 3 years' experience; field adhesion tests (ASTM C1521 hand-pull) at the mock-up house for each exterior substrate pairing.
- **1.06 Delivery, Storage, and Handling:** Store 40–80 degrees F; use before the expiration date.
- **1.07 Warranty:** Manufacturer's 20-year (silicone) / 5-year (polyurethane) material warranty; Contractor's 2-year workmanship.

Part 2 — Products

- **2.01 Manufacturers:** Sika (Sikaflex); Tremco (Dymonic 100, Spectrem); Dow (DOWSIL 790/795/786); Pecora (890NST, 898NST sanitary); GE Silicones; or approved equal.
- **2.02 Materials:** Exterior siding/trim/masonry: ASTM C920 Type S, Grade NS, Class 25 (or 35), Use NT/M/A/O, paintable polyurethane or hybrid (e.g., Dymonic 100, Sikaflex-15LM) — or, where a non-paintable joint is acceptable, neutral-cure silicone Class 50 in matching color; horizontal slab and landing joints: ASTM C920 Type S, Grade P, Class 25, Use T self-leveling polyurethane; sanitary: ASTM C920 mildew-resistant 100 percent silicone, Use NT; interior trim: ASTM C834 siliconized acrylic latex; acoustical/air-seal: non-hardening synthetic rubber; backer rod ASTM C1330 Type C (closed-cell or bicellular) sized 25 percent larger than the joint.
- **2.03 Performance Criteria:** Joint width 1/4 in minimum at siding-to-trim where sealed (1/8 in field gap per 07 46 46 is sealant-filled without backer); width-to-depth 2:1, depth 1/4–1/2 in; ± 25 percent movement (Class 25) at siding/trim, ± 50 percent at masonry-to-siding transitions; no three-sided adhesion (bond breaker at shallow joints).
- **2.04 Finishes:** Exterior colors matched to adjacent siding/trim/masonry per scheme (custom color where standard colors do not match); sanitary sealant white or clear; interior trim caulk painted.
- **2.05 Accessories:** Primers per the manufacturer's substrate chart (fiber cement cut edges, masonry, PVC); masking tape; bond-breaker tape.

Part 3 — Execution

- **3.01 Examination:** Verify joints are dry, clean, and within width tolerance; verify backer rod is required (joints > 3/8 in).
- **3.02 Preparation:** Clean per ASTM C1193; prime where required; install backer rod or bond-breaker tape; mask adjacent finishes.
- **3.03 Installation:** Tool concave with a wet tool within the working time; slab joints: fill to 1/8 in below the surface with self-leveling sealant over backer; sanitary joints at all changes of plane in tile (09 30 13) and at fixture-to-counter joints; do not seal weeps, flashed butt joints, or the bottom edge of lap siding.
- **3.04 Field Quality Control:** ASTM C1521 hand-pull at 5 locations on the mock-up house per substrate; re-test any failed pairing after primer change.
- **3.05 Cleaning and Protection:** Remove excess immediately with the manufacturer's solvent; protect from traffic until cured (24–72 hours).

[REVIEW REQUIRED] Sealant types and colors are generic pending the Architect's joint schedule; confirm siding-manufacturer sealant requirements for the factory-finish warranty.

Division 08 — Openings

Section 08 14 00 — Wood Doors (with 08 16 13 Fiberglass Entry Doors)

Part 1 — General

- **1.01 Section Includes:** Fiberglass (basis of design) or wood exterior entry doors, 3'-0" × 6'-8" × 1-3/4 in, at front, rear/patio, and garage-to-house locations, with zero-threshold sill systems; interior molded or flush wood doors 3'-0" × 6'-8" (32-in minimum clear opening) with lever hardware; door frames, weatherstripping, and hardware.
- **1.02 Related Sections:** 03 30 00 (landing elevation); 06 10 00 (rough openings, garage separation); 07 27 00 (sill pans, flashing); 07 92 00; 09 91 13 / 09 91 23 (finishes); 26 50 00 (porch light at entry).
- **1.03 References:** 2024 IRC (GA 2026) R311.2 (egress door: 32-in clear, 78-in height), R311.3.1 (landing at the required egress door **not more than 1-1/2 in below the top of the threshold; 7-3/4 in permitted where the door does not swing over the landing** — the design uses ≤ 1/2 in), R302.5.1 (garage-to-house door: solid wood 1-3/8 in, solid/honeycomb steel, or 20-minute rated; **self-closing and self-latching**), R308.4 (safety glazing in doors and sidelites), R609 (exterior doors: testing/labeling); ICC A117.1-2017 §404.2.3 (32-in clear), §404.2.4 (thresholds ≤ 1/2 in, beveled 1:2), §404.2.6 (lever hardware 34–48 in AFF), §404.2.8 (≤ 5 lbf interior opening force); AAMA/WDMA/CSA 101/I.S.2/A440-22 (NAFS) for exterior doors; WDMA I.S. 1A (interior architectural wood flush doors) / I.S. 6A (molded); ANSI/BHMA A156.2 (locks), A156.1 (hinges), A156.4 (closers); NFRC 100/200; ENERGY STAR Version 7.0 (doors).
- **1.04 Submittals:** Door schedule; product data with NFRC labels and DP rating; threshold/sill pan detail at the zero-step landing; hardware schedule with finish; interior door style sample.
- **1.05 Quality Assurance:** Installer trained by the door manufacturer; the first entry door installed is the mock-up (water test the sill).
- **1.06 Delivery, Storage, and Handling / 1.07 Warranty:** Store doors upright, covered, in a conditioned space; fiberglass entry doors limited lifetime (glass 20-year, finish 10-year); interior doors 5-year; hardware 10-year mechanical / 5-year finish; Contractor 2-year.

Part 2 — Products

- **2.01 Manufacturers:** Entry: Therma-Tru (Fiber-Classic / Smooth-Star), Masonite (Belleville / Barrington), ProVia (Signet), JELD-WEN (Smooth-Pro), Pella; interior: Masonite, JELD-WEN, Steves & Sons; sills: Endura Products, Pemko; hardware: Schlage, Kwikset (Halo/SmartKey), Emtek; or approved equal.
- **2.02 Materials:** Entry doors: fiberglass skins (smooth for schemes 1 and 2; fir-grain for scheme 3 stain-grade look), polyurethane foam core, composite (rot-proof) bottom rail and jambs, factory-hung in a composite/primed frame; glazing: tempered, low-E insulated, decorative or clear per elevations with sidelite where shown; **threshold: low-profile "ADA" sill, ≤ 1/2 in total height with 1:2 bevel, with a multi-point or adjustable sill cap, integral sill pan/end dams, and a high-performance door bottom (double bulb)** — the covered 6-ft porch and sloped landing are the primary water defense; rear door: same, or 6'-0" sliding/hinged patio door with a low-profile track (≤ 1/2 in) per plan; garage-to-house: 20-minute fire-rated fiberglass or 1-3/8-in solid-core wood, with self-closing hinges and a self-latching lockset (R302.5.1), weatherstripped; interior: 1-3/8-in molded hollow-core (solid-core at bedrooms and baths for acoustics) 2-panel shaker, primed; pocket doors (with lever-style edge pulls) where plans show, no barn doors on accessible routes.
- **2.03 Performance Criteria:** Exterior doors NAFS Performance Grade 30 minimum (DP 30, VERIFY against the adopted wind speed); NFRC U ≤ 0.20 (opaque) / ≤ 0.25 (glazed) meeting ENERGY STAR v7.0 door criteria (VERIFY current values); air leakage ≤ 0.5 cfm/ft² for swinging doors (IECC R402.4.3); clear opening 32 in minimum at every door; lever hardware; 5-lb maximum interior opening force; no thresholds at interior doors (09 65 00 continuous flooring).
- **2.04 Finishes:** Entry door exterior: scheme 1 deep charcoal (illustrative #3B3F44), scheme 2 near-black (#2B2B2B) or natural stain, scheme 3 deep green (#2F4F3E) — as selected; interior face and all interior doors semi-gloss white (09 91 23); hardware satin nickel or matte black per scheme.
- **2.05 Accessories:** Ball-bearing hinges 4 × 4 in (3 per door; 4 at 8-ft doors); privacy sets at baths/bedrooms, passage elsewhere; door stops; peephole/door viewer at 43 and 60 in (two heights) or a video doorbell; smart deadbolt (keypad) option; weatherstripping (Q-lon) and corner seals; sill pans (07 27 00).

Part 3 — Execution

- **3.01 Examination:** Verify rough opening size and square; **verify the landing is within 1/2 in of the interior floor and slopes away 2 percent** before setting the sill.
- **3.02 Preparation:** Install sill pan with end dams and back dam sealed to the subfloor/slab; shim points at hinges and strike.
- **3.03 Installation:** Set entry doors per ASTM E2112 and the manufacturer: bed the sill in two beads of sealant over the pan, plumb and square with 1/8-in even reveal, foam (low-expansion) the perimeter, flash head and jambs to the WRB; adjust sill cap for a compressed sweep seal; interior doors plumb with 1/8-in reveals, hardware at 36 in centerline (levers 34–48 in AFF), strikes adjusted for self-latching at the garage door.
- **3.04 Field Quality Control / 3.05 Cleaning and Protection:** Spray test the entry sill at the mock-up house (AAMA 502 field test optional); verify each garage door self-closes and latches from a 6-in open position; protect doors with cardboard until final

paint; remove labels except NFRC/fire labels.

[REVIEW REQUIRED] Life-safety adjacent (egress, garage separation, safety glazing) and the zero-threshold sill (well inside the R311.3.1 limits but demanding a drawn water-management detail) — the Architect shall detail and the AHJ shall confirm.

Section 08 36 13 — Sectional Doors (Carriage-Style Insulated Garage Door)

Part 1 — General

- **1.01 Section Includes:** One 16'-0" × 7'-0" insulated steel sectional overhead door per cottage, carriage-house ("long-panel" recessed) design with optional top-row windows and decorative hardware; tracks, torsion springs, weatherseals; belt-drive opener with photo-eyes and battery backup.
- **1.02 Related Sections:** 03 30 00 (garage apron, zero-step slab); 06 10 00 (header, jamb backing); 07 46 46 (trim); 07 62 00 (head flashing); 26 00 00 (opener circuit, not in this outline).
- **1.03 References:** 2024 IRC (GA 2026) R609 (garage doors tested per ASTM E330 or ANSI/DASMA 108 and labeled), R309 (garage floor slope to the door or to a drain); ANSI/DASMA 102 (sectional door specifications), ANSI/DASMA 108 (wind load), DASMA 116 (impact — not required); UL 325 (openers: entrapment protection, photo-eyes); ASTM A653 (galvanized steel); ASTM C1363 (thermal; DASMA TDS 163 R-value method).
- **1.04 Submittals:** Product data with wind-load label, R-value per DASMA TDS 163, window and hardware options, color chips; opener data with UL 325 listing.
- **1.05 Quality Assurance:** Installer IDEA-certified (Institute of Door Dealer Education and Accreditation) or manufacturer-authorized; spring counterbalance rated 25,000 cycles minimum (HOA durability).
- **1.06 Delivery, Storage, and Handling / 1.07 Warranty:** Deliver after siding and painting; store sections flat; door limited lifetime (steel skin) / 10-year (hardware, insulation delamination) / 3-year (paint finish); opener 10-year motor, 5-year parts (belt lifetime); Contractor 1-year (springs adjusted).

Part 2 — Products

- **2.01 Manufacturers:** Clopay (Coachman); Amarr (Classica); Overhead Door (Courtyard / Carriage House 5900 series); Wayne Dalton (9700 series); C.H.I. Overhead Doors (5983); openers: LiftMaster (8550W/87504), Genie, Chamberlain; or approved equal.
- **2.02 Materials:** Three-layer door: 24–27-ga hot-dip galvanized steel skins (embossed carriage design, ~2-in polyurethane foam core), 1-3/8 to 2-in section thickness, steel end stiles; **R-value 12 minimum (R-16+ preferred) per DASMA TDS 163**; 2-in galvanized track, 14-ga brackets, nylon rollers with sealed ball bearings (quiet operation for 55+ homes), torsion springs, bottom aluminum retainer with U-shaped vinyl seal, vinyl jamb weatherseal; optional insulated (double-glazed) top-row windows with SDL grilles matching the house windows; decorative black handles/hinges per scheme.
- **2.03 Performance Criteria:** Wind-load rated per ANSI/DASMA 108 for the site's design pressure (115 mph Exposure B, 1-story — VERIFY; typically ± 20–25 psf) and labeled; air infiltration seal on all four sides; opener 3/4-hp DC belt drive with battery backup, photo-eyes at 6 in, auto-reverse (UL 325), soft start/stop, Wi-Fi/MyQ-type app control, wall console with lock and light, two remotes, exterior keypad at 48 in AFF; door opening force (manual) ≤ 25 lb with springs balanced.
- **2.04 Finishes:** Factory baked-on polyester (or PVDF where offered) in scheme 1 warm white, scheme 2 white or sage (VERIFY availability; otherwise paint per 09 91 13 with the manufacturer's approved system), scheme 3 warm white or almond — as selected.
- **2.05 Accessories:** Perimeter PVC stop molding with seal; exterior trim (07 46 46); safety pull-cord, emergency release; optional "Torquemaster"-type enclosed springs; low-headroom kit only where trusses require.

Part 3 — Execution

- **3.01 Examination:** Verify opening 16'-0" × 7'-0" ± 1/4 in, jambs plumb, headroom ≥ 12 in (or per opener), backroom ≥ 8 ft, and a level slab at the door line consistent with the zero-step apron (garage slab may slope to the door per R309.1 — VERIFY the apron detail).
- **3.02 Preparation:** Install 2×6 jamb and header backing (06 10 00); install center bearing plate backing.
- **3.03 Installation:** Per DASMA and the manufacturer's instructions; sections level, tracks plumb, springs wound and balanced so the door stays in place at any height; opener rail level, force settings and travel limits set, photo-eyes aligned, wall console mounted per the manufacturer's UL 325 instructions (typically ≥ 5 ft AFF, out of children's reach) with the keypad, remotes, and app control providing seated-reach operation (A117.1 §308 — VERIFY the AHJ's position on a lower console); weatherseal compressed 1/2 in.
- **3.04 Field Quality Control / 3.05 Cleaning and Protection:** Verify the UL 325 reversal with a 2×4 test and the photo-eye test; verify the wind-load label matches the submittal; wipe panels; leave the operating manual and spring-cycle rating with the HOA closing package; no painting of weatherseals or rollers.

[REVIEW REQUIRED] Wind-load design pressure and label class must be confirmed against the adopted wind speed; door color availability by scheme is unverified.

Section 08 53 13 — Vinyl Windows (Fiberglass Alternate — 08 54 13)

Part 1 — General

- **1.01 Section Includes:** Single-hung or double-hung vinyl (uPVC) windows (fiberglass or composite as an approved alternate), 1-over-1 (schemes 2 and 3) and 2-over-2 simulated-divided-lite (scheme 1), with insulating low-E glass, integral nailing flanges, screens, and hardware; fixed and awning units where shown; mulled assemblies; egress-sized bedroom units.
- **1.02 Related Sections:** 06 10 00 (rough openings, headers); 07 27 00 (sill pans, flashing, WRB laps); 07 46 46 (trim); 07 92 00; 09 91 13 (exterior trim paint, not the window); 12 24 00 window shades (not in this outline).
- **1.03 References:** 2024 IRC (GA 2026) R310 (emergency escape and rescue openings: 5.7 SF net clear, 24-in height, 20-in width, sill $\leq$ 44 in AFF, at every sleeping room), R308.4 (safety glazing near doors, tubs/showers, and large low panes), R312.2 (window fall protection — not triggered at one story), R609 (exterior windows: NAFS performance and labeling, installation per the manufacturer and flashing per R703.4); 2015 IECC (GA) Table R402.1.2 (U-0.35 / SHGC-0.27) and §R402.4.3 (air leakage $\leq$ 0.3 cfm/ft²); AAMA/WDMA/CSA 101/I.S.2/A440-22 (NAFS-22); AAMA 303 (vinyl profiles); NFRC 100/200/400; ENERGY STAR Version 7.0 (Windows, Doors, and Skylights); ASTM E2112 (installation); ICC A117.1-2017 §309 (operable parts $\leq$ 5 lbf, 15–48 in reach).
- **1.04 Submittals:** Window schedule with NFRC U/SHGC/VT, NAFS PG rating, egress net-clear-opening calculation for each bedroom unit, SDL bar profile, exterior/interior color samples, screen and hardware samples; installation detail set (sill pan, jamb, head) coordinated with 07 27 00.
- **1.05 Quality Assurance:** Windows from a single manufacturer; installer trained (AAMA InstallationMasters or manufacturer program); mock-up: one bedroom window in the envelope mock-up house, water-tested (AAMA 502, optional).
- **1.06 Delivery, Storage, and Handling / 1.07 Warranty:** Store upright, off the ground, covered, out of direct sun; vinyl frame limited lifetime, insulated glass 20-year (seal failure), hardware 10-year, exterior color (dark laminates/paint) 10-year, transferable to subsequent owners; Contractor 2-year.

Part 2 — Products

- **2.01 Manufacturers:** Vinyl: Pella (250 Series), Andersen (100 Series Fibrex composite), MI Windows and Doors (V3000 / EnergyCore), Cornerstone/Ply Gem (1500 Series), JELD-WEN (V-4500), Milgard (Trinsic); fiberglass alternate: Marvin (Elevate/Essential), Pella (Impervia), Milgard (Ultra); or approved equal.
- **2.02 Materials:** Multi-chamber extruded uPVC (AAMA 303) with fusion-welded corners, or fiberglass/composite alternate; single-hung (double-hung at the Owner's option) with tilt-in sash, block-and-tackle balances, cam lock(s) at $\leq$ 5 lbf; glazing: double-pane, 3/4-in IGU, low-E (spectrally selective, e.g., Cardinal LoE-366 or equal), argon fill, warm-edge spacer, tempered where R308.4 applies; **SDL grilles: 7/8-in exterior/interior applied bars with a spacer bar in the airspace (2-over-2 on scheme 1; 1-over-1 no grilles on schemes 2 and 3);** full or half fiberglass-mesh screens (aluminum frames); integral nailing fin, J-channel not used (trimmed per 07 46 46).
- **2.03 Performance Criteria:** NAFS Performance Class R or LC, PG 30 minimum (DP 30 — VERIFY with the adopted wind speed and the sizes on the elevations); **NFRC U-factor $\leq$ 0.28 and SHGC $\leq$ 0.23** as the design basis — this meets ENERGY STAR v7.0 for both the South-Central and Southern zones (Gwinnett County's ENERGY STAR zone to be confirmed on the ENERGY STAR climate-zone finder) and exceeds the Georgia prescriptive U-0.35 / SHGC-0.27; VT $\geq$ 0.45; air leakage $\leq$ 0.3 cfm/ft² (IECC R402.4.3); egress at bedrooms per R310 (a 3'-0" $\times$ 5'-0" single-hung typically provides ~5.0–5.7 SF; confirm each unit — casement or a larger unit may be needed); sill heights 30–36 in AFF for seated views, with locks within a 15–48 in reach.
- **2.04 Finishes:** Exterior: white (schemes 1 and 2) or bronze/black laminate (scheme 3), as selected; interior white; hardware white or matching; screen frames match exterior.
- **2.05 Accessories:** Mull kits with structural reinforcement for paired units; factory-applied fin sealant; snap-in grilles are not acceptable (SDL specified); window-opening control devices not required (one story).

Part 3 — Execution

- **3.01 Examination:** Verify rough opening +1/2 in width / +1/4 in height (or per manufacturer), square within 1/8 in, and sill pan/WRB prepared per 07 27 00.
- **3.02 Preparation:** Apply sealant at the fin (jambs and head, not the sill unless the manufacturer requires a weep gap); position shims at 1/4 points.
- **3.03 Installation:** Per ASTM E2112 Method A1 (fin over WRB with self-adhered jamb and head flashing, WRB lapped over the head flashing) and R609; fasten fins at 8–12 in o.c.; foam (low-expansion) the perimeter; install exterior trim with a 1/8-in sealed gap to the frame; verify sash operates and locks with $\leq$ 5 lbf.

- **3.04 Field Quality Control / 3.05 Cleaning and Protection:** Egress net-clear-opening verified on one installed unit per type; NFRC labels left on until inspection, then removed; protect from mortar and paint; clean glass with non-abrasive cleaner; replace scratched IGUs.

[REVIEW REQUIRED] Life-safety adjacent (egress and safety glazing); PG rating, ENERGY STAR zone, and the egress sizing of every bedroom window require the Architect's schedule.

Division 09 — Finishes

Section 09 29 00 — Gypsum Board

Part 1 — General

- **1.01 Section Includes:** Interior gypsum board on walls and ceilings; moisture- and mold-resistant board at baths, laundry, and kitchen wet walls; garage/house separation board (fire-taped); cement board or glass-mat backer at tile; Level 4 finish (Level 5 at the great room ceiling where lit by clerestory/wash light, optional); corner beads and trims; the ceiling-plane air barrier.
- **1.02 Related Sections:** 06 10 00 (framing, blocking, fireblocking); 07 21 00; 07 27 00 (airtight drywall); 09 30 13 (tile backer); 09 91 23 (paint); 10 28 13 (grab bars fasten through the board into blocking).
- **1.03 References:** 2024 IRC (GA 2026) R702.3 (gypsum board), Table R702.3.5 (application — 1/2-in board on 24-in o.c. ceilings only with restrictions; see 2.02), R702.4 (tile backers at tub/shower walls: cement, fiber-cement, or glass-mat gypsum), R302.6 (garage separation: 1/2-in gypsum on the garage side of walls and ceilings; 5/8-in Type X where habitable space above — not applicable), R302.11 (fireblocking); ASTM C1396 (gypsum board), C1178 (glass-mat gypsum), C1325 (cementitious backer), C840 (application and finishing), C475 (joint materials), C1002 (screws), C1047 (trims); GA-214 (Levels of Gypsum Board Finish); ASTM D3273 (mold resistance, score 10).
- **1.04 Submittals:** Product data; finish-level plan; garage separation detail; texture sample (if any).
- **1.05 Quality Assurance:** Installer with 5 years' experience; sample room (the first cottage's primary bath and great room) approved for finish level before proceeding.
- **1.06 Delivery, Storage, and Handling / 1.07 Warranty:** Deliver after dry-in; store flat in conditioned space, off the floor; do not install above 90 percent RH; Contractor 2-year (no nail pops or cracked joints, one repair visit at 11 months).

Part 2 — Products

- **2.01 Manufacturers:** Georgia-Pacific Gypsum (ToughRock, DensArmor Plus, DensShield — Atlanta, GA); USG (Sheetrock, Mold Tough, Durock); National Gypsum (Gold Bond, XP, PermaBase); CertainTeed; or approved equal.
- **2.02 Materials:** Walls: 1/2-in ASTM C1396 regular board; ceilings on 24-in o.c. trusses: **5/8-in regular or 1/2-in sag-resistant "ceiling board"** (Table R702.3.5 — 1/2-in regular board is not permitted at 24 in o.c. under water-based texture); baths, laundry, kitchen sink wall: 1/2-in moisture/mold-resistant (ASTM C1396 + D3273 score 10; paperless glass-mat preferred in showers' dry zones); tile backer at showers and tub walls: 1/2-in cement board (C1325) or glass-mat water-resistant gypsum backer (C1178, with bonded waterproofing per 09 30 13); garage side of separation walls/ceiling: 1/2-in regular (5/8-in Type X optional for the HOA's insurance benefit), joints taped (fire-taped); joint compound setting-type at first coat, ready-mix finish; paper-faced metal or vinyl corner bead; bullnose at the Owner's option (not at tile or door casings).
- **2.03 Performance Criteria:** Level 4 finish per GA-214 at all painted surfaces (Level 5 optional where noted); ceiling board fastened at 12 in o.c. field, 7 in perimeter (screws 1-1/4 in); walls 16 in o.c.; no board within 1/2 in of the slab (sealed gap, 07 27 00); air barrier: gypsum ceiling sealed to top plates and around all penetrations (IECC Table R402.4.1.1).
- **2.04 Finishes:** Smooth (no texture) walls and ceilings — a light knockdown texture on ceilings only at the Owner's option; no popcorn; painted per 09 91 23.
- **2.05 Accessories:** Control joints (C1047) at ≤ 30 ft in long ceilings; resilient channel not required; acoustical sealant at bedroom/bath partitions (optional); access panels at sprinkler valves/plumbing (11-in × 14-in, paintable).

Part 3 — Execution

- **3.01 Examination:** Verify framing is dry (≤ 19 percent MC), blocking is complete (photo log per 06 10 00), the pre-drywall energy/insulation inspection has passed, and rough-in inspections are signed.
- **3.02 Preparation:** Install backer at tile areas first; install ceiling-plane air-sealing gaskets at top plates; mark grab-bar blocking lines on the framing edge for the tile setter and accessory installer.
- **3.03 Installation:** Per ASTM C840 and GA-216: ceilings first, perpendicular to trusses with floating interior angles; walls horizontal, joints staggered; screw-attach (no nails) in ceilings; garage separation continuous with taped joints and fire-rated sealant at penetrations; finish Level 4 (three coats over tape, sanded); cement board joints with alkali-resistant mesh and

thin-set.

- **3.04 Field Quality Control / 3.05 Cleaning and Protection:** Inspect the finish under a raking light at 45 degrees (side-lit) before priming; verify garage separation continuity above the ceiling line; vacuum sanding dust; protect tile backer from water until the membrane is applied.

[REVIEW REQUIRED] Life-safety adjacent (garage separation, fireblocking); ceiling board type on 24-in o.c. trusses and the Level 5 extent to be confirmed by the Architect.

Section 09 30 13 — Ceramic Tiling

Part 1 — General

- **1.01 Section Includes:** Porcelain floor tile at baths and laundry; ceramic/porcelain wall tile at curbless (roll-in) showers to the ceiling and at tub surrounds where a tub option is elected; kitchen backsplash tile; sloped mortar beds, bonded waterproofing, linear drains (drain body by 22 40 00), grout, and transitions flush with LVP (no thresholds).
- **1.02 Related Sections:** 03 30 00 (1-1/2-in slab recess at showers); 07 92 00 (sealant at changes of plane); 09 29 00 (backer); 09 65 00 (flush transition); 10 28 13 (grab bars through tile into blocking); 22 40 00 (linear drain, valve, hand-held shower).
- **1.03 References:** ANSI A108 / A118 / A136.1 (installation and materials), ANSI A137.1 (tile), ANSI A326.3 (DCOF); TCNA Handbook 2025 — methods **B422C (curbless shower, bonded waterproofing over slab recess)**, F125A (crack isolation), W244 (backer-board walls), EJ171 (movement joints); ASTM C373, C1028 (superseded — use A326.3), C1027 (abrasion); ICC A117.1-2017 §608.2.2 (roll-in shower 30 × 60 in minimum; no curb; ≤ 1/2-in beveled entry) and §608.6 (hand-held shower), §303 (changes in level); 2024 IPC (GA 2026) §417.5.2 (shower floor slope 1/4 in per ft to the drain — **VERIFY the section number in the 2024 edition**) and §417.4 (shower compartment 900 in² minimum).
- **1.04 Submittals:** Tile samples (two per type/color), grout and sealant color charts, membrane and mortar data, shower layout with drain position and slope, DCOF report for floor tile.
- **1.05 Quality Assurance:** Tile setter CTEF Certified Tile Installer (or 5 years' documented experience); mock-up: the first cottage's primary shower flood-tested for 24 hours before tile.
- **1.06 Delivery, Storage, and Handling / 1.07 Warranty:** Deliver in original cartons with dye lot; store dry above 50 degrees F; tile manufacturer 10-year; membrane manufacturer system warranty 10-year (where the setting-system manufacturer supplies all components); Contractor 2-year (no cracking, debonding, grout failure, or leaks).

Part 2 — Products

- **2.01 Manufacturers:** Tile: Daltile (Mohawk), Florida Tile, Crossville, MSI, Emser; setting systems: Schluter Systems (Kerdi/Kerdi-Line), Laticrete (Hydro Ban), Custom Building Products (RedGard, Prolite), Mapei; or approved equal.
- **2.02 Materials:** Bath floors: porcelain, ANSI A137.1 P1 (impervious, ≤ 0.5 percent absorption), 12 × 24 in matte or 2 × 2 in mosaic; shower floors: 2 × 2 in porcelain mosaic (or 1 × 1 in) for slope conformance; shower walls: 12 × 24 in porcelain or 3 × 6 / 4 × 12 in ceramic (A137.1 W or P) to the ceiling; backsplash: 3 × 6 in ceramic (or as selected); mortar: ANSI A118.15 (large format) / A118.4; grout: A118.7 high-performance cementitious, or A118.3 epoxy at shower floors; bonded waterproofing A118.10 (sheet or liquid, with pre-formed corners and drain integration); crack-isolation A118.12 over the slab at bath floors; sloped mortar bed A108.1A (or pre-sloped foam tray with linear-drain kit); linear drain: stainless, 32–48 in, wall-side or entry-side placement per plan, ASME A112.6.3 / A112.18.2 (VERIFY drain listing) — provided under 22 40 00.
- **2.03 Performance Criteria:** DCOF ≥ 0.42 (A326.3, interior wet) at all bath and shower floors; shower floor slope 1/4 in per ft (≥ 2 percent) to the drain within a 36 × 60 in minimum roll-in footprint; **shower entry: zero curb, ≤ 1/4 in vertical or ≤ 1/2 in beveled change in level (A117.1 §303)**; bath floor tile finished flush (± 1/8 in) with adjacent LVP by setting the tile assembly in the recessed slab; movement joints at the perimeter and at changes of plane (07 92 00) per EJ171; 95 percent mortar coverage in wet areas; lippage ≤ 1/32 in (rectified) per A108.02.
- **2.04 Finishes:** Colors as selected per scheme from the manufacturer's full range (warm gray/greige floors; white or soft-white walls; sanded grout color matched); matte finishes only on floors.
- **2.05 Accessories:** Schluter-type metal edge trim at outside corners and tile-to-drywall edges; recessed niche (pre-formed, waterproofed) and a 17–19-in-high built-in bench or fold-down seat (10 28 13) in every shower; grab-bar locations pre-drilled with diamond bits into blocking; pre-formed curbless shower tray systems permitted as an alternate to site-sloped beds.

Part 3 — Execution

- **3.01 Examination:** Verify the slab recess (1-1/2 in) and drain rough-in elevation allow the full slope to the drain with the tile finishing flush at the entry; verify backer installation, blocking, and the slab's moisture (ASTM F2170 RH or per the membrane manufacturer) and flatness (1/8 in in 10 ft for large format).
- **3.02 Preparation:** Form and cure the sloped bed (or set the pre-sloped tray); install the bonded waterproofing continuous from the drain flange up the walls to 72 in minimum (full height in the roll-in zone), with corners and seams per the manufacturer;

flood test 24 hours (ASTM D5957); install crack-isolation at bath floors.

- **3.03 Installation:** Per ANSI A108 and the TCNA method: back-butter large format; joint width 1/8 in (3/16 in at mosaics); align shower-wall courses to the floor grid; hold sealant (not grout) at all changes of plane and at the tile/LVP joint if not flush; grout after 24 hours; seal cementitious grout after cure.
- **3.04 Field Quality Control / 3.05 Cleaning and Protection:** Water test the finished shower (fill to the entry line, verify drainage with no ponding > 1/8 in); lift one tile per shower floor to verify coverage; clean haze within 24 hours; protect floors with breathable covering until Substantial Completion.

[REVIEW REQUIRED] Curbless-shower assembly, slab recess depth, drain type, and the flush LVP transition are interdependent details the Architect must draw; confirm IPC 2024 section numbers.

Section 09 65 00 — Resilient Flooring (Luxury Vinyl Plank)

Part 1 — General

- **1.01 Section Includes:** Rigid-core (SPC or WPC) luxury vinyl plank throughout living areas, kitchen, bedrooms, hall, closets, and den — continuous with no interior thresholds or transition strips at doorways; flush transitions to tile (09 30 13) and to the garage/entry door thresholds; resilient/wood base by 06 20 00.
- **1.02 Related Sections:** 03 30 00 (slab flatness, moisture, curing compound compatibility); 06 20 00 (base and shoe); 09 30 13; 08 14 00 (door undercuts, thresholds).
- **1.03 References:** ASTM F3261 (rigid-core resilient plank — Standard Specification); ASTM F710 (preparing concrete floors); ASTM F2170 (in-situ RH) and F1869 (calcium chloride); ASTM E648 (critical radiant flux, Class I ≥ 0.45 W/cm²) and E662 (smoke ≤ 450); ASTM D2047 (static COF ≥ 0.5 , dry); ANSI/RFCI; FloorScore / CDPH Standard Method v1.2 (VOC); ICC A117.1-2017 §302 (firm, stable, slip-resistant), §303 (no change in level > 1/4 in); 2024 IRC (GA 2026) R311.3 (doorway floor levels).
- **1.04 Submittals:** Product data with wear-layer thickness, ASTM F3261 classification, locking-system run limits, FloorScore certificate, 3 full-length plank samples per color, layout plan with maximum run lengths and expansion provisions.
- **1.05 Quality Assurance:** Installer certified by the manufacturer; mock-up: the first cottage's great room/kitchen run reviewed for the flush tile transition and end-joint stagger.
- **1.06 Delivery, Storage, and Handling / 1.07 Warranty:** Acclimate in the conditioned house 48 hours at 65–85 degrees F; store flat; manufacturer's limited lifetime residential wear warranty (25-year minimum), waterproof warranty; Contractor 2-year (no gapping, peaking, or hollow spots).

Part 2 — Products

- **2.01 Manufacturers:** Shaw Floors (Floorté / COREtec — Dalton, GA); Mohawk (SolidTech — Calhoun, GA); Mannington (Adura Max/Rigid); AHF Products (Armstrong Flooring); or approved equal.
- **2.02 Materials:** Rigid-core plank, ASTM F3261, 5.5–8 mm total thickness with attached acoustic pad, **wear layer ≥ 20 mil**, ceramic-bead or urethane finish, 7 × 48 in (or 9 × 60 in) planks, beveled micro-edge, angle-tap or drop-lock joint; waterproof core; phthalate-free, FloorScore certified.
- **2.03 Performance Criteria:** Class I (E648); SCOF ≥ 0.5 (D2047); indentation ≤ 0.010 in (ASTM F1914, 250-lb static load — 55+ walkers/wheelchairs); maximum continuous run per the manufacturer (typically 50–60 ft) — both plans fit within a single expansion zone so no doorway T-moldings are required; perimeter expansion 3/8 in concealed under base/shoe; **transitions: flush ($\leq 1/8$ in) to tile by slab recess, or a 1/4-in-maximum beveled reducer where flush is not achieved; no transition strips at interior doorways**; slab moisture ≤ 85 percent RH (F2170) or per the manufacturer; pH ≤ 9 ; flatness 3/16 in in 10 ft (F710).
- **2.04 Finishes:** Light-to-medium warm wood tones (e.g., "natural oak", "greige oak") as selected per scheme; low-gloss; high LRV contrast with base and door trim (white) for low-vision wayfinding.
- **2.05 Accessories:** Manufacturer's reducer/end-cap only at the garage door threshold and patio door; stainless or aluminum flush "Schluter Reno"-type profile at tile where needed; underlayment not required (attached pad); repair kit left with the HOA.

Part 3 — Execution

- **3.01 Examination:** Verify slab moisture, pH, flatness, and that curing compounds are removed or compatible; verify tile areas are set to the flush elevation; verify doors are hung so undercut clearance is 1/2 in over the finished floor.
- **3.02 Preparation:** Grind high spots and fill lows with cementitious patch (F710); sweep and vacuum; snap layout lines parallel to the longest wall or the main sightline from the entry; plan end-joint stagger ≥ 8 in.
- **3.03 Installation:** Floating per the manufacturer: 3/8-in perimeter gap at all walls, cabinets, and fixed objects (cabinets set on top of the floor is NOT permitted — cut around islands); run planks continuously through doorways; tap joints fully; cut neatly around the tile edge to a straight sealed joint; install base and shoe over the gap with no fasteners into the planks.

- **3.04 Field Quality Control / 3.05 Cleaning and Protection:** Walk test for hollow spots and peaking; check transitions with a straightedge ($\leq 1/8$ in); clean with the manufacturer's neutral cleaner; protect with breathable paper (no plastic) and hardboard at traffic paths until Substantial Completion.

[REVIEW REQUIRED] Product family (SPC vs. WPC), wear layer, and the flush-transition detail are Architect selections; confirm run limits against each plan's longest diagonal.

Section 09 91 13 — Exterior Painting (Three Color Schemes)

Part 1 — General

- **1.01 Section Includes:** Field painting of fiber-cement siding and trim (where not factory-finished), cellular PVC trim (where painted), porch ceilings, wood doors, exposed flashings/gutters where color-matched, and cedar staining; touch-up of factory-finished siding; color schedule for the three schemes.
- **1.02 Related Sections:** 06 20 00 (cedar, trim); 07 46 46 (factory finish alternative); 07 92 00 (paintable sealants); 08 14 00 / 08 36 13 (factory finishes); 09 91 23.
- **1.03 References:** MPI Architectural Painting Specification Manual (current edition) and MPI Approved Products List: MPI #6 (exterior latex primer, wood), MPI #3 (alkali-resistant primer, cementitious), MPI #15 (exterior latex, low sheen, G3–4) for siding, MPI #11 (exterior latex semi-gloss, G5) for trim and doors, MPI #33 (exterior semi-transparent stain) for cedar; ASTM D3359 (adhesion), D4442 (moisture); SSPC/PDCA P1–P3; 40 CFR 59 Subpart D (VOC — Georgia has no stricter AIM rule; use ≤ 100 g/L where available).
- **1.04 Submittals:** Color schedule per scheme (field, trim, door, porch ceiling, gutters, garage door), 8 × 10-in drawdowns of each color, product data, stain sample on the specified cedar; 4 × 8-ft mock-up per scheme on the model cottages.
- **1.05 Quality Assurance:** Painter with 5 years' exterior residential experience; one manufacturer for all coatings; mock-ups become the standard for sheen and coverage.
- **1.06 Delivery, Storage, and Handling / 1.07 Warranty:** Store 50–90 degrees F; no application below 50 degrees F, above 90 degrees F, or when rain is forecast within 24 hours; manufacturer's limited lifetime (premium acrylic); Contractor 3-year (no peeling, blistering, or chalking beyond ASTM D4214 No. 8).

Part 2 — Products

- **2.01 Manufacturers:** Sherwin-Williams (Duration / Emerald Exterior, WoodScapes stain); Benjamin Moore (Aura Exterior, Arborcoat); PPG (Manor Hall / Timeless, ProLuxe stain); or approved equal.
- **2.02 Materials:** 100 percent acrylic exterior latex, self-priming over factory-primed fiber cement (spot-prime cut edges with alkali-resistant acrylic primer); acrylic semi-gloss for trim, doors, and PVC (light-reflectance value ≥ 55 on PVC to limit heat distortion, per the trim manufacturer); cedar: penetrating oil/alkyd or waterborne semi-transparent stain, "natural cedar" tone, with a UV-blocking clear topcoat optional; galvanized/aluminum: acrylic bonding primer + finish where painted.
- **2.03 Performance Criteria:** Dry film thickness 1.5 mils per finish coat minimum, two finish coats (one on factory-finished touch-up); moisture in fiber cement ≤ 15 percent and cedar ≤ 15 percent before coating (D4442); sheen: siding low sheen (G3), trim/doors semi-gloss (G5), porch ceiling low sheen; door color coats on both faces; HOA repaint cycle planned at 8–10 years (factory ColorPlus extends to 15).
- **2.04 Finishes — color schedule (hex values are illustrative placeholders; final colors by the Architect from the manufacturers' full ranges):**
 - **Scheme 1 "Warm White":** field board-and-batten warm white (#F3EFE6); trim, corners, fascia, and gutters soft white (#FAF8F3); front door deep charcoal (#3B3F44); porch ceiling pale sky (#DCE5E8); cedar posts/brackets natural stain; stone piers/water table gray-tan ledge; roof charcoal.
 - **Scheme 2 "Sage":** field lap sage (#8E9C86); trim warm white (#F6F3EC); front door near-black (#2B2B2B) or natural stain; porch ceiling warm white; cedar painted trim color or natural (Architect's option); brick water table red-brown blend with buff mortar; roof weathered wood.
 - **Scheme 3 "Clay/Beige":** field lap clay-beige (#CDB89A); trim cream (#F3EEE2); front door deep green (#2F4F3E); porch ceiling cream; cedar natural stain; stone buff/tan ledge; roof driftwood/brown.
- **2.05 Accessories:** Primers per substrate; caulk per 07 92 00 (painted); masking; touch-up kits (one quart of each color per cottage delivered to the HOA).

Part 3 — Execution

- **3.01 Examination:** Verify sealants are cured, siding clearances are correct, cedar is dry, and factory-finish touch-ups are complete; verify no visible flashing gaps remain to be "painted over".
- **3.02 Preparation:** Wash dust; sand PVC glossy spots; spot-prime nail heads, cut edges, and any bare substrate; back-prime replaced trim; mask masonry, windows, and stone.

- **3.03 Installation:** Brush-and-roll or spray-and-back-roll two finish coats; stain cedar two coats wiped, end grain saturated; paint doors on all six sides; do not paint weatherseals, weeps, labels, vents, or rollers; keep a wet edge on lap siding for full courses.
- **3.04 Field Quality Control / 3.05 Cleaning and Protection:** DFT spot checks (Tooke gauge) at 3 locations per elevation on the mock-up; cross-hatch adhesion (D3359) on PVC and fiber cement at the mock-up; remove masking within 24 hours; clean overspray from stone, glass, and roofing; leave touch-up kits and the color schedule with the HOA.

[REVIEW REQUIRED] Colors are placeholders; MPI numbers should be confirmed against the current MPI manual; factory-finish (ColorPlus) vs. field paint is an Owner cost/maintenance decision that changes this section's scope.

Section 09 91 23 — Interior Painting

Part 1 — General

- **1.01 Section Includes:** Priming and painting of gypsum board walls and ceilings, interior wood/MDF trim and doors, garage interior (walls and ceiling), closets, and interior side of exterior doors; low-VOC finishes; high-contrast trim for low-vision wayfinding.
- **1.02 Related Sections:** 06 20 00 (trim); 08 14 00 (doors); 09 29 00 (Level 4 substrate); 07 92 00 (interior caulk).
- **1.03 References:** MPI Architectural Painting Specification Manual: MPI #50 (latex primer sealer for gypsum), MPI #39 (latex primer for wood), MPI #53 (interior latex flat, G1) ceilings, MPI #52 (interior latex, G3 eggshell) walls, MPI #54 (interior latex semi-gloss, G5) or MPI #147 (waterborne alkyd, G5) trim and doors; MPI X-Green / GS-11 / CDPH v1.2 (VOC and emissions); GA-214 (finish level); ASTM D4828 (scrub resistance); IES RP-28 (lighting for older adults — contrast guidance).
- **1.04 Submittals:** Color schedule and 8 × 10-in drawdowns; product data with VOC content; sheen samples on the 09 29 00 sample room.
- **1.05 Quality Assurance:** Painter with 3 years' experience; sample room approval before proceeding; single manufacturer.
- **1.06 Delivery, Storage, and Handling / 1.07 Warranty:** Store 50–90 degrees F; apply at 60–85 degrees F with ventilation; manufacturer's limited lifetime; Contractor 1-year with a touch-up visit at 11 months.

Part 2 — Products

- **2.01 Manufacturers:** Sherwin-Williams (ProMar 200 Zero VOC / Duration Home / Emerald Urethane Trim); Benjamin Moore (Ultra Spec 500 / Regal Select / Advance); PPG (Speedhide Zero / Breakthrough); or approved equal.
- **2.02 Materials:** Gypsum primer (MPI #50, high-build where Level 4 flashing is a risk); walls: acrylic latex eggshell (G3), washable, ≥ 400 cycles scrub (D4828) — baths, laundry, and kitchen: mildew-resistant "kitchen and bath" satin; ceilings: flat (G1) ceiling paint; trim and doors: waterborne alkyd/urethane-modified semi-gloss (G5); garage: primer + one coat flat/eggshell (walls) and flat (ceiling); VOC ≤ 50 g/L (flat/eggshell), ≤ 100 g/L (semi-gloss).
- **2.03 Performance Criteria:** One primer coat plus two finish coats on new gypsum; two finish coats on trim over primer; DFT per MPI; **visual contrast: door/trim color vs. wall color LRV difference ≥ 30 points** (low-vision aid); no sprayed "back-rolled-once" single-coat walls.
- **2.04 Finishes:** Walls warm off-white (illustrative "Alabaster"-type, LRV ~80); ceilings white flat; trim and doors bright white semi-gloss; one accent wall per plan optional; garage walls light gray.
- **2.05 Accessories:** Painter's caulk (07 92 00); masking; touch-up kit (one quart per color) per cottage.

Part 3 — Execution

- **3.01 Examination:** Verify Level 4 finish, dust-free surfaces, gypsum moisture < 12 percent, and that HVAC is operating for cure.
- **3.02 Preparation:** Sand and fill trim; caulk trim-to-wall joints after primer; remove hardware and cover plates; mask floors (09 65 00) with breathable protection.
- **3.03 Installation:** Prime, inspect under raking light, touch up defects, then two finish coats (spray-and-back-roll walls, brush trim); cut clean lines at ceiling; paint doors flat-laid or hung with hardware removed; paint closet interiors and behind fixtures.
- **3.04 Field Quality Control / 3.05 Cleaning and Protection:** Inspect at 45-degree raking light; correct holidays and flashing; reinstall hardware; remove masking; ventilate 48 hours before flooring protection is removed.

[REVIEW REQUIRED] Generic colors and MPI numbers pending the Architect's interior finish schedule.

Division 10 — Specialties

Section 10 14 00 — Signage (Monument Entry Sign)

Part 1 — General

- **1.01 Section Includes:** One ground-mounted monument entry sign, ≤ 32 SF of sign face, at the Arcade Road entrance: masonry base and cap matching the homes, dimensional letters "Cottages at Arcade Springs", ground-mounted shielded LED illumination, and footing. (Assigned to Division 10 rather than 13 34 00 Fabricated Engineered Structures because it is a signage assembly; the sign is permitted separately per the Lilburn checklist §4.q.)
- **1.02 Related Sections:** 03 30 00 (footing); 04 20 00 / 04 40 00 (brick or stone cladding by scheme — stone is the default at the entry); 26 50 00 (sign lighting); 32 93 00 (planting bed); 33 00 00 utilities/sight distance (not in this outline).
- **1.03 References:** Lilburn Zoning Ordinance 2023-603 sign regulations (subdivision/community entrance sign: permitted area, height, setback, illumination — **VERIFY the article and the 32-SF/height limits at the pre-application conference**; sign permit and possible sign easement per the Site Development Plan Review Checklist §4.q); Gwinnett County DOT driveway permit sight-distance triangle (no obstruction > 2.5 ft in the triangle — **VERIFY**); ASCE 7-22 (wind on the freestanding wall/sign — **VERIFY** the adopted edition); ASTM C1364 (cast stone); ASTM B209 / B26 (aluminum plate / castings); UL 48 and UL 879 (illuminated signs, where applicable); IES/IDA Model Lighting Ordinance (shielding).
- **1.04 Submittals:** Shop drawings with dimensions, area calculation, letter style/size, materials, footing, and illumination; masonry and cap samples; sign permit application drawings.
- **1.05 Quality Assurance:** Sign fabricator licensed in Georgia with 5 years' experience; masonry by the 04 series mason; footing by a Georgia PE if the sign exceeds 6 ft or 32 SF.
- **1.06 Delivery, Storage, and Handling / 1.07 Warranty:** Protect letters and cap in crates; masonry per Division 04; letters 10-year (finish), LED fixtures 5-year, masonry 2-year workmanship.

Part 2 — Products

- **2.01 Manufacturers:** Letters: Gemini Incorporated (cast aluminum or flat-cut), Matthews Architectural Products, ASI Signage Innovations; HDU letters/panels: Coastal Enterprises (Precision Board), Sign Foam (SignArt); fabrication by a local sign contractor; or approved equal.
- **2.02 Materials:** Reinforced CMU core on a 3,000-psi footing 12 in below grade (frost) sized for wind; ledge/stacked stone (04 40 00) or brick (04 20 00) cladding; cast-stone cap (C1364) sloped with drip; sign face: 1/2-in cast-aluminum or 1/4-in flat-cut aluminum letters, 8–10-in cap height, pin-mounted with 1/2-in standoffs on a cast-stone or HDU panel (15-lb HDU, primed and painted); sign area ≤ 32 SF measured per the ordinance's method (**VERIFY**: face vs. panel); overall height ≤ 6 ft above grade (**VERIFY**).
- **2.03 Performance Criteria:** Located outside the Gwinnett DOT sight triangle and outside the public right-of-way, within (or adjacent to) the 10-ft landscape strip along Arcade Rd per the site plan (**VERIFY** setback); wind-resistant per ASCE 7 for the wall and letters; letter contrast ≥ 70 percent with the background; illumination by two ground-mounted, fully shielded LED floods ≤ 3000K (2700K preferred), aimed at the face only, with ≤ 0.1 fc spill at the property line and photocell/timer control (dark-sky); no internal illumination, no exposed bulbs, no changeable copy.
- **2.04 Finishes:** Letters: satin black or dark bronze (schemes 1–3 neutral); cap buff or gray per the default stone scheme; panel color matching the trim warm white.
- **2.05 Accessories:** Stainless pins and epoxy anchors; conduit sleeve through the footing; address numerals (4535–4541 range) on a separate address sign per the Gwinnett Fire Marshal (4-in minimum, contrasting); landscape bed with irrigation stub.

Part 3 — Execution

- **3.01 Examination:** Verify the surveyed location against the R/W, sight triangle, landscape strip, utilities (call 811), and the sign permit.
- **3.02 Preparation:** Excavate and pour the footing with anchor dowels and conduit; cure 7 days.
- **3.03 Installation:** Lay CMU core and cladding per Division 04; set the cap in full mortar with sealant joints; mount letters with a template, level, on standoffs; install and aim lights at night with the Architect present; plant the bed per 32 93 00.
- **3.04 Field Quality Control / 3.05 Cleaning and Protection:** Verify the sign area and height against the permit; night aiming check for glare from Arcade Rd; clean masonry (no acid) and letters; protect until planting is complete.

[REVIEW REQUIRED] Sign area, height, setback, and illumination limits are unverified against the Lilburn sign article; a separate sign permit (and possibly a sign easement) is required.

Section 10 28 13 — Toilet Accessories (Bath Accessories and Grab Bars)

Part 1 — General

- **1.01 Section Includes:** Grab bars (installed, or "blocked-for" with locations recorded) at every toilet, shower, and tub; fold-down shower seats; shower slide bars (by 22 40 00); mirrors; robe hooks; towel bars/rings; toilet-paper holders; medicine cabinets; shower niches (09 30 13); the grab-bar blocking record for each cottage.
- **1.02 Related Sections:** 06 10 00 (blocking); 09 29 00; 09 30 13 (tile penetrations); 22 40 00 (fixtures, hand-held showers); 12 36 00 (vanity tops).
- **1.03 References:** ICC A117.1-2017 §604.5 (grab bars at water closets: side wall 42 in long, 12 in max from the rear wall; rear wall 36 in), §608.3 (roll-in shower: 24 in max from the entry along the back wall, full length of side walls per the variant), §608.4 (fold-down seat), §609 (grab bars: 1-1/4 to 2-in diameter, 1-1/2-in clearance, 33–36 in AFF, 250-lb load in any direction), §603.3 (mirrors: bottom edge ≤ 40 in AFF), §308 (reach ranges 15–48 in); ASTM A666 (stainless Type 304); ASTM F446 (grab bars) is withdrawn — structural loading is cited from A117.1 §609.8; the IRC has no grab-bar requirement (voluntary universal-design program).
- **1.04 Submittals:** Accessory schedule with mounting heights; product data; grab-bar layout per bath (one drawing per plan) showing the blocking zone; blocking photo log (06 10 00) delivered to the HOA.
- **1.05 Quality Assurance:** Installed by the finish carpenter; pull-test one installed grab bar per cottage to 250 lb.
- **1.06 Delivery, Storage, and Handling / 1.07 Warranty:** Deliver boxed after painting; grab bars 10-year (finish and structural), accessories 5-year, mirrors 5-year (silvering); Contractor 2-year.

Part 2 — Products

- **2.01 Manufacturers:** Grab bars and seats: Bobrick, Bradley, ASI American Specialties (commercial-grade), Moen Home Care and Kohler (residential designer grade in matching finishes); mirrors/medicine cabinets: Kohler, Roborn, Jensen; or approved equal.
- **2.02 Materials:** Grab bars 1-1/4- or 1-1/2-in diameter, 18-ga Type 304 stainless (satin or peened) or designer finishes matching the plumbing trim (matte black, satin nickel), concealed-flange mounting with #14 stainless screws into 06 10 00 blocking (no hollow-wall anchors); fold-down shower seat, phenolic or teak slats on a stainless frame, 250-lb rated, 17–19 in AFF; mirrors 1/4-in float glass (ASTM C1503) with polished edges or framed, bottom edge ≤ 40 in AFF; robe hooks ≤ 48 in AFF; toilet-paper holder 7–9 in in front of the toilet, 15–48 in AFF; towel bars at 48 in AFF.
- **2.03 Performance Criteria:** Every toilet: blocking for a 42-in side bar and a 36-in rear bar (install at closing at the Owner's election; at minimum install the side bar); every shower: blocking for the full back-wall and side-wall bars per §608.3, an 18-in vertical bar at the entry, and the fold-down seat; every tub (where the tub option is chosen): §607.4 layout; all bars 250-lb rated; contrast with wall tile where the buyer elects color bars.
- **2.04 Finishes:** Match the 22 40 00 trim finish per bath; satin stainless where designer finishes are unavailable.
- **2.05 Accessories:** Stainless mounting hardware; escutcheon sealant (07 92 00) at tile; blocking labels ("GRAB BAR BLOCKING 33–36 in") photographed before drywall.

Part 3 — Execution

- **3.01 Examination:** Verify blocking locations from the 06 10 00 photo log; verify tile is cured and finish painting is complete.
- **3.02 Preparation:** Lay out from the fixture centerlines, not the walls; drill tile with diamond bits; apply sealant to penetrations.
- **3.03 Installation:** Mount grab bars 33–36 in AFF to the top of the gripping surface, level, with 1-1/2 in clear to the wall and no obstructions above within 12 in; mount the seat per §608.4; mount mirrors and accessories at the scheduled heights (accessories in reach ranges, never over the grab-bar zone).
- **3.04 Field Quality Control / 3.05 Cleaning and Protection:** 250-lb pull test on one bar per cottage; verify heights with a story pole; clean fingerprints; deliver the per-lot blocking record and the grab-bar layout to the HOA for future retrofits.

[REVIEW REQUIRED] Voluntary program: the Owner shall decide which bars are installed at closing versus blocked-for; confirm current BHMA/ASTM designations.

Division 12 — Furnishings

Section 12 36 00 — Countertops

Part 1 — General

- **1.01 Section Includes:** Quartz (engineered stone) countertops at kitchens and baths (basis of design), with solid-surface as an approved alternate at baths; undermount sink cutouts; backsplashes (4-in matching or full-height tile per 09 30 13); a 34-in-high accessible work surface option in each kitchen and a 34-in lavatory height at one bath.

- **1.02 Related Sections:** 06 20 00 / 06 41 00 casework (not in this outline); 07 92 00 (sealant at walls and sinks); 09 30 13 (tile backsplash); 22 40 00 (sinks, faucets).
- **1.03 References:** ISFA (International Surface Fabricators Association) Quartz Surfacing Fabrication Standards and Solid Surface Fabrication Standards; NSF/ANSI 51 (food-zone materials); ASTM C880 (flexural strength, engineered stone) and ASTM C1353 (abrasion); GREENGUARD Gold (emissions); ICC A117.1-2017 §1003.12 (Type A kitchen work surface: 30 in wide, 34 in max AFF, knee space) and §606.3 (lavatory rim ≤ 34 in) — voluntary; OSHA 29 CFR 1926.1153 (respirable crystalline silica — quartz fabrication).
- **1.04 Submittals:** 6 × 6-in samples per color; shop drawings with seam locations, sink cutouts, overhangs, and support brackets; edge profile sample; fabricator's silica-control program statement.
- **1.05 Quality Assurance:** Fabricator certified by the slab manufacturer (for warranty) with CNC and water-jet fabrication and wet-cutting only; field-measure after cabinets are set; mock-up: the model cottage kitchen.
- **1.06 Delivery, Storage, and Handling / 1.07 Warranty:** Transport on A-frames on edge; store indoors; quartz manufacturer's limited lifetime residential (15-year minimum) warranty; solid surface 10-year; fabricator 2-year (seams, cracks, delamination).

Part 2 — Products

- **2.01 Manufacturers:** Quartz: Cambria (US-made), Caesarstone, Cosentino (Silestone), MSI (Q Premium); solid surface: DuPont/Corian Design (Corian), LX Hausys (HI-MACS), Wilsonart (Solid Surface); or approved equal.
- **2.02 Materials:** Quartz: 3 cm (1-1/4 in) slabs, ≥ 90 percent quartz aggregate in polymer resin, NSF/ANSI 51, GREENGUARD Gold, matte/honed or polished; solid surface: 1/2-in sheet on a built-up edge (1-1/2 in) per ISFA; edges eased (1/8-in radius) or 1-1/4-in square eased — no sharp ogee profiles; backsplash: 4-in matching quartz at baths, tile at kitchens (09 30 13); sinks undermount (22 40 00) with polished cutouts.
- **2.03 Performance Criteria:** Seams ≤ 1/16 in, color-matched epoxy, none within 6 in of a sink cutout; overhangs ≤ 10 in unsupported (steel brackets beyond); kitchen counter height 36 in AFF standard with one 30-in-wide × 34-in section (or a lowered island end) where the plan allows; bath vanity tops at 34 in AFF with a removable-front base cabinet option for knee space; heat resistance per the manufacturer (trivets required — note in the HOA manual); no visible substrate at edges.
- **2.04 Finishes:** Light, low-veining neutrals (illustrative "white with soft gray veining" for kitchens; "warm white" or "light greige" for baths) as selected per scheme; matte finish preferred for glare control in 55+ kitchens.
- **2.05 Accessories:** Sink clips and epoxy; plywood sub-top at dishwasher and sink runs; steel support brackets at overhangs; silicone sealant (07 92 00) at the wall and the sink rim; cutouts for faucets, soap dispensers, and cooktops per the appliance schedule.

Part 3 — Execution

- **3.01 Examination:** Verify cabinets are level within 1/8 in in 10 ft, plumb, and fastened; verify sink and appliance rough-ins.
- **3.02 Preparation:** Template with digital or physical templates after cabinets are set; confirm seam locations with the Architect; shim cabinets, never the top.
- **3.03 Installation:** Set on a continuous bead of silicone on the cabinet rails; level; join seams with color-matched adhesive and seam-setter clamps; mount undermount sinks with clips and silicone; install backsplashes with a 1/8-in sealed joint at the top; cut in place only with wet, HEPA-vacuumed tools.
- **3.04 Field Quality Control / 3.05 Cleaning and Protection:** Straightedge check for flatness and overhang; water-fill test of sink seals; clean with the manufacturer's cleaner; cover with corrugated board until the punch list; provide care instructions to the HOA/homeowner.

[REVIEW REQUIRED] Quartz vs. solid surface by room and the accessible-counter option are Owner decisions; confirm the appliance schedule for cutouts.

Division 22 — Plumbing

Section 22 40 00 — Plumbing Fixtures (fixtures and trim only)

Part 1 — General

- **1.01 Section Includes:** Water closets (comfort height), lavatories and faucets, kitchen sink and faucet, roll-in shower valves and trim with hand-held showers on slide bars, linear drains, optional tub with hand-held shower, laundry box, hose bibbs, water-heater and piping excluded (22 10 00 / 22 30 00 not in this outline).
- **1.02 Related Sections:** 03 30 00 (shower recess, rough-ins); 09 30 13 (tile, drains, waterproofing); 10 28 13 (grab bars, seats); 12 36 00 (sinks in tops); 21 13 00 NFPA 13D (not in this outline).

- **1.03 References:** 2024 International Plumbing Code with Georgia Amendments (2026) — Chapter 4 fixtures, §417 showers (900 in² compartment, 1/4 in/ft floor slope — VERIFY section numbers), §424.3 (shower valves: ASSE 1016 pressure-balance or thermostatic, 120 degrees F maximum), §607 (hot-water temperature); Georgia Water Stewardship Act (O.C.G.A. §8-2-3): high-efficiency fixtures — water closets ≤ 1.28 gpf, lavatory faucets ≤ 1.5 gpm, showerheads ≤ 2.0 gpm (VERIFY current values); EPA WaterSense; ASME A112.19.2/CSA B45.1 (vitreous china), A112.19.3 (stainless sinks), A112.18.1/CSA B125.1 (faucets and showers), ASME A112.6.3 (floor and trench drains); ASSE 1016/A112.18.1 (shower valves), ASSE 1070 (point-of-use temperature limiting at lavatories, optional); ICC A117.1-2017 §604 (water closet height 17–19 in to seat top, centerline 16–18 in from the side wall), §606 (lavatories: rim ≤ 34 in, knee clearance, insulated traps), §608 (roll-in shower controls 38–48 in AFF, hand-held on a 59-in hose with a slide bar), §309.4 (lever/loop handles ≤ 5 lbf, no grasping); NSF/ANSI 372 (lead-free).
- **1.04 Submittals:** Fixture schedule with model numbers, flow rates, WaterSense labels, finishes; rough-in drawings for the linear drain and valve heights; hand-held slide-bar mounting height.
- **1.05 Quality Assurance:** Plumber licensed in Georgia; fixtures from single manufacturers per category for finish matching; the model cottage's primary bath is the mock-up for valve, slide-bar, and drain positions.
- **1.06 Delivery, Storage, and Handling / 1.07 Warranty:** Store fixtures boxed in the garage until setting; china limited lifetime (defects), faucets/valves limited lifetime (function) and 10-year (finish), drains 5-year; Contractor 2-year.

Part 2 — Products

- **2.01 Manufacturers:** Kohler; TOTO; American Standard (LIXIL); faucets/showers: Moen, Delta, Kohler; linear drains: Schluter (Kerdi-Line), Infinity Drain, QuickDrain (Oatey); sinks: Elkay, Kohler, Blanco; or approved equal.
- **2.02 Materials:** Water closets: vitreous china, elongated, **comfort/chair height (17–19 in to the top of the seat)**, 1.28 gpf (WaterSense, MaP ≥ 800 g), soft-close slow-close seat, with rear-wall and side-wall grab-bar blocking (06 10 00); lavatories: undermount vitreous china (or integral solid surface at the alternate) with single-lever faucet, 1.2–1.5 gpm, lever handle; kitchen: 18-ga stainless undermount single bowl (30-in) with a pull-down single-lever faucet 1.8 gpm; showers: pressure-balance or thermostatic valve (ASSE 1016) with lever handle set at 38–48 in AFF near the entry, **hand-held shower on a 24-in slide bar (rated as a grab bar where possible) with a 59-in hose and a fixed head via a two-way diverter**, ≤ 2.0 gpm; linear drain 32–48 in stainless with tile-in or brushed grate; optional 60-in acrylic/porcelain-enameled tub (ASME A112.19.1) with a lever-handle valve and hand-held; laundry box with lever ball valves; frost-free hose bibbs (ASSE 1019) with vacuum breakers; insulated P-traps at open-knee lavatories.
- **2.03 Performance Criteria:** All operable parts lever/loop-type, ≤ 5 lbf; faucet handles single-lever; water closet centerline 18 in from the side wall (A117.1 §604.2, 16–18 in) and 60-in turning circle preserved; shower control reachable from outside the spray; hot water ≤ 120 degrees F at showers/tubs (ASSE 1016) and ≤ 110 degrees F at lavatories if ASSE 1070 valves are elected; flow rates per the Georgia Water Stewardship Act and WaterSense.
- **2.04 Finishes:** White china; faucet/shower/accessory finishes per scheme (satin nickel or matte black), matching 10 28 13.
- **2.05 Accessories:** Angle stops (quarter-turn lever), supply lines (braided stainless), escutcheons, wax-free toilet seals, drain assemblies with cleanouts, dishwasher air gap or high loop per the IPC (GA) — VERIFY.

Part 3 — Execution

- **3.01 Examination:** Verify rough-in dimensions (water closet centerline, valve height, drain location and elevation for the 09 30 13 slope), and that tile/waterproofing is complete before trim-out.
- **3.02 Preparation:** Set the linear drain flange level with the membrane per the 09 30 13 system; protect finishes during tile work.
- **3.03 Installation:** Per the IPC (GA) and the manufacturers: water closets on flanges set at the finished floor, caulked at the base except the rear; lavatories with clearances per §606; shower trim with the slide bar fastened into 06 10 00 blocking (not into tile alone); hand-held hose without kinks; aerators installed; hose bibbs with backflow protection.
- **3.04 Field Quality Control / 3.05 Cleaning and Protection:** Verify flow rates and WaterSense labels against the schedule; test the ASSE 1016 limit stop at 120 degrees F; leak test all connections; clean fixtures with non-abrasive cleaner; protect china until turnover.

[REVIEW REQUIRED] Life-safety adjacent (scald protection); IPC 2024/GA section numbers, Georgia fixture flow limits, and drain listings are unverified; the plumbing engineer/designer shall complete the schedule.

Division 26 — Electrical

Section 26 50 00 — Lighting (fixtures only)

Part 1 — General

- **1.01 Section Includes:** Interior LED luminaires (recessed downlights, surface/flush ceiling fixtures, vanity lights, under-cabinet task lights, closet lights, night-lights), porch and garage exterior wall lanterns, driveway/garage-door lights, and the private-lane and pathway post-top lighting; lamps and drivers; controls (dimmers, occupancy/vacancy sensors, photocells, timers). Wiring, panels, and branch circuits are in 26 05 00 / 26 24 00 (not in this outline).
- **1.02 Related Sections:** 07 21 00 (IC-rated fixtures in insulated ceilings); 07 27 00 (air-tight housings); 08 36 13 (opener light); 10 14 00 (sign lighting); 32 16 13 (pole bases); 32 93 00 (light spill on the buffer).
- **1.03 References:** 2023 National Electrical Code with Georgia Amendments (2026) — Article 410 (luminaires), 210.70 (required lighting outlets); 2015 IECC (GA) §R404.1 (≥ 75 percent high-efficacy lamps — VERIFY the GA-amended percentage; specify 100 percent LED); UL 1598 (luminaires), UL 8750 (LED equipment); ENERGY STAR (luminaires) / DLC (exterior); IES RP-28-20 (Lighting and the Visual Environment for Older Adults and the Visually Impaired — design basis for illuminance and glare); IES TM-15 (BUG ratings); IDA-IES Model Lighting Ordinance (MLO); IES RP-8 (roadway/pedestrian lighting — Class P local residential); Lilburn Site Development Plan Review Checklist §5.f (submit a subdivision street-lighting plan and pole/fixture specifications — VERIFY the lighting standards that apply outside the Highway 29 Overlay); 2024 IRC (GA 2026) R303 (light and ventilation — interior lighting outlets).
- **1.04 Submittals:** Fixture schedule with photometrics (IES files), CCT/CRI, BUG ratings for all exterior fixtures, lane-lighting point-by-point calculation (average, minimum, max/min, and spill at the property lines/buffer), pole and base details, control schedule; samples of decorative exterior lanterns.
- **1.05 Quality Assurance:** Electrical contractor licensed in Georgia; lane-lighting photometric plan sealed by a Georgia PE or prepared by a lighting-certified designer (LC); night-time aiming and spill check with the Architect before acceptance; decide utility-owned (Georgia Power/Walton EMC lease) vs. HOA-owned lane lighting before design (VERIFY the serving utility).
- **1.06 Delivery, Storage, and Handling / 1.07 Warranty:** Store fixtures boxed and dry; luminaires 5-year (LED and driver), exterior poles/finish 5-year, decorative lanterns 3-year; Contractor 2-year.

Part 2 — Products

- **2.01 Manufacturers:** Interior: Lithonia Lighting / Juno / Halo (Acuity — Conyers, GA), Cooper Lighting Solutions (Signify — Peachtree City, GA), WAC Lighting, Progress Lighting, Kichler; exterior lanterns: Hinkley, Kichler, Progress, Troy; lane post-tops: Lithonia (Acuity), Cooper (Signify), Hubbell (Kim/Beacon), Sternberg; or approved equal.
- **2.02 Materials:** All lamps and integrated LEDs 3000K CCT (2700K permitted at bedrooms and exterior lanterns), CRI ≥ 90 interior / ≥ 80 exterior, dimmable (TRIAC/ELV or 0–10 V) with compatible listed dimmers; recessed downlights 4-in or 6-in canless wafer or IC-rated air-tight (ASTM E283 ≤ 2.0 cfm) housings with wet-location trims at showers; surface fixtures with diffusers (no exposed LEDs/bare lamps — glare control per RP-28); vanity: vertical side sconces or a wide diffused bar at 66 in AFF; under-cabinet linear LED at all kitchen counters; closets: surface LED (NEC 410.16 clearances); night-lights: plug-in or switched low-level LED (≤ 1 fc) in the hall and each bath, on a photocell; exterior lanterns: full-cutoff/fully shielded, opaque top, ≤ 3000 K, ≤ 800 lm at porches; garage exterior: shielded downlights $\leq 1,000$ lm on a photocell/motion; lane: 12–14-ft aluminum poles on concrete bases with full-cutoff post-top or "acorn-style with a full-cutoff optic" LED luminaires, 2,000–3,500 lm, 3000K, BUG $\leq$ B1-U0-G1, photocell with an astronomic timer and 50 percent dimming after 11 p.m.; pathway/bollard lights at the green and mail kiosk, ≤ 500 lm, fully shielded.
- **2.03 Performance Criteria:** Interior illuminance per IES RP-28 (roughly twice conventional residential levels): kitchen counters 50–75 fc task, bath vanity 30–50 fc at the face (vertical), general living 20–30 fc, hall/stairs (none) 10–20 fc, closets 10 fc; uniformity $\leq 3:1$ in circulation; exterior: lane average 0.4–0.6 fc, min 0.1 fc, max/min $\leq 10:1$ (RP-8 Class P local, VERIFY Lilburn's standard), spill ≤ 0.1 fc at the R-1 property lines and buffer, zero uplight (U0), all exterior luminaires fully shielded (MLO); porch lights on a switch and a photocell option; occupancy/vacancy sensors at closets, laundry, garage, and baths (fan interlock); rocker/toggle switches at 42–48 in AFF, lighted where useful; all fixtures ENERGY STAR or DLC listed where categories exist.
- **2.04 Finishes:** Exterior lanterns matte black (schemes 1 and 3) or oil-rubbed bronze (scheme 2), as selected; lane poles black; interior trims white or matching hardware.
- **2.05 Accessories:** Compatible dimmers (listed pairings), photocells, astronomic timers, pole bases (32 16 13), in-line fuses at poles, surge protection for lane circuits, spare drivers (5 percent) to the HOA.

Part 3 — Execution

- **3.01 Examination:** Verify ceiling insulation depth for IC/air-tight housings, blocking for lanterns and vanity lights, and pole-base locations clear of sidewalks (≥ 2 ft from the back of curb and out of the sidewalk).
- **3.02 Preparation:** Coordinate fixture locations with the reflected ceiling plans and the sprinkler layout (13D heads); set pole bases with conduit and grounding before paving.
- **3.03 Installation:** Per NEC 410 and the manufacturers: recessed housings sealed to the ceiling gypsum (07 27 00); lanterns on gasketed blocks with the top of the fixture at 66–72 in AFF; poles plumb, luminaires level, photocells facing north; label the

dimmer/driver pairing at each panel; program timers.

- **3.04 Field Quality Control / 3.05 Cleaning and Protection:** Night check: light-meter readings at 10 lane points and at the R-1 property lines against the photometric plan; verify CCT with a meter on a sample of fixtures; verify no uplight; clean lenses and remove labels; deliver O&M data and spare drivers to the HOA.

[REVIEW REQUIRED] Lilburn lighting/light-trespass standards outside the Overlay, the serving utility, and IECC high-efficacy percentage as amended are unverified; the lane photometric plan is a required submittal.

Division 32 — Exterior Improvements

Section 32 12 16 — Asphalt Paving (Private Lane, Guest Bays, and Driveways)

Part 1 — General

- **1.01 Section Includes:** Hot-mix asphalt pavement, graded aggregate base, and subgrade preparation for the 22-ft-wide private lane (in the 31-ft tract; 36-ft tract where widened), hammerhead turnarounds (60-ft legs, 20 ft wide), guest parking bays, the mail-kiosk pull-off, and the Arcade Road entrance apron within the county R/W (per the Gwinnett DOT driveway permit); residential driveways (asphalt or concrete per 32 16 13 at the Owner's option); pavement markings (stop bar, guest-space striping).
- **1.02 Related Sections:** 32 16 13 (curb and gutter, driveway aprons); 26 50 00 (lane lighting); 31 23 00 earthwork and 33 40 00 storm drainage (not in this outline); 32 17 23 pavement markings (included here).
- **1.03 References:** Lilburn Site Development Plan Review Checklist §4.s ("all components of private streets and alleys must meet minimum standards for public street") and §7.i (minimum section for parking areas and drives: **4 in GAB and 2 in Type E or F**), §7.k (street grades ≤ 12 percent; 12–15 percent require an as-graded survey), §7.l (horizontal curves per Gwinnett County standards); **Lilburn Development Regulations (Code Appendix B) — VERIFY the public-street structural section; not retrievable online;** Gwinnett County UDO §900-70.4.A (new local and minor collector streets: **≥ 8 in graded aggregate base, 2 in of 19 mm Superpave intermediate course, 1.25 in of 9.5 mm Superpave Type II surface course**) and §900-70.2.C (fill compacted to 95 percent, top 12 in to 100 percent, of maximum dry density) — used as the presumptive public-street standard; Gwinnett County Standard Drawings (pavement structural sections, Std. 401); GDOT Standard Specifications 2021: Section 310 (GAB), 400 (HMA construction), 412 (tack), 828 (HMA mixes), 815 (GAB material), 820 (asphalt cement PG 64-22 / PG 67-22 as GDOT-specified); ASTM D698 (standard Proctor), D1557 (modified Proctor) — VERIFY which the county uses; ASTM D2950 / D6938 (density); ASTM D6927 (Marshall — not used, Superpave); Lilburn Zoning Ordinance 2023-603 Table 4.2 (pervious paving allowed; front-yard parking discouraged).
- **1.04 Submittals:** Geotechnical pavement recommendations (CBR/subgrade), GAB source certification (GDOT-approved), HMA job-mix formulas (GDOT-approved 19 mm and 9.5 mm Type II), density test reports, plan of the typical section (lane, turnaround, bays, driveways), marking layout.
- **1.05 Quality Assurance:** Paving contractor GDOT-prequalified or with 5 years' subdivision street experience; independent geotechnical testing agency for subgrade, GAB, and HMA density (Division 01 45 00); proof-roll witnessed by the Engineer and the City inspector.
- **1.06 Delivery, Storage, and Handling / 1.07 Warranty:** HMA delivered ≥ 275 degrees F (mix temperature per the JMF), covered trucks; no paving below 50 degrees F ambient (surface course) or on wet base; Contractor 2-year (no rutting > 1/4 in, raveling, cracking, or base failure), matching the City's maintenance-bond period for public streets (VERIFY, since the lane is private and HOA-maintained).

Part 2 — Products

- **2.01 Manufacturers (HMA suppliers):** GDOT-approved plants — e.g., C.W. Matthews Contracting, E.R. Snell Contractor, Pittman Construction / Vulcan Materials; or approved equal.
- **2.02 Materials:** Subgrade: residual soil or engineered fill compacted per §900-70.2.C; GAB: GDOT Section 815 crushed stone; HMA: GDOT Section 828 Superpave 19 mm (intermediate) and 9.5 mm Type II (surface), PG 64-22 or PG 67-22 binder per GDOT, recycled asphalt pavement (RAP) ≤ 25 percent (or per GDOT); tack coat GDOT 412; pavement markings: waterborne traffic paint (GDOT 652) or thermoplastic at the stop bar; pervious concrete/pavers at guest bays as an Owner option (Table 4.2).
- **2.03 Performance Criteria — sections: Private lane, turnarounds, and the entrance (public-street standard, presumptive): 8 in GAB + 2 in 19 mm Superpave + 1.25 in 9.5 mm Type II (3.25 in HMA total) — VERIFY against the Lilburn Development Regulations at the pre-application meeting; guest bays and the mail pull-off: 6 in GAB + 2 in 9.5 mm Type II (exceeds the checklist's 4-in/2-in minimum); driveways (if asphalt): 4 in GAB + 2 in 9.5 mm Type II (checklist §7.i minimum); compaction: subgrade 95/100 percent, GAB 100 percent of ASTM D698 (or D1557 — VERIFY), HMA 92–96**

percent of theoretical maximum density (Gmm) per GDOT 400; geometry: 22-ft face-to-face of curb (no parking), design speed 15 mph, 2 percent crown, grades ≤ 12 percent (target ≤ 8 percent), 25-ft minimum centerline radius at turnarounds, hammerheads 60-ft legs $\times$ 20 ft; cross-slope 2 percent; ride: $\leq 1/4$ in in 10 ft; edge: 24-in curb and gutter (32 16 13); the pavement section shall accommodate a 75,000-lb fire apparatus (IFC 2024 App. D102.1 — VERIFY the Gwinnett Fire requirement for a private lane).

- **2.04 Finishes:** Dense-graded surface; markings: 4-in white guest-space lines, 24-in white stop bar at Arcado Rd, "PRIVATE DRIVE — NO PARKING" signage (Gwinnett Fire) per the site plan.
- **2.05 Accessories:** Geotextile separation fabric (GDOT Section 881, type per the geotechnical engineer) where the geotechnical engineer requires over soft residual clays; underdrains at the swale crossing ($u \approx 700$ –1,000) if groundwater is encountered; traffic-rated utility frames set flush with the surface course.

Part 3 — Execution

- **3.01 Examination:** Verify all underground utilities (sewer, water, storm, electric, lane-lighting conduit) are installed, tested, and backfilled; verify subgrade elevations and proof-roll acceptance (no pumping under a loaded tandem truck); verify curb and gutter is complete and cured.
- **3.02 Preparation:** Scarify, moisture-condition, and compact the subgrade; remove unsuitable material and replace with GAB or engineered fill per the geotechnical engineer; place GAB in ≤ 6 -in lifts to grade $\pm 1/2$ in and prime or keep moist until paving; sweep; apply tack.
- **3.03 Installation:** Per GDOT Section 400 and the JMF: intermediate course 2 in compacted (single lift), surface course 1.25 in compacted, placed with a paver (no hand-spreading except at edges), rolled with breakdown/intermediate/finish passes; joints staggered ≥ 12 in from the intermediate course; longitudinal joints at lane center; match curb-and-gutter flow line within $1/4$ in above (never below) the gutter lip; hand-work around drainage structures; markings after 14 days' cure.
- **3.04 Field Quality Control / 3.05 Cleaning and Protection:** Density tests: subgrade and GAB one per 200 ft per lift, HMA cores or nuclear gauge one per 1,000 SY per lift; thickness verified by cores (one per 1,000 SY); straightedge; open to construction traffic only after the intermediate course is placed, and place the surface course after most home construction to avoid construction damage (phased with the two-phase plan); sweep and protect from spills; repair any damage before the surface course.

[REVIEW REQUIRED] *The public-street structural section in the Lilburn Development Regulations could not be retrieved; the Gwinnett UDO §900-70 section is presumptive. The fire-apparatus load requirement for the private lane, the Proctor method, and GDOT section numbers require confirmation by the civil engineer.*

Section 32 16 13 — Concrete Curbs, Gutters, Sidewalks, and Driveways

Part 1 — General

- **1.01 Section Includes:** 24-in concrete curb and gutter (6-in vertical face) along the lane; curb ramps with detectable warnings at the Arcado Road sidewalk and at internal crossings; 5-ft sidewalks (2 ft behind the back of curb; 6 ft where street trees are planted); driveway aprons; concrete driveways and porch/patio walks (where concrete is selected); the mail kiosk pad; pickleball court slab is excluded (32 18 23 by the court contractor — not in this outline); light-pole bases.
- **1.02 Related Sections:** 03 30 00 (house slabs, landings — the walk shall meet the porch landing flush); 26 50 00 (pole bases); 32 12 16; 32 92 00 / 32 93 00 (strip between curb and walk).
- **1.03 References:** Lilburn Site Development Plan Review Checklist (5-ft sidewalks on all roads; 2 ft from the back of curb, 6 ft where street trees; curb ramps and crosswalks at crossings; pole locations); **Lilburn Development Regulations (Appendix B) — VERIFY curb type and sidewalk thickness;** Gwinnett County Standard Drawings (24-in curb and gutter; sidewalk; driveway apron; curb ramp) and UDO §900 — presumptive; GDOT Standard Specifications 2021 Section 441 (concrete curb, gutter, sidewalk, driveways) and Section 500 (Class A concrete, 3,000 psi); GDOT Standard Details 9032B (curb and gutter) and A3 (curb ramps) — VERIFY; ASTM C94, C150, C33, C260 (air entrainment), C309 (curing), D1751 (expansion joint filler), A1064 (WWR where used); U.S. Access Board PROWAG (2023 final rule, 36 CFR 1190) §R304 (curb ramps), §R302 (pedestrian access routes: 2 percent cross slope, 5 percent running or match the roadway grade), §R305 (detectable warning surfaces) — applied by policy on the private lane, mandatory at the Arcado Road public sidewalk; ICC A117.1-2017 §402–403 (accessible routes: ≤ 5 percent running, ≤ 2 percent cross, ≥ 36 in clear, $1/4$ -in max vertical change) — the design basis for lane-to-porch routes.
- **1.04 Submittals:** Mix design; jointing plan; curb-ramp details with detectable-warning product data; typical sections (curb and gutter, sidewalk, driveway apron); pole-base detail (26 50 00).
- **1.05 Quality Assurance:** Flatwork contractor with 5 years' subdivision experience; ACI-certified flatwork finisher on site; the first 100 ft of curb and 50 ft of sidewalk are the mock-up; slope verification with a smart level at every ramp and landing before acceptance.

- **1.06 Delivery, Storage, and Handling / 1.07 Warranty:** Concrete placed within 90 minutes of batching; no placement when ambient < 40 degrees F without cold-weather protection; Contractor 2-year (no scaling, spalling, cracking > 1/16 in, or settlement > 1/4 in).

Part 2 — Products

- **2.01 Manufacturers (ready-mix):** Thomas Concrete; Argos USA; Heidelberg Materials; detectable warnings: ADA Solutions, Armor-Tile, Tufile (cast-in-place composite or cast-iron); or approved equal.
- **2.02 Materials:** Concrete GDOT Class A, 3,000 psi minimum (3,500 psi at driveways and aprons), air-entrained 5–7 percent (ASTM C260; weathering "moderate"), 4-in maximum slump, 3/4-in maximum aggregate; curb and gutter: 24-in-wide, 6-in vertical face, 6-in gutter thickness (Gwinnett/GDOT standard profile — no rolled/mountable curb, which defeats curb ramps and channelization); sidewalk: 5 ft wide × 4 in thick (6 in at driveway crossings), broom finish; driveways/aprons: 6 in thick (4 in at private walks), broom finish, with 1/2-in expansion joints at the curb and the garage slab; light-pole bases: 24-in diameter × 48-in-deep (or per 26 50 00), 3,000 psi, with anchor bolts and conduit; expansion joint filler ASTM D1751 1/2 in; curing compound ASTM C309 Type 1-D; detectable warning surfaces: 24 in deep × full ramp width, contrasting color (federal yellow or brick red vs. gray concrete).
- **2.03 Performance Criteria:** Curb ramps at every crossing: 1:12 (8.33 percent) maximum running slope, 2 percent maximum cross slope, 4-ft × 4-ft landings ≤ 2 percent, flush (0 in) at the gutter, detectable warnings; sidewalks 5 ft minimum clear (no obstructions: poles, hydrants, mailboxes outside the walk), ≤ 2 percent cross slope, running slope matching the lane (≤ 5 percent preferred; the lane's ≤ 12 percent grade limit governs and steeper segments are permitted where they match the road grade per PROWAG); **walks from the lane sidewalk to each porch: ≤ 5 percent running, ≤ 2 percent cross, ≥ 36 in (48 in preferred) wide, meeting the porch landing with ≤ 1/4-in vertical change (zero-step route);** driveway aprons with the sidewalk carried across at ≤ 2 percent cross slope (the apron slope occurs between the curb and the walk); joints: sidewalk contraction joints at 5 ft, expansion at 50 ft and at structures; curb contraction joints at 10 ft, expansion at 50 ft and at radii returns; curb ramps and pole bases doweled to the curb.
- **2.04 Finishes:** Medium broom perpendicular to travel; tooled edges 1/4-in radius; curb face steel-troweled then lightly broomed; decorative (exposed aggregate/colored) walks at the village green only with the Director's approval (checklist §5.b analog — VERIFY).
- **2.05 Accessories:** Weep pipes through the curb for lot drainage where the grading plan requires; steel plate at storm inlets (curb inlet throats by 33 40 00); 2-ft × 8-ft amenity pads every 300 ft along the Arcade Road frontage if required (checklist §5.b is written for the Overlay — VERIFY applicability); mail-kiosk pad 6 in thick with a 5-ft clear approach.

Part 3 — Execution

- **3.01 Examination:** Verify subgrade compaction (95 percent) and 4-in GAB or compacted stone base under curb and gutter (2 in of stone under walks where the geotechnical engineer requires); verify forms/slipform stringline to the design flow line ± 1/4 in; verify ramp and landing layout against the accessible-route plan.
- **3.02 Preparation:** Set forms or slipform; place expansion joints and dowels; wet the base; install detectable-warning setting frames; stake pole bases from the 26 50 00 layout.
- **3.03 Installation:** Place, consolidate, screed, and float; broom finish; tool joints within 12 hours or saw within 24; cure 7 days with compound or wet cover; strip forms after 24 hours and backfill within 7 days; set detectable warnings in fresh concrete; slope every ramp and landing with a 4-ft smart level before the concrete sets and correct while plastic.
- **3.04 Field Quality Control / 3.05 Cleaning and Protection:** Cylinders per 50 cy; slope audit of every ramp, landing, and porch approach recorded on a checklist (a slope > 2.0 percent cross or > 8.33 percent ramp is rejected and replaced); flow-line water test after the surface course; keep construction traffic off walks; protect curb from asphalt tack overspray; repair scaling by replacement, not patching.

[REVIEW REQUIRED] Curb profile, sidewalk thickness, and ramp details are presumptive Gwinnett/GDOT standards pending the Lilburn Development Regulations; the accessible-route slopes on the lots with cross-slope (NE to SW) need the civil engineer's grading plan.

Section 32 31 13 — Chain Link Fences and Gates (Pickleball Court Enclosures)

Part 1 — General

- **1.01 Section Includes:** Black vinyl-coated chain-link fencing, 10 ft high, enclosing each 30-ft × 60-ft pickleball pad (two courts, each a 20-ft × 44-ft playing area), with accessible pedestrian gates, a maintenance gate, and optional sound-attenuating and wind-screen panels; fence posts and footings; court surfacing, nets, and posts are by 32 18 23 (not in this outline).
- **1.02 Related Sections:** 03 30 00 / 32 16 13 (court slab and approach walk — by the court contractor); 26 50 00 (no court lighting is proposed — daylight play only; VERIFY); 32 93 00 (screen planting between the courts and adjacent R-1 lots); 10 14 00 (rules signage).

- **1.03 References:** Lilburn Zoning Ordinance 2023-603 — accessory structure and fence height/setback provisions (**VERIFY: a 10-ft sport-court fence may exceed the residential fence height limit and require a specific approval or condition; keep the courts out of the 20-ft buffer and ≥ 5 ft from it — checklist " ≥ 5 -ft parking setback from buffers" analog**); ASTM F668 (PVC-coated chain-link fabric, Class 2b fused-and-bonded), ASTM F1043 (framework strength, Group IC) and ASTM F1083 (Schedule 40 pipe), ASTM F626 (fittings), ASTM F900 (gates), ASTM F567 (installation practice), ASTM F1183 (aluminum alternate — not used), ASTM A123/A153 (galvanizing); USA Pickleball Official Rulebook §2 (court dimensions: 20 × 44 ft, recommended minimum 30 × 60 ft playing surface, 34-in net at center) and USA Pickleball / ASBA Pickleball Courts: Construction & Maintenance Manual (fence heights, windscreens, acoustic fencing guidance); ICC A117.1-2017 §404 (gates on accessible routes: 32-in clear, lever/loop hardware, ≤ 5 lbf, no threshold).
- **1.04 Submittals:** Fence layout with post spacing and gate locations; product data for fabric, framework, fittings, gates, hardware, and acoustic/windscreen panels (with wind-load data); footing detail; color sample.
- **1.05 Quality Assurance:** Fence contractor with 5 years' sports-fence experience; the first court's fence reviewed before the second; note that pickleball noise (≈ 70 dBA at 100 ft) is the most common 55+/neighbor complaint — the amenity block location, orientation, acoustic panels, and HOA hours (e.g., 8 a.m.–8 p.m.) are the mitigation, offered as a voluntary zoning condition.
- **1.06 Delivery, Storage, and Handling / 1.07 Warranty:** Store fabric rolls on end, framework off the ground; PVC coating 15-year (chalk/peel), framework 10-year, gates and hardware 5-year, acoustic panels 10-year; Contractor 2-year.

Part 2 — Products

- **2.01 Manufacturers:** Chain link systems: Master Halco, Merchants Metals, Ameristar (ASSA ABLOY) — ornamental alternate at the entry/green; acoustic panels: Acoustifence (Acoustiblok), Sound Fighter Systems, Kinetics Noise Control (VERIFY sports-fence products); windscreens: Aer-Flo, Putterman; or approved equal.
- **2.02 Materials:** Fabric: 9-ga core (0.148 in) with 6-ga finished PVC-coated (Class 2b fused and bonded) black, 1-3/4-in mesh (ball containment; 2-in acceptable), knuckled top and bottom selvage (no barbs), 10-ft height at the end (baseline) fences and full 10 ft at the sides (4-ft side fences optional at the Architect's choice — full height specified for containment and screen mounting); framework ASTM F1043 Group IC (or F1083 Sch. 40), galvanized and black PVC/polyester coated: terminal/corner/gate posts 3-in OD (10-ft fence, set 42 in deep), line posts 2-1/2-in OD at ≤ 10 ft o.c., top and bottom rails 1-5/8-in OD, mid rail at 5 ft; fittings F626 black; tension wire 7-ga at the bottom rail if no bottom rail; gates F900: two 4-ft-wide single swing pedestrian gates per pad (≥ 32 in clear, self-closing with a lever/loop latch operable ≤ 5 lbf, no drop rod at the walking surface, 1/2-in maximum threshold — ideally 0), one 6-ft double maintenance gate per pad; hardware black; optional windscreens (open-mesh, ≤ 70 percent blockage, with wind-load relief) and optional mass-loaded vinyl acoustic panels (≥ 1 lb/SF, STC ≥ 28) on the sides facing the nearest R-1 lots — post spacing and footings sized for the panel wind load if installed.
- **2.03 Performance Criteria:** Wind: framework and footings designed for 115 mph Exposure B (VERIFY) with screens/panels attached (typical: 12-in-dia × 42-in footings at line posts, 16-in × 48-in at terminals — Engineer to confirm); fabric taut without sag (≤ 1 in deflection under a 50-lb pull at mid-span); bottom of fabric ≤ 2 in above the court surface; gate clear opening 32–36 in; no sharp edges (knuckled selvage; bolt ends peened); court pad drains at 1 percent minimum away from the fence (by 32 18 23); fence located ≥ 5 ft from the buffer and outside all setbacks (VERIFY per the Zoning Ordinance).
- **2.04 Finishes:** All components black (PVC-coated fabric; polyester-powder or PVC-coated framework and fittings); hardware black.
- **2.05 Accessories:** Post caps, rail ends, tension bars and bands, truss rods at gate posts; rules/hours sign (10 14 00) at each gate; bench pads and a shade structure by others; ball-containment "kick plate" (1-ft-high solid panel) at the bottom of the end fences optional.

Part 3 — Execution

- **3.01 Examination:** Verify the court layout, pad edges, and gate locations against the amenity plan and the accessible route from the clubhouse/mail kiosk; verify setbacks and the buffer line from the survey.
- **3.02 Preparation:** Call 811; auger footings; set posts plumb in 3,000-psi concrete (crowned at grade); cure 3 days before stretching.
- **3.03 Installation:** Per ASTM F567: rails and mid-rail, then fabric stretched with a come-along and tied at 12 in o.c. on rails and 24 in on posts (aluminum ties, black), knuckled ends; gates hung plumb with hinges adjusted for self-closing and latch alignment; screens/panels with grommets at 12 in o.c. and tie-wraps rated for UV, leaving relief gaps per the manufacturer.
- **3.04 Field Quality Control / 3.05 Cleaning and Protection:** Plumb ($\pm 1/4$ in in 10 ft) and height checks; gate operation and latch force test; touch up coating scratches with the manufacturer's paint; remove spoil and concrete splatter from the court pad before surfacing.

[REVIEW REQUIRED] Fence height versus the Lilburn residential fence limit, wind design with screens, and the noise-mitigation package are unverified and should be raised at the pre-application conference.

Section 32 92 00 — Turf and Grasses (Sod)

Part 1 — General

- **1.01 Section Includes:** Topsoil placement and fine grading; warm-season sod on all lots (front, side, and rear yards outside the buffer and preserved woods), the village green, lane strips, and pond embankments above the normal pool; temporary and permanent seeding of disturbed areas outside sod limits (by 31 25 00 erosion control — coordinate); initial establishment; irrigation by 32 84 00 (not in this outline).
- **1.02 Related Sections:** 32 16 13 (walk edges); 32 93 00 (beds and buffer supplement); 31 25 00 (erosion and sediment control — Georgia "Green Book"), 33 40 00 (pond grading) — not in this outline.
- **1.03 References:** Georgia Crop Improvement Association (GCIA) Certified Sod program; TPI (Turfgrass Producers International) Guideline Specifications to Turfgrass Sodding; UGA Cooperative Extension turfgrass recommendations for the Georgia Piedmont; ASTM D5268 (topsoil for landscaping); Manual for Erosion and Sediment Control in Georgia (current ed.) for temporary cover; Lilburn Site Development Plan Review Checklist (landscape strip and buffer planting; tree protection); Gwinnett County outdoor water-use rules (irrigation schedules — VERIFY).
- **1.04 Submittals:** Sod source and GCIA certificate per variety; topsoil analysis (UGA soil test: pH, organic matter, texture) and amendments; sod layout by variety; establishment schedule; the HOA maintenance plan (mowing height, fertilization, irrigation).
- **1.05 Quality Assurance:** Landscape contractor licensed in Georgia with a GCLP (Georgia Certified Landscape Professional) on staff; sod cut ≤ 24 hours before laying (36 in cool weather); mock-up: the model-cottage lots.
- **1.06 Delivery, Storage, and Handling / 1.07 Warranty:** Deliver sod on pallets, keep shaded and moist, do not stack > 24 hours; 90-day establishment period with replacement of any area not 95 percent covered and rooted; Contractor 1-year on grading and drainage (no ponding > 24 hours).

Part 2 — Products

- **2.01 Sources:** GCIA-certified Georgia sod farms (e.g., Super-Sod / Patten Seed, NG Turf, Legacy Turf Farms — Georgia growers); or approved equal.
- **2.02 Materials:** Full-sun lots, green, and lane strips: 'TifTuf' hybrid bermudagrass (drought-tolerant, GCIA certified) or 'Zeon' / 'Emerald' zoysiagrass (finer texture, shade-tolerant to ~4 hours sun — preferred at front yards for a cottage character); shaded rear yards under retained trees and buffer edges: turf-type tall fescue blend (Georgia-recommended cultivars) sod, or convert to mulched native groundcover beds (32 93 00) where sun < 4 hours; topsoil: ASTM D5268, sandy loam to loam, pH 6.0–6.5, ≥ 3 percent organic matter, 4 in placed at all sodded areas (6 in at the green); starter fertilizer per the soil test; lime per the soil test (Piedmont clays are typically acidic).
- **2.03 Performance Criteria:** Finish grade 1 in below walks and curbs (sod thickness), sloping ≥ 2 percent away from foundations for 10 ft (IRC R401.3: 6 in in 10 ft) — **at zero-step entries, achieve fall with swales alongside the landing rather than by raising the landing**; no slopes > 3:1 in sodded areas (use 32 93 00 plantings or fescue with erosion netting on steeper pond banks); sod strips staggered, butted, no gaps > 1/4 in, rolled, watered within 30 minutes; establishment: 95 percent cover and rooted (cannot be lifted) at 90 days; mowing height per variety (bermuda 1–1.5 in, zoysia 1.5–2 in, fescue 3–3.5 in).
- **2.04 Finishes:** Uniform variety per street frontage (no mixed varieties on adjacent lots); mulch (32 93 00) at all bed edges with a spade-cut edge.
- **2.05 Accessories:** Sod staples on slopes > 4:1; erosion-control blanket (Georgia Green Book Type) under fescue on pond banks; temporary irrigation where the permanent system is not yet live.

Part 3 — Execution

- **3.01 Examination:** Verify rough grades ± 0.1 ft, positive drainage, utility trenches settled, and the buffer/tree-protection fencing in place; verify the soil test results are in hand.
- **3.02 Preparation:** Rip compacted subsoil 6 in (except within tree drip lines); spread and rake topsoil; incorporate lime/fertilizer; fine grade; water to settle; remove stones > 1 in and debris.
- **3.03 Installation:** Lay sod on moist soil within 24 hours of cutting, perpendicular to slopes, staggered joints, tight seams, rolled, and watered to 4-in depth; keep off for 2 weeks; first mowing when rooted; install warm-season sod March–October (dormant sod November–February accepted only with the Architect's approval and extended establishment); fescue September–November or February–April.
- **3.04 Field Quality Control / 3.05 Cleaning and Protection:** Inspect at 30, 60, and 90 days for cover, weeds, and drainage; replace failed areas; clean walks and curbs of soil daily; protect with temporary fencing at the model lots until turnover to the HOA with the maintenance plan.

[REVIEW REQUIRED] Variety selection depends on the lot-by-lot sun exposure after clearing; irrigation and water-use rules are unverified; the erosion-control sequencing is by the civil engineer's ES&PC plan.

Section 32 93 00 — Plants (Buffer Supplement, Landscape Strip, Street Trees, and Foundation Plantings)

Part 1 — General

- **1.01 Section Includes:** Supplemental evergreen and canopy planting within the 20-ft perimeter buffer where the existing vegetation is sparse (no clearing or grading in the buffer); the 10-ft landscape strip along Arcade Road; lane street trees; screen planting at the pickleball courts, ponds, and mail kiosk; foundation and porch plantings at each cottage (HOA-maintained); native/adapted Georgia Piedmont species; mulch, staking, and a 1-year maintenance/warranty period. Irrigation by 32 84 00; tree protection by 01 56 39 / 31 10 00 (not in this outline).
- **1.02 Related Sections:** 10 14 00 (sign bed); 26 50 00 (light spill and pole/tree conflicts); 32 16 13 (root barriers at walks); 32 31 13 (court screen); 32 92 00 (topsoil, sod edges).
- **1.03 References:** Lilburn Zoning Ordinance 2023-603 §313 (buffers: undisturbed, supplemented to an effective screen — **VERIFY the planting standard and the "opaque within 3 years" language**) and Table 4.1 (20-ft buffer abutting R-1 for cottage homes); Lilburn Site Development Plan Review Checklist: 10-ft landscape strip along all public R/W with planting at the stated rate ("for each 10-foot LS strip install 1 tree and 1 shrub" per unit length — VERIFY the rate), tree protection for trees > 5-in caliper in buffers and strips, street-tree list (Willow Oak, Overcup Oak, Nuttall Oak, Pin Oak, Shumard Oak, Lacebark Elm, Japanese Zelkova — written for the Overlay; used as the accepted list), 4-in caliper at 6 ft from the back of curb for street trees along the public R/W; ANSI Z60.1-2014 (American Standard for Nursery Stock); ANSI A300 Part 6 (planting and transplanting); ISA BMPs; Georgia Exotic Pest Plant Council (GA-EPPC) Category 1 invasive list (prohibited); UGA Extension "Native Plants for Georgia" (Parts I–IV) — the species basis; Georgia Department of Agriculture nursery certification.
- **1.04 Submittals:** Planting plan with a schedule (botanical/common name, size, quantity, root condition, spacing); nursery certificates and source list (Georgia-grown preferred); soil test and amendments; mulch sample; the buffer inventory (existing trees > 5 in caliper, canopy gaps) that drives the supplement quantities; the HOA maintenance plan.
- **1.05 Quality Assurance:** Landscape contractor with a GCLP; plants inspected at the nursery or on delivery for Z60.1 conformance; ISA-certified arborist to mark buffer gaps and supervise work within tree drip lines; the model-cottage frontage is the foundation-planting mock-up.
- **1.06 Delivery, Storage, and Handling / 1.07 Warranty:** Deliver plants within 48 hours of digging (B&B) or in containers; protect from wind and sun; keep root balls moist; plant within 3 days of delivery; 1-year warranty with one replacement of dead or defective plants at the end of the period; 90-day establishment maintenance (watering, weeding, resetting) included.

Part 2 — Products

- **2.01 Sources:** Georgia Green Industry Association member nurseries (e.g., Bold Spring Nursery, Hackney Nursery, Select Trees, Saunders Brothers — regional growers); or approved equal.
- **2.02 Materials — species palette (Georgia Piedmont natives and proven adapted cultivars; final selections by the Landscape Architect):**
 - Canopy (buffer supplement and green): *Quercus alba* (white oak), *Q. phellos* (willow oak), *Q. shumardii* (Shumard oak), *Q. lyrata* (overcup oak), *Nyssa sylvatica* (black gum), *Liriodendron tulipifera* (tulip poplar, buffer only), *Acer rubrum* (red maple) — 2-1/2–3-in caliper B&B in the buffer; 4-in caliper at the Arcade Road strip per the checklist.
 - Evergreen screen (buffer gaps, courts, ponds, kiosk): *Ilex opaca* (American holly), *Juniperus virginiana* (eastern red cedar, sunny edges), *Magnolia virginiana* (sweetbay), *Magnolia grandiflora* 'Little Gem' (adapted cultivar), *Ilex vomitoria* (yaupon, adapted), *Osmanthus americanus* (devilwood) — 6–8 ft height B&B, double staggered rows at 8–10 ft o.c. where the buffer is open.
 - Understory/ornamental: *Cercis canadensis* (redbud), *Cornus florida* (dogwood, part shade), *Amelanchier arborea* (serviceberry), *Chionanthus virginicus* (fringetree), *Oxydendrum arboreum* (sourwood), *Hamamelis virginiana* (witch hazel) — 1-1/2–2-in caliper or 6–8 ft.
 - Shrubs: *Hydrangea quercifolia* (oakleaf hydrangea), *Itea virginica* 'Henry's Garnet', *Ilex glabra* (inkberry, 'Shamrock'), *Callicarpa americana* (beautyberry), *Fothergilla gardenii*, *Viburnum dentatum* / *V. nudum*, *Clethra alnifolia*, *Rhododendron canescens* (Piedmont azalea, shade), *Ilex verticillata* (winterberry, wet edges) — 3- and 7-gal.
 - Groundcovers/perennials: *Carex pensylvanica*, *Chasmanthium latifolium* (shade), *Muhlenbergia capillaris* (muhly grass), *Rudbeckia fulgida*, *Echinacea purpurea*, *Amsonia hubrichtii*, *Packera aurea*, *Phlox divaricata* — 1-gal and plugs; pond banks above the pool: native seed mix (Georgia Green Book) with *Juncus*, *Carex*, *Panicum virgatum*.
 - Lane street trees (in the 5-ft lawn strip only where the 36-ft tract allows a ≥ 6-ft strip; otherwise in lot front yards): *Q. phellos*, *Q. shumardii*, *Ulmus parvifolia* 'Allee' (lacebark elm), *Zelkova serrata* — 3-in caliper, 40 ft o.c. average, ≥ 6 ft from the back of curb,

≥ 10 ft from lane lights and ≥ 15 ft from sewer laterals.

- **Prohibited:** GA-EPPC Category 1 species — *Ligustrum* spp., *Nandina domestica*, *Pyrus calleryana* (Bradford pear), *Elaeagnus* spp., *Lonicera japonica*, *Hedera helix*, *Wisteria sinensis*, *Vinca* spp.; *Lagerstroemia* (crape myrtle) is adapted and permitted outside the buffer.
- **2.03 Performance Criteria:** Buffer supplement quantity: planted only in gaps mapped by the arborist to achieve a continuous 6-ft-high visual screen within 3 growing seasons (≈ 1 evergreen per 8–10 lineal ft of gap plus 1 canopy tree per 30–40 ft — VERIFY against §313); landscape strip: 1 canopy tree (4-in cal.) and shrubs per the checklist rate per 10 ft (VERIFY); foundation plantings: low (≤ 30 in mature) shrubs under windows, no thorny or high-pollen species at porches, ≥ 3 ft clear of the siding for the 6-in siding-to-grade clearance and HVAC access, ≥ 18 in clear of walks; sight triangles at the Arcade Road entrance kept ≤ 30 in high (VERIFY the Gwinnett DOT triangle); all plants Z60.1 with a well-developed root system, no circling roots, single leader on canopy trees; planting soil per the soil test; mulch 3 in (pine straw or shredded hardwood) held 3 in from trunks; no mulch volcanoes.
- **2.04 Finishes:** Bed lines spade-cut; steel edging optional at the green; tree grates not used; root barriers (18-in deep) at street trees within 5 ft of walks.
- **2.05 Accessories:** Staking only where needed (remove at 1 year); trunk guards (deer/mower); tree-watering bags for the first season; slow-release fertilizer tablets; rodent/deer repellent as needed; tree-protection fencing (orange, at the drip line) for retained trees and the buffer during construction (checklist requirement).

Part 3 — Execution

- **3.01 Examination:** Verify the buffer inventory and gap map, tree-protection fencing, finish grades (32 92 00), irrigation stubs, and utilities (811); verify the lane lights, laterals, and sight triangles are staked.
- **3.02 Preparation:** Lay out and obtain the Landscape Architect's approval of locations; dig holes 2–3 times the root-ball width and no deeper than the ball (root flare at or 1 in above grade); no digging within the drip line of retained buffer trees except by hand for the supplement plants; amend backfill only per the soil test (no wholesale replacement).
- **3.03 Installation:** Per ANSI A300 Part 6: remove wire baskets (top 2/3) and burlap; set plumb; backfill in lifts with water; form a 3-in saucer; mulch; stake only in windy exposures; prune only dead/broken branches; water thoroughly and weekly (1 in) through the first season; plant trees October–March preferred (containers year-round with irrigation).
- **3.04 Field Quality Control / 3.05 Cleaning and Protection:** Inspect at planting, 30 days, and the end of the 1-year period; replace failures at the next planting season; remove tags and stakes at 1 year; keep beds weed-free through establishment; the HOA's landscape contract takes over at turnover with the plan and warranty documentation.

[REVIEW REQUIRED] Buffer supplement rate, landscape-strip rate, street-tree placement in the 31-ft tract, and species suitability per the post-clearing sun/soil conditions require the Landscape Architect and the Lilburn buffer standard (§313 / Development Regulations).

Summary and review flags

Specifications written: 30 sections across 11 divisions. Sections flagged **[REVIEW REQUIRED]**: 30 of 30 (every section is outline-level with generic manufacturers; the reasons cluster as follows).

1. **Code edition change (all sections):** Georgia's mandatory codes changed on 1 January 2026 to the **2024 IRC / IBC / IPC / IMC / IFGC with 2026 Georgia Amendments, the 2023 NEC (2026 GA Amendments), and the 2024 IFC** — not the 2018 editions assumed in the task brief and in FACTS.md §2 ("IFC 2018 App. D"). Section numbers cited follow the 2018–2024 numbering and must be re-verified against the 2024 texts. The **energy code remains the 2015 IECC with Georgia Amendments (2023)**: Zone 3 wall R-13 (not R-20/R-13+5), ceiling R-38 (R-30 alternative), fenestration U-0.35/SHGC-0.27, ≤ 5 ACH50 — the design (R-20 walls, U ≤ 0.28 / SHGC ≤ 0.23 windows, ≤ 3 ACH50 target) exceeds these minimums.
2. **Life-safety-adjacent sections (Architect/AHJ confirmation):** 08 14 00 (egress door, garage separation door — self-closing/self-latching, safety glazing, zero-threshold water management vs. IRC R311.3), 08 36 13 (garage-door wind-load label, UL 325 opener), 08 53 13 (bedroom egress net-clear openings, safety glazing, PG rating), 09 29 00 (garage/house separation, fireblocking, ceiling board on 24-in trusses), 07 31 00 (Class A, wind attachment), 22 40 00 (ASSE 1016 scald protection), 03 30 00 / 06 10 00 (all structural values pending the geotechnical report and a Georgia PE).
3. **Local standards not retrievable — VERIFY at the Lilburn pre-application conference:** the **Lilburn Development Regulations (Appendix B) public-street structural section** (32 12 16 uses Gwinnett UDO §900-70: 8 in GAB + 2 in 19 mm + 1.25 in 9.5 mm Superpave as presumptive; the checklist's 4-in GAB / 2-in Type E-F minimum applies to drives and parking), curb profile and sidewalk thickness (32 16 13), the §313 buffer-supplement planting standard and the landscape-strip planting rate (32 93 00), the sign article's area/height/setback/illumination limits for a ≤ 32-SF monument sign and the separate sign permit (10 14 00), the residential fence-height limit vs. a 10-ft sport-court fence and the pickleball noise package (32 31 13),

lighting/light-trespass standards outside the Highway 29 Overlay and the serving utility (26 50 00), and Gwinnett Fire's apparatus-load requirement for the private lane (32 12 16).

4. **Zero-step entry detail set (03 30 00, 04 40 00, 07 46 46, 08 14 00, 32 16 13, 32 92 00):** the 1/2-in maximum rise at every door is reconciled with IRC wood/siding clearances by stopping siding on the masonry water table (stone may be 1/2 in above a walking surface on the same foundation, IRC R703.12.1), sloping landings 2 percent, draining with swales rather than raised landings, and specifying ADA-type sills under a 6-ft porch roof; the Architect must draw these five details before the elevations are finalized.
5. **Owner selections still open:** factory ColorPlus vs. field paint (07 46 46 / 09 91 13); manufactured vs. natural stone (04 40 00); SPC vs. WPC plank (09 65 00); quartz vs. solid surface by room (12 36 00); which grab bars are installed at closing vs. blocked-for (10 28 13); utility-leased vs. HOA-owned lane lighting (26 50 00); sod variety by lot exposure (32 92 00); ENERGY STAR climate zone for Gwinnett County (08 53 13). All colors and hex values in this document are illustrative placeholders.

Sources

- Georgia DCA, "Current State Minimum Codes for Construction" — <https://dca.georgia.gov/community-assistance/construction-codes/current-state-minimum-codes-construction> (accessed 2026-08-28: 2024 IRC/IBC/IPC/IMC/IFGC/ISPC with 2026 GA Amendments; 2023 NEC; 2024 IFC; 2015 IECC with GA Supplements and Amendments, 2023); DCA announcement "New Georgia Codes and Amendments — Effective January 1, 2026" — <https://dca.georgia.gov/announcement/2025-12-09/new-codes-jan-2026>
- 2015 IECC with Georgia Amendments, Zone 3 values — Insulation Institute summary https://insulationinstitute.org/wp-content/uploads/2023/02/GA_Code_2023_v2.pdf and UpCodes, Georgia State Minimum Standard Energy Code 2015, Chapter RE 4 — https://up.codes/viewer/georgia/iecc-2015/chapter/RE_4/re-residential-energy-efficiency
- Gwinnett County UDO §900-70 Street Construction Standards — http://gwinnettcountry.elaws.us/code/coor_apxa_ch900_sec900-70; City of Lilburn Site Development Plan Review Checklist — <https://www.cityoflilburn.com/DocumentCenter/View/1888> (local copy data/site-dev-plan-review-checklist.txt)
- Lilburn Zoning Ordinance 2023-603 — <https://www.cityoflilburn.com/DocumentCenter/View/3664> (excerpts in data/ordinance-excerpts.md); project facts: status/cottage-submission-2026-08-28/FACTS.md §2, §3, §4, §6 (2026-08-28)
- Product and industry standards cited by designation in each section (ASTM, ANSI, TCNA, SMACNA, MPI, DASMA, NAFS, ICC A117.1-2017, IES RP-28, ASTM F3261, GDOT Standard Specifications 2021) — editions to be confirmed by the specifier of record.

Disclaimer: This is an AI-generated analysis for preliminary planning purposes. All findings must be verified by a licensed professional before use in design, permitting, or regulatory submissions.

title	HOA, HOPA and Covenant Outline — The Cottages at Arcado Springs
date	2026-09-03
address	4535, 4537, 4539 & 4541 Arcado Rd SW, Lilburn, GA 30047 — Land Lot 123, 6th District, Gwinnett County (PINs R6123 033 / 015 / 014 / 162)
status	DRAFT — NOT SEALED, NOT LEGAL ADVICE. An outline of a legal instrument prepared by the owner-applicant. Every substantive paragraph flagged [ATTORNEY REVIEW REQUIRED] must be drafted, redlined and issued by a Georgia land-use or community-association attorney before recording.
generator	tools/hoa_budget.py (all budget quantities and totals; outputs data/hoa-budget-derived.json and verify/hoa-budget-output.md)

13 — HOA, HOPA and Covenant Outline

What this document is. The City of Lilburn is being asked to rezone R-1 → R-2 in reliance on eighteen voluntary conditions (data/voluntary-conditions.md, offered under Lilburn Zoning Ordinance, Ord. 2023-603, §1003-4(5)). Five of those conditions — the 55+ restriction, the 20-ft buffer, the fire-sprinkler offer, the stream-buffer preservation and the HOA maintenance obligation — are promises about how the community will be *operated for the next fifty years*, and a zoning condition alone does not operate anything. The instrument that does is a recorded Declaration of Covenants, Conditions, Restrictions and Easements. This document is the outline of that Declaration: what each article must do, who performs each obligation, on what cycle, on what record, and what it costs. It ends with a line-item annual budget for the association at full build-out.

What this document is not. It is not the Declaration, and no sentence in it should be recorded as written. It is an owner-prepared outline whose purpose is (a) to let the Planning Director and the City Attorney see that the voluntary conditions have a real enforcement mechanism behind them, and (b) to give the community-association attorney a complete scope so the drafting is one engagement rather than three.

Conventions.

- **[ATTORNEY REVIEW REQUIRED]** marks every paragraph where operative legal language must be drafted by counsel. It is not a formality: the paragraphs so marked are the ones that decide whether the covenant is enforceable.
- **VERIFY** marks a fact or a decision that is still open.
- Statutory citations carry the code and section. Ordinance citations carry the ordinance number and section ("Lilburn Zoning Ordinance 2023-603 §402-1"). Code citations carry the edition ("IFC 2024 (GA) Appendix D103.4"). Nothing here asserts legal compliance; the strongest statement made is "appears consistent with".
- Areas are stated in SF with the area type; dimensions use architectural notation (20' - 0").
- Every dollar figure in §8 is computed by tools/hoa_budget.py from data/layout.json and data/plans.json; no figure is typed in by hand.

1. The enforcement map — which instrument carries which promise

The eighteen voluntary conditions in data/voluntary-conditions.md are reproduced verbatim in the Letter of Intent (docs/03 §11). This table says where each one actually lives after the hearing.

#	Voluntary condition	Zoning condition (City-enforced)	Declaration article	Plat / other recorded instrument
1	Age restriction (HOPA 55+)	Yes — the operative promise	Art. 4	Plat note; deed reference
2	Maximum units (≤ 47; 41 shown)	Yes	Art. 2 (property subject)	Final plat lot count
3	Height — 1 story, ridge ≤ 24 ft	Yes	Art. 8	—
4	Buffers — 20 ft undisturbed, easement	Yes	Art. 5	Buffer easement on plat; deed reference
5	Fire sprinklers (NFPA 13D, offered voluntarily)	Yes, with the release mechanic	Art. 7	—
6	Single access, no gate	Yes	Art. 6	Plat (one entrance)
7	No Arcado Road driveways	Yes	Art. 8	Plat note — no access easement to Arcado
8	Architectural standards, colours, no wall signs	Yes	Art. 8	—

#	Voluntary condition	Zoning condition (City-enforced)	Declaration article	Plat / other recorded instrument
9	Stream buffer preserved in a common tract	Yes	Art. 5 §5.9	Plat — 25/50/75-ft lines, creek-woods tract
10	HOA maintenance of lane, walks, ponds, buffers, yards	Yes	Art. 6	Plat note (private maintenance) per Site Development Plan Review Checklist §4.o
11	Lighting — full cut-off ≤ 3000 K, no court lighting	Yes	Art. 6, Art. 8	Street-lighting plan at LDP
12	Construction hours	Yes — construction-phase only	Art. 8 (declarant + owners)	—
13	Tree survey and preservation	Yes — LDP-stage	Art. 5, Art. 6	Tree preservation plan
14	Entry sign — one monument ≤ 32 SF	Yes	Art. 6, Art. 8	Sign easement if required (checklist §4.q)
15	Phasing at u ≈ 1,000 ft	Yes	Art. 2 (annexation)	Supplemental Declaration for Phase 2
16	Sanitary sewer — gravity only, no private pumping	Yes — the hard one	Art. 6 §6.8 (a prohibition, not an obligation)	Off-site sewer easement, recorded
17	Level-1 TIS, sight-distance certification, DOT permit	Yes — pre-LDP	—	—
18	Sidewalks and transit cooperation	Yes	Art. 6	Plat

Conditions 12, 15, 16 and 17 are **developer** obligations that are performed and exhausted before the last home is sold; they belong in the zoning conditions and should not be written as perpetual covenants running against 41 individual homeowners. Conditions 1, 4, 5, 8, 10 and 11 are **perpetual operating** obligations and must be in the Declaration, because that is the only document that binds the 41st owner in 2076. **[ATTORNEY REVIEW REQUIRED]** — the allocation in the last two columns is a drafting judgement and should be confirmed against the conditions as finally adopted by the City Council.

2. The instrument set and the Georgia legal framework

2.1 Documents to be produced

#	Instrument	Recorded?	Who prepares	When
1	Declaration of Covenants, Conditions, Restrictions and Easements	Yes — Gwinnett County Superior Court deed records	Community-as sociation attorney	Before the first lot is conveyed and before the first building permit (voluntary condition)
2	Articles of Incorporation, The Cottages at Arcade Springs Homeowners Association, Inc.	Filed with the Georgia Secretary of State	Attorney	Before the Declaration
3	Bylaws	No (attached as an exhibit)	Attorney	With the Declaration
4	Architectural and Landscape Guidelines	No — Board-adopted, amendable	Declarant, then ACC	With the Declaration
5	HOPA Policies and Procedures (the §100.306 "published policies")	No — published and posted	Attorney + manager	With the Declaration
6	Occupancy Certificate, Age Affidavit and Biennial Survey forms	No	Attorney	With the Declaration
7	Buffer Easement (within the Declaration, plus a plat easement and a deed recital)	Yes	Attorney + surveyor	With the plat
8	Storm Water BMP Maintenance Agreement with as-built hydrology, volume certification, maintenance schedule, access easement, inspection report and 18-month maintenance bond	Yes — "Originals to be recorded by applicant with Clerk of Court"	PE + attorney	Site Development Plan Review Checklist §2.g.3, at the Certificate of Development Conformance stage
9	Off-site sanitary sewer easement (Phase 2)	Yes	Attorney	Before the Phase-2 LDP (voluntary condition 16)
10	Supplemental Declaration annexing Phase 2	Yes	Attorney	At the Phase-2 final plat
11	Consent and Subordination of any mortgagee	Yes	Lender's counsel	With the Declaration

2.2 Georgia framework

- **Georgia Property Owners' Association Act, O.C.G.A. §44-3-220 et seq. (Article 6, §§44-3-220 – 44-3-235).**

The Declaration should **expressly submit the property to the POA Act**. The Act is opt-in; a declaration that does not say so creates a common-law association with materially weaker remedies. Three consequences matter here:

1. **Perpetual covenants.** O.C.G.A. §44-3-234 disapplies the twenty-year limitation that O.C.G.A.

§44-5-60 otherwise places on covenants restricting land to specified uses. A 55+ restriction and a buffer easement that expire in 2046 would be worthless to the City; submission to the Act is what makes them perpetual. **This is the single most important drafting decision in the instrument.**

2. **Automatic statutory lien.** O.C.G.A. §44-3-232 gives the association a lien for assessments without

separately filing it, together with late charges, interest, costs of collection and reasonable attorney's fees actually incurred. The section also carries limits the budget must respect: no foreclosure unless the lien is at least \$2,000.00; the lien lapses four years after the assessment first became due; and a written statement of amounts due must be furnished within five business days of a request or the lien is extinguished as to the requesting party.

3. **Amendments bind non-voting owners.**

- **Georgia Nonprofit Corporation Code, O.C.G.A. §14-3-101 et seq.** — the association is a Georgia nonprofit membership corporation; annual registration with the Secretary of State; a registered agent; director indemnification, which the D&O policy in §8 backs.

- **Federal Fair Housing Act, 42 U.S.C. §3601 et seq.,** and the housing-for-older-persons exemption at **42 U.S.C. §3607(b)(2)(C)**, implemented at **24 CFR Part 100, Subpart E, §§100.300–100.308.**

- **Georgia Fair Housing Act, O.C.G.A. §8-3-200 et seq.** — the state analogue, enforced by the Georgia Commission on Equal Opportunity. **VERIFY with counsel which state section carries the older-persons exemption and whether its terms differ from the federal rule;** the Declaration must satisfy both.

- **Federal statutes that override covenants** and must therefore be written into the use restrictions rather than discovered later: the Freedom to Display the American Flag Act of 2005 (P.L. 109-243) and the FCC over-the-air-reception-devices rule, 47 C.F.R. §1.4000 (satellite antennas).

[ATTORNEY REVIEW REQUIRED] — §2.2 states the framework the outline assumes. Counsel must confirm every section number, the current text of the POA Act after the most recent General Assembly session, and whether any 2025–2026 amendment affects rental restrictions, fining procedure or reserve disclosure.

3. Article 4 — Housing for Older Persons (the 55+ program)

3.1 The exemption claimed, and the arithmetic

The community claims the "55 or older" exemption from the familial-status provisions of the Fair Housing Act under 42 U.S.C. §3607(b)(2)(C) and 24 CFR §100.304. Under 24 CFR §100.304(a)(2) that exemption requires the community to satisfy **all three** of §100.305 (80 percent occupancy), §100.306 (intent) and §100.307 (verification of occupancy). 24 CFR §100.304(b) confirms that a subdivision "governed by a common set of rules, regulations or restrictions" — here, the Declaration — is a "housing facility or community" for this purpose, and expressly includes property "governed by a homeowners' association".

Input	Value	Source
Dwelling units at full build-out	41	data/layout . json metrics.lots (2026-09-03 layout; the count fell from 43 when every lot was screened against all three digitised stream reaches and the basins were re-sized on 7.54 disturbed acres)
Statutory threshold	80 % of occupied units with at least one occupant 55 or older	24 CFR §100.305(a)
0.80×41	32.8 units	arithmetic
Fractional-unit rule — the fraction counts into the required number	round up	24 CFR §100.305(g)
Minimum qualifying units at full occupancy	33 of 41	derived (data/dev-summary-derived . json hopa_units_needing_55plus)
Cushion available for non-qualifying occupancy	8 units	$41 - 33$ (hopa_units_without_55plus_cap)

Input	Value	Source
Covenant cap on Board-granted Exception Occupancies (recommended)	6 units	leaves a 2-unit margin for error and for units in transition
New-construction grace	the 80 % test does not apply "until at least 25 percent of the units are occupied" — $25\% \times 41 = 10.25 \rightarrow$ 11 units	24 CFR §100.305(d)

What the drop from 43 units changed. The required number of qualifying units falls from 35 to **33**; the cushion of units that need not house a person 55 or older is **unchanged at 8**, because $0.80 \times 43 = 34.4 \rightarrow 35$ (cushion 8) and $0.80 \times 41 = 32.8 \rightarrow 33$ (cushion 8). The recommended Exception-Occupancy cap of six units therefore still leaves the same two-unit margin, and no covenant mechanic in this article needs to change — only the number stated in it.

Voluntary condition 1 was conformed on 2026-09-03 and now reads "**at least 33 of the 41 units** ... leaving at most 8 units that need not be", and requires the figure to be recomputed if the unit count changes (data/voluntary-conditions.md condition 1, generated from docs/02 §9). At the condition-2 ceiling of 47 units the figure becomes $0.80 \times 47 = 37.6 \rightarrow$ **38 of 47**, and the Declaration must state it that way — as a formula tied to the then-current unit count, not as a frozen number.

3.2 The occupancy covenant

[ATTORNEY REVIEW REQUIRED]. Outline of the operative provisions:

- 1. Community intent.** The Property is intended and operated as housing for persons 55 years of age or older within the meaning of 42 U.S.C. §3607(b)(2)(C) and 24 CFR §§100.304–100.308, and every Owner takes title subject to that intent.
 - 2. The rule stated affirmatively.** Every occupied Dwelling shall have at least one Occupant who is 55 years of age or older ("a Qualifying Occupant"). The covenant states 100 percent, not 80 percent; the 20 percent cushion is not a right of any Owner, it is administrative headroom held by the Association.
 - 3. Exception Occupancies.** The Board may grant a written, revocable Exception Occupancy for a Dwelling without a Qualifying Occupant only (a) while the number of Exception Occupancies plus vacant non-qualifying units is at or below the cap in §3.1 (six units at 41 dwellings), (b) for a stated term not exceeding twelve months, renewable, and (c) on written findings recorded in the minutes. The Board **shall** deny any request that would take the community below the §100.305(a) threshold, and no estoppel, waiver or course of dealing creates a right to renewal.
 - 4. Minimum age of other occupants.** No person under the age of 19 shall be a permanent Occupant.
- [ATTORNEY REVIEW REQUIRED]** — a community that qualifies under §100.304 is exempt from the familial-status provisions, and 24 CFR §100.306(d) confirms a qualifying community *may* nonetheless allow families with children; but the exemption does not touch the disability, race, colour, religion, sex, national origin or Georgia-law protections, and a flat minor-occupancy ban must be drafted with an express reasonable-accommodation carve-out and a guest policy (a suggested guest limit is 60 days in any twelve-month period).
- 5. No discrimination on any other ground**, and a statement that the Association will make reasonable accommodations and permit reasonable modifications under 42 U.S.C. §3604(f).

3.3 Who verifies, how often, and what record is kept

24 CFR §100.307(b) requires the community to "develop procedures for routinely determining the occupancy of each unit", including whether at least one occupant is 55 or older, and §100.307(c) requires those procedures to be updated "at least once every two years". The Declaration and the HOPA Policies should assign that work concretely:

Who	Role	Instrument
The Board of Directors	Adopts and annually re-adopts the HOPA Policies and Procedures; grants or denies Exception Occupancies; is the "authorized representative" that certifies in writing under 24 CFR §100.308(b)(2)–(3)	Declaration Art. 4; Bylaws
The HOPA Administrator — a named officer of the Association, in practice the community manager under the management agreement	Collects and files Occupancy Certificates and age documentation; maintains the Occupancy Register; runs the biennial survey; issues transfer eligibility letters; reports to the Board quarterly	HOPA Policies §2; management agreement scope
Counsel	Reviews the survey form, the affidavit and the published policies on a three-year cycle	Budget line, §8
The Owner	Delivers a completed Occupancy Certificate and age documentation at every closing, at every lease, and within 30 days of any change in occupancy	Declaration Art. 4

The record. The Association maintains an **Occupancy Register** — one row per Dwelling: unit address, lot number, the name of every Occupant, whether at least one Occupant is 55 or older, the date and type of the document relied on, the date of the most recent survey, and whether an Exception Occupancy is in force. From that register the Association derives, at each 31 March of an odd-numbered year, the **Occupancy Summary**: occupied units, qualifying units, the percentage, and the §100.305(a)/(g) minimum for the then-current unit count.

Acceptable documentation is that listed in 24 CFR §100.307(d): a driver's licence, birth certificate, passport, immigration card, military identification, "comparable official document", or "a certification in a lease, application, affidavit, or other document signed by any member of the household age 18 or older" asserting that at least one occupant is 55 or older. 24 CFR §100.307(e) makes any form adequate if it contains "specific information about current age or date of birth". The Association's practice should be to **record the document type and expiry, not to retain a photocopy of the identity document**, to limit its data-breach exposure — **[ATTORNEY REVIEW REQUIRED]**, because §100.307(a) also requires the community to be able to produce verification "in response to a complaint", and counsel must decide the retention rule that satisfies both.

Refusal to co-operate. 24 CFR §100.307(g) permits the community to rely on government records or public documents, prior forms or applications, or a statement signed by a person with personal knowledge, where an occupant refuses to complete the survey. The HOPA Policies should adopt that hierarchy verbatim rather than treating a refusal as a fatal gap.

Cycle. Occupancy Certificates at closing and at each lease; a 30-day change-of-occupancy report; the **Biennial Occupancy Survey mailed on or before 1 February and closed on 31 March of each odd-numbered year**, which satisfies §100.307(c) with a margin; the Occupancy Summary and the §100.308(b)(2) certification adopted at the annual meeting; the summary "available for inspection upon reasonable notice and request" as §100.307(i) requires; records retained for not less than seven years.

3.4 Published policies and demonstrated intent (24 CFR §100.306)

24 CFR §100.306(a) requires published policies and procedures that "demonstrate the intent" to operate as 55+ housing, and lists the factors HUD weighs: the manner in which the community is described to prospective residents (a)(1); advertising (a)(2); lease provisions (a)(3); "written rules, regulations, covenants, deed or other restrictions" (a)(4); the maintenance and consistent application of the procedures (a)(5); the actual practices of the community (a)(6); and "public posting in common areas of statements describing the facility or community as housing for persons 55 years of age or older" (a)(7). The Declaration should therefore require:

- The recorded Declaration itself (factor (a)(4)) — a plat note and a recital in every deed.
- A **posted statement** in the clubhouse and at the mail kiosk (factor (a)(7)), in the form set out in the HOPA Policies.
- **Marketing discipline:** every advertisement, sign, website page and brochure describes the community as "housing for persons 55 years of age or older" (factor (a)(2)). 24 CFR §100.306(b) states that phrases such as "adult living" or "adult community" are **inconsistent** with the intent to operate as 55+ housing; voluntary condition 1 already prohibits them. The Declaration should bind the declarant, the builder, every listing agent and the Association to that vocabulary. This is the single most common way communities lose the exemption, and it costs nothing to avoid.
- **Consistent application** (factors (a)(5) and (a)(6)): the Board "shall", not "may", enforce Article 4; a failure to enforce is itself a breach of the zoning condition and of the Declaration.

3.5 Transfer, resale and leasing

[ATTORNEY REVIEW REQUIRED]. Outline:

1. **Notice before closing.** A selling Owner gives the Association written notice not fewer than 30 days before closing, with the purchaser's name and the proposed Occupants.
2. **Purchaser's package.** The purchaser delivers a signed Occupancy Certificate, age documentation per §100.307(d), and an acknowledgement of Articles 4, 5, 7 and 8 (55+, buffer, sprinklers, architectural and parking rules).
3. **Association's response within 10 business days:** an **Occupancy Eligibility Letter** (or a written refusal stating the ground), issued together with the statement of amounts due that O.C.G.A. §44-3-232 requires within five business days of request.
4. **No right of first refusal, no approval of the purchaser.** The Association verifies *occupancy*, not the buyer; a right of first refusal would complicate lending and is not needed for HOPA.
5. **A closing that proceeds without the letter is a violation** enforceable against the Owner by fine,

injunction and lien, but **does not void the deed or affect the title** — a non-forfeiture clause is essential for title insurance and for secondary-market eligibility.

6. **Leasing.** Minimum term 12 months; no short-term or transient rental; the tenant household must include a Qualifying Occupant unless the Board grants an Exception Occupancy; the lease is subordinate to the Declaration and the Association may enforce directly against the tenant; a leased-unit cap (a suggested 15 percent — six Dwellings at 41, since $0.15 \times 41 = 6.15$) protects both HOPA administration and lender eligibility. **[ATTORNEY REVIEW REQUIRED]** — rental restrictions in Georgia are sensitive to how and when they are adopted; imposing them in the original declarant-drafted Declaration, before any lot is conveyed, is far safer than amending them in later.

3.6 Inherited units, surviving occupants and hardship

This is the question the City will ask and the question every buyer's family will ask.

Event	Rule the Declaration should adopt
The sole Qualifying Occupant dies or moves permanently, and a spouse or co-Occupant under 55 remains	The surviving co-Occupant, if he or she was in lawful occupancy on that date, may remain for life , counted as an Exception Occupancy against the cap in §3.1. This is the humane rule and the market expects it.
A Dwelling passes by will or intestacy to an heir under 55	The heir may own the Dwelling — the Declaration restricts occupancy, not ownership — but may not occupy it. The heir shall, within 12 months of the date of death (the Board may extend once by 6 months for good cause), either convey the Dwelling or lease it to a household with a Qualifying Occupant.
An Owner requires a live-in caregiver under 55	Permitted as a reasonable accommodation on written request with medical certification; the caregiver is not an Owner and acquires no occupancy right that survives the Owner.
Hardship (illness, deployment, job loss, a house that will not sell)	The Board may grant a time-limited Exception Occupancy within the cap, on written findings.
The cap is full	The Board shall refuse; the applicant's remedy is to sell or lease. The Declaration should say plainly that the 80 percent test is a community asset that no individual owner may consume.

The Association's obligation on learning of a death or a change in occupancy is to update the Occupancy Register within 30 days and to notify the Owner or personal representative in writing of the twelve-month clock. **[ATTORNEY REVIEW REQUIRED]** — the interaction of these rules with Georgia probate practice, with a surviving spouse's year's support (O.C.G.A. §53-3-1 et seq.), and with a lender in possession after foreclosure must be drafted expressly.

3.7 Violation and remedy

Notice of violation → 30-day cure period → hearing before the Board on request → daily fine at a rate fixed in the Bylaws → suspension of amenity privileges (never of access to the Dwelling or of the yard-care service the Owner pays for) → action for injunction and specific performance → costs and attorney's fees actually incurred → the O.C.G.A. §44-3-232 lien for the sums charged. For a violation that puts the exemption itself at risk — an occupancy that would take the community below 33 qualifying units — the Declaration should authorise the Association to seek **immediate injunctive relief without exhausting the fine schedule**. **[ATTORNEY REVIEW REQUIRED]** — fining authority and the required notice-and-hearing procedure must track the POA Act and the Bylaws exactly, or the fines are uncollectable.

3.8 The good-faith certification and the City's annual letter

24 CFR §100.308 gives a person a defence against personal liability for civil money damages where he had actual knowledge that the community "has, through an authorized representative, asserted in writing that it qualifies", and where the community has certified "in writing and under oath or affirmation" that it satisfies the 55+ requirements. The Declaration should therefore require:

- an **annual written certification under oath**, adopted by the Board and signed by the President as the authorized representative (§100.308(b)(3)), retained in the records and furnished on request to any Owner, purchaser, lender, listing agent or closing attorney; and
- an **annual HOPA certification letter to the City of Lilburn Planning Director**, on or before 30 April, attaching the Occupancy Summary for the most recent survey, so that the City can confirm voluntary condition 1 is being performed without having to ask. This costs the association one letter a year and is the cheapest possible answer to "how would the City ever know?".

4. Article 5 — the buffer easement (the instrument the lot yield rests on)

4.1 Why this article exists

Lilburn Zoning Ordinance 2023-603 **§313(1)** provides that buffer requirements "supersede these minimum required yards". Reading §313(1) together with Table 4.1 (cottage home: minimum lot size 3,000 SF, minimum lot width 50 ft, **minimum lot depth 100 ft, all uses**; 20-ft buffer abutting R-1 for dwelling types other than detached single-family) is what allows a 20-ft undisturbed buffer to lie **inside** the rear 20 ft of a lot instead of in a separate tract. The cross-section is:

*100.0 ft lot depth + 35.9 ft minimum lane tract + 100.0 ft lot depth = **235.9 ft** against a strip width of **236.0 ft** measured across the first lot row ($u \approx 255$ ft) — a margin of about **0.1 ft (1-1/4 in)**.*

(Lane tract width 35.9 ft minimum / 46.1 ft maximum from data/layout.json metrics.lane_tract_width_ft_min and _max; the 236.0-ft strip width is the cross-strip distance between the southwest and northeast boundary courses of data/layout.json boundary_ring at the mid-station of Lot 1, and the strip widens to about 241.3 ft at the last lot row. **The margin is tighter than the 0.6 ft this document reported at 43 lots** — the 2026-09-03 layout carries a 35.9-ft minimum lane tract rather than 35.4 ft. The RLS boundary survey governs, and if the survey returns a width below 235'-11" at any lot row the section must be re-cut before the plat. Decimal feet are used in this paragraph because that is how the dimension is carried in the layout data; the plat will state it in feet and inches.)

If the buffer cannot lie inside the lots, lot depth falls to roughly 82 ft, below the Table 4.1 minimum of 100 ft, and every lot needs a §1005 variance. **The buffer easement is therefore not a landscaping detail; it is the consideration the applicant offers for the §313(1) reading, and it must be drafted as if the whole 41-lot plan depends on it, because it does.**

Quantities: **41,000 SF (0.94 ac) of buffer easement inside lots** — 41 lots × 50'-0" × 20'-0" (data/layout.json metrics.buffer_easement_on_lots_sf) — plus the buffer where it runs through common tracts. This 41,000 SF is **not** counted in the **141,089 SF (3.24 ac; 33.8 percent of the 9.577-ac GIS area, 34.3 percent of the 9.44-ac deeded area)** of common open space shown on Sheet C-2.0; it is additional.

4.2 The grant

[ATTORNEY REVIEW REQUIRED]. Outline:

1. **Reservation, not later grant.** The Declarant **reserves** the Buffer Easement over the rear 20'-0" of each Lot at the time the Declaration is recorded, before any Lot is conveyed, so that no Owner ever takes title free of it and no joinder of Owners is ever needed.
2. **Area.** "The Buffer Easement Area" = that portion of each Lot lying within 20'-0" of the rear Lot line as shown and dimensioned on the recorded final plat, together with the buffer strips within the Common Property. The plat, not the Declaration's prose, fixes the geometry.
3. **Grantees.** The Association **and the City of Lilburn**, the City taking as a grantee and third-party beneficiary with an **independent right of enforcement** that does not depend on the Association acting first, and that survives any failure, dissolution or insolvency of the Association.
4. **Character.** Perpetual; running with the land; binding on and benefiting successors; ****not extinguished by merger**** of the dominant and servient estates; not subject to abandonment by non-use; not terminable by the expiration of any covenant term (see O.C.G.A. §44-3-234, §2.2 above).
5. **Form question — [ATTORNEY REVIEW REQUIRED].** Counsel must choose between (a) a ****restrictive covenant** and easement in gross in favour of the City, **and (b) a conservation easement** under the Georgia Uniform Conservation Easement Act, O.C.G.A. §44-10-1 to §44-10-8, **which requires a qualified holder (a governmental body or a charitable organisation — the City can qualify) and carries its own consequences for enforcement, tax treatment, amendment and the holder's monitoring duty. Option (a) is the lighter instrument and is what voluntary condition 4 contemplates ("a recorded buffer/conservation easement"); option (b) is harder to dissolve and reads more strongly to a hearing body. Do not leave the phrase "buffer/conservation easement" undefined in a recorded document**** — the two are different instruments.

4.3 Prohibitions

Within the Buffer Easement Area, no Owner, Occupant, guest, contractor, the Association or the Declarant shall:

1. cut, clear, mow to turf, girdle, poison, top or remove any tree, shrub or understory vegetation, except as §4.4 permits;
2. grade, fill, excavate, stockpile, or alter drainage or existing contours;
3. erect, place or maintain any **structure** — including a shed, greenhouse, playset, gazebo, pergola, deck, patio, pool, hot tub, firepit, HVAC condenser, generator, satellite antenna, clothesline, flagpole footing or dog run;
4. erect or maintain any **fence**, wall or hedge planted as a fence;

5. place any **impervious surface**, walkway, paver, gravel or artificial turf;
6. store vehicles, boats, trailers, building materials, firewood, compost or yard waste;
7. discharge roof drains, sump pumps, irrigation or pool water into the area;
8. apply herbicide (other than targeted treatment of invasive species under an approved plan), or dump grass clippings, leaves or soil;
9. install lighting other than the shielded fixtures approved under voluntary condition 11;
10. take any other action inconsistent with the "undisturbed" character the buffer must have to satisfy Lilburn Zoning Ordinance 2023-603 Table 4.1 and §313(1).

4.4 What is permitted

- **Approved pond and utility encroachments** not exceeding 50 ft in perpendicular width, as voluntary condition 4 and the Site Development Plan Review Checklist allow, as shown on the approved plans.
- **The single buffer reduction disclosed in the Letter of Intent §5** — the southwest curb return of the entrance, where the 20'-0" buffer is reduced by 2'-0", to a residual width of **18'-0"**, at the return's tangent point on the Arcade Road right-of-way line. The plan area of the encroachment is **0.1 SF** (data/layout.json metrics.sw_curb_return_buffer_encroachment_ft = 2.0 and metrics.sw_curb_return_buffer_encroachment_sf = 0.1; 15'-0" southwest return radius). **This supersedes the 17'-11" this document previously carried.** Voluntary condition 4 was conformed on 2026-09-03 and now states the same figure — "the buffer narrows to a minimum of 18'-0" instead of 20'-0", a reduction of 2.0 ft measured along the right-of-way affecting 0.1 SF of pavement" — so the recorded easement, the zoning condition and the drawing will say one thing. **Letter of Intent §5 must be checked against that wording before filing;** a document and a drawing that state different residual widths is exactly the contradiction the completeness audit called a filing defect. No other reduction is authorised, and the Declaration should say so, so that the recorded instrument and the zoning conditions cannot drift apart.
- **Removal of a dead, diseased or hazardous tree** on the written opinion of an ISA-certified arborist, with replacement at not less than one tree for one within the same growing season.
- **Invasive-species management** under a written plan approved by the Board (English ivy, privet, kudzu, wisteria, autumn olive), by hand or by targeted treatment, with no soil disturbance.
- **Supplemental evergreen planting** to thicken the screen where existing vegetation is sparse (voluntary condition 4), at the rate used for the buffer concept: approximately **one evergreen per 8–10 lineal ft of mapped gap plus one canopy tree per 30–40 lineal ft** (docs/12 §32 93 00 ¶2.03).
- **Access by the Association, the City and utility providers** for inspection, maintenance and emergency work.

4.5 Marking, plat notes and disclosure

- **Permanent buffer markers** — cast or engraved posts reading "BUFFER EASEMENT — DO NOT DISTURB — CITY OF LILBURN" — set by the Declarant at every point where the buffer line meets a side Lot line and at intervals not exceeding 50 ft, and maintained by the Association in perpetuity. Markers are what make the covenant enforceable against a new owner with a chainsaw who never read the Declaration.
- **Plat note** on the final plat, dimensioning the 20'-0" easement on every Lot and reciting the Declaration book and page and the zoning case number.
- **Deed recital** in every deed from the Declarant and a covenant requiring the same recital in every subsequent deed.
- **Resale disclosure:** the Buffer Easement is described in the Occupancy Eligibility Letter package (§3.5) and in the Association's resale certificate, with a plat excerpt for the specific Lot.

4.6 Enforcement and the re-planting remedy

[ATTORNEY REVIEW REQUIRED]. The remedy, not the prohibition, is what a land-use attorney will test:

1. **Inspection.** The Association shall inspect the entire Buffer Easement Area not less than **annually** and after any land-disturbing activity on a Lot, and shall photograph and log each inspection.
2. **Notice and cure.** Written notice of violation to the Owner; **10 days** to stop active disturbance;

90 days to complete restoration.

3. **Restoration standard** (stated in the Declaration so that no negotiation is needed later): replant to a density of not less than **one canopy tree of 2-in caliper minimum per 400 SF disturbed** and **one evergreen of 6-ft minimum height per 8 lineal ft of disturbed buffer frontage**, restore the ground layer and any altered grade, and warrant the replacement plantings for **24 months** (the Site Development Plan Review Checklist §6.m requires a 12-month warranty of new planting; the covenant doubles it because this planting is a remedy, not an installation).

4. **Self-help.** If the Owner does not perform, the Association **shall** enter, perform the work, and charge the cost, plus 15 percent administration and attorney's fees actually incurred, to the Owner as a **specific assessment** secured by the O.C.G.A. §44-3-232 lien.

5. **The City's independent remedy.** The City may enforce the easement directly against the Owner and against the Association — by citation for violation of the zoning condition, by withholding permits and certificates of occupancy on the affected Lot, and by action for injunctive relief — without first demanding that the Association act.

6. **No waiver.** Failure to enforce in any instance is not a waiver as to any other, and no violation is cured by lapse of time.

7. **Reserve funding.** The Landscape renovation and buffer re-planting component in the reserve schedule (§8.3) is the fund that stands behind items 3 and 4 when the violator is unknown or a storm is the cause.

4.7 Amendment lock

No amendment, release, relocation, subordination or termination of Article 5, or of the corresponding plat easement, shall be effective without the **prior written consent of the City of Lilburn**, and any purported amendment without that consent is void. The same lock applies to Article 4 (55+), Article 7 (sprinklers) and the provisions of Articles 6 and 8 that carry a zoning condition. **[ATTORNEY REVIEW REQUIRED]** — the lock must be drafted so that it binds a future 67 percent supermajority of owners under the POA Act amendment provisions, which is precisely the case the City is worried about.

4.8 If the City requires the buffer in a separate tract

The fallback is already identified in FACTS.md §3: 50'-0" × 82'-0" lots (4,100 SF, above the 3,000-SF Table 4.1 minimum) with a concurrent §1005 variance on lot depth. The Declaration should be drafted so that this conversion is possible without re-platting every Lot — a reserved right in the Declarant, during the period of Declarant control, to convey the Buffer Easement Area into a Common Property tract and to record a corresponding plat revision, at the Declarant's cost. **[ATTORNEY REVIEW REQUIRED]**.

4.9 Stream buffer (voluntary condition 9)

The stream head and its 50-ft undisturbed buffer and 75-ft impervious setback sit in a **20,000-SF common "creek woods" tract** owned by the Association, not in any Lot. The Declaration should carry the same prohibition list as §4.3, the same City third-party enforcement, and the plat should show the 25-ft state buffer under O.C.G.A. §12-7-6(b)(15) and the 50-ft and 75-ft city lines separately, as voluntary condition 9 requires. No Owner has any right to clear, mow, garden or build in the creek woods.

5. Article 6 — common property, the private lane, stormwater and landscape

5.1 What the Association owns and maintains

Common Property tract	Area	Association obligation
Lane tract (private street, 22'-0" pavement in a 35.9–46.1-ft tract, widening to 60'-0" through the amenity block)	71,085 SF	Own in fee; maintain pavement, curb, walk, signs, striping, lighting, drainage
Front amenity tract (clubhouse, village green, 2 pickleball courts, guest parking bay, mail-kiosk bay, mail kiosk, entry sign, 10-ft landscape strip)	46,777 SF	Own, operate, insure, maintain
Pocket green (u 530–580)	10,000 SF	Own, mow, maintain
Pocket green (u 1,080–1,130)	10,000 SF	Own, mow, maintain
Terminus green + rear buffer	9,312 SF	Own, mow, maintain
Pond 1 tract (dry detention / water quality)	20,000 SF	Own; inspect, mow, clean, remove sediment
Pond 2 tract (dry detention / water quality)	20,000 SF	Own; inspect, mow, clean, remove sediment

Common Property tract	Area	Association obligation
Creek woods (stream head + 50/75-ft buffers), preserved	20,000 SF	Own; leave undisturbed; invasive management only
Stream-setback open-space tract (SW side, u 1,230–1,280) — unlotted, no impervious	5,000 SF	Own; leave undisturbed. New tract: this is the lot that was removed on 2026-09-03 when every lot was screened against all three digitised stream reaches
Common open space total	141,089 SF = 3.24 ac = 33.8 % of the 9.577-ac GIS area (34.3 % of 9.44 ac dedeed)	the eight tracts enumerated above sum exactly to the published metric (data/layout.json open_space_summary), so the 201-SF residual this document reported at 43 lots is closed
Buffer easement within the 41 Lots (additional, not counted above)	41,000 SF	Enforce Art. 5; inspect annually
Yard maintenance on 41 Lots (voluntary condition 10)	≈ 1,294 SF of maintained yard per Lot (Plan A 1,365 SF / Plan B 1,220 SF)	Mow, edge, trim, blow, prune, mulch, leaf removal

(Areas from data/layout.json open_space_tracts and metrics; per-Lot maintained yard = 5,000 SF lot – per-Lot impervious from data/plans.json lot_siting – 1,000 SF of buffer easement.)

The Declaration must grant the Association an **easement over each Lot** for the yard-care service, for inspection of the buffer and the sprinkler system, and for maintenance of anything the Owner fails to maintain, on reasonable notice except in an emergency.

5.2 The private lane — no dedication, and the standard

- The lane is **private in perpetuity**. The plat carries the note the Site Development Plan Review Checklist §4.o and §4.s require — new streets labelled "Private", public and private maintenance responsibilities noted on the concept, preliminary and final plats — and a statement that the City and Gwinnett County accept no maintenance obligation.

- Lilburn Zoning Ordinance 2023-603 **§319** requires every lot to abut at least 30 ft on a public street "or an approved private street", and the checklist requires that "all components of private streets and alleys must meet minimum standards for public street". **The Association's maintenance covenant is what keeps 41 lots legally served.** A failure to maintain is not merely a nuisance; it undermines the lot frontage.

- Maintenance standard to be stated in the Declaration: keep the pavement in good repair; crack-seal on a cycle not exceeding **7 years**; resurface on a cycle not exceeding **20 years** or when the pavement condition index falls below 70, whichever is first; maintain 13'-6" of vertical clearance by pruning limbs over the travelled way; keep the three hammerhead turnarounds and the entrance clear at all times; maintain "NO PARKING — FIRE LANE" signs on both sides (IFC 2024 (GA) Appendix D103.6) and the authority to tow.

- No gates** (voluntary condition 6); no speed humps or other obstruction that would impede fire apparatus without the written concurrence of the Gwinnett County Fire Marshal.

5.3 Sidewalks

14,619 SF of 5-ft walk in the lane tract and on the entry drive (data/layout.json metrics.sidewalk_sf; a walk on the northeast side of the whole lane, a second walk on the southwest side from u = 530 ft where the strip first gives that half-section its full 11 + 2 + 5 = 18 ft), plus the Arcade Road frontage walk (voluntary condition 18). Maintenance: repair any vertical displacement of 1/4 in or more; maintain cross-slope; keep clear of vegetation; replace panels lifted by street-tree roots (the reserve schedule funds 25 percent panel replacement per 20-year cycle). The community is a 55+ community and the walks are its principal amenity; the standard should be written as an accessibility standard, not a cosmetic one.

5.4 Stormwater — the ponds, the recorded agreement and the sediment cycle

- Ownership.** Both ponds sit on their own tracts owned by the Association, as the Site Development Plan Review Checklist §11.i.1 requires ("Locate detention pond on its own parcel owned by developer or a property owner's association", citing Development Regulations 8.2.4.g).

- The recorded agreement.** The **Storm Water BMP Maintenance Agreement** — with as-built hydrology, volume certification, maintenance schedule, access easement, inspection report and an 18-month BMP maintenance bond — is recorded with the Clerk of Superior Court by the applicant (checklist §2.g.3). The Declaration must (a) recite it by book and page, (b) obligate the Association to perform it, and (c) prohibit any amendment of the Declaration that would leave it unperformed.

- Access.** The 30-ft BMP access easement for a residential project, with a 15-ft graded access road at not

more than 20 percent grade, kept cleared and grubbed (checklist §11.i.2 and .4), is a permanent Association obligation — including the obligation not to plant it closed.

- **Inspection cycle.** A written inspection by a qualified professional **annually**, and after any storm producing more than 2 in of rainfall in 24 hours; the report is filed with the City if the City asks and retained for not less than seven years. Budget line, §8.2: 2 ponds × \$900/yr.
- **Sediment cycle.** Forebay hand-cleaning **annually to every 5 years**, and whenever accumulated sediment occupies more than 50 percent of the forebay volume; **full sediment removal from the pond bottom when accumulated sediment reduces the design storage volume by 10 percent, and in any event on a cycle not exceeding 10 years.** The basins were enlarged on 2026-09-03, when the disturbed area was recomputed at **7.54 acres** (it had been sized on 4.4): design storage is now **33,129 cf (Pond 1, 52 × 180 ft basin)** and **44,734 cf (Pond 2, 64 × 180 ft basin)** — **77,863 cf total**, at 6-ft depth and 3:1 side slopes (data/layout.json ponds, stormwater) — so the 10-percent trigger is roughly **3,300 cf** of sediment in Pond 1 and **4,500 cf** in Pond 2. Against a screening requirement of **75,410 cf** (7.541 disturbed ac × the Gwinnett County Stormwater Management Manual rate of 10,000 cf per disturbed acre, with the runoff-reduction volume taken as zero at concept stage), **77,863 cf provided is 103 percent of required**; if the 15,552-cf RRv credit in data/dev-summary-derived.json is allowed the requirement falls to 59,858 cf. The design PE's GCSWMM routing governs. The reserve schedule funds sediment removal at \$18,000 per pond on a 10-year cycle, and the enlargement raises the volume the association will one day have to dredge by about 76 percent over the 44,218 cf this document carried at 43 lots.
- **Receiving water.** Discharge is to the existing swale and stream corridor only, never onto an adjoining yard (docs/03 §6). The receiving stream is the Jackson Creek headwaters (GAR030701030315), on Georgia's 2024 §303(d) list, which is why the covenant should also prohibit the Association and Owners from discharging pool water, pressure-washing effluent or fertiliser-laden runoff into the buffer or the ponds.

5.5 Landscape, street trees and yard care

- **Yard care on every Lot** (voluntary condition 10): mowing, edging, trimming and blowing on the schedule in §8.2; shrub pruning, pre-emergent and mulch twice a year; leaf removal. The Owner keeps the right to plant within ACC guidelines but not to interfere with the service or to alter irrigation.
- **Street trees** at 40 ft on centre — **73 street trees** at full build-out, plus **10 trees and 10 shrubs** in the 10-ft landscape strip along the 237.62-ft Arcade Road frontage (checklist §6.j: one tree and one shrub per 25 lineal ft; $237.62 \div 25 = 9.5 \rightarrow 10$) and **2 parking-bay trees** (checklist §6.k: one tree per seven spaces, 12 spaces $\rightarrow 2$). **85 trees in total** (tools/hoa_budget.py, from data/layout.json). **The street-tree count rose with the lane, not with the lot count** — it is measured along the travelled way, which is unchanged at 1,754 ft: one row at 40 ft on centre for the full length ($1,754 \div 40 = 43$) plus a second row over the reach that carries a walk on both sides, which now begins at $u = 530$ ft rather than the $u \approx 1,160$ assumed at 43 lots ($(1,754 - 530) \div 40 = 30$). $43 + 30 = 73$ **street trees**, against 57 before (data/layout.json lane.widen_both_sidewalks_from_u = 530.0).
- **Warranty and replacement.** New planting is warranted by the Owner or contractor for 12 months from the certificate of occupancy, or a maintenance bond is provided (checklist §6.m); after the warranty the Association carries a replacement allowance (§8.2, 3 percent of the tree stock per year).
- **No removal.** No Owner may remove or top a street tree or a common-area tree; replacement is at the Association's direction and at the Owner's cost where the Owner caused the loss.

5.6 The reserve study requirement

The Declaration should **require** — not merely permit — that the Association:

1. obtain a **reserve study before the first conveyance of a Lot**, prepared by an independent reserve analyst, listing every component, its replacement cost, useful life and remaining life;
2. **update it at least every three years**, and after any casualty or major capital work;
3. **fund reserves at not less than the straight-line contribution** the current study computes, which is the method used in §8.3;
4. keep reserves in a **segregated account**, not available for operating expenses without a Board resolution recorded in the minutes and repayment within 24 months; and
5. disclose the reserve balance, the funding percentage and the study date in every resale certificate.

Georgia does not compel a property owners' association to fund reserves. That is exactly why the covenant must, and why a City that is being asked to rely on a private association to maintain a 1,754-ft street and two detention ponds for the next half-century should want to see this paragraph.

5.7 Utilities and lighting

Full-cutoff fixtures at 3000 K or less, mounted at not more than 14 ft with house-side shields facing the adjoining R-1 lots, and **no lighting on the pickleball courts** (voluntary condition 11). The Association pays the energy and maintains the fixtures (§8.2: 16 fixtures). **VERIFY** whether the fixtures are association-owned or supplied on the Georgia Power outdoor-lighting tariff; the answer moves one operating line and one reserve component.

5.8 Sanitary sewer — what the Association must *not* own

Voluntary condition 16 forecloses the option that would otherwise have been the association's largest and least predictable expense: "no private pump station, private force main, private gravity sewer, grinder pump or low-pressure sewer system shall be proposed or built", following the Gwinnett County Department of Water Resources *Standard Policy for Private Developments: Condominiums, Townhomes and Subdivisions* (rev. 9/2018), which allows private facilities "only ... for commercial properties under single ownership within a development". The Declaration should carry that prohibition forward as a covenant, so that a future board cannot solve a service problem by taking a pump station onto the association's books. Every dwelling connects by gravity to a Gwinnett County main, and the county owns and maintains it.

The consequence for the budget is stated plainly in §8.5: this decision removes an O&M exposure that the 2026-08-29 adversarial review priced at **\$30,000–\$60,000 per year** — but it also creates the Phase-2 risk that §8.5 prices, because if the off-site gravity easement and the DWR capacity certification are not obtained, Phase 2 is not built and the association is 24 lots carrying the fixed costs of a 41-lot community.

6. Article 7 — the NFPA 13D sprinkler covenant

6.1 What the sprinklers are, in code terms

State the posture precisely, because it changed on 2026-09-03 when the voluntary conditions were revised, and the package audit at `audit-2026-09-03/completeness.md` item M12 still describes the older position:

- At 41 dwellings the two-access requirement of **IFC 2024 Appendix D107.1**, as amended by

Ga. Comp. R. & Regs. 120-3-3-.04 (which raises the trigger to more than 120 dwelling units), **is not engaged**. The sprinklers are therefore **not** a required mitigation for the single point of access.

- The lane is a **1,754-ft dead end**, which under **IFC 2024 (GA) Appendix D103.4** (dead ends over 750 ft)

requires the fire code official's special approval. The mitigation package presented for that approval is the one in `docs/08 Memo E`: three 120-ft hammerhead turnarounds, a 22-ft clear lane with enforced no parking, hydrant spacing, and — offered voluntarily — sprinklers in every home.

- The sprinklers also halve the required fire flow under **IFC 2024 Appendix B §B105.1** (500 gpm with sprinklers against 1,000 gpm without), which matters on a ≈1,754-ft dead-end water main.

- Voluntary condition 5 offers the sprinklers **voluntarily** and provides a release: if the Gwinnett County

Fire Marshal states in writing that they are unnecessary, the condition may be released by the Fire Marshal's written concurrence and City staff's approval without a further public hearing.

The design **appears consistent with** IFC 2024 (GA) Appendix D only if the Fire Marshal accepts the D103.4 special-approval package; nothing here asserts otherwise.

6.2 The covenant

[ATTORNEY REVIEW REQUIRED]. Outline:

1. **Installation.** Every Dwelling shall be equipped with an automatic residential sprinkler system designed and installed to **NFPA 13D**; the clubhouse is sprinklered as the building code requires.
2. **Non-removal.** No Owner or Occupant shall remove, disable, disconnect, cap, paint over, obstruct or render inoperative any sprinkler head, pipe, control valve, gauge or alarm; shall not close the control valve except for repair; and shall not modify the system except by a licensed contractor with prior ACC consent and a copy of the revised hydraulic calculation delivered to the Association.

3. **Maintenance.** The Owner shall maintain and test the system in accordance with NFPA 13D and the manufacturer's instructions, keep the space around each head clear, maintain heat in every space containing piping, and repair any damage promptly at the Owner's cost.
4. **Verification.** The Owner shall deliver a signed sprinkler-condition certificate to the Association **every two years**, on the same cycle as the biennial HOPA occupancy survey so that one mailing serves both. The Association has a right of entry on 48 hours' notice to inspect, and in an emergency without notice.
5. **Disclosure at resale.** The system, this Article and the maintenance obligation are disclosed in the resale package under §3.5.
6. **Remedy.** Notice, 30-day cure, self-help repair by the Association at the Owner's cost as a specific assessment secured by the O.C.G.A. §44-3-232 lien, and injunctive relief. A disabled system is a life-safety violation and should not be left to the fine schedule alone.
7. **Amendment lock and release.** Article 7 may not be amended or deleted except in the manner voluntary condition 5 provides — the Fire Marshal's written concurrence plus City staff approval — and never by owner vote alone.
8. **Insurance credit.** The Association should require Owners to notify their carriers; the sprinklers also support the association's own property and liability rating (§8.2).

7. Article 8 — architectural control, use and parking

7.1 The Architectural Control Committee

Three members, appointed by the Declarant during the period of Declarant control and by the Board thereafter; written application; decision within 30 days; **no application is deemed approved by the passage of time** — silence is a denial with a right of appeal to the Board — so that an inadvertent lapse can never authorise a second storey or a structure in the buffer. **[ATTORNEY REVIEW REQUIRED]**. Written Architectural and Landscape Guidelines, amendable by the Board, subordinate to the Declaration and to the zoning conditions.

7.2 The three colour schemes and the distribution rule

Voluntary condition 8 fixes the palette; docs/12 §09 91 13 ¶2.04 and data/plans.json materials_by_scheme fix the content:

Scheme	Field	Base	Trim	Roof
1 — Warm White	fiber-cement board-and-batten, warm white	stacked/ledge stone water table (30 in) and porch piers	soft white; 8×8 cedar posts and brackets	architectural asphalt shingle, weathered wood / charcoal
2 — Sage	fiber-cement lap, 6-in exposure, sage green	brick veneer water table (30 in), red-brown blend	soft white; 8×8 cedar posts	architectural asphalt shingle, charcoal
3 — Clay	fiber-cement lap, 6-in exposure, clay/beige	ledge stone water table, buff/tan	cream	architectural asphalt shingle, driftwood/brown

Distribution rule. No scheme shall be used on more than **40 percent of the Dwellings** — at 41 Dwellings that is $0.40 \times 41 = 16.4$, i.e. **16 Lots**; the distribution drawn in docs/09 §4.2 puts the largest scheme on 14 — and **no two adjacent Lots** and **no two Lots directly opposite one another across the lane** shall carry the same scheme. The scheme is assigned at closing, recorded in the Association's records, and may be changed only to another approved scheme with ACC consent and repainting of the whole exterior. The Declarant delivers one quart of each colour per cottage to the Association as touch-up stock (docs/12 §09 91 13 ¶2.05). Repaint cycle 8–10 years (docs/12 §09 91 13 ¶2.03).

7.3 Massing and additions

- **One story only.** No second storey, no habitable attic, no dormer or roof alteration creating habitable space above the first floor, no increase in ridge height. Maximum ridge 24'-0" above finished grade (voluntary condition 3); as drawn on Sheets A-2.1 and A-2.2, Plan A "The Springbrook" is **17'-10" above the finished floor / 18'-6" above finished grade** and Plan B "The Laurel" is **19'-0" / 19'-8"** to the main gable ridge, and the clubhouse on Sheet A-3.0 is **20'-10" / 21'-6"** (finished floor 0'-8" above grade; the ridge heights on the plan sheets govern).
- **No accessory dwelling unit** of any kind — no garage apartment, basement apartment, guest cottage or separately rentable space, whether or not separately metered.
- **Detached accessory structures:** not more than one per Lot, not exceeding 120 SF, not exceeding 10 ft in

height, set back at least 5'-0" from any Lot line (Site Development Plan Review Checklist §4.m), with a 5-ft landscape strip (checklist §6.l), **and never within the Buffer Easement Area** (§4.3).

- **Garage recess preserved.** The front wall of the garage shall remain recessed ****not less than 5'-0"**

behind the front wall of the dwelling**, as Lilburn Zoning Ordinance 2023-603 Table 4.2 requires for front-loaded garages. No Owner shall enclose, convert or extend the garage forward, and the garage shall remain available for the parking of two vehicles. Garage doors shall be kept closed except when in use.

- **Front porch** not less than 6'-0" deep, covered, retained (voluntary condition 8); zero-step entries retained.

- **No wall signs on any dwelling** (voluntary condition 8). The community's only sign is the single monument entry sign, ≤ 32 SF and ≤ 6'-0" high, externally lit, permitted separately (voluntary condition 14).

- Solar collectors: permitted only on rear-facing roof planes, flush-mounted, with ACC consent.

[ATTORNEY REVIEW REQUIRED] — VERIFY whether any Georgia statute limits an association's authority over solar installations before the guideline is written as a prohibition.

- Antennas and flags: the guidelines must accommodate 47 C.F.R. §1.4000 (over-the-air reception devices) and the Freedom to Display the American Flag Act of 2005.

7.4 Parking — the rule the fire code depends on

- **No parking on the lane at any time.** The lane pavement is 22'-0" wide; IFC 2024 (GA) Appendix D103.6 requires "NO PARKING — FIRE LANE" signs on both sides of a 20–26-ft fire apparatus access road, and the 22-ft width is only defensible to the Fire Marshal if it is genuinely kept clear. The Association shall maintain the signs and **shall** enforce, including by towing at the vehicle owner's expense after posted notice. **[ATTORNEY REVIEW REQUIRED]** — Georgia's private-property towing requirements (signage content, placement, notice and the towing company's licensing) must be satisfied or the tow is unlawful.

- **On-Lot parking.** Two garage spaces plus two driveway spaces per Lot; the garage door stands **26'-8"** from the lot line so that two cars park on the driveway without overhanging the sidewalk (`data/plans.json` `lot_siting.garage_door_from_lot_line_ft` = 26.67). Total on-Lot parking **164 spaces** (4 × 41) against the **82** required by Lilburn Zoning Ordinance 2023-603 Table 8.1 (2 per dwelling unit) — `data/layout.json` `metrics.parking_required` and `parking_provided_on_lot`.

- **Guest parking.** 12 spaces at the amenity block (8 guest + 4 at the mail kiosk); a 72-hour limit, no storage parking, no overnight commercial vehicles.

- **Prohibited:** boats, recreational vehicles, trailers, commercial vehicles over 3/4 ton or bearing commercial lettering, inoperable or unregistered vehicles, and vehicle repair other than emergency repair, anywhere on a Lot outside a closed garage.

7.5 Other use restrictions to be drafted

Residential use only; no home occupation with customer or employee traffic; leasing rules per §3.5; refuse containers stored inside the garage except on collection day; no clothes lines visible from the lane; pets limited in number and under leash, with waste rules; nuisance, noise and light provisions; no fences on any Lot except a decorative fence not exceeding 4'-0" in height approved by the ACC and never within the Buffer Easement Area (checklist §4.n); holiday decoration timing; and the construction-period rules that carry voluntary condition 12 (exterior construction 7:00 AM–7:00 PM Monday–Friday, 8:00 AM–5:00 PM Saturday, none on Sundays or federal holidays).

8. Article 9 — assessments, and the annual budget

8.1 Basis and method

This is the answer to the question the package could not answer before: **what does this association actually cost to run?** The budget below is built from the bottom up. Every quantity is read at run time from `data/layout.json` and `data/plans.json` by `tools/hoa_budget.py`; every unit price is a named planning allowance with the benchmark it came from; the reserve contribution is computed component by component from replacement cost divided by useful life. Nothing is a percentage of a guess.

Method notes.

- **Straight-line (component) reserve funding**, with **no** interest earnings and **no** inflation offset

assumed. That is deliberately conservative: interest on the reserve balance would reduce the required contribution, and inflation would increase it, and the two are left to cancel until a real reserve analyst models them (\$5.6).

- **5 percent operating contingency**, applied to the operating subtotal only, because this is a first-year budget for a community that does not yet exist and no line has a quotation behind it.
- **2 percent delinquency allowance**, applied to the sum of operating, contingency and reserves. The O.C.G.A. §44-3-232 lien makes eventual recovery likely but the budget cannot spend a receivable.
- Costs are **2026 dollars at full build-out**, all 41 Dwellings complete and occupied. §8.4 handles the build-out years and §8.5 the downside cases.
- Every line is marked **VERIFY**: none of these prices is a quotation. Three bids per contract before the first budget is adopted.

8.2 Quantities the budget is built on

Reproduced verbatim from `verify/hoa-budget-output.md`, the raw output of `tools/hoa_budget.py`, re-run 2026-09-03 against the 41-lot layout.

Quantity	Value	Source
Lots	41	layout.json metrics.lots
Plan mix	21 A / 20 B	layout.json lots
Lane length (entrance to terminus)	1,754 ft	metrics.lane_length_ft
Lane + entry pavement	48,430 SF	metrics.pavement_sf (polygons compute 48,430 SF)
Sidewalk	14,619 SF	metrics.sidewalk_sf (polygons compute 14,619 SF)
Lane tract, less pavement and walk (verge)	8,036 SF	tract – pavement – sidewalk
Front amenity tract	46,777 SF, of which 8,262 SF built/paved	metrics + amenity polygons
Greens (3 tracts)	29,312 SF	layout.json greens
Pond tracts	40,000 SF (2 ponds, 20,970 SF of basin)	open_space_tracts, ponds
Creek woods (preserved)	20,000 SF	open_space_tracts
Buffer easement inside lots	41,000 SF	metrics.buffer_easement_on_lots_sf
Maintained common landscape	115,863 SF = 2.66 ac	verge + amenity soft + greens + pond tracts
Undisturbed natural area	61,000 SF = 1.40 ac	creek woods + buffer easement
Maintained yard per lot	Plan A 1,365 SF / Plan B 1,220 SF (avg 1,294 SF)	5,000 SF – plans.json lot_siting impervious – 1,000 SF buffer
Street trees / total trees	73 / 85	40 ft o.c. + frontage strip + parking bay
Light fixtures	16	150 ft o.c. + 4 at the amenity block (assumption)

8.3 Line-item annual budget — full build-out, 41 lots

Line	Quantity (from data/layout.json / data/plans.json)	Unit cost	Annual
1 Grounds			
Lot yard care — mow, edge, trim, blow	41 lots × 34 visits	\$22/lot/visit	\$30,668
Lot beds — pruning, pre-emergent, mulch, leaf removal	41 lots × 2 visits	\$120/lot/yr	\$4,920
Common-area turf and beds (lane verge, greens, amenity, pond tracts)	2.66 ac	\$4,500/ac/yr	\$11,970
Buffer easement and creek-woods stewardship (undisturbed areas)	1.40 ac	\$900/ac/yr	\$1,260
Street- and strip-tree pruning (3-year cycle, annualised)	85 trees ÷ 3	\$65/tree	\$1,842
Tree, shrub and turf replacement (3 % of stock per year)	85 trees × 3 %	\$450/tree	\$1,148
Irrigation operation, backflow testing and repair (amenity block)	1 system	\$1,400/yr	\$1,400
2 Lane and stormwater			
Lane sweeping and gutter cleaning	2 events/yr	\$650/event	\$1,300

Line	Quantity (from data/layout.json / data/plans.json)	Unit cost	Annual
Pavement crack fill and pothole patching (operating)	48,430 SF	\$0.05/SF/yr	\$2,422
Fire-lane and traffic signs, pavement markings	lane inventory	\$900/yr	\$900
Storm inlet, pipe and outlet-structure cleaning	1 vacuum-truck day	\$1,400/yr	\$1,400
Pond annual inspection and report to the City	2 ponds	\$900/pond/yr	\$1,800
Pond routine maintenance (racks, forebay, rip-rap, erosion)	2 ponds	\$1,300/pond/yr	\$2,600
Lane and common lighting — energy and fixture maintenance	16 fixtures × 12 mo	\$23/fixture/mo	\$4,416
Snow and ice event allowance	0.5 event/yr	\$600/yr	\$600
3 Amenity			
Clubhouse electricity	2,400 SF	\$1.85/SF/yr	\$4,440
Clubhouse water and sewer, amenity irrigation water	1 meter	\$2,600/yr	\$2,600
Clubhouse gas	1 service	\$700/yr	\$700
Clubhouse janitorial and supplies	104 visits/yr	\$65/visit	\$6,760
Clubhouse repairs, HVAC contract, pest, alarm and NFPA 13 sprinkler ITM	1 building	\$3,900/yr	\$3,900
Internet, cameras, access control, Knox box	1 building	\$1,700/yr	\$1,700
Pickleball court cleaning, nets and screens	2 courts	\$900/yr	\$900
Mail kiosk cleaning, lighting and locks	1 kiosk	\$500/yr	\$500
4 Administration			
Professional community management	41 doors × 12 mo	\$22/door/mo	\$10,824
HOPA age-verification program (biennial survey, transfer certificates, records)	41 units ÷ 2 yr + 3 transfers	\$18/unit + \$150/transfer	\$1,719
Insurance — property, GL, D&O, crime, umbrella	41 units + clubhouse	\$13,500/yr	\$13,500
Financial review and Form 1120-H	1/yr	\$1,600/yr	\$1,600
Legal — counsel and covenant enforcement	allowance	\$2,500/yr	\$2,500
Reserve study (update every 3 years, annualised)	1 ÷ 3 yr	\$3,500/study	\$1,167
Banking, software, portal, postage, meetings, state registration	1/yr	\$1,900/yr	\$1,900
Operating subtotal			\$123,356
Operating contingency, 5 %			\$6,168
Reserve contribution (schedule below)			\$40,451
Delinquency allowance, 2 % of assessments			\$3,399
TOTAL ANNUAL BUDGET			\$173,374
Per lot per year (41 lots)			\$4,229
Per lot per month			\$352

Reserve component schedule (the reserve contribution above is the sum of the annual column):

Component	Qty	Unit	Unit cost	Replac ement cost	Life (yr)	Annual	Basis
Lane and entry-drive asphalt — mill and overlay	48,430	SF	\$4	\$193,720	20	\$9,686	overlay \$3.50–5.50/SF and mill-and-overlay \$4.50–8.00/SF in 2026 (Buildcalchub; Wright Construction; Paving Estimator); \$4.00 assumes a single mobilisation for the whole lane
Lane crack sealing and surface treatment	48,430	SF	\$0.30	\$14,529	7	\$2,076	crack seal + fog/seal coat between overlays
Concrete sidewalk — 25 % panel replacement per cycle	3,655	SF	\$9	\$32,895	20	\$1,645	4-in walk removal and replacement; 25 % of panels per 20-year cycle (root heave, settlement, ADA cross-slope corrections)
Curb, gutter and valley-gutter repair allowance	1,754	LF	\$4	\$7,016	25	\$281	10 % of the curb line replaced per 25-year cycle at ~\$40/LF

Component	Qty	Unit	Unit cost	Replacement cost	Life (yr)	Annual	Basis
Detention pond sediment removal	2	pond	\$18,000	\$36,000	10	\$3,600	dry basins of 0.24 ac each holding 77,863 cf between them (2026-09-03 re-sizing); Southeast stormwater dredging \$25–75/CY and \$25,000–75,000 for a half-acre HOA pond (Angi 2026; AUE Land; Landshore 2026) — the allowance predates the enlargement and is to be re-priced before the first reserve study
Pond outlet structures, trash racks, rip-rap, emergency spillway	2	pond	\$12,000	\$24,000	30	\$800	outlet control structure and headwall rehabilitation
Storm sewer pipe and structure rehabilitation	1,754	LF of lane	\$68.50	\$120,149	50	\$2,403	lane storm drainage — pipe, inlets and junction boxes at end of service life, priced per foot of lane so the Phase-1 case scales
Clubhouse roof — architectural asphalt shingles	3,000	SF	\$6.50	\$19,500	25	\$780	clubhouse footprint × 1.25 slope factor; tear-off and replacement
Clubhouse HVAC (2 split systems)	2	each	\$11,000	\$22,000	15	\$1,467	3-ton systems, replacement including duct repairs
Clubhouse exterior repaint	1	LS	\$9,000	\$9,000	8	\$1,125	docs/12 \$09 91 13 2.03 sets the HOA repaint cycle at 8–10 years
Clubhouse interior — flooring, paint, kitchen, restrooms, furnishings	1	LS	\$60,000	\$60,000	15	\$4,000	\$25/SF of clubhouse for finishes and FF&E
Clubhouse water heater, appliances, fitness equipment	1	LS	\$14,000	\$14,000	12	\$1,167	allowance
Pickleball court resurfacing	2	court	\$8,000	\$16,000	6	\$2,667	acrylic resurface of a 30' × 60' pad; 5–8-year cycle
Pickleball fencing	360	LF	\$28	\$10,080	20	\$504	4-ft black vinyl-coated chain link on the pad perimeter
Lane and amenity light poles and fixtures	16	each	\$3,800	\$60,800	25	\$2,432	14-ft full-cutoff post-top ≤3000 K (voluntary condition 11); assumes association-owned fixtures — if they are on the Georgia Power tariff this component drops out
Mail kiosk — cluster box units and shelter	1	LS	\$28,000	\$28,000	25	\$1,120	4 USPS-approved CBUs at ~\$2,600 plus a 16' × 10' shelter (checklist \$4.r, 'protection from the elements')
Monument entry sign, wall and landscape lighting	1	LS	\$28,000	\$28,000	20	\$1,400	≤32-SF monument sign (voluntary condition 14), masonry base, external lighting
Site furnishings — benches, tables, trash receptacles, kiosk board	1	LS	\$16,000	\$16,000	15	\$1,067	village green and pocket greens
Irrigation system replacement (amenity block)	1	LS	\$18,000	\$18,000	20	\$900	controller, valves, heads and mainline
Landscape renovation and buffer re-planting fund	1	LS	\$20,000	\$20,000	15	\$1,333	periodic renovation of common beds and the supplemental evergreen buffer planting; also the fund behind the re-planting remedy in Article 6 of the declaration
Total annual reserve contribution						\$40,451	straight-line component method, no interest or inflation offset

Optional line not included in the total above. A bundled residential solid-waste contract — one hauler, one truck a week on a private lane, instead of 41 separate subscriptions — would cost about **\$22.00 per home per month = \$10,824 per year**, taking dues to **\$374 per lot per month**. Whether to buy refuse collection as a community is a Board decision, and **VERIFY** whether the City's residential collection arrangement serves private streets at all.

8.4 What each Owner pays, and how the money is collected

Item	Amount	Note
Annual assessment per Lot	\$4,229	\$173,374 ÷ 41 Lots; every Lot is 5,000 SF, so an equal per-Lot share is the defensible allocation

Item	Amount	Note
Monthly assessment per Lot	\$352	\$4,229 ÷ 12, of which \$250.72 operating (\$123,356 ÷ 41 ÷ 12), \$82.22 reserves (\$40,451 ÷ 41 ÷ 12), \$12.54 contingency (\$6,168 ÷ 41 ÷ 12) and \$6.91 delinquency allowance (\$3,399 ÷ 41 ÷ 12)
Initial capital contribution at each closing, including resales	2 × the monthly assessment = \$704	paid to reserves and working capital; not an advance of dues
Special assessment	Board may levy up to one monthly assessment per year without a vote; anything larger requires member approval	[ATTORNEY REVIEW REQUIRED] — the threshold and the vote must track the POA Act
Late charge, interest, costs and attorney's fees actually incurred	per the Bylaws	O.C.G.A. §44-3-232
Statement of amounts due on request	within 5 business days , or the lien is extinguished as to the requesting party	O.C.G.A. §44-3-232

What the drop from 43 lots to 41 did to the dues. The total annual budget fell only **\$931**, from \$174,305 to **\$173,374** — 0.53 percent — while the number of Lots paying it fell 4.65 percent. The arithmetic, line by line:

Component	43 lots	41 lots	Change	Why
Operating — the four per-Lot lines (yard care, lot beds, management, HOPA administration)	\$50,413	\$48,131	–\$2,282	these are the only operating lines that scale with the Lot count
Operating — everything else	\$74,607	\$75,225	+\$618	tree pruning and replacement (69 → 85 trees, +\$564) and pavement crack fill (46,466 → 48,430 SF, +\$99), less buffer stewardship (1.45 → 1.40 ac, –\$45)
Operating subtotal	\$125,020	\$123,356	–\$1,664	
Reserve contribution	\$39,616	\$40,451	+\$835	asphalt overlay and crack seal on 1,964 SF more pavement (+\$478) and sidewalk panel replacement on 3,655 SF of walk instead of 2,860 (+\$358)
Contingency (5 %) and delinquency allowance (2 %)	\$9,669	\$9,567	–\$102	percentages of the lines above
Total annual budget	\$174,305	\$173,374	–\$931	

The denominator fell 4.65 percent and the numerator fell 0.53 percent, so the dues rose \$14, from \$338 to \$352 per Lot per month — a 4.1 percent increase, and \$4,229 instead of \$4,054 a year. This is the plainest statement of what the two lost lots cost, and it should be disclosed rather than buried: two fewer homes now carry a slightly longer-walked, slightly wider-paved lane and two basins that hold 77,863 cf instead of 44,218. (The enlarged basins do not themselves raise the budget — sediment removal is priced per pond, not per cubic foot — but they will cost more to dredge than the \$18,000-per-pond allowance assumes, which is open item 12 in §10.)

Build-out years. Until all 41 Lots are conveyed the assessments do not cover the budget, because the common property is complete long before the last home sells. Two structures are conventional and the Declaration must pick one — **[ATTORNEY REVIEW REQUIRED]**:

- **(a) Declarant deficit funding (recommended).** The Declarant pays the difference between assessments actually collected and actual expenses, month by month, until the last Lot is conveyed or Declarant control ends. Simple, transparent, and it protects the reserve schedule from day one.
- **(b) Reduced assessment on unsold Lots.** The Declarant pays 25 percent of the full assessment on Lots it owns. Cheaper for the Declarant, but it starves reserves in exactly the years when the asphalt is quietly aging, and buyers' lenders scrutinise it.

Whichever is chosen, the **reserve contribution should be funded at 100 percent from the first month**, and the Declarant should deliver the initial reserve study before the first closing (§5.6).

8.5 Sensitivities the Board and the City should see

Case	Annual budget	Per Lot per month	Why
Full build-out, 41 Lots — the published figure	\$173,374	\$352	§8.3

Case	Annual budget	Per Lot per month	Why
Phase 2 not built (voluntary condition 16 not satisfied) — 24 Lots carrying the amenity, the entry, one pond and the front 1,000 ft of lane	\$129,291	\$449	the same generator run at phase1; Phase 1 is Lots 1–10 and 15–28, Phase 2 the remaining 17
Full build-out with bundled refuse collection	\$184,198	\$374	\$8.3 optional line
Pavilion and green instead of a 2,400-SF clubhouse	≈ \$143,000–\$151,000	≈ \$291–\$307	the clubhouse carries \$20,100 of operating cost (electricity \$4,440 + water and sewer \$2,600 + gas \$700 + janitorial \$6,760 + repairs, HVAC, pest, alarm and sprinkler ITM \$3,900 + internet, cameras and access control \$1,700) and \$8,539 of reserve funding (roof \$780 + HVAC \$1,467 + repaint \$1,125 + interior and FF&E \$4,000 + water heater, appliances and fitness \$1,167); a pavilion removes most but not all of it. Hand-computed from \$8.3, not a generator run. A design decision, not a budgeting one — and it is the cheapest single lever on the dues figure

The **Phase-2 case is the important one**. Voluntary condition 16 is honest about it — if the off-site gravity easement and the Gwinnett DWR capacity certification are not obtained, Phase 2 is not built. The association then has 24 Dwellings paying for a clubhouse, two pickleball courts, an entry drive, a monument sign, a mail kiosk and a detention pond that were all sized for 41. **\$449 per month is a marketable number for a 55+ cottage only if it is disclosed before the first contract, not discovered after**. The Declaration and the sales documents should carry the two-column budget, not one.

8.6 How this compares, and what the package should now carry

Community	Dues	Comparability
Annsbury Park, Lilburn (140 R-2 cottage lots)	≈ \$104/mo	no yard care , public streets, minimal amenity
Cottages at Noble Village, Lilburn (55+)	\$93–\$280/mo	Lilburn's only 55+ product; smaller amenity
Saddlebrook, Snellville (55+, ≈130 homes)	\$192–\$257/mo	detached 55+, yard care, larger community to spread fixed costs
The Cottages at Arcado Springs (41 Lots)	\$352/mo	yard care on every Lot, a 1,754-ft private street the association owns and repaves, two detention ponds, a clubhouse, HOPA administration
Soleil Summit Chase, Snellville (55+, 265 homes)	\$360–\$389/mo	full amenity, but 265 homes carrying it

The figure is where the arithmetic says it should be: above the communities with public streets or no yard care, below the large-amenity community, and — the point a 41-lot community cannot escape — **high per home because 41 homes is a small denominator for a private street and two ponds**. The 2026-09-03 loss of two lots made that worse, not better, by \$14 a month (\$8.4).

Reconciliation with the rest of the package.

Source	Figure	Disposition
docs/02-develop ment - summary .md §8.2, as first written	"Estimated dues \$200–\$350/month (VERIFY by budget)"	Superseded . A placeholder carrying an explicit instruction to replace it with a budget. This is that budget.
docs/02-develop ment - summary .md §8.2, as revised 2026-09-03	the \$200–\$350 range is " not supported by a budget"; a 1,754-ft private lane, two ponds, a 2,400-SF clubhouse, yard care for every Lot, insurance, management and the HOPA programme are "more consistent with \$300–\$400 per lot per month "; "a line-item budget is to be published in the HOA/HOPA covenant outline rather than a range"	Answered . The revision was right to retire the range and right about the band. The line-item figure it asks for is \$352 . docs/02 §8.2 should now read " \$352 per Lot per month at full build-out (41 Lots), 2026 dollars — \$173,374 per year, computed line by line in docs/13 \$8 by tools/hoa_budget .py; \$449 per month if Phase 2 is not built. "
2026-08-29 adversarial review, s tatus/plan - judg ment - 2026-08-29 /judges/cottage - plan - critic .md	"≈ \$200k/yr = \$300–400/lot/mo"	Confirmed in band, refined . The bottom-up total is \$173,374, about 13 percent below the reviewer's order-of-magnitude \$200k, and \$352 sits inside the \$300–400 range. The reviewer's underlying point stands: a \$352 assessment on a cottage priced at \$420,000–\$460,000 (data/dev-summary-derived.json price_lo / price_hi) is roughly \$1,000 a month of carrying cost before taxes and insurance, and it is a real constraint on absorption.
docs/10 - submitt al - checklist - and - roadmap .md	cites this section rather than restating a number	Correct as written; keep it that way.

One figure, everywhere: \$352 per Lot per month at full build-out (41 Lots); \$449 if Phase 2 is not built. Any document in this package that states a dues range should be corrected to that pair, with the year (2026 dollars) attached, because a dues figure

without a date is a promise no one can keep.

9. Recording sequence, declarant control and transition

9.1 Sequence

Step	What is recorded or filed	When	Why in this order
1	Nothing. Declaration outline (this document) and draft HOPA Policies accompany the Letter of Intent	With the rezoning application	Shows the City the enforcement mechanism exists before it votes
2	Zoning conditions adopted by the Mayor and City Council	At the Council hearing	The conditions are the source; the covenants implement them
3	Title consolidation. The four parcels are conveyed to a single Declarant entity — PINs R6123 033 and 015 (Awad) and R6123 014 and 162 (Mendez and Roblero de Leon)	Before the land disturbance permit	A declaration can only be recorded by every record owner, or by one owner after the assemblage closes. This is a gate, not a formality.
4	Articles of Incorporation filed with the Georgia Secretary of State	Before the Declaration	The Association must exist to take the Common Property
5	Final plat , creating 41 Lots and the Common Property tracts, dimensioning the 20'-0" buffer easement on every Lot, labelling the lane "Private", showing the BMP access easement, the stream buffers and the maintenance-responsibility note (checklist §4.o, §4.s)	After LDP approval	The Declaration describes Lots by plat reference, so the plat is recorded first
6	Declaration of Covenants, Conditions, Restrictions and Easements , with the Bylaws attached, submitting the property to the POA Act, and any mortgagee's Consent and Subordination	Immediately after the plat — and, per the voluntary condition, before the first building permit and before the first Lot is conveyed	Every Owner then takes title subject to Articles 4, 5, 7 and 8; no joinder is ever required
7	Storm Water BMP Maintenance Agreement with as-built hydrology, volume certification, maintenance schedule, access easement, inspection report and 18-month bond, recorded with the Clerk of Superior Court	At the Certificate of Development Conformance stage — "at least 3 weeks prior to Final Plat or 1st Certificate of Occupancy" (checklist §2.g)	The Declaration recites it by book and page
8	Off-site sanitary sewer easement for the Phase-2 gravity connection	Before the Phase-2 land disturbance permit (voluntary condition 16)	Phase 2 cannot proceed without it
9	Supplemental Declaration annexing Phase 2 (or, if condition 16 is not satisfied, no annexation and the Phase-2 area stays wooded or is re-platted as open space)	At the Phase-2 final plat	The Declaration must reserve the annexation right and must work if the right is never exercised
10	Deeds, each reciting the Buffer Easement and the 55+ restriction	At each closing	Factor (a)(4) of 24 CFR §100.306

9.2 Who the Declarant is

[ATTORNEY REVIEW REQUIRED] and VERIFY. As of 2026-09-03 title is split: R6123 033 and R6123 015 are held by Awad, and R6123 014 and R6123 162 by Mendez and Roblero de Leon. The Declaration must be executed by a single Declarant that holds all four parcels, or by all record owners jointly. The cleanest structure is a Georgia limited liability company formed to take title at the assemblage closing, which then becomes Declarant. The rezoning application itself faces the same signature problem (docs/07), and it should be solved once, for both.

9.3 Declarant control and transition

- **Class B membership** in the Declarant, terminating on the **earliest** of: (a) conveyance of **75 percent** of the Lots to Owners other than the Declarant; (b) **7 years** after the first Lot is conveyed; or (c) the Declarant's written election. **[ATTORNEY REVIEW REQUIRED]** — the trigger must also work in the Phase-2 downside case, where 41 Lots may never exist; express the percentage against Lots *created*, not Lots *planned*.
- **Turnover package** delivered at transition: minute books, the Occupancy Register and every HOPA record, the reserve study and reserve account, as-built drawings and the recorded BMP Maintenance Agreement, all warranties and bonds, the sprinkler documentation for every Dwelling, contracts, insurance policies, and a **transition financial review** by an independent accountant at the Declarant's cost.
- **Declarant covenant.** The Declarant remains liable for deficits accrued during its control and for the completion of the Common Property, and those obligations survive transition.

9.4 What the City should make a condition, and what it should leave to the covenants

Make it a zoning condition (City-enforced, amendable only by the Council)	Leave it to the covenants (Association-enforced, amendable by owners)
55+ operation under 42 U.S.C. §3607(b)(2)(C) and 24 CFR §§100.304–100.308, with the annual letter to the Planning Director	The survey form, the affidavit wording, the fine schedule, the guest policy
Maximum 47 Dwellings; detached single-family only	Lot-by-lot plan assignment
One story, ridge ≤ 24 ft	Interior finishes, landscaping choices
The 20-ft buffer, held in a recorded easement naming the City, with the City's consent required for any amendment	The re-planting species list, the invasive-management plan
The single disclosed buffer reduction at the southwest curb return, and no other	—
NFPA 13D in every Dwelling, releasable only on the Fire Marshal's written concurrence	The biennial owner certificate mechanics
One entrance, no gate, no Arcade Road driveways	Guest-parking time limits, towing contractor
Stream buffer preserved in a common tract	Trail or seating in the greens
Private lane and all common property maintained by a mandatory association that may not be dissolved without the City's consent	Dues, budget, reserve percentage, contractor selection
Gravity sewer only, no private pumping; capacity certification before Phase 2	—
Recording of the Declaration before the first building permit	Everything else

The distinction that matters: **the City should hold the promises it would want to enforce if the association failed**, and nothing else. Conditions that reach into the association's housekeeping are unenforceable in practice and give the Council a false sense of control.

10. Attorney redline checklist and open items

For counsel, in order of consequence.

- 1. Submit the property to the POA Act** (O.C.G.A. §44-3-220 et seq.) expressly, and confirm that O.C.G.A. §44-3-234 gives the 55+ restriction and the buffer easement perpetual duration notwithstanding O.C.G.A. §44-5-60. If this is wrong, everything else is decoration.
- 2. Choose the buffer instrument** — restrictive covenant and easement in gross, or a conservation easement under O.C.G.A. §44-10-1 to §44-10-8 — and draft the City's third-party enforcement right and the amendment lock so that they survive a future 67 percent owner vote (§4.2, §4.7).
- 3. Draft the HOPA package to the four sections:** §100.304 (the exemption), §100.305 (80 percent, the fractional-unit rule, the 25-percent new-construction start), §100.306 (published policies, posting, advertising vocabulary), §100.307 (procedures, documentation, the two-year update, the summary available for inspection). Confirm the Georgia Fair Housing Act analogue (O.C.G.A. §8-3-200 et seq.) adds nothing further.
- 4. The under-19 permanent-occupancy rule** and its reasonable-accommodation carve-out (§3.2 item 4).
- 5. Inheritance, surviving spouse and the twelve-month sell-or-lease clock** against Georgia probate practice and year's support (§3.6).
- 6. Leasing restrictions** adopted in the original Declaration rather than by later amendment (§3.5 item 6).
- 7. Fining and towing procedure** — notice, hearing, and Georgia's private-property towing requirements (§3.7, §7.4).
- 8. Non-forfeiture clause** so that a transfer in breach of Article 4 is a personal violation and never a title defect (§3.5 item 5).
- 9. Mortgagee protections** and secondary-market eligibility (Fannie Mae, FHA and VA project standards for an age-restricted PUD), which interact with the leasing cap, the declarant-control period and the reserve funding level.
- 10. Declarant identity and joinder** (§9.2) — the four-parcel title split is a gate for both the application and the Declaration.
- 11. The Phase-2 contingency** written into Articles 2, 6 and 9 so that a 24-Lot community is governable and the budget in §8.5 is disclosed to buyers.

12. Reserve covenant (§5.6) — a Georgia association is not required to fund reserves; this Declaration must require it.

Open items — VERIFY before recording.

#	Item	Who resolves it
1	Which O.C.G.A. section carries the Georgia Fair Housing Act older-persons exemption, and whether its terms differ from 24 CFR Part 100 Subpart E	Counsel
2	Whether the lane lighting is association-owned or on the Georgia Power outdoor-lighting tariff (moves one operating line and one reserve component)	Declarant + Georgia Power
3	Whether the City's residential solid-waste arrangement serves private streets	City of Lilburn
4	Gwinnett DWR confirmation that the Phase-2 gravity route is certifiable (voluntary condition 16)	Declarant + DWR
5	The Gwinnett County Fire Marshal's written position on the IFC 2024 (GA) Appendix D103.4 special approval, and whether the voluntary sprinkler condition is released	Declarant + Fire Marshal
6	Every unit price in §8 replaced by three written quotations before the first budget is adopted	Declarant + manager
7	The initial reserve study, by an independent analyst, before the first conveyance	Declarant
8	Insurance quotations — the \$13,500 allowance is a benchmark midpoint, not a bound quote	Declarant's broker
9	Confirmation that voluntary condition 5's sprinkler release mechanic is acceptable to the City Attorney as a condition that can be lifted administratively	City Attorney
10	Whether the City will accept the annual HOPA certification letter (§3.8) as the reporting mechanism for condition 1	Planning Director
11	CLOSED 2026-09-03. data/voluntary-conditions.md (generated from docs/02 §9) was conformed to the 41-lot layout: condition 1 now reads 33 of 41 with a cushion of 8 (§3.1), condition 2 "41 shown", condition 4 the 18'-0" residual buffer (§4.4), condition 5 "at 41 units", condition 8 the settled sheet register, condition 9 the new stream-setback tract, and condition 15 the u ≈ 980-ft phase line with 24 and 17 lots. Re-read the conditions against this outline before the Declaration is drafted — this document's Article 4, 5, 7 and 8 outlines are written to the conformed text	Owner + counsel
12	Pond sediment-removal allowance. The reserve schedule prices dredging at \$18,000 per pond, a figure set when the basins held 44,218 cf between them. They now hold 77,863 cf at 6-ft depth (§5.4). Re-price before the first reserve study is adopted	Declarant + reserve analyst
13	Whether data/layout.json metrics.pavement_sf (48,430 SF) should be published as the pavement quantity for reserve purposes, given that it includes the 2,070 SF of guest and mail-kiosk bays and the 184 SF of entrance curb returns as well as the lane, entry drive and hammerhead legs	Declarant + PE

11. Status

DRAFT — NOT SEALED. This outline is owner-prepared. It is not legal advice, it is not a declaration, and no part of it should be recorded. The Declaration, Articles, Bylaws and HOPA Policies must be drafted, and this outline redlined, by a Georgia land-use or community-association attorney; the reserve study must be prepared by an independent reserve analyst; the stormwater maintenance schedule must be sealed by a Georgia Professional Engineer; and every unit price in §8 must be replaced by a quotation before the first annual budget is adopted.

***Disclaimer:** This is an AI-generated analysis for preliminary planning purposes. All findings must be verified by a licensed professional before use in design, permitting, or regulatory submissions.*

title	Site Context Brief — Cottages at Arcade Springs
date	2026-08-28
address	4535, 4537, 4539 & 4541 Arcade Rd SW, Lilburn, GA 30047 (Gwinnett County; PINs R6123 033 / 015 / 014 / 162; Land Lot 123, 6th District; approx. 33.87985, -84.11968)
skill	site-planner (as:environmental-analysis, as:mobility-analysis, as:demographics-analysis, as:site-history)
status	DRAFT — preliminary planning context; not a sealed document

Site Context Brief — Cottages at Arcade Springs

Project: R-2 rezoning (City of Lilburn, Ord. 2023-603 §402) for a ~44-home, one-story, 55+ (HOPA) detached cottage community on the 9.44-acre four-parcel Arcade Road assemblage. **Purpose of this document:** the site analysis that should inform the first design conversation — climate, hazards, soils, access, transit, the people who live here, the neighbors, and the planning climate — before zoning envelope, programming or pricing. It does not analyze the zoning envelope (see FACTS.md §3 and the ordinance excerpts) and it does not re-price the project (see the 2026-08-25 ledger). **How to read it:** FACTS.md (2026-08-28) is the single source of truth for geometry, topography, the stream, sewer inverts, zoning text and program decisions; nothing here contradicts it. Every number below carries a source and a vintage; items that could not be sourced are marked **Not available** rather than estimated. Straight-line distances are labeled as such; drive distances are free-flow OSRM routing. Four sub-analyses were run in parallel on 2026-08-28 (environmental, mobility, demographics, site history) and are integrated in §7–8.

1. Site identity

The Arcade Springs assemblage is a 9.44-acre, four-parcel "bowling alley" — 236 ft wide at Arcade Rd, 246 ft at the rear, roughly 1,740 ft deep — that runs N28.7°W off a 13,800–15,300-vpd county collector between two very different neighbors: the 1970–71 brick split-levels of King David Manor (no HOA; 2026 FMV median \$349,200) on its northeast line and the 2007–2016 stone-and-shingle estates of Legends at Parkview (HOA; median \$693,500) on its southwest line, with 1970s Nantucket closing the rear. Two houses (1972 and 1987) sit at the front; the rear two-thirds is Piedmont hardwood on eroded Appling–Pacolet clays over gneiss, falling northeast-to-southwest across a mid-strip swale, a 952–958-ft ridge, and a 921-ft pocket where an unnamed headwater of Jackson Creek reaches 30 ft inside the property line. It is Zone X, Seismic Design Category B, Radon Zone 1 and Climate Zone 3A — an ordinary metro-Atlanta building site with one extraordinary dimension: it is a single-frontage strip in a city whose 2024 plan and 2026 draft plan both say, in effect, "single-family only south of the tracks, and match what is there." The site sits inside one of Lilburn's oldest and most owner-occupied census tracts (504.61: 35.5 % aged 55+, 90 % owner-occupied, 67 % of homes built before 1980 — ACS 2020–2024) and is one of the few large infill parcels left on this stretch of Arcade Rd. What makes it distinct is the pairing: a physically constrained sliver that can only ever be one lane of houses, next to a resident population that has no one-story alternative within the city limits — the single 55+ product ever built in Lilburn (Cottages at Noble Village) sold out.

2. Key numbers at a glance

Metric	Value	Source	Vintage
Assemblage	9.44 ac deeded (9.58 GIS); 236–246 ft × ~1,740 ft; only frontage Arcade Rd, ≈237 ft	FACTS §1 (Gwinnett GIS)	2026-08-28
Zoning / plan	R-1 → requested R-2; 2024 Comp Plan "Established Residential"; 2026 draft plan almost certainly "Suburban-Low" — single-family cottages "Appropriate"; "Do not further extend sewer in this area"	FACTS §3; 2026 CPA draft pp. 51–54, 72–73	2026-06-16 draft
Prior case	RZ-2025-01 (MU, 104 condos) tabled 2025-08-28, never heard → no §1003-11 12-month bar apparent	PC minutes	2025-08-28
Flood / seismic / radon	Zone X (panel 13135C0114F, 2006-09-29); SDC B; EPA Radon Zone 1	FEMA NFHL; USGS; GA Radon Program	live / 2023 / 2024
Climate	CZ 3A; 49.7 in rain; HDD 2,767 / CDD 1,882; design DB 21.7 °F / 93.7 °F; dew point 74.2 °F	NCEI 1991–2020; ASHRAE 2021	—
Soils	HSG B Appling/Pacolet/Gwinnett over most of the site; HYB = Helena (HSG D, wet) at the front SW corner; ARE = Ashlar (rock 22–40 in) under both pond sites ; no hydric units	NRCS SDA GA135	2025-09-02
Receiving water	Jackson Creek headwaters — 303(d) Not Supporting (bacteria, bio; urban runoff)	GA EPD 2024 305(b)/303(d)	2024
Walk / Bike Score	5 / 18	Walk Score	2026-08-28

Metric	Value	Source	Vintage
Transit	Ride Gwinnett Route 25 stop at Killian Hill @ Arcado, 0.30 mi ; inside $\frac{3}{4}$ -mi ADA paratransit band; no Sunday service; no microtransit in Lilburn	Ride Gwinnett GTFS 2026-07-02	2026
Arcado Rd	Minor collector, 35 mph, 14,800 AADT (2017) ; left-turn lane not warranted (< 75 lots); ISD 390 ft; access spacing 244 ft (Arcadia Place opposite the NE frontage end)	GDOT via ARC; Gwinnett DOT 2007; UDO \$900	2017 / 2007 / 2014
Fire / EMS	First-due Station 3, 2.9 mi / 6 min; 1,700-ft dead end > IFC D103.4's 750 ft	Gwinnett GIS; OSRM	2026-08-28
Tract 504.61	35.5 % aged 55+; 25.7 % 65+; 90.1 % owner-occupied; 66.9 % built pre-1980; median value \$301,500 (wide MOE)	ACS 2020–2024	2024
ZCTA 30047	70,121 pop; 25.8 % 55+; median HH income \$88,652; 76.0 % owner-occupied; median value \$367,300; ≈8,500 owner-householders 55+	ACS 2020–2024	2024
Lilburn city	15,390 pop (16,369 est. 2025); 25.0 % 55+; 48.2 % Hispanic; 5.43 mills proposed 2026 + new \$40k senior exemption	ACS 2020–2024; PEP; City of Lilburn	2024–2026
Adjoining values	Legends at Parkview 14 abutting lots FMV \$579,100–\$894,600 (median \$693,500); King David Manor 13 lots \$292,800–\$412,500 (median \$349,200); Nantucket 3 lots \$308,400–\$417,000	Gwinnett 2026 digest	2026-01-01
Comparable 55+	Cottages at Noble Village (Lilburn, 1,225–1,450 sf) sold out; resales \$342,500–\$414,900; Soleil Summit Chase (Snellville) mid-\$500s; Saddlebrook ≈\$447k	listing sites	2026-08-28
Market	Lilburn ZHVI \$401,137 (–2.6 % YoY, 2026-06); Redfin Lilburn median \$380,000 (2026-03); FMLS Atlanta detached 4.4 months supply (2026-07)	Zillow; Redfin; FMLS	2026
Neighbor opposition (2025)	Petition 1,025 signatures; six objections — traffic, capacity, air/noise, stormwater , character, spot-zoning	Change.org	2025-08
Site houses / basis	4541: 1972; 4535: 1987 (4,267 sf); 2026 FMV all four \$1,267,600; recorded purchases 2023–24 total \$1,572,500	Gwinnett 2026 digest; listings	2026

3. Environment and climate

Verified geometry, topography, stream, floodplain and sewer facts are in FACTS.md §1–2 and are not restated except where extended. Climate normals are for Atlanta Hartsfield (USW00013874, ~24 mi SW); the nearest Gwinnett station (Norcross, USC00096407, ~7 mi WNW) publishes precipitation normals only.

3.1 Climate (NOAA/NCEI 1991–2020 U.S. Climate Normals; ASHRAE 2021 Handbook of Fundamentals)

Metric	Value	Source	Vintage / accessed
Annual mean max / min / avg temperature	71.9 / 53.2 / 62.6 °F	NCEI normals, USW00013874	1991–2020; accessed 2026-08-28
Annual precipitation	49.71 in (Atlanta); 54.73 in (Norcross)	NCEI normals (Norcross USC00096407)	1991–2020
Wettest / driest months	Jul 5.27 in / Apr 3.36 in — no dry season	NCEI monthly normals	1991–2020
Annual snowfall	2.9 in	NCEI monthly normals	1991–2020
Heating / cooling degree days (base 65)	2,767 / 1,882 (NCEI); 2,578 / 1,969 (ASHRAE)	NCEI; ASHRAE station sheet 722190	1991–2020; ASHRAE POR 1994–2019
July / January mean max–min	89.1–71.3 °F / 52.3–34.3 °F	NCEI monthly normals	1991–2020
ASHRAE heating design dry-bulb	99.6 %: 21.7 °F; 99 %: 26.4 °F	ASHRAE 722190 sheet	2021 HOF
ASHRAE cooling design DB / MCWB	0.4 %: 93.7 / 73.8 °F; 1 %: 91.6 / 73.6 °F; dew point 0.4 %: 74.2 °F	ASHRAE 722190 sheet	2021 HOF

Metric	Value	Source	Vintage / accessed
Record high / low (airport, GHCN-Daily)	106 °F (2012-06-30) / -8 °F (1985-01-21)	NCEI daily summaries	1930–2026-08-24
Climate zone	IECC / ASHRAE 3A (Gwinnett County); Köppen Cfa	ASHRAE 169-2013 Addendum a county table	2020
Prevailing wind	NW Nov–May (27–32 % of hours Dec–Feb); W Jun–Jul; E Aug–Oct; mean 8.6 mph; 6.6 % calm	NCEI hourly normals, wind	1991–2020
Solar noon altitude (computed: 90° – 33.88° ± 23.44°)	Jun 21: 79.6°; equinox: 56.1°; Dec 21: 32.7°	arithmetic	computed
Sunrise / sunset azimuth (computed)	Jun 21: 61° / 299° (14.3 h day); Dec 21: 119° / 241° (9.7 h)	arithmetic	computed

Orientation reading. The lane runs N28.7°W. Houses on the NE side of the lane face SW (front azimuth ≈241°); houses on the SW side face NE (≈61°). In June the sun sets at azimuth 299° and is still at 25° altitude at 5 pm solar time, so the **SW-facing front porches of the NE-side lots and the SW-facing rear patios of the SW-side lots take the hot western sun**; a roof overhang cannot block sun that low. NE-facing fronts (SW-side lots) get morning sun and shaded afternoons — the better porch orientation. Winter noon sun at 32.7° gives useful passive gain to every SW/SE wall.

3.2 Hazards

Hazard	Finding	Source	Vintage/ accessed
Flood	Zone X — Area of Minimal Flood Hazard ; FIRM panel 13135C0114F , effective 2006-09-29 . Confirms FACTS §2. The headwater stream is not a FEMA-mapped watercourse; the local 50 + 25-ft buffer, not a BFE, governs the rear pocket.	FEMA NFHL layer 28 query; panel layer 3	NFHL live; accessed 2026-08-28
Seismic (ASCE 7-22, Risk Cat. II, Site Class D assumed)	SS 0.25 g, S1 0.094 g, SDS 0.21 g, SD1 0.13 g → Seismic Design Category B ; PGA_M 0.12 g	USGS design-values service	2023 NSHM; accessed 2026-08-28
Radon	Gwinnett is one of four Georgia counties in EPA Radon Zone 1 (highest potential). ZIP-level test statistics: Not available (UGA map is graphic-only).	Georgia Radon Program fact sheet, 2024 ; EPA Georgia map	2024 / 1993
Tornado	12 tornado segments in Gwinnett County 1950–2024 (max F/EF2; 2 deaths, 18 injuries); nearest recorded track ≈2 mi (EF0, 2020-10-10)	SPC tornado database ; NWS Peachtree City severe climatology	1950–2024
Ice storm benchmark	Feb 11–13, 2014: ¼ to over ½ in ice accumulation near Atlanta	NWS FFC event page	2014
Air quality (ozone, 2015 8-hr NAAQS)	Atlanta 7-county area incl. Gwinnett: Marginal nonattainment redesignated to maintenance effective 2022-11-16; 2023–2025 area design value 0.073 ppm (above the 0.070 standard); Gwinnett monitor 131350002: 0.067 ppm	EPA Green Book, Georgia ; EPA ozone design values 2023–2025 ; Federal Register 2022-17846	Green Book to 2026-07-31
Termite	Georgia is in the IRC "very heavy" termite infestation probability zone	2021 IRC Fig. R301.2(7) , via PNNL BASC	2021 IRC
Wildfire	Not a factor in the sources consulted; not assessed further	—	—

3.3 Soils and geology (NRCS Soil Data Access, survey area GA135; Gwinnett GIS soils layer)

The seven map units flagged in FACTS §2 resolve as follows. Locations are from 54 point samples of the Gwinnett GIS soils layer on a 100 × 80-ft grid in site-local coordinates (u along the strip from Arcade Rd), so boundaries are approximate.

Unit	Series (NRCS name)	Slope	HSG	Drainage	Depth to rock	Shrink-swell (max LEP)	Hydric	NRCS limitation ratings (septic / dwellings / local roads)	Where on site
AmC2	Appling sandy loam, 6–10 %, moderately eroded	6–10 %	B	Well	none < 2 m	2.1 (low)	No	Somewhat limited / Not limited / Very limited (low strength)	Front half, u 0–800 (lane and NE side)
ApB	Appling–Hard Labor complex, 2–6 % (Appling 60 %, Hard Labor 25 %)	2–6 %	B (Hard Labor C)	Well / moderately well; Hard Labor seasonal water table ≈32 in	none	1.5	No	Hard Labor: very limited septic, somewhat limited dwellings (wetness)	Band at u ≈400–500
PfC2	Pacolet sandy loam, 6–10 %, moderately eroded	6–10 %	B	Well	none	1.8	No	Somewhat limited / Not limited / Somewhat limited	Ridge and NE flank, u 800–1,400; includes the stream-head pocket
GeB2	Gwinnett clay loam, 2–6 %, eroded	2–6 %	B	Well	paralithic 40–60 in	1.5	No	Somewhat limited / Not limited / Somewhat limited	NW corner only (u ≈1,700, SW)
GgE2	Gwinnett loam, 10–25 %, eroded	10–25 %	B	Well	paralithic ≈50 in	1.5	No	Very limited for all (slope)	Rear, u 1,400–1,700
ARE	Ashlar–Rion–Waterree complex, 10–25 %	10–25 %	B	Well	Ashlar: lithic bedrock 22–40 in; Waterree: paralithic 20–40 in	1.5	No	Very limited for all (slope, depth to rock)	Mid swale u 800–1,100 (lane and SW side = Pond 1) and u ≈1,500 SW (Pond 2)
HYB	Helena sandy loam, 2–6 %	2–6 %	D	Moderately well; seasonal high water table ≈24 in	none	3.5 (moderate)	No	Very limited for septic, dwellings, roads and excavations (wetness)	Front SW corner, u 0–200, v ≈ –200 — the amenity block, entry pavement and sewer tie-in area

Sources: [NRCS Soil Data Access](#) (tables mapunit, muaggatt, component, corestrictions, chorizon, cointerp; mukeys 639990, 639929, 639972, 639948, 639953, 640009, 639955; data as of 2025-09-02); [Gwinnett GIS soils layer 1](#); [1967 Soil Survey of Gwinnett County \(archive\)](#). Accessed 2026-08-28.

Findings that extend FACTS:

- **No map unit on site is rated hydric by NRCS.** FACTS §2 asked whether HYB was a hydric floodplain unit at the stream; it is not — HYB is Helena sandy loam (HSG D, water table ≈2 ft, moderate shrink-swell) and it sits at the **front SW corner**, under the planned amenity block and the sewer tie-in, not at the rear pocket. Expect undercut or moisture-conditioned fill, slab moisture detailing and careful pavement subgrade there.
- **Shallow gneiss (ARE, rock at roughly 2–3.5 ft)** underlies exactly the two proposed pond sites and the mid-swale lane crossing — rock-excavation risk for Pond 1, Pond 2 and any deep sewer trench at u 800–1,100. Rock probes before committing detention volumes.
- The "2" suffix (AmC2, PfC2, GeB2, GgE2) marks eroded phases with clayey kaolinitic subsoil exposed: highly erodible during grading, poor infiltration for runoff-reduction practices, and "very limited" for local roads (low strength) on Appling.
- Elsewhere HSG B, low shrink-swell, no water table: good ground for slab-on-grade cottages with conventional footings.
- Bedrock is biotite gneiss / mica schist / amphibolite under a saprolite mantle ([Georgia Geologic Survey 1999 via Macrostrat](#); [USGS HA 730-G](#)).

3.4 Hydrology, receiving waters and buffers

- FACTS §2 topography and stream facts stand (921.6–958.2 ft NAVD88; ridge u 1,080–1,200; low pocket 921.6 ft at u 1,384; headwater stream entering the SW line at u 1,356–1,392; second branch just outside the SW line at u 868–1,038).
- Receiving waters: the site drains SW into the **Jackson Creek** basin (Gwinnett major-basin layer), USGS HUC-12 **030701030307 "Garner Creek–Yellow River"**, Yellow River → Ocmulgee. Sources: [Gwinnett GIS basins, layer 7](#); [USGS WBD](#). Accessed 2026-08-28.
- **Impairment:** Georgia EPD 2024 Integrated 305(b)/303(d) list (EPA-approved 2024-09-03): *Jackson Creek, headwaters to Sweetwater Creek* (GAR030701030315, Gwinnett, 8.4 mi) — **Not Supporting, Categories 4a/5, Bacteria and Bio F (fish community), source: urban runoff, TMDL priority 2025**; Camp Creek headwaters–Jackson Creek (GAR030701030314) 4a Bacteria; Garner Creek (GAR030701030326) 4a Bacteria/Bio F. Source: [GA EPD 2024 305\(b\)/303\(d\) Excel](#).

- Buffer law: O.C.G.A. §12-7-6(b)(15) — 25-ft state buffer "as measured horizontally from the point where vegetation has been wrested by normal stream flow" ([GA EPD Field Guide, Sept 2017](#)); Metro North Georgia Water Planning District Model Stream Buffer Protection Ordinance — 50-ft undisturbed buffer from top of bank plus 25-ft impervious setback (75 ft), sewer easements ≥ 25 ft from top of bank ([MNGWPD](#)). Lilburn applies the 50 + 25 (FACTS §2). Field delineation of "top of bank" at the stream head is the controlling survey item for the rear lots.

3.5 Trees and ecology

- Lilburn Code of Ordinances Ch. 109 Art. V (Buffers, Landscaping and Vegetation; codified through Ord. 2026-665): residential subdivisions must submit a tree preservation plan meeting **16 tree density units (TDU) per acre** (§109-168(3), §109-169), calculated exclusive of zoning-buffer acreage; existing trees ≥ 3 in DBH credited per Table B; **preserved trees ≥ 12 in earn up to 4 $\times$ credit**, plus 50 % in required yards or floodplain; replacement trees per Table C (2-in 0.5 TDU ... 4-in 0.7 TDU); plans by a landscape architect, urban forester or arborist (§109-177). Lilburn has **no "specimen tree" definition** — the term appears only in the rezoning environmental question (Ord. 2023-603 §1003). Source: [Municode, Lilburn Ch. 109 Art. V](#) (accessed 2026-08-28).
- County tree-canopy percentage from a primary source: Not available. USFWS IPaC species screen: not run.
- The wooded rear (u > 1,300: stream head, 75-ft setback, GgE2 slopes) absorbs the 16-TDU requirement at multiplied credit while screening Nantucket Dr — a triple asset.

3.6 Design implications for one-story 55+ cottages (Climate Zone 3A)

- **Cooling-first envelope:** dehumidification drives comfort more than sensible load (0.4 % dew point 74.2 °F); R-49 attic with radiant-barrier decking, tight envelope, low-SHGC glazing on SW/W walls; heat pumps sized to the 93.7 / 73.8 °F design day.
- **Porches:** the 6-ft depth in FACTS §4 is the floor; make west-exposed porches (SW-facing fronts on NE-side lots, rear patios on SW-side lots) 8 ft deep with vertical screening and canopy trees; put the primary outdoor room on the NE side wherever the plan allows.
- **Foundations:** slab-on-grade with zero-step entries suits the HSG B soils and removes crawlspace humidity; add capillary break, sub-slab vapor retarder and a passive radon stack (Zone 1). Plan for undercut on HYB (front SW corner) and rock in ARE at the ponds and mid-swale lane.
- **Termite (very heavy):** treated sills, 6-in clearance to grade, borate-treated framing or shields, no foam-to-grade contact, pre-treatment.
- **Ice-storm resilience:** underground electric in the lane, generator-ready clubhouse (warming center), limb-clear zone over the lane, ice-tolerant street trees.
- **Erosion / stormwater:** eroded kaolinitic subsoils on a NE→SW cross-slope demand phased clearing and immediate stabilization; infiltration-based runoff reduction will test poorly on clay subsoil — favor bioretention with underdrains; confirm the rock line at both ponds before fixing detention volumes; the impaired receiving stream (Jackson Creek, urban runoff) makes the water-quality volume and the preserved creek woods a substantive, not cosmetic, part of the application narrative.

4. Mobility and access

Position check (Gwinnett GIS parcels layer 7 + OpenStreetMap, accessed 2026-08-28): Arcade Rd runs from US-29 in downtown Lilburn southeast to Killian Hill Rd, then southwest past the site to Rockbridge Rd (3.69 mi). At the site the road runs NE–SW and the assemblage lies on its **northwest** side, extending N28.7°W (consistent with FACTS §1). Killian Hill Rd is 0.3 mi NE; Rockbridge Rd 0.7 mi SW; downtown Lilburn 2.0 mi by road. **Arcadia Place (a Lilburn city street) meets Arcade Rd about 60 ft from the assemblage's NE front corner**, i.e. opposite the NE end of the 237-ft frontage. Drive distances/times are OSRM routing on OpenStreetMap (free-flow); straight-line distances are from OSM/Gwinnett GIS coordinates.

4.1 Walkability

Metric	Value	Source	Vintage / accessed
Walk Score (4541 Arcade Rd SW)	5 / 100 — "Car-Dependent"	Walk Score	2026-08-28
Bike Score	18 / 100 — "Somewhat Bikeable"	same	2026-08-28
Transit Score	Not available (not published for the address or the city)	Walk Score Lilburn	2026-08-28
Sidewalk on Arcade Rd at the frontage	Not verified — no public Gwinnett sidewalk inventory found; OSM tags none	Gwinnett GIS hub ; OSM	2026-08-28

Metric	Value	Source	Vintage / accessed
Nearest signal	Arcado Rd @ Killian Hill Rd, 0.3 mi NE (no marked crossings mapped)	OSM Overpass	2026-08-28
Nearest mapped crosswalks	Arcado Rd @ Cole Dr signal, 0.5 mi SW; Arcado @ Rockbridge Rd SW signal, 0.7 mi	OSM Overpass	2026-08-28

4.2 Transit

Item	Value	Source	Vintage / accessed
Nearest fixed route	Ride Gwinnett Route 25 "GTC – Stone Mountain" stops at "Killian Hill Road at Arcado Road": stop 2528 (toward Stone Mountain) 0.30 mi straight-line from the site point; stop 2578 (toward Gwinnett Place Transit Center) 0.31 mi. Walk ≈0.3 mi along Arcado Rd to the Killian Hill signal	Ride Gwinnett GTFS, feed 2026-07-02; Route 25 schedule	feedvalid 2026-07-02 → 2026-12-26
Route 25 corridor and service	GTC → Pleasant Hill Rd → US-29 → Lilburn (Church St @ Library, Main St) → Killian Hill Rd → Five Forks Trickum Rd → Rockbridge Rd → Stone Mountain; 54 stops. Weekday 17 trips, 50-min headway, 6:36 am–7:56 pm (SB) / 6:51 am–8:11 pm (NB); Saturday hourly; no Sunday service. Launched 2023-08-26	GTFS stop_times; Gwinnett news 2023-08-29	feed 2026-07-02
Route 30 "Lilburn – GTC"	Nearest stop Lawrenceville Hwy @ US Postal, 1.5 mi straight-line; ~30-min weekday headway	GTFS; Route 30 schedule	feed 2026-07-02
2026 local network	Routes 10A/10B, 20, 25, 30, 35, 40, 50, 70; Mon–Sat only; Route 45 discontinued 2026-06-27	Routes & schedules	2026-08-28
ADA para transit	Curb-to-curb within ¾ mile of local fixed routes , both trip ends; Mon–Sat, fixed-route hours; certification by application. Site ≈0.3 mi from Route 25 → inside the buffer	Paratransit information	2026-08-28
Microtransit	Snellville, Lawrenceville, Norcross zones only; Lilburn not covered	Microtransit	2026-08-28
2024-11-05 transit referendum	Failed: 53.45 % No (217,007) / 46.55 % Yes (188,977) — 1 % sales tax, 30 yr, ~75 projects incl. countywide microtransit. No dedicated funding for expansion	11Alive ; WSB-TV	2024-11
Gwinnett Senior Services transportation	Age 60+ (or spouse), county residents: assisted door-through-door (medical and non-medical, mileage fee); unassisted curb-to-curb for scheduled non-emergency medical (income-based cost share); senior-center transport. Residents register individually; HOPA residents aged 55–59 are not eligible. Nearest senior center: Centerville (3025 Bethany Church Rd, Snellville) 7.0 mi / 13 min; Lawrenceville 11.7 mi; Norcross distance Not available	Senior Services	2026-08-28
Xpress commuter (ATL/GRTA)	Network restructured 2025-06-16 (27 → 15 routes). Indian Trail P&R (1900 Indian Trail Lilburn Rd NW), 5.6 mi / 11 min: Route 415 (→ Emory) and 416 (→ Downtown), M–F peak only, \$3; Hewatt Rd P&R (2331 Hewatt Rd), 5.2 mi / 11 min: Route 419 → Downtown/Midtown, \$4	Xpress corridor changes; 415; 416; 419	2025-09
MARTA rail	Doraville 12.6 mi / 22 min; Indian Creek 12.1 mi / 25 min	OSRM/OSM	2026-08-28

4.3 Roads and traffic

Metric	Value	Source	Vintage / accessed
Arcado Rd classification	Gwinnett GIS ROADCLASS Minor Collector , county-owned and maintained, inside Lilburn; GDOT functional class 6U Urban Minor Collector	Gwinnett GIS streets, layer 6 ; ARC GDOT counts layer	2026-08-28
Posted speed	35 mph , US-29 to Rockbridge Rd; 25-mph school zones at Killian Hill Christian School and Arcado ES (speed cameras)	Gwinnett Speed Zone Ordinance list (GDOT-approved 2025-09-10)	2025-09
Arcado Rd AADT	GDOT station 135-6689 (between Killian Hill Rd and US-29, 0.58 mi NE of the site): 14,800 (2017) ; 14,500 (2016); 14,200 (2015); 13,300 (2014); trucks 2.2 %. No GDOT station on the Killian Hill–Rockbridge segment. Post-2017 values: Not available (check TADA manually). Consistent with FACTS \$1's 13,800–15,300 vpd	ARC Open Data, GDOT AADT 2008–2017; GDOT TADA	2017

Metric	Value	Source	Vintage / access ed
Killian Hill Rd	GDOT 4U Urban Minor Arterial; station 135-6356 (N of Arcado) 17,200 (2016) ; station 135-0358 (S of Arcado) 20,400 (2017) . Posted 45 mph through the Arcado intersection	same ARC layer; speed-zone list	2017 / 2025
Arcado @ Killian Hill intersection	Signalized. 2017 SPLOST F-1262-03 "Killian Hill Road at Arcado Road Intersection Improvements" — Concept Complete , Tier I, no construction date	2017 SPLOST project handout (as of May 2025)	2025-05
Killian Hill Rd widening	2017 SPLOST F-1262, Church St → Arcado Rd, 2 → 5 lanes — Concept Complete (unchanged since May 2024); limits stop at Arcado Rd; F-1233-02 matching funds (county \$1,400,000 / city \$328,395) listed as a potential county/city project. Lilburn CID once cited \$25M and construction "in 4–5 years" (page now 404). Sidewalk/turn-lane scope and schedule: Not available	same handout; Gwinnett DOT SPLOST program (May 2024) ; Lilburn CID (404)	2025-05
2023 SPLOST nearby	F-1482 Arcado Rd @ CSX crossing (concept; near downtown); M-1353 Paxton Ln @ Killian Hill Rd (design, construction late 2027); F-1498 Killian Hill Rd Trickum MS access (complete). No Arcado Rd sidewalk or road project at the site	2023 SPLOST handout (as of Sept 2025)	2025-09
Gwinnett Destination 2050 CTP	Executive summary names no Arcado Rd project; sidewalk/bike needs "prevalent throughout the county"	CTP executive summary	2024-06
Driveway / access standards (county road)	Gwinnett DOT right-of-way permit. Gwinnett UDO §900-40 Table 900.2 intersection sight distance 390 ft at 35 mph (445 ft at 40; 500 ft at 45; 2-lane, 7.5-s gap, AASHTO Case B1); Table 900.3 access spacing 244 ft centerline-to-centerline at 35 mph ; Table 900.1 minor-collector R/W 60–80 ft	UDO §900-40 ; UDO §900-10 ; ROW permits	UDO 2014
Left-turn-lane warrant	Gwinnett DOT <i>Criteria and Guidelines for Left Turn Lanes</i> (2007-06-20), Table 1 (residential): 2-lane road, 30–35 mph, ADT ≥ 6,000 → left-turn lane required at ≥ 75 lots. 44 lots → not warranted . Counts must be < 2 years old or a developer 24-hr count	Gwinnett DOT left-turn criteria	2007

4.4 Regional access (OSRM / OpenStreetMap, free-flow, accessed 2026-08-28)

Destination	Drive
US-29 @ Killian Hill Rd / @ Arcado Rd–Main St	1.9 mi, 5 min / 2.0 mi, 4 min
Ronald Reagan Pkwy west terminus @ US-29	3.3 mi, 8 min
I-85 exit 102 (Beaver Ruin) / exit 101 (Indian Trail)	6.1 mi, 11 min / 6.2 mi, 13 min
US-78 @ East Park Place; I-285 exit 39 @ US-78	5.7 mi, 13 min; 11.2 mi, 18 min
Downtown Lilburn (City Hall, 340 Main St)	2.0 mi, 5 min
Piedmont Eastside Medical Center (Snellville)	7.7 mi, 17 min
Northside Hospital Gwinnett (Lawrenceville)	10.9 mi, 20 min
Emory Decatur / Emory Hillandale	14.6 mi, 22 min / 16.6 mi, 31 min
Grocery: Kroger 4155 Lawrenceville Hwy / Publix 4045 Five Forks Trickum Rd / Walmart 4004 Lawrenceville Hwy / Publix 375 Rockbridge Rd	2.1 mi / 2.4 mi / 2.6 mi / 3.9 mi
Pharmacy: Killian Hill Pharmacy, 1025 Killian Hill Rd SW	2.2 mi, 5 min
Urgent care: Eastside Urgent Care, 976 Killian Hill Rd SW	2.0 mi, 5 min
Gwinnett Fire Station 3, 4394 Five Forks Trickum Rd — first-due per GIS station-boundary layer	2.9 mi, 6 min (Station 25, 3575 Lawrenceville Hwy 3.5 mi; Station 2, 12 Harmony Grove Rd 4.1 mi)
Lilburn Police HQ, 4600 Lawrenceville Hwy	2.4 mi, 5 min

Hospitals and fire stations from Gwinnett GIS GC_Main layers 22/23/29; amenities from OSM Overpass; [Lilburn Police](#). Gwinnett Fire & Emergency Services publishes ISO 2/2X (2024-05-01), CFAI accreditation and 107,658 calls (2025) but **no response-time statistic — Not available** ([GFES about](#)).

4.5 Trails and bicycle

Camp Creek Greenway: 4.2 mi paved multi-use trail (ADA; sidewalk and boardwalk) linking Lilburn City Park, Lions Club Park and Killian Hill Rd; parking at City Park, Lions Club Park and Velva Ct ([City of Lilburn](#)). Nearest mapped segment 0.9 mi straight-line / 1.4 mi drive; City Park trailhead (76 Main St NW) 1.4 mi / 4 min. **No trail or bike lane exists or is programmed on Arcado Rd or Killian Hill Rd** (OSM; SPLOST lists) — although the 2026 draft plan's Suburban-Low policy calls for a Killian Hill sidepath (§6.5).

Gwinnett Countywide Trails Master Plan: Camp Creek Greenway is a Signature Trail; nearby active items are the Norcross-to-Lilburn Trail at Indian Trail/Beaver Ruin (2017 SPLOST F-1083) and a Southwest Gwinnett Safe Access Study node at Five Forks Trickum @ Killian Hill ([Trails in the works](#)).

4.6 Site-access implications

- **One full-movement driveway** on a 2-lane, 35-mph minor collector at ~14,800 vpd (2017). A left-turn lane is **not warranted** (44 lots < 75) and ~13–14 peak-hour trips fall below the 21-trip TIS threshold (FACTS §3). A right-in/right-out is neither required nor desirable on an undivided road (it would push U-turns to the Killian Hill and Cole Dr signals).
- **Entrance alignment:** Arcadia Place enters Arcado Rd opposite the NE end of the frontage. FACTS §4 places the entrance "centered on the frontage," which would sit roughly 175 ft from Arcadia's centerline — short of the 244-ft spacing in UDO Table 900.3. Either align the lane with Arcadia Place as a four-leg intersection or shift it SW to ≥ 244 ft; settle with Gwinnett DOT and the city at the pre-app.
- **Sight distance:** 390 ft each way at 35 mph (445 ft if DOT applies a 40-mph design speed) on a curved frontage — field survey required.
- **Dead-end lane ~1,700 ft (IFC 2018 App. D103.4, > 750 ft):** no secondary access is possible — the rear abuts Nantucket Dr lots and both long sides abut R-1 lots with no stub streets. Mitigation stays NFPA 13D sprinklers, a 22-ft clear lane and T-turnarounds. First-due fire/EMS is Station 3 at 2.9 mi / 6 min free-flow.
- **Pedestrian link to transit:** the only route to the Route 25 stops is ~0.3 mi along Arcado Rd with no verified sidewalk. Build the 5-ft frontage sidewalk (Lilburn checklist) and ask the county/city to close the gap to Killian Hill — a cheap, credible "transit-served 55+ community" argument since the site is inside the ADA paratransit band.
- School bus: not applicable (HOPA).

5. Demographics and market

Geography (Census Geocoder, accessed 2026-08-28): the site is in Census Tract **504.61** (GEOID 13135050461), Lilburn city (place FIPS **46356**), Gwinnett County (13135), ZCTA 30047. All ACS figures are **2020–2024 5-year** estimates pulled from the data.census.gov table API on 2026-08-28. Tract 504.61 is small (2020 count 2,655; ACS population 2,997 ±771) — its figures are indicative only and carry wide margins of error.

5.1 Headline table

Metric	ZCTA 30047	Lilburn city	Gwinnett County	Tract 504.61	Source (ACS 2020–2024 unless noted)
Population	70,121	15,390	979,864	2,997 (±771)	DP05_0001E
2020 Census / 2010 Census	69,172 / 59,660 ¹	14,502 / 11,596 (+25.1 %)	957,062 / 805,321 (+18.8 %)	2,655 / n.a.	DHC 2020 P1; SF1 2010 P1
Population estimate, July 1 2025	n.a.	16,369	1,018,099	n.a.	Census PEP V2025
Median age	37.5	35.7	36.0	42.4	DP05_0018E
Age 55+ share	25.8 %	25.0 %	23.2 %	35.5 %	DP05_0013/14/24PE
Age 65+ share	14.2 %	16.6 %	11.3 %	25.7 %	DP05_0024PE
Households with a person 65+	30.3 %	37.3 %	23.8 %	53.1 %	DP02_0015PE
Householder 65+ living alone (% of households)	7.1 %	11.9 %	6.2 %	7.6 %	DP02_0009PE + 0013PE
Households / mean household size	21,727 / 3.21	5,065 / 3.01	327,043 / 2.98	1,152 / 2.60	DP02_0001E / 0016E
Median household income (2024 \$)	\$88,652	\$63,581	\$87,890	\$65,000 (±\$37,244)	DP03_0062E
Households earning ≥ \$100k	45.5 %	30.3 %	44.1 %	39.4 %	DP03_0059–61PE
Households with Social Security / retirement income	29.0 % / 21.9 %	35.1 % / 22.1 %	22.4 % / 17.8 %	51.5 % / 43.6 %	DP03_0066 / 0068PE
Poverty rate	15.0 %	22.9 %	10.8 %	1.0 %	DP03_0128PE
Housing units / owner-occupied	22,443 / 76.0 %	5,291 / 61.4 %	340,962 / 66.3 %	1,186 / 90.1 %	DP04_0001E / 0046PE
Median value, owner-occupied	\$367,300	\$329,700	\$380,900	\$301,500 (±\$78,844)	DP04_0089E
Median gross rent	\$1,850	\$1,778	\$1,810	\$2,455	DP04_0134E
1-unit detached share	84.6 %	71.3 %	70.6 %	98.8 %	DP04_0007PE

Metric	ZCTA 30047	Lilburn city	Gwinnett County	Tract 504.61	Source (ACS 2020–2024 unless noted)
Built before 1980	33.4 %	25.6 %	13.9 %	66.9 %	DP04_0022–26PE
Owner units without a mortgage	34.8 %	36.2 %	29.8 %	29.6 %	DP04_0092PE
Zero-vehicle households	2.3 %	3.5 %	3.8 %	1.0 %	DP04_0058PE
Owner-householders aged 55–64 / 65+	23.4 % / 28.0 %	16.4 % / 39.5 %	23.2 % / 23.1 %	9.2 % / 55.5 %	S2502_C04_014–017E
Disability rate, 65–74 / 75+	16.8 % / 46.6 %	13.3 % / 67.9 %	19.3 % / 44.6 %	12.2 % / 81.7 %	S1810_C03_017 / 018E
Bachelor's or higher (25+)	39.9 %	27.9 %	39.0 %	39.0 %	DP02_0068PE
Hispanic / NH White / NH Black / NH Asian	29.0 / 30.0 / 17.7 / 17.8 %	48.2 / 21.0 / 14.2 / 13.5 %	23.7 / 30.6 / 27.8 / 13.2 %	31.2 / 34.6 / 22.3 / 7.5 %	DP05_0090 / 96 / 97 / 99PE
Foreign-born / non-English at home	31.7 / 49.1 %	35.2 / 61.2 %	28.4 / 38.1 %	16.0 / 44.0 %	DP02_0094 / 0114PE
Labor-force participation / ACS unemployment	61.4 / 4.2 %	63.7 / 5.1 %	68.6 / 4.2 %	75.4 / 5.3 %	DP03_0002 / 0009PE
Drove alone / carpool / work-from-home / transit	67.2 / 14.0 / 16.3 / 0.7 %	61.8 / 24.9 / 10.8 / 0.1 %	66.7 / 11.8 / 18.4 / 0.5 %	43.0 / 33.9 / 21.9 / 0.4 %	DP03_0019–24PE
Mean travel time to work	33.5 min	31.9 min	32.8 min	28.1 min	DP03_0025E

¹ 2010 and 2020 ZCTA boundaries differ; not strictly comparable. Sources: [data.census.gov table API \(DP02/03/04/05, S0101, S1810, S2502, DHC 2020 P1/P12, SF1 2010 P1\)](https://data.census.gov/tables/API/DPO2/03/04/05/S0101,S1810,S2502,DHC2020/P1/P12,SF12010/P1); [Census Geocoder](#); [Census PEP V2025 cities and counties](#); [ACS variable dictionary](#). All accessed 2026-08-28.

Lilburn 2020 Decennial age structure (DHC P12): 55+ = 3,685 (25.4 %); 65+ = 1,961 (13.5 %) → ACS 2020–2024 65+ = 16.6 % (2,558 ±531). The city is aging faster than the county.

Industry mix of employed residents (DP03): ZCTA 30047 — education/health 19.1 %, retail 13.3 %, professional/management 13.2 %, construction 12.2 %; Lilburn city — construction 19.6 %, retail 16.0 %, accommodation/food 12.0 %; Gwinnett — education/health 18.8 %, professional 14.9 %, retail 11.9 %.

5.2 Senior-housing demand

- **Local move-down pool** (S2502 × owner-occupied units): owner-householders aged 55+ ≈ 8,500 in ZCTA 30047 (51.4 % of 16,508 owner units), ≈ 1,740 in Lilburn city (55.9 % of 3,110), ≈ 100,000 in Gwinnett (46.3 % of 216,729). One-third of the ZIP's owners carry no mortgage (equity for an all-cash cottage purchase). 33 % of the ZIP's housing and 67 % of the site tract's was built before 1980 — 1970s two-story stock (King David Manor, built ~1970) with no one-level alternative inside the city.
- **Regional forecast:** ARC Series 17 (Feb 2024) — 21-county region 7.9 M by 2050; residents 75+ rise from under 5 % (2020) to about 12 % (926,700) by 2050; Gwinnett 957,062 → 1,200,534 (+25 %) by 2050 ([Rough Draft Atlanta, 2024-02-14](#); [ARC forecasts](#)). Earlier ARC series (AJC, 2019-10-11): Gwinnett 55+ from 177,717 (19.8 %, 2015) to 506,169 (34.1 %, 2050) (AJC). Gwinnett-specific 65+ counts from the Series 17 dashboard: Not available.
- Gwinnett 2045 Unified Plan (adopted 2024-02-20) projects >500,000 new residents by 2045; 65+ figures: Not available ([Gwinnett 2045 Unified Plan](#)). Gwinnett issued RFP RP054-24 (2024-10-09) for a county-wide 60+ senior-services needs assessment (RFP).
- **National product benchmarks** (NAHB Eye on Housing, July 2026, from Census SOC): 47,000 age-restricted starts in 2025 (3.45 % of all starts), 30,000 single-family; median age-restricted SF price **\$523,000**, average 2,500 sf, \$166/sf, average lot 0.19 ac ([NAHB, 2026-07](#)); 17 % of NAHB builders build age-restricted product ([NAHB, 2025-10](#)). NAHB/Wells Fargo HMI: 35 in Aug 2026, below 40 for 15 straight months ([NAHB, 2026-08](#)). NAHB 55+ HMI: Not available (page 404).

5.3 Competitive set (builder/listing sites, accessed 2026-08-28; commercial sources)

Community	City (distance)	Builder	Product	Size (sf)	Price	Years	HOA
Cottages at Noble Village , 4817 Church St NW	Lilburn (~2 mi, downtown)	Seed Capital Investments	gated 55+ 2 BR / 2 BA cottages, zero-step (one listing site calls them attached/condo — verify tenure)	1,225–1,450	opened "\$380s+ / from \$399,900"; resales \$342,500–\$375,000; current list \$414,900; sold out (Jan 2026)	2020–2023	\$93–\$280/mo
Soleil Summit Chase, 3465 Rosebud Rd SW	Snellville	Patrick Malloy Communities	265 detached ranch, gated, clubhouse/pool/pickleball	1,418–2,818	mid-\$500s–high-\$600s (avg ≈ \$570k)	2024–	\$389/mo
Saddlebrook	Snellville	Century Communities	130 detached ranch, gated	1,704–3,098	avg ≈ \$446,600	2017–	n.a.

Community	City (distance)	Builder	Product	Size (sf)	Price	Years	HOA
Gladstone Landing	Snellville	Paran Homes	66 attached TH, master-on-main	1,940–2,029	avg ≈ \$355,800	2019–2022	\$175–\$200/mo
Kelly Preserve	Loganville	n.a.	85 detached, gated	n.a.	\$300k–\$400k	2023–	n.a.
Courtyards at Bailey Farms	Dacula	n.a.	128 detached	n.a.	\$400k–\$700k	2023–	n.a.

Sources: [Cottages at Noble Village \(Jome\)](#), [Homes by Marco](#); [Soleil Summit Chase \(55places\)](#), [PMC pre-sale](#); [Saddlebrook](#); [Gladstone Landing](#); [55places Snellville](#), [55places Lawrenceville](#). Tucker/Stone Mountain 55+ listings: Not available. **No detached 55+ community exists inside Lilburn today; the only Lilburn 55+ product (Noble Village) sold out.**

5.4 Housing market (commercial sources; Zillow/Redfin blocked direct fetch, figures from their indexed pages)

- Zillow ZHVI: Lilburn city \$401,137 (–2.6 % YoY, ~23 days to pending, through 2026-06-30); ZIP 30047 \$386,338 (–0.5 % YoY); Gwinnett \$408,823 (–2.6 % YoY, ~34 days to pending, 2026-07-31) — [Zillow Lilburn](#), [Zillow 30047](#), [Zillow Gwinnett](#).
- Redfin: Lilburn median sale \$380,000 (March 2026, –10.1 % YoY), \$183/sf (+5.2 %), 65 days on market; Gwinnett \$421,000 (three months to June 2026, –1.0 %) — [Redfin Lilburn](#), [Redfin Gwinnett](#).
- Supply: Lilburn months of supply — Not available. Atlanta MSA July 2026 (Georgia MLS): median \$405,000 (+2.5 % YoY), 4.82 months ([GAMLS](#)); FMLS Greater Atlanta July 2026 detached: median \$475,000, 4.4 months, 38 DOM, 98.5 % sale-to-list; attached \$355,180, 6.5 months ([FMLS via Simply List Atlanta, 2026-08-13](#)).
- **Legends at Parkview** (abutting SW): built 2007–2012, 3,300–7,000 sf, 5–7 BR; sales \$625,000 (Jan 2024, 4,286 sf), \$645,000 (Sep 2024, 3,433 sf), \$665,000 (May 2023), \$875,000 (Apr 2022, 5,415 sf); ≈ \$165/sf; HOA "\$760–\$800" (period not stated); lot sizes Not available — [Homes by Marco](#), [neighborhoods.com](#). Consistent with the FACTS \$1 digest range of \$580k–\$895k.
- **King David Manor** (abutting NE): built ~1970, average 1,468 sf (listings 1,482–2,587 sf), no HOA; closed sales \$345,000–\$422,000, median \$383,500 (dates not shown); 206 King David Dr \$332,500 (Aug 2022, 1,700 sf) — [neighborhoods.com](#), [Zillow 206 King David Dr](#).
- **Price positioning for a 1,444–1,600 sf one-story cottage**: Noble Village 1,225–1,450 sf resold at \$342,500–\$375,000 (≈ \$260–\$290/sf); Soleil's 1,418-sf plan sits in a mid-\$500s community; Saddlebrook (1,704 sf+) ≈ \$446k; national age-restricted SF median \$523,000 at 2,500 sf. A \$400–\$450k cottage here would sit between Noble Village and Soleil and above the ZIP's \$367,300 median owner-occupied value. (The project ledger of 2026-08-25 carries the pro-forma; this brief does not re-price.)

5.5 Property tax — a decision-relevant selling point

- 2025 millage (Gwinnett Tax Commissioner, updated 2025-08-26): county general fund 6.95; fire 3.20; recreation 1.00; GCPS M&O 18.70 (adopted 2025-08-13) + school bond 1.45 — county + school inside Lilburn = 31.6 mills; city billed separately ([millage rates](#); [GCPS FY2026 millage](#)). 2026: county general fund held at 6.95 (adopted 2026-08-04) ([Gwinnett news](#)); 2026 school and city rates not final.
- City of Lilburn: 4.43 mills 2015–2025; **5.430 mills proposed for 2026**, adoption at a special called meeting 2026-08-31 (pending as of today); city exemptions: S1R homestead \$10,000, L5 senior (65+) \$10,000, plus an **additional \$40,000 senior exemption beginning 2026** ([Lilburn Taxes](#); [Lilburn news flash](#); [Gwinnett Daily Post, 2025-02-24](#)).
- Gwinnett senior exemptions: **L5A — age 65+, household income ≤ \$124,648 (2026) → 100 % exemption from school M&O and bond (20.15 mills in 2025) on the home and up to one acre**, plus \$10,000 county / \$7,000 recreation; VOE freezes county assessed value for any homesteaded owner ([available exemptions](#); [senior exemption FAQ](#)).
- Illustrative (derived; verify with the Tax Commissioner): \$400,000 cottage at 40 % assessment = \$160,000; non-senior with S1R ≈ \$5,400/yr (2025 rates, county + school + city); owner 65+ with L5A + city L5 + the 2026 \$40k ≈ \$2,300/yr — roughly a 55–60 % cut, and the L5A income cap covers most retirees.

5.6 Employment and economy

- BLS LAUS, Gwinnett County (series LAUCN131350000000003): June 2026 **3.4 %** (preliminary); May 2.9 %; Apr 2.6 %; Georgia 3.6 %, U.S. 4.4 % ([BLS API](#); [GDOL Gwinnett profile, Jul 2026](#)). June 2026 labor force 529,962.
- QCEW 2025 annual average (GDOL): 390,090 jobs; avg weekly wage \$1,314; retail 12.4 %, health care 11.9 %, local government 9.1 %.
- Largest employers (Partnership Gwinnett; vintage not stated): GCPS 19,375; Gwinnett County 8,142; Northside Hospital Gwinnett 3,800 (Lawrenceville); Publix 2,800; **Piedmont Eastside Medical Center 1,200 (Snellville — nearest hospital)** ([Partnership Gwinnett](#)). Gwinnett Chamber major-employer list: Not available (page 404).

5.7 Schools

HOPA 55+ yields zero students (FACTS §2). Cluster: Camp Creek ES / Trickum MS / **Parkview HS — Niche 2026 overall "A", #9 public high school in Gwinnett, #52 in Georgia, 3,216 students, 19:1 (Niche; GCPS profile)**. GCPS capacity data: Not available. The cluster's reputation matters less to the buyer than to the neighbors — it underpins Legends at Parkview values and is why "zero students" is worth stating in the letter of intent.

6. Site history and neighborhood context

Method note: Gwinnett qPublic returned HTTP 403 to every fetch on 2026-08-28; parcel data below is from the Gwinnett Tax Assessor 2026 digest (assessment date 2026-01-01, already extracted to data/owners-found.json) and listing portals. Distances are **straight-line** from the site centroid (33.87985, -84.11968) using OpenStreetMap/Nominatim coordinates unless stated.

6.1 The four parcels

Parcel	Fact	Source	Vintage
4541 Arcado (R6123 033)	Brick split-level built 1972 , 3 BR / 2.5 BA, ~2,053 sf (Zillow ≈1.7k); listing says public sewer; sold 2024-09-06 for \$622,500 combined with 4539 (DB 61414/683); 2026 FMV \$285,000 (land \$60,000 + dwelling \$225,000); 2024 tax \$624	Movoto 4541 ; 2026 digest	accessed 2026-08-28
4539 (015)	Vacant 2.00 ac; 2026 FMV \$87,200 (land only); same 2024 sale	2026 digest	2026-01-01
4537 (014)	"Residual Land" 2.00 ac; 2026 FMV \$6,800 ; sold 2023-07-20 for \$950,000 combined with 4535 (DB 60727/25)	2026 digest	2026-01-01
4535 (162)	Built 1987 , 4,267 sf, 3–4 BR / 4–5 BA, full finished basement, 4.45 ac; 2026 FMV \$888,600 (land \$246,300 + dwelling \$642,300); 2024 tax \$10,098. Listed 2022 at \$1,095,000 → \$975,000; listed 2026-02-12 at \$1,700,000 → \$1,499,000 (2026-03-12) → expired 2026-06-14 (MLS 7718389). Septic vs sewer "not specified" — VERIFY	Movoto 4535 ; 2026 digest	accessed 2026-08-28
Combined 2026 FMV, four parcels	\$1,267,600 (assessor); combined recorded purchase prices 2023–2024: \$1,572,500	2026 digest	2026-01-01
Zoning history on the parcels	Only RZ-2025-01 (R-1 → MU; 104 condos + 24,000 sf commercial). Tabled 2025-08-28 by the Planning Commission for a defective legal ad; never re-advertised; absent from every 2025–2026 PC and Council agenda scanned through PC 2026-08-27. No staff report, PC recommendation or Council vote exists → no denial, so Ord. 2023-603 §1003-11's 12-month bar is not apparently triggered . Formal disposition (withdrawn vs dormant) still needs written confirmation from the Planning Director, since §1003-10 bars withdrawal after advertisement	PC agenda 2025-08-28 ; PC minutes item 4709 ; 2026 agendas _01222026-683 ... _08272026-712	accessed 2026-08-28
Environm ental history	GA EPD Hazardous Site Inventory / UST searches: no Arcado Rd or Killian Hill Rd entries — none found ; no industrial legacy. 4541 on public sewer per listing; 4535 septic status unknown	GA EPD HSI	accessed 2026-08-28
Adjoining R-2 "Arcado Road Tow nhomes"	County zoning layer: TYPE R2, jurisdiction Lilburn, record created 2014-05, modified 2025-02; Gwinnett DWR sewer project S2018D0315 "Arcado Road Townhomes" (the outfall in FACTS §2). Rezoning case number/year: Not available . (The only Arcado Rd townhome rezoning in the press — 255 Arcado Rd, 26 TH, approved 4-0 on 2016-05-10 — geocodes ~0.44 mi NE and is probably a different parcel; VERIFY on the city zoning map)	Gwinnett GIS zoning/sewer layers; AJC 2016	accessed 2026-08-28

6.2 Adjoining subdivisions — character and values

Subdivision	Era / builder	Product	Values (2026 digest FMV of abutting lots; recent sales)	Source
Legends at Parkview (SW side; plat 118/187)	GIS subdivision record 2005-04; sewer permit B2007D0060; homes built 2007–2016; builder Not available	3,433–7,000 sf two-story "Tudor and Traditional," stone/shingle exteriors; HOA (registered agent Liberty Community Management)	14 abutting lots FMV \$579,100–\$894,600 (median \$693,500) ; sales \$612k–\$665k (2023–2024); listings \$625k–\$685k	Homes by Marco ; 2026 digest
King David Manor (NE side; plat S/159, Unit 1)	1970–71; no HOA	Brick split-levels and ranches, 1,482–2,609 sf, lots 0.45–0.76 ac	13 abutting lots FMV \$292,800–\$412,500 (median \$349,200) ; 266 King David Dr \$350,000 (2023-07-27); 206 King David Dr \$332,500 (2022-08); 4531 Arcado (front corner) \$350,300, same owner since 1989	Zillow 206 King David Dr ; 2026 digest

Subdivision	Era / builder	Product	Values (2026 digest FMV of abutting lots; recent sales)	Source
Nantucket (NW end; plats 1/268, 3/73, 4/157)	1975–79; no HOA	1970s traditional / ranch	3 abutting lots FMV \$308,400–\$417,000; 4520 Nantucket Dr \$278,400 (2021-05-14)	Zillow/Trulia; 2026 digest

The site is the seam between a 1970s brick neighborhood on one long side and a 2007–2016 stone-and-shingle estate subdivision on the other; the proposed one-story cottages will be the lowest-massed buildings on the strip, and their values would land between King David Manor and Legends at Parkview.

6.3 Civic and community uses (straight-line distances, OSM/Nominatim)

- **Schools:** Trickum MS 0.46 mi N; Parkview HS (998 Cole Dr SW) 1.44 mi S; Camp Creek ES on the Cole Dr campus ≈1.3 mi (a conflicting geocode at "100 Arcade Rd" 0.81 mi N — VERIFY address); Killian Hill Christian School 0.67 mi; Providence Christian Academy 1.73 mi.
- **Churches:** Iglesia Bautista (St Marks Dr) 0.17 mi; Camp Creek Primitive Baptist (est. 1823) 0.54 mi; Living Way Vietnamese Alliance 0.55 mi; Killian Hill Baptist 0.66 mi; Parkview Christian 0.78 mi; Salem Missionary Baptist (c. 1834) 1.27 mi; First Baptist Lilburn (1840) 1.44 mi; Lilburn Christian Church (314 Arcade Rd) 1.50 mi.
- **Civic:** Lilburn City Park (76 Main St) 0.95 mi; Camp Creek Greenway nearest segment 0.97 mi; City Hall + Gwinnett County Public Library Lilburn branch (340 Main St; opened 2016-10-18, ~19,000 sf library) 1.39 mi; Wynne-Russell House 1.69 mi; Lilburn Activity Building (788 Hillcrest Rd, former library, Gwinnett Parks community rooms) 2.11 mi; BAPS Shri Swaminarayan Mandir (460 Rockbridge Rd NW) 2.46 mi.
- **Senior services:** Sunrise Senior Living (3997 Five Forks Trickum Rd) 1.83 mi; Gwinnett senior centers — Centerville (3075 Bethany Church Rd, Snellville) ≈6 mi, Lawrenceville (225 Benson St) 9.75 mi, Norcross (5030 Georgia Belle Ct) distance Not available; Gwinnett Senior Services Center, 567 Swanson Dr, Lawrenceville.
- **Hospitals:** Piedmont Eastside Medical Center (Snellville) 5.64 mi; Northside Hospital Gwinnett (Lawrenceville) 8.2 mi; Emory Decatur 11.2 mi.
- **Grocery / pharmacy clusters:** Killian Hill Rd / Five Forks Trickum Rd — Publix (4045 Five Forks Trickum) 1.68 mi, Killian Hill Pharmacy 1.79 mi; US-29 / Pleasant Hill — Walmart Supercenter (4004 Lawrenceville Hwy) 1.68 mi, Kroger (4155 Lawrenceville Hwy) 1.72 mi, ALDI 2.0 mi; Rockbridge Rd / Killian Hill — Ingles + CVS 1.96 mi; Rockbridge / US-29 — Publix 2.55 mi, Walgreens 2.44 mi, CVS 2.57 mi.

Sources: OpenStreetMap Overpass + Nominatim (accessed 2026-08-28); [AJC, new Lilburn City Hall and library](#); [Gwinnett news, Lilburn Activity Building](#).

6.4 Architectural character and historic resources

- Lilburn: McDaniel settlement; Seaboard Air Line rail 1890; renamed 1896 for Lilburn Trigg Myers; incorporated 1910; downtown fire 1920; charter dissolved 1939, re-established 1955 ([Wikipedia](#); [Old Town Lilburn history](#)).
- **Wynne-Russell House (1826, Plantation Plain, NRHP 1977)** is the city's only NRHP property, 1.69 mi away ([AJC](#)). The **Old Town Overlay** (design guidelines; Design Review Board created by Ord. 2026-670, adopted 2026-04-13) is the city's only historic/design overlay; the county overlay layer returns **zero features at the site** — no historic district or design overlay touches Arcade Rd ([Ord. 2026-670](#)).
- "Arcado" name origin: Not available (Arcado ES, 1981, was named for the pre-existing road).
- Arcade Rd housing stock: 1965–1987 houses on 0.5–4-ac lots (4423 Arcade 1965; 4541 1972; 4535 1987); King David Manor brick split-levels and ranches (1970–71); Nantucket (1975–79); Legends at Parkview stone/shingle two-story (2007–16). A Table 4.2 "compatibility" argument should rest on one-story massing (lower than every neighbor), brick or stone water tables and porch piers echoing King David Manor's brick and Legends' stone, gable roofs, recessed garages and the 20-ft perimeter buffer.

6.5 Planning context

2024 Comprehensive Plan (ARC-prepared, adopted Feb 2024; [PDF](#)), p. 37 "Established Residential": *"The area is characterized by single-family houses on larger lots, and the street networks are typically curvilinear. There is little vacant land or opportunity for redevelopment, so the existing pattern is expected to continue; however infill parcels offer opportunities for some variety... Higher density development will generally not be encouraged in the area, as other parts of the City already offer a range of housing types."* Other signals: p. 10 — 64 % of survey participants favored "Horizontal Density Increase (Accessory Dwelling Units, **Cottage Courts**, Minimum Lot Requirement Reductions, etc.)" over vertical density; 59 % said Lilburn has the "Right Amount of Senior Housing" and 59 % "Need(s) Less Medium- and High-Density Housing." p. 27 — steering committee's strongest support: live/work, **cottage homes**, ADUs, duplexes. p. 47 work item **H.03: "Incentivize accommodations for aging population and elements of**

visitability (accessible living, masters on main) within all medium and high density residential districts." SWOT (pp. 99–101): "Affordable housing for seniors is needed"; "seniors are moving away"; "No sidewalks down Killian Hill."

2026 Comprehensive Plan Amendment — DRAFT 6.16.26 (TSW, 110 pp; draft; plan page) — not in FACTS; material. The 2024 plan carried the 2019 character-area map forward for lack of consensus; the amendment replaces it with a six-class Future Development Map (Core-High/Medium/Low, Suburban-High/Medium/Low; map p. 52): *"the future development approach recognizes that the Lilburn community wants to retain the character of the neighborhoods south of the railroad tracks. As such, almost all properties—aside from existing commercial and light industrial uses—received the lowest intensity designation"* (p. 51, verified from the PDF text). The site is 0.83 mi **south** of the CSX line, so it is almost certainly **Suburban-Low** (pp. 72–73, verified from the PDF text on 2026-08-28): *"Suburban-Low consists of the remaining residential properties south of the railroad tracks. Its current character includes large-lot single-family, particularly where sewer is lacking. There are some newer single-family developments on smaller lots, but they still exhibit suburban form. Of all designations, Suburban-Low is expected to experience the least change. Any new development is likely to be a one-for-one replacement of an existing house."* Target density: 1–3 stories. Its stoplight chart (p. 73) lists **Single-Family Cottages and Single-Family as "Appropriate"**; rowhouses and quad/tri/duplexes "Appropriate with Considerations — Only at major intersections (e.g., Arcade Road and Killian Hill Road)"; mixed use (vertical and horizontal), multifamily, office, retail, hotel and industrial "Not Appropriate." Other policies (p. 73): **"Do not further extend sewer in this area"**; "Encourage new developments to match existing design character"; "Redesign Killian Hill Road to include a sidepath." Zoning-code strategies (p. 73): "Require new development to orient towards the main corridors and/or regional trails"; "Improve street/block connectivity"; **"Allow for a modest variety of housing types, including single-family cottages and infill"**; "Incorporate site design standards to create a scale of development compatible with existing patterns"; "Require sidewalks." The land-use table (p. 54) defines Single-Family Cottages as *"Small footprint single-family houses typically 1,200 square feet or less. Can be standalone, around a courtyard, or within mixed residential areas"* — the 1,444–1,600-sf plans read as plain "Single-Family," also "Appropriate." Ch. 4 (p. 31) calls Arcade Rd a major corridor with "high speeds, wide crossings... limited pedestrian and bicycle safety." Status: PC close-out hearing 2026-06-25; Council transmittal to DCA 2026-07-13; adoption pending (not on the 2026-08-10 agenda); a **UDO rewrite** is under way (code diagnostic 2026-07-07). Engagement comments in the appendix include "No more density in Parkview Cluster" and "Don't build one more house/townhome/apt community until school population/overcrowding is addressed." **VERIFY the site's Future Development Map class and whether the UDO alters R-2 cottage standards before filing.**

Precedent cases (all accessed 2026-08-28):

- **RZ-2024-01** (4413/4423 Arcade Rd at Killian Hill; MU, 23 TH + commercial, 6.15 du/ac): staff recommended denial for plan non-conformance; **Council denied 2024-06-10** (staff report; minutes).
- **RZ-2014-02 / Ord. 2014-469** (MU, 14.32 ac at Killian Hill/Arcado; opposed by a "Lilburn Rezoning Opposition" petition citing traffic, tree loss, cut-through, schools) → built as The Preserve at Killian Hill townhomes; **Phase 3 MCP approved 2023-03-13 for 24 detached cottage lots** (3,500-sf lots, 40-ft widths, 1,400 sf min) + 14,000 sf commercial (Patch 2014; Preserve Ph. 3 staff report).
- **Annsbury Park** (70 Killian Hill Rd, beside Trickum MS; Stanley Martin): **140 "cottage lot" R-2 subdivision on 27 ac**, 2,067–3,241 sf, \$450k–\$539k, HOA \$104/mo, under construction.
- **Cottages at Noble Village** (Main St roundabout): 55+ one-level 2 BR / 2 BA cottages, ~68 units on 8.5 ac, built 2020–23, sold out; resales \$342,500–\$414,900.
- Downtown pipeline: First & Main (DDA, 21 TH + commercial, site work since 2025-11), Rangewater 269 apartments, six-story Porchline, Main Street Townes (80 rental TH), Young St 68 TH (SUP-2022-02), Downtown Master Concept charrette 2026-02-16/20. RZ-2026-01 (city-initiated, 249 Poplar, OI → R-1) and LAN-26-01 (annexation of 114 Arcade Rd) in June 2026 (2026 agendas).
- **Moratoria:** townhome/MU application moratorium Jan–Apr 2020 (AJC); commercial-building moratorium Res. 2026-01 (2026-02-09) ended 2026-04-13; Res. 2026-10 (2026-08-10) 90-day moratorium on non-residential in the Old Town Overlay. **None affects residential rezonings.** Ord. 2025-657 (2025-10-13) amended §602/§603/Art. 14; "Single-family (cluster-cottage, creative lot configuration)" remains **P** in R-2 (Ord. 2025-657).
- **Gwinnett 2045 Unified Plan** (adopted 2024-02-20): west-Gwinnett "Daily Communities" include "Lilburn Halo," "Parkview," "Killian Hill"; sewer single-family areas are "Traditional+" (townhomes, plexes, cottages, SF appropriate); the R-100 land 314 ft away: class Not available (2045 Unified Plan). The Killian Hill Small Area Plan (adopted 2025-07-22) covers the US-78 node ~4+ mi south, not this site.

6.6 Commercial activity

Lilburn Main St / Old Town (1.4 mi): library, city hall, small storefronts, restaurants, actively redeveloping. US-29 / Pleasant Hill (1.7–2.0 mi): the real grocery and big-box node (Walmart Supercenter, Kroger, ALDI, Nam Dae Mun). Killian Hill / Five Forks Trickum (1.7–1.8 mi): Publix, independent pharmacy, Sunrise Senior Living. Rockbridge / US-29 (2.4–2.6 mi): Publix, CVS, Walgreens, BAPS traffic. The 2026 draft plan calls US-29 "aging commercial properties, large surface parking areas" ripe for mixed

use, and Uptown (US-29/Killian Hill) "not experiencing as much redevelopment demand as Old Town."

6.7 Neighbor sentiment and sensitivities

1. **Petition against RZ-2025-01** — "Protect Our Neighborhoods" (Change.org, started 2025-08-20): **1,025–1,026 signatures**, no updates as of 2026-08-28. Six objections: traffic and fatal accidents on Arcade Rd; school, utility and EMS capacity; air quality, noise and stress "particularly affecting children and seniors"; **stormwater runoff, erosion, flooding**; loss of "quiet, single-family homes" character; spot-zoning precedent ([petition](#)). A 55+ HOPA detached plan at 4.7 du/ac answers traffic, schools and character directly (0 students; ~13–14 PM peak trips) but **not stormwater** — that is where the rebuttal must be strongest, and no pond may discharge toward Village Green Ct.
2. **Organized abutters**: Legends at Parkview has an HOA (registered agent Liberty Community Management); King David Manor and Nantucket have **no HOA**, so those neighbors appear individually.
3. **Council record**: the last Arcade/Killian Hill MU-with-townhomes case was denied (2024-06-10) after a staff denial recommendation; both prior Arcade MU attempts drew petitions. Cottage-lot **R-2** cases, by contrast, are being approved and built (Annsbury Park 140 lots; Preserve Ph. 3 24 cottage lots) — the correct precedents to cite.
4. **Timing**: a 2026-09-25 filing lands mid-adoption of the 2026 plan amendment and the UDO rewrite. Suburban-Low's green light for single-family cottages is an asset; "Do not further extend sewer in this area" is a liability for the rear-phase off-site sewer line and should be pre-negotiated with staff and Gwinnett DWR.

Gaps: qPublic blocked (no deed detail beyond the digest; 4535 septic/sewer unconfirmed); Legends at Parkview builder; "Arcade Road Townhomes" case number; site's exact class on the draft Future Development Map; Gwinnett 2045 class for the nearby R-100 land; Camp Creek ES address conflict; road (vs straight-line) distances not computed.

7. Synthesis — connecting the four analyses

7.1 Opportunities

1. **The plan language has moved toward this product, not away from it.** The 2024 Comprehensive Plan's "Established Residential" text (p. 37) already allowed "infill parcels ... some variety," the 2024 survey favored horizontal density ("Cottage Courts") 64:36, and the 2026 draft amendment's Suburban-Low chart rates Single-Family Cottages and Single-Family "Appropriate" while directing the zoning code to "allow for a modest variety of housing types, including single-family cottages and infill" (draft pp. 54, 73). A one-story detached cottage plan at 4.7 du/ac is the one residential product on the strip that is plan-consistent under both the adopted plan and the draft — the 2025 MU filing and the 2024 MU case at Killian Hill were not.
2. **The buyer lives next door.** Tract 504.61 is 35.5 % aged 55+, 90 % owner-occupied and 67 % pre-1980 housing; ZIP 30047 holds roughly 8,500 owner-householders aged 55+ (ACS 2020–2024), a third of ZIP owners carry no mortgage, and the only 55+ product ever built in Lilburn (Cottages at Noble Village, 1,225–1,450 sf) sold out with resales at \$342k–\$415k. King David Manor's 1970 split-levels are exactly the houses the target buyer is leaving. This is a move-down market inside a single school cluster, which also makes "zero students" a credible, checkable promise to the Parkview-cluster neighbors.
3. **Tax exemptions do the marketing.** Gwinnett's L5A (65+, income ≤ \$124,648 in 2026) removes the 20.15-mill school levy entirely, and Lilburn adds a \$40,000 senior exemption in 2026 — roughly a 55–60 % cut on a \$400k cottage (derived; verify). Few Gwinnett 55+ communities sit inside a city with its own senior exemption.
4. **The physical constraints line up with the program.** The stream head, its 75-ft setback, the GgE2 slopes and the wooded rear ($u > 1,300$) are unbuildable for lots but perfect for the 16-TDU/acre tree requirement (preserved ≥12-in trees earn up to 4× credit), the ≥20 % open-space target and a walking loop — the three things a 55+ HOA wants. Zone X, SDC B, HSG B soils over most of the front 1,300 ft and low shrink-swell favor cheap slab-on-grade, zero-step construction.
5. **Precedent is on the R-2 cottage side.** Annsbury Park (140 R-2 cottage lots by Trickum MS, Stanley Martin, under construction) and The Preserve at Killian Hill Phase 3 (24 cottage lots on 3,500-sf, 40-ft lots, approved 2023-03-13) are the cases to cite; both are on Killian Hill Rd within 1.5 mi and both have smaller lots than the 50 × 100-ft basis here.
6. **The refiling clock is not running.** RZ-2025-01 was tabled on 2025-08-28 for a defective ad and never heard, so no §1003-11 denial bar exists (confirm in writing). The 2025 petition's 1,025 signatures were against a 104-condo MU plan; three of its six objections (traffic, schools, single-family character) are structurally answered by a HOPA detached plan.

7.2 Constraints

1. **"Do not further extend sewer in this area."** The 2026 draft plan's Suburban-Low policy (p. 73) collides with the rear-phase sewer options in FACTS §2 (an off-site 8-in line to the Legends at Parkview main, or a lift station). It does not affect the front ~1,000 ft, which drains by gravity to the existing "Arcade Road Townhomes" outfall manhole (invert 927.13). The lots beyond the $u \approx 1,100$ –1,200 ridge (the FACTS §4 Phase 2) are the ones at risk; the plan-consistency argument for Phase 2 must be made at the pre-app, not discovered at the hearing. Note also that the draft's own text expects "one-for-one replacement of an

existing house" in Suburban-Low — 44 for 2 is a change in kind, and the letter of intent should own that.

2. **Rock and wet ground sit exactly where the engineering wants to dig.** ARE (Ashlar, gneiss at 22–40 in) underlies both proposed ponds (u 800–1,100 and u ≈1,500) and the mid-swale lane; HYB (Helena, HSG D, water table ≈24 in, moderate shrink-swell) underlies the front SW corner where the amenity block, entry pavement and sewer tie-in go. The 2025 hydro basis (10,000 cf/disturbed acre) may not be achievable in rock at the assumed depth. Order rock probes and infiltration tests before the LDP grading plan is drawn.
3. **Stormwater is the neighbors' unanswered objection.** The receiving stream (Jackson Creek headwaters, GAR030701030315) is on Georgia's 2024 303(d) list for bacteria and biology from urban runoff; the site's eroded kaolinitic clays are highly erodible and infiltrate poorly; the cross-slope falls toward Village Green Ct; and "stormwater runoff, erosion, flooding" was objection #4 in the 2025 petition. The design cannot rely on infiltration-based runoff reduction and must show, in the application, that no pond discharges toward the Legends lots.
4. **Sun and heat on the wrong side.** Half the houses (the NE-side lots) front SW into the June sunset (az 267–299°, 25° altitude at 5 pm), in a climate with a 0.4 % design dew point of 74.2 °F and 1,882 cooling degree days; a 6-ft porch will be uninhabitable on summer afternoons. Porch depth, screening and shade trees on that side are a design cost, not an option.
5. **Single frontage, single access, 1,700-ft dead end.** No secondary emergency access is physically possible (FACTS §4); IFC 2018 App. D103.4 makes dead ends over 750 ft a special-approval item; NFPA 13D sprinklers are the accepted mitigation but add cost per unit. Arcade Rd is a 35-mph, 2-lane minor collector at 14,800 vpd (GDOT station 135-6689, 2017 — the latest retrievable). A left-turn lane is not warranted (44 lots is under the 75-lot trigger in Gwinnett DOT's 2007 criteria), but the entrance must clear 390 ft of intersection sight distance each way on a curved frontage, and a frontage-centered entrance would sit ~175 ft from Arcadia Place — under the 244-ft access spacing in Gwinnett UDO Table 900.3 — so the lane either aligns with Arcadia Place or shifts SW. The Killian Hill Rd widening (F-1262, Church St → Arcade Rd, 2 → 5 lanes) and the Arcade/Killian Hill intersection project (F-1262-03) are both only "Concept Complete" with no construction date, so no relief from through-traffic can be assumed in the plan horizon.
6. **Car-dependent by design.** Zero-vehicle households are 1–2 % in the tract and ZIP; transit use to work is under 1 %. Walk Score is 5. The one transit asset is real but thin: Ride Gwinnett Route 25 stops at Killian Hill Rd @ Arcade Rd, 0.30 mi away (weekday 50-min headways, Saturday hourly, no Sunday), which puts the site inside the ¾-mile ADA paratransit band — but the walk is along Arcade Rd with no verified sidewalk. Gwinnett Senior Services transportation is 60+ only, Lilburn has no microtransit zone, and the November 2024 transit referendum failed 53–47, so nothing better is funded. Parking at 2/unit + guest bays is not negotiable downward here the way it would be near transit; the site's walkability story rests on the internal loop and on the (unbuilt) Arcade sidewalk/Killian Hill sidepath, not on destinations.
7. **The hearing lands during a plan and code rewrite.** A 2026-09-25 filing meets PC on 10/22 and Council on 11/9 while the 2026 Comprehensive Plan Amendment is between DCA transmittal (2026-07-13) and adoption and the UDO diagnostic (2026-07-07) is under way. Staff may ask that the case wait for the new code; the risk is delay more than denial.
8. **Neighbors are organized on one side and individual on the other.** Legends at Parkview has an HOA (Liberty Community Management); King David Manor and Nantucket do not. Outreach has to be two different campaigns.

7.3 Cross-cutting connections

- **Environmental × history:** The 1970 King David Manor lots were built before the 1972 Clean Water Act stormwater regime and Lilburn's 1999 tree ordinance; the site's clay subsoil, the 303(d) listing and a petition that named runoff mean the stormwater exhibit will be read more closely than the elevations.
- **Demographics × history:** A 35 %-55+ tract with no HOA on two of three sides is also the tract most likely to supply the petition's signers *and* the cottages' buyers. The same households.
- **Environmental × mobility:** The stream pocket and the ridge that blocks gravity sewer are at the far end of the only lane; Phase 2 is the expensive phase on every axis — sewer, rock, fire access length and plan policy. Sequence the application so Phase 1 (the front ~1,000 ft of lane that drains by gravity) stands alone.
- **Mobility × demographics:** Low car-free share and the L5A exemption argue for a buyer who drives, owns outright and stays — which is also the resident profile that generates the fewest peak trips (0.30/DU) and no students. Say so in the §1003-7 criteria.

8. Recommendations for the design team

1. **Take five questions to the pre-application conference and get the answers in writing before anything is drawn to scale.** (a) Disposition of RZ-2025-01 — tabled, not denied, so no §1003-11 bar (Ord. 2023-603); (b) the site's class on the 2026 draft Future Development Map and how "Do not further extend sewer in this area" (draft p. 73) is read against a Phase 2 line into an already-sewered subdivision versus an on-site lift station; (c) whether staff will hear the case before the UDO rewrite lands; (d) the cottage-home buffer reading (20 ft under Table 4.1; §313(1) coincidence with the rear yard); (e) with Gwinnett

DOT, whether the entrance must align with Arcadia Place or clear 244 ft from it. Items (b), (c) and (e) are new since FACTS was written.

2. **Design Phase 1 to stand alone and make the application about Phase 1.** Every hard constraint clusters at the rear: the ridge that blocks gravity sewer, the stream head and its 75-ft setback, the GgE2 slopes, the rock under Pond 2, the far end of a 1,700-ft dead end, and the draft plan's sewer policy. The front ~1,000 ft drains by gravity to the existing outfall manhole (invert 927.13), sits on HSG B Appling soils and — by derivation from FACTS §4 (1,000 ft of lane, 50-ft lots both sides, less the amenity block at u 20–220 and Pond 1 at u 760–940 on the SW side) — holds on the order of 25 lots plus the amenity block; verify against the site plan. Show Phase 2 as a future phase with its sewer route unresolved rather than letting it carry the case.
3. **Spend the first engineering dollars in the ground, not on renderings.** In addition to the boundary/topographic survey and sewer-invert survey already planned: rock probes at both pond sites and the mid-swale lane crossing (ARE, gneiss at 22–40 in); test pits and an undercut plan at the front SW corner (HYB Helena, water table ≈24 in) where the amenity block, entry pavement and sewer tie-in go; infiltration tests (expect poor on eroded kaolinitic clays — plan bioretention with underdrains); a field delineation of top-of-bank at the stream head; and a certified-arborist inventory of ≥12-in trees in the rear woods for the 16-TDU/acre credit. These decide whether the 2025 hydro basis (10,000 cf per disturbed acre) is achievable and where the lot line at the rear really falls.
4. **Make stormwater and the creek woods the centerpiece exhibit, and run two separate neighbor campaigns.** The receiving stream is 303(d)-listed for urban runoff; the cross-slope falls toward Village Green Ct; and "stormwater runoff, erosion, flooding" was the one 2025 objection a HOPA detached plan does not answer by itself. Show, in the application, that no pond outfalls toward the Legends lots, that the WQv is treated, and that the stream pocket is a recorded conservation area. Then brief the Legends at Parkview HOA (Liberty Community Management) as an organization, and King David Manor and Nantucket — which have no HOA — house by house; the same 55+ households that signed the petition are the cottages' buyers.
5. **Let the climate and the access constraints shape the house and the lane.** NE-side lots front the June sunset (azimuth 267–299°, 25° altitude at 5 pm): give those fronts 8-ft porches with vertical screening and canopy trees, low-SHGC glazing on W/SW walls, and put the primary outdoor room on the NE side of every plan that allows it. Build every house slab-on-grade, zero-step, radon-ready (Zone 1), termite-detailed (very heavy zone) and NFPA 13D-sprinklered (the dead-end mitigation); put electric underground and make the clubhouse generator-ready (2014 ice-storm benchmark). Build the 5-ft frontage sidewalk and ask the county/city to close the ~0.3-mi gap to the Route 25 stop at Killian Hill — the cheapest credible "transit-served 55+" claim available. Keep 2 spaces per unit plus guest bays; at Walk Score 5 and 1–2 % car-free households, parking is not a negotiating chip here.

9. Not available / to verify

- Post-2017 GDOT AADT for Arcade Rd (station 135-6689) — check TADA manually; a developer 24-hr count will be required for the driveway permit in any case.
- Sidewalk presence along the Arcade Rd frontage and to Killian Hill Rd; Walk Score Transit Score; Gwinnett fire/EMS response-time statistics.
- Killian Hill Rd widening (F-1262) scope and schedule beyond "Concept Complete"; whether it adds sidewalks or turn lanes at Arcade.
- The site's exact class on the 2026 draft Future Development Map (inferred Suburban-Low from p. 51 text and location south of the CSX line); whether the UDO rewrite changes R-2 cottage standards; adoption date of the plan amendment.
- Formal disposition of RZ-2025-01 (tabled vs withdrawn vs dormant) — written confirmation from the Planning Director.
- 4535 Arcade Rd septic vs sewer status; qPublic deed detail (site returned 403); Legends at Parkview builder; the adjoining "Arcade Road Townhomes" R-2 case number and year; Camp Creek ES address conflict; Norcross Senior Center distance.
- Gwinnett-specific 65+ forecasts from ARC Series 17; Lilburn months-of-supply; NAHB 55+ HMI; Gwinnett tree-canopy percentage from a primary source; ZIP-level radon test statistics; monthly temperature normals for a Gwinnett station.
- USFWS IPaC species screen (not run); Georgia's adopted ASCE 7 edition (SDC B expected under either).
- Property-tax illustration in §5.5 is derived from published 2025 millage and 2026 exemptions; verify with the Gwinnett Tax Commissioner and the City of Lilburn once the 2026 rates are adopted (city rate hearing 2026-08-31).

10. Principal sources

Every figure above carries its own link; the primary sources are listed once here. All accessed 2026-08-28 unless a vintage is given.

Project facts: [status/cottage-submission-2026-08-28/FACTS.md](#) (Gwinnett GIS parcels/zoning/streets/sewer/streams/floodplain/soils; USGS 3DEP; Lilburn Ord. 2023-603; 2026 Application Instructions; 2024

Comprehensive Plan).

Environment: NCEI U.S. Climate Normals 1991–2020, Atlanta USW00013874 and Norcross USC00096407; ASHRAE 2021 HOF station 722190; ASHRAE 169-2013 Add. a; FEMA NFHL; USGS ASCE 7-22 design values; Georgia Radon Program fact sheet 2024; SPC tornado database 1950–2024; NWS Peachtree City severe-weather climatology and Feb 2014 ice storm; EPA Green Book, Georgia; NRCS Soil Data Access; USGS WBD; GA EPD 2024 305(b)/303(d); GA EPD state-waters buffer field guide 2017; MNGWPD model stream-buffer ordinance; Lilburn Code Ch. 109 Art. V (Municode).

Mobility: Walk Score; Ride Gwinnett GTFS (feed 2026-07-02), routes & schedules, paratransit, microtransit; Gwinnett Senior Services; Xpress; ARC Open Data — GDOT AADT 2008–2017; Gwinnett speed-zone ordinance list (2025-09); 2017 SPLOST project handout (May 2025); 2023 SPLOST project handout (Sept 2025); Gwinnett DOT left-turn-lane criteria (2007); Gwinnett UDO §900-40; Camp Creek Greenway; Gwinnett trails in the works; OpenStreetMap (Overpass, Nominatim, OSRM).

Demographics and market: data.census.gov ACS 2020–2024 5-year (DP02/03/04/05, S0101, S1810, S2502); Census Geocoder; Census PEP V2025; ARC population forecasts; NAHB age-restricted housing characteristics (2026-07); Zillow Lilburn; Redfin Lilburn; Gwinnett Tax Commissioner millage and exemptions; City of Lilburn taxes; BLS LAUS; GDOL Gwinnett profile; Niche — Parkview HS; 55+ community listings (55places, Jome, Homes by Marco).

History and planning: Gwinnett Tax Assessor 2026 digest (data/owners-found.json); 2024 Comprehensive Plan; 2026 Comprehensive Plan Amendment DRAFT 6.16.26; Lilburn Agenda Center (PC and Council 2025–2026); RZ-2024-01 staff report and Council minutes 2024-06-10; Preserve at Killian Hill Ph. 3 staff report; Ord. 2025-657; Ord. 2026-670; Gwinnett 2045 Unified Plan; Change.org petition, RZ-2025-01; AJC, Gwinnett Daily Post and Patch coverage linked in §6; listing portals (Movoto, Zillow, Redfin, Homes by Marco).

Disclaimer: *This is an AI-generated analysis for preliminary planning purposes. All findings must be verified by a licensed professional before use in design, permitting, or regulatory submissions.*

title	Submittal Checklist and Roadmap — Cottages at Arcade Springs
date	2026-09-03
address	4535, 4537, 4539 & 4541 Arcade Rd SW, Lilburn, GA 30047 — Land Lot 123, 6th District, Gwinnett County (PINs R6123 033 / 015 / 014 / 162)
status	DRAFT — owner-prepared roadmap; costs and lead times are planning allowances to be replaced by quotes; items marked NEEDS PROFESSIONAL are DRAFT until sealed
generator	SUPERSEDED — tools/roadmap.py wrote the 2026-08-29 v1 of this file and has not been maintained since. It still asserts 43 lots and 4.56 du/ac and aborts, and if that assert were removed it would overwrite this document with the v1 text (no items 20–25, renderings/ described as empty). Do NOT run it. This file is maintained by hand against data/layout.json, data/plans.json and data/dev-summary-derived.json; the checkers that police it are listed at item 26.

10 — Submittal Checklist and Roadmap

Application: R-1 → R-2 (Medium-Density Residential), Lilburn Zoning Ordinance 2023-603 §402, for **41 one-story detached cottage homes** (55+ HOPA, voluntary condition) on **9.44 ac deeded** (9.577 ac by GIS; the RLS survey governs) — **4.34 du/ac** on the deeded acreage (4.28 du/ac on the GIS acreage), against the R-2 maximum of 8. **Lot count, corrected 2026-09-03:** the site-plan generator was re-run against the defect list in audit-2026-09-03/site-geometry.md and **the yield fell from 43 lots to 41** — one lot fell inside a stream buffer once every lot was screened against all three digitised reaches (the slot at u 1,230–1,280 on the south-west side is now a 5,000-SF stream-setback open-space tract), and one more was given up to enlarge the detention basins, which had been sized on 4.4 disturbed acres when the real figure is 7.54. Every number in this document is read from the corrected data. **Governing lists:** City of Lilburn 2026 Application Instructions — Rezoning/SUP/CIC, items (1)–(11) (data/application-instructions-2026.txt); Lilburn Zoning Ordinance 2023-603 §1003-4 items 1–6, §1003-6 notice, §1003-10 withdrawal, §1003-11 refiling bar (data/ordinance-excerpts.md). **Numbers:** data/layout.json and data/plans.json, recomputed by tools/dev_summary_calcs.py → data/dev-summary-derived.json and by tools/hoa_budget.py → data/hoa-budget-derived.json; external facts re-verified 2026-09-03 in audit-2026-09-03/external-facts.md. **Sheet numbers are those of the register in tools/transmittal.py SHEETS and no others** — verify/check_sheet_refs.py fails on any number not in it. Local coordinates **u** (along the strip from the Arcade Rd R/W corner, N 28°43' W) and **v** (across, + toward the NE line) are those of FACTS.md §1.

How to read the status column. READY DRAFT = the package contains a complete owner-prepared draft; NEEDS PROFESSIONAL = a licensed Georgia RLS / PE / architect / attorney must produce or seal it before filing; NEEDS SIGNATURE = the content is drafted and only signatures and notarization remain; TO BUILD = the item is identified and not yet produced; VERIFY = a fact or a decision outside the package data must be settled first. Cost and lead-time ranges are the planning allowances supplied with the 2026-08-29 roadmap brief (survey \$8–18k / 3–6 weeks; MCP seal \$8–25k; TIS \$6–15k; attorney \$7.5–25k; renderings \$3–15k); **none is a quote — VERIFY each with a written proposal.**

1. Checklist — the 11 Instruction items, the §1003-4 items, and the practical extras

#	Item (s ource)	What it is	What THIS package provides (file)	Who must p roduce / seal	Status	Est. cost	Lead time
1	Applicat ion Form (Instr. item 1; §1003- 4.2)	The 7-page city form (DocumentCenter/View/193): applicant, owners, property, proposed development, attachments list; typed or black ink	docs/01-application-fo rm-data.md — every field with its entry and basis (case RZ-2026-__ left blank)	Owner (applica nt) tran scribes; attorne y reviews the two- status rider (owner of 033/ 015 + contract purchas er of 01 4/162)	READY DRAFT — owner-status rider VERIFY (executed contract for 014/162). The form offers three status boxes — Owner's Agent / Property Owner / Contract Purchaser	\$0	1 day to transcri be

#	Item (source)	What it is	What THIS package provides (file)	Who must produce / seal	Status	Est. cost	Lead time
2	Application Fee (Instr. item 2; §1003-4.1)	Non-refundable fee, check payable to City of Lilburn. Rezoning, 5.0–9.9 ac = \$1,250 under both the superseded FY2025-2026 schedule and the FY2026-2027 Fee Schedule adopted by Resolution No. 2026-08 on 2026-08-31 , which is the operative authority (§1003-4.1: fee "as established by resolution of the City Council")	Amount, tier, authority and payee stated in docs/01-application-form-data.md	Owner	READY DRAFT — the adopted FY26-27 schedule is days old and not yet posted on the city site; confirm the amount at the counter on filing day (question P8 in §3)	\$1,250	At filing
3	Standards Governing Exercise of the Zoning Power (Instr. item 3; §1003-7 criteria A–F)	The form's six rezoning criteria; two lines each on the form, answers attached	docs/04-standards-governing-zoning-power.md — criteria (A)–(F), 290–300 words each (tools/check_docs.py verifies the count)	Owner; attorney review recommended (the form cites the 2011 Resolution and §1003-8 — legacy citations, noted in the document)	READY DRAFT	in attorney allowance	Done; 1 wk attorney review
4	Conflict of Interest Form (Instr. item 4; O.C.G.A. §36-67A-3)	Campaign-contribution / gift disclosure (\$250+ to any Lilburn official in the prior 2 years) and certification; due within 10 days of filing (10/05 if filed 09/25; 11/09 if filed 10/30)	docs/07-conflict-of-interest-and-certifications.md §§1–2 (content; declarations left to the signer). The roster is confirmed: Mayor Johnny Crist; Council Christina van Maanen (P1), Scott Batterton (P2), Michael Hart (P3), Emil Powella (P4); Planning Commission James Hampton (Chair), Beanie Danos (Vice Chair), Peter Sirbu, Michael Delashmit, Kenneth Stephenson	Owner/applicant signs; attorney co-signs if representing (§36-67A-1(1)) makes the attorney an "applicant")	NEEDS SIGNATURE — the contribution history is a fact only the signer can declare	\$0	Same day as filing
5	Notarized Signatures (Instr. item 5; §1003-4.2)	Applicant certification + property-owner certification for each owner of record: Awad (033, 015); Mendez and Roblero de Leon (014, 162) — "an attachment, if multiple owners"	docs/07-conflict-of-interest-and-certifications.md §§3–6 (who signs for which parcel; signing schedule; cooperation covenant)	Owners + applicant before a Georgia notary; attorney prepares the consent or contract	NEEDS SIGNATURE — hard acceptance gate : without the 014/162 owners' signatures or written consent the application is not filed by an owner, agent or consenting contract purchaser (§1003-2) and the Director may refuse to process it (§1003-3.2)	notary nominal (VERIFY)	1 sitting; schedule the Mendez / Roblero signing in week 3

#	Item (source)	What it is	What THIS package provides (file)	Who must produce / seal	Status	Est. cost	Lead time
6	Letter of Intent (Instr. item 6; §1003-4.5)	Describes the request and its justification; must state whether any buffer reduction is requested ; lists voluntary conditions	docs/03-letter-of-intent.md — states the one buffer reduction requested (the southwest entrance curb return, where the 20-ft buffer is reduced by 2'-0" to a residual 18'-0" over a plan area of 0.1 SF — data/layout.json metric s.sw_curb_return_buffer_encroachment_ft/_sf), and reproduces the 18 voluntary conditions verbatim from data/voluntary-conditions.md	Owner signs; a attorney reviews the conditions and the HOPA language	READY DRAFT — attorney review; phone / e-mail / contract-status brackets VERIFY. Voluntary condition 4 was conformed to 18'-0" on 2026-09-03; check that §5's buffer-reduction paragraph states the same 18'-0" and the same 0.1-SF area — a letter and a drawing that disagree on the residual width is the document-versus-drawing contradiction the completeness audit called a filing defect	in attorney allowance	1 wk review
7	Legal Description (Instr. item 7; §1003-4.3)	Typed metes-and-bounds description of only the property to be rezoned	docs/05-legal-description-DRAFT.md — GIS-fabric metes and bounds with per-segment bearings and distances (tools/legal.py, verify/legal-output.md); marked DRAFT	Georgia RLS — the sealed survey description replaces the draft	NEEDS PROFESSIONAL	in survey allowance	3–6 wks (with item 9)
8	Site Plan (Instr. item 8; §1003-4.6)	"Include one (1) full size copy drawn to scale and email a pdf electronic file to returner@cityoflilburn.com" — property lines, streets and R/W, buffers, setbacks, buildings, driveways, parking, lots, floodplain, adjoining zoning and owners, existing and proposed topography, landscaping, drainage	Sheet C-2.0 drawings/mcp-sheet.svg / .png (Master Concept Plan, ARCH D 36 × 24 in at 1" = 60'), mcp-web.svg, mcp-front.png, mcp-rear.png; Sheet C-1.0 drawings/existing-conditions.svg / .png; full-size PDF at submission/C-2-0_master-concept-plan_36x24.pdf; data in data/layout.json (41 lots , travelled way 1,754 ft , common open space 141,089 SF = 3.24 ac = 33.8 % of the GIS area / 34.3 % of the deeded area in eight tracts, impervious 184,148 SF = 44.1 % of the GIS area / 44.8 % of the deeded area , ponds 33,129 / 44,734 cf at 6-ft depth = 77,863 cf against a 75,410-cf screening requirement on 7.54 disturbed acres)	Georgia PE (civil) re-draws on the RLS base and seals; landscape architect for the planting and buffer plan (may follow at §1106 site & design review — VERIFY)	NEEDS PROFESSIONAL — the concept plan has been re-issued (turnarounds at stations 594.5 / 1,164.5 / 1,744.5 ft, 6-ft pond basins, real house footprints, entrance curb returns drawn, all three stream reaches screened); existing topography is USGS 3DEP and approximate. Nine C-series sheets in the register are still unissued: C-0.0, C-2.1, C-2.2, C-2.3, C-2.4, C-3.0, C-4.0, C-6.0, C-7.0 and C-8.0	\$8,000 –25,000	2–4 wks after the survey
8a	Reductions of the site plan	The 2026 Instructions require one full-size copy and a PDF by e-mail — they do not require five 8.5" × 11" reductions; that requirement comes from the older application form	11 × 17 reductions and a 3-panel 8.5" × 11" key-plan series (a straight 36 × 24 → 8.5 × 11 shrink is ≈ 1" = 204' and illegible)	Owner	TO BUILD — bring five reductions as a courtesy, but do not represent them as an Instructions requirement	printing nominal	1 day

#	Item (source)	What it is	What THIS package provides (file)	Who must produce / seal	Status	Est. cost	Lead time
9	Boundary Survey (Instr. item 9; §1003-4.4)	Survey plat sealed by a registered land surveyor; may be combined with the site plan if the plan carries the survey information. §1003-4.4 lets the Director accept a Tax Assessor map "where no survey is available" — do not rely on it	None — data/site-parcels-2240.json holds the GIS rings the RLS can start from; drawings/existing-conditions.svg shows what the survey must confirm	Georgia RLS (boundary, acreage, the 234.7–246.4-ft width at every lot row, sewer inverts, stream top of bank, Arcade Rd sight distance)	NEEDS PROFESSIONAL — the width decides the 50' × 100' lot section against the 50' × 82' fallback, and the margin is now 0.1 ft : 100 + 35.9 (minimum lane tract) + 100 = 235.9 ft against a 236.0-ft strip at the first lot row	\$8,000 –18,000	3–6 wks — the critical-path item
10	List of Adjoining Property Owners (Instr. item 10; §1003-6.3 mailing)	Names and mailing addresses of all adjoining owners from the Tax Assessor; the City mails hearing letters to them	docs/06-adjoining-property-owners.md — 31 privately owned tax parcels plus the Gwinnett County Arcade Rd right-of-way = 32 notice recipients (tools/adjoiners.py, verify/adjoiners-output.md)	Owner — re-run the extract on the filing date	READY DRAFT — re-pull at filing. Ask the Director to reconcile the County digest against the City tax records §1003-6.3 names, so no notice is defective	\$0	1 hour
11	Architectural Renderings and Elevations (Instr. item 11)	An elevation or rendering of each side of each structure visible from the street , to scale or in proper perspective, with colours and materials of all structures and roofing and the location and size of wall signs	Complete in the package: drawings/plan-a-elev.svg/.png (Sheet A-2.1 , Plan A, four elevations at 3/16" = 1'-0"), drawings/plan-b-elev.svg/.png (Sheet A-2.2 , Plan B), drawings/plan-colors.svg/.png (Sheet A-2.3 , three colour schemes and the material schedule), drawings/amenity-sheet.svg/.png (Sheet A-3.0 , clubhouse, mail kiosk and monument entry sign — plans, elevations and details); floor-plan sheets A-1.1 / A-1.2 and typical-lot sheets A-4.0 / A-4.1 ; twelve illustrative perspectives in renderings/ (R-01 ... R-12) with renderings/PROMPTS.md; and the item-11 narrative at docs/09-architectural-character-elevations-and-renderings.md, which states that no wall signs are proposed on any structure	Architect reviews and seals	READY DRAFT — every structure the applicant proposes now has scaled elevations of every visible side; see item 11a for what is left	\$3,000 –15,000 (the architect's scope is now review and seal rather than origination — VERIFY)	2–4 wks

#	Item (source)	What it is	What THIS package provides (file)	Who must produce / seal	Status	Est. cost	Lead time
11a	Clubhouse, mail kiosk and entry-sign elevations	Item (11) applies to every structure the applicant plans to erect. The 2,400-SF (60'-0" × 40'-0") clubhouse is the building most visible from Arcade Rd, and the monument sign is the "sign" the instruction names	BUILT. drawings/amenity-sheet.svg (Sheet A-3.0 , ARCH C 24 × 18): clubhouse floor plan, roof plan and four elevations (2,400 SF under roof = 2,000 SF enclosed + a 400-SF covered porch; plate 10'-0" + 0'-10" heel; 40'-0" span at 6:12 → ridge 20'-10" above the finished floor / 21'-6" above finished grade , under the 24'-0" of condition 3; 114.8 ft from the Arcade Rd R/W; occupant load 71, three exits); mail kiosk 16'-0" × 10'-0", 4:12 hip, ridge 11'-7" above grade, 3 CBUs = 48 compartments + 6 parcel lockers, with the checklist \$4.r weather-protection note; monument sign with its area computation — whole face 8'-0" × 3'-9" = 30.0 SF against the 32-SF voluntary limit, overall height 3'-9" against the 6'-0" limit, 19.0 ft from the R/W, and the 390-ft departure sight triangle drawn with the sign 20.1 ft clear of the nearer sight line and wholly outside it	Owner-produced at concept level; architect seals	ISSUED — no longer blocking. Three items remain, none of them an elevation: the sheet still prints "A-3" rather than A-3.0 in its title block, still cross-refers to "Sheet C-1" for the site plan (the register makes that C-2.0 / C-2.3), and still carries a note reading "the 43 dwellings are sprinklered under NFPA 13D". Correct tools/amenity.py and regenerate	in the architect allowance	Done
12	Development Summary (voluntary; § 1003-4.6 "other information as reasonably required by the Director")	Site data report in the §736 order so staff sees the same content as in 2025	docs/02-development-summary.md (tools/dev_summary_calcs.py → data/dev-summary-derived.json)	Owner; PE confirms the engineering figures	READY DRAFT — the calculator was re-run 2026-09-03 against the 41-lot layout; verify/check_dev_summary.py tests the document against the data	\$0	Done
13	Technical memorandum (voluntary — support for §1003-7(4) "burden some use of streets, utilities, schools")	Traffic, sewer gravity and route, water, stormwater, fire, schools, environmental screen	docs/08-technical-memoranda.md memos A–G	Georgia PE supervised A–E; GCPS and GA EPD correspondence superseede F–G	READY DRAFT (feasibility screens only)	in PE allowance	With item 8

#	Item (source)	What it is	What THIS package provides (file)	Who must produce / seal	Status	Est. cost	Lead time
14	Voluntary zoning conditions (§1003-4.5 "special conditions voluntarily made")	18 conditions: 1 age restriction (HOPA, 33 of 41 units), 2 maximum 47 units, 3 height, 4 buffers including the single reduction requested , 5 fire sprinklers (voluntary offer), 6 single access, 7 no Arcade Rd driveways, 8 architectural standards, 9 stream buffer, 10 HOA maintenance, 11 lighting, 12 construction hours, 13 tree survey, 14 entry sign, 15 phasing, 16 sanitary sewer (gravity only), 17 traffic impact study and access management, 18 sidewalks and transit	data/voluntary-conditions.md, reproduced verbatim in docs/03 §11 and docs/02 §9	Owner offers; attorney reviews enforceability (the HOPA covenant is recorded before the first permit)	READY DRAFT — attorney review. Conformed to the 41-lot layout on 2026-09-03: condition 1 now reads "at least 33 of the 41 units" ($0.80 \times 41 = 32.8$, and 24 CFR §100.305(g) rounds the fraction in), leaving a cushion of 8; condition 2 "41 shown"; condition 4 the residual buffer of 18'-0" , a 2.0-ft reduction affecting 0.1 SF; condition 5 "at 41 units"; condition 8 the settled sheet register (A-1.1, A-1.2, A-2.1, A-2.2, A-2.3, A-3.0, A-4.0, A-4.1); condition 9 the new unlotted stream-setback tract at u 1,230–1,280; condition 15 the u ≈ 980-ft phase line with 24 lots in Phase 1 and 17 in Phase 2 , and the removal of the existing 4535 Arcade Rd dwelling, which stands on the lane alignment about 1,200 ft back. Condition 3's ridge figures (17'-10" Plan A, 19'-0" Plan B above the finished floor; 18'-6" and 19'-8" above grade) match the issued sheets — an earlier version of this row said the sheets drew 19'-4"; they do not. 19'-4" is what data/plans.json still prints, and that file is what needs conforming (docs/09 §5.3 item G-1)	in attorney allowance	1 wk
15	Sewer capacity certification (practical extra; Gwinnett DWR form Rev. 07/2023; condition 16)	DWR's written confirmation that the receiving system can accept 10,670 gpd average daily flow ($41 \times 250 \text{ gpd} = 10,250 \text{ gpd} + 420 \text{ gpd}$ for the 2,400-SF clubhouse, per the <i>DWR Wastewater Flow Estimation Guidelines</i> Rev. 10/2022; peak $\times 4.0 = 42,680 \text{ gpd} = 29.6 \text{ gpm}$; Phase 1 6,420 gpd, Phase 2 4,250 gpd). Phase 1 by gravity to the on-site manhole (invert 927.13 at u ≈ 272, v ≈ -224 per GIS); Phase 2 by the ≈ 178-ft off-site gravity tie to the Legends at Parkview main (invert 919.58)	docs/08-technical-memoranda.md Memo B; docs/02-development-summary.md §5.2	Georgina PE signs and seals the request; submit to DWR Capacity Certification @GwinnettCounty.com	NEEDS PROFESSIONAL — file on the form's " Pre-Rezoning " request type; inverts are GIS attributes until surveyed	no fee stated on the form (VERIFY with DWR)	10 business days minimum, plus 20 more if analysis or flow monitoring is required
16	DOT access coordination (practical extra; condition 17)	Gwinnett DOT (Arcade Rd is county-maintained): entrance in the SW third of the frontage (v = -190), separation 251.2 ft along the Arcade Rd centerline (250.7 ft chord) from the Arcadia PI centerline against the 244-ft minimum of Gwinnett UDO §900-40.5 Table 900.3; 390-ft intersection sight distance (§900-40.4 Table 900.2), drawn on Sheet A-3.0 for the entry sign; one driveway per 400 ft of frontage (§900-40.5(C)) → one; left-turn-lane warrant 75 lots (GCDOT <i>Criteria and Guidelines for Left Turn Lanes</i> , 6/20/2007, Table 1) → not warranted at 41	docs/08 Memo A; docs/02 §6.2; docs/11-site-content-brief.md	Georgina PE requests the pre-application determination; RLS measures sight distance	VERIFY — written DOT determination on the study level and the turn lane; the driveway permit is an LDP-stage item	\$0 now (the study is priced at item 20)	2–3 wks for a written response (VERIFY)

#	Item (source)	What it is	What THIS package provides (file)	Who must produce / seal	Status	Est. cost	Lead time
17	School letter (practical extra; §1003-7(4))	GCPS planning's projection of students (the July 2025 letter projected +57 for 104 condominiums); a HOPA 55+ community generates 0	docs/08 Memo F; docs/02 §7; 2025 letter status/rz-2025-01/4817_school-impact-analysis.pdf	City requests from GCPS after filing; owner asks the Director at pre-application to request it early	VERIFY — who requests it and when; the HOPA covenant's minor-occupancy clause (attorney)	\$0 expected (VERIFY)	2–4 wks after the City's request (VERIFY)
18	Community record (practical extra; supports §1003-7(1) and the hearing)	Evidence that neighbours were informed: public site updated from the 2025 MU concept to this proposal, comment-board log, neighbour-meeting notices and sign-in sheets for King David Manor / Legends at Parkview / Nantucket, sign photographs, tear sheets	docs/03-letter-of-intent.md (commitment to meet before the PC hearing); public/cottages.html (unlisted concept study); §2.2 below (notice audit)	Owner	VERIFY — the public-site update is not yet made; public/cottages/submission/ is a stale mirror and must be rebuilt before it can serve as the record	printing and postage nominal	Weeks 5–8
19	Concurrent variance (§1005) — fallback only	Lot-depth variance to 82 ft if staff requires the 20-ft buffer in a separate tract instead of the §313(1) easement-in-lot basis (criteria §1005-3.2 a–g: exceptional narrowness). Needs a third, shallower house plan — the buildable depth on an 82-ft lot is 41'-0" and neither Plan A (51'-10") nor Plan B (57'-6") fits. Sheet C-2.4 in the register is the fallback layout exhibit and is not yet drawn	Not drafted — the design basis avoids it (Q2 below)	Owner/attorney	VERIFY at pre-application — draft only if staff rejects the §313(1) reading	\$500 ZBA hearing fee (+\$25 each additional variance on the same property), FY26-27 schedule	Same cycle as the rezoning if staff allows

#	Item (source)	What it is	What THIS package provides (file)	Who must produce / seal	Status	Est. cost	Lead time
20	Level 1 Traffic Impact Study (Gwinnett DOT Traffic Impact Study Guidelines, 2023; condition 17) — NEW, and required	The Guidelines state "A TIS must be submitted concurrently with all zoning applications and land disturbance development," and exempt only "single-family detached and townhome residential development consisting of no more than seven lots or units. " At 41 lots the project is not exempt. At 0–20 projected peak-hour trips — the ITE 11th ed. LUC 251 band at 41 units is 10.7–16.0 PM peak trips, 12.3 at the FACTS rate , and 152–177 daily — the level is Level 1 (document-only) : location description, existing and proposed land use, trip generation, access-management review, adjacent access spacing upstream and downstream, intersection sight distance, connectivity and circulation, functional classification, posted speed and future identified projects. Operations analysis, turn-lane warrants and signal warrants begin at Level 2 and are not required	docs/08 Memo A already contains every Level 1 element; it must be re-issued as a standalone, PE-signed deliverable	Georgia PE / traffic engineer	TO BUILD — file with the application	\$6,000 –\$15,000 (the roadmap brief's TIS allowance; a document-only Level 1 study should price at the bottom of that range — VERIFY with a written proposal)	3–5 wks — start in week 1
20a	New 24-hour traffic counts on Arcadio Rd	The same Guidelines require counts "less than two years old" from GCDOT or GDOT and, failing that, that "the developer shall be responsible for obtaining new, 24-hour counts, from an approved contractor, as directed by GCDOT." The package's count is GDOT station 135-6689, 2017 — nine years old and ineligible	Nothing — the 14,800 AADT figure in every document is the 2017 count and must be labelled as such until replaced	GCDOT-approved count contractor	TO BUILD — precedes the Level 1 study	not in package data — VERIFY with a quote	schedule in week 1; fieldwork is short but the queue is not (VERIFY)
21	Cover sheet and sheet index	The Site Development Plan Review Checklist §1.e.1 is the house standard staff reads: project name and use, address / District / Land Lot / PINs in the title block, owner and 24-hour contact, every professional's name and contact, sheet index , site location map, zoning and use data block, and a blank "ZONING CASE NO. RZ-2026-____" line	docs/00-transmittal-and-index.md performs the transmittal function; there is no cover sheet on the drawing set	Owner drafts; PE seals with the set	TO BUILD	\$0	1 day

#	Item (source)	What it is	What THIS package provides (file)	Who must produce / seal	Status	Est. cost	Lead time
22	Existing Conditions sheet	§1003-4.6 asks for "existing roads and streams, flood plains, existing and proposed buildings and structures ... areas of existing vegetation" — elements the concept plan carries only as text	drawings/existing-conditions.svg / .png — boundary and the four interior parcel lines, 2-ft contours, both existing houses and their accessory structures drawn in place with the "to be removed" note, the tree line, the stream with its 25 / 50 / 75-ft lines, the existing 8-in sewer and 20-ft easement with GIS inverts, the water main and hydrant, the Arcade Rd R/W, adjoining parcels with PIN, owner and zoning, and the FEMA note. Only 4541 Arcade Rd stands near the frontage (footprint centred about u 203, v -174); 4535 Arcade Rd sits about 1,200 ft back, at about (u 1,202, v -114), on the ridge and directly on the proposed lane alignment — a Phase-2 demolition standing on the lane, not a frontage structure (FACTS.md §2, corrected 2026-09-03)	Owner-prepared; RLS seals the survey version	ISSUED as Sheet C-1.0 and bound at submission/sheets/C-1-0_existing-conditions.pdf. Two corrections remain: the SVG title block still reads C-0 and its cross-reference still calls the master concept plan C-1 — under the settled register those are C-1.0 and C-2.0 . Correct tools/existing_conditions.py and regenerate	\$0 (in the survey allowance when sealed)	Done
23	docs/09 — Architectural Character, Elevations and Renderings	The item-11 narrative binding the sheets together: every proposed structure × which sides are street-visible × the sheet that shows them; the colour and material schedule; the roof-material statement; the sign schedule (one monument sign, drawn at 30.0 SF and 3'-9" against voluntary limits of 32 SF and 6'-0"; no wall signs)	docs/09-architectural-character-elevations-and-renderings.md	Owner; architect confirms	READY DRAFT — re-issued 2026-09-03 at 41 lots: §1.1 now records the sheet register as settled, §2.3 and §2.4 record the clubhouse, mail kiosk and monument sign as issued on A-3.0 , §4.2 re-cuts the colour-scheme distribution for 41 lots (14 / 14 / 13, largest 34.1 % against the 40 % cap), and §7.3 lists all twelve perspectives as generated	\$0	Done
24	docs/13 — HOA, HOPA and covenant outline	The recorded declaration is the enforcement mechanism for voluntary conditions 1, 4, 5, 9, 10 and 16, and the City will ask for it: the HOPA age-verification programme (≥ 80 % of occupied units, i.e. 33 of 41 under 24 CFR §100.305(a) with the fractional-unit rule of §100.305(g), leaving a cushion of 8, with published policies, biennial verification and the 24 CFR §100.306(b) marketing-language bar); the buffer/conservation easement language that makes the §313(1) reading work; lane and stormwater maintenance; the sprinkler-maintenance covenant; and a line-item 41-lot HOA budget in place of the "\$200–\$350/month" range	docs/13-hoa-hopa-and-covenant-outline.md, with every budget quantity computed by tools/hoa_budget.py from data/layout.json and data/plans.json into data/hoa-budget-derived.json and verify/hoa-budget-output.md. The published figure is \$352 per Lot per month at full build-out (41 Lots), \$173,374 per year; \$449 per month if Phase 2 is not built. The 43 → 41 correction raised the dues \$14 a month, because the budget fell only 0.5 % while the number of Lots paying it fell 4.7 % (docs/13 §8.4)	Owner drafted; attorney reviews	READY DRAFT — attorney review	in attorney allowance	Done

#	Item (source)	What it is	What THIS package provides (file)	Who must produce / seal	Status	Est. cost	Lead time
25	PDF binder	Instruction item (8) requires a PDF by e-mail, and the counter copy is easier to review bound and bookmarked	submission/RZ-2026_cottages-at-arcado-springs.pdf (the whole submittal in Instruction-item order), submission/C-2-0_master-concept-plan_36x24.pdf (full size, for the e-mail), and submission/sheets/ + submission/docs/ (per-sheet and per-document PDFs), built by tools/binder.py	Owner	READY DRAFT — rebuild after every document or sheet change. tools/binder.py ORDER already uses the settled register and now includes A-3-0 ; the eight sheets currently bound are A-1-1, A-1-2, A-2-1, A-2-2, A-2-3, A-3-0, C-1-0 and C-2-0. Two gaps: the binder order omits A-4.0 / A-4.1 (the typical-lot sheets, which are issued), and the numbers printed inside amenity-sheet.svg and existing-conditions.svg still disagree with their file names (items 11a and 22)	\$0	Minutes to rebuild
26	Package self-checks (practical extra; not filed)	Four scripts test the documents against the data rather than against numbers frozen into the scripts. They are the answer to the failure that tabled RZ-2025-01: a document and a drawing that disagree	verify/check_sheet_refs.py — pulls every "Sheet X-N.N" reference out of every file in docs/ and tests it against the register in tools/transmittal.py SHEETS and against the files actually in drawings/; fails on any number the register has never heard of, and on any citation of an unissued sheet that the sentence does not flag as unissued. verify/check_dev_summary.py — tests docs/02 against data/layout.json and data/dev-summary-derived.json (about thirty numbers), plus the FACTS.md §6 house rules (the disclaimer block and marker, the ban on the certainty verb, no pre-2024 IFC edition). tools/check_docs.py — the same house rules for docs/03 and docs/04, plus the criteria word counts and the verbatim condition text. tools/hoa_budget.py — recomputes the whole HOA budget from the layout and fails if docs/13 does not carry the totals it computes	Owner runs all four before every rebuild of the binder	ONE FAILING, as of 2026-09-03. tools/hoa_budget.py PASS (41 lots; docs/13 carries every total it computes). tools/check_docs.py PASS (docs/03 and docs/04; criteria at 290–298 words; layout 41 lots, 4.34 du/ac). verify/check_dev_summary.py PASS (31 data-driven values in docs/02 against data/layout.json and data/dev-summary-derived.json). verify/check_sheet_refs.py FAIL — it resolves 109 references and rejects five, all in docs/03 §§3, 4, 5 and the attachment list, which cite C-6.0, C-2.3 (twice), C-2.4 and C-8.0 as though those sheets were in hand. They are in the register and are not drawn, so each sentence must either say so (the phrasing the script accepts, and the phrasing docs/02 and the voluntary conditions already use) or the sheet must be built before filing. docs/00, docs/02, docs/09, docs/10 and docs/13 contribute no failures. Re-run all four and record the result at the go/no-go	\$0	Seconds

Package files not on the city's list but part of the record: FACTS.md (basis of everything), docs/11-site-context-brief.md, docs/12-outline-specifications.md, the 2026-09-03 audits in audit-2026-09-03/, verify/ outputs and tools/ generators. Nothing in this package is sealed; every drawing and description carries the DRAFT label the house rules require.

2. Filing calendar and legal-notice audit

2.1 Two cycles — printed schedule only; both must be re-confirmed

The dates below are the Planning Commission Hearing Schedule printed in the 2026 Application Instructions. As of 2026-09-03 the city's public-meetings calendar publishes only **September 2026** (Council 9/14, ZBA 9/23, PC 9/24); **no October, November or December 2026 Planning Commission or Council meeting has been calendared**, and the schedule itself carries the footnote **"*Dates subject to change — special called meetings are common to avoid conflicts with national holidays." Confirm every date below with the Planning Director before relying on it.**

Milestone	Cycle A (target)	Cycle B (fallback)	Rule
Pre-application conference with the Planning Director (Reid Turner, 770-279-3715, reidturner@cityoflilburn.com)	by Fri 09/18/2026 — request it the week of 08/31	by Fri 10/23/2026	Instructions p. 1: "one week prior to submittal is highly recommended"
Filing deadline (340 Main St; fee \$1,250; one full-size site plan + the PDF by e-mail)	Fri 09/25/2026	Fri 10/30/2026	Instructions p. 2 schedule
Conflict-of-interest disclosure due	Mon 10/05/2026	Mon 11/09/2026	O.C.G.A. §36-67A-3(B): within 10 days of filing
Legal ad out (Gwinnett Daily Post, the official legal organ)	Tue 09/29/2026	Tue 11/03/2026	§1003-6.1: 15–45 days before each hearing; Instructions: ≥ 15 days before PC, ≥ 30 before Council
Letters mailed and sign posted	Wed 10/07/2026	Wed 11/11/2026	§1003-6.2/.3: sign and letters 15–45 days before the hearing
Planning Commission public hearing (7:30 PM, 340 Main St; work session 7:00 PM)	Thu 10/22/2026 — not yet calendared, confirm	Thu 11/26/2026 — Thanksgiving Day; the schedule flags it "subject to change"; not yet calendared, and the replacement date is UNRESOLVED	4th Thursday
City Council public hearing (7:30 PM; work session 6:30 PM)	Mon 11/09/2026 — not yet calendared, confirm	Mon 12/14/2026 — not yet calendared, confirm	2nd Monday
Last day to withdraw	before the Council legal ad (the 09/29 ad covers both hearings — effectively before 09/29)	before 11/03	§1003-10: no withdrawal once the Council hearing is advertised; no fee refund once scheduled
Day counts (computed)	ad → PC 23 days ; ad → Council 41 days ; sign and letters → PC 15 days (the bare minimum)	ad → PC 23; ad → Council 41; sign and letters → PC 15 days (bare minimum)	all inside the 15–45-day window — but a one-day slip of the sign or the letters makes the PC notice defective

Where the work stands on Thursday 2026-09-03 — week 1, day 4. The twelve-week path in §5.2 counts week 1 from Monday 08/31, so the go/no-go is fifteen days away. Done: the concept plan re-issued at 41 lots with all three stream reaches screened and the basins re-sized on 7.54 disturbed acres; Sheets C-1.0, C-2.0, A-1.1, A-1.2, A-2.1, A-2.2, A-2.3, A-3.0, A-4.0 and A-4.1 issued (ten of the register's twenty-one); all twelve perspectives generated; docs/00, docs/09 and docs/13 written; the PDF binder rebuilt; the four checkers in place. **Not yet started, and all four are on the critical path:** the RLS survey has not been ordered, the attorney has not been engaged, the pre-application conference has not been booked, and the 24-hour traffic counts have not been scheduled. The survey's 3–6-week clock has therefore not begun, which is what decides Cycle A against Cycle B.

Which cycle? Cycle A is reachable only if the RLS delivers the sealed survey and legal description within about 3½ weeks of a week-1 order (08/31 → ≈ 09/24), the 014/162 owner consent is executed by then, and the Level 1 study and the clubhouse/sign sheet are done. **Go/no-go on Fri 09/18/2026** — add "elevation set complete (including item 11a)", "Level 1 TIS delivered" and "PDF binder rebuilt" to its criteria. If any is missing, do not file on the Director's §1003-4.4 discretion; move to Cycle B (10/30). Cycle B's own risk is that its PC hearing falls on Thanksgiving Day and the replacement date is not yet published — **ask the City Clerk for the Oct–Dec calendar before choosing.**

A note on timing that cuts against waiting. The 2026 Comprehensive Plan Amendment was transmitted to DCA by Council action on 2026-07-13, the Regional Commission's 40-day report window closed on or about 2026-08-23, and adoption can be calendared for **2026-09-14, 2026-10-12 or 2026-11-09**. Its Suburban-Low policy "Do not further extend sewer in this area" is therefore likely to be adopted law by either cycle's Council hearing. Two of the five Planning Commissioners — Beanie Danos and Kenneth Stephenson — sat on the Amendment's steering committee. Phase 1 extends no sewer at all; that is the point to lead with, and voluntary condition 16 binds it.

2.2 Legal-notice audit — why and how

RZ-2025-01 was **tabled on 2025-08-28** because of "a clerical error in the legal advertisement for the public hearing via the Gwinnett Daily Post advertisement and on-site sign advertisement" (PC minutes, [status/rz-2025-01/2025-08-28_PC-minutes-item4709.pdf](#)); it was never heard. The City drafts and places the notice (§1003-6), but the applicant bears the delay, so audit every notice element before and after it runs:

Step	When (Cycle A)	What to check / keep	Rule
N1. Ask the Director for the draft ad text before it goes to the Gwinnett Daily Post	09/25–09/28	Time, place and purpose; property location (all four addresses: 4535, 4537, 4539, 4541 Arcado Rd SW; PINs R6123 033/015/014/162; Land Lot 123, 6th District); acreage 9.44; present zoning R-1 and proposed zoning R-2 ; both hearing dates and times; case number	§1003-6.1 lists these elements for applicant-initiated cases
N2. Confirm publication and buy two tear sheets ; obtain the publisher's affidavit copy from the City	09/29	Date of issue, page, wording identical to N1; file under verify/notice/	Instructions p. 2 item 5 (the Gwinnett Daily Post is the official legal organ)
N3. Count days from the publication date to each hearing	09/29	PC: 23 (must be 15–45); Council: 41 (must be ≥ 30 and ≤ 45)	§1003-6.1; Instructions
N4. Photograph the sign the day it is posted, date-stamped, showing the sign text and the Arcado Rd frontage; re-photograph weekly and after storms	10/07, then weekly to 11/09	Sign text: case number, both hearing dates and times, place; posted "in a conspicuous location on the property or in the public right-of-way along frontage"; replace immediately if damaged and re-photograph	§1003-6.2: 15–45 days before the hearing
N5. Count days from posting to the PC hearing	10/07	15 days — the minimum; if the sign goes up on 10/08 or later, tell the Director the same day (the remedy is to re-notice for the next cycle, not to hold a defective hearing)	§1003-6.2
N6. Verify the mailing list against docs/06 -adjoining-property-owners.md (32 recipients) and ask for a copy of the mailed letter and the mailing date	10/07	Letters go to the owner of record and all abutting owners at the tax-record address, ≥ 15 days before PC. §1003-6.3 says "as shown by City tax records"; our list is the County digest, as Instruction item 10 directs — ask the Director to reconcile the two	§1003-6.3(a)–(b)
N7. Check the agenda posting for the PC and Council meetings and that the case appears under the right number, address and zoning	10/15, 11/02	Screenshot the agenda with the URL and date	Instructions p. 2 items 4 and 6
N8. Attend both hearings in person (or by counsel)	10/22, 11/09	"The applicant must be represented" at both	Instructions p. 2
N9. Keep the whole record (N1–N8) in one folder and give the attorney a copy before each hearing	continuous	If the notice is defective and the City tables the case, no refiling bar applies (tabling is not a denial under §1003-11), but the cycle is lost	§1003-10, §1003-11

For Cycle B substitute: ad 11/03 (PC 23 days, Council 41 days), sign and letters 11/11 (15 days to PC), hearings 11/26 (subject to change) and 12/14.

3. Pre-application agenda — Reid Turner, Planning Director (one week before filing)

Bring: drawings/mcp-sheet.png (C-2.0), drawings/existing-conditions.png (C-1.0), Sheets A-2.1 / A-2.2 / A-2.3 / A-3.0, docs/02-development-summary.md §§1–4, data/voluntary-conditions.md, the 2025 PC minutes. Record the answers in verify/preapp-notes.md and update FACTS.md the same day.

#	Question (precise)	Why it matters / what we propose	If the answer is unfavourable
Q1	Table 4.1 (R-2) lists the buffer abutting R-1 as 0 ft for "detached single-family dwellings" and 20 ft for "all other allowed dwelling types." Does staff read a fee-simple detached cottage home on a 50' × 100' lot as a detached single-family dwelling (0 ft) or as another dwelling type (20 ft)?	We design to the 20-ft buffer and offer it as condition 4 either way; the reading decides whether the buffer is a requirement or a voluntary offer	None — the plan already provides 20 ft
Q2	§313(1): "Buffer requirements ... supersede these minimum required yards." May the 20-ft buffer coincide with the 20-ft rear yard of each perimeter lot , held in a recorded buffer/conservation easement in favour of the HOA and the City? Our arithmetic is 100 + 35.9 + 100 = 235.9 ft against a 236.0-ft strip width at the first lot row — a 0.1-ft (1-1/4-in) margin, tighter than the 0.5 ft this document reported at 43 lots because the corrected lane tract is 35.9 ft at its narrowest, not 35.4. Note that §313(1) speaks of buffers "established by this article," and Article 3 sets no buffer widths; §318 imports the district buffers by reference	This single reading carries the 50' × 100' lot module and therefore the whole lot yield	Lots become 50' × 82' (4,100 SF > 3,000) with a concurrent \$1005 variance on the 100-ft lot depth and a third, shallower house plan — ask whether the variance can run in the same cycle (ZBA fee \$500)
Q3	Open space — asked and answered; confirm only. Development Regulations §5.9 is "Recreation Areas," and it applies to single-family detached subdivisions of 50 acres or more . At 9.44 acres it is not triggered, so our 33.8 % of common open space (34.3 % of the deeded acreage) is voluntary. Please confirm, and tell us who "the Department" and "the Board of Commissioners" are in §5.9's county-derived text when the case is a city case	Closes an item the package previously carried as unresolved; the 141,089 SF of common tracts in eight named HOA tracts is then an affirmative selling point, not a compliance question	None expected

#	Question (precise)	Why it matters / what we propose	If the answer is unfavourable
Q4	Private lane standard. For an "approved private street" under §319, is the accepted section a 35.9–46.1-ft tract with 22-ft pavement, a 2-ft strip and a 5-ft sidewalk (both sides beyond $u = 530$ ft, where the strip first gives the southwest half-section its full $11 + 2 + 5 = 18$ ft; the tract widens to 60 ft through the amenity block so the 18-ft-deep guest and kiosk bays lie inside it), curb and gutter, 15-mph design speed, no on-street parking? Which department reviews the lane — the City engineer or Gwinnett DOT? And is a sidewalk required on the southwest side for $u < 530$, where 6 of the 14 SW lots front a reach with no walk on their side?	The section sets the lot-depth arithmetic in Q2	A wider public-standard section in a 236-ft strip pushes lot depth below 100 ft → the Q2 fallback
Q5	Table 4.2: culs-de-sac "not permitted except in the most unusual circumstances" , and block widths and lengths not exceeding 600 feet . With a single public frontage and no abutting street at the rear, does a 1,754-ft lane ending in hammerhead (T) turnarounds satisfy the rule? And how does staff measure a block — between street intersections, which is how we report it (594.5 / 570.0 / 580.0 ft , the spacing of the three hammerhead T-intersections at stations 594.5 / 1,164.5 / 1,744.5 ft, every one inside the 600-ft rule and inside the 750-ft IFC dead-end trigger) or between the open-space breaks (the lot-frontage runs, 300 / 500 / 550 ft)?	The strip cannot connect through — Nantucket Dr lots close the NW end. We would rather move the turnaround than argue the measurement	If a circular bulb is required at the terminus, the rear lot count drops by 1–3 from 41 and the terminus green absorbs the bulb
Q6	Stream buffer. Does the City require a field study to establish whether and where a regulated stream begins — the ordinance's spring/seep test or the 25-acre drainage-area point, which lies downstream of this property — before top of bank is located? Who delineates for the City? Every lot, dwelling, driveway, pond and pavement is now screened against all three digitised reaches, not only the on-site one, at the digitised centreline plus a 5.0-ft top-of-bank allowance — so everything is held at ≥ 80 ft for impervious and ≥ 55 ft for disturbance: nearest lot 80.6 ft , nearest dwelling 106.2 ft , nearest driveway 102.6 ft , pond top of bank 80.2 ft , lane pavement 80.5 ft (data/layout.json metrics.stream_clearances_ft)	Condition 9 promises the field study first and the top-of-bank line second. The lot that failed the screen has already been dropped: the slot at u 1,230–1,280 on the southwest side is unlotted and becomes a 5,000-SF stream-setback open-space tract, which is one of the two lots that took the yield from 43 to 41	If the field-delineated top of bank falls farther into the site than the 5.0-ft allowance assumes, the next lots at risk are those nearest the 80.6-ft clearance; the yield stays inside the 40–47 range the conditions offer
Q7	Sewer. Phase 1 uses the existing 8-in main on the property and extends nothing. For Phase 2, a private pump station, private force main or private gravity sewer is not available to us : Gwinnett DWR's <i>Standard Policy for Private Developments: Condominiums, Townhomes and Subdivisions</i> (rev. 9/2018) allows those "only ... for commercial properties under single ownership within a development," a county-owned station fails <i>Developer Pump Station Standards</i> (WSR-24) §1.3.1(A) because gravity is 178 ft away rather than "more than 5,000 feet down gradient," and DWR publishes no grinder-pump or low-pressure-sewer standard at all. The only viable Phase-2 route is a ≈ 178-ft off-site gravity tie to the existing Legends at Parkview main. Does the City read the Suburban-Low policy "Do not further extend sewer in this area" as barring a 178-ft connection to an existing main in the same basin?	This is now a policy-interpretation question, not an engineering choice. Condition 16 binds Phase 2 to a gravity connection only, bars every private pumping alternative, and provides that Phase 2 is not built at all if the easement or the DWR certification is not obtained	Phase 1 (24 lots) still stands alone. If the answer is that the policy bars the tie, the rear 17 lots are not built and the Phase-2 area stays wooded or is re-platted as open space — say so at the hearing rather than discovering it later
Q8	HOPA 55+ as a zoning condition. Will the City accept and enforce a condition that the community operate as housing for older persons under 42 U.S.C. §3607(b)(2)(C) / 24 CFR §100.300–.308 — at 41 units, 33 units must have a resident 55 or older ($0.80 \times 41 = 32.8$; §100.305(g) rounds the fraction in), leaving a cushion of 8 — with the declaration recorded before the first building permit? Does staff want the declaration text with the application, and does the 0-student school finding depend on it?	Condition 1 is the basis for the traffic (LUC 251) and school (0 students) findings	If the City will not enforce an age condition, the covenant still binds the HOA; traffic would be restated at LUC 210 (≈ 40 PM trips) and the school letter would count students
Q9	Fire marshal referral. Will the City refer the plan to the Gwinnett County Fire Marshal at the rezoning stage or only at LDP? We ask for special approval of a 1,754-ft dead end at 22-ft pavement under IFC 2024 as modified by Ga. Comp. R. & Regs. 120-3-3-.04 , Appendix D103.4 (over 750 ft), and we note that Georgia's D107.1 sets the two-access trigger at more than 120 dwelling units , so at 41 units it is not engaged. NFPA 13D sprinklers in every home (condition 5) are offered voluntarily . Can we get the Fire Marshal's concurrence letter before the PC hearing?	A written concurrence removes the largest technical objection at the hearing, and the sprinkler offer is a negotiating chip for the 22-ft width, not a required mitigation	Georgia's D107.1 Exception 3 lists alternatives (increased widths, enhanced radii, D103.6 signage, an additional non-vehicular all-weather access, or the Fire Code Official's written statement), and O.C.G.A. §25-2-12(e)(4) provides a waiver with a 30-day deemed-approval clock
Q10	Disposition of RZ-2025-01 and the §1003-11 bar. Please confirm in writing that RZ-2025-01 was tabled on 2025-08-28 for a defective legal ad, was never heard and was not denied , so the 12-month refiling bar does not apply — and whether it should be formally withdrawn under §1003-10 before this application is accepted	A denial would bar filing for 12 months; a dormant application could be treated as pending	If staff deems it pending, withdraw it in writing before filing (permitted until the Council ad — none ever ran for Council)

#	Question (precise)	Why it matters / what we propose	If the answer is unfavourable
Q11	Which Future Development class do PINs R6123 033 / 015 / 014 / 162 fall in — Suburban-Low or Suburban-Medium? The Future Development Map (p. 52) is a raster with no parcel labels. Single-family cottages and single-family are "Appropriate" in both, but the sewer policies are opposite: Suburban-Low says "Do not further extend sewer in this area" and Suburban-Medium says "Coordinate the expansion of sewer infrastructure to fill gaps"	If any part of the assemblage is Suburban-Medium, the Q7 objection weakens materially	Ask for the class in writing either way
Q12	When will the Comprehensive Plan Amendment be adopted, and will it be adopted before this case is heard? Transmittal was approved 2026-07-13; the Regional Commission's report window closed about 2026-08-23; adoption is available 9/14, 10/12 or 11/9/2026	The application is written to stand either way, but the hearing strategy differs	Plan for the Amendment to be adopted law at the Council hearing
Q13	Does the City adopt Gwinnett DOT's TIS Guidelines for a city zoning case, and what study level does it want? The Guidelines are written for unincorporated Gwinnett under UDO §270-20.8; we are preparing a Level 1 study and commissioning new 24-hour counts because the 2017 GDOT count fails the two-year rule	Determines whether the study is filed with the application or later	If the City wants more than Level 1, the study moves onto the critical path and Cycle B becomes the realistic filing
Q14	Does the pending UDO / zoning-code rewrite change R-2 cottage standards, and will staff ask this case to wait for it? The code diagnostic was before a steering committee on 2026-07-07 and no draft text exists	A request to wait would cost a cycle or more	Ask for it in writing, and ask whether a pending-ordinance doctrine would be applied
Q15	Fee and form mechanics. Does the FY2026-2027 Fee Schedule (Resolution No. 2026-08, adopted 2026-08-31) apply to a 9/25 or 10/30 filing, and is \$1,250 the amount to write the check for? Is the clubhouse entered on the form as "1 accessory community building" under NON-RESIDENTIAL? Does the Director want the site-plan PDF before the filing date for a completeness check under §1003-3.1?	The adopted schedule is days old and not yet posted; the rezoning tier is unchanged at \$1,250 in both schedules, but the counter is the authority	Bring a blank check and the fee schedule printout

Also ask, not counted among the fifteen: the Development Regulations text for sidewalks and the landscape strip on a collector; whether the City requests the GCPS letter and the DWR certification itself; and the current sign or notice fee, if any.

4. What changed versus the 2025 filing (RZ-2025-01)

Topic	RZ-2025-01 (filed July 2025; tabled 2025-08-28)	This application (2026)	Source
Program	"Arcado Springs Mixed-Use Community": 104 condominium units (~1,700 SF each) + 13 commercial buildings, 24,400 SF retail/service/office	41 one-story detached cottage homes on fee-simple 50'-0" × 100'-0" lots (Plan A 1,406 SF / Plan B 1,492 SF conditioned; 21 A and 20 B), 55+ HOPA, clubhouse 2,400 SF, 2 pickleball courts; no commercial	4814_development-summary-report.pdf; data/plans.json
Zoning requested	R-1 → MU (Mixed-Use, §736)	R-1 → R-2 (§402); use "Single-family (cluster-cottage, creative lot configuration)" = P in §602 (no SUP)	ordinance excerpts
Comprehensive Plan	MU on an "Established Residential" parcel; the 2026 Amendment's Suburban-Low chart lists Mixed Use, Multifamily and Retail as Not Appropriate	Single-family cottages and single-family = Appropriate in Suburban-Low (and in Suburban-Medium); Established Residential → R-1, R-2 per §203. The Amendment was transmitted to DCA 2026-07-13 and is not yet adopted	2026 CPA p. 73; 2024 plan printed p. 35
Acreage	"Approximately 9.00-acre" total	9.44 ac deeded (0.99 + 2.00 + 2.00 + 4.45); 9.577 ac by GIS; the RLS survey governs	Gwinnett digest; FACTS.md §1
Legal description	Narrative: "extends eastward for approximately 1,250 linear feet"; sides labelled north = Village Green Ct, south = King David Dr; no bearings or distances	GIS-derived metes and bounds with per-segment bearings and distances (the strip runs N 28°43' W, 1,721.7–1,757.0 ft deep, 234.7–246.4 ft wide; King David Manor on the NE , Legends at Parkview on the SW) — DRAFT until the RLS seals	4818_legal-description_9.00ac.pdf; docs/05-legal-description-DRAFT.md
Site plan basis	MCP dated 8/17/2025 on an unsealed boundary; the program failed the Table 4.6 MU mix as drawn	Generator on the GIS parcel fabric in surveyable local coordinates (1 unit = 1 ft); every lot tested for boundary, buffer, stream setback, frontage ≥ 30 ft, ≥ 3,000 SF and ≥ 50 ft wide; to be re-drawn on the RLS base and sealed by a PE	4813_master-concept-plan.pdf; data/layout.json

Topic	RZ-2025-01 (filed July 2025; tabled 2025-08-28)	This application (2026)	Source
Use mix	Residential + retail/service/office; bike lanes; park-to-park easements	Residential only; greens, creek woods, ponds, the stream-setback tract and the amenity tract = 141,089 SF = 3.24 ac = 33.8 % of the GIS area / 34.3 % of the deeded area in eight common tracts that sum exactly to that figure, of which 13,960 SF is building or pavement (clubhouse roof, court pads, mail kiosk, amenity walks and the hammerhead legs inside the greens), leaving 127,129 SF = 30.5 % green-only	data/layout.json open_space_summary; data/dev-summary-derived.json
Density	104 DU / 9.44 ac ≈ 11.0 du/ac (MU)	4.34 du/ac on 9.44 ac deeded (4.28 on the 9.577-ac GIS acreage) against R-2's maximum 8 and R-1's 4 — 41 ÷ 9.44	layout.json metrics.density_du_ac_deeded/_gis
Buffers	Not stated in the LOI	20-ft undisturbed buffer on all three shared lines , in a recorded easement under §313(1), with one reduction requested — the southwest entrance curb return, where the buffer is reduced by 2'-0" to a residual 18'-0" over a plan area of 0.1 SF . Plus 41,000 SF of buffer easement inside the 41 lots, not counted in the open space above	data/layout.json metrics.sw_curb_return_buffer_encroachment_ft/_sf; data/voluntary-conditions.md 4; docs/03 §5
Traffic	No TIS filed; order-of-magnitude ≈ 2,030 daily / ≈ 214 PM (LUC 220 + 822)	ITE 11th ed. LUC 251: 152–177 daily, 10.7–16.0 PM peak (12.3 at the FACTS rate of 0.30 per unit; 8.2–13.9 AM) ≈ 5.8 % of the 2025 concept's 213.8 PM trips — and a Level 1 TIS is required and will be filed , with new 24-hour counts	data/dev-summary-derived.json trips_251_*; docs/08 Memo A
Schools	GCPS letter (July 2025): +57 students (Camp Creek ES +25, Trickum MS +13, Parkview HS +19) into a cluster with two over-capacity schools	0 students (HOPA covenant, 33 of 41 units must house a resident 55 or older); request a fresh GCPS letter	4817_school-impact-analysis.pdf; Memo F
Impervious / stormwater	Hydrology sheet: 235,388 SF impervious (55.4 %), 3 basins, 74,167 cf detention by the 10,000-cf/ac rule	184,148 SF = 44.1 % of the GIS area / 44.8 % of the deeded area — a 21.8 % reduction on the 2025 sheet's 235,388 SF; disturbed area 7.54 ac (it had been sized on 4.4, which is why the basins were enlarged and a lot was lost); required detention 75,410 cf on the 10,000-cf-per-disturbed-acre screening rate with the runoff-reduction volume taken as zero, or 59,858 cf if the 15,552-cf RRV credit is allowed; 2 dry ponds hold 77,863 cf as drawn at 6-ft depth (33,129 + 44,734) = 103 % of the no-credit requirement ; WQv 18,659 cf ; the PE's GCSWMM routing governs	data/layout.json stormwater_impervious_summary; docs/02 §5.3
Sewer	Capacity request filed 2025 (stale-dated)	Phase 1 gravity to the existing on-site main — no extension ; Phase 2 gravity only , a ≈ 478-ft line with ≈ 178 ft off site, no private pump station or grinder pumps (DWR policy), DWR certification to be re-requested for 10,670 gpd (41 × 250 gpd + 420 gpd clubhouse; Phase 1 6,420 gpd, Phase 2 4,250 gpd)	data/dev-summary-derived.json adf_total_gpd; Memo B; condition 16
Conditions offered	None recorded in the LOI	18 voluntary conditions , including 55+ HOPA (33 of 41 units), a 47-unit cap, one story with a 24-ft ridge, a 20-ft buffer with one disclosed 2-ft reduction, gravity-only sewer, a Level 1 TIS with new counts, and NFPA 13D sprinklers offered voluntarily — all conformed to the 41-lot layout on 2026-09-03 (item 14)	data/voluntary-conditions.md
Notice / process	Tabled for a defective Gwinnett Daily Post ad and sign; never heard; no denial → no §1003-11 bar (VERIFY in writing)	Notice audit N1–N9 (§2.2); attend both hearings; no withdrawal after the Council ad	PC minutes 2025-08-28

5. Budget roll-up and 12-week critical path

5.1 Budget (planning allowances — VERIFY each with a quote)

Line	Low	High	Lead time	Note
Application fee — Rezoning, 5.0–9.9 ac	\$1,250	\$1,250	at filing	FY2026-2027 Fee Schedule, Resolution No. 2026-08, adopted 2026-08-31 (Ord. 2023-603 §1003-4.1: the fee is "as established by resolution of the City Council"). The tier is unchanged from FY25-26. Non-refundable; check payable to City of Lilburn. Confirm at the counter — the schedule is days old and not yet posted
Boundary survey + sealed metes-and-bounds legal description (GA RLS); add the sewer-invert survey and Arcade Rd sight distance to the scope	\$8,000	\$18,000	3–6 wks	Planning allowance — VERIFY with two RLS quotes; the invert and sight-distance add-ons may price separately

Line	Low	High	Lead time	Note
Master Concept Plan sealed by a Georgia PE (site plan on the survey base; preliminary lane, sewer and pond design; DWR capacity request)	\$8,000	\$25,000	2–4 wks after the survey	Planning allowance — scope is the rezoning concept plan only, not LDP construction drawings
Level 1 Traffic Impact Study (required — Gwinnett DOT 2023 Guidelines; the exemption is 7 lots, not 21 trips)	\$6,000	\$15,000	3–5 wks	No longer a \$0 low case. A Level 1 study is document-only — no operations analysis, no turn-lane warrant — so it should price at the bottom of the allowance. VERIFY with a written proposal
New 24-hour traffic counts on Arcado Rd (GCDOT-approved contractor)	VERIFY	VERIFY	schedule in week 1	The 2017 GDOT count (station 135-6689, 14,800 AADT) fails the Guidelines' two-year currency rule. Not priced in package data — obtain a quote; excluded from the totals below
Attorney — purchase contract or owner consent for R6123 014 and 162, LOI and conditions review, HOPA covenant outline, notice audit, hearing representation	\$7,500	\$25,000	start week 1; through Council	Planning allowance — VERIFY with an engagement letter; hearing representation drives the high end
Architectural elevations and renderings (Instruction item 11)	\$3,000	\$15,000	2–4 wks	Sheets A-1.1, A-1.2, A-2.1, A-2.2, A-2.3, A-3.0 , A-4.0 and A-4.1 and twelve perspectives are already in the package as owner-prepared illustrative work, and item 11a is issued, so the architect's scope is now review and seal only — re-quote on that reduced scope, which should price at or below the bottom of the range
Total of priced lines	\$33,750	\$99,250		Low = survey, PE and architect at the bottom of their ranges with a bottom-of-range Level 1 study; high = every line at the top with full attorney representation

Contingent land-use fees (FY2026-2027 schedule) — not in the totals, but budget for them:

Contingency	Fee	When it bites
Applicant-requested tabling after the public hearing is advertised	50 % of the original application fee = \$625	Exactly the risk in the Cycle A / Cycle B decision above — if the survey slips after the ad runs, tabling is not free
Waiver of Development Regulations	\$300	Private-street or lane-width relief (Q4)
Modification of Development Regulations	\$200 per modification requested	Same
ZBA public hearing (variance)	\$500 , plus \$25 for each additional variance on the same property	The \$1005 lot-depth variance fallback (Q2)

Not priced and excluded from the totals (VERIFY): the Gwinnett DWR sewer capacity certification (no fee is stated on the Rev. 07/2023 form — confirm with DWR, 678-376-7026); Gwinnett DOT access and driveway-permit pre-coordination (no fee expected at the rezoning stage; the permit fee comes later); the GCPS school-impact letter (requested by the City; no applicant fee expected); a notary for four certifications; stream top-of-bank field delineation (may fold into the survey or a separate environmental scope); geotechnical rock probes at the two pond sites, which should be ordered before the pond volumes are fixed; printing (one full-size site plan, 11 × 17 reductions, courtesy 8.5 × 11 sets, tear sheets); and the community record (website update, neighbour-meeting notices, sign photographs). Later-stage costs (\$1106 site & design review, LDP at \$40/disturbed ac + NPDES \$80/ac on **7.54 disturbed acres**, the DOT driveway permit, the tree survey, construction documents) are outside this rezoning budget.

5.2 Twelve-week critical path (week 1 = Monday 08/31; Cycle A with the Cycle-B branch)

Wk	Dates	Tasks	Owner	Gate / deliverable
1	08/31–09/04	Done by Thu 09/03: the concept plan re-issued at 41 lots (turnarounds at stations 594.5 / 1,164.5 / 1,744.5 ft, 6-ft pond basins holding 77,863 cf, real house footprints, entrance curb returns, all three stream reaches screened) with <code>tools/dev_summary_calcs.py</code> and <code>tools/hoa_budget.py</code> re-run behind it; Sheet A-3.0 drawn , which closes item 11a; Sheet C-1.0 issued; all twelve perspectives generated; docs/00, docs/09 and docs/13 written; the voluntary conditions conformed; the binder rebuilt. Still to do this week: order the RLS survey (boundary, acreage, the 234.7–246.4-ft width at every lot row, sewer inverts at the u ≈ 272 and Legends manholes, stream top of bank, Arcado Rd sight distance) — the 3–6-week clock has not started, and it is the critical path; engage the attorney (014/162 contract or written consent; Declaration drafting from docs/13); request the pre-application date; brief the PE (concept-plan seal scope, DWR capacity request on the "Pre-Rezoning" type) and the traffic engineer (Level 1 study + order the 24-hour counts)	Owner	Signed RLS proposal with a delivery date; pre-app booked; counts scheduled
2	09/07–09/11	(Labor Day 09/07.) Pre-application conference — the fifteen questions of §3; record answers in <code>verify/preapp-notes.md</code> ; update <code>FACTS.md</code> . PE submits the DWR sewer capacity request; PE/RLS request the DOT access determination. Attorney reviews the LOI, the 18 conditions and the criteria responses. Ask the Director to request the GCPS letter and to confirm the Oct–Dec hearing calendar	Owner, PE, attorney	Pre-app notes; DWR and DOT requests logged; hearing dates confirmed

Wk	Dates	Tasks	Owner	Gate / deliverable
3	09/14–09/18	Survey fieldwork complete (fast case). Re-run tools/siteplan.py on the survey rings and the pre-application answers, then re-run tools/dev_summary_calcs.py, tools/hoa_budget.py and all four checkers behind it — if the surveyed width moves, the lot count moves again and every document must be re-conformed ; PE drafts the sealed concept plan; architect reviews and seals A-1.1 through A-4.1; traffic engineer delivers the Level 1 study. Attorney redlines docs/13 (\$10 checklist). Schedule the notary sitting with the Mendez / Roblero owners	RLS, PE, architect, traffic engineer	Draft sealed concept plan; sealed elevation set; Level 1 study
4	09/21–09/25	Go/no-go 09/18 : sealed survey and legal description in hand, 014/162 consent executed, concept plan sealed, elevation set sealed (item 11a is already issued), Level 1 study delivered, all four checkers in item 26 passing , PDF binder rebuilt → assemble the package (items 1–26), notarize (4 certifications), re-pull the adjoining-owner list, print one full-size copy and the reductions, e-mail the PDF. File Fri 09/25/2026	Owner	Filed application, receipt, case number
5	09/28–10/02	N1 review the draft ad text before 09/29; N2–N3 tear sheets and day counts on 09/29. Update the public website from the 2025 MU concept to this proposal. Set neighbour-meeting dates	Owner, attorney	Notice folder started
6	10/05–10/09	Conflict-of-interest disclosure by 10/05 . N4–N6 on 10/07: photograph the sign, verify the mailing list, count 15 days. Neighbour meeting 1 (King David Manor). Follow up the DWR and DOT responses	Owner	COI filed; sign photograph; mailing confirmation
7	10/12–10/16	Answer staff-report questions within 48 hours; deliver the Fire Marshal concurrence request; neighbour meeting 2 (Legends at Parkview, Nantucket). N7 agenda check. Weekly sign photograph	Owner, PE	Responses to staff; meeting sign-in sheets
8	10/19–10/23	PC work session 7:00 PM and Planning Commission hearing Thu 10/22/2026 7:30 PM (confirm the date) — applicant present with counsel; present the conditions, traffic (12.3 PM trips and the Level 1 study), schools (0), buffers (20 ft with the one 2-ft entrance reduction to 18'-0") and sewer (the existing main, no extension in Phase 1)	Owner, attorney	PC recommendation
9	10/26–10/30	Revise the conditions or plan per the PC recommendation (no unit-count increase). Cycle-B branch : if the go/no-go failed in week 4, this is the Fri 10/30/2026 filing with the completed survey (the 6-week case delivers ≈ 10/11)	Owner, PE	Revised condition set; or the Cycle-B filing
10	11/02–11/06	Council work-session materials; re-photograph the sign (still ≥ 15 days before Council — posted 10/07 = 33 days). Cycle B: N1–N3 on 11/03	Owner	Council packet
11	11/09–11/13	City Council hearing Mon 11/09/2026 7:30 PM (confirm the date) — decision. Cycle B: COI by 11/09; N4–N6 on 11/11	Owner, attorney	Ordinance / conditions of approval
12	11/16–11/20	Record the conditions; order the LDP-stage work (geotechnical rock probes at both pond sites, tree survey, stream delineation if not in the survey, DWR route design); instruct counsel to draft the Declaration, Articles, Bylaws and HOPA Policies from docs/13. Cycle B: PC 11/26 (subject to change) → Council 12/14 (weeks 13–15)	Owner	Post-approval task list

Critical path: RLS survey (3–6 wks) → PE re-draw and seal (2–4 wks) → filing. At the fast end (3 + ½ wk) Cycle A holds; at the slow end (6 + 2 wks) only Cycle B does. **As of 2026-09-03 the survey has not been ordered, so the fast end already means the whole 3-week case with no slack.** The Level 1 traffic study (3–5 wks, counts first) runs alongside and must finish before the go/no-go; item 11a is off the path, having been issued on 2026-09-03; the attorney, DWR, DOT and GCPS items remain off it.

Statements of consistency in this document are preliminary: each "appears consistent with" finding in the package rests on GIS geometry, USGS topography, ordinance excerpts and the 2026-09-03 external re-verification, and is to be confirmed by the sealed survey, the PE's plan and the Planning Director's answers at the pre-application conference.

Disclaimer: This is an AI-generated analysis for preliminary planning purposes. All findings must be verified by a licensed professional before use in design, permitting, or regulatory submissions.